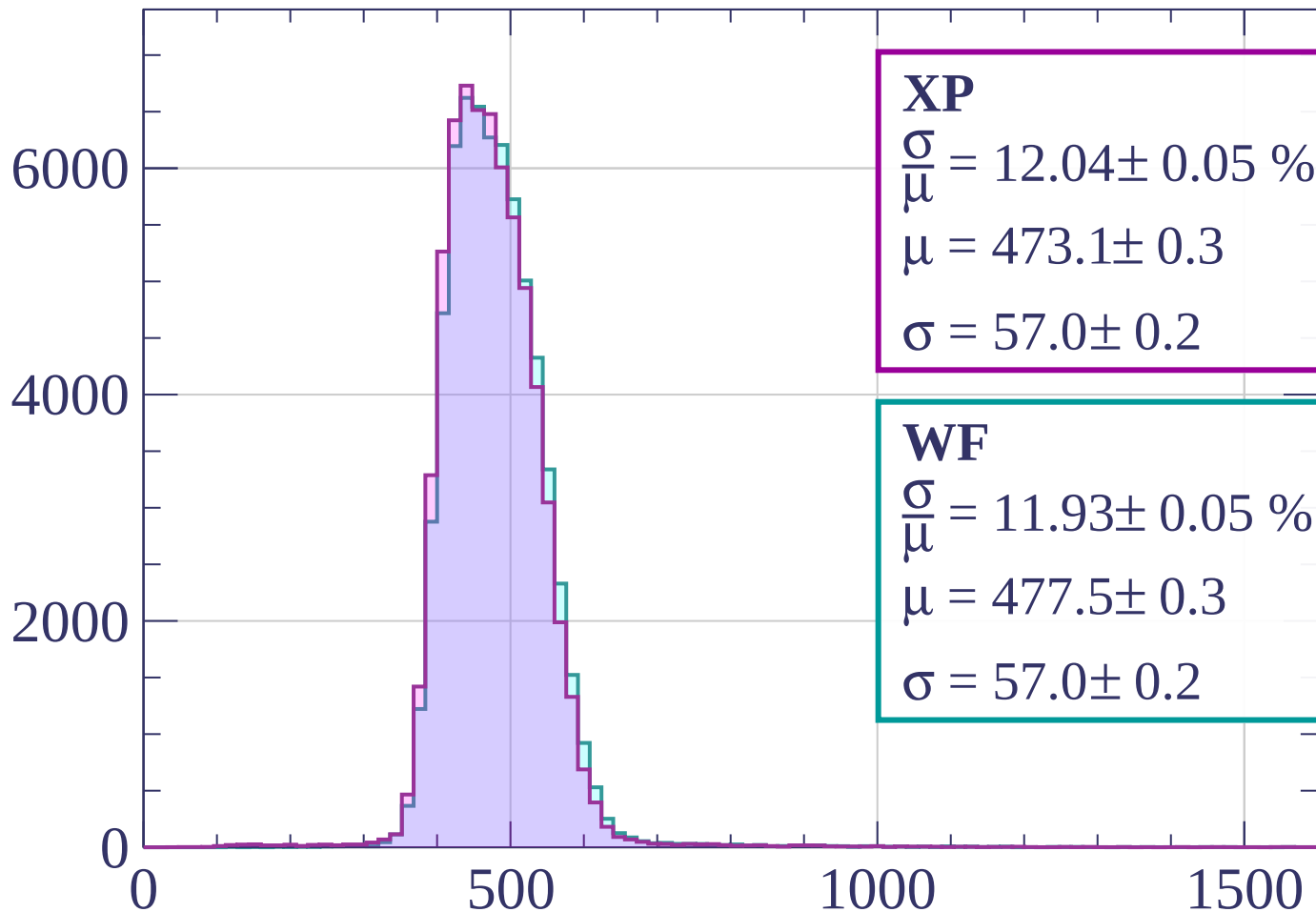


Count



XP

$$\frac{\sigma}{\mu} = 12.04 \pm 0.05 \%$$

$$\mu = 473.1 \pm 0.3$$

$$\sigma = 57.0 \pm 0.2$$

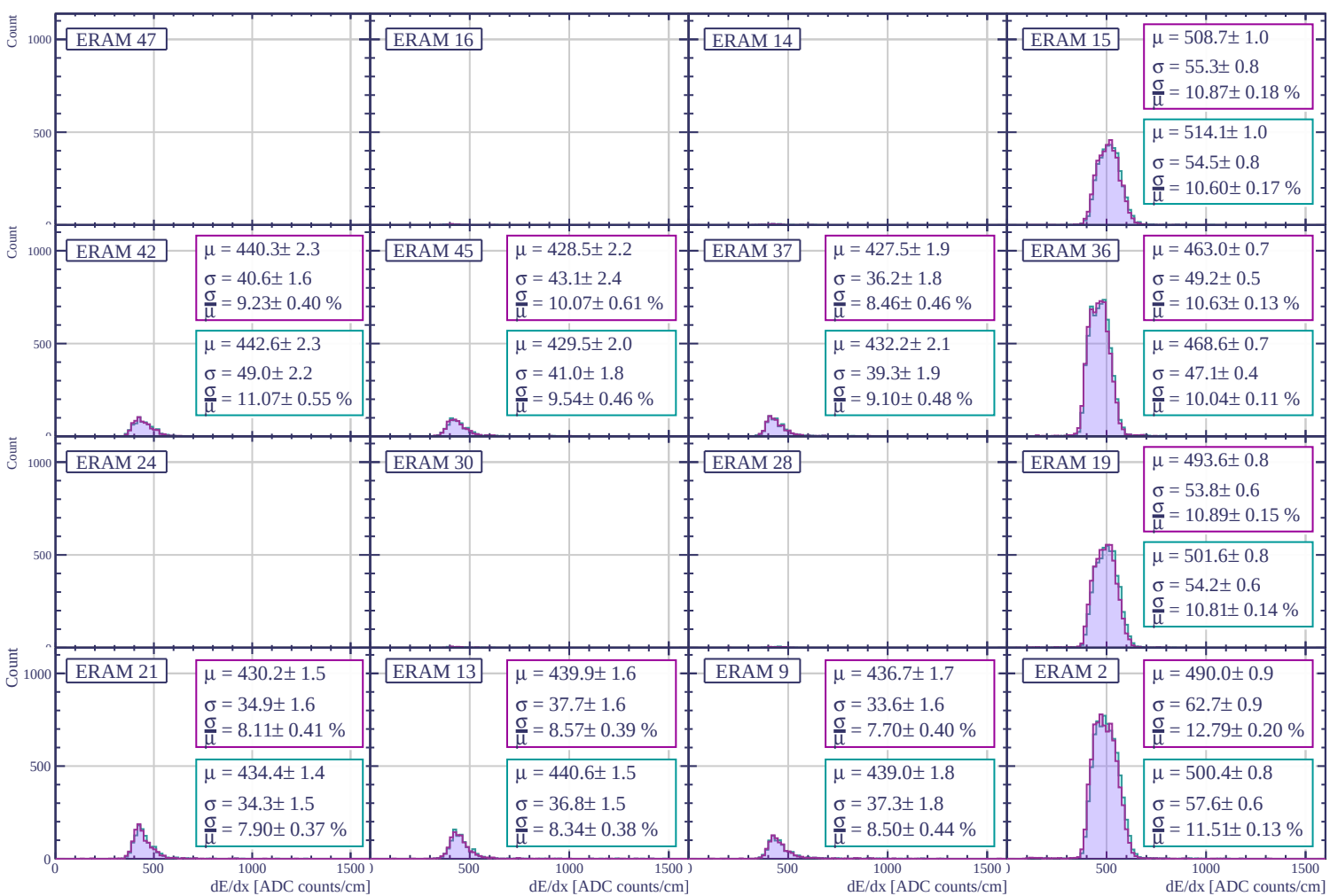
WF

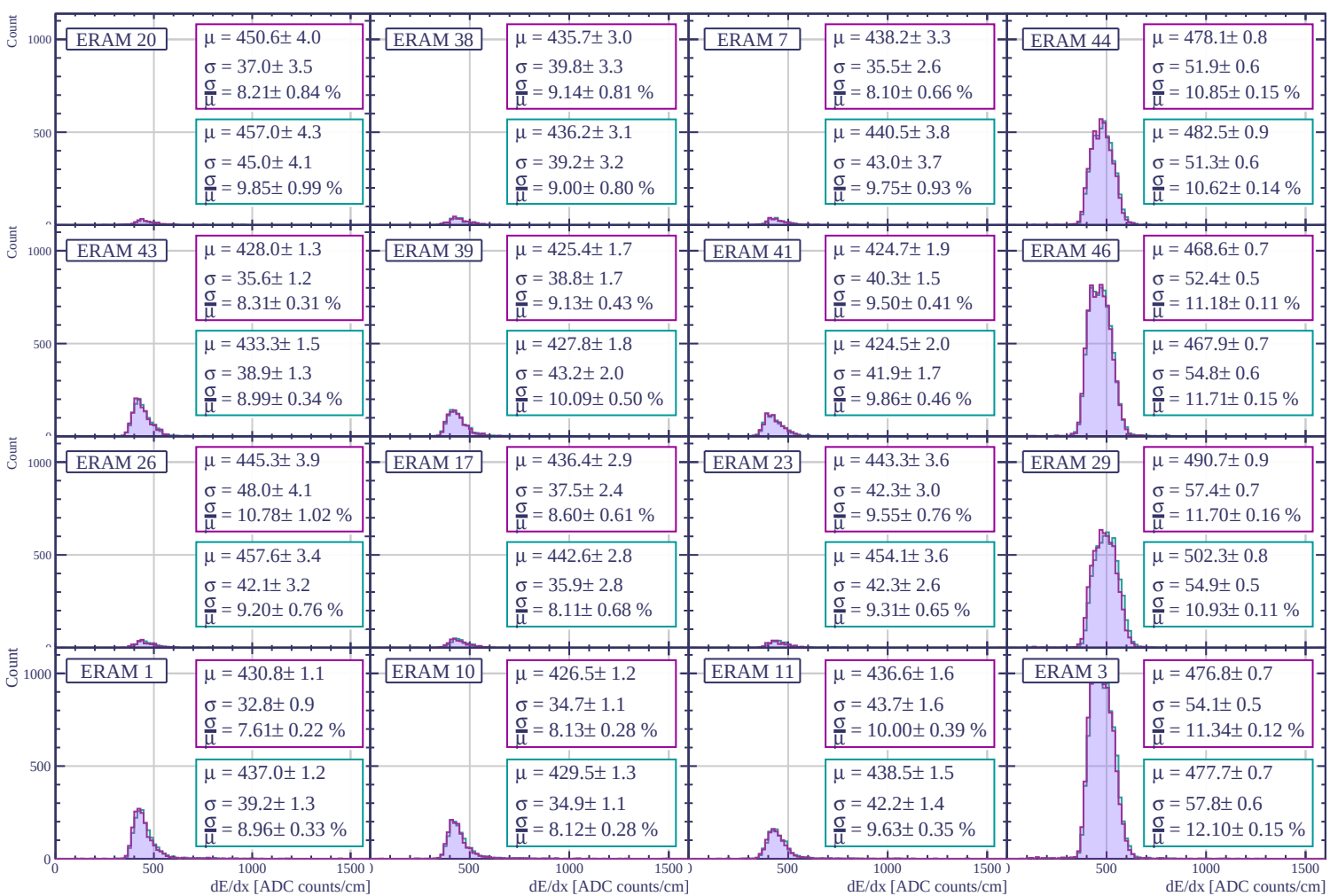
$$\frac{\sigma}{\mu} = 11.93 \pm 0.05 \%$$

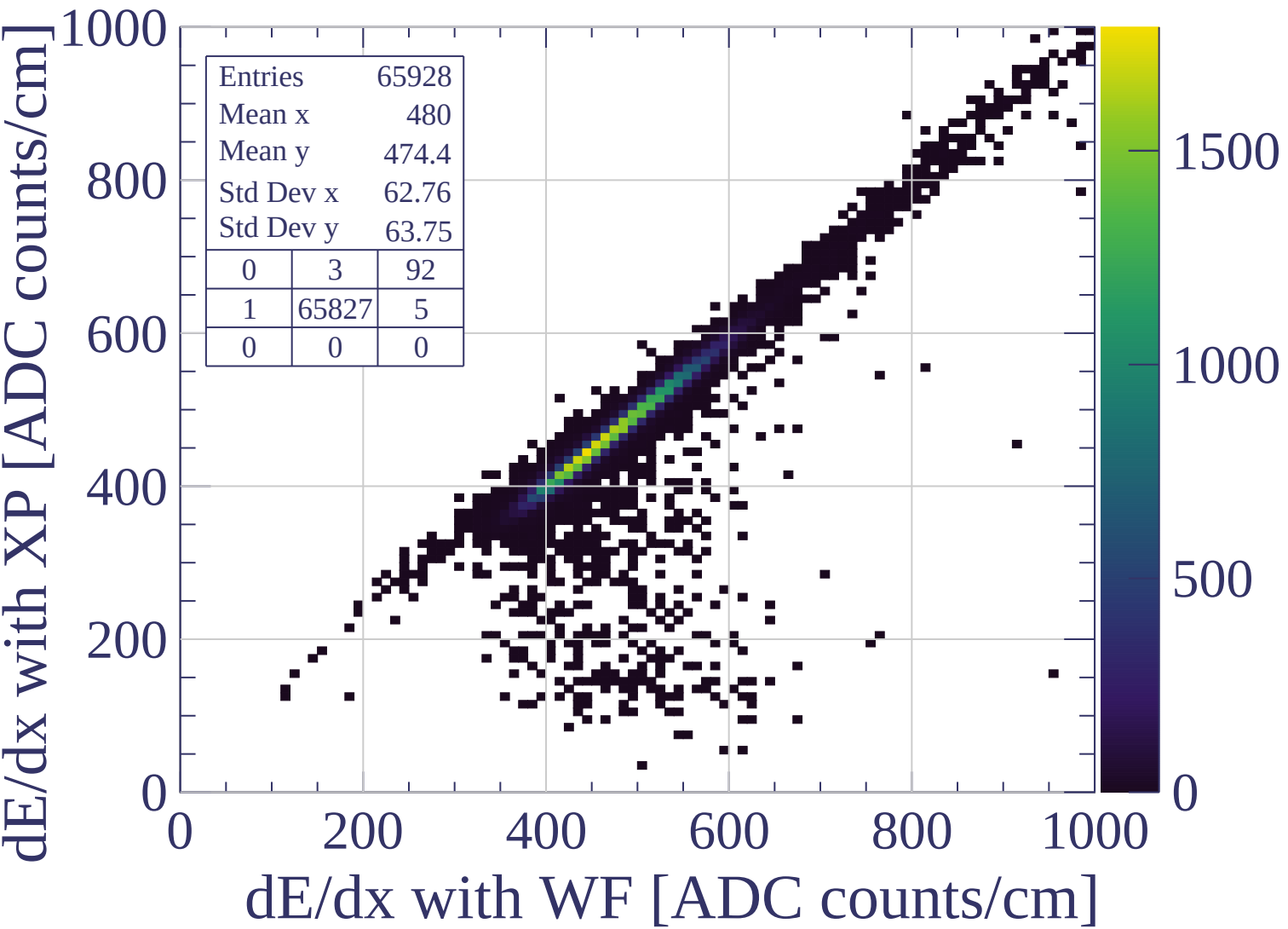
$$\mu = 477.5 \pm 0.3$$

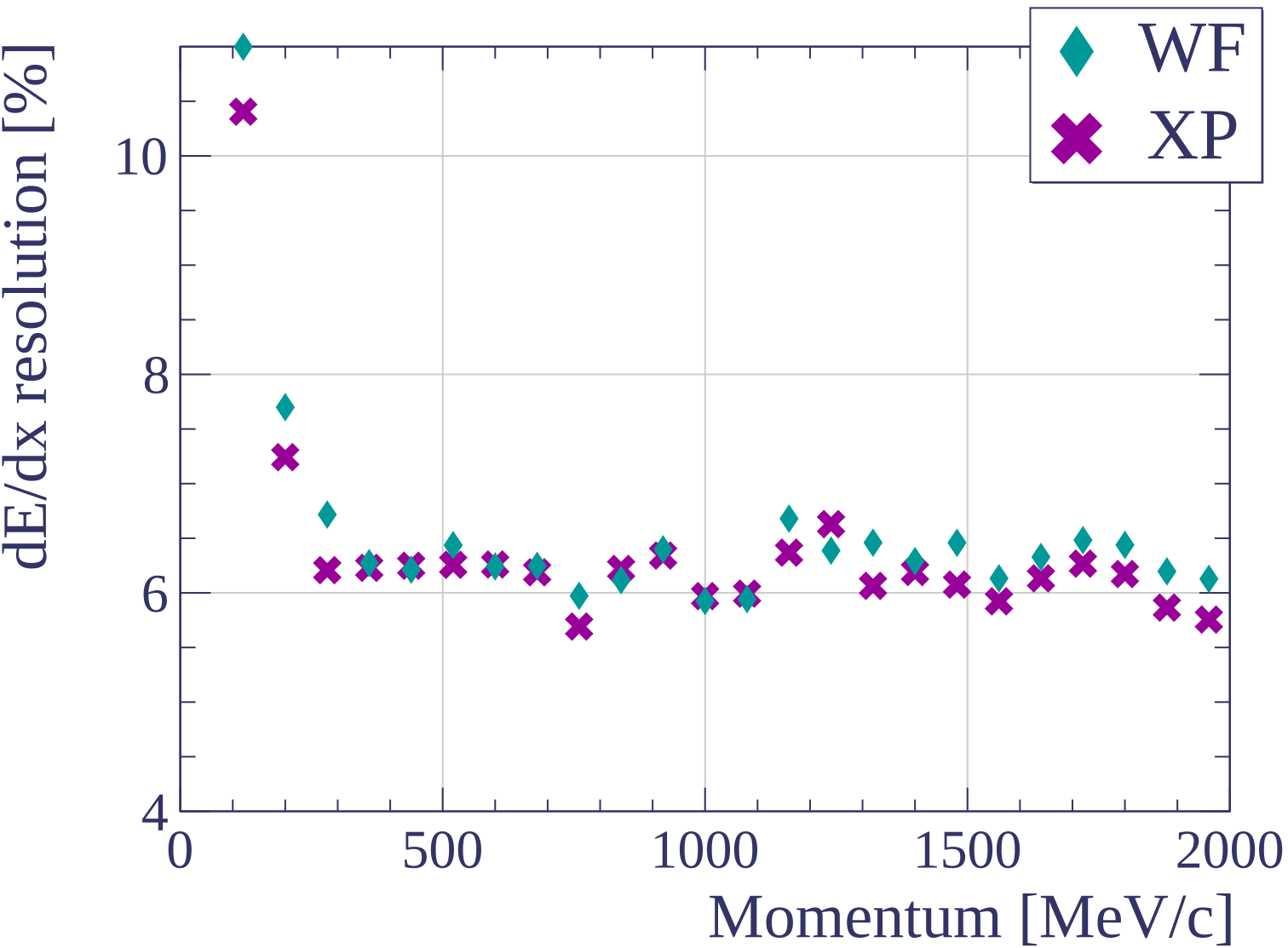
$$\sigma = 57.0 \pm 0.2$$

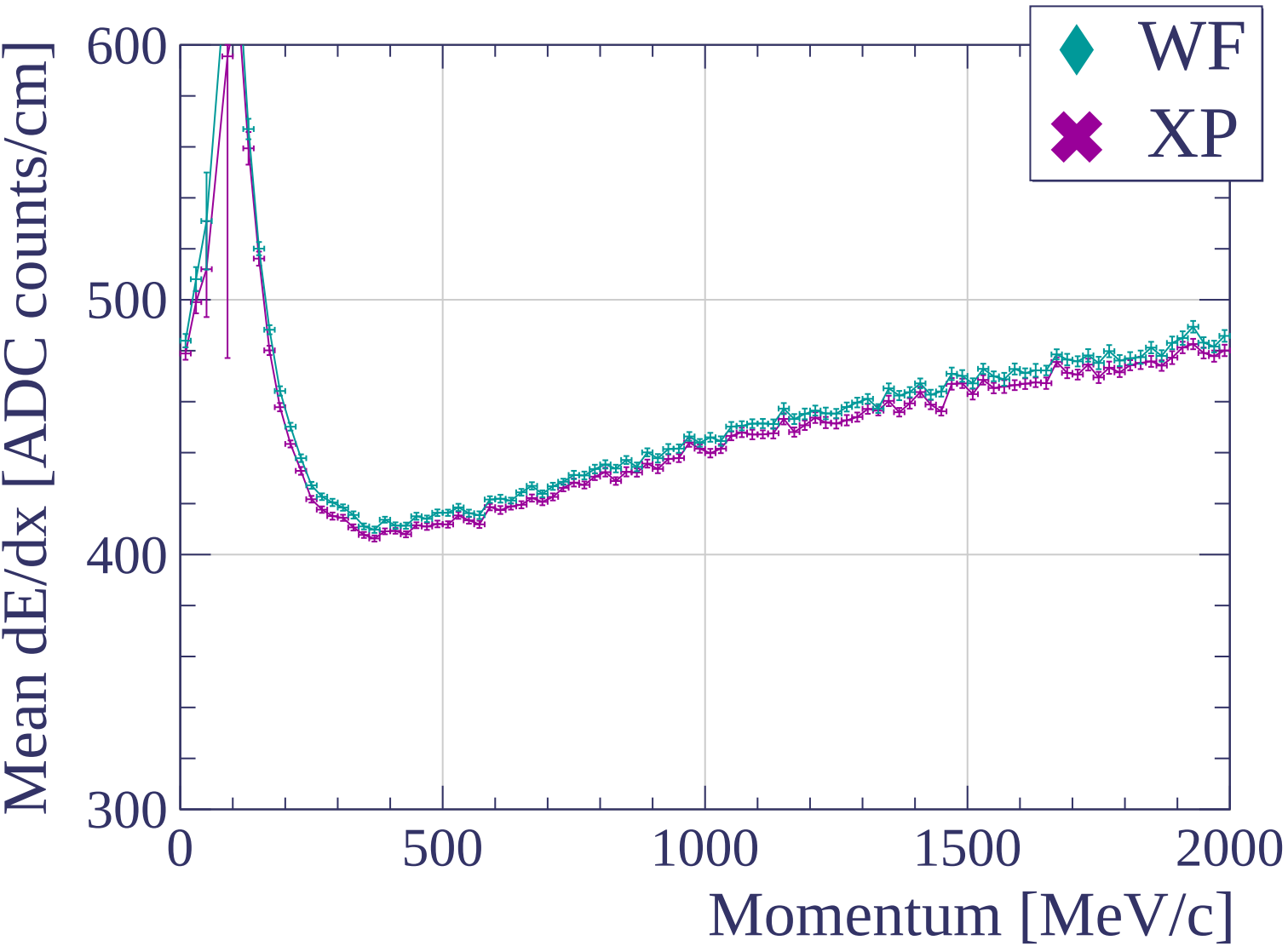
dE/dx [ADC counts/cm]

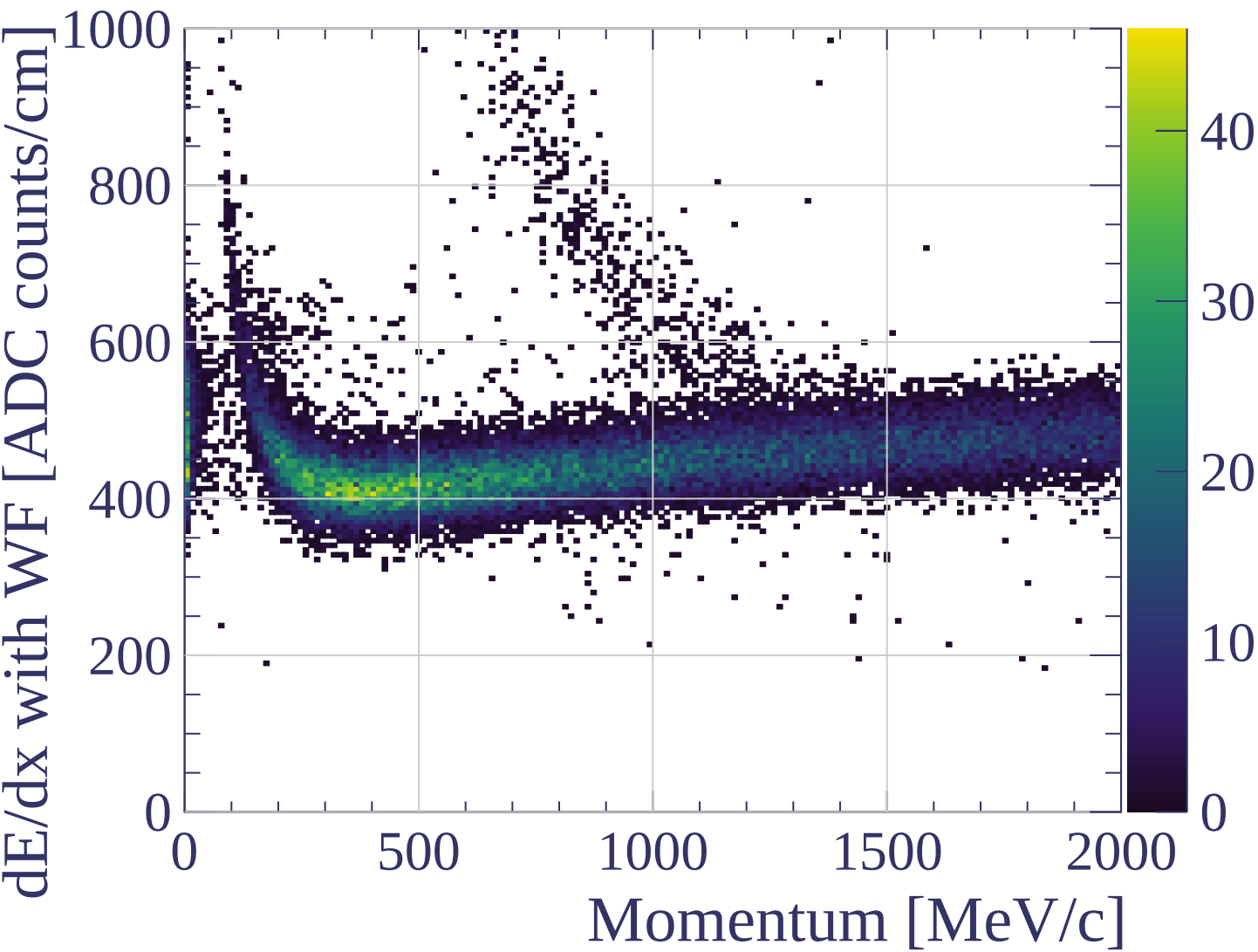


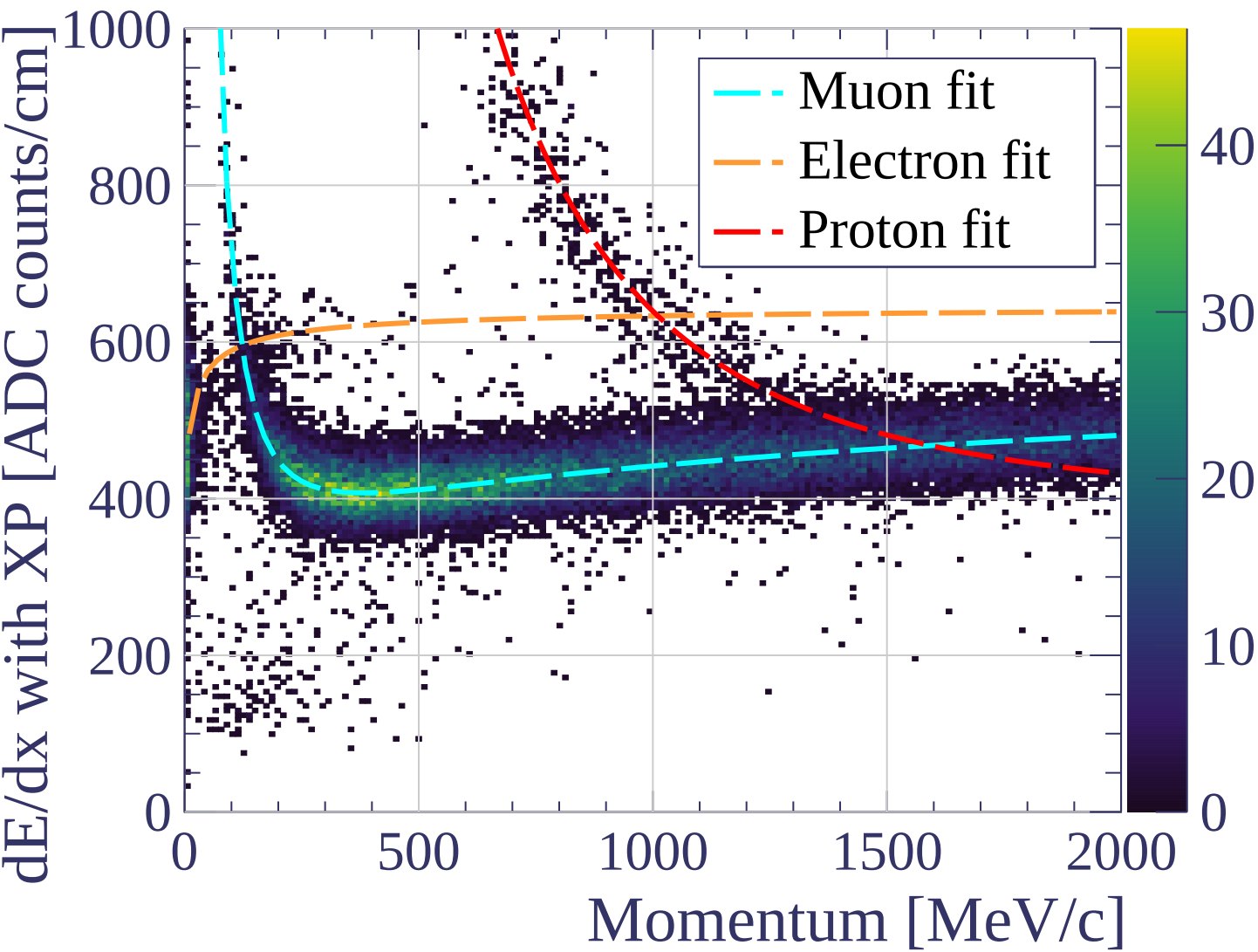




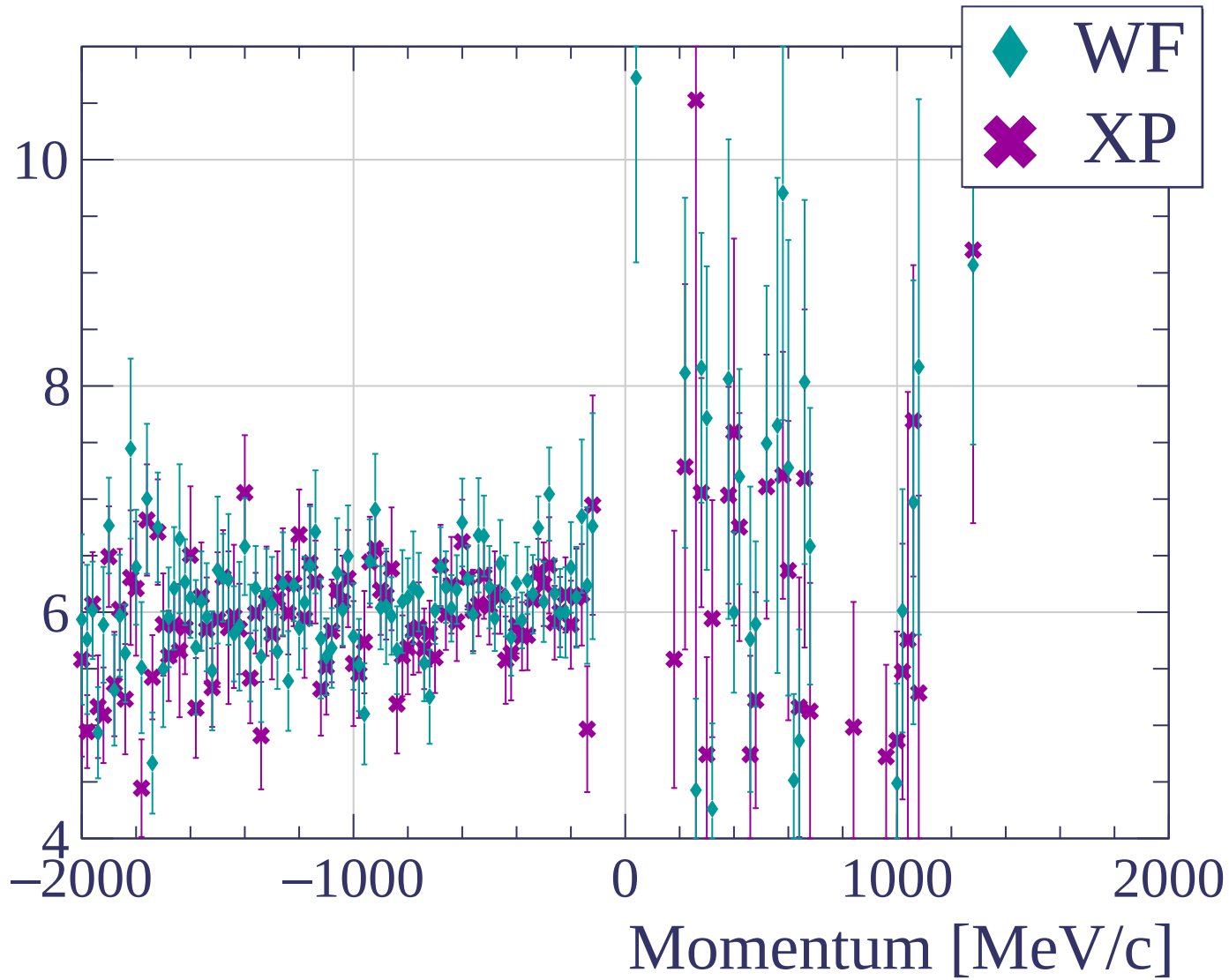


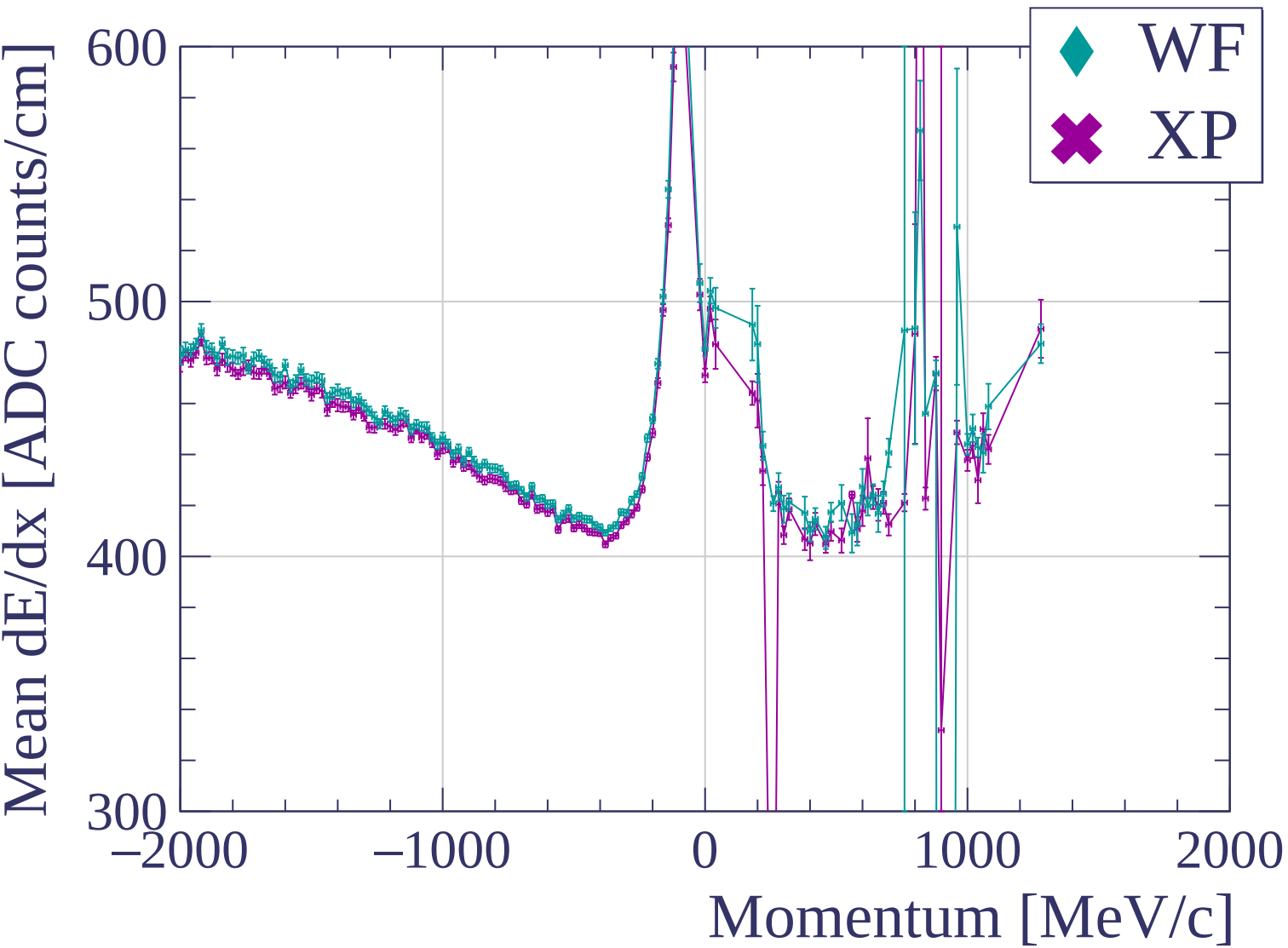


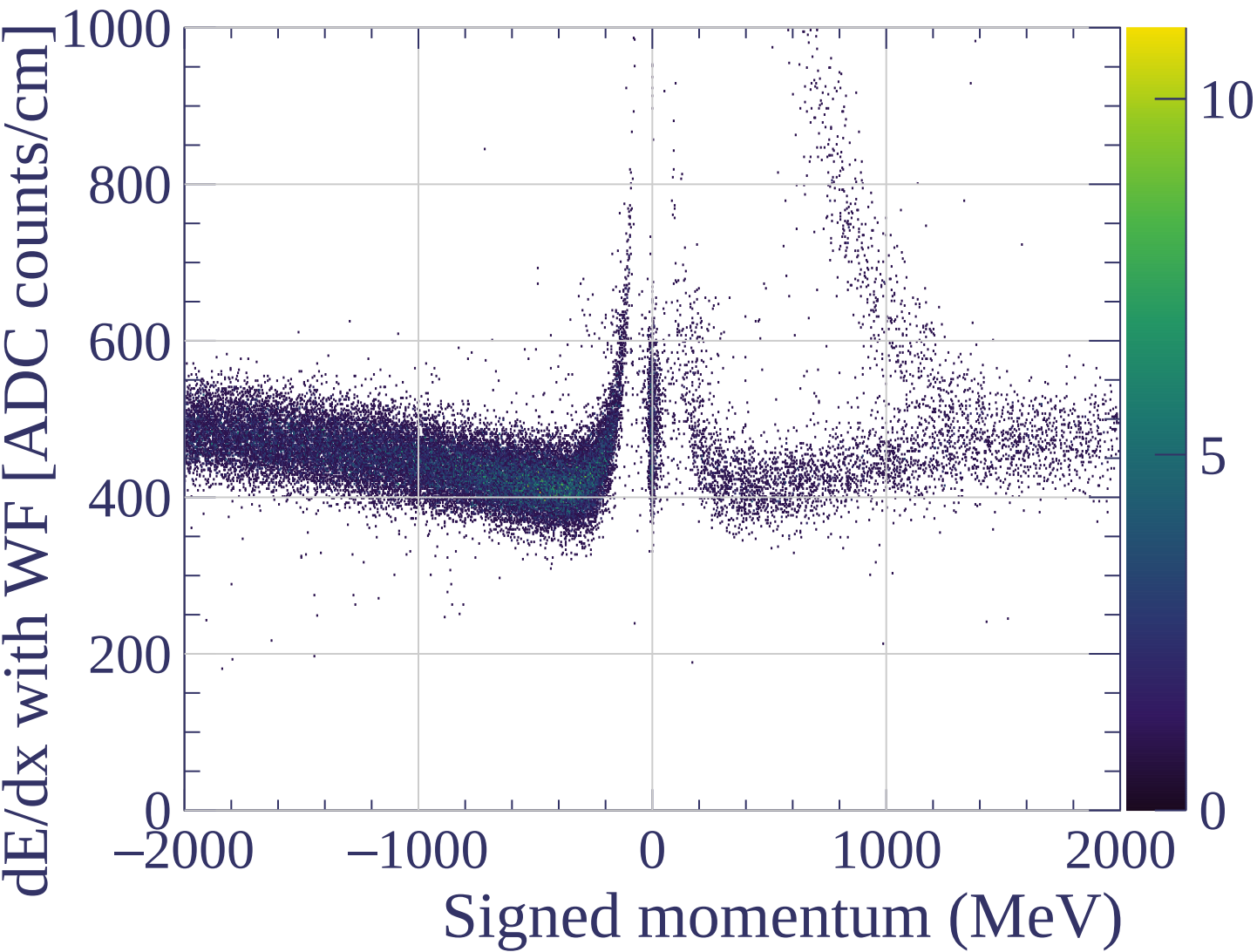


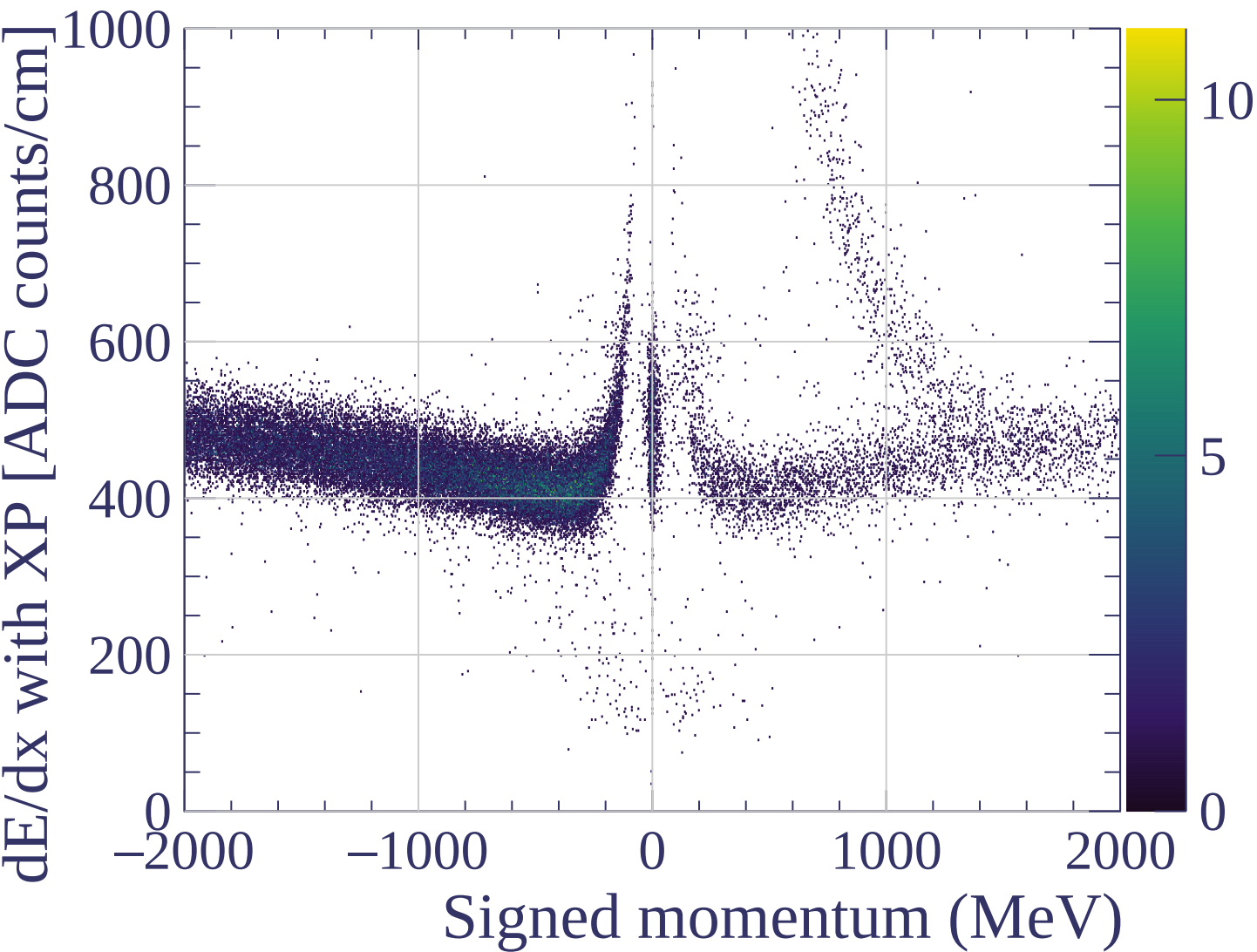


dE/dx resolution [%]

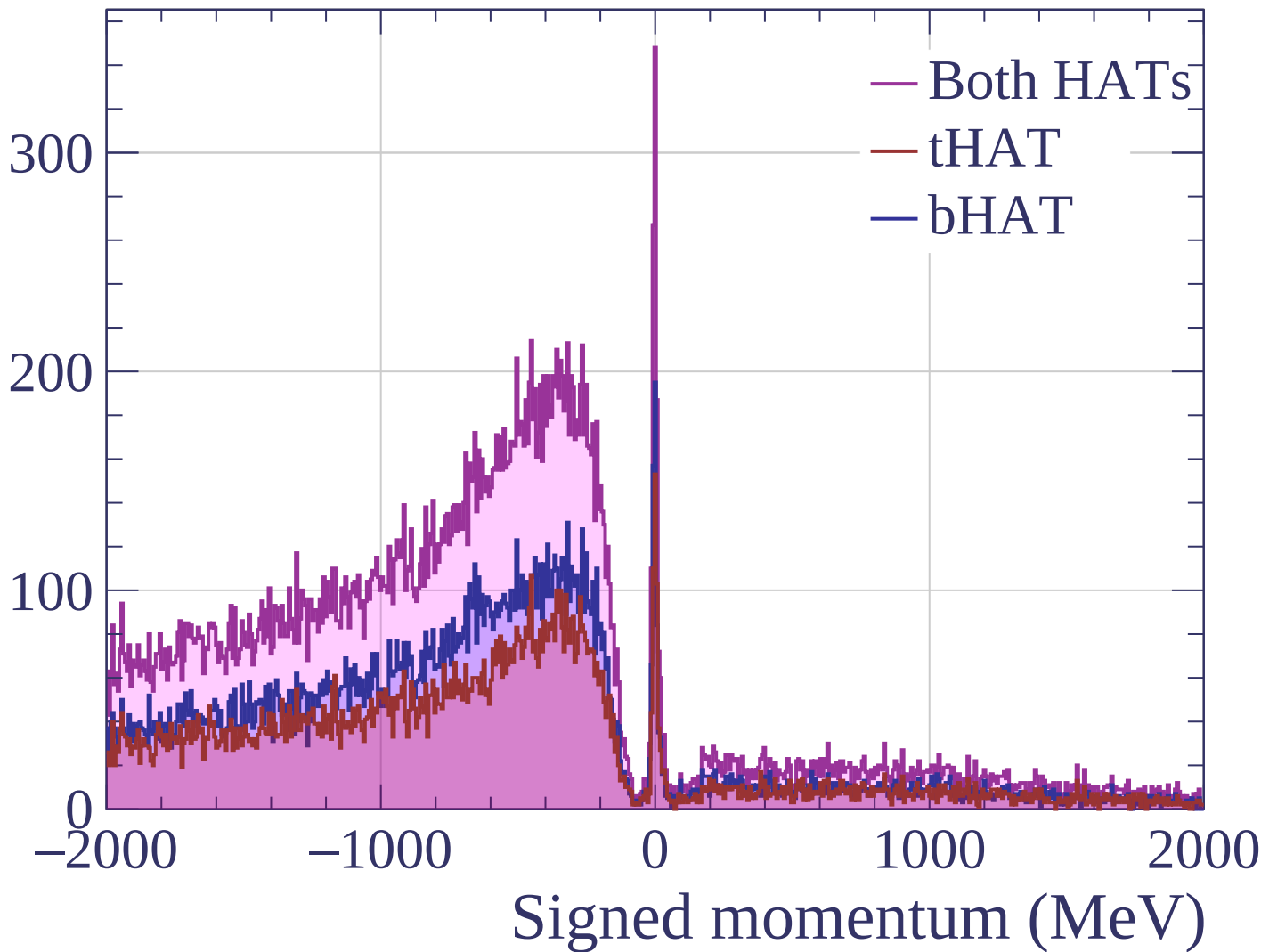


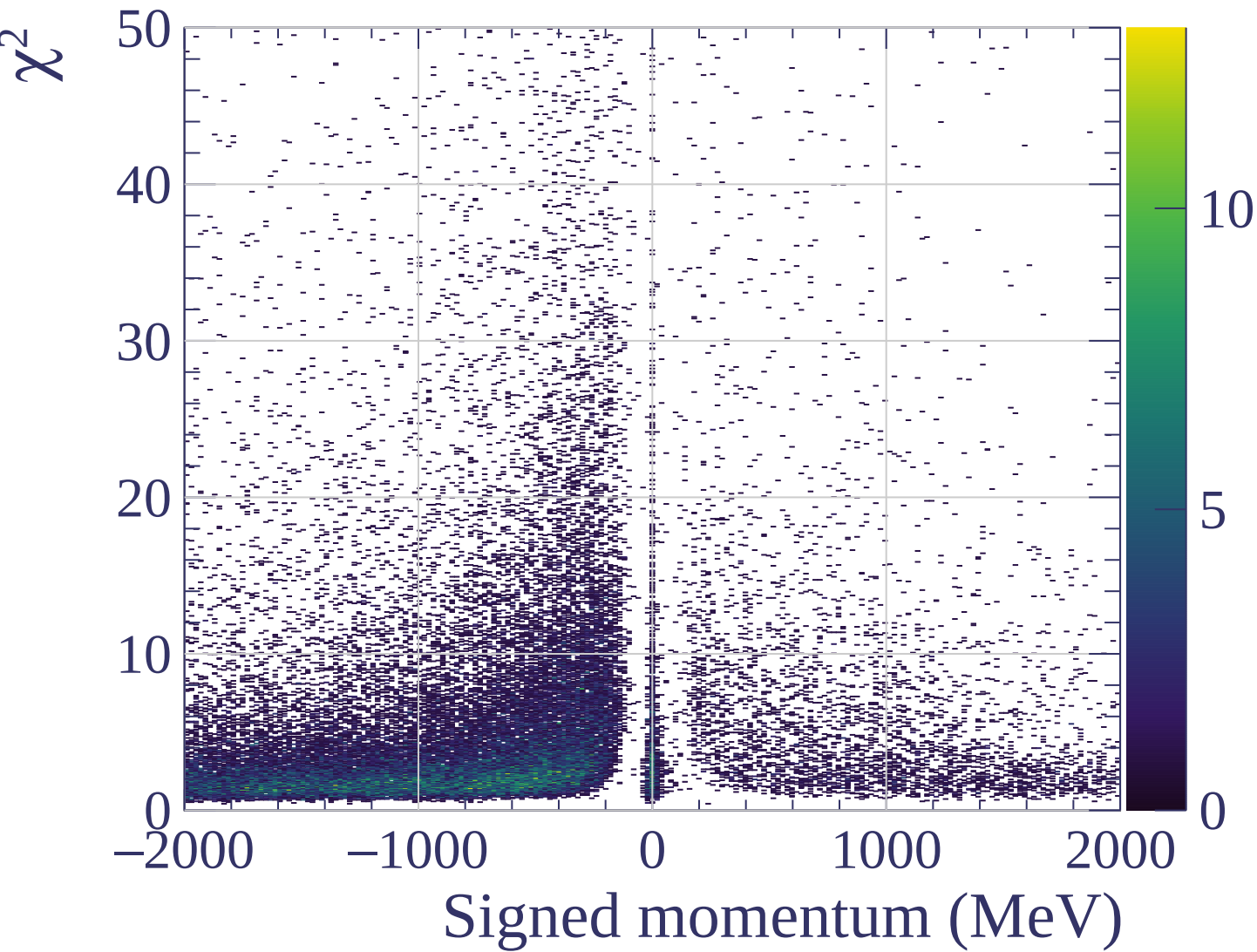


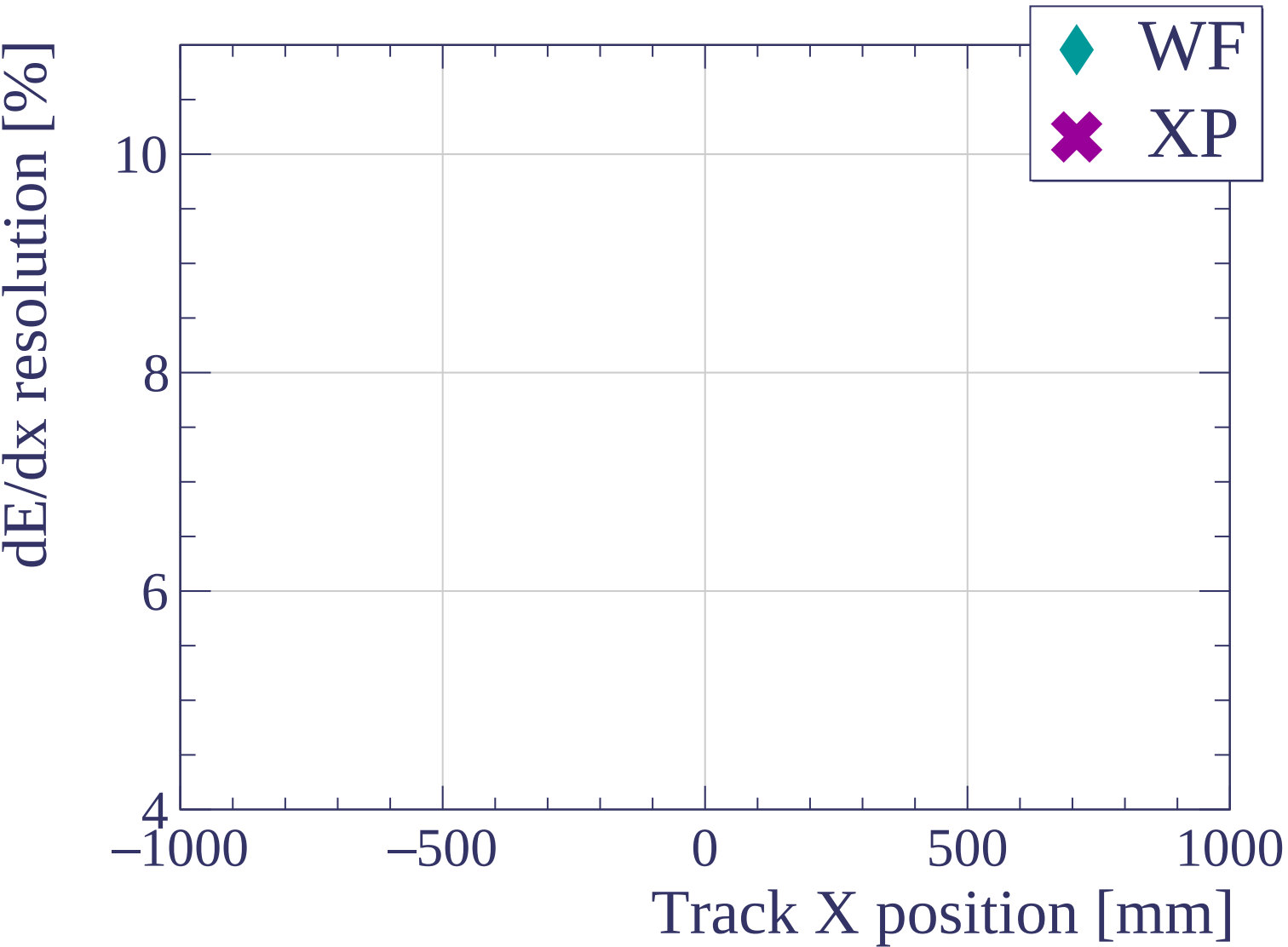


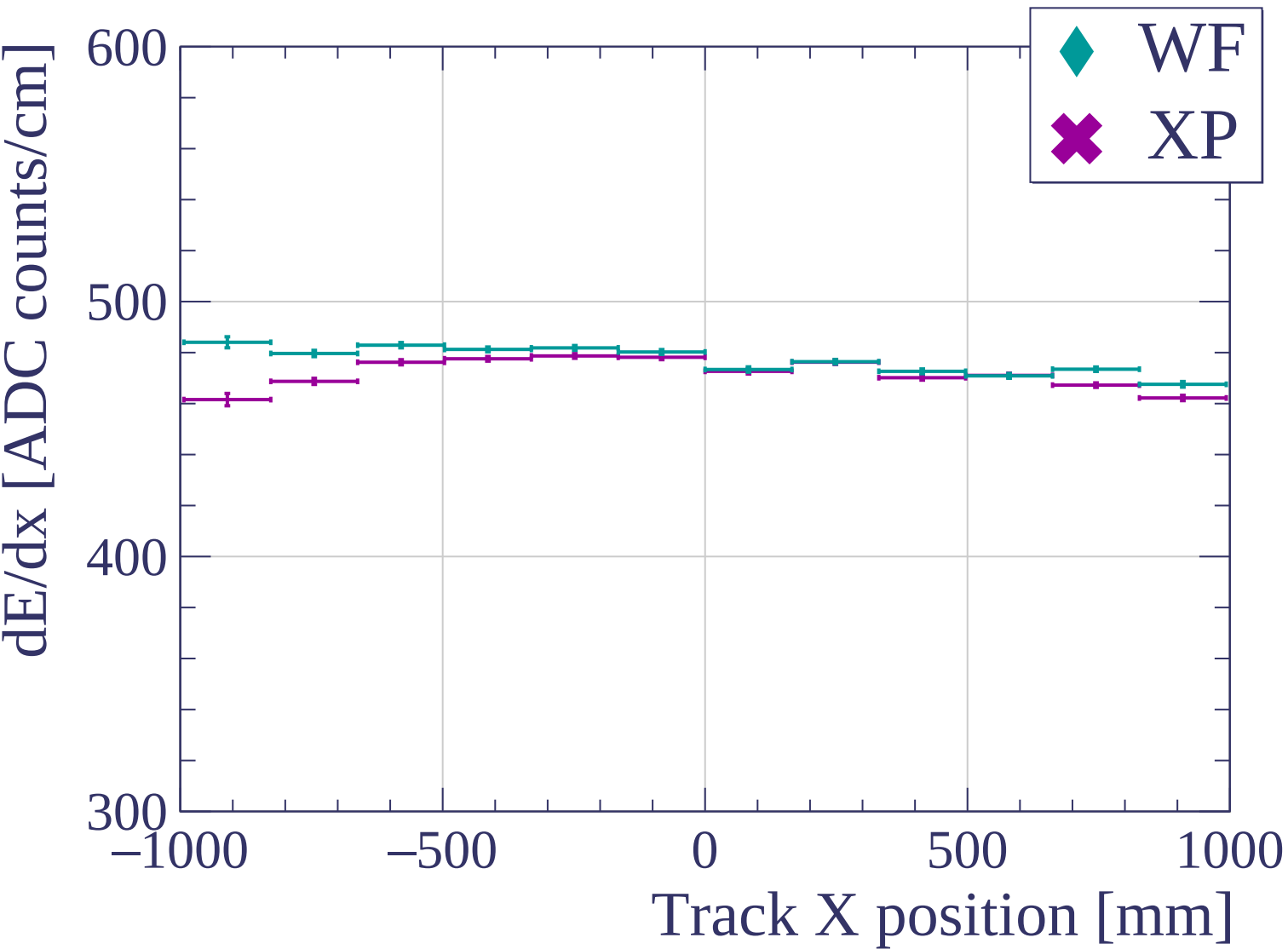


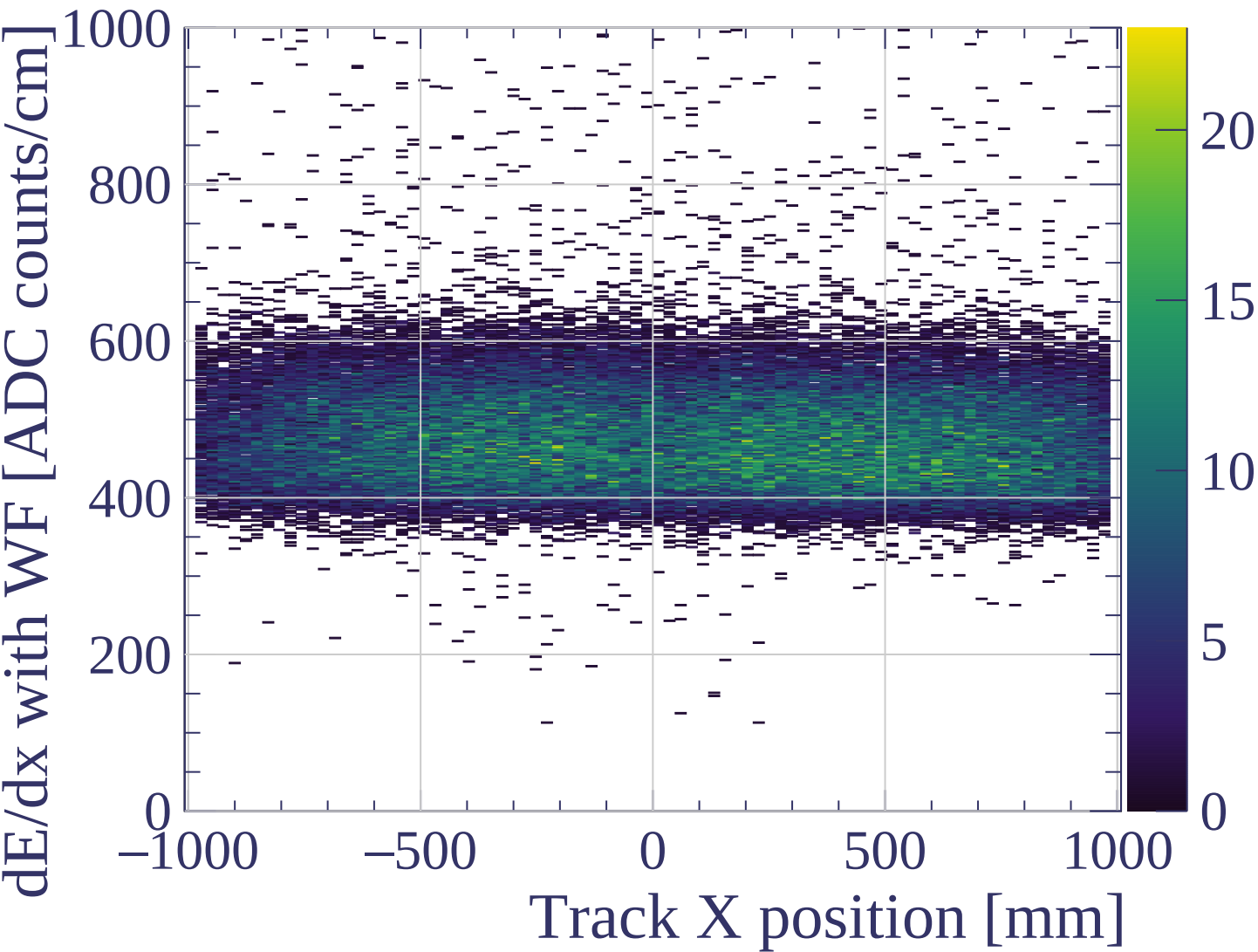
Count

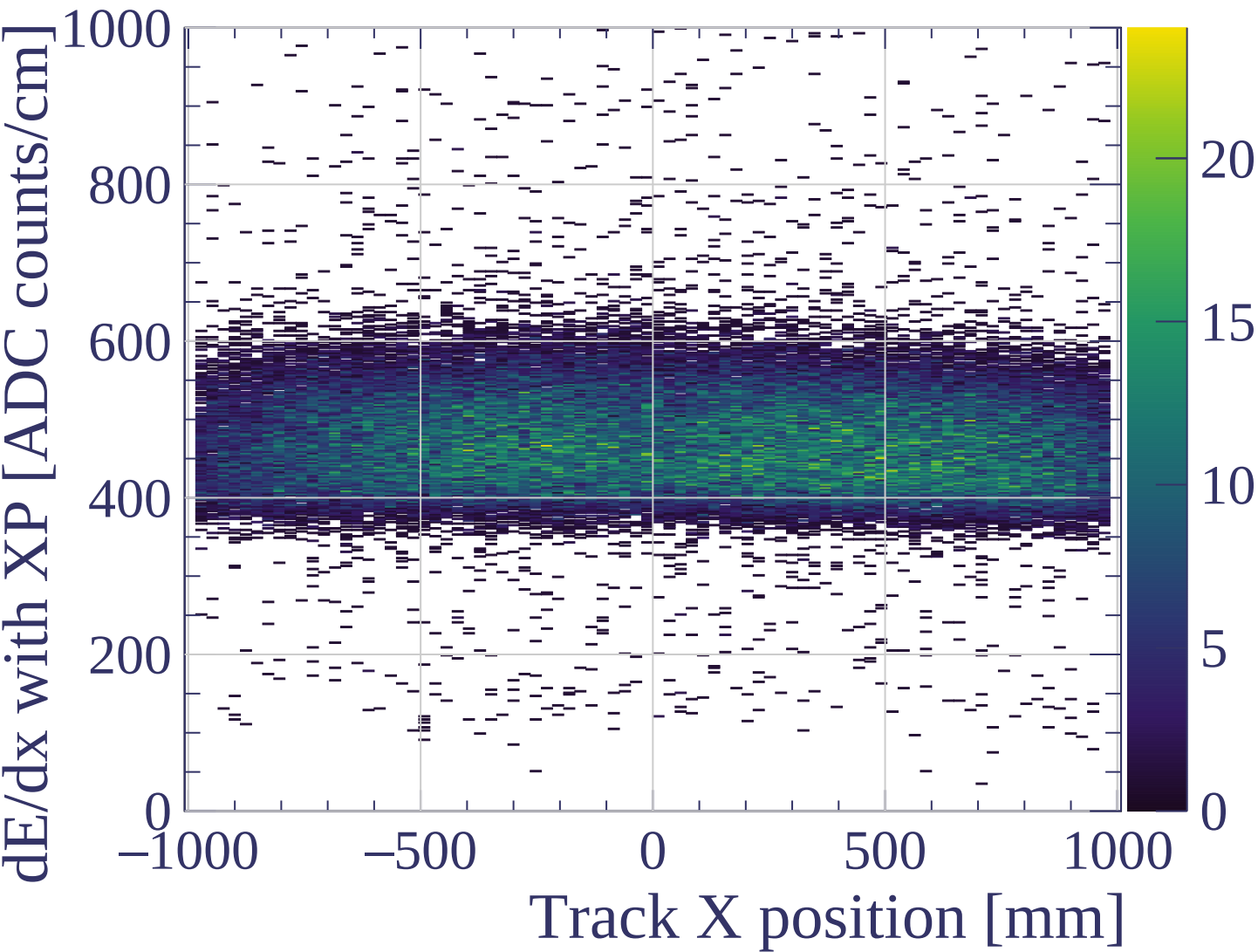




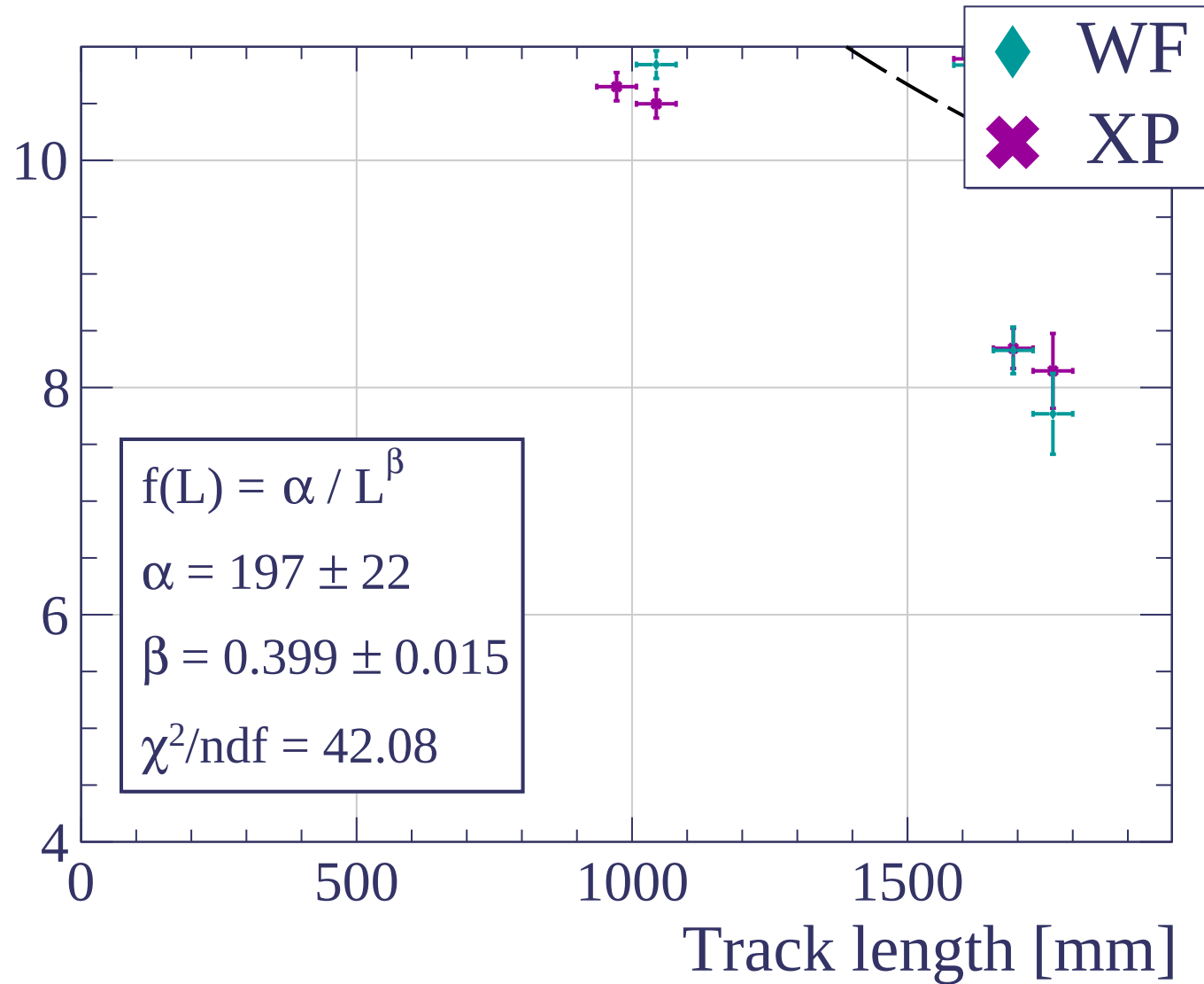


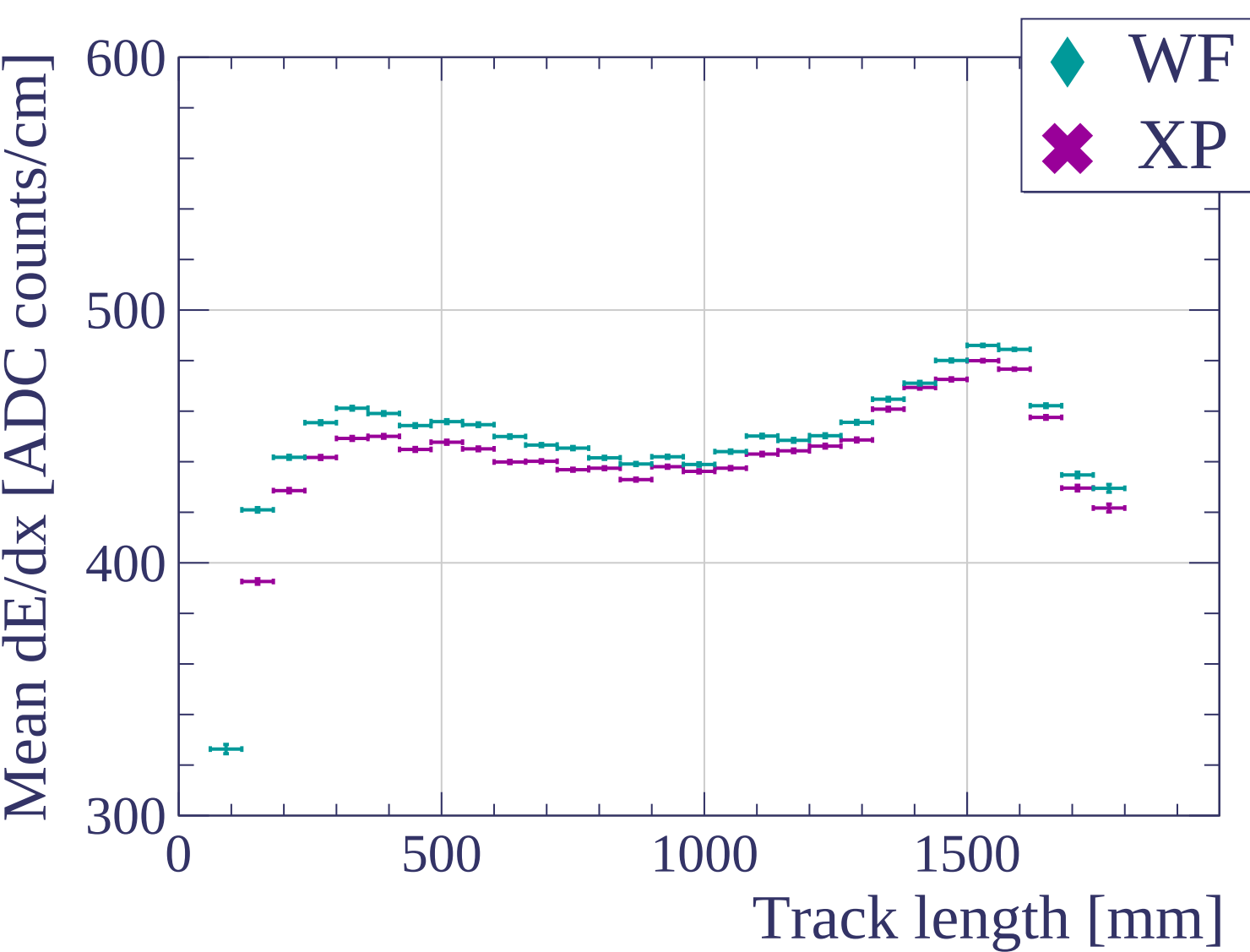


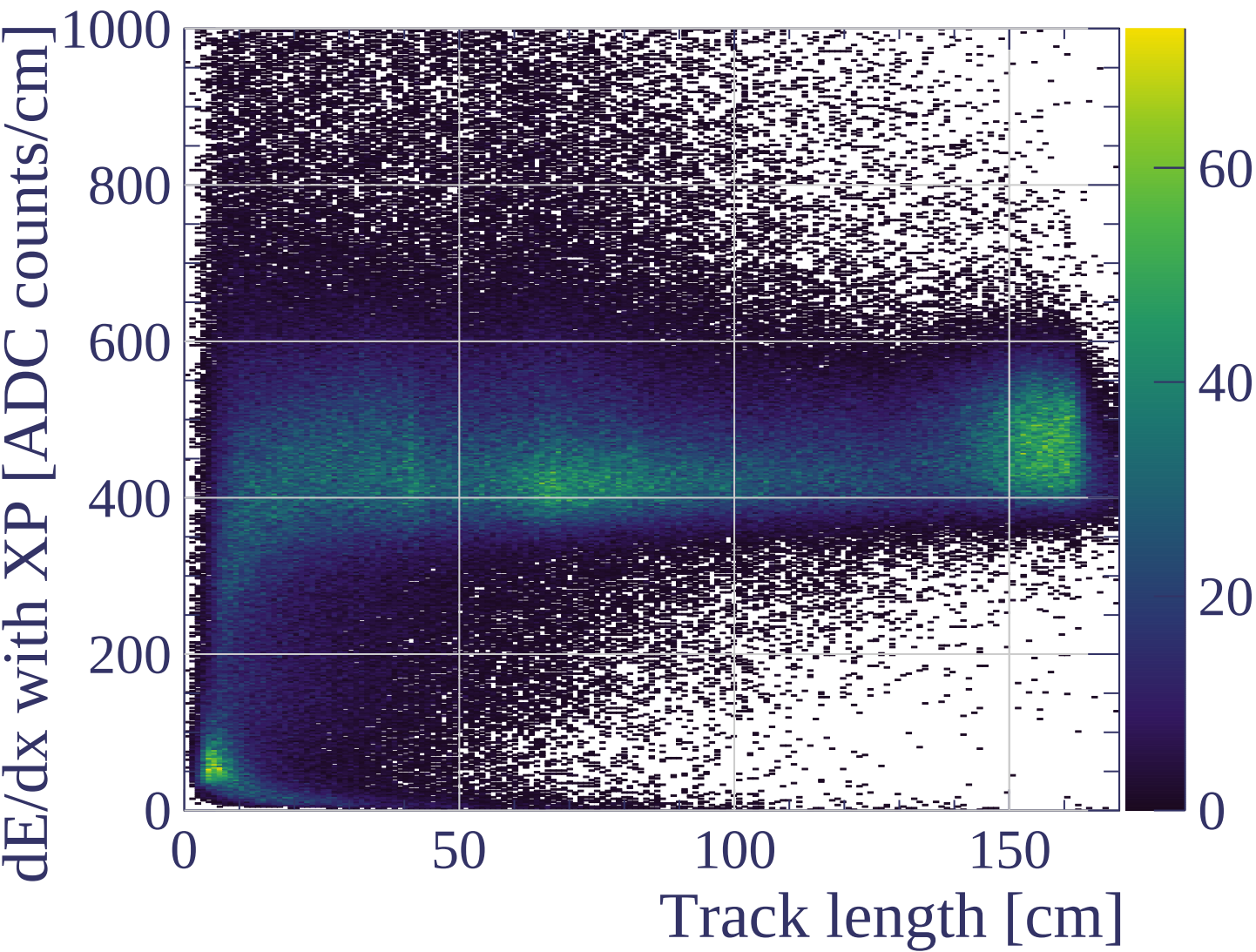


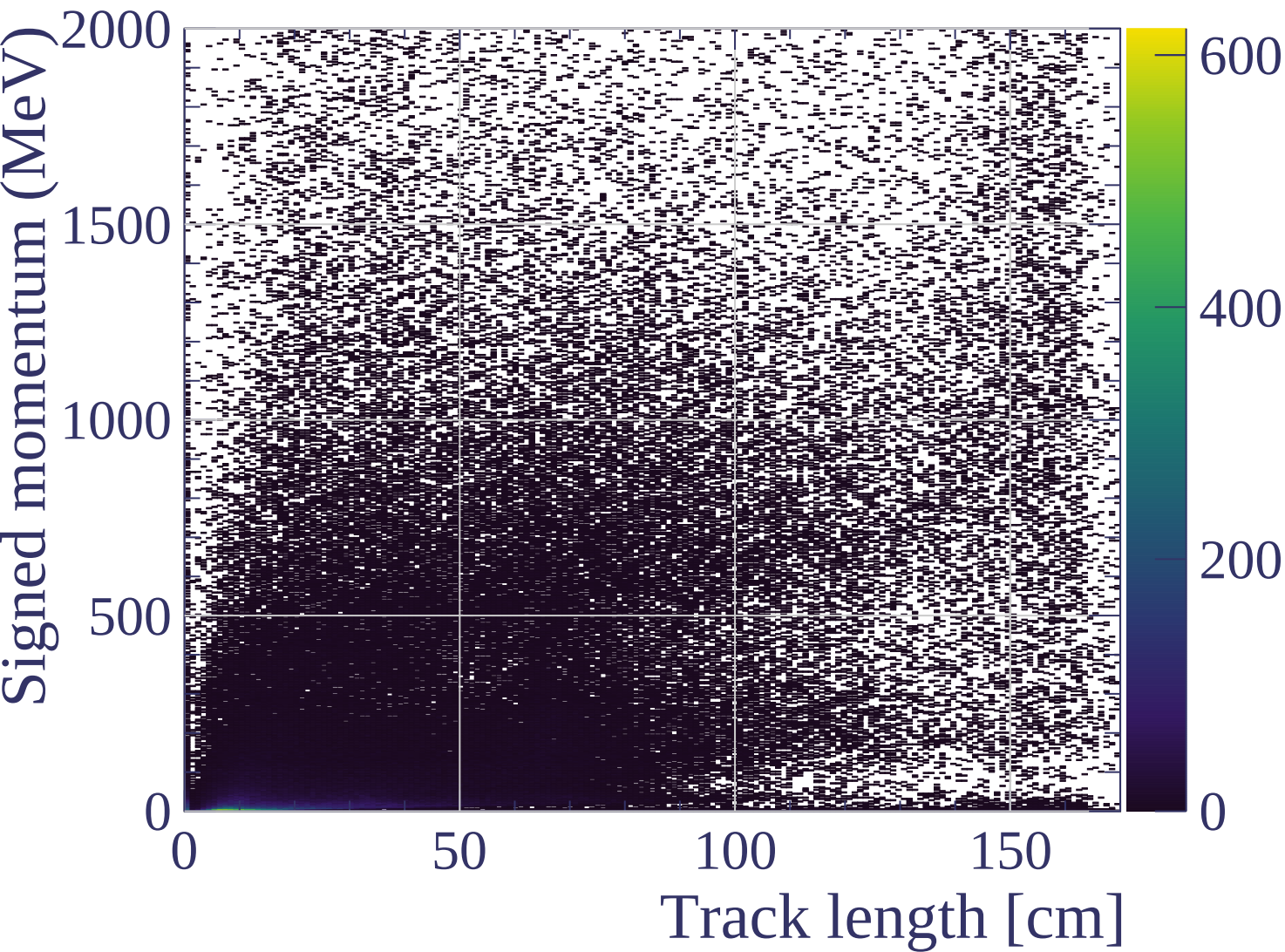


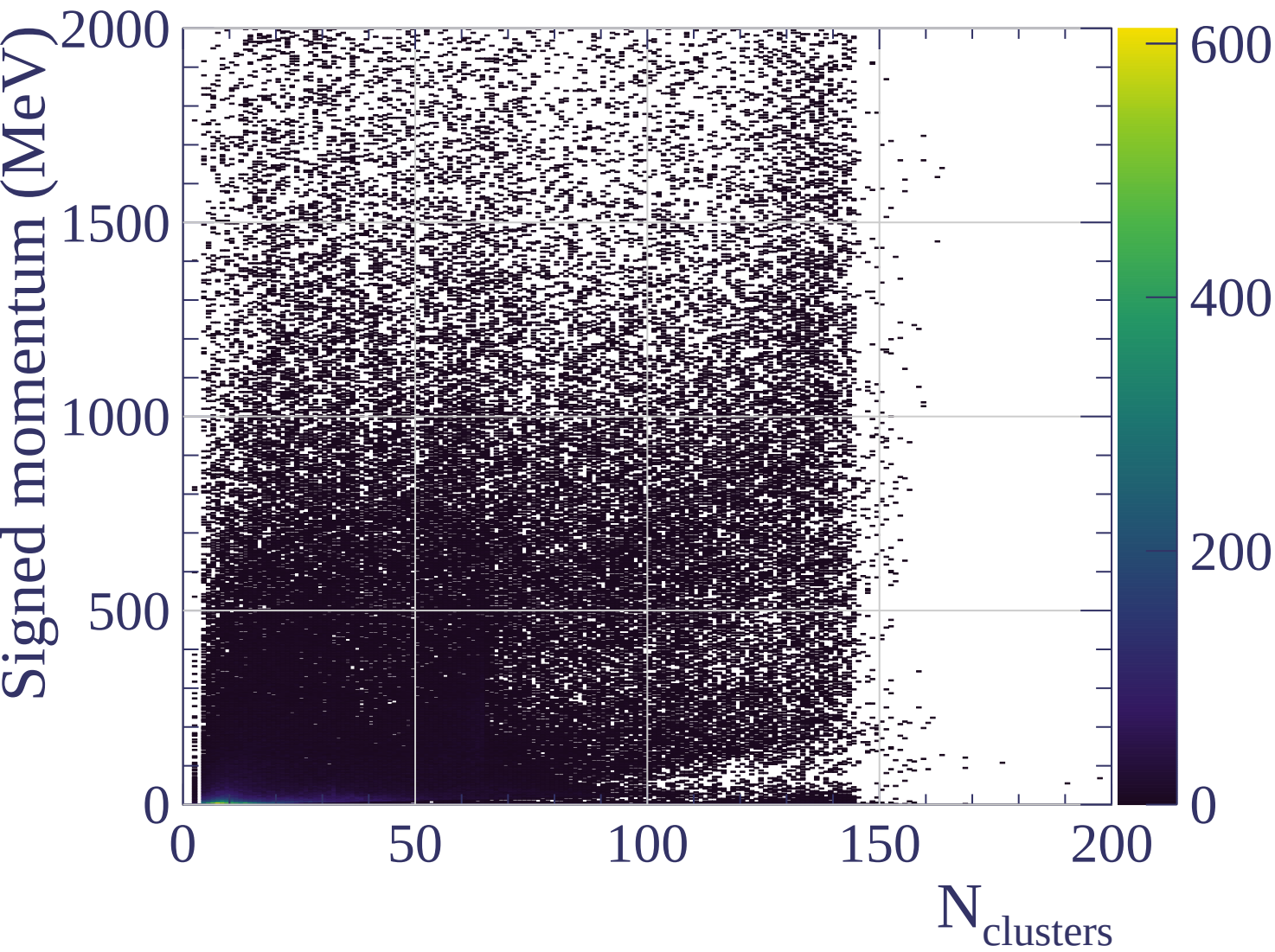
dE/dx resolution [%]



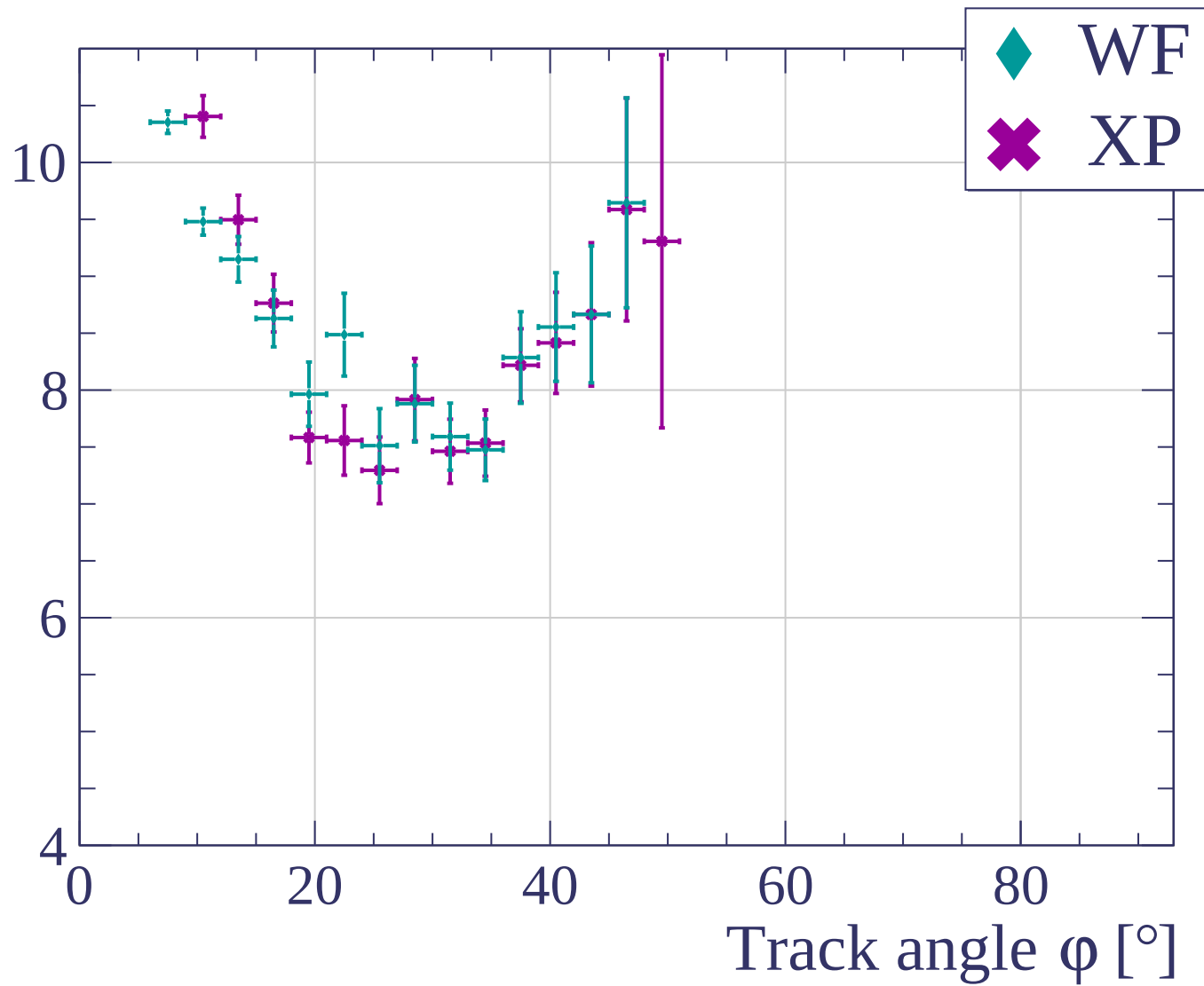


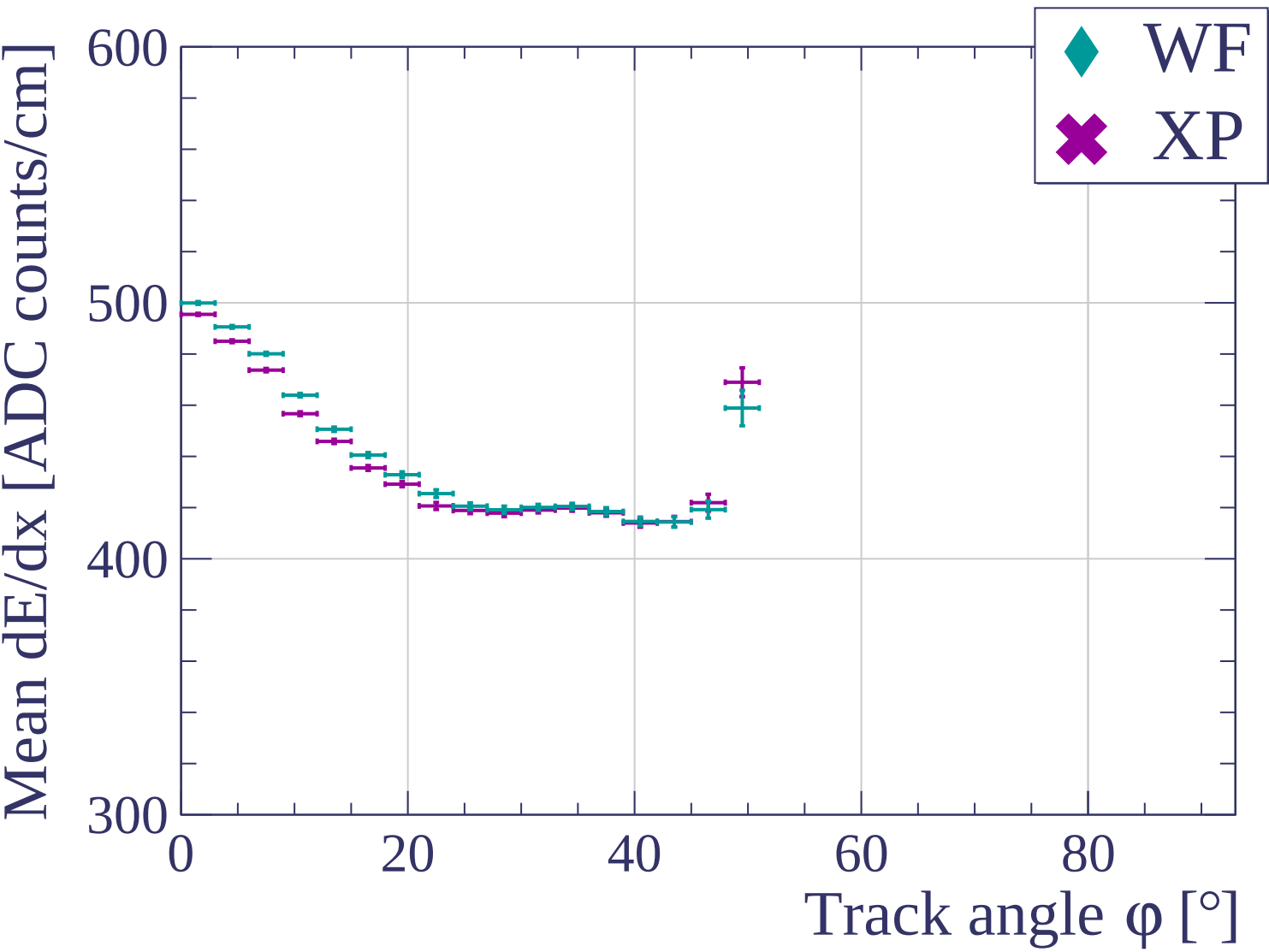




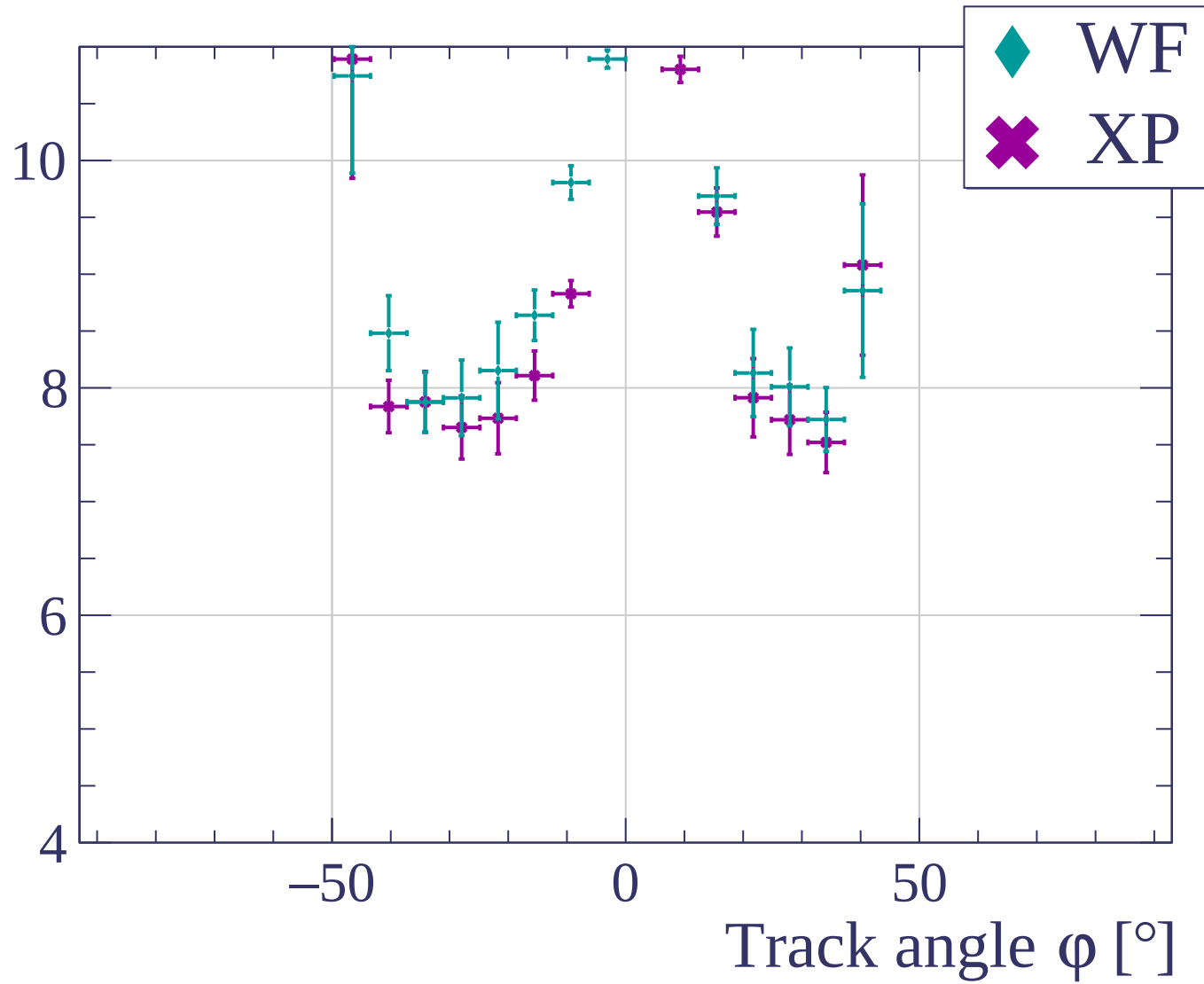


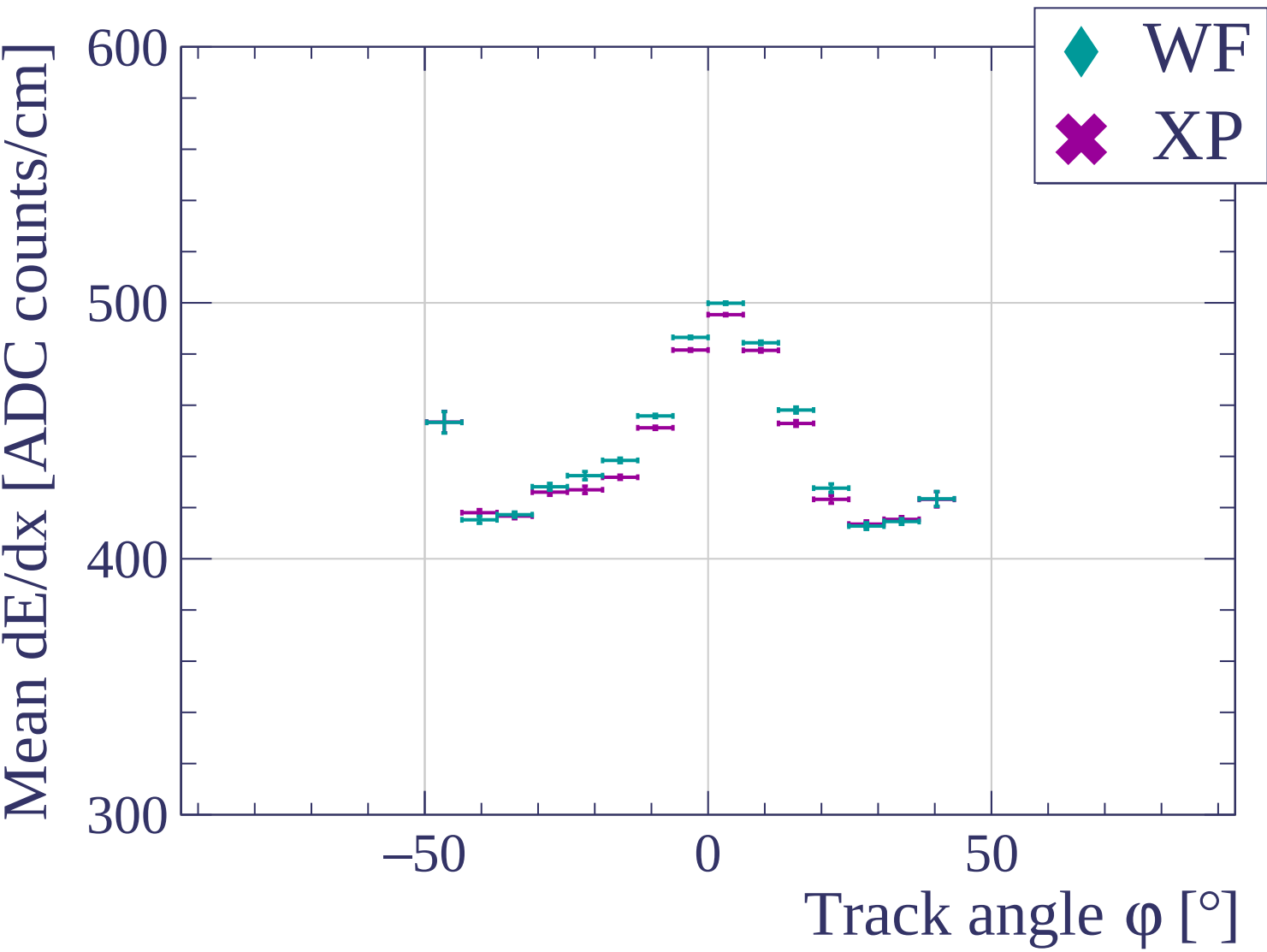
dE/dx resolution [%]

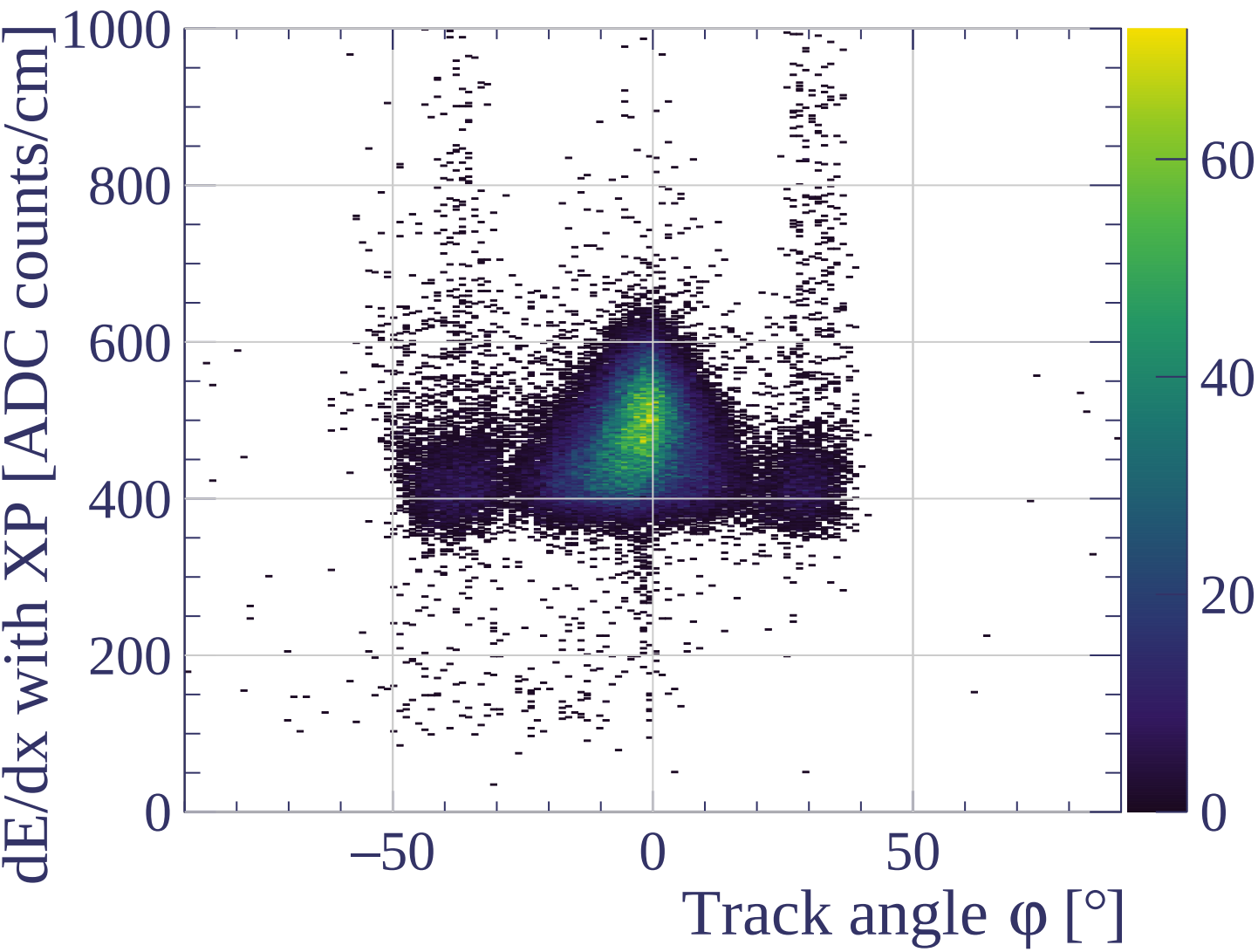




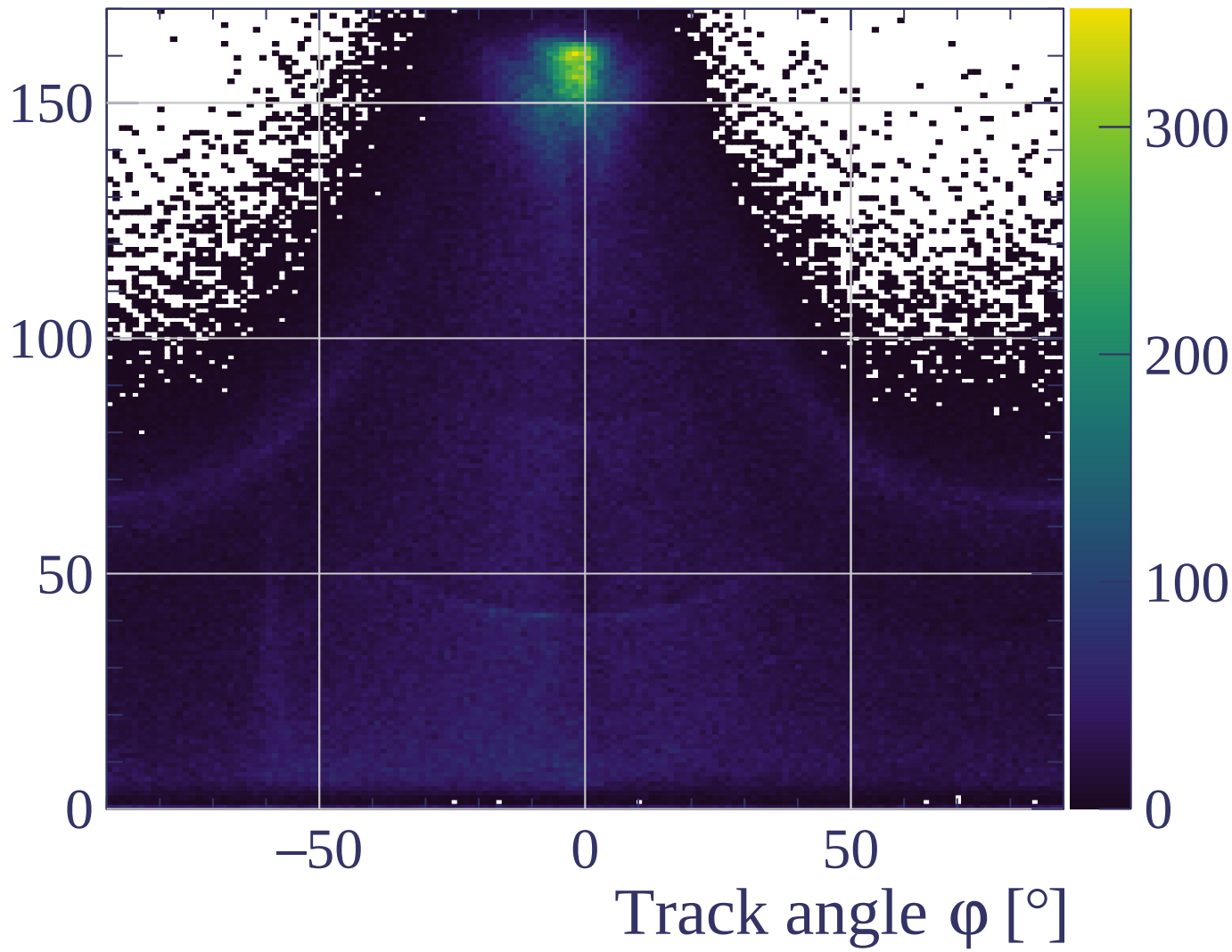
dE/dx resolution [%]

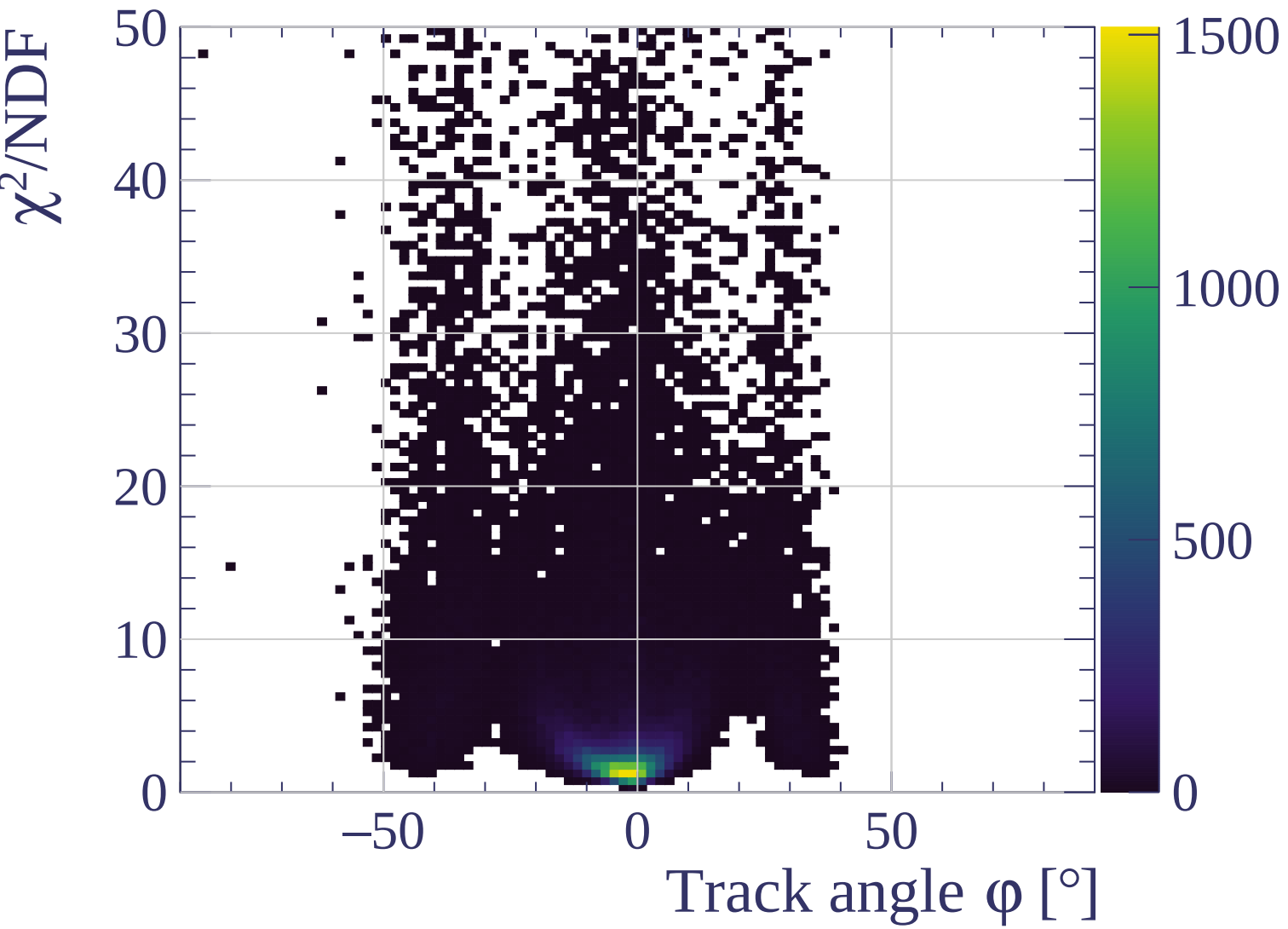


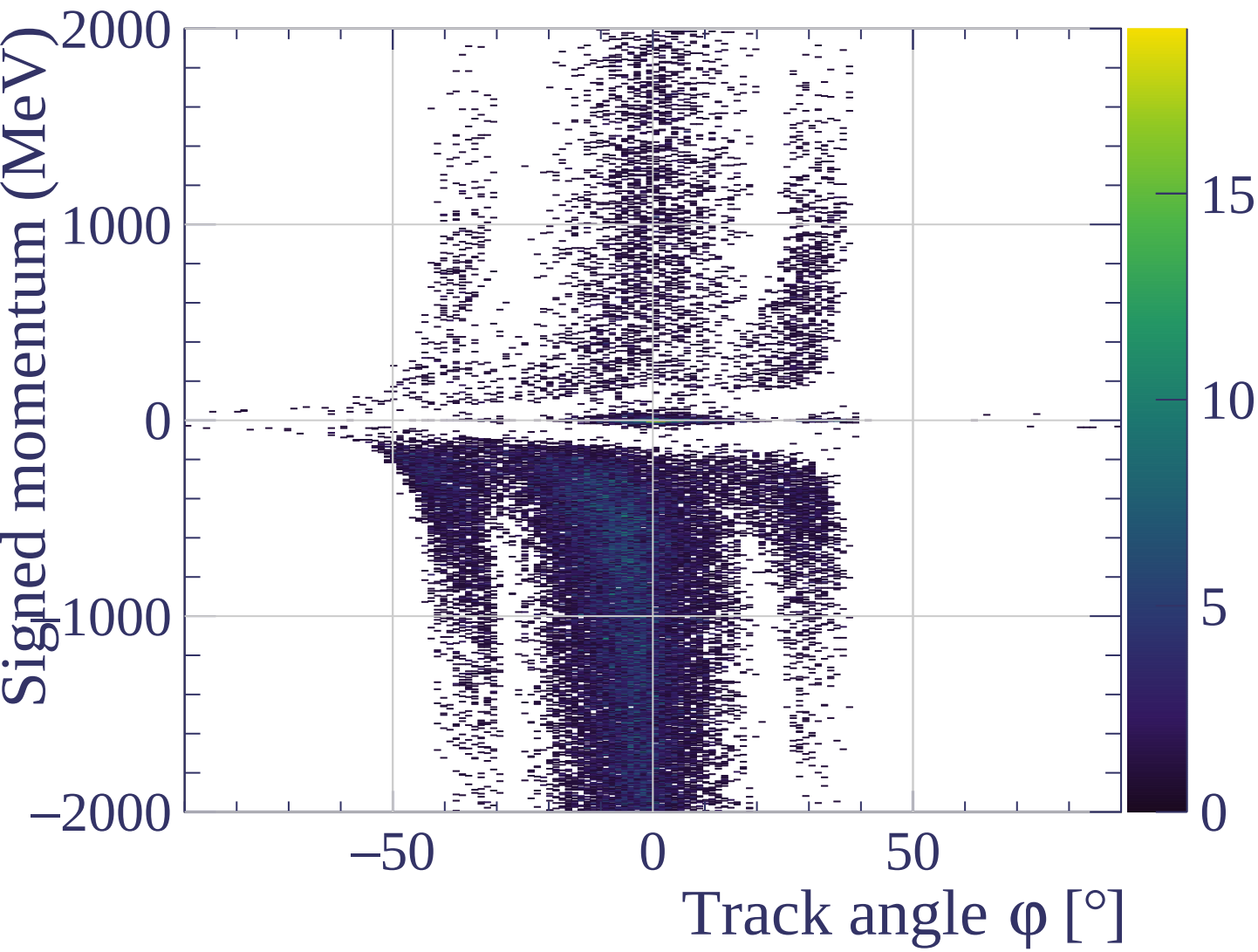


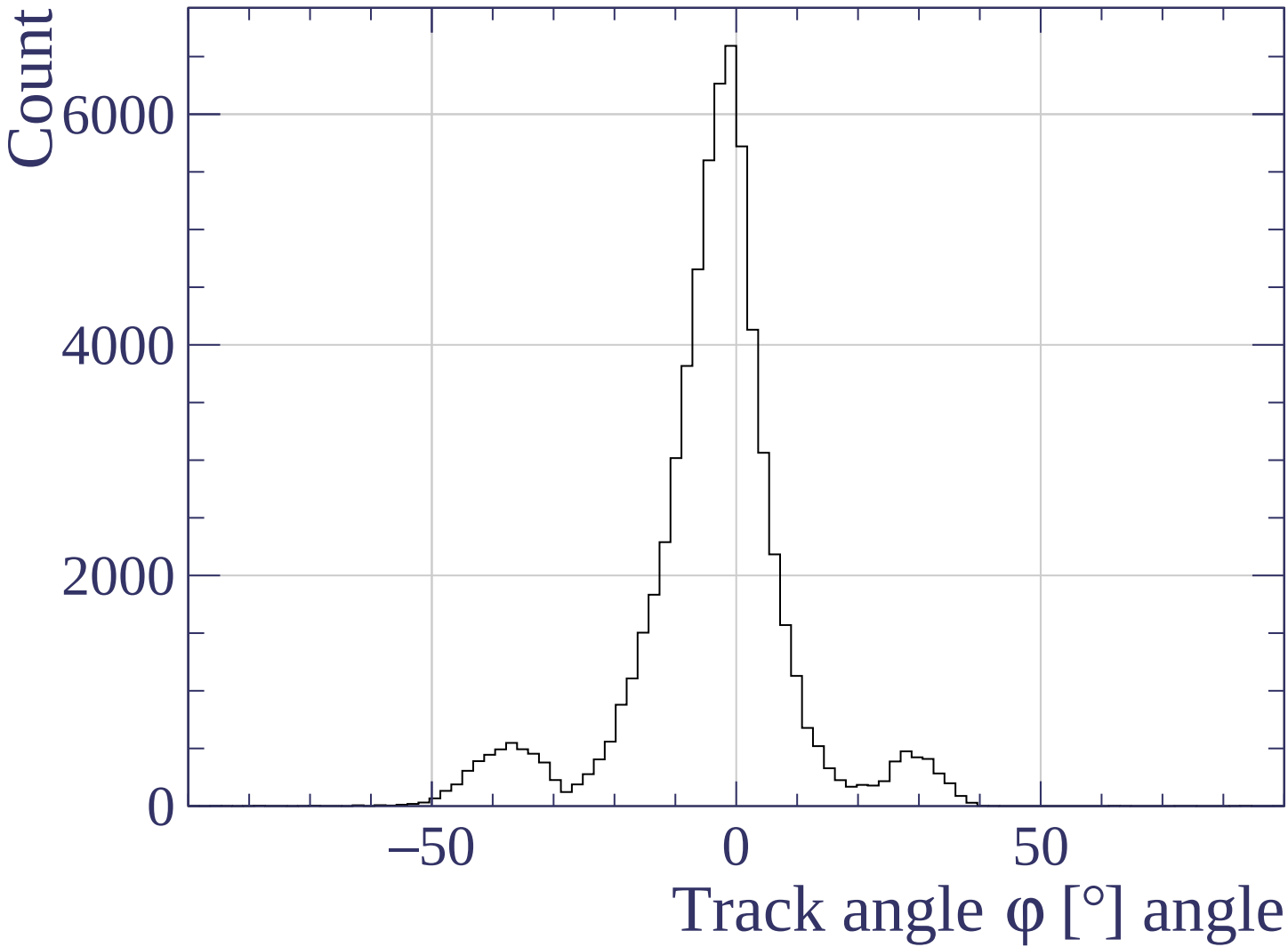


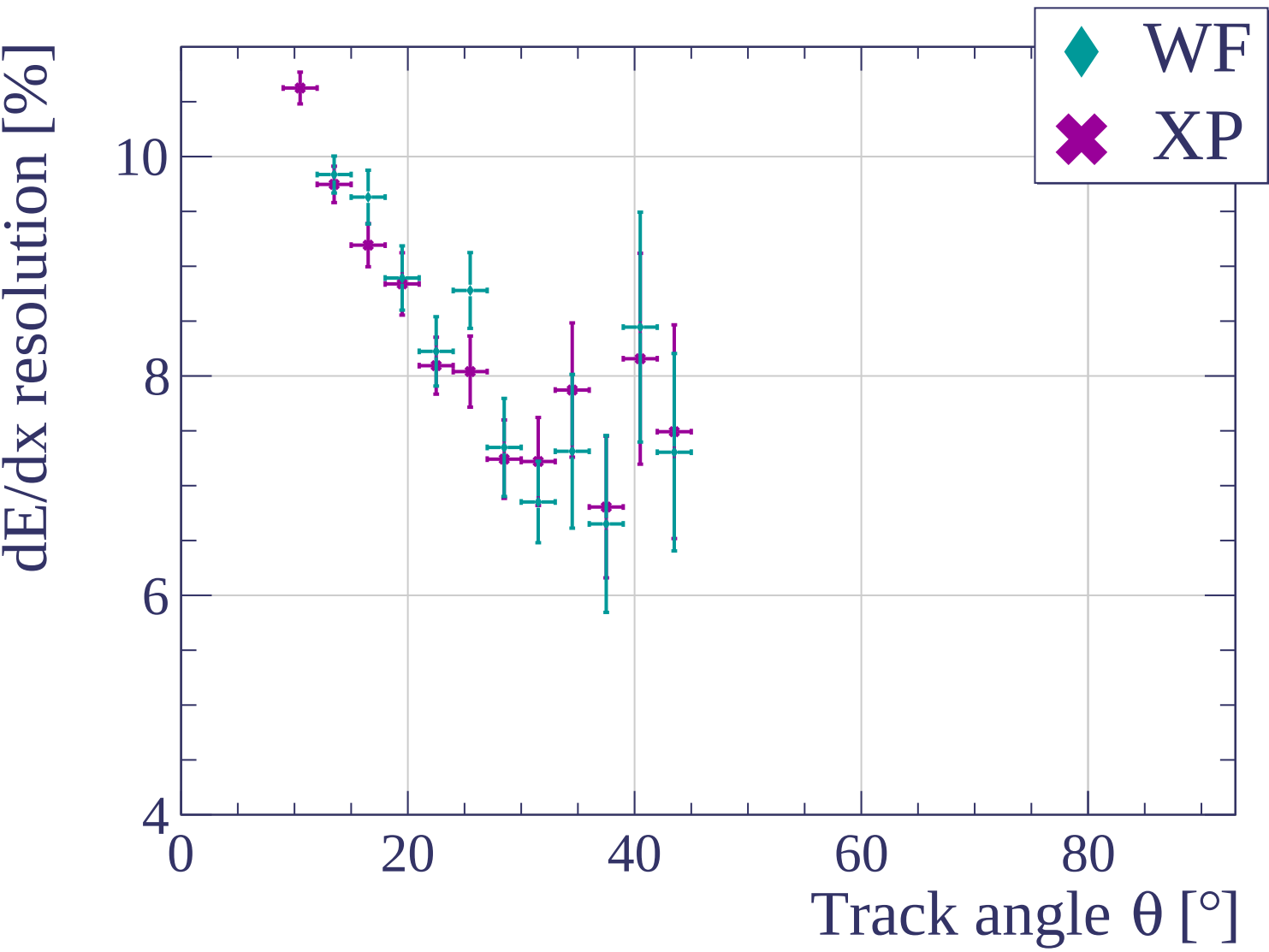
Track length [cm]

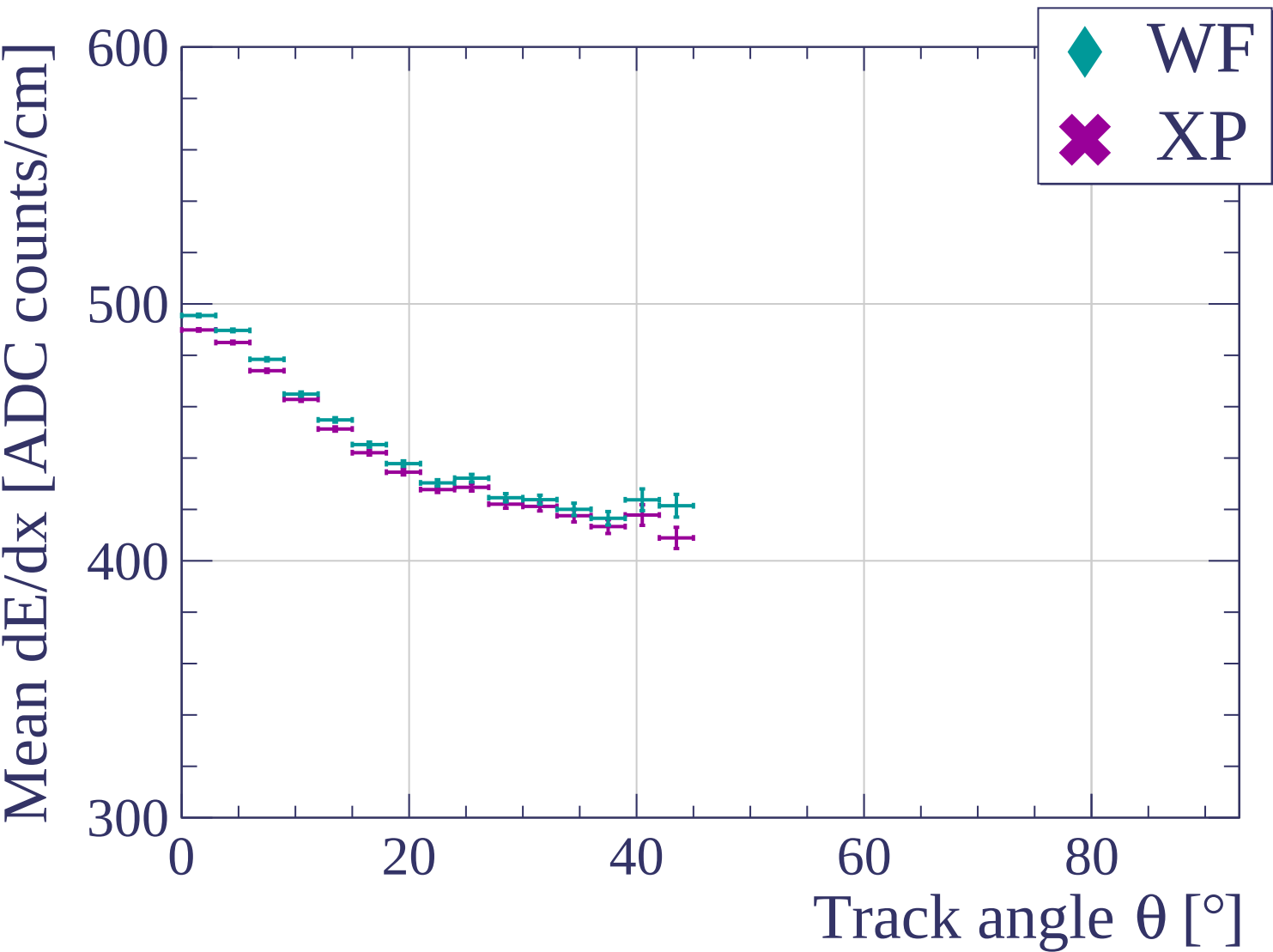




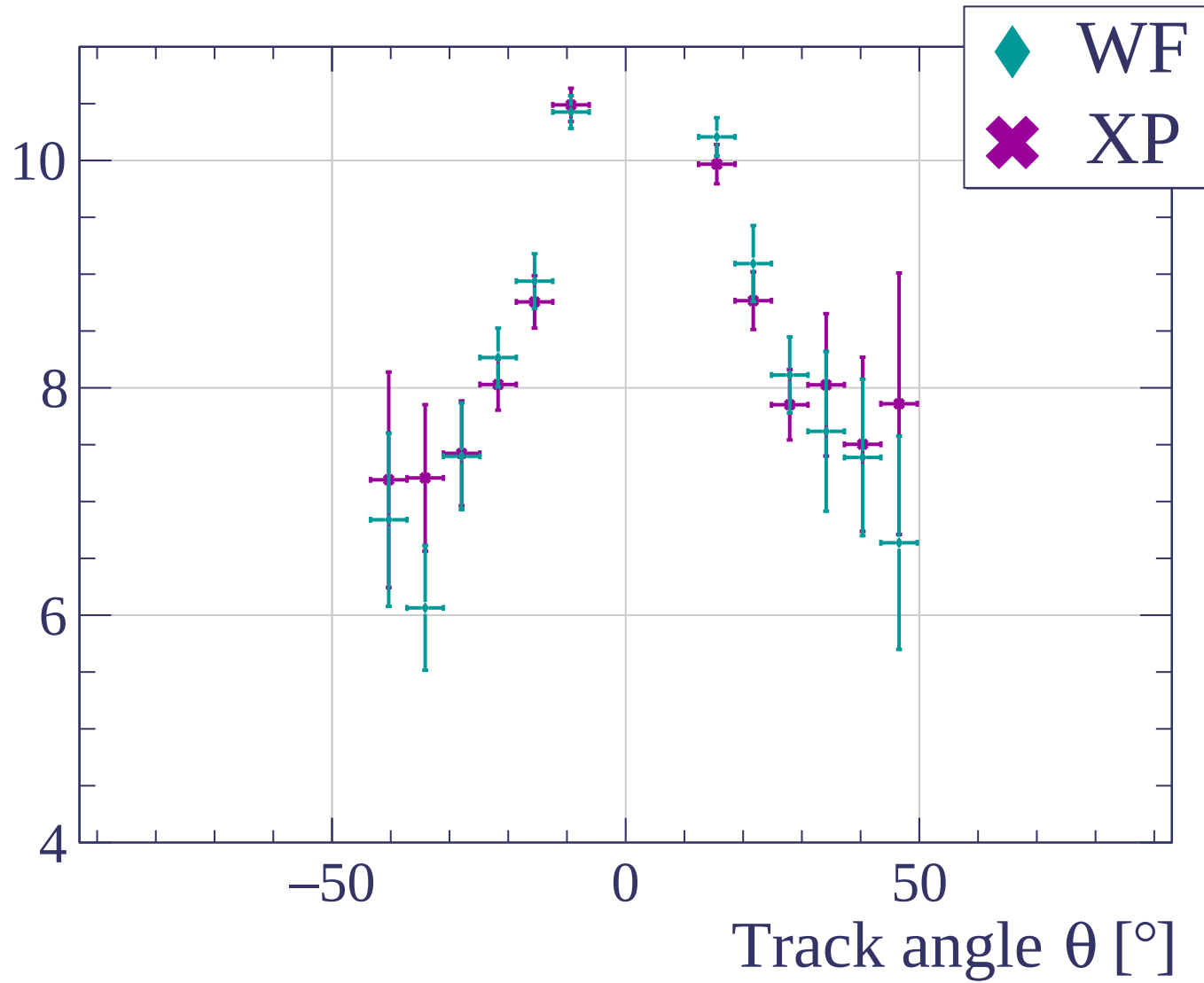


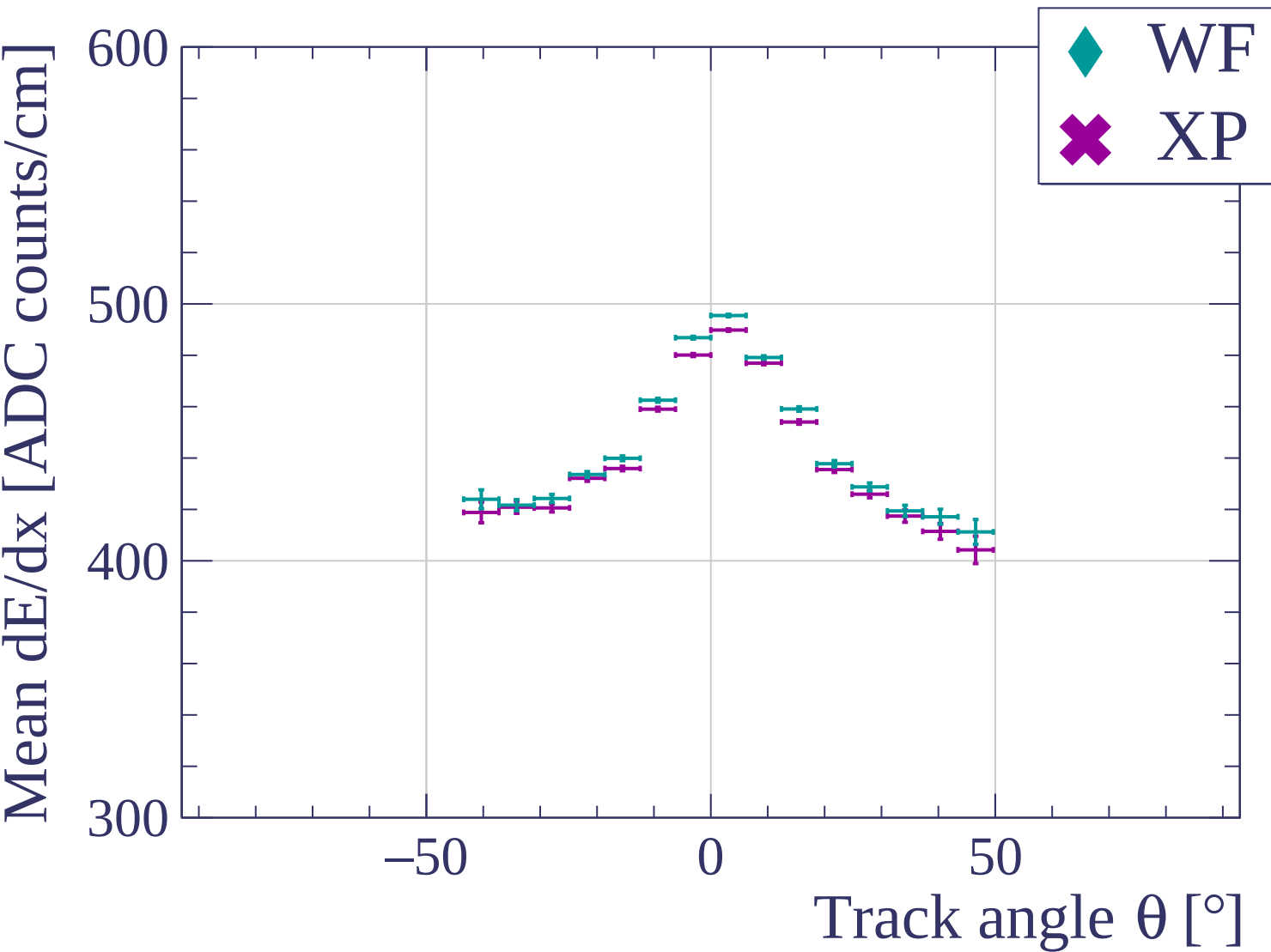


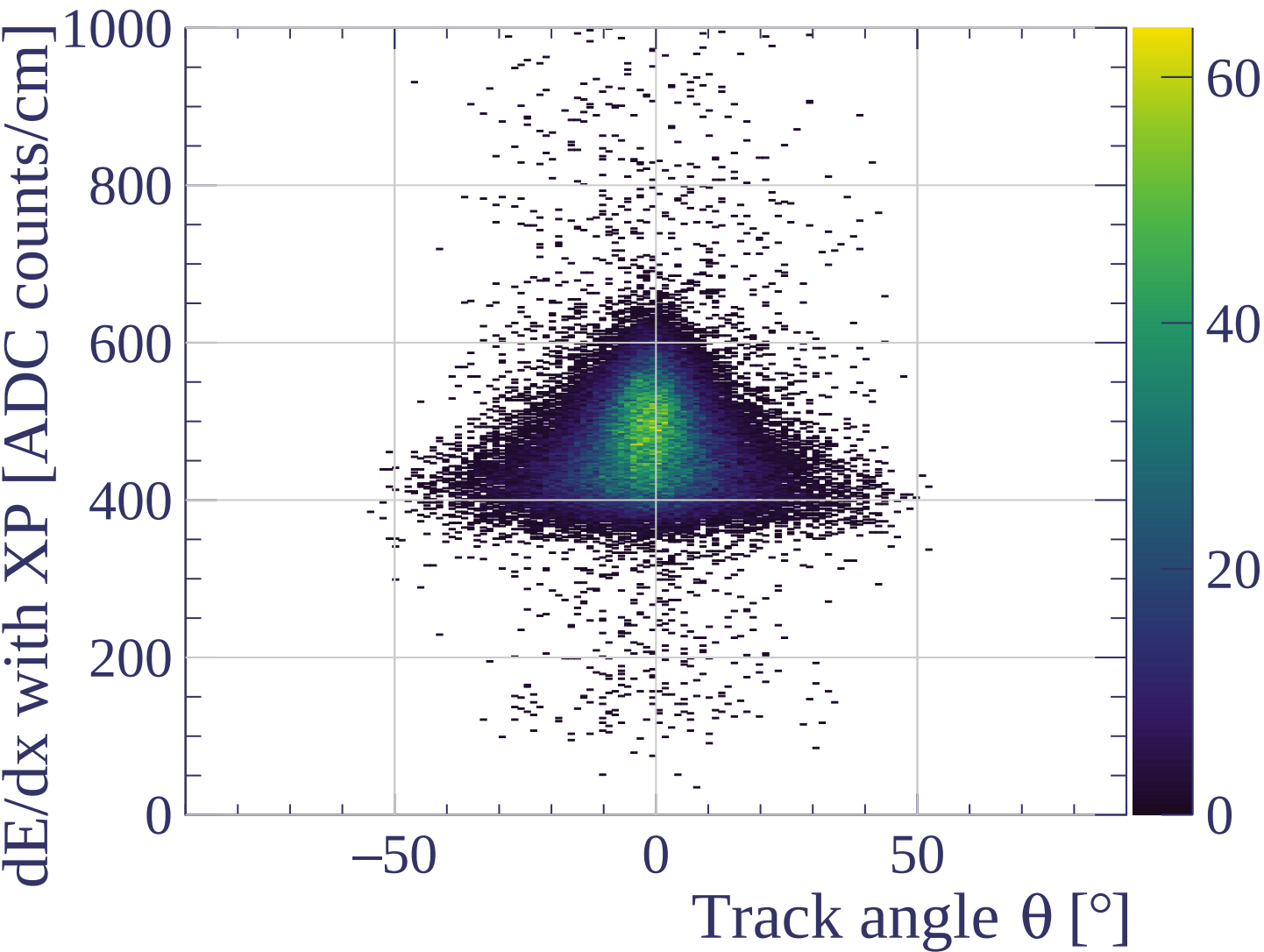




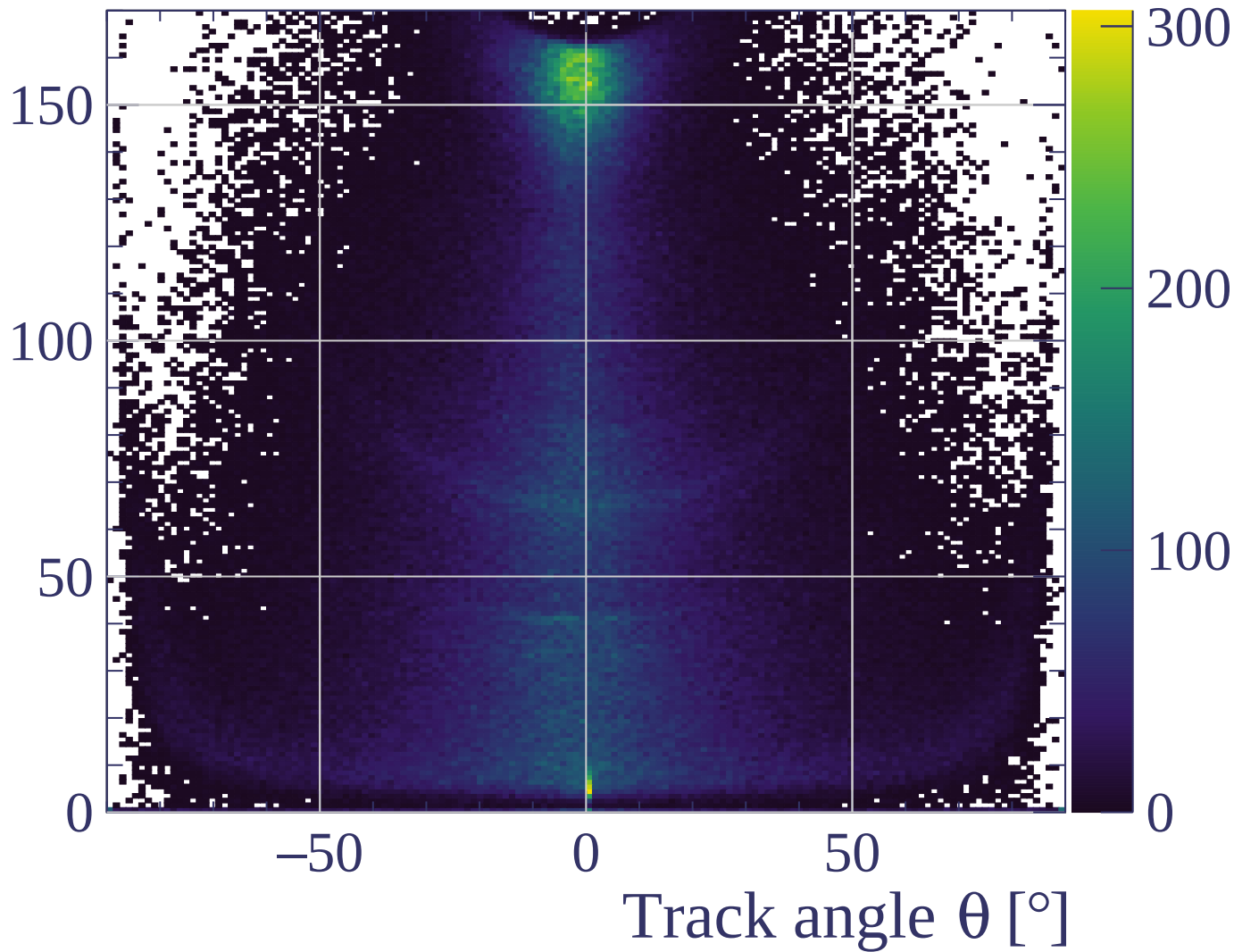
dE/dx resolution [%]

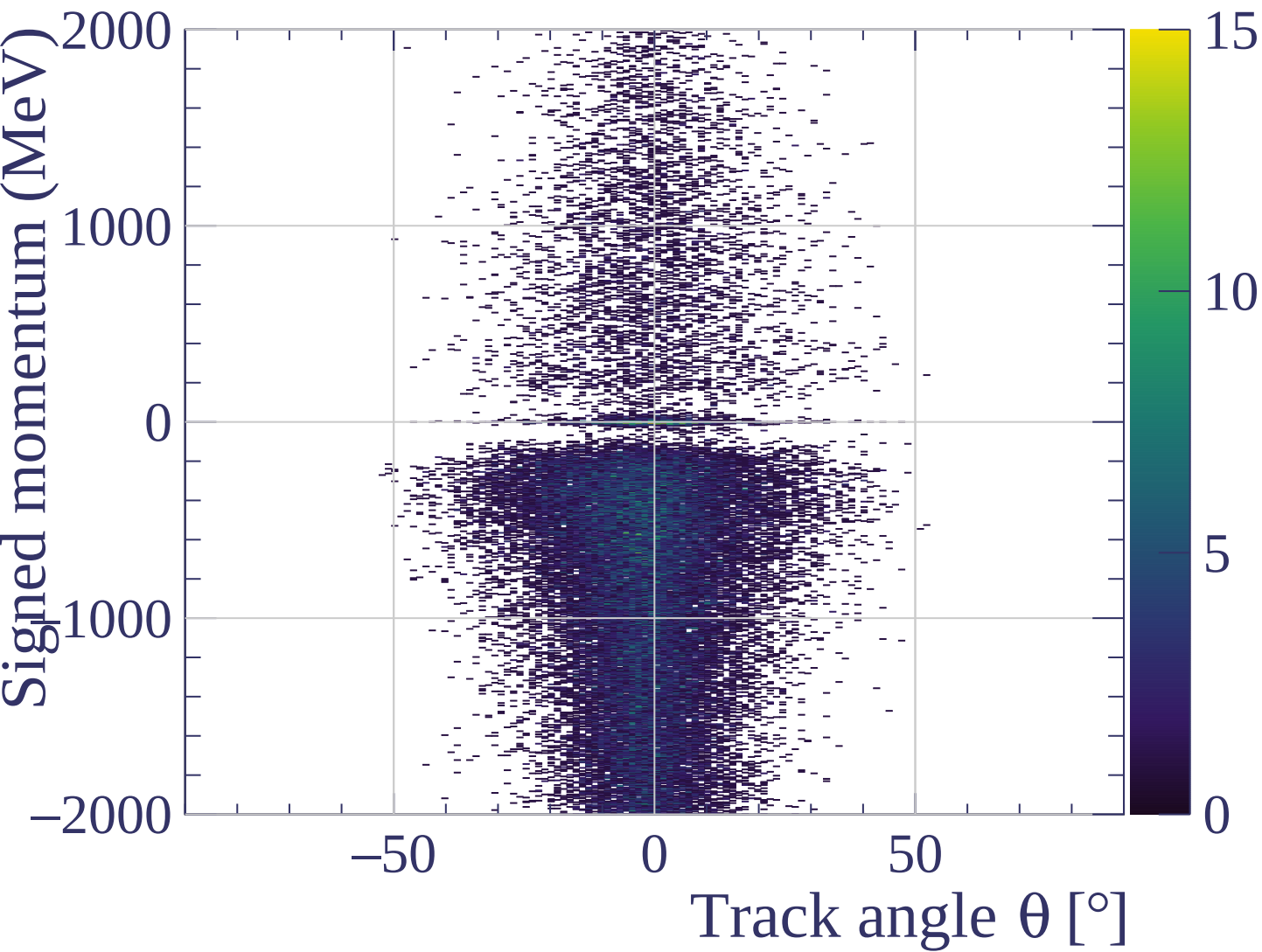


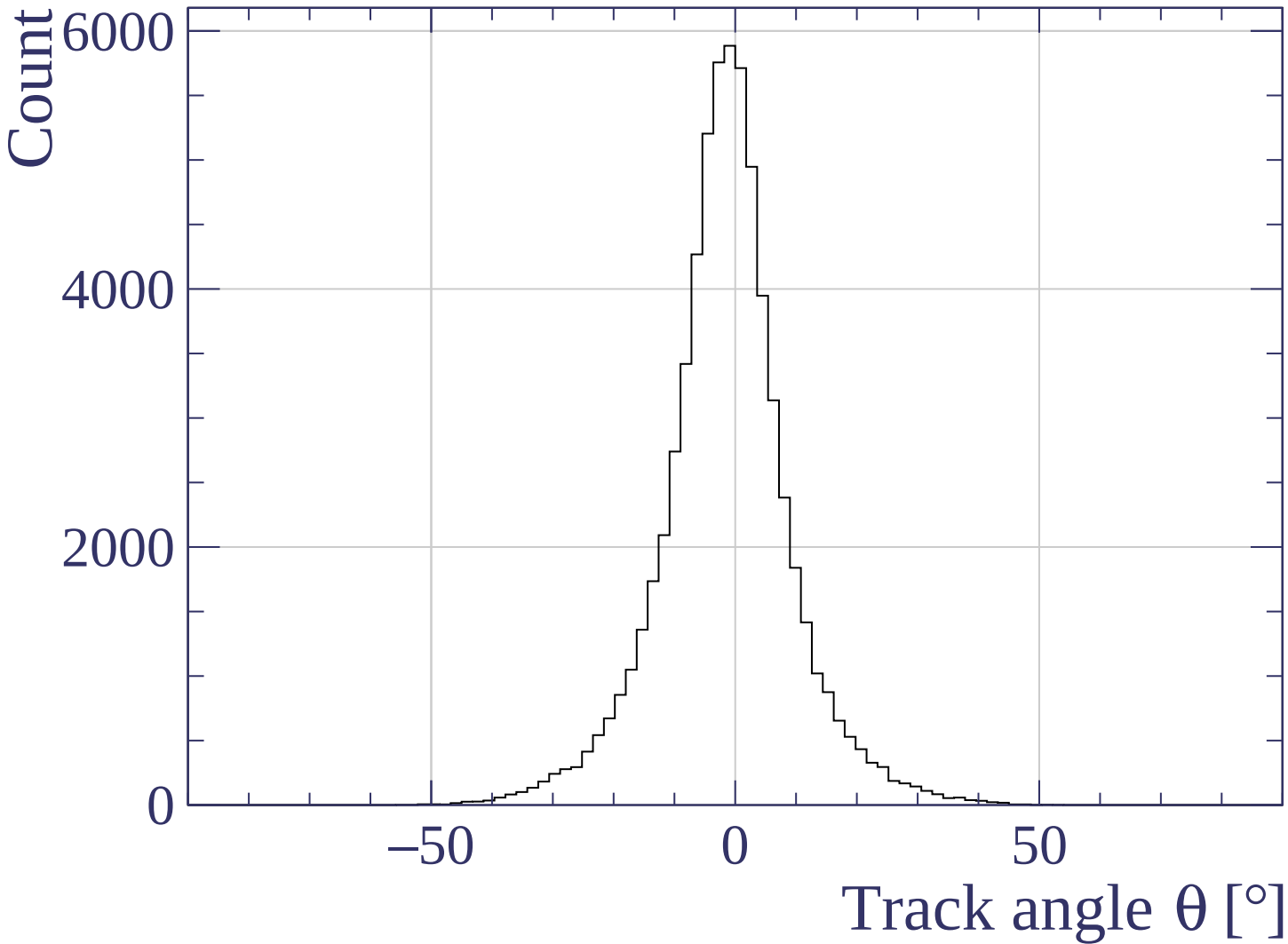




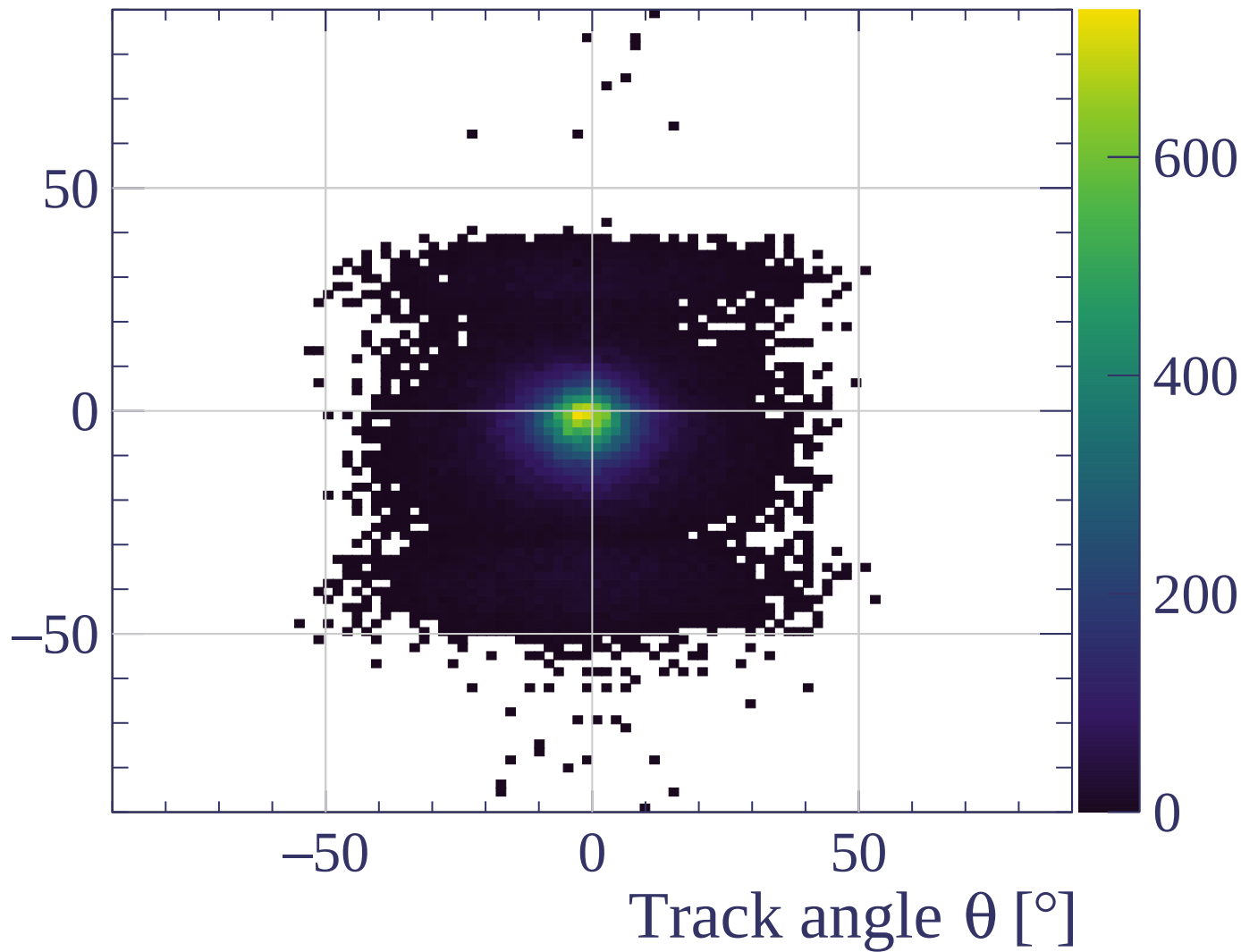
Track length [cm]



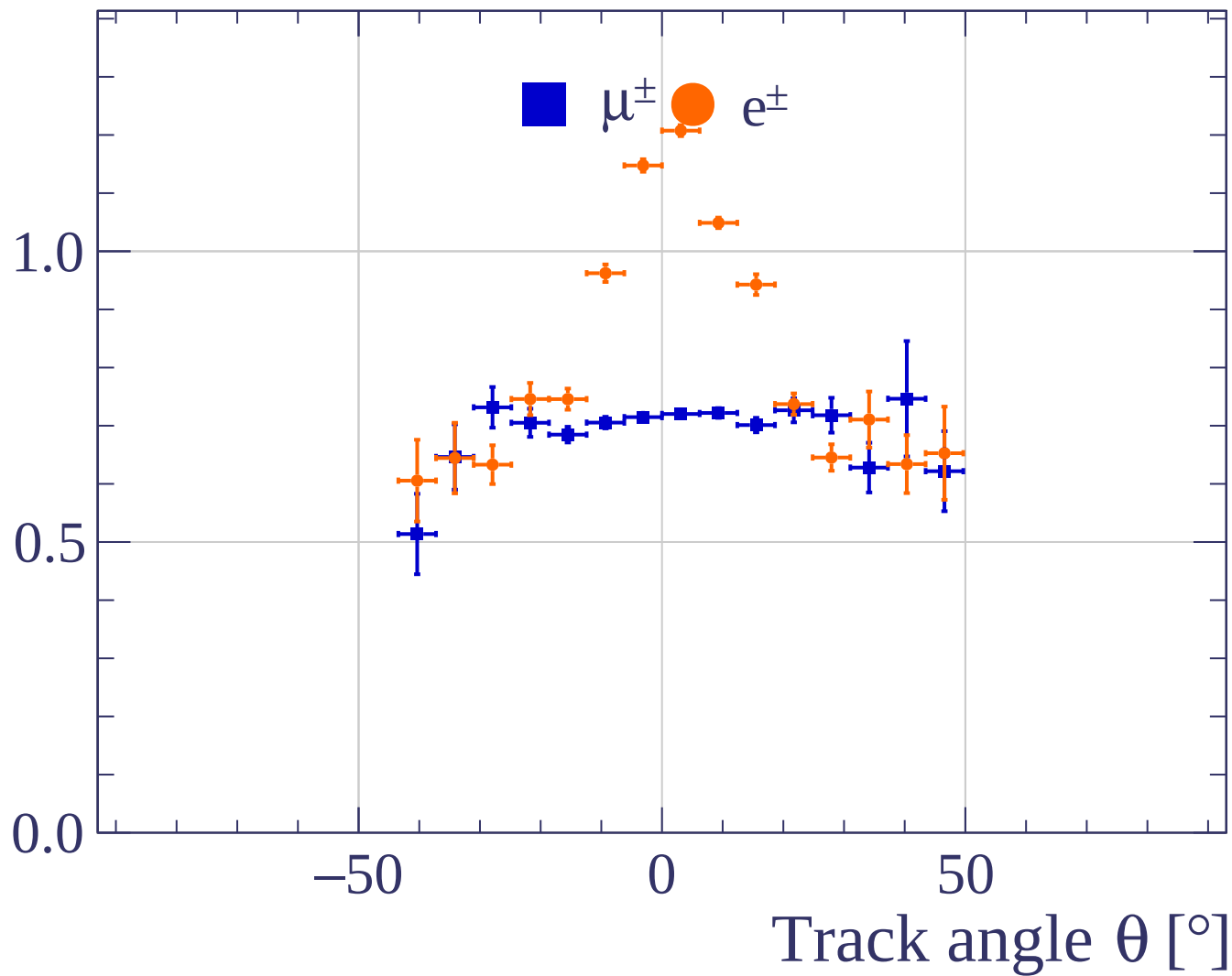




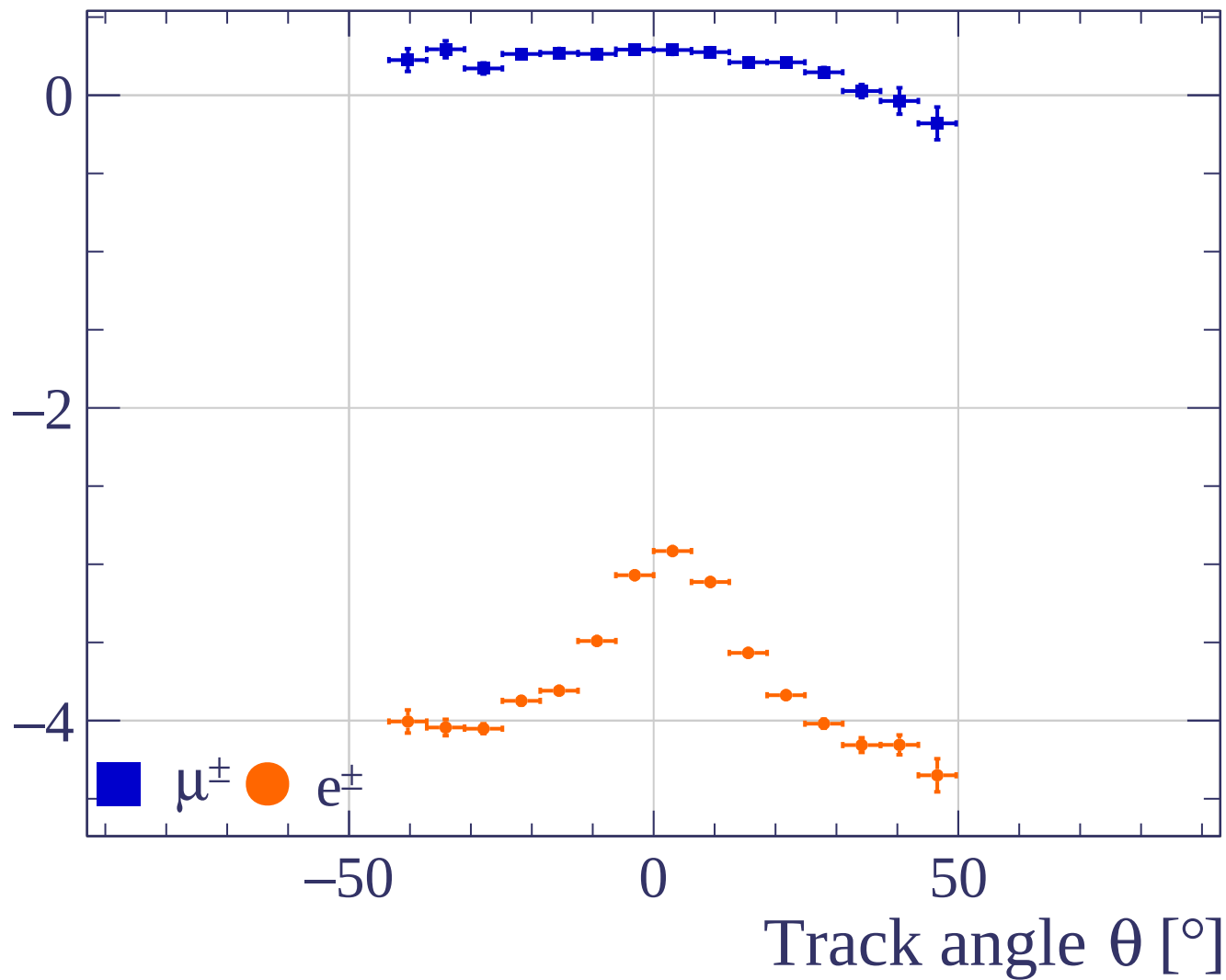
Track angle ϕ [°]



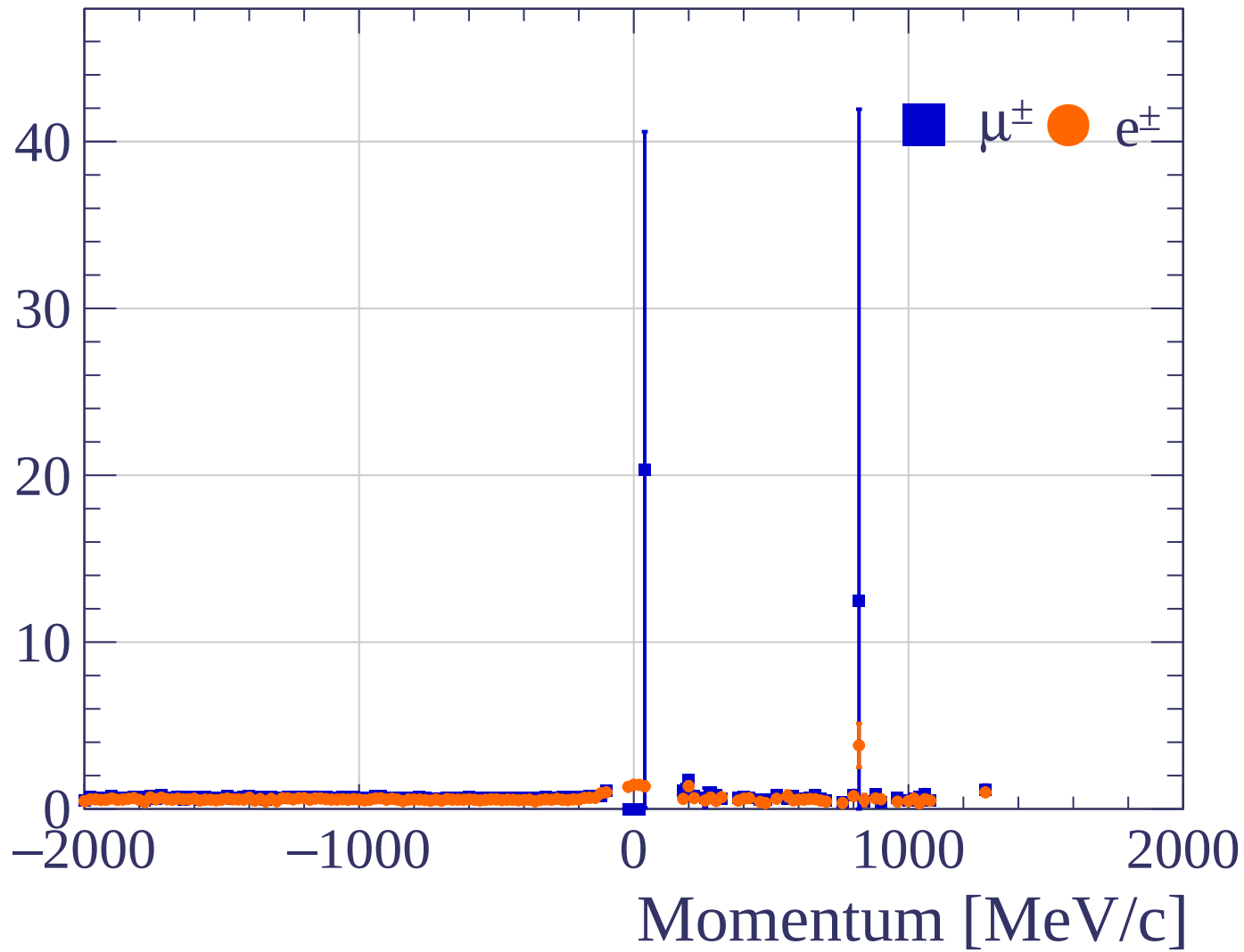
Pulls standard deviation



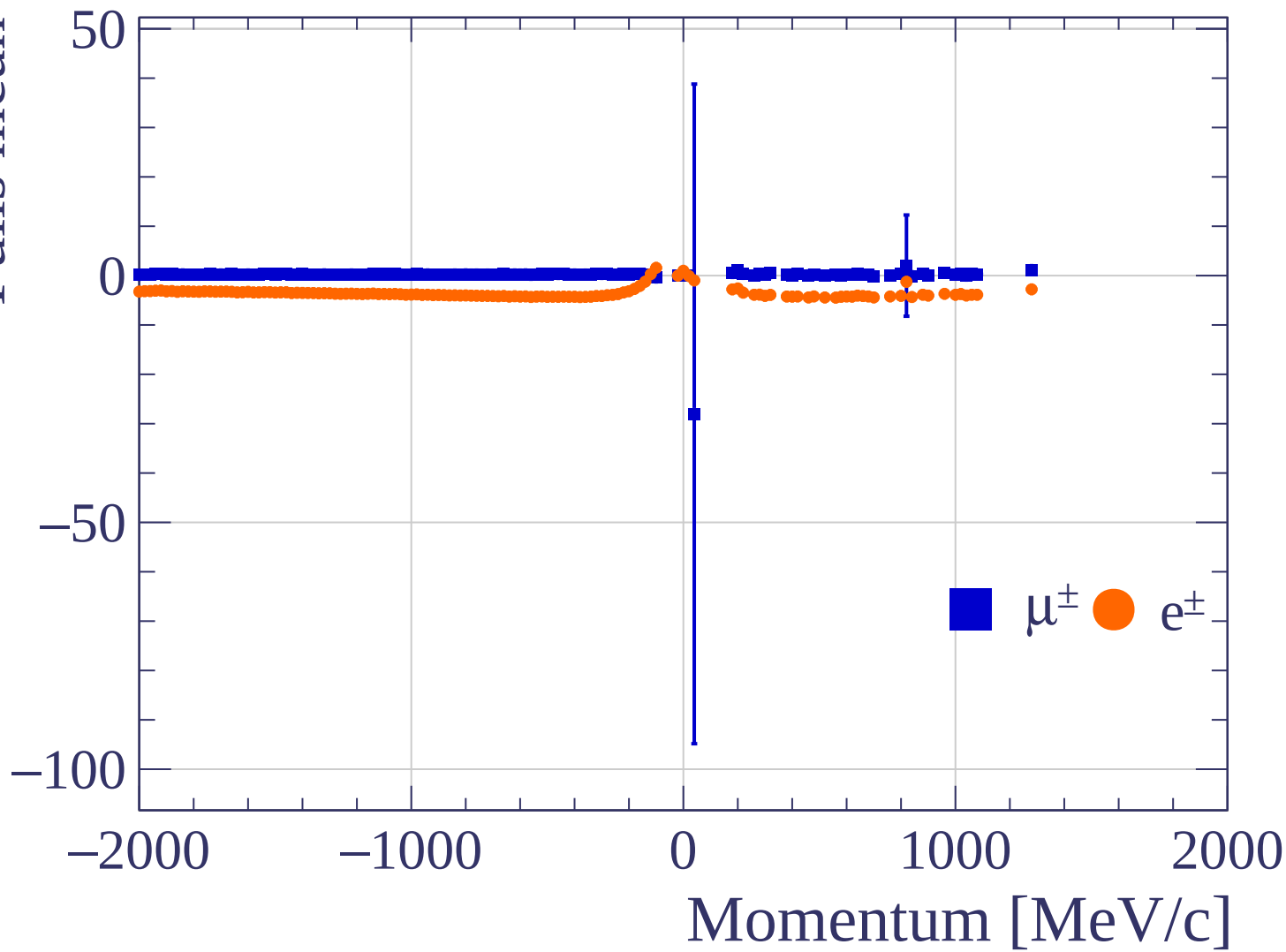
Pulls mean



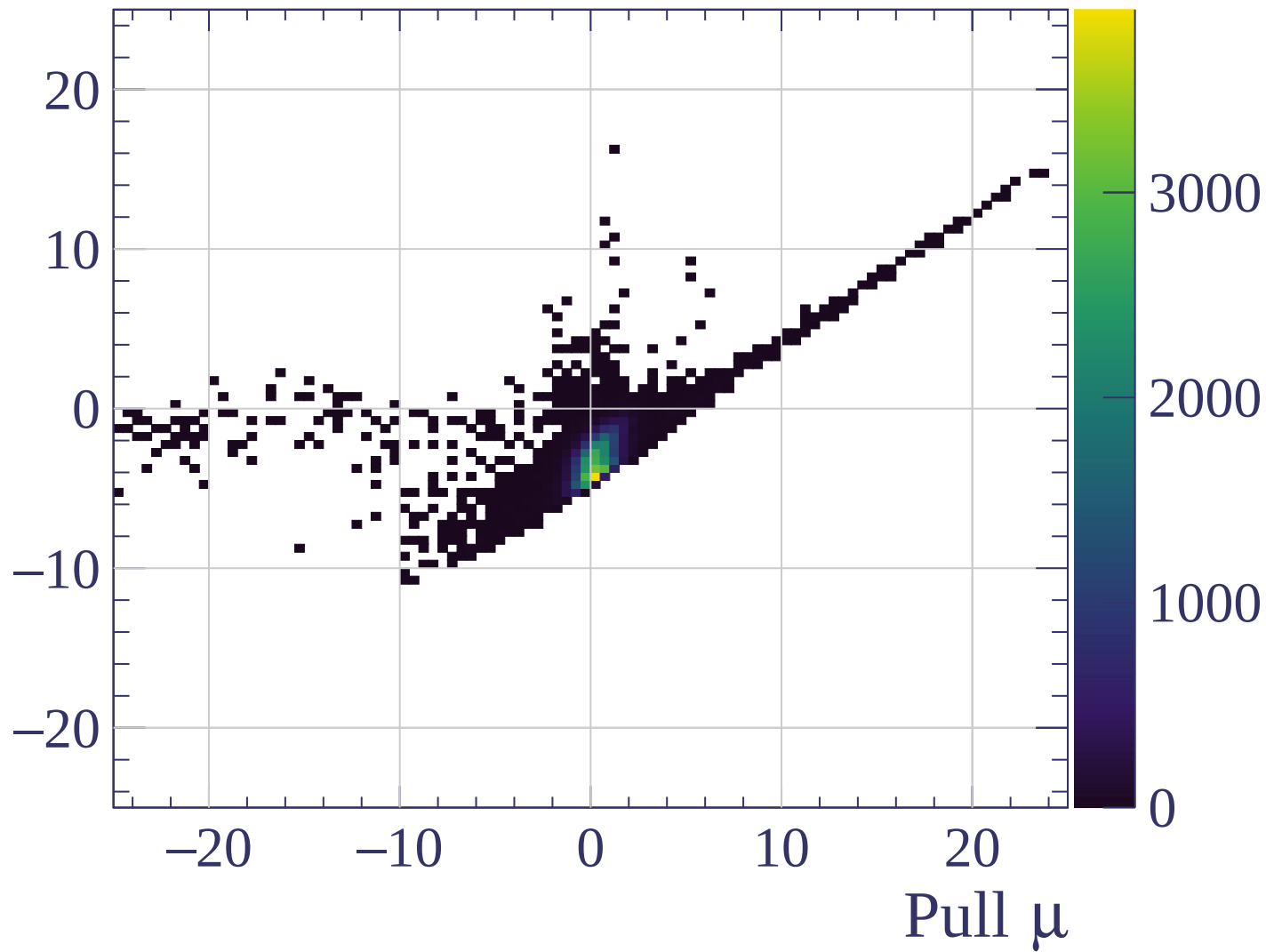
Pulls standard deviation

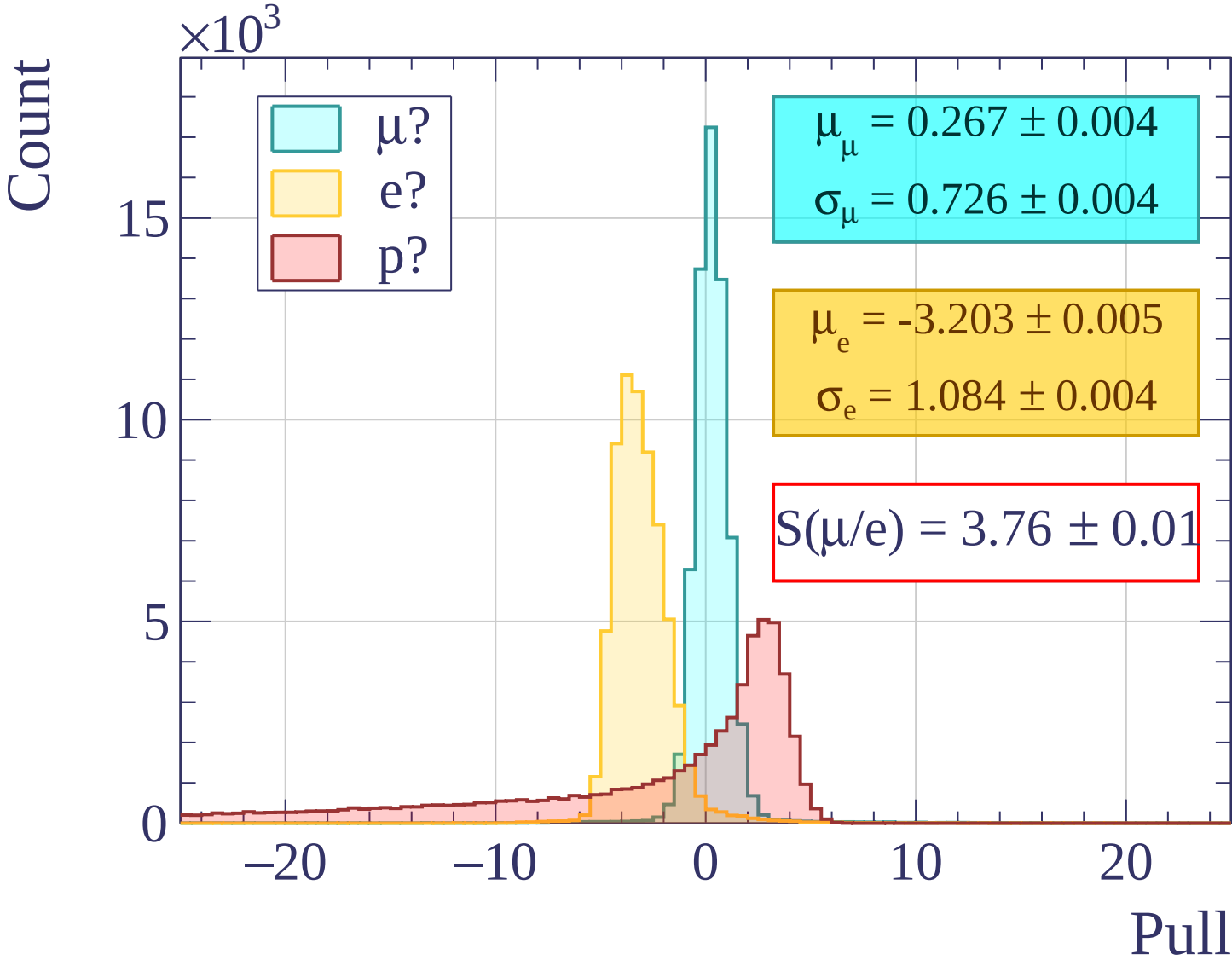


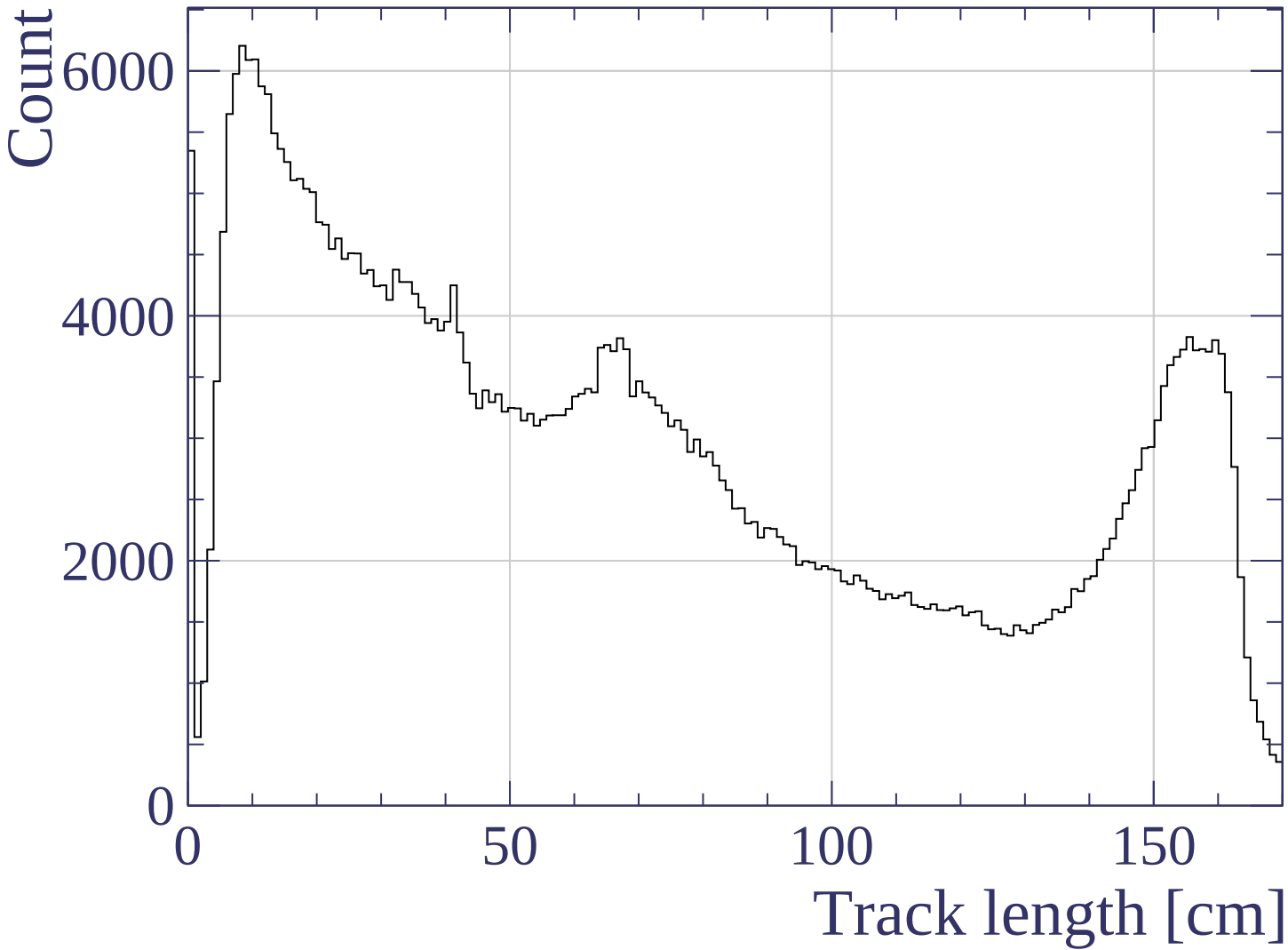
Pulls mean

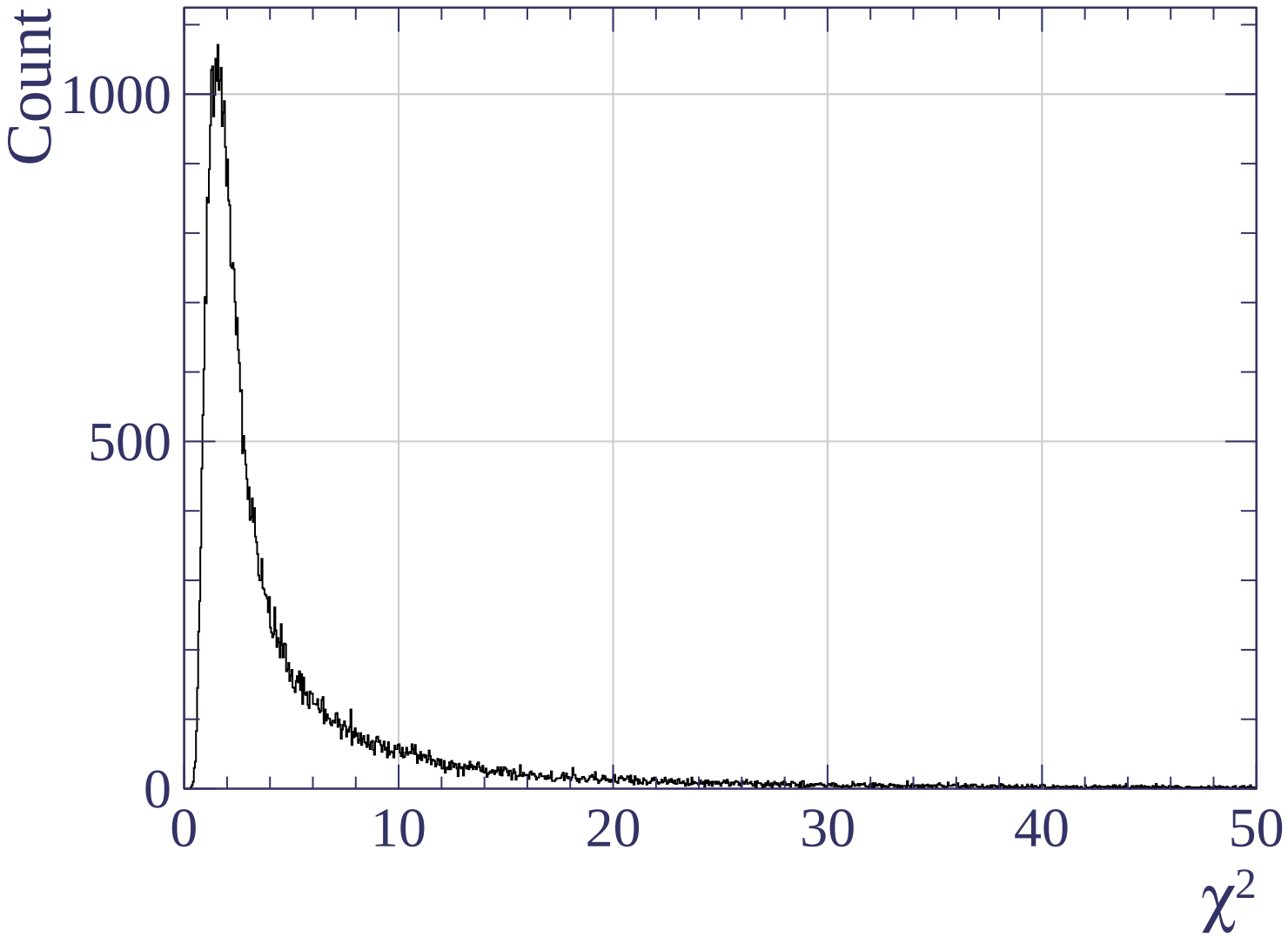


Pull e

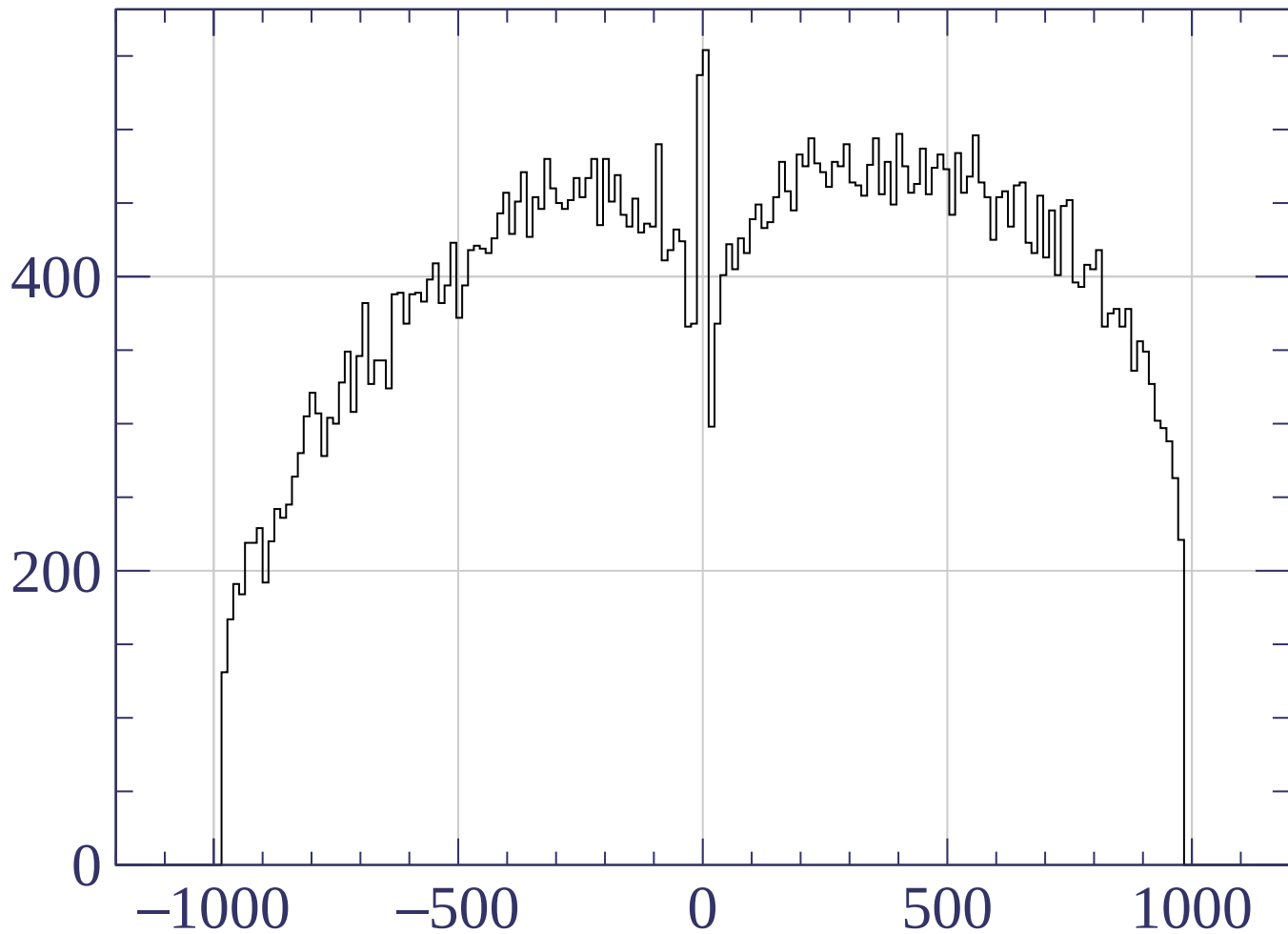




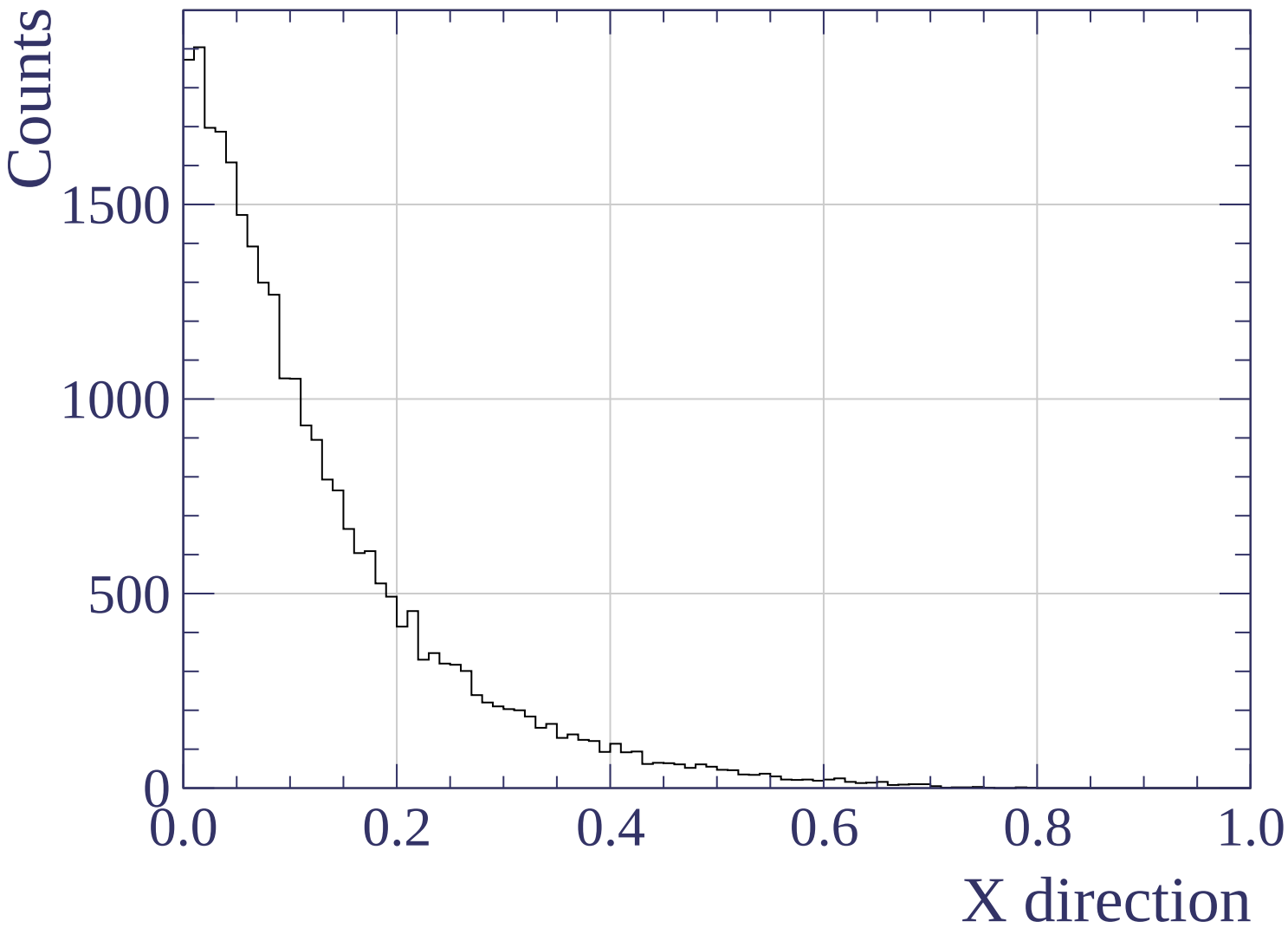


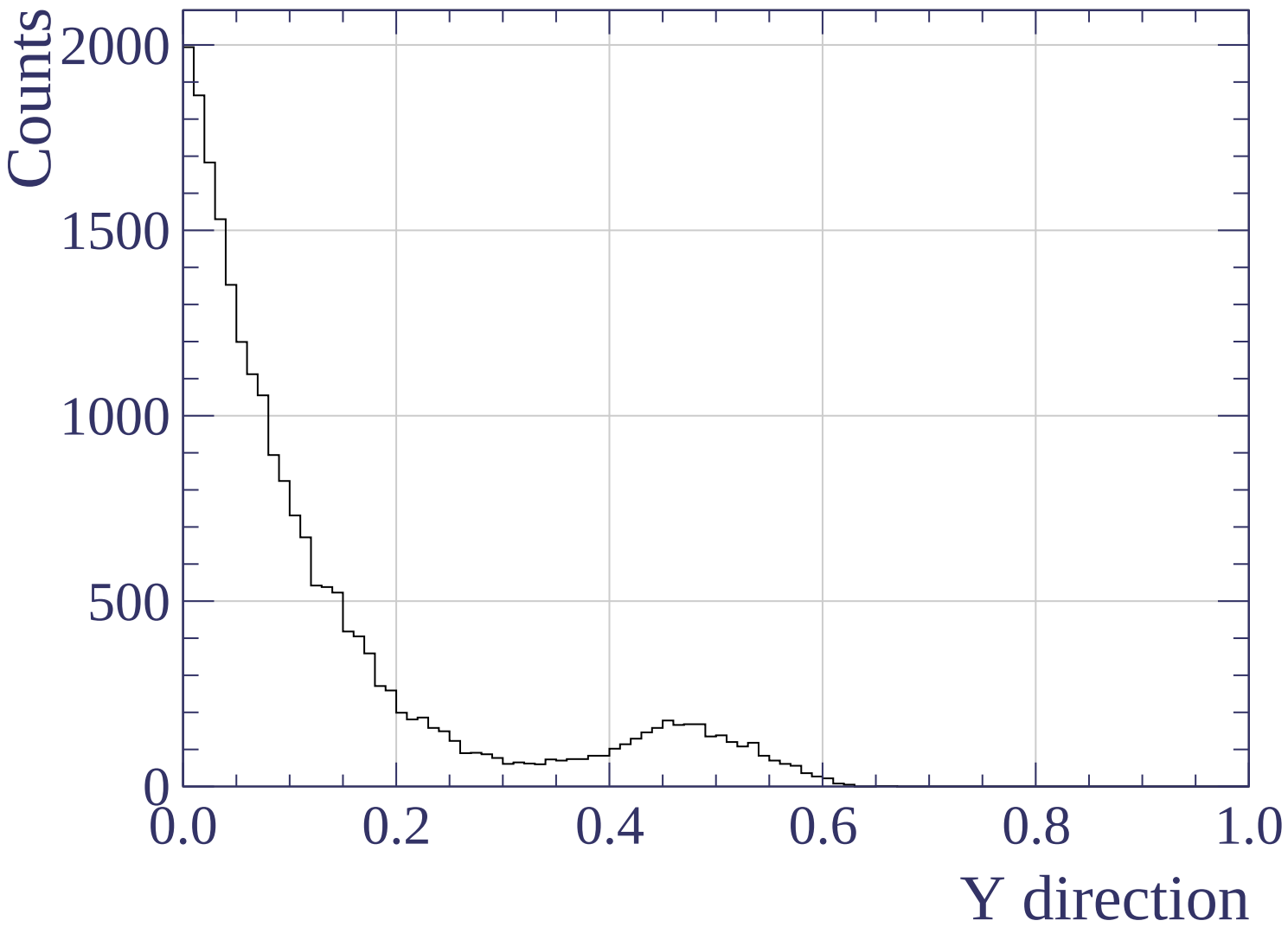


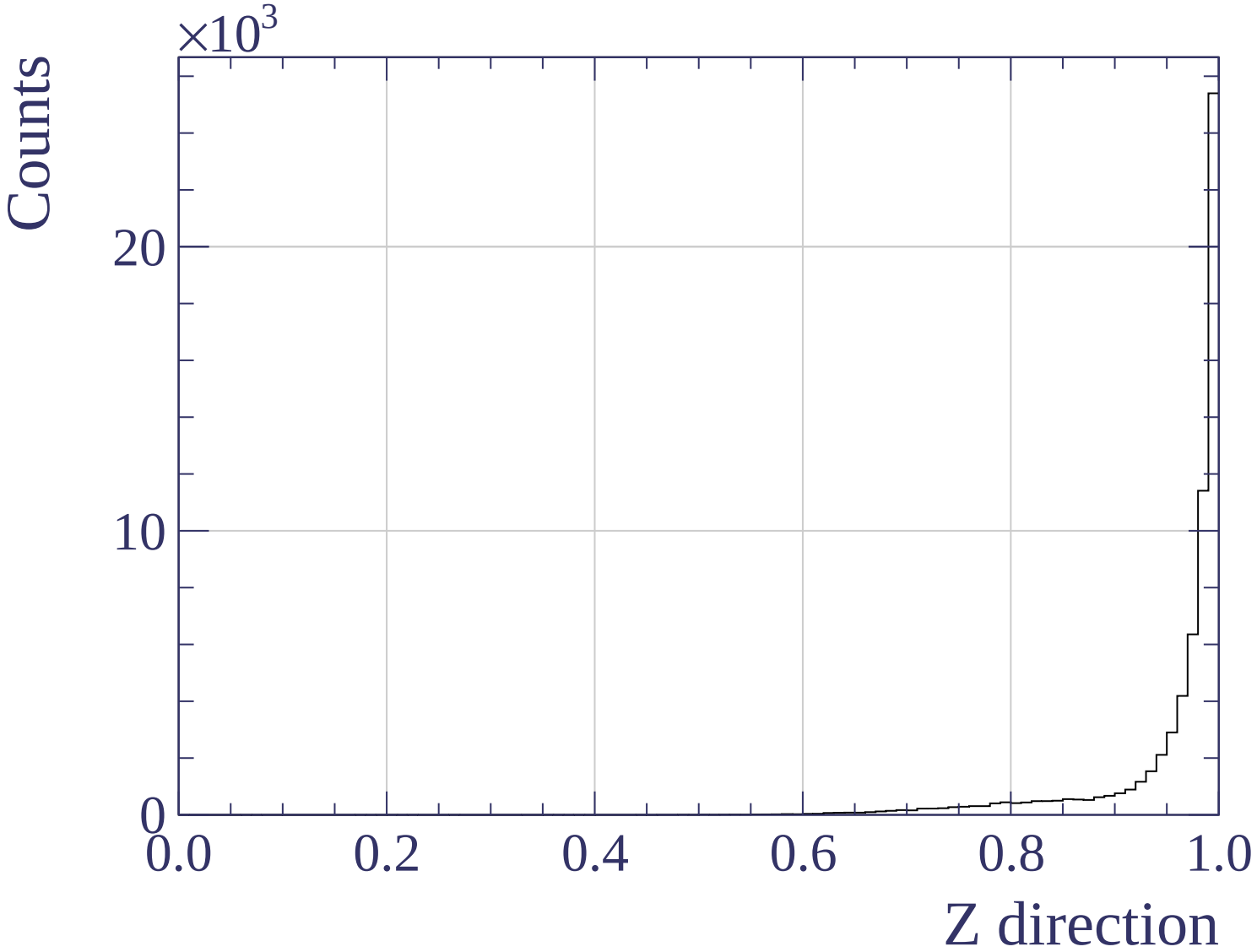
Count

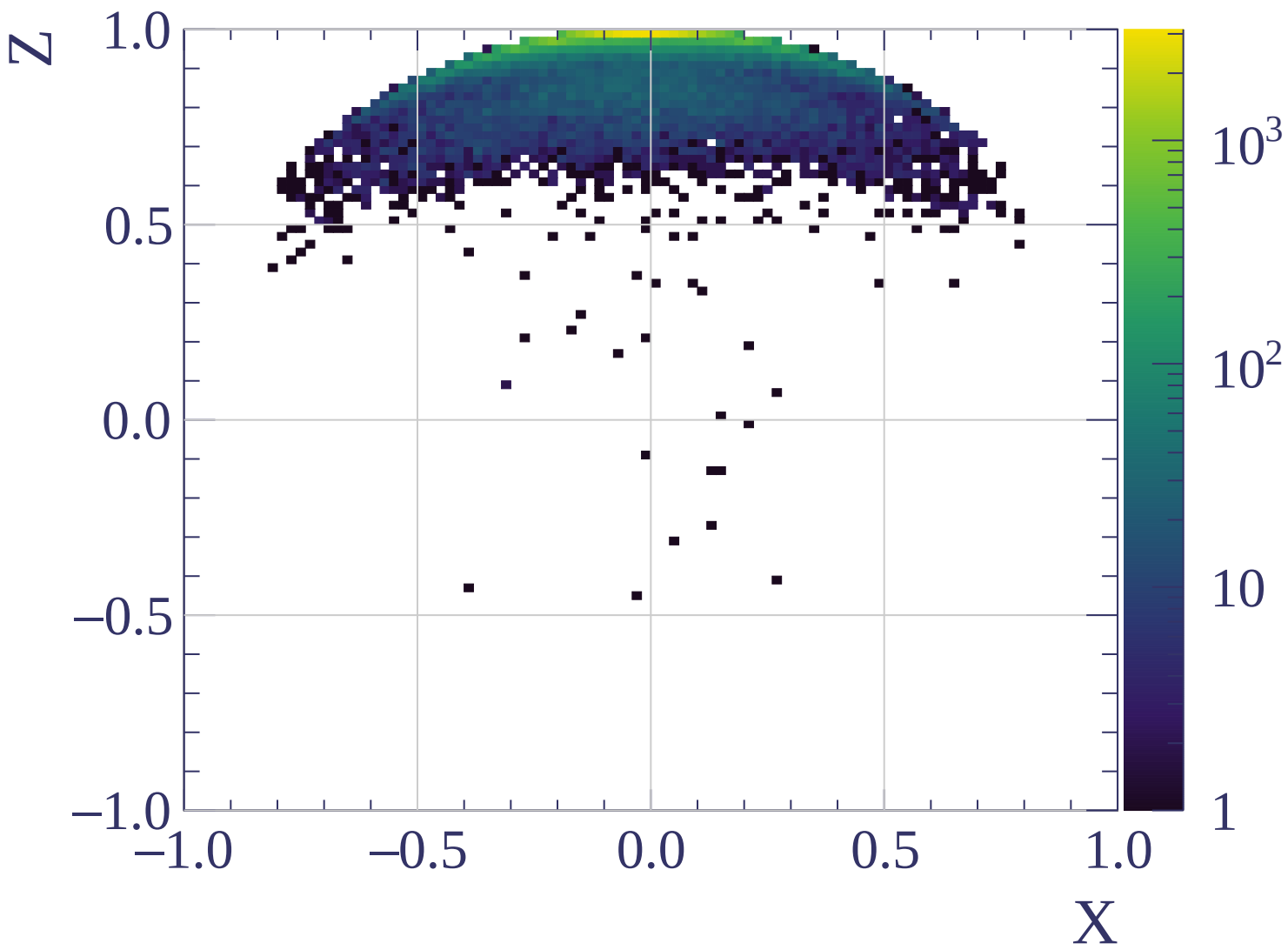


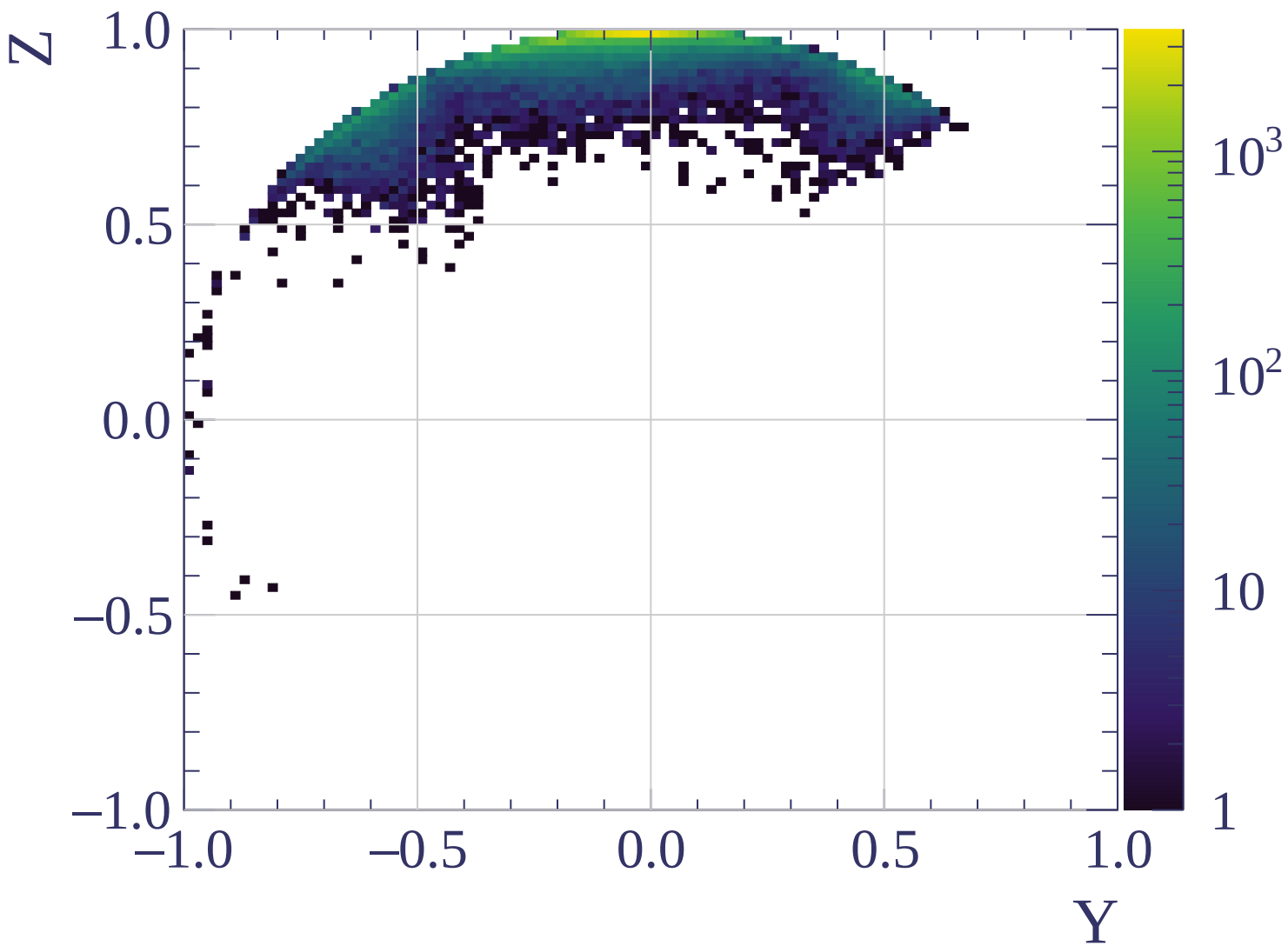
Track X position [mm]

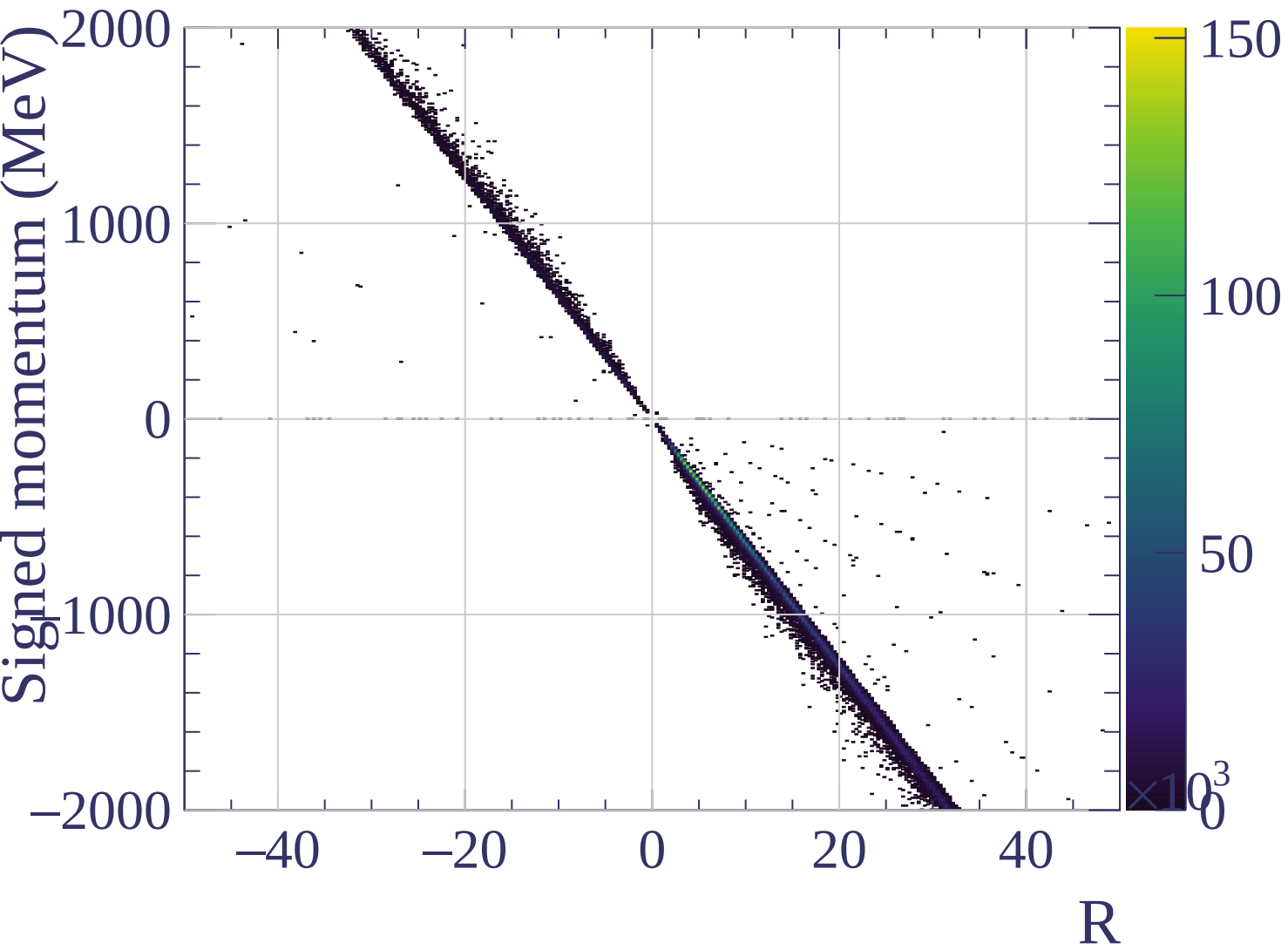


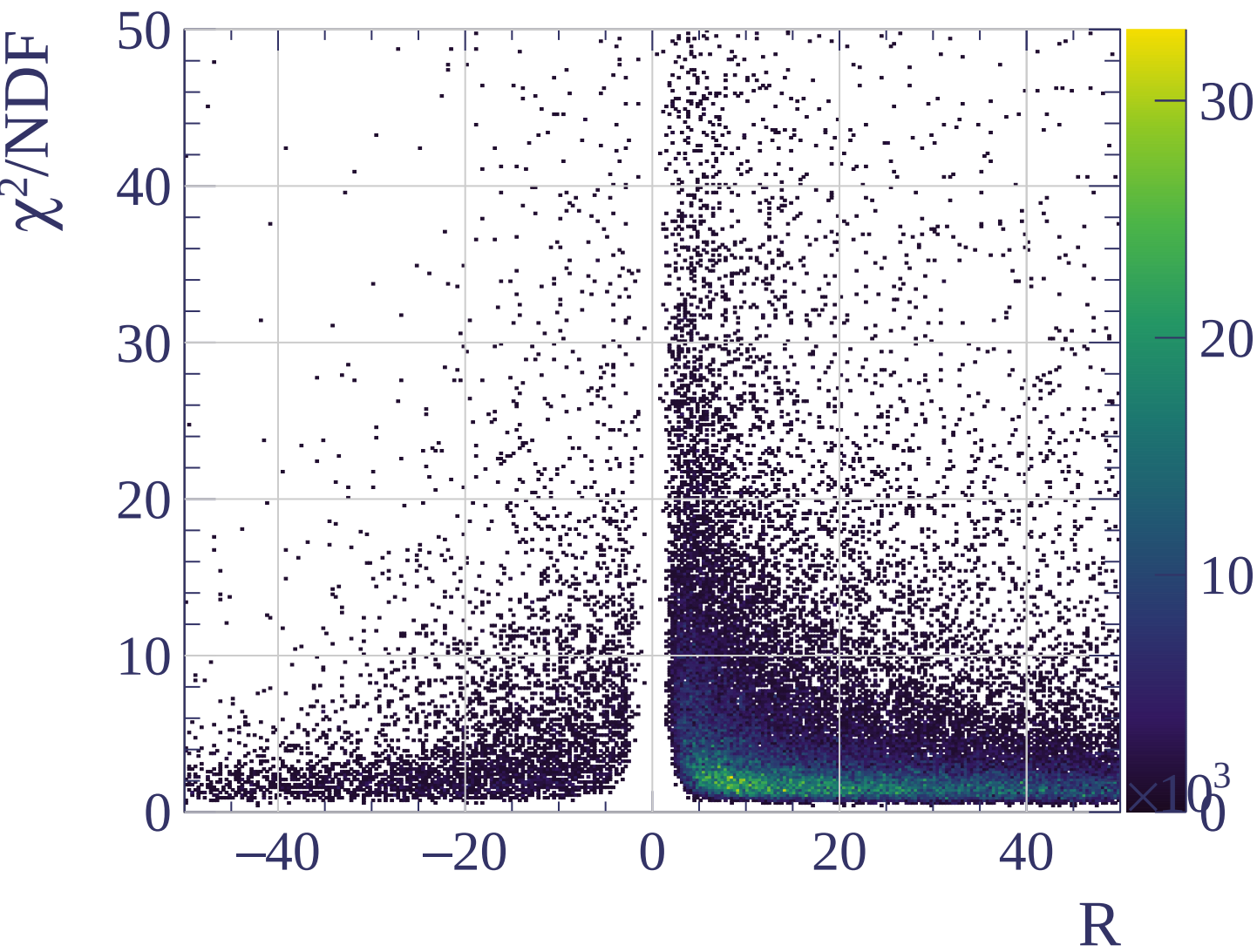




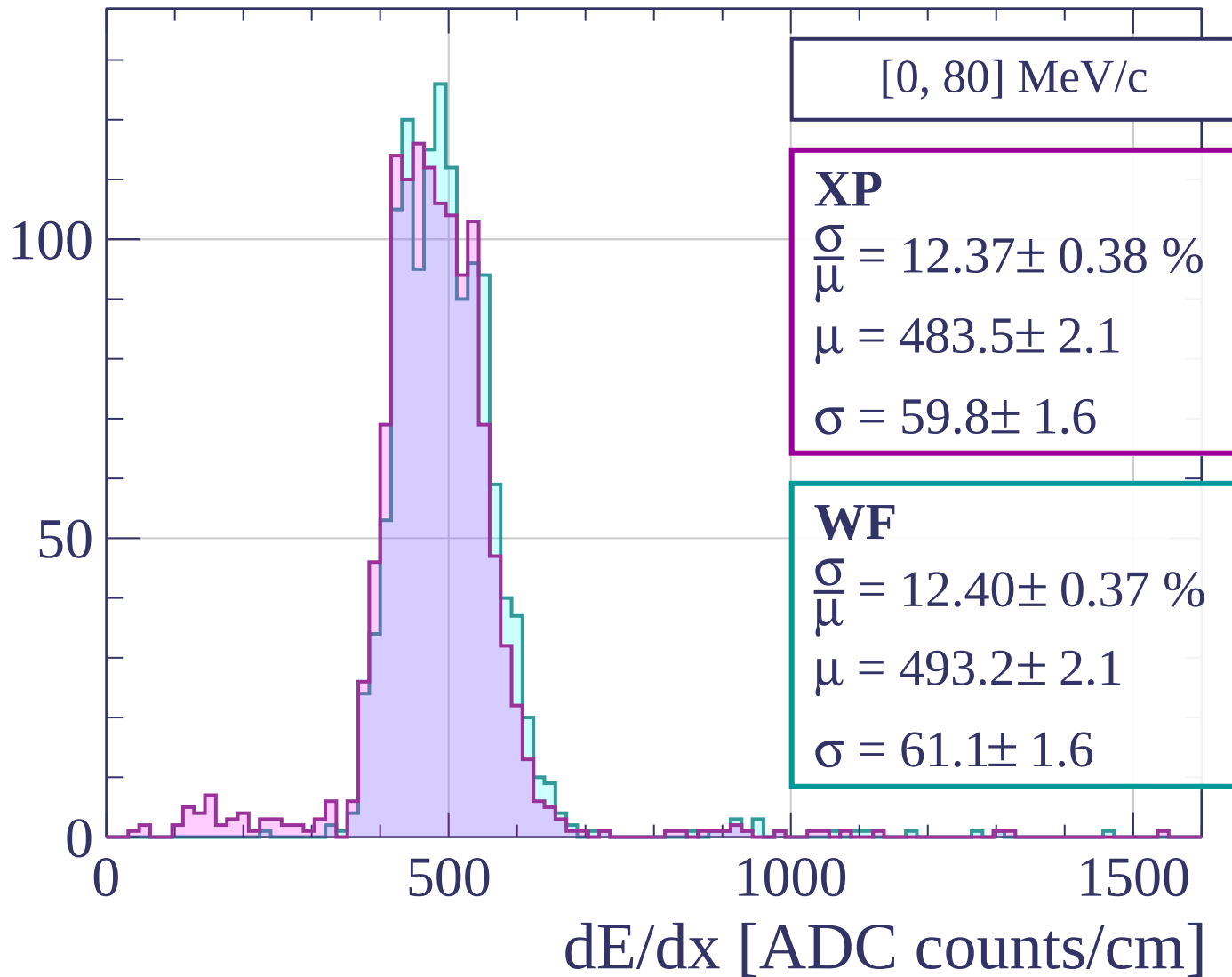




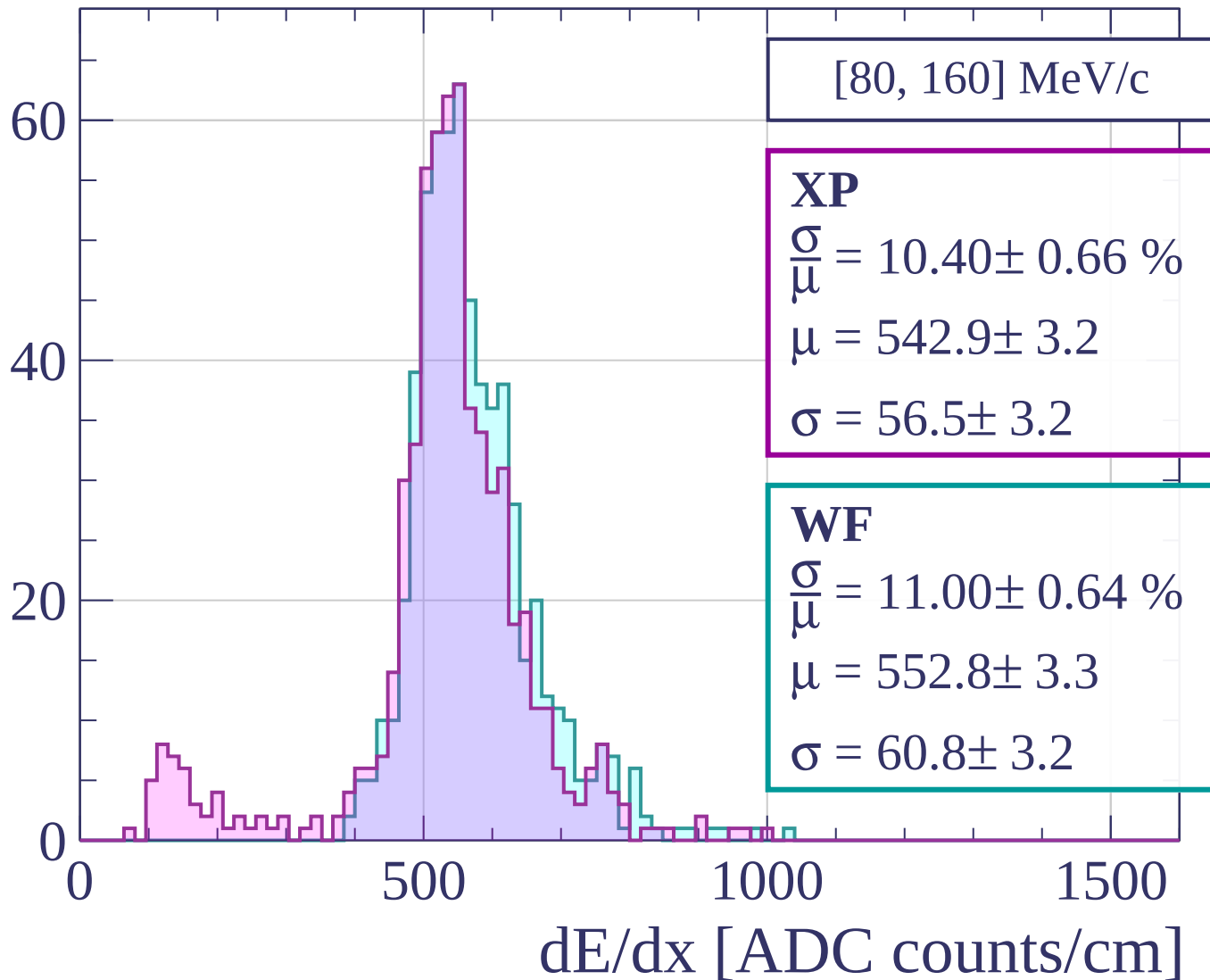


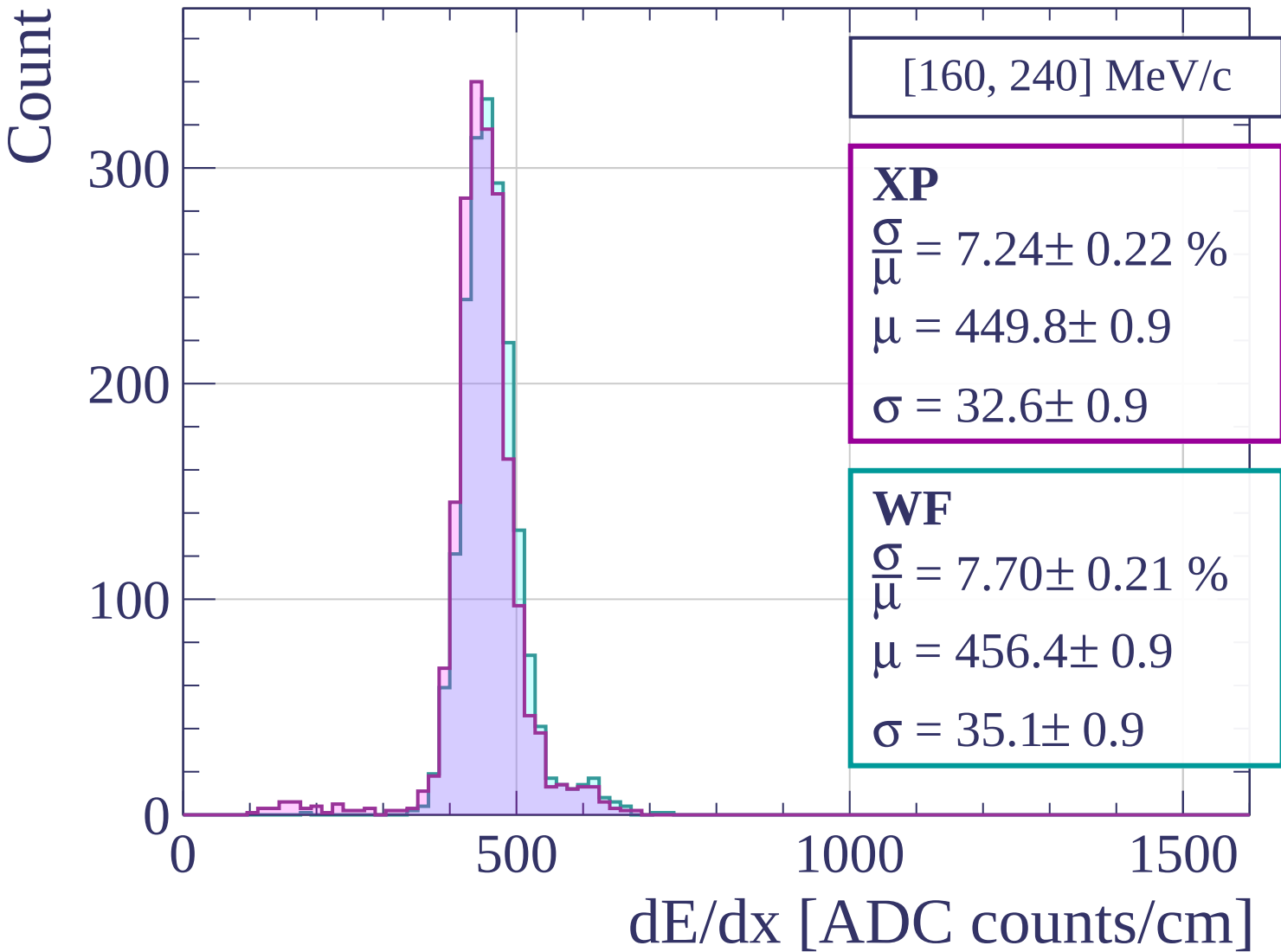


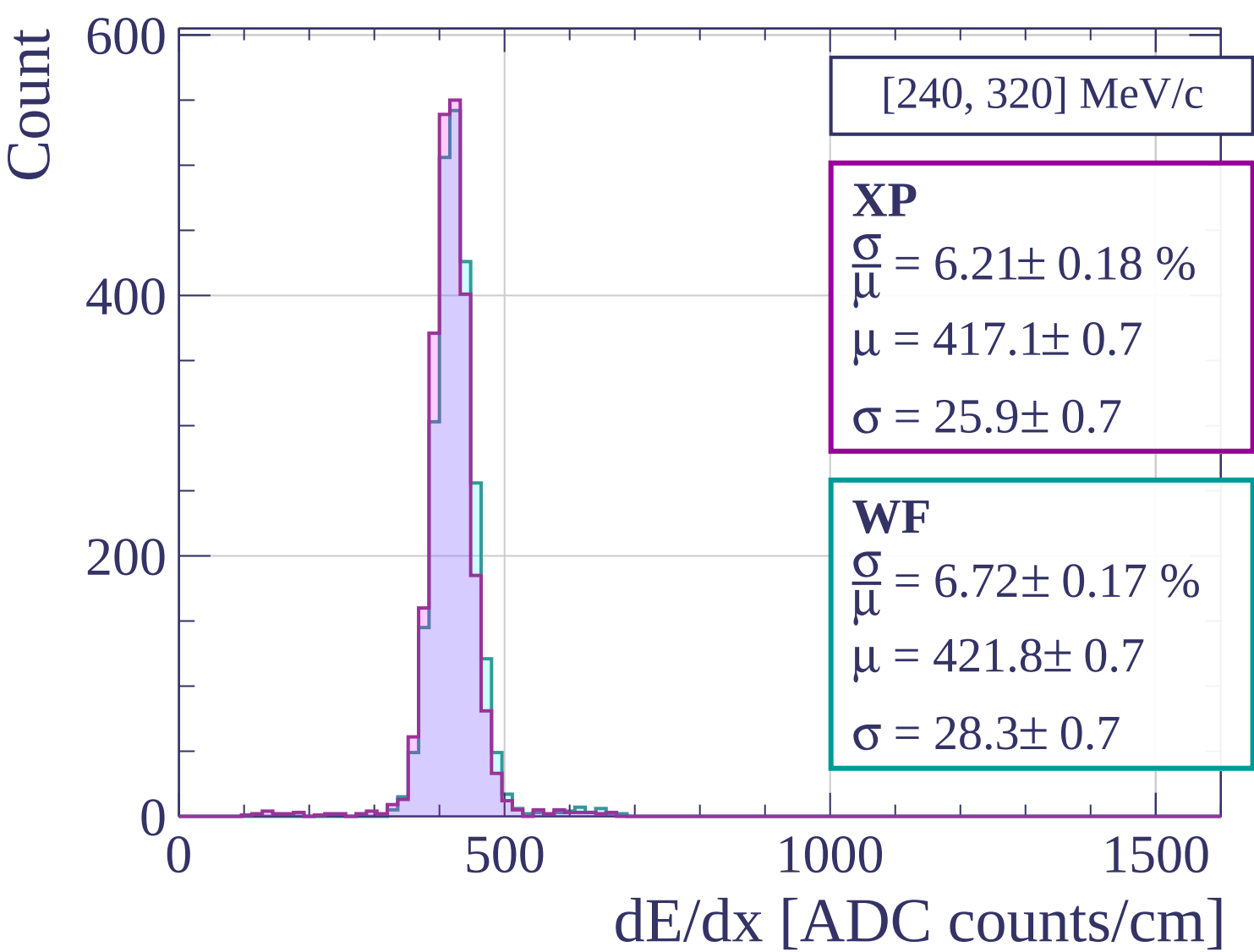
Count



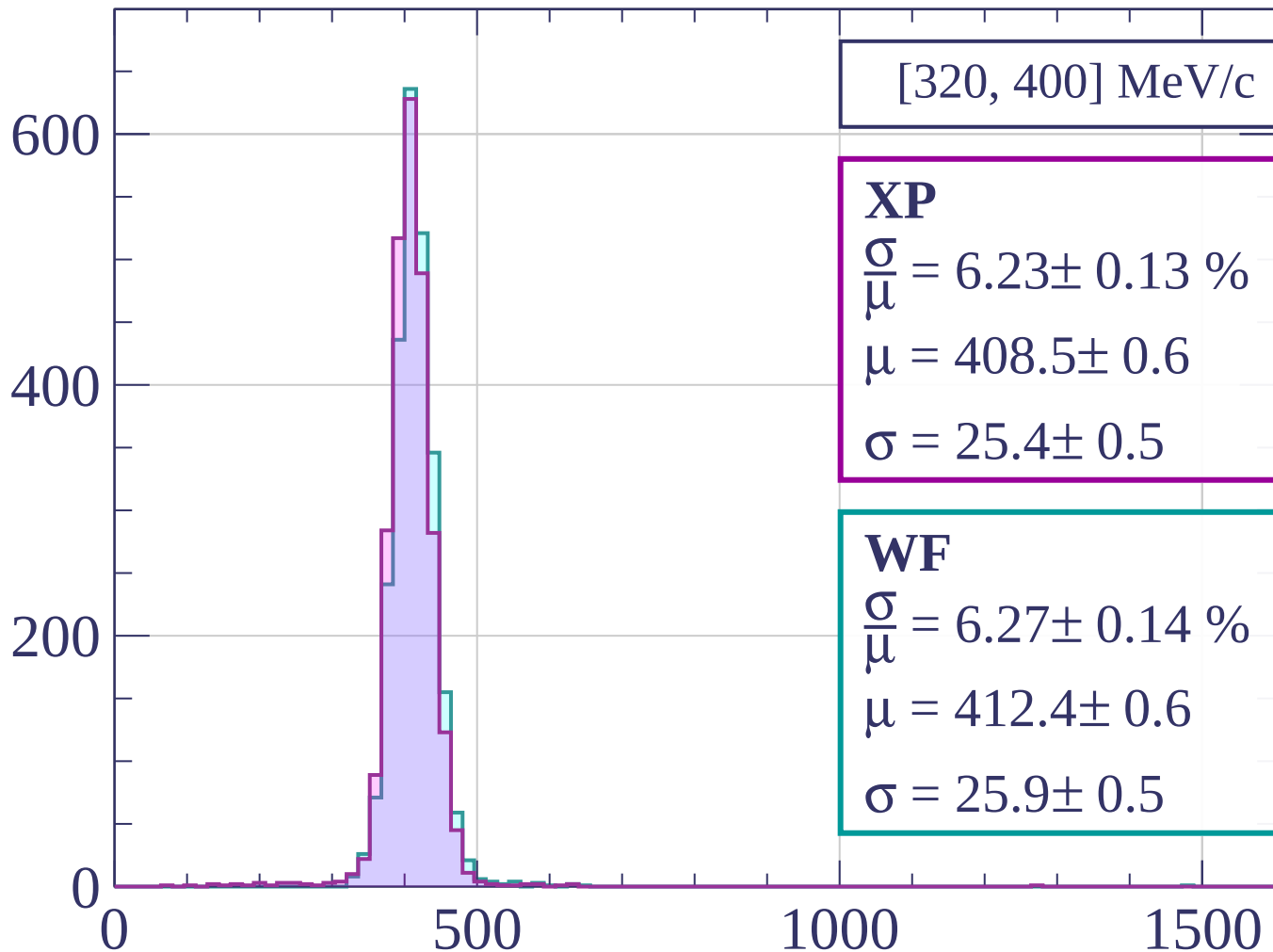
Count





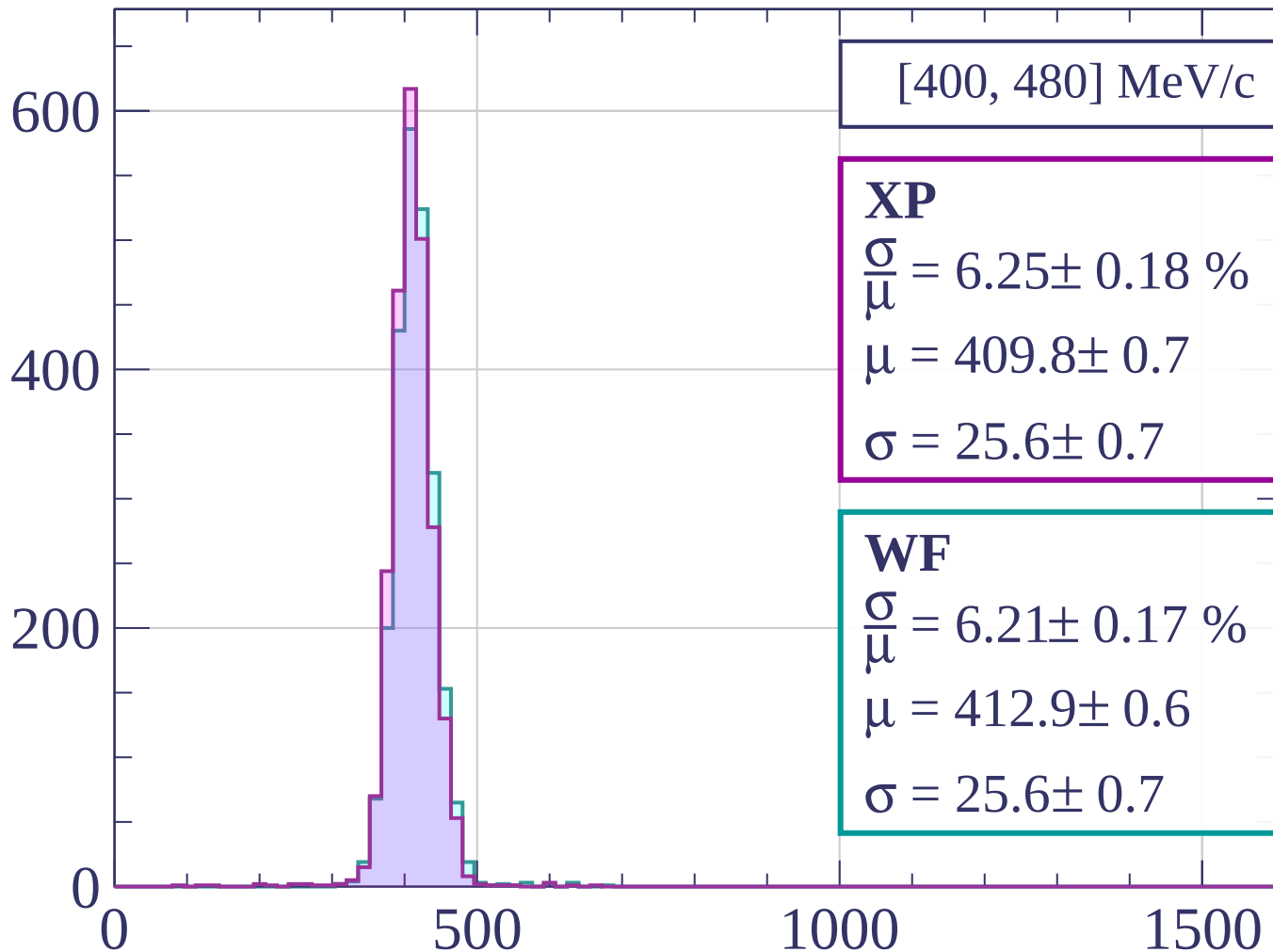


Count



dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count

[480, 560] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.26 \pm 0.18 \%$$

$$\mu = 412.9 \pm 0.7$$

$$\sigma = 25.8 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 6.44 \pm 0.20 \%$$

$$\mu = 416.7 \pm 0.7$$

$$\sigma = 26.8 \pm 0.8$$

400

200

0

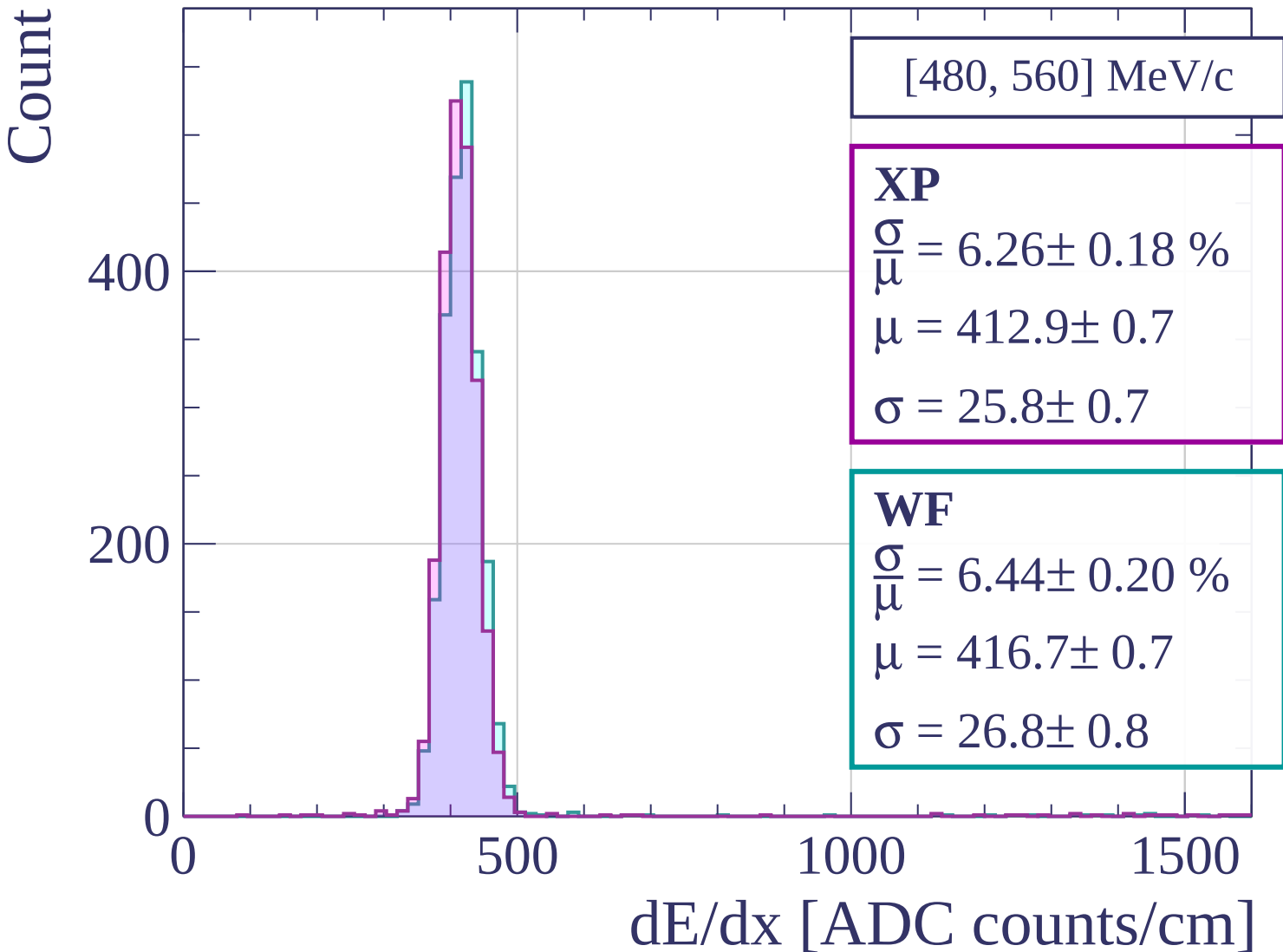
0

500

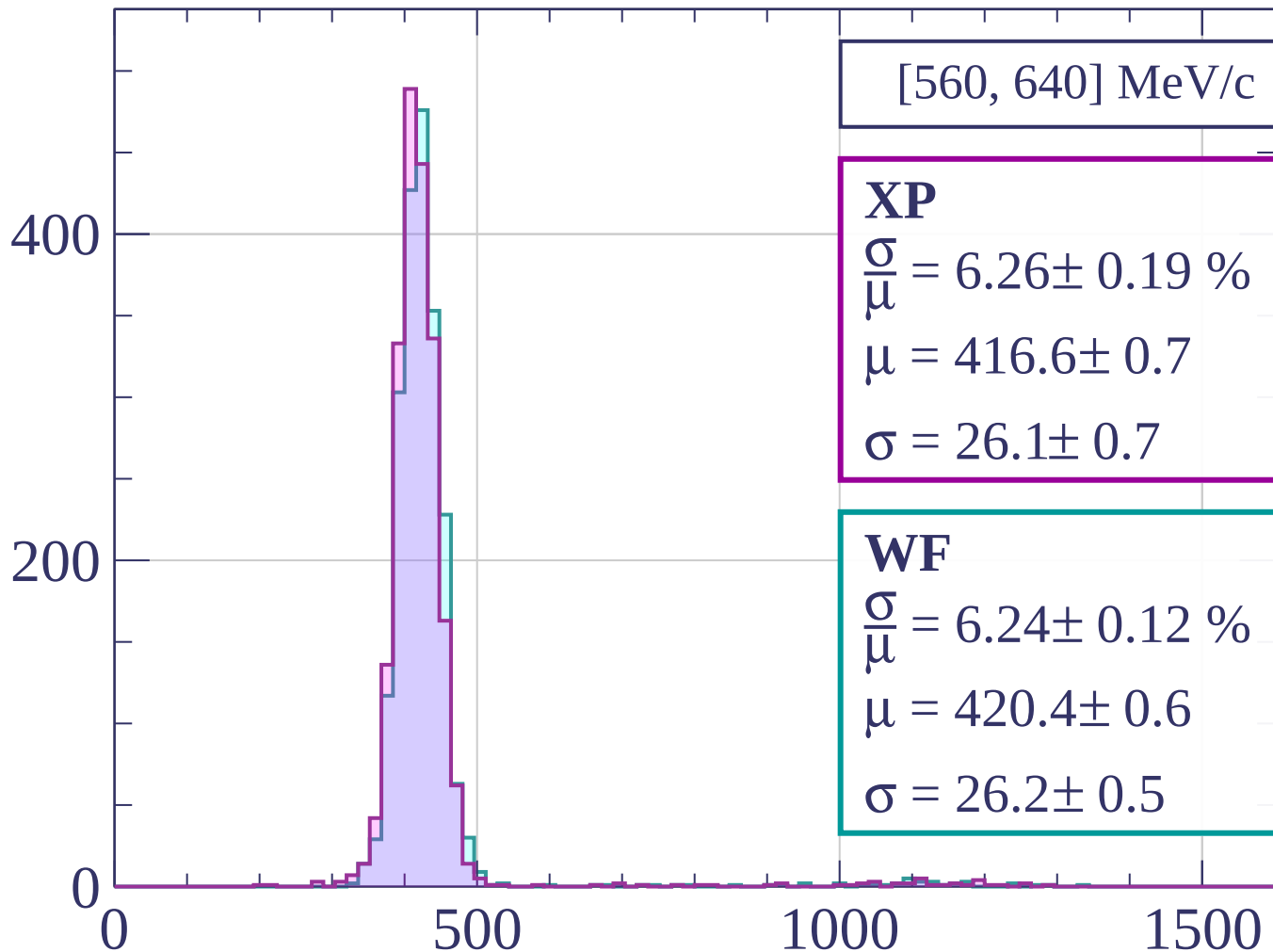
1000

1500

dE/dx [ADC counts/cm]



Count



dE/dx [ADC counts/cm]

Count

[640, 720] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.19 \pm 0.16 \%$$

$$\mu = 421.7 \pm 0.7$$

$$\sigma = 26.1 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 6.25 \pm 0.16 \%$$

$$\mu = 425.4 \pm 0.7$$

$$\sigma = 26.6 \pm 0.6$$

400

200

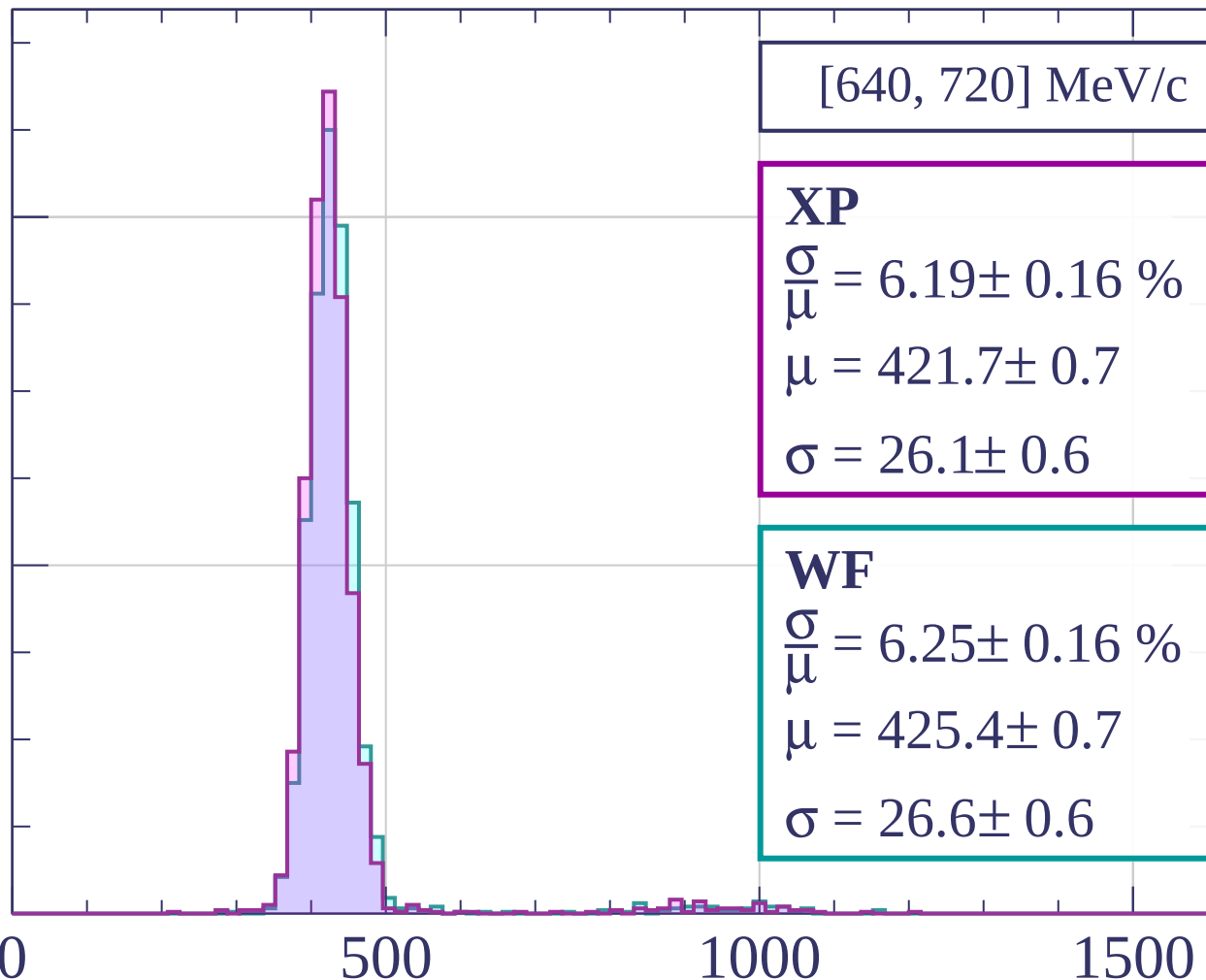
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[720, 800] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.69 \pm 0.17 \%$$

$$\mu = 427.9 \pm 0.7$$

$$\sigma = 24.4 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 5.97 \pm 0.20 \%$$

$$\mu = 430.7 \pm 0.8$$

$$\sigma = 25.7 \pm 0.8$$

400

300

200

100

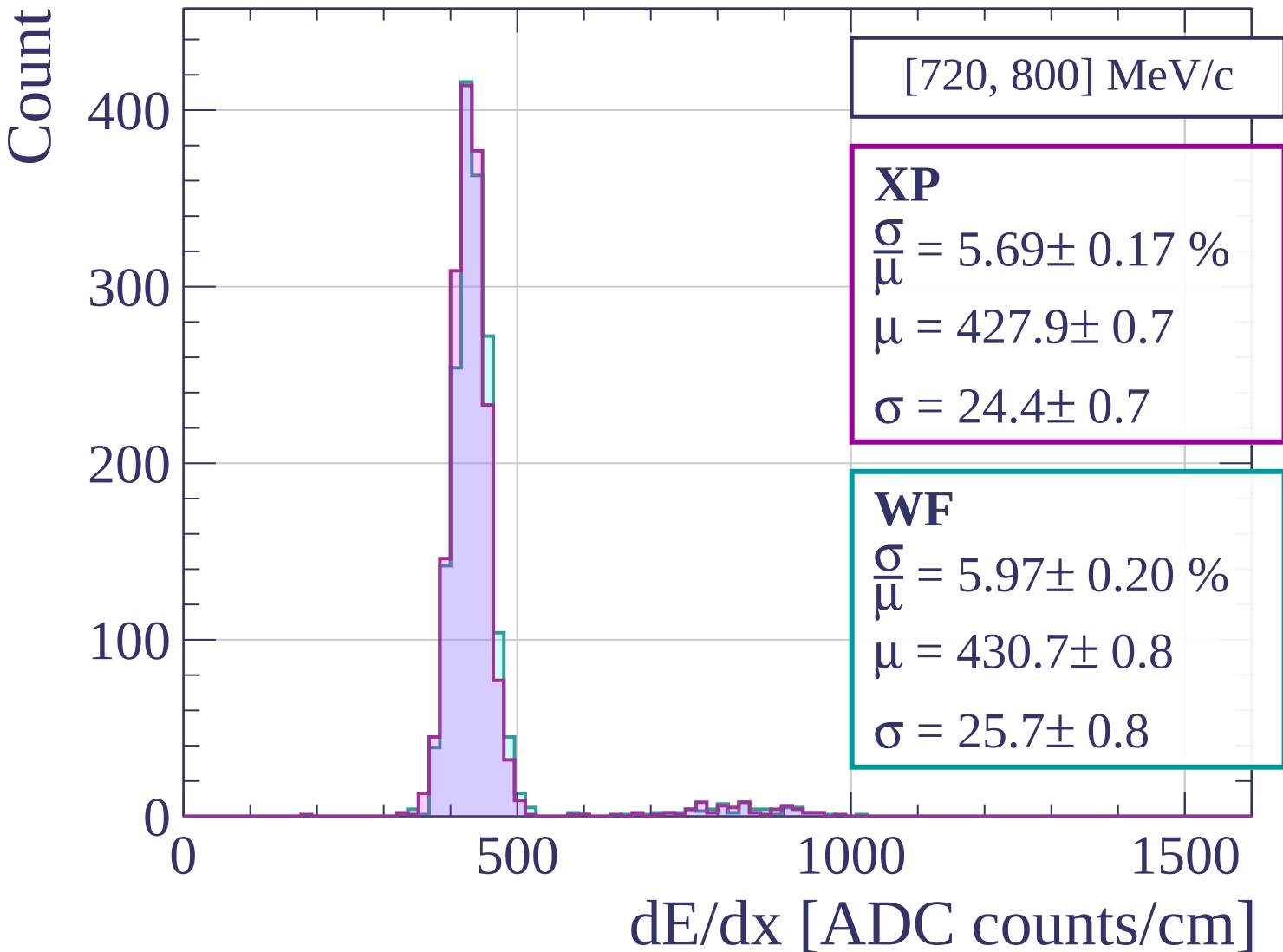
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[800, 880] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.22 \pm 0.24 \%$$

$$\mu = 431.3 \pm 0.9$$

$$\sigma = 26.8 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 6.12 \pm 0.18 \%$$

$$\mu = 435.2 \pm 0.8$$

$$\sigma = 26.6 \pm 0.7$$

300

200

100

0

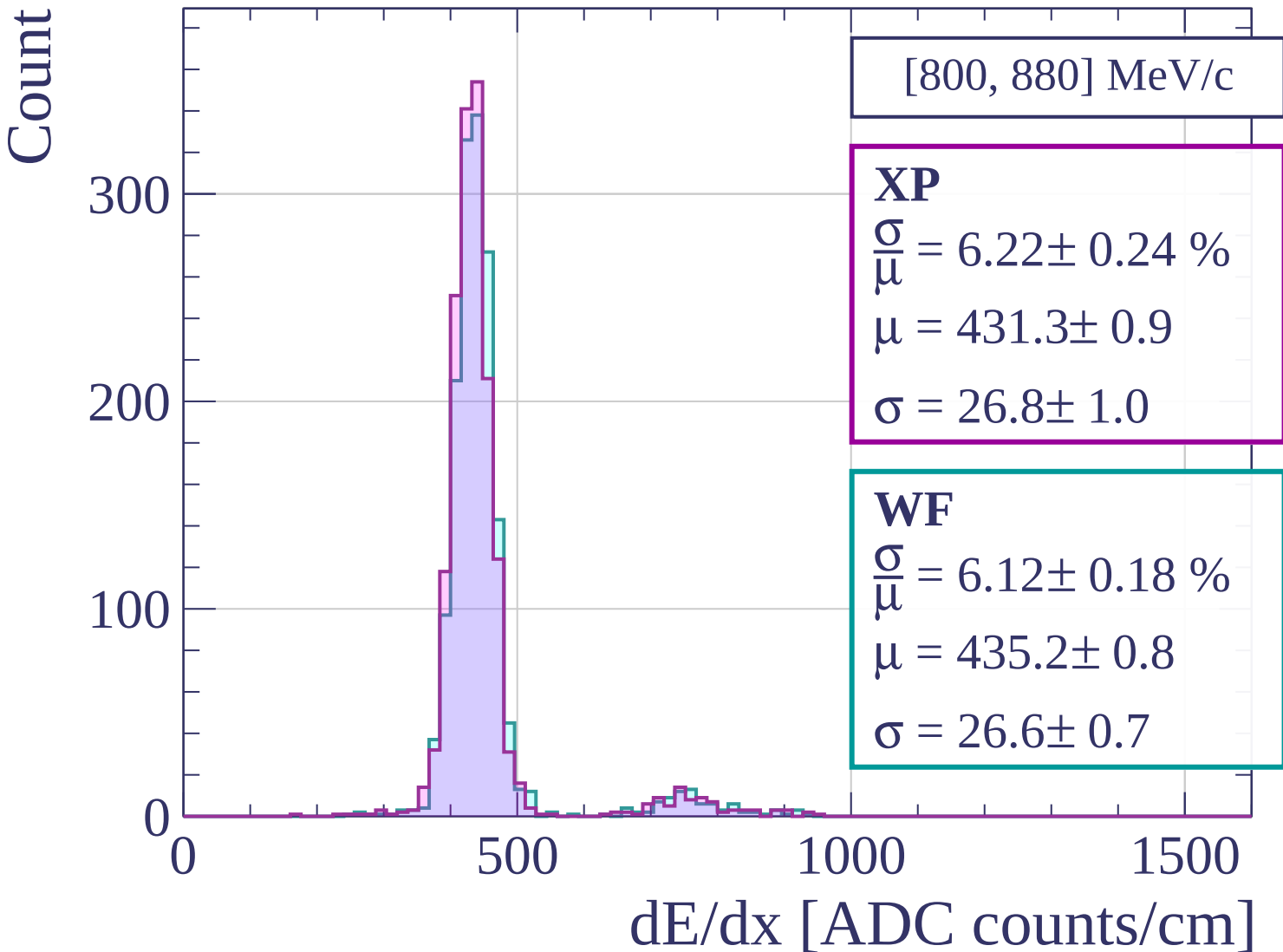
0

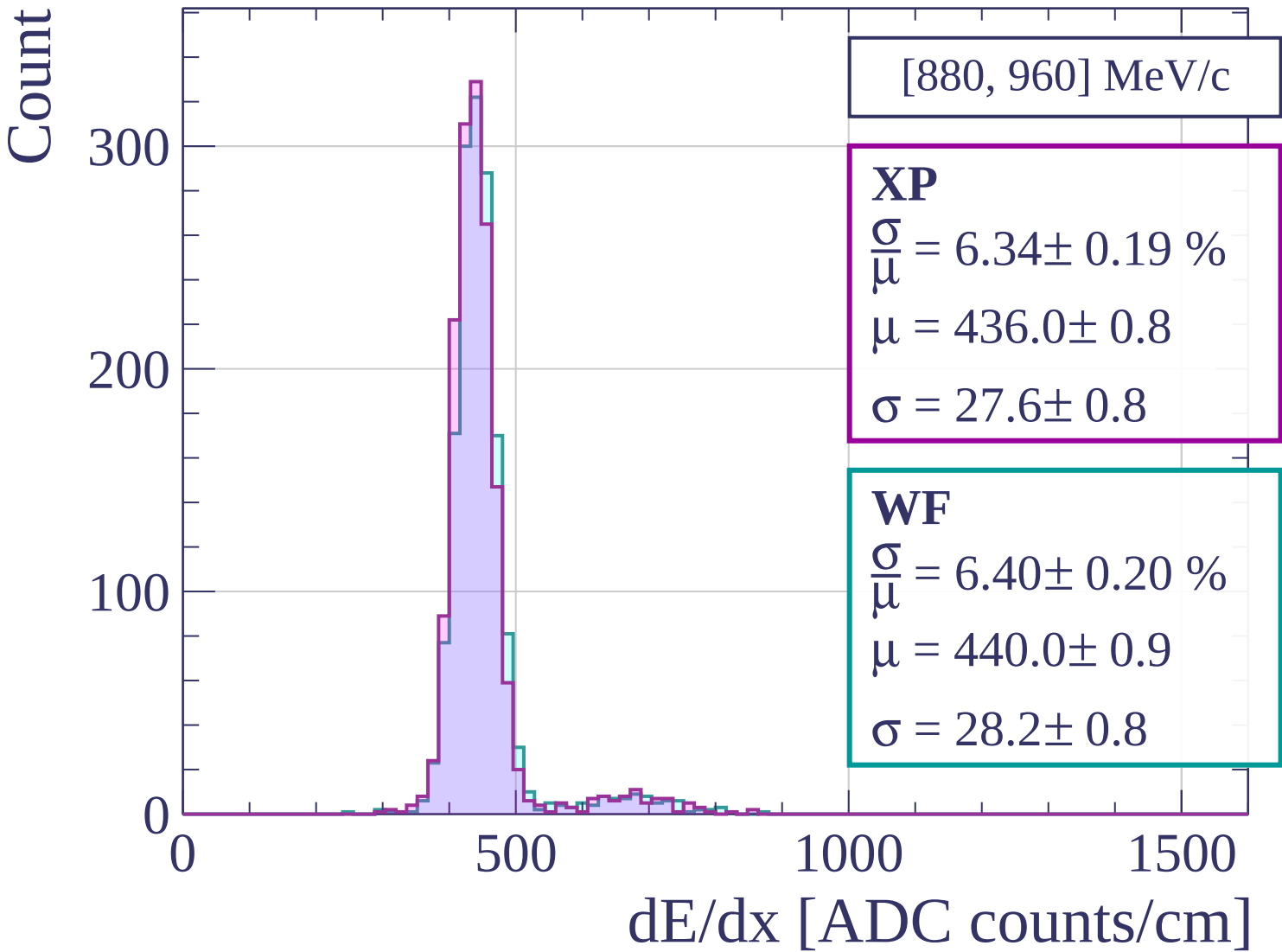
500

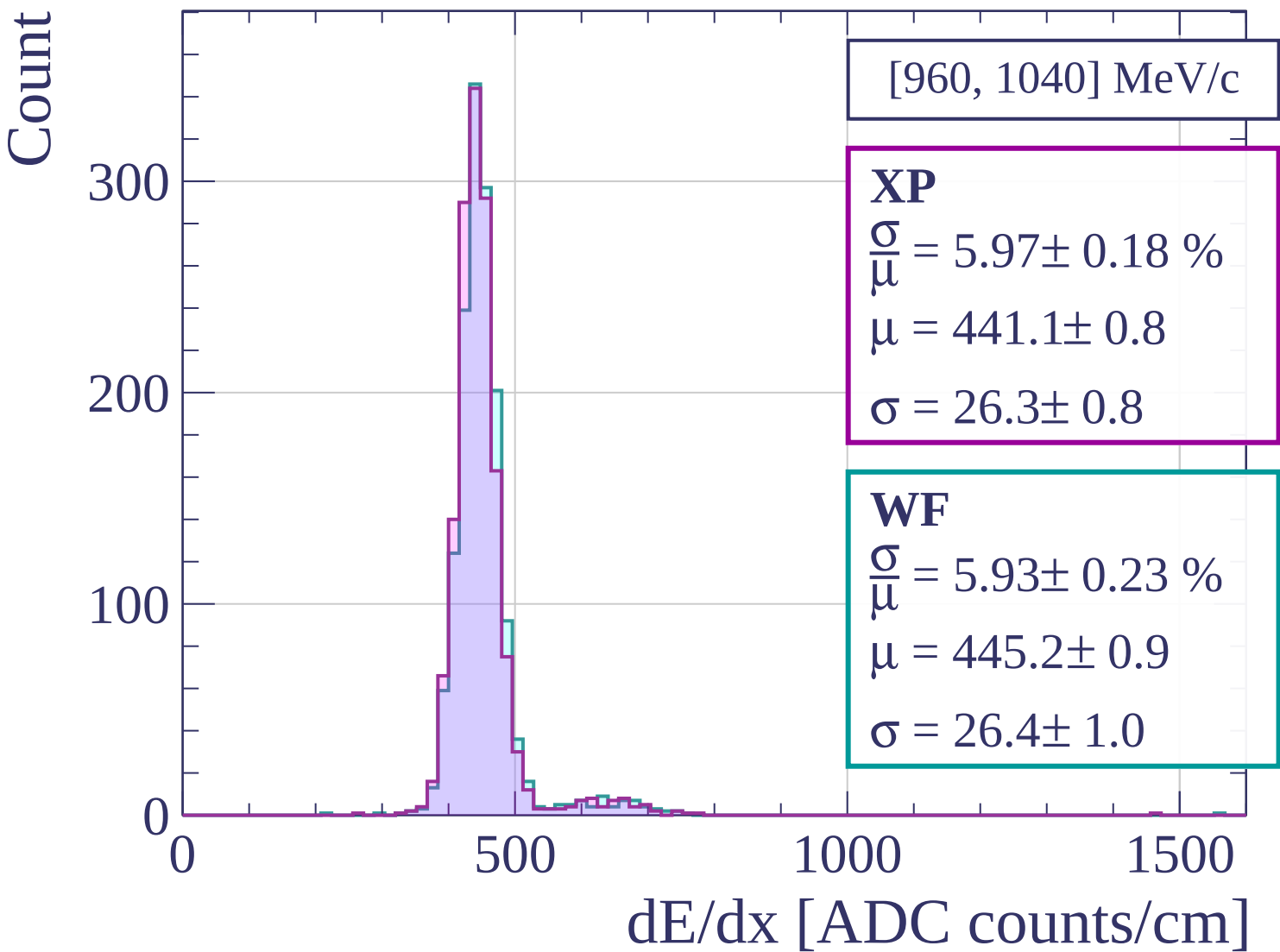
1000

1500

dE/dx [ADC counts/cm]







Count

[1040, 1120] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.99 \pm 0.25 \%$$

$$\mu = 446.8 \pm 0.9$$

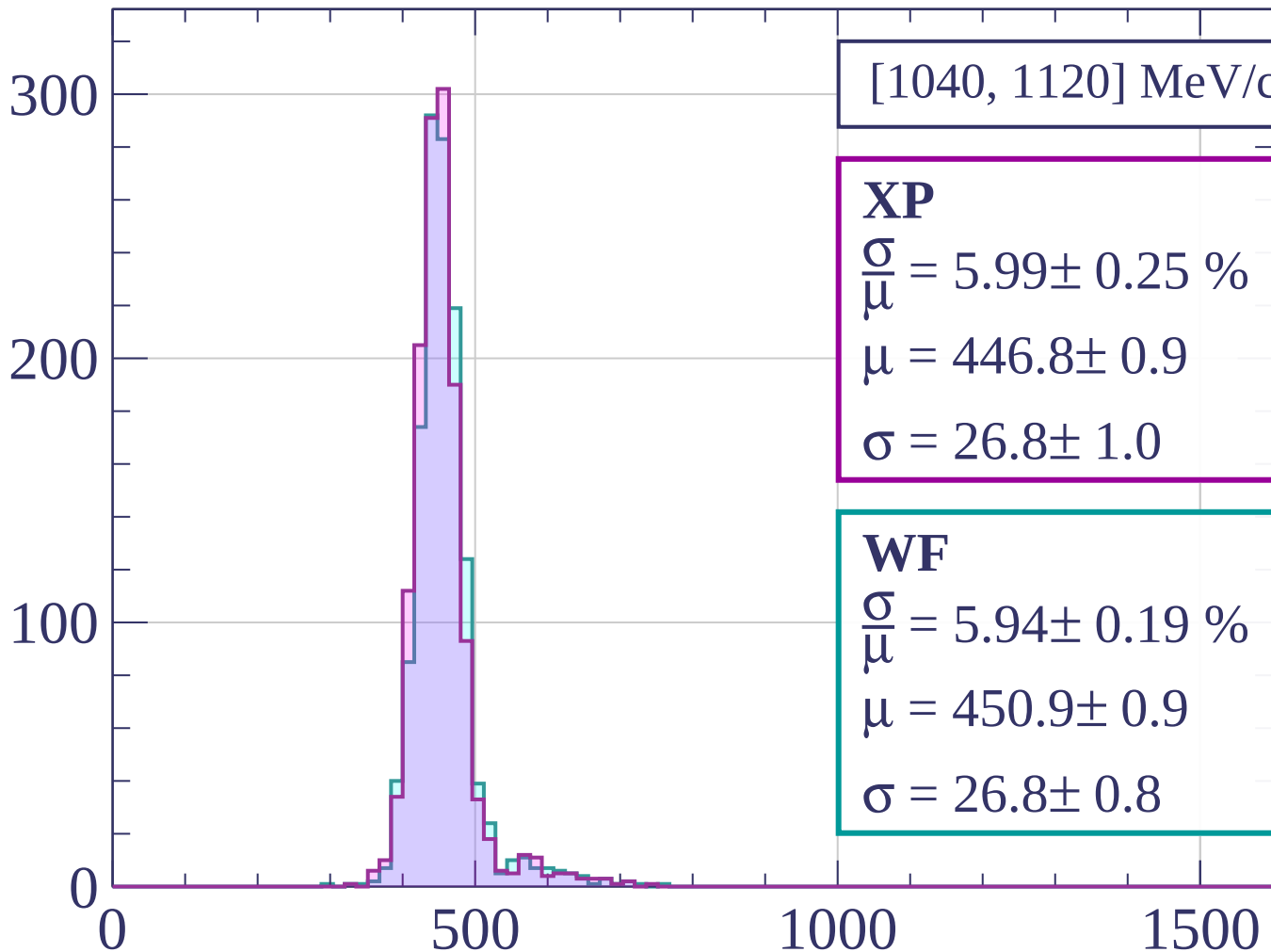
$$\sigma = 26.8 \pm 1.0$$

WF

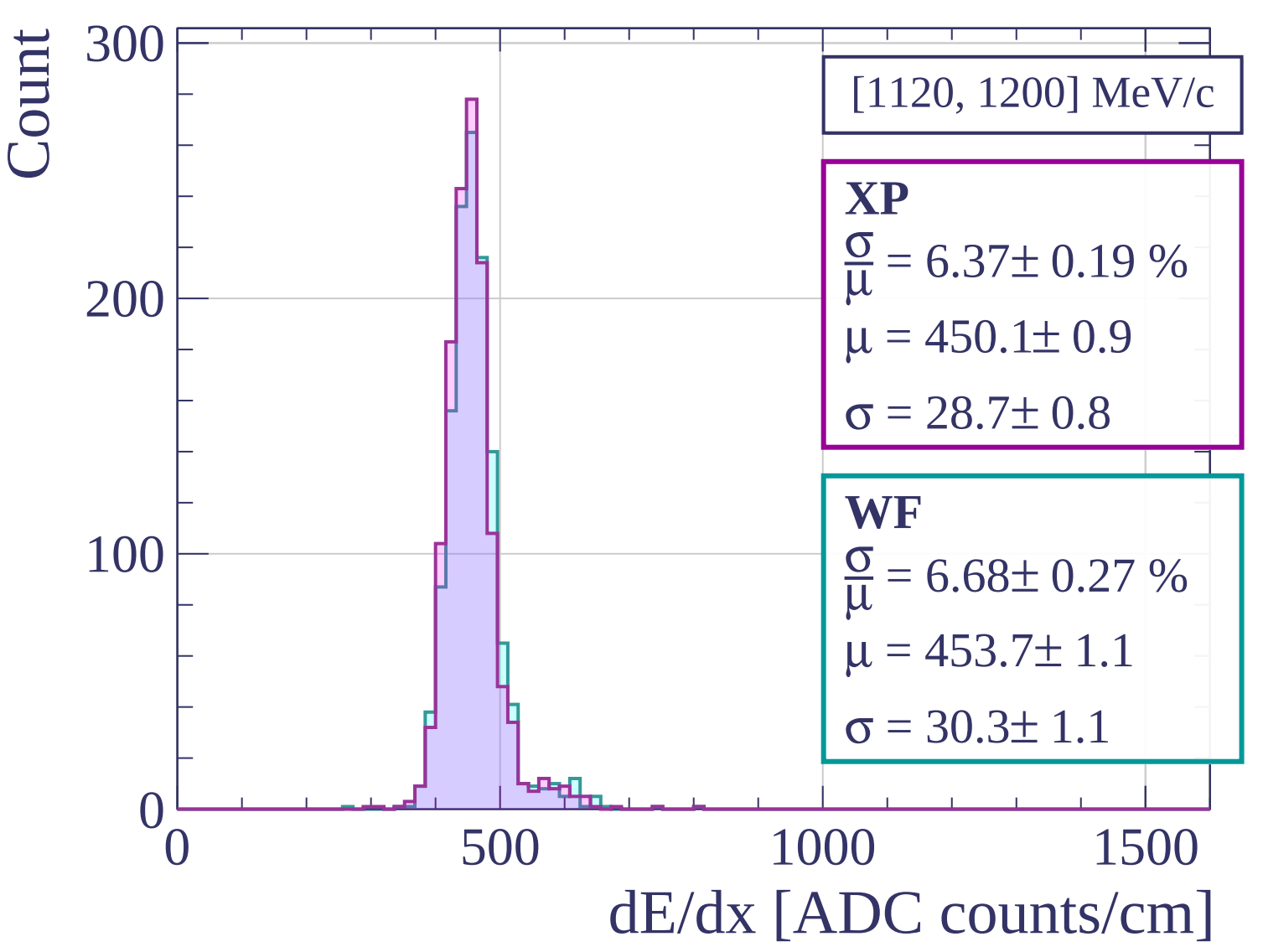
$$\frac{\sigma}{\mu} = 5.94 \pm 0.19 \%$$

$$\mu = 450.9 \pm 0.9$$

$$\sigma = 26.8 \pm 0.8$$



dE/dx [ADC counts/cm]



Count

[1200, 1280] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.63 \pm 0.25 \%$$

$$\mu = 452.4 \pm 1.0$$

$$\sigma = 30.0 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 6.39 \pm 0.24 \%$$

$$\mu = 456.1 \pm 1.0$$

$$\sigma = 29.1 \pm 1.0$$

200

100

0

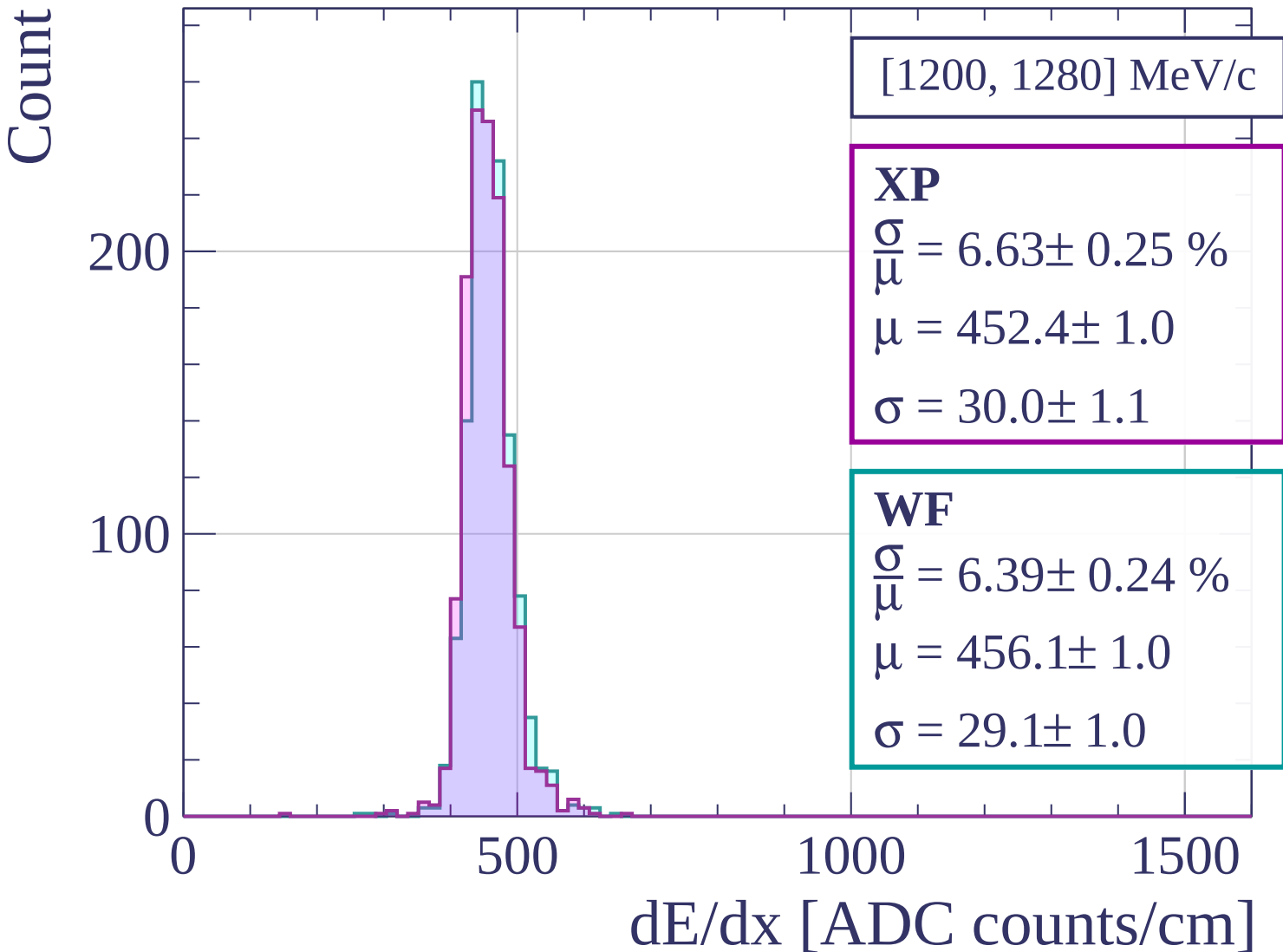
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[1280, 1360] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.06 \pm 0.22 \%$$

$$\mu = 456.8 \pm 0.9$$

$$\sigma = 27.7 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 6.46 \pm 0.21 \%$$

$$\mu = 461.9 \pm 1.0$$

$$\sigma = 29.8 \pm 0.9$$

200

100

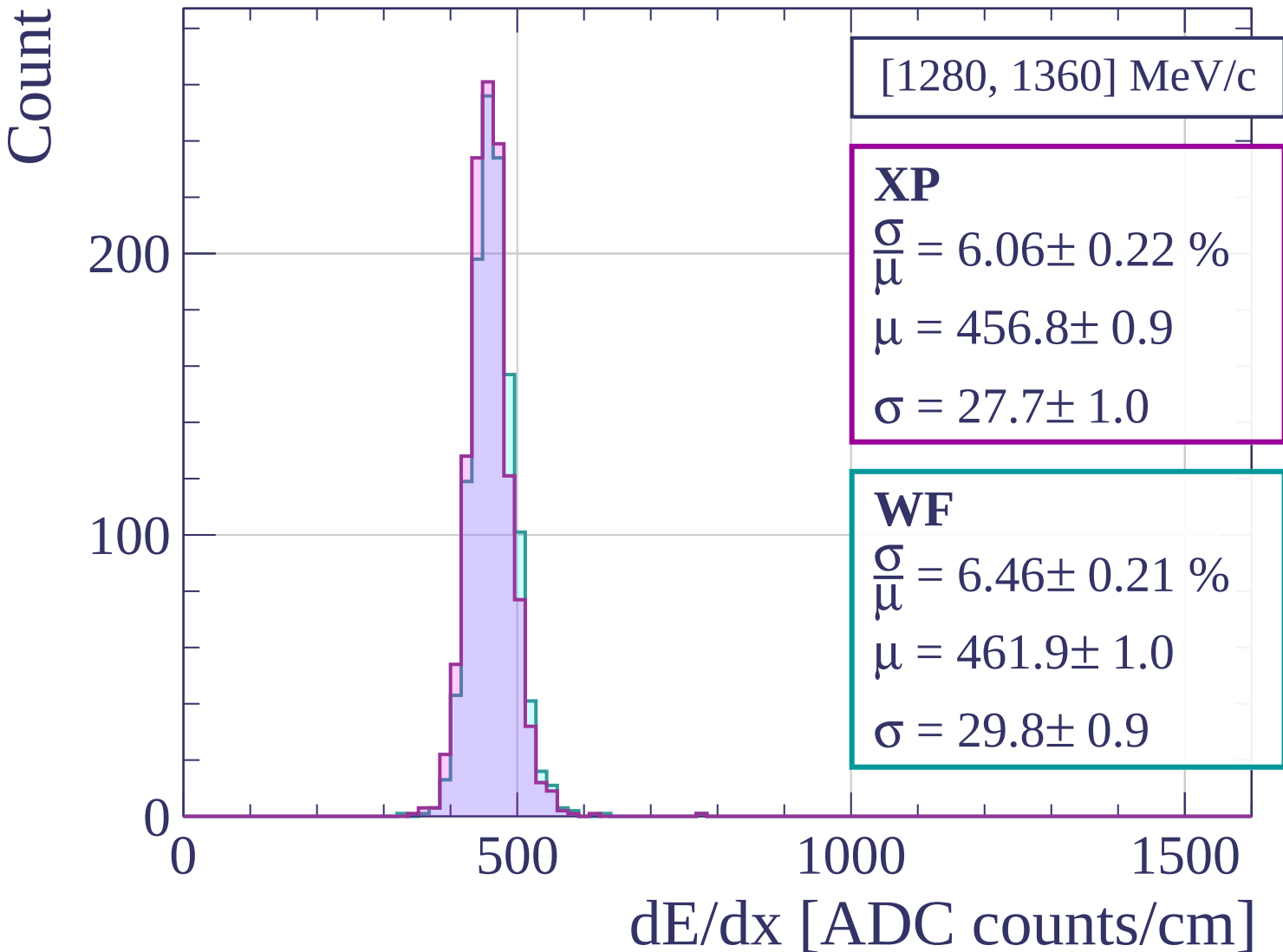
0

500

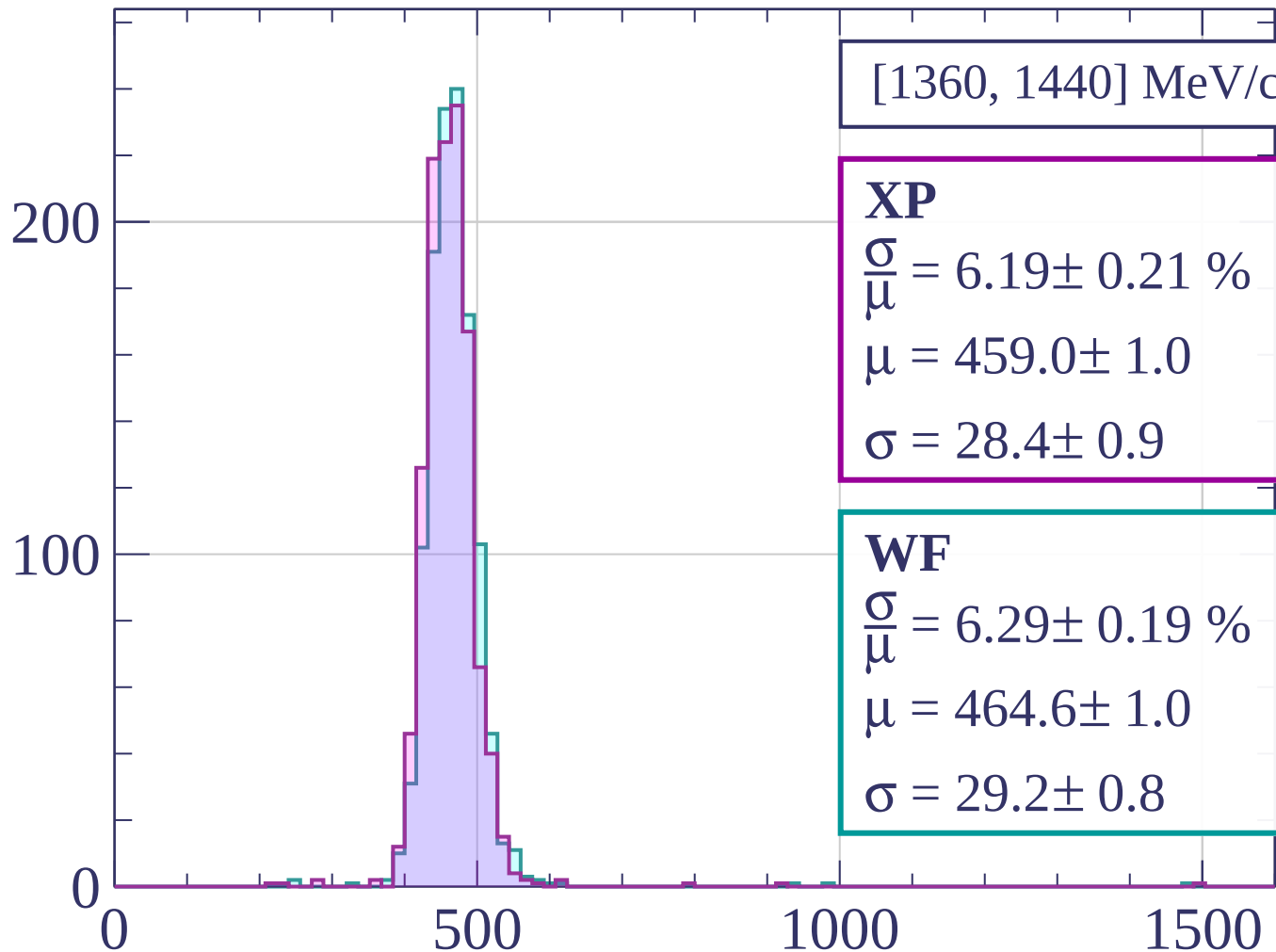
1000

1500

dE/dx [ADC counts/cm]



Count



dE/dx [ADC counts/cm]

Count

[1440, 1520] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.08 \pm 0.20 \%$$

$$\mu = 464.4 \pm 1.0$$

$$\sigma = 28.2 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 6.46 \pm 0.28 \%$$

$$\mu = 468.2 \pm 1.2$$

$$\sigma = 30.2 \pm 1.3$$

200

100

0

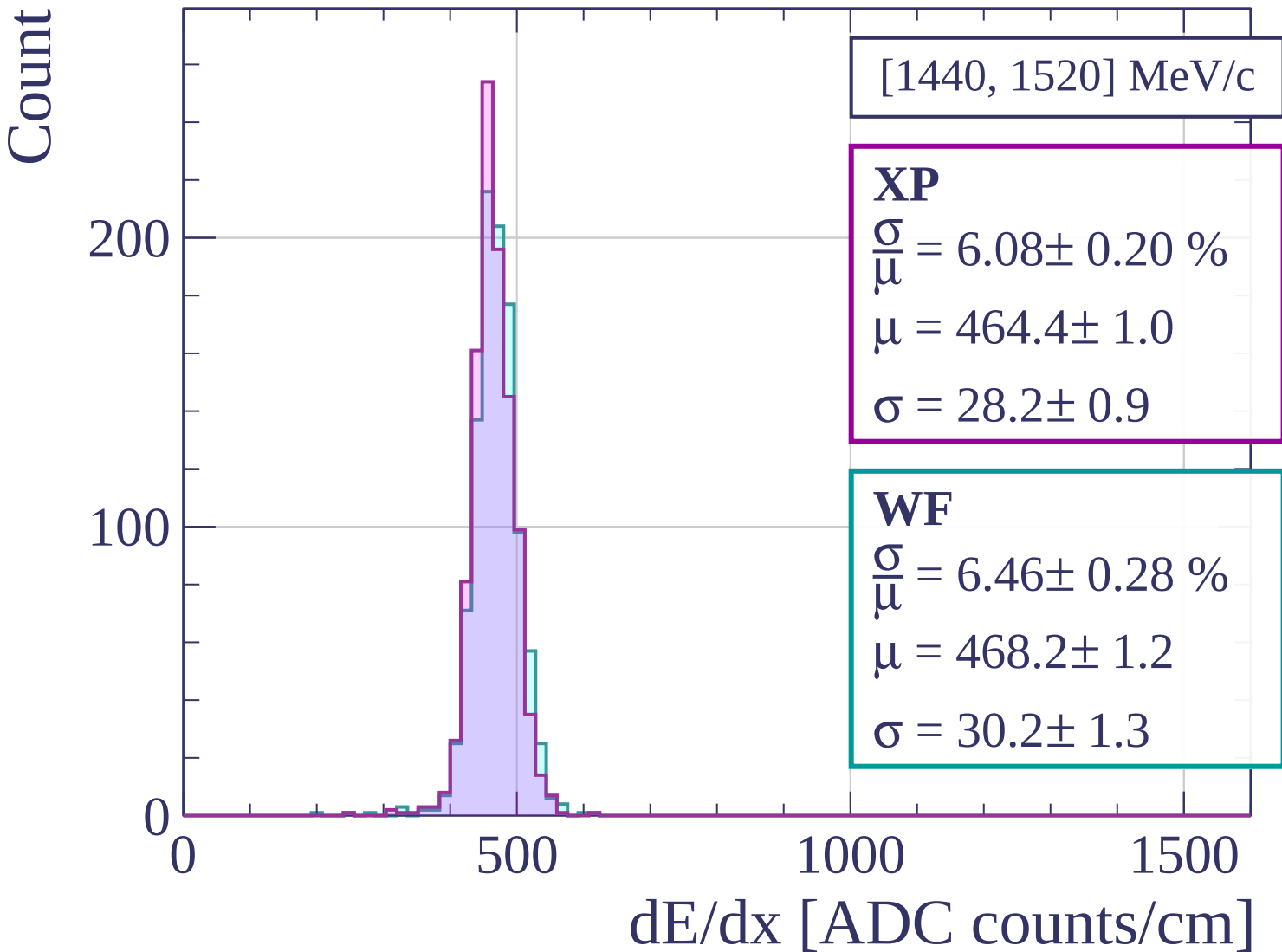
0

500

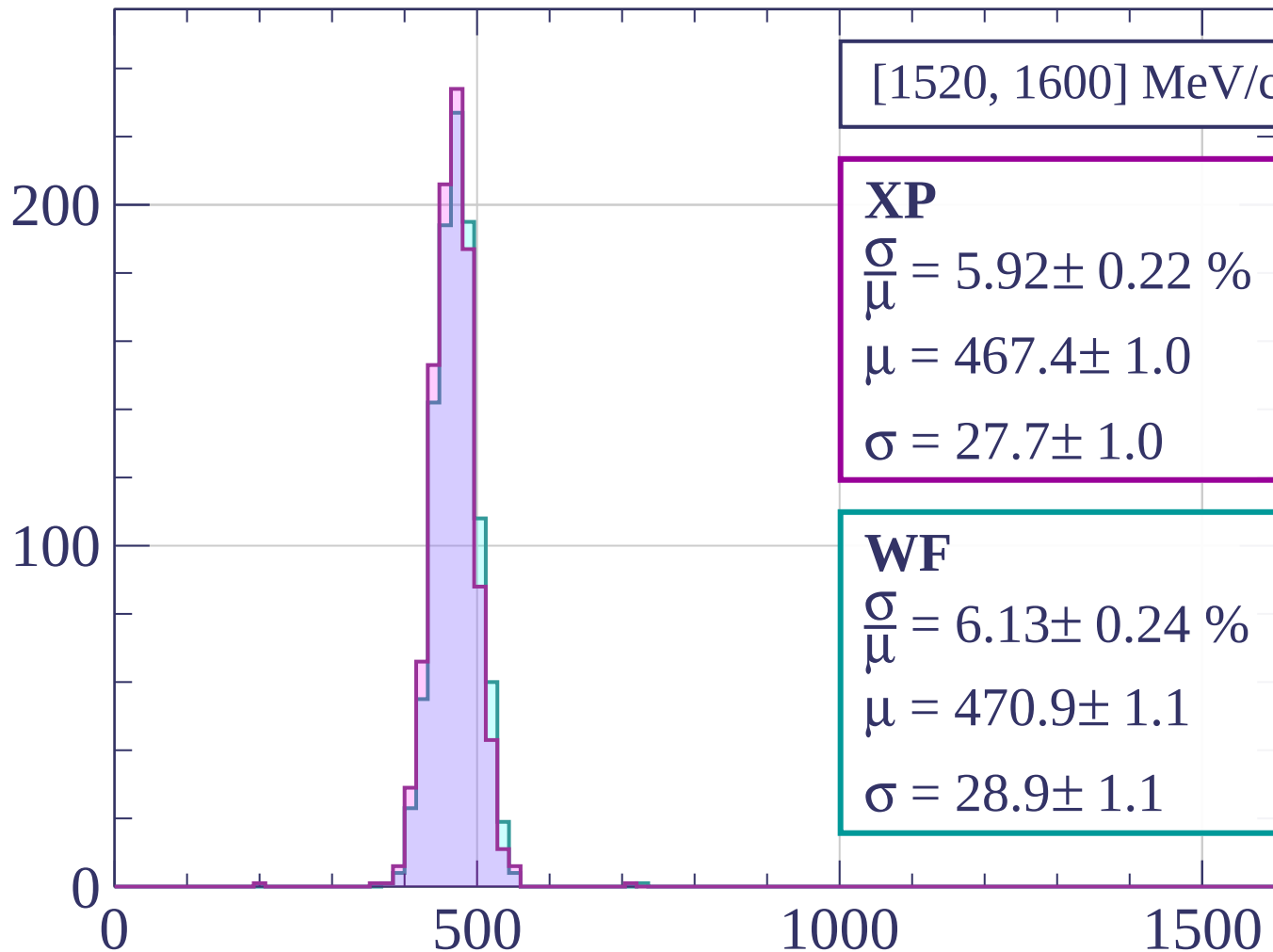
1000

1500

dE/dx [ADC counts/cm]



Count



dE/dx [ADC counts/cm]

Count

[1600, 1680] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.13 \pm 0.24 \%$$

$$\mu = 469.1 \pm 1.0$$

$$\sigma = 28.8 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 6.33 \pm 0.26 \%$$

$$\mu = 473.2 \pm 1.1$$

$$\sigma = 29.9 \pm 1.2$$

200

100

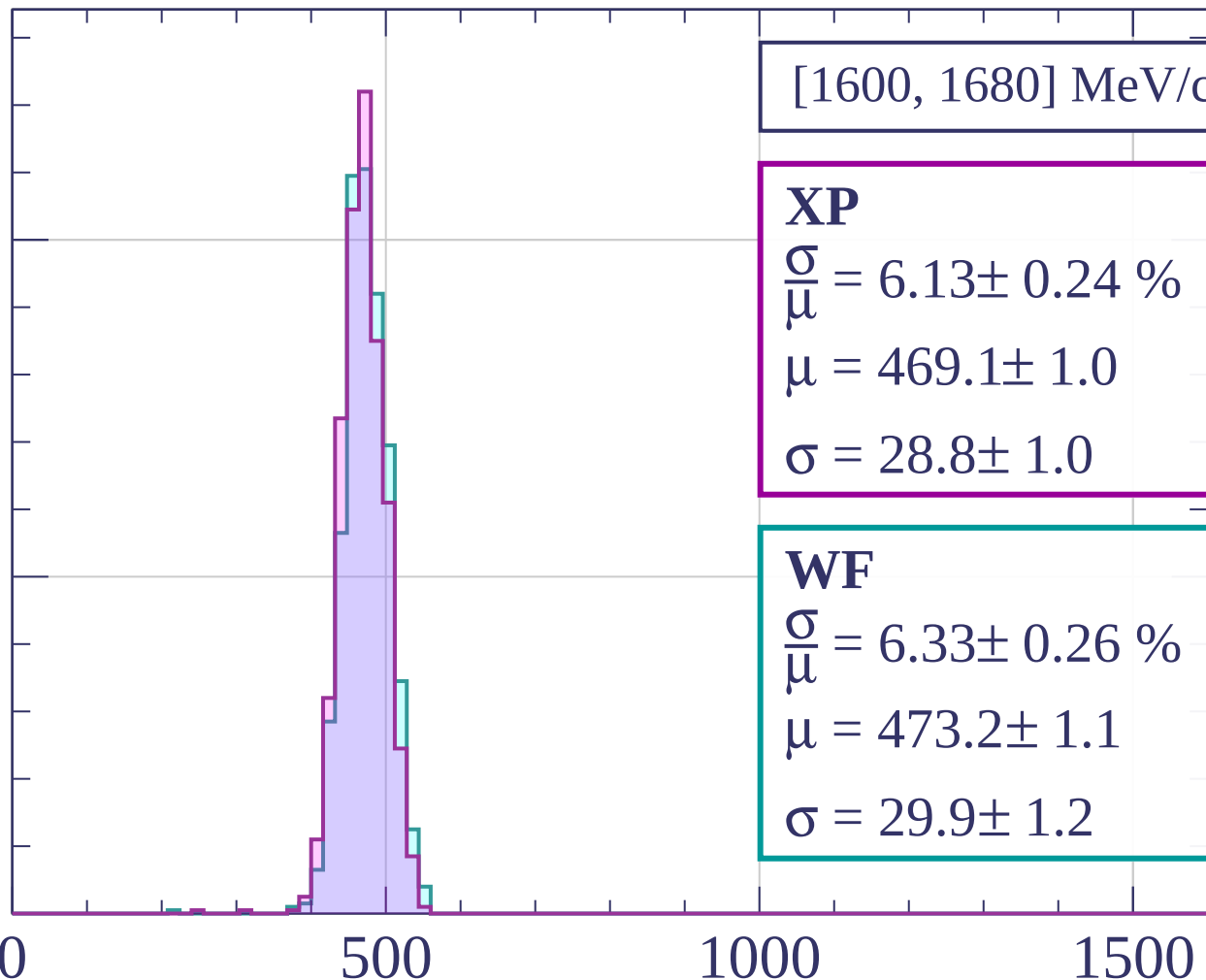
0

500

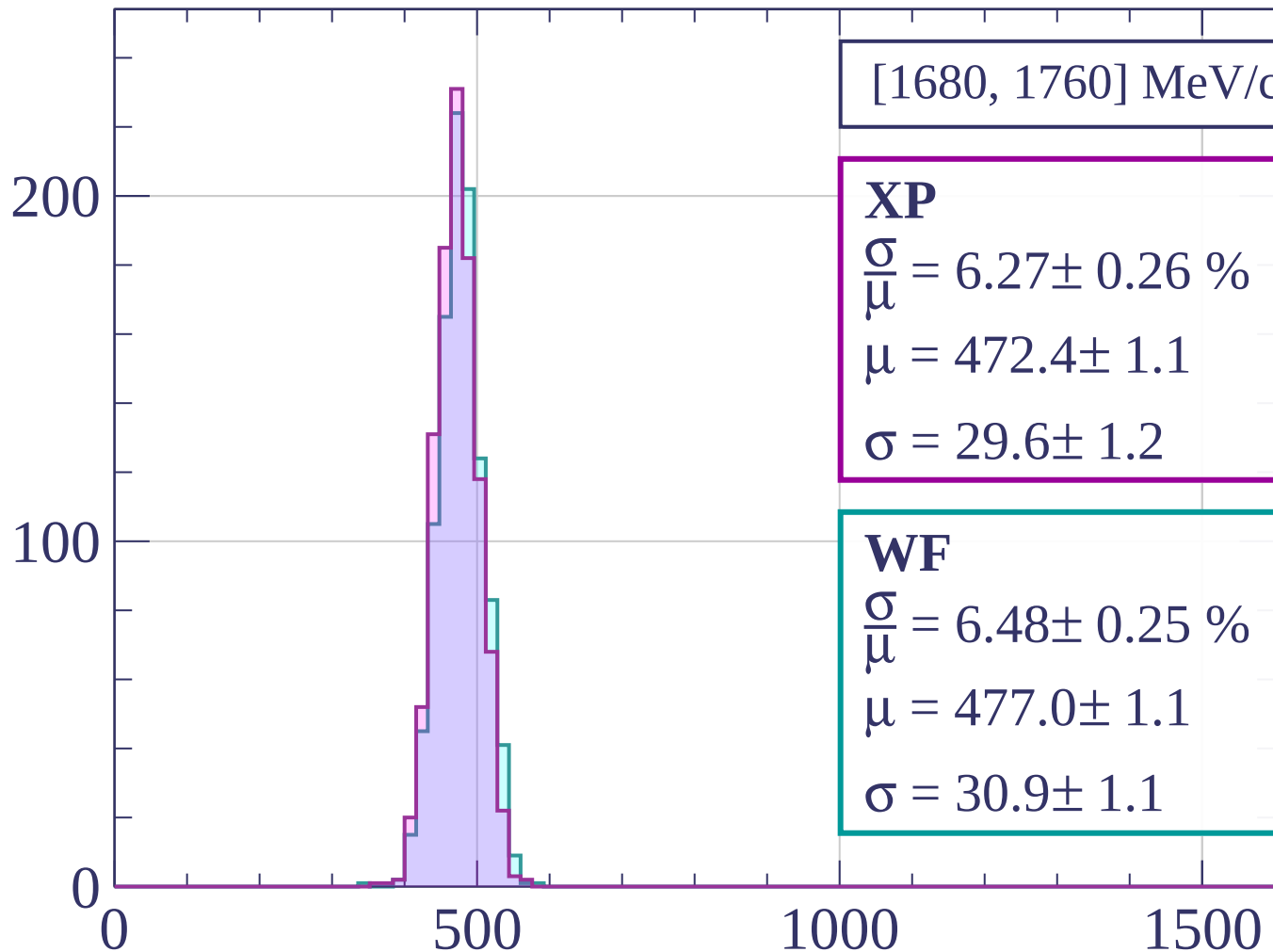
1000

1500

dE/dx [ADC counts/cm]



Count



[1680, 1760] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.27 \pm 0.26 \%$$

$$\mu = 472.4 \pm 1.1$$

$$\sigma = 29.6 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 6.48 \pm 0.25 \%$$

$$\mu = 477.0 \pm 1.1$$

$$\sigma = 30.9 \pm 1.1$$

dE/dx [ADC counts/cm]

Count

[1760, 1840] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.17 \pm 0.29 \%$$

$$\mu = 473.5 \pm 1.2$$

$$\sigma = 29.2 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 6.44 \pm 0.27 \%$$

$$\mu = 478.6 \pm 1.2$$

$$\sigma = 30.8 \pm 1.2$$

200

150

100

50

0

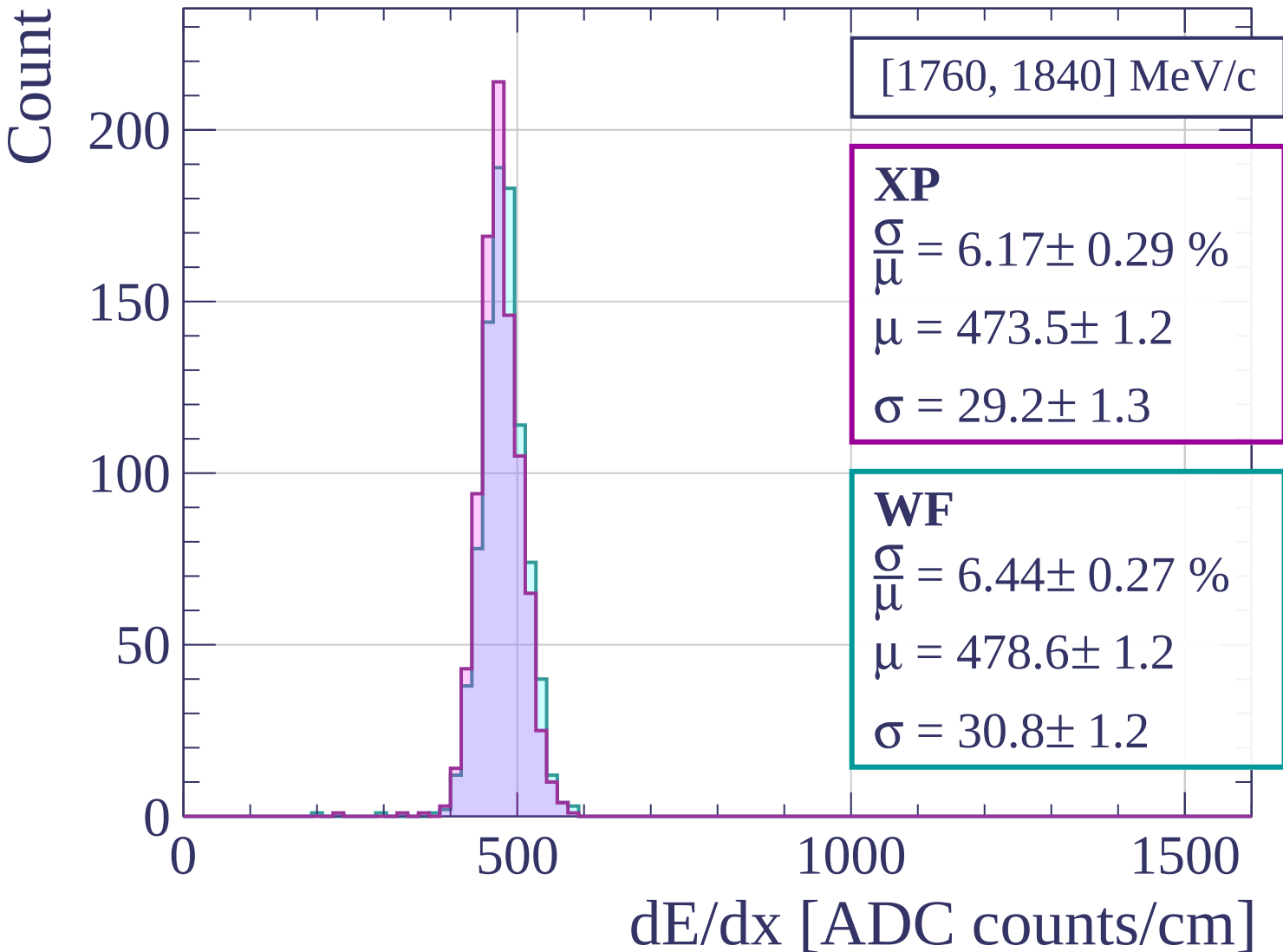
0

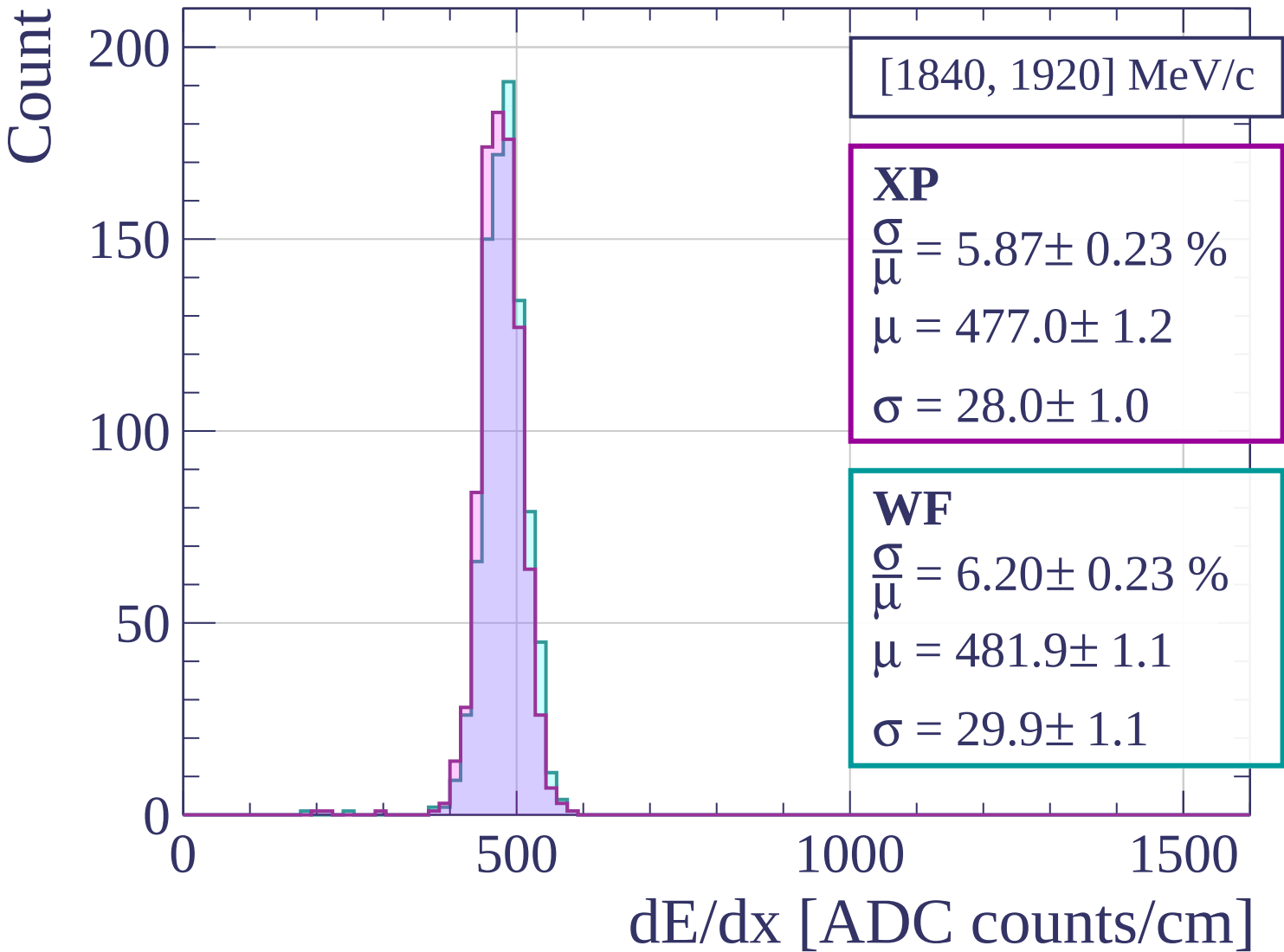
500

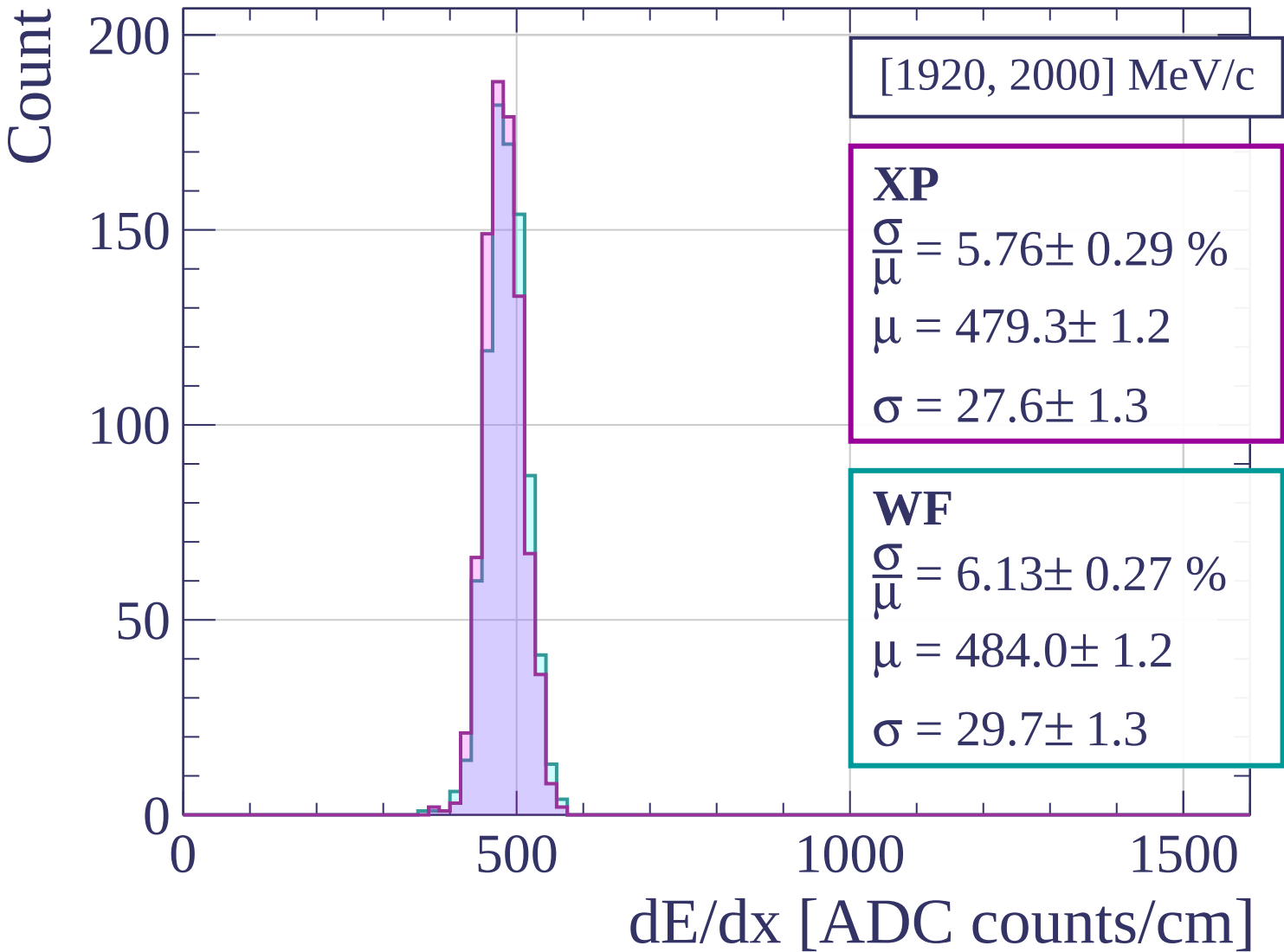
1000

1500

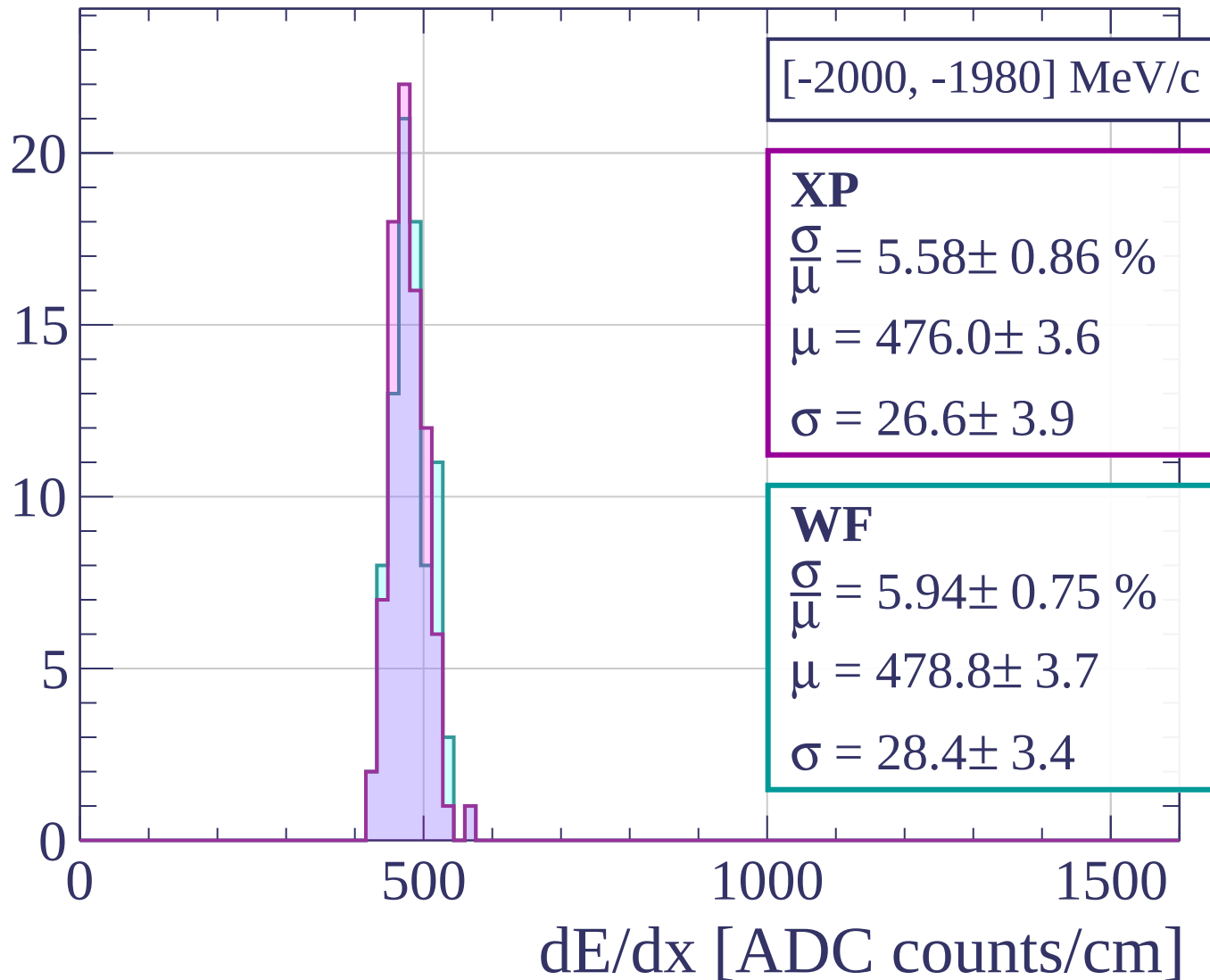
dE/dx [ADC counts/cm]



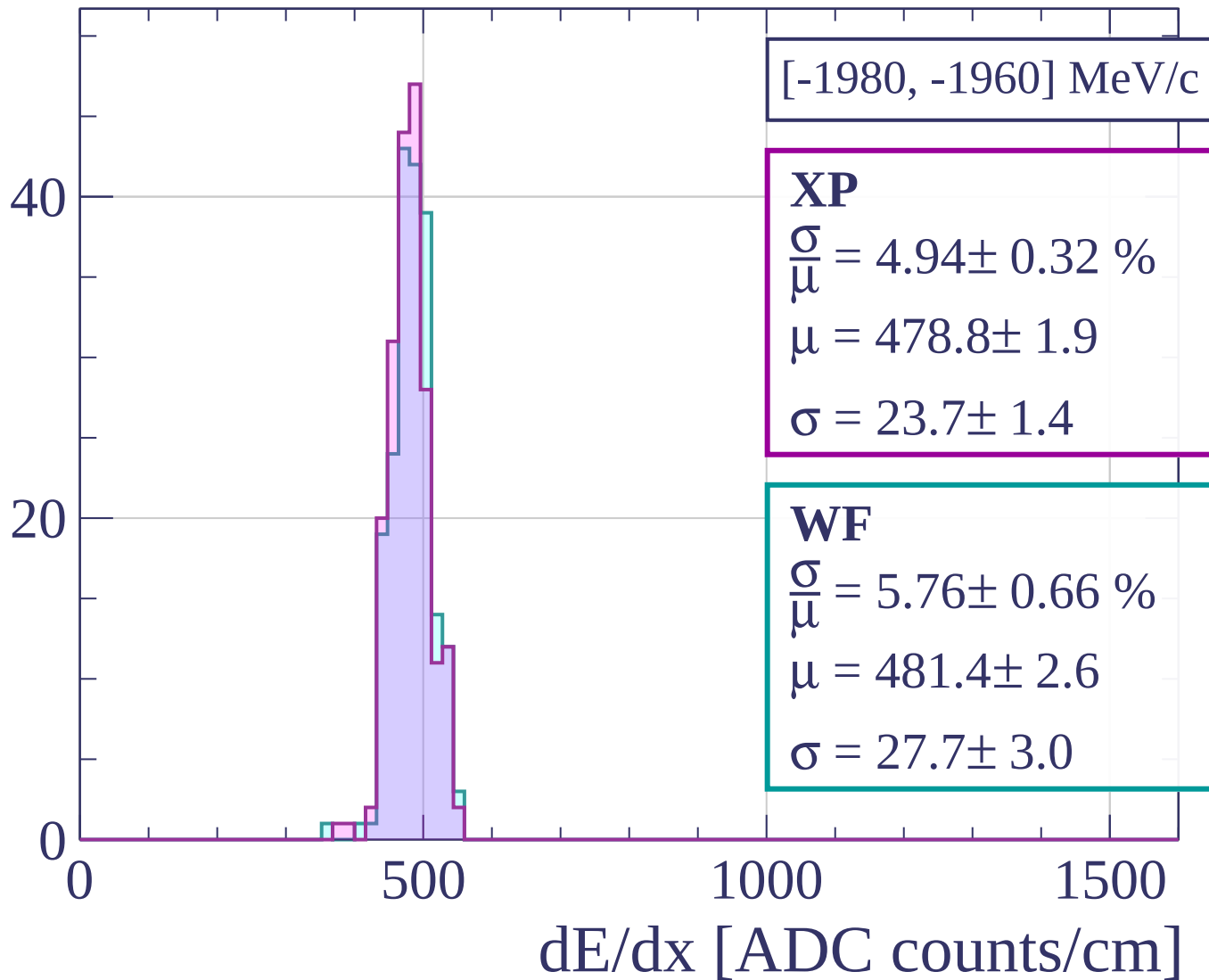




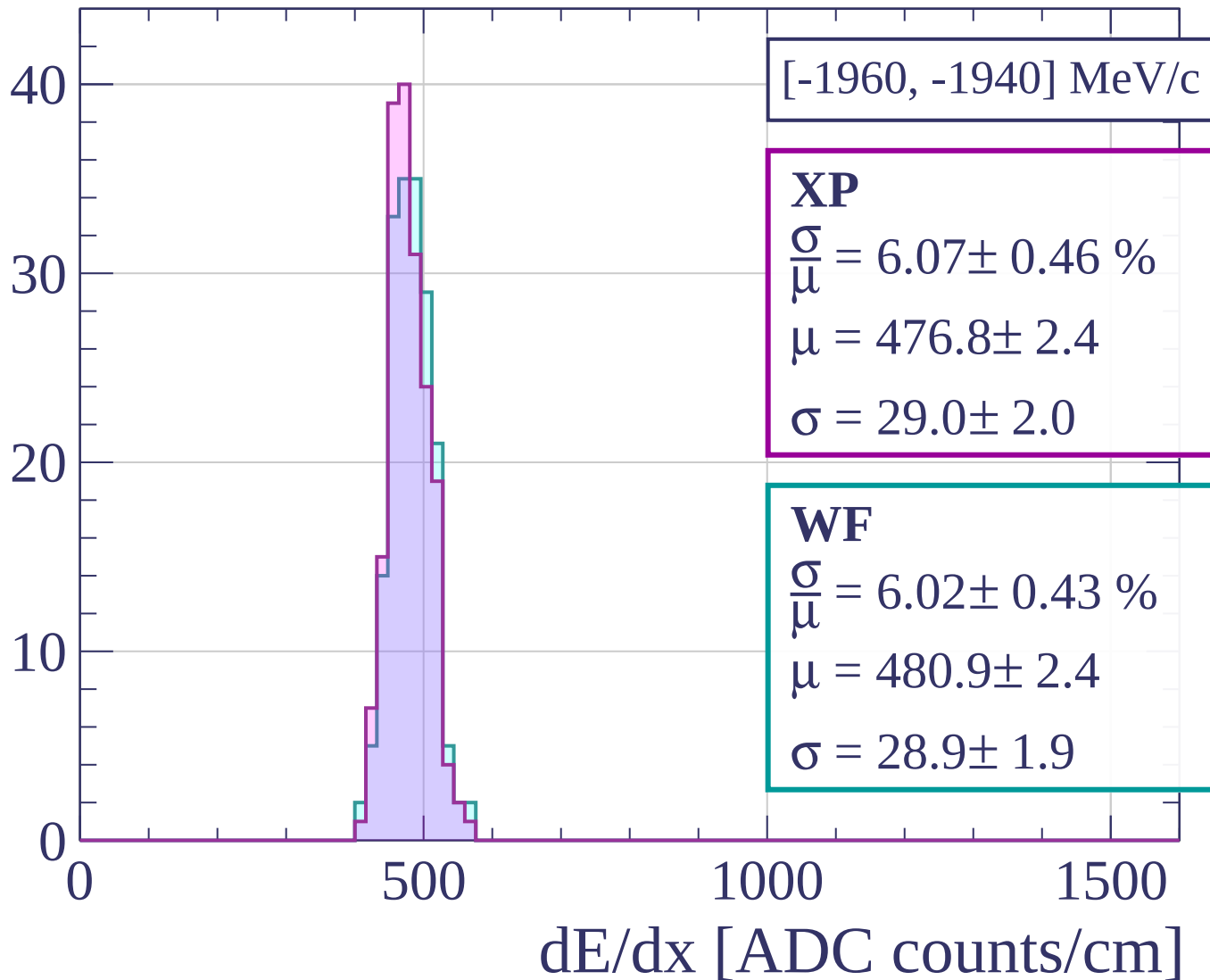
Count



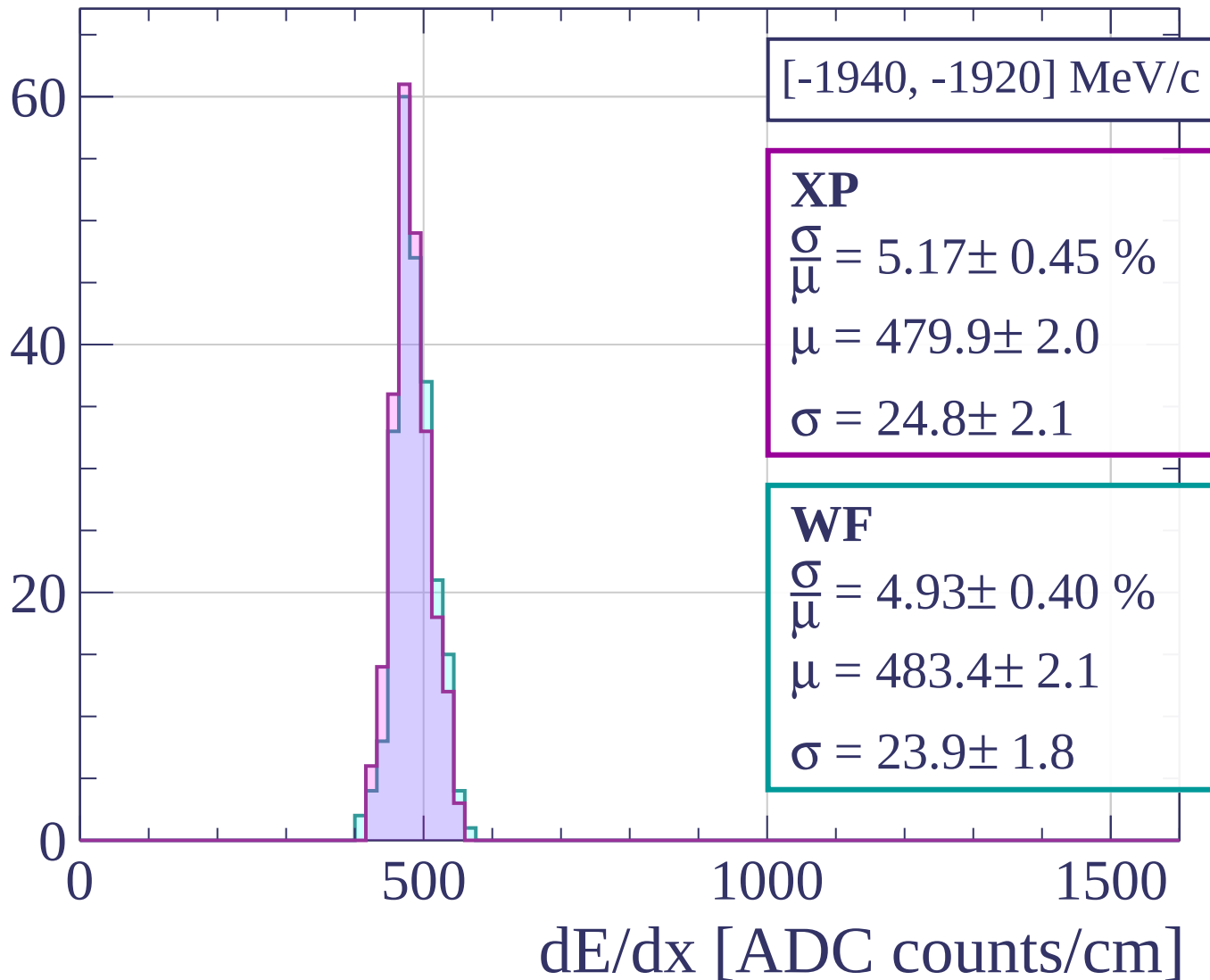
Count



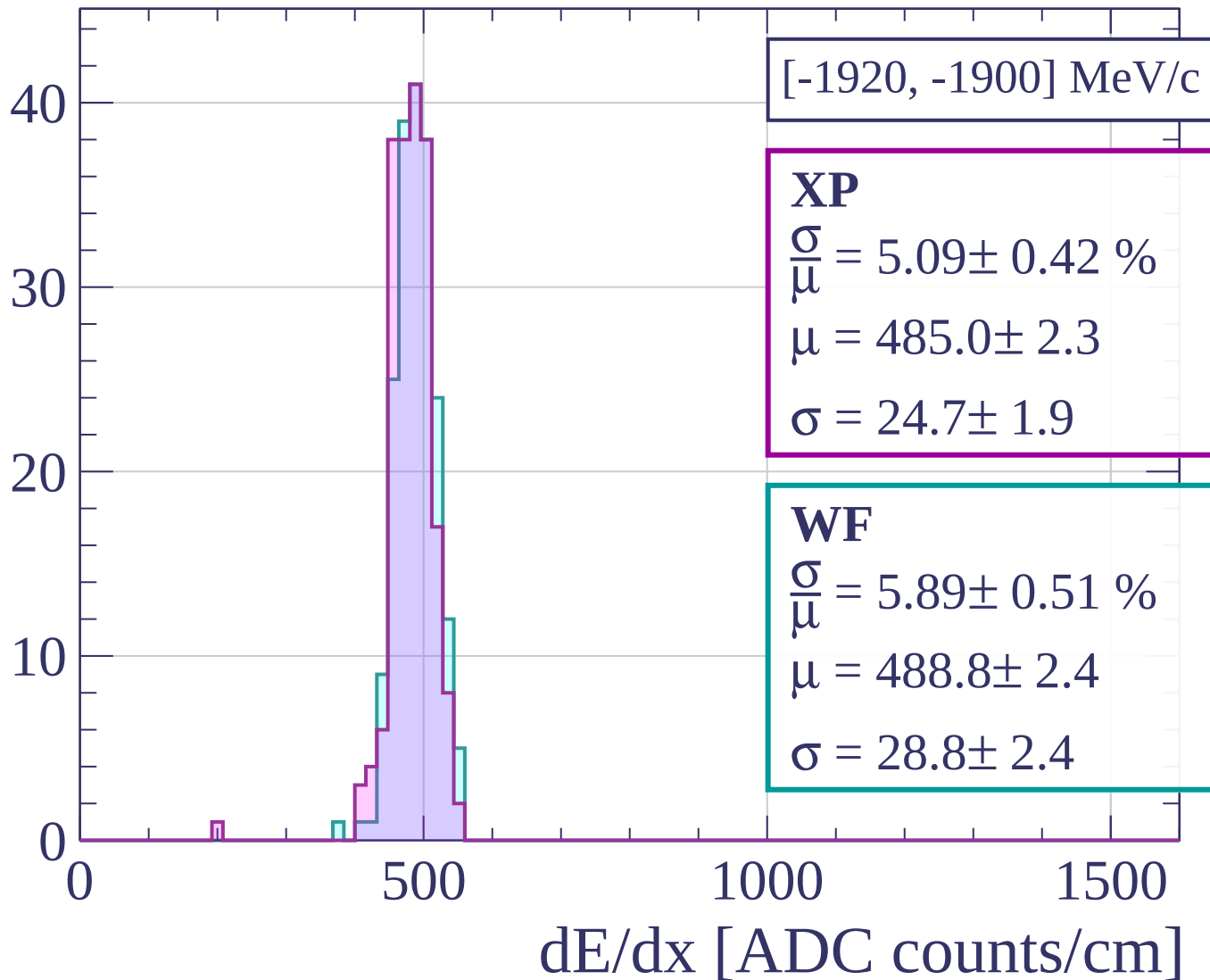
Count



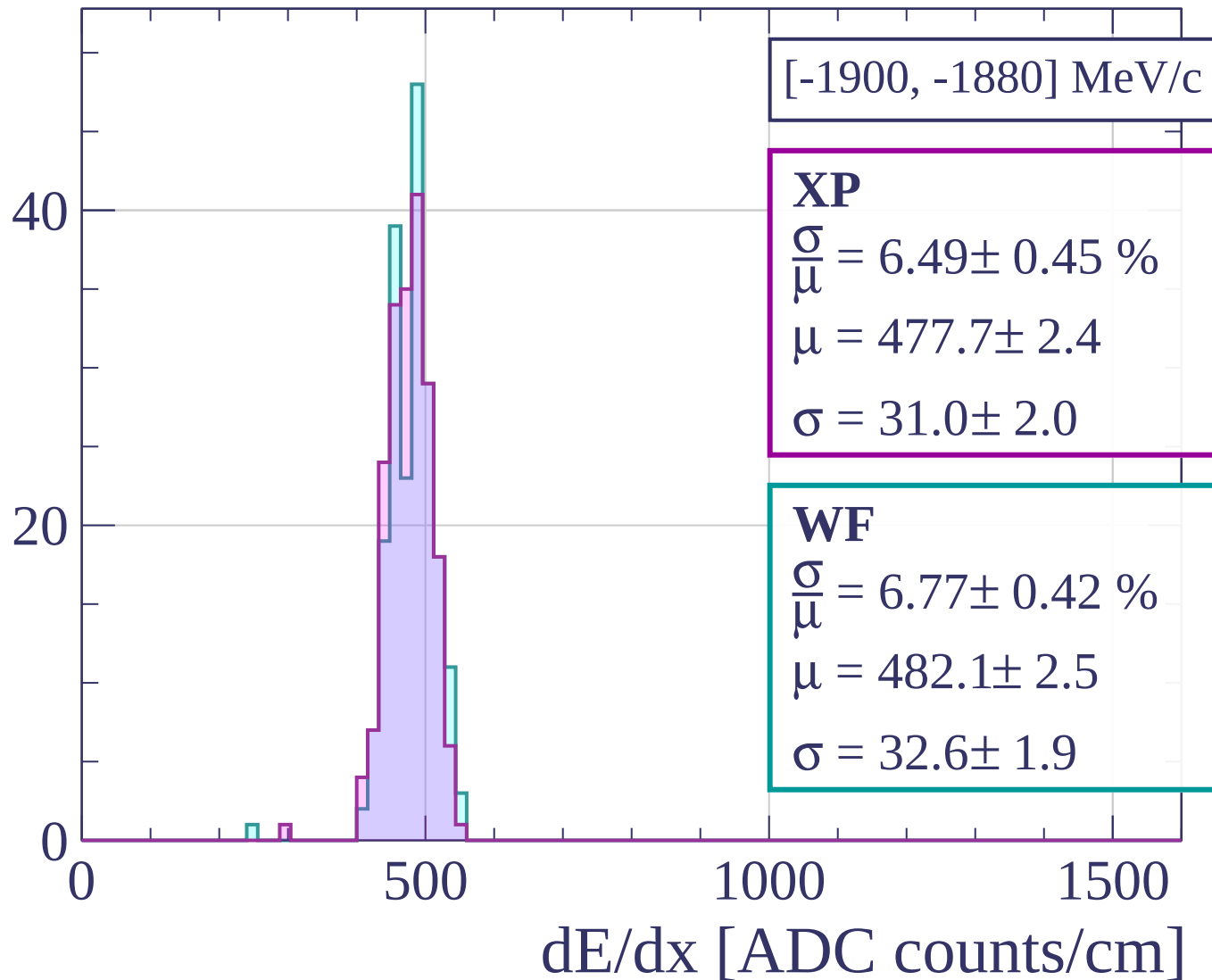
Count



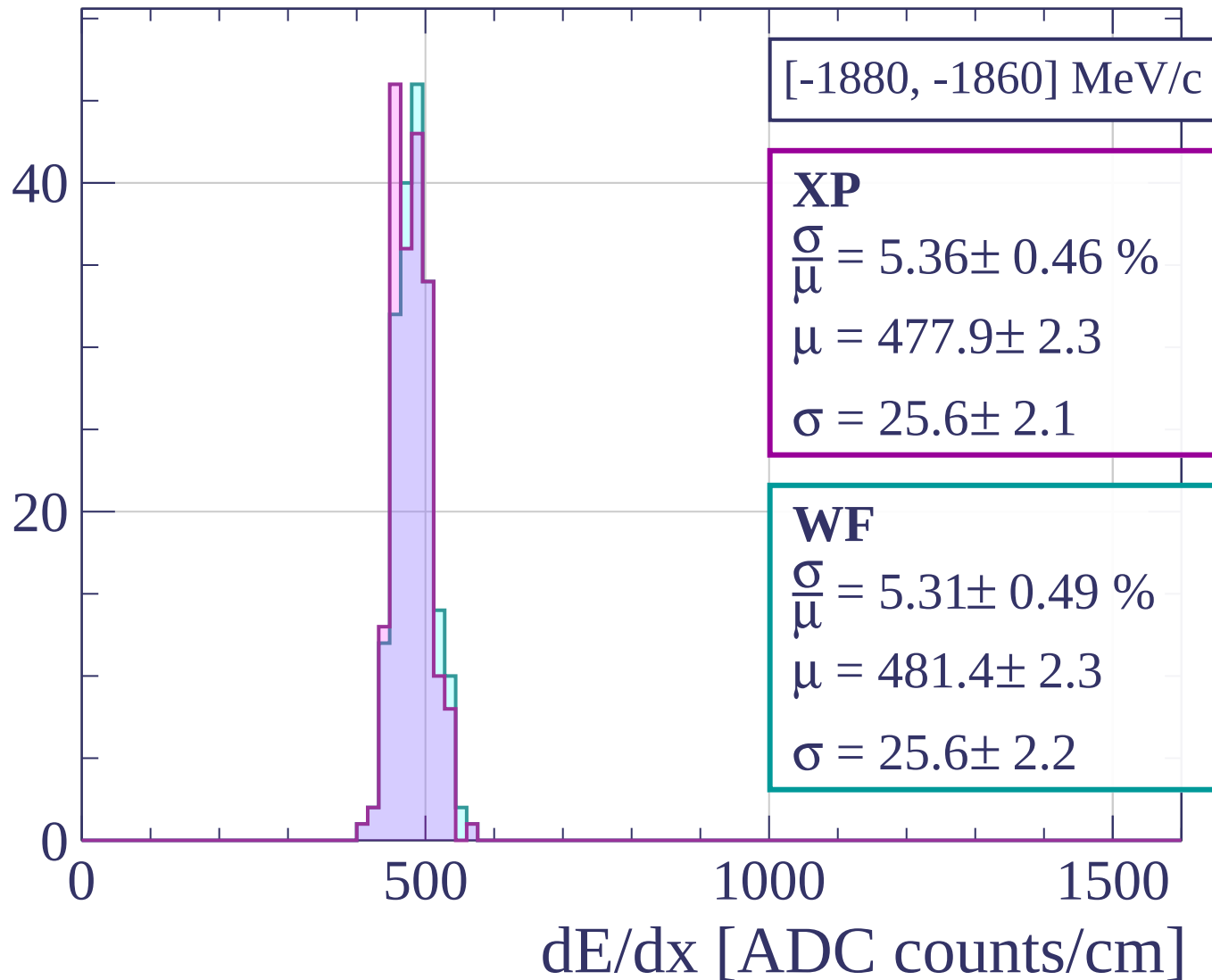
Count



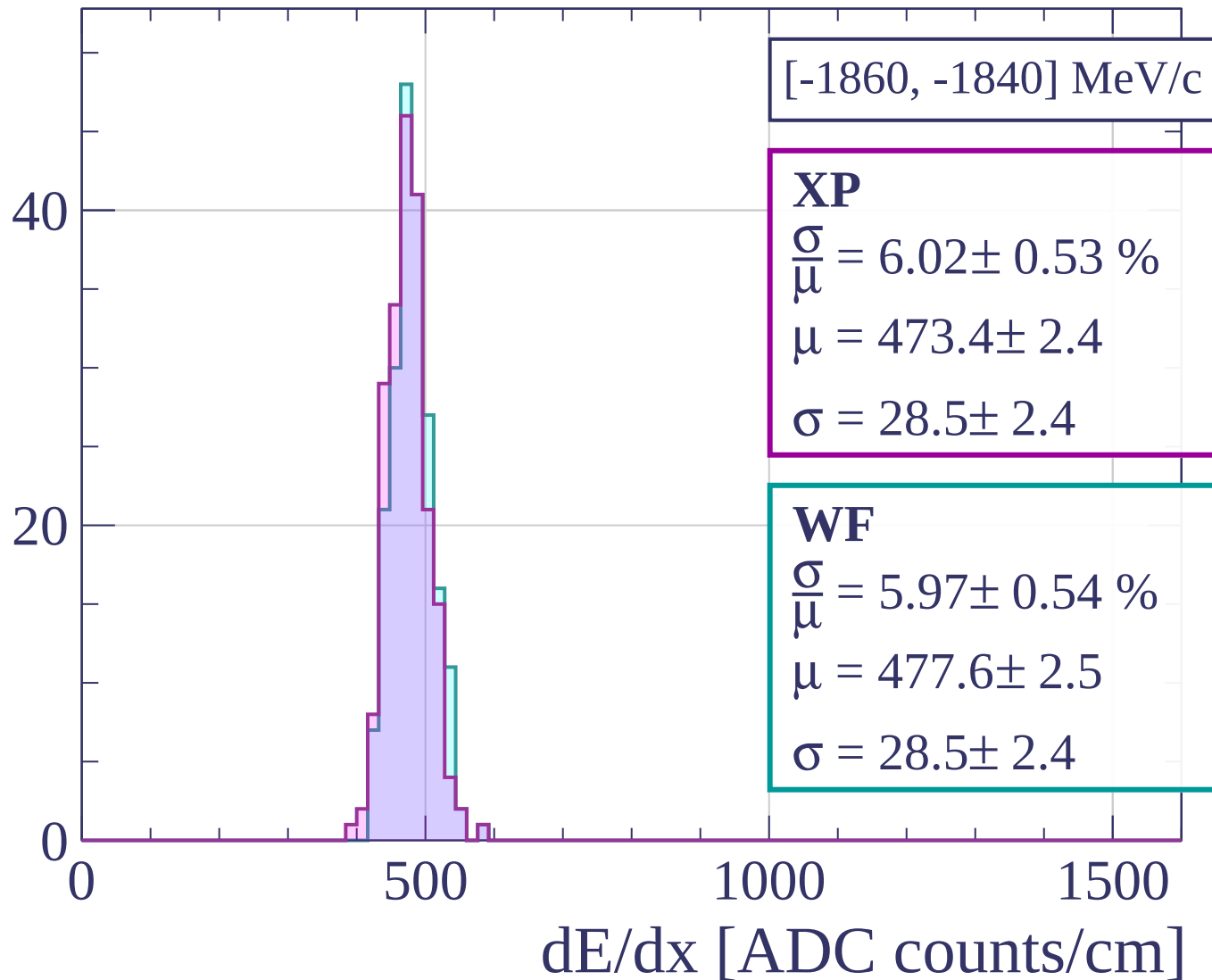
Count



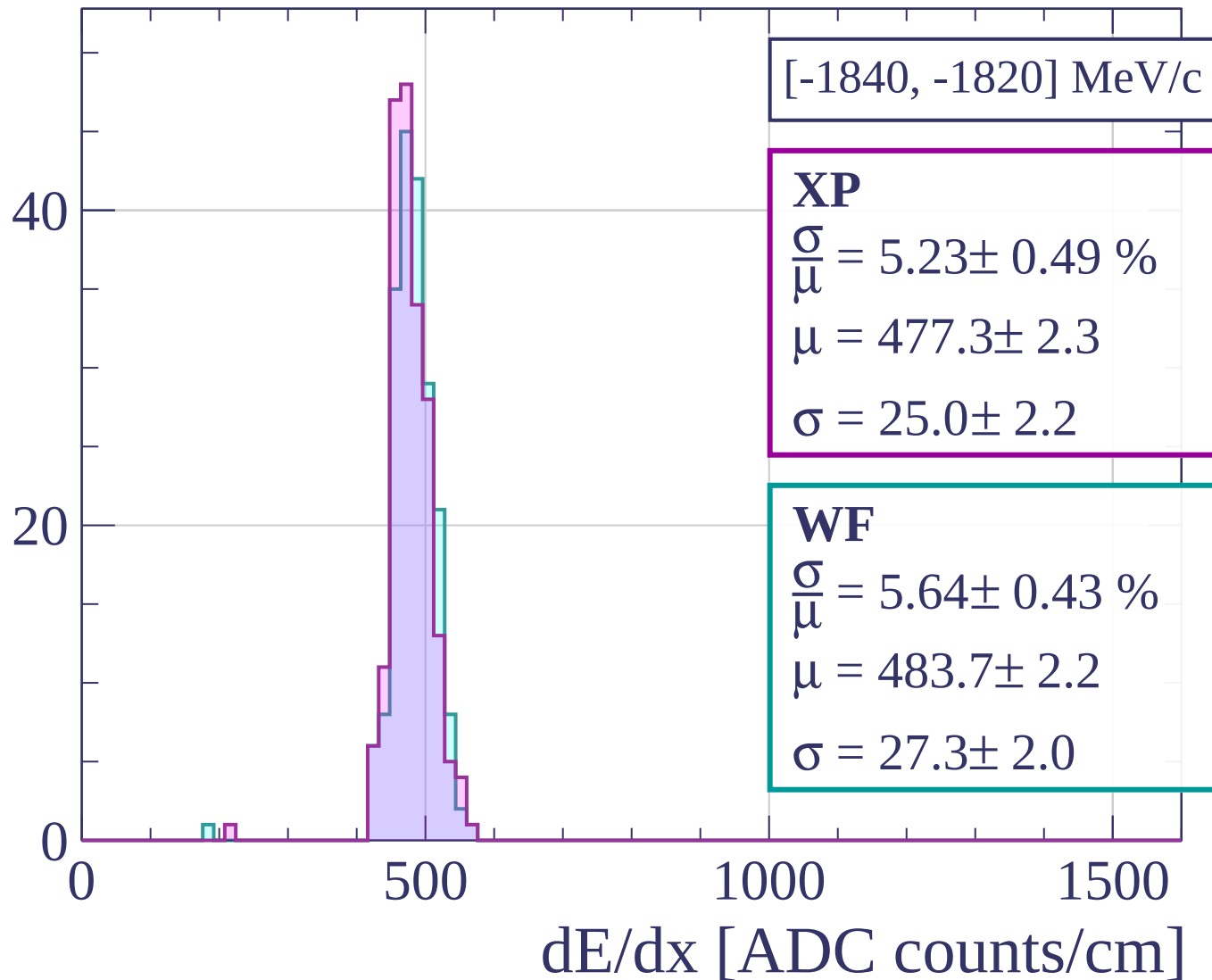
Count



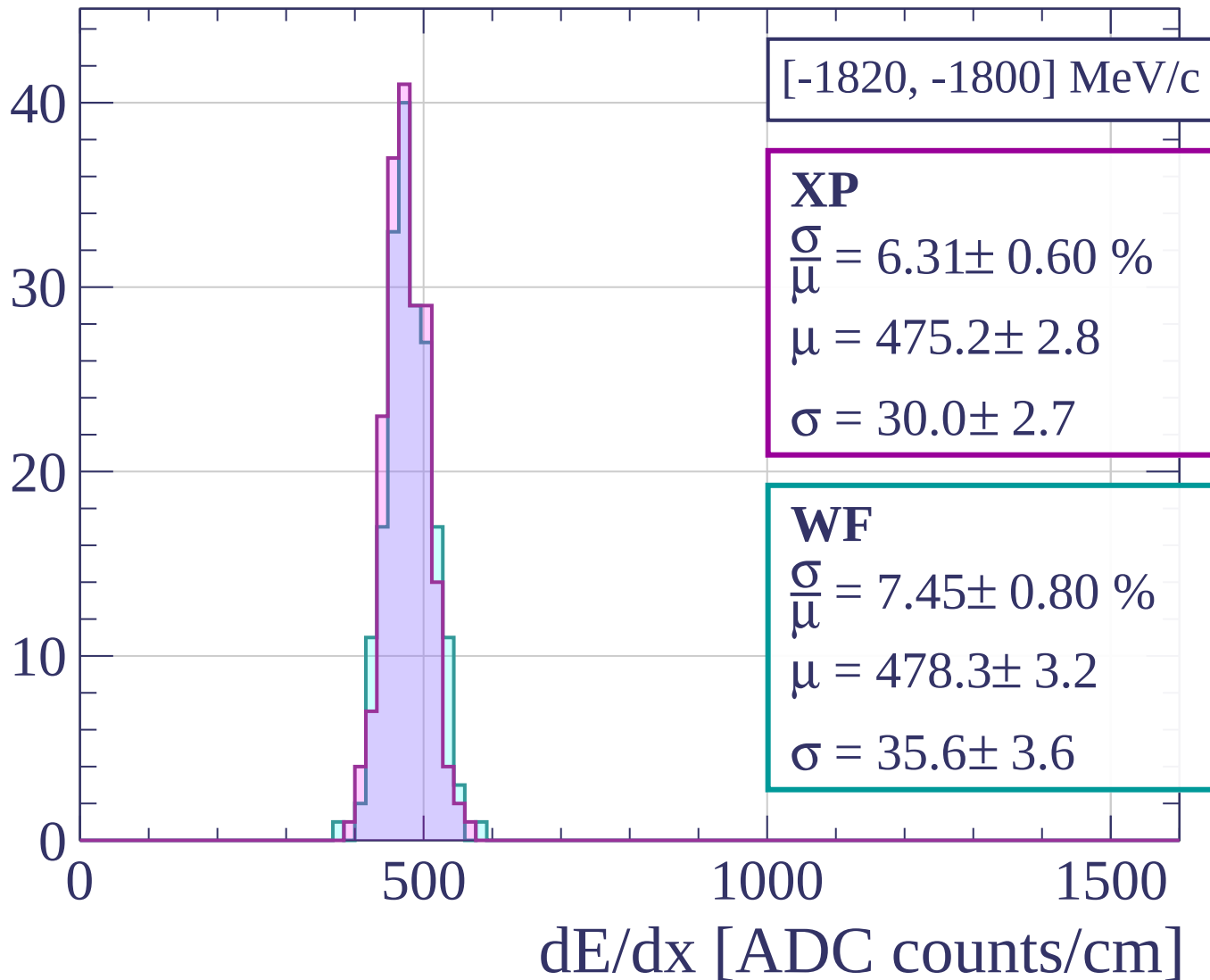
Count



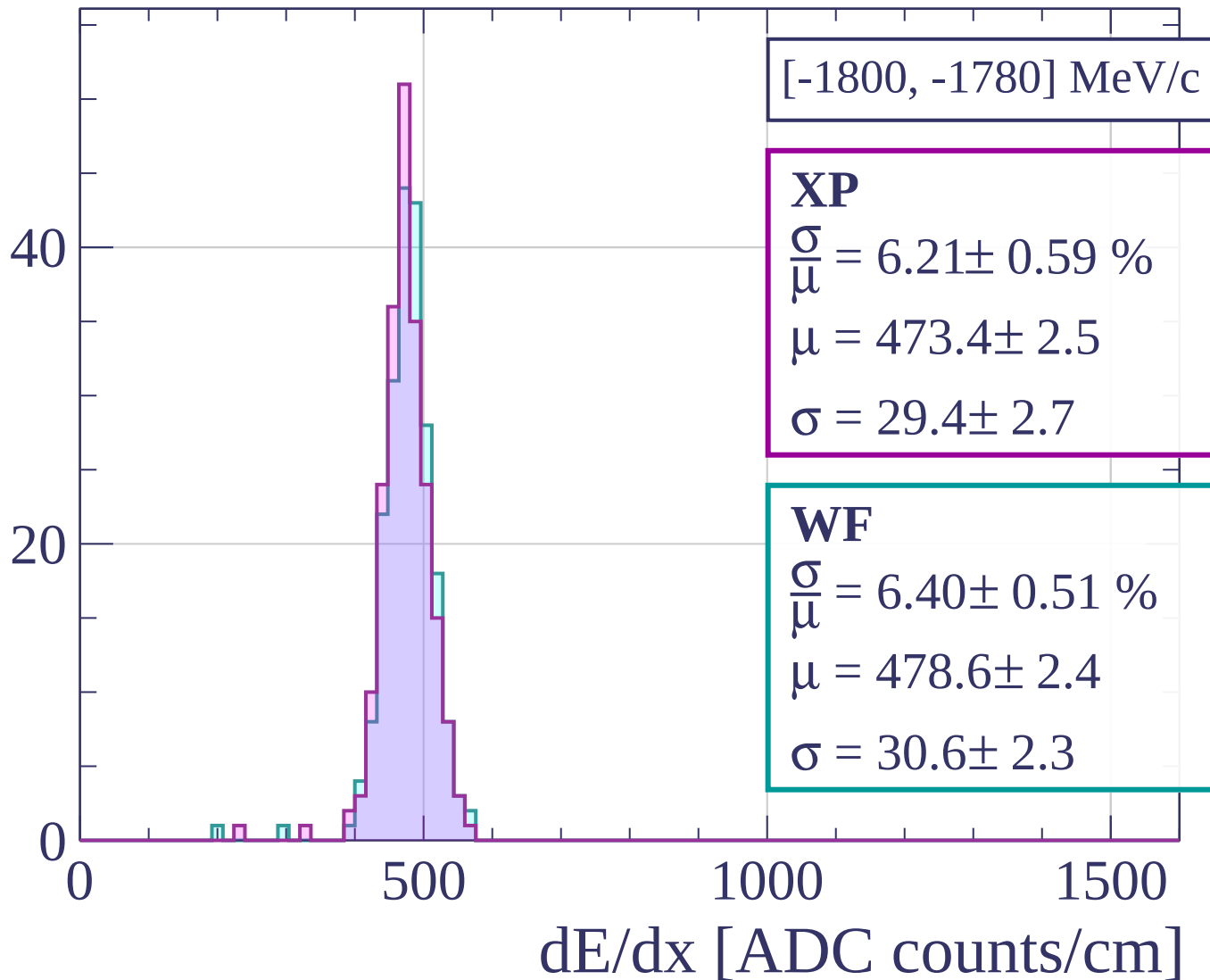
Count



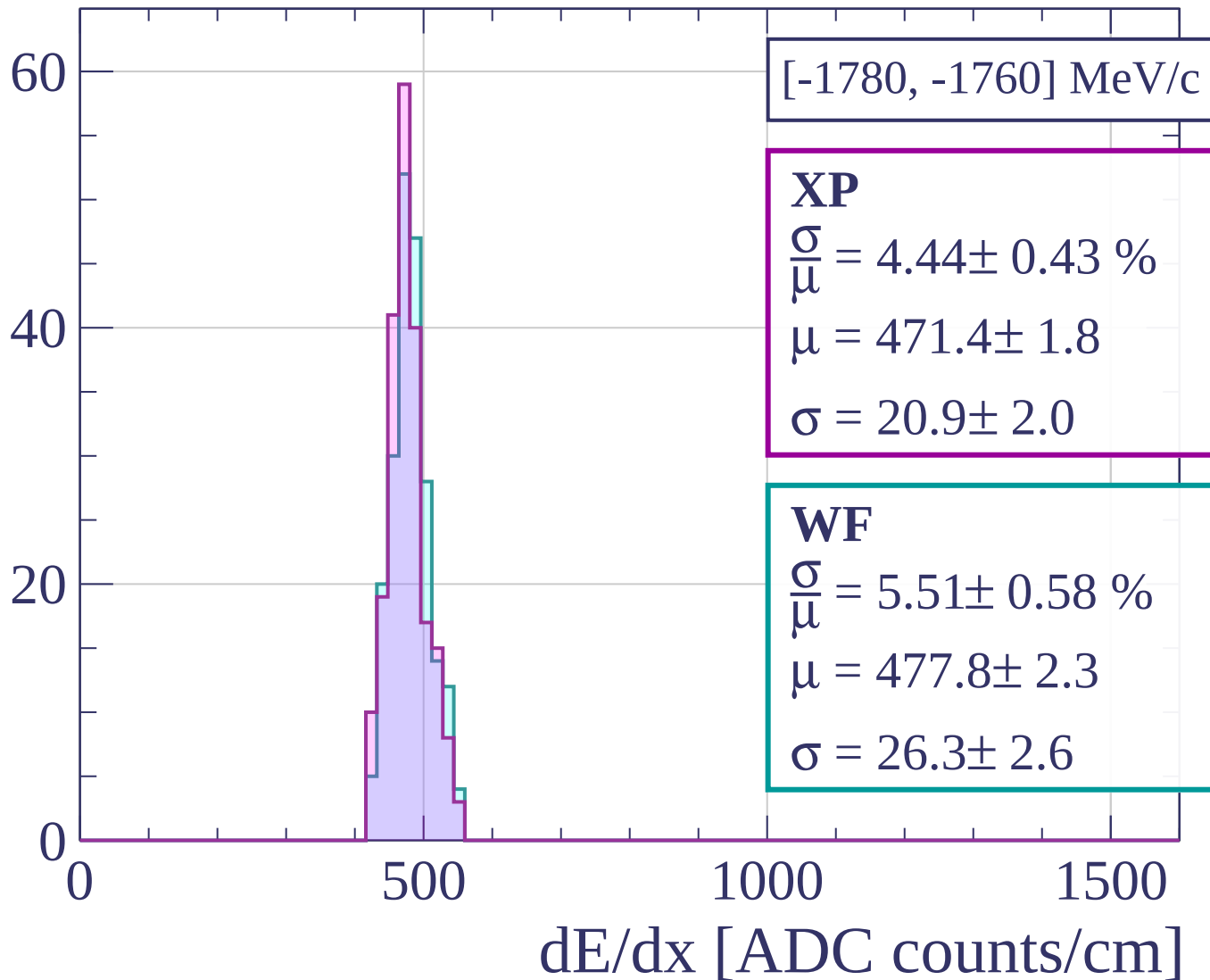
Count



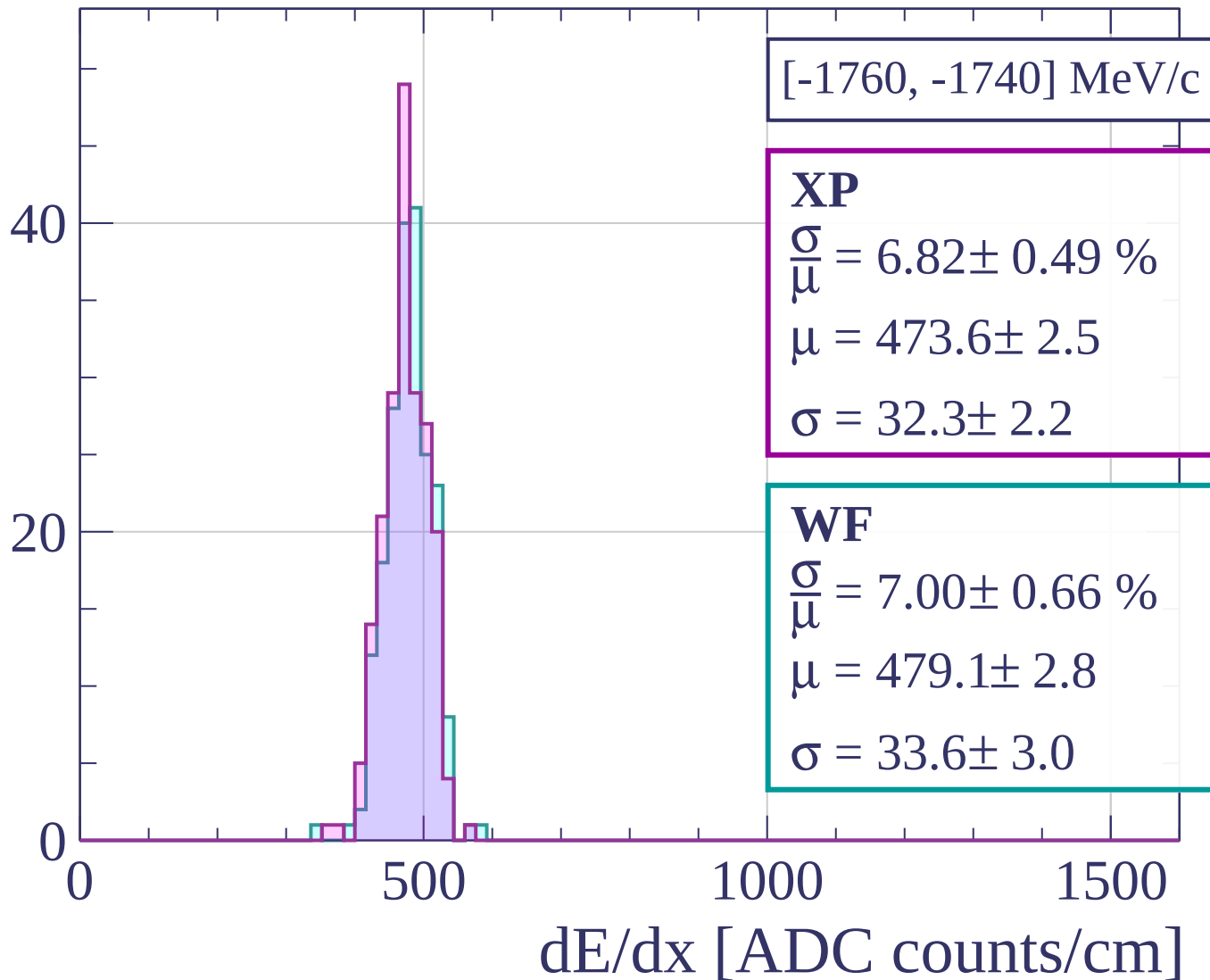
Count



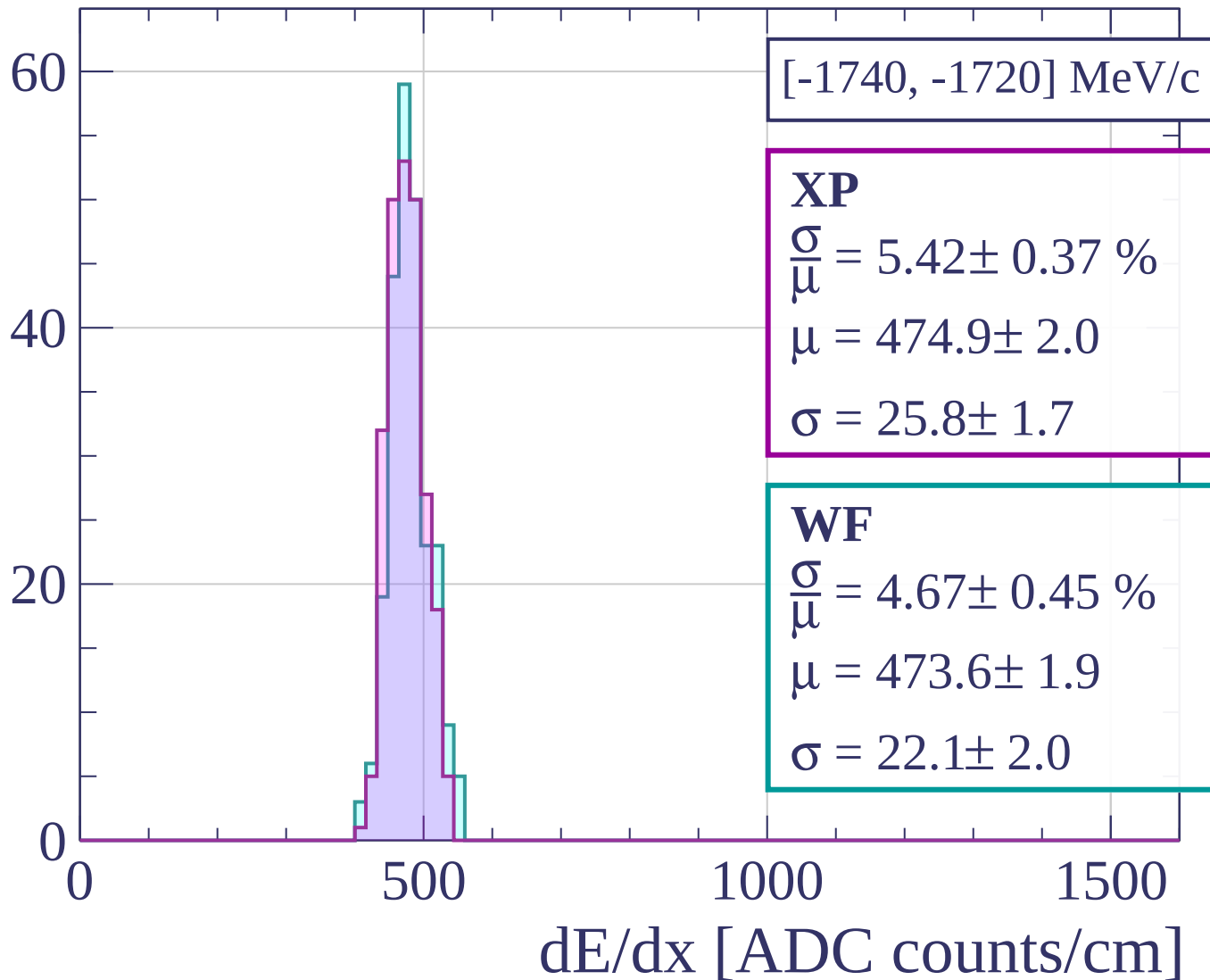
Count



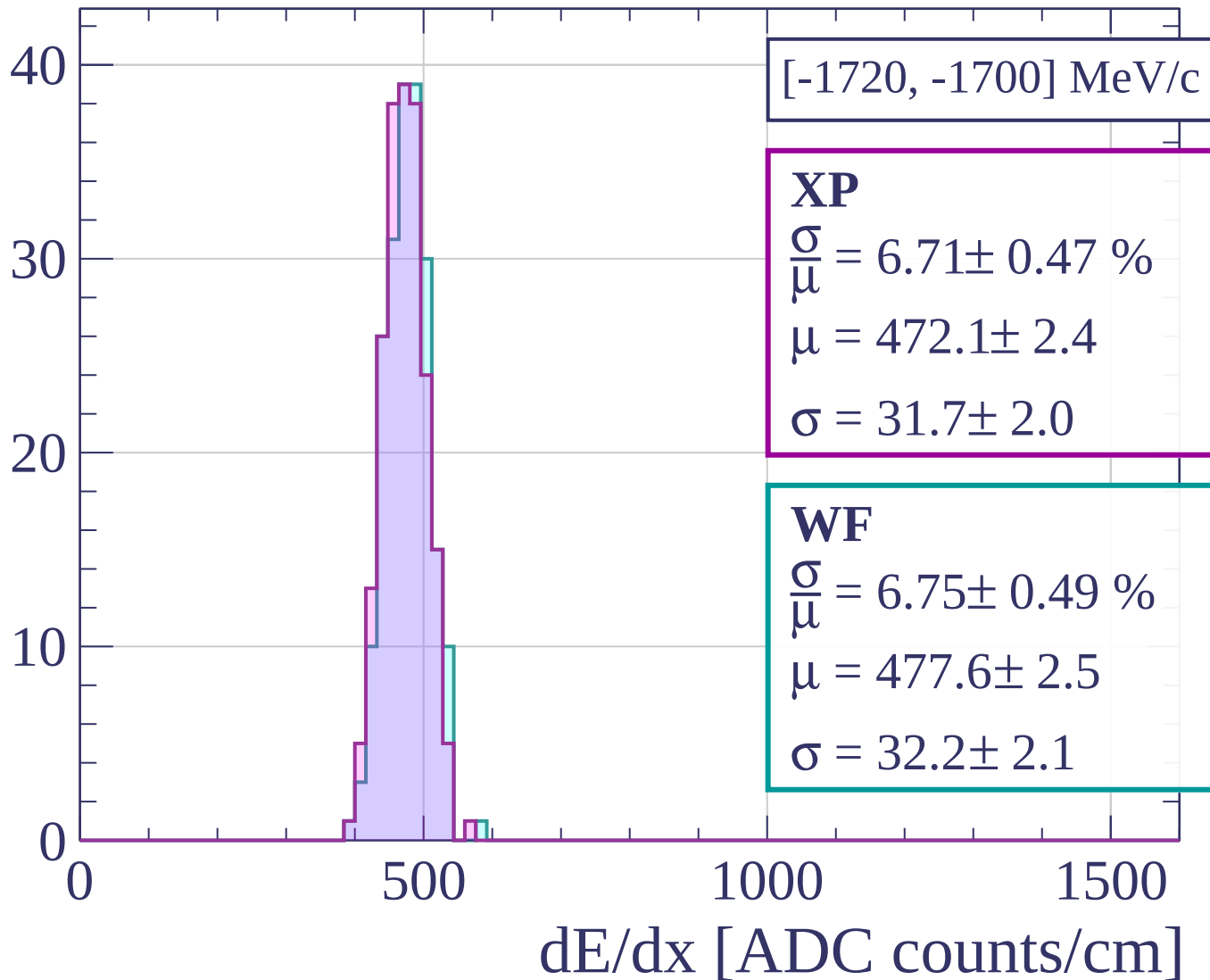
Count



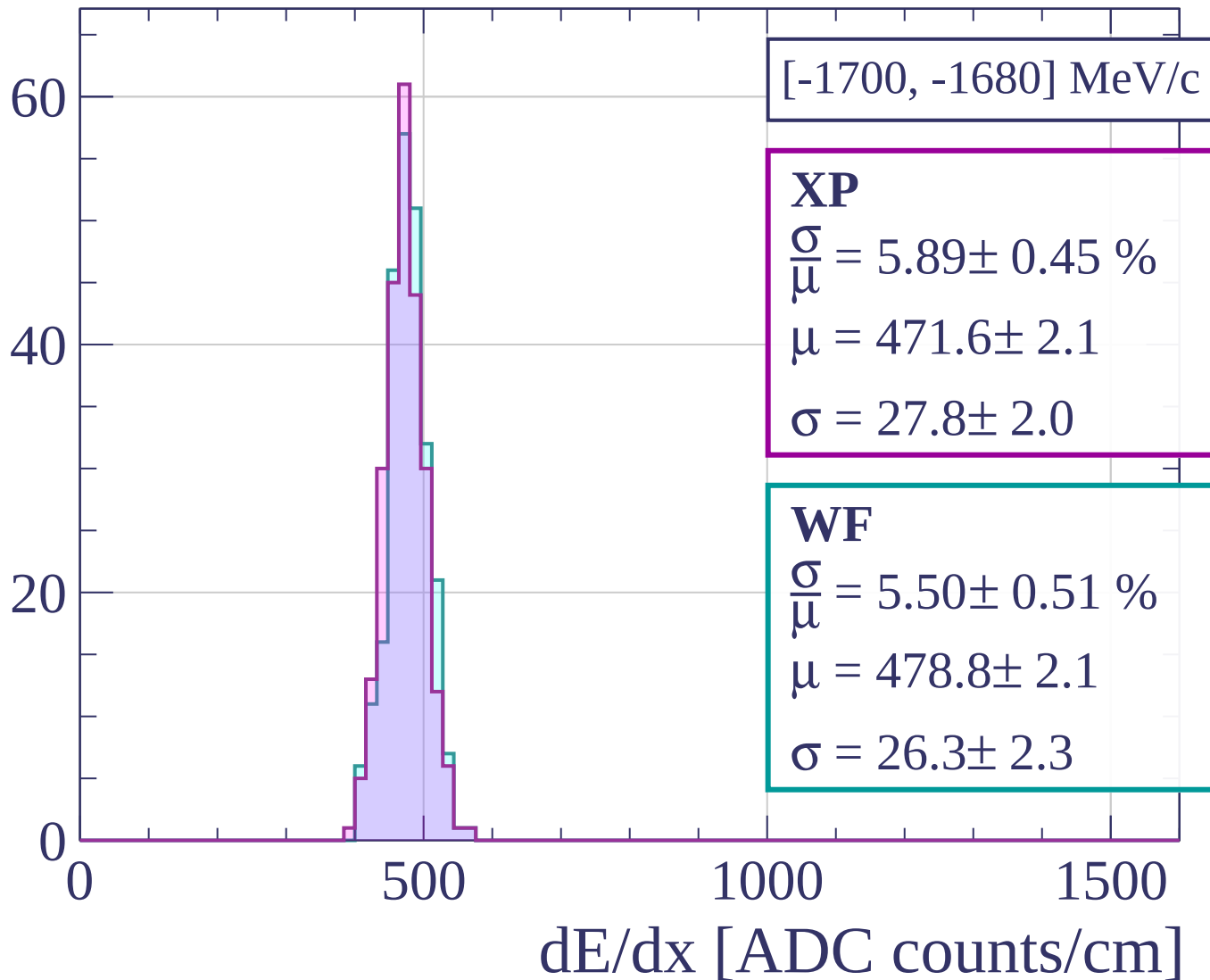
Count



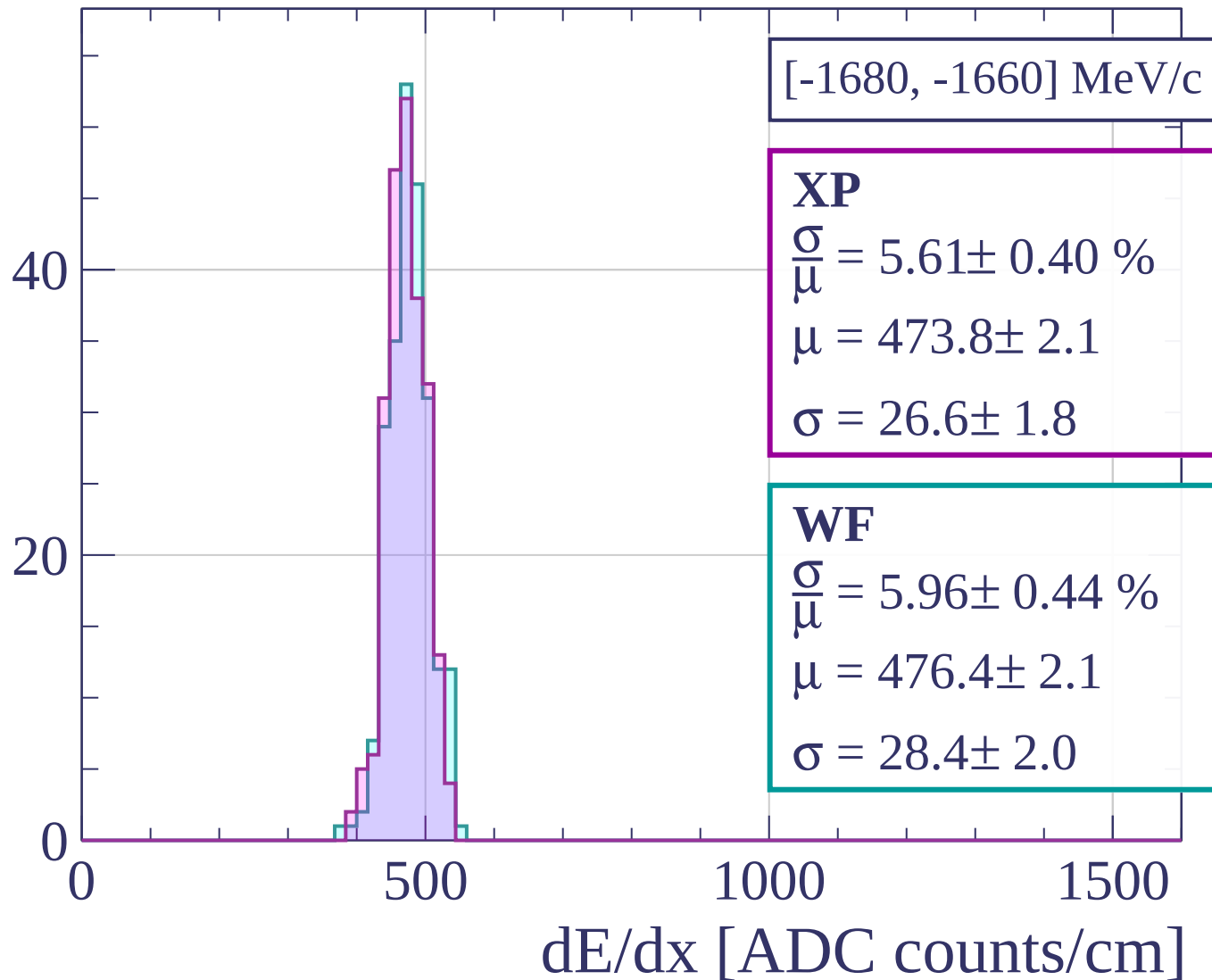
Count



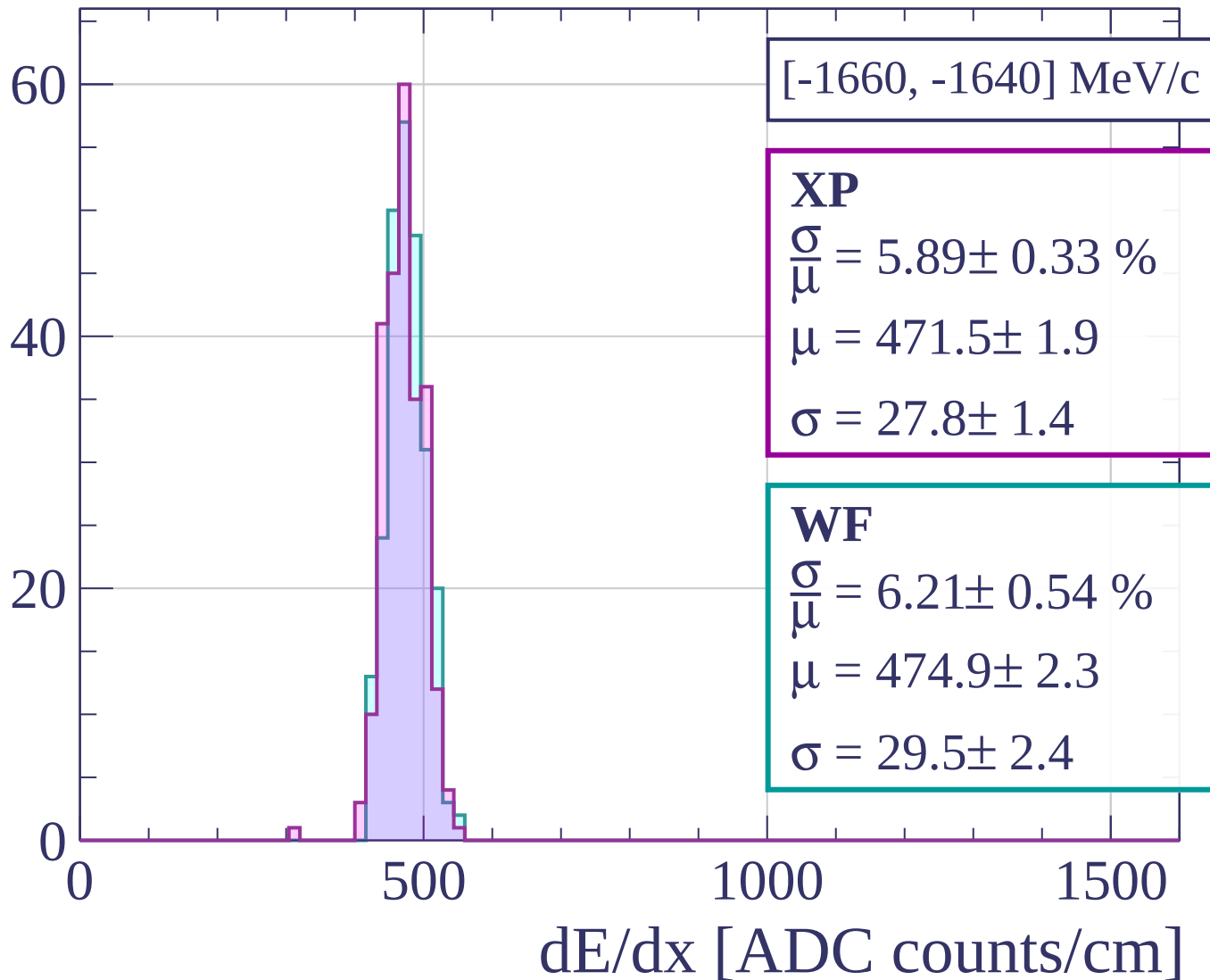
Count



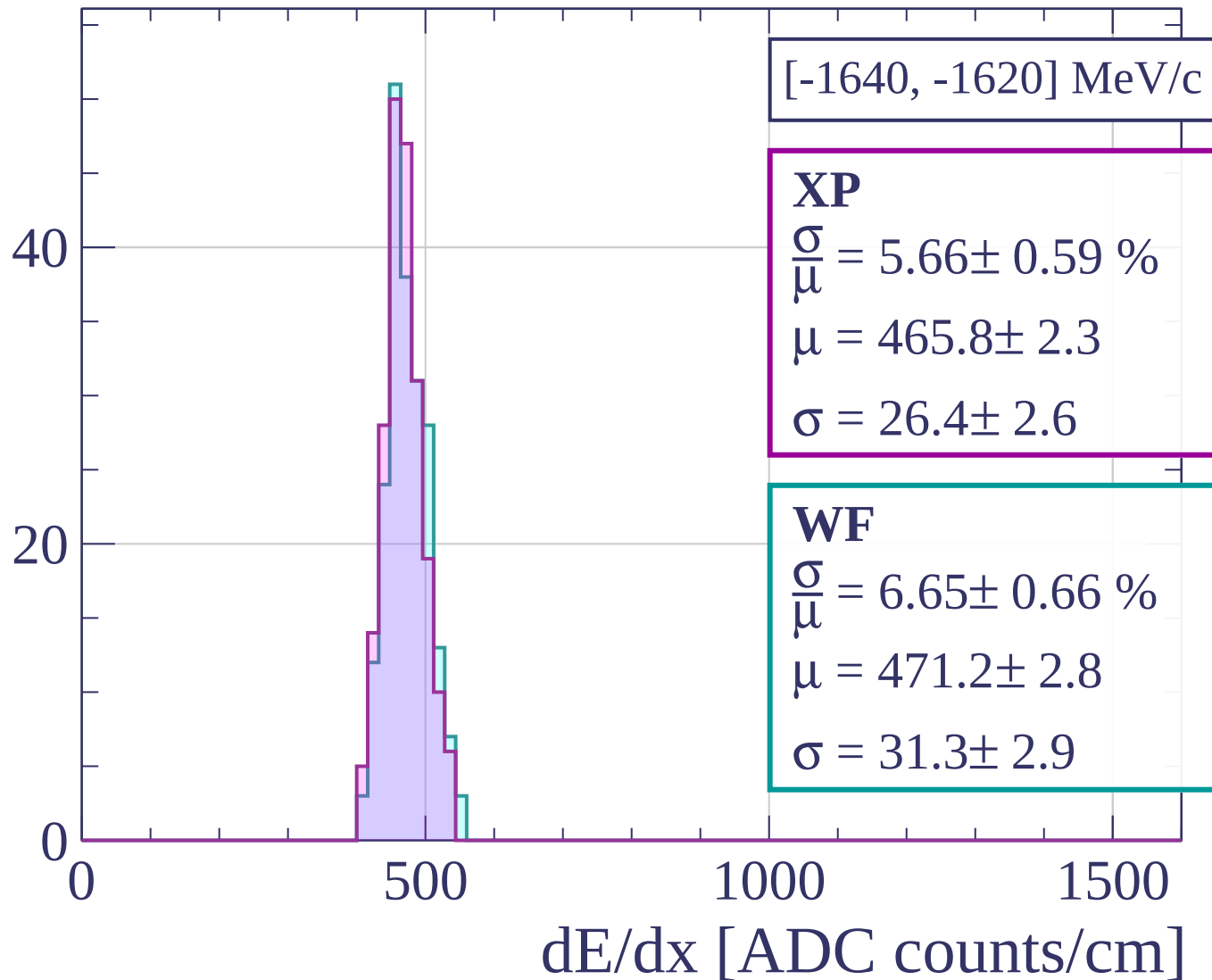
Count



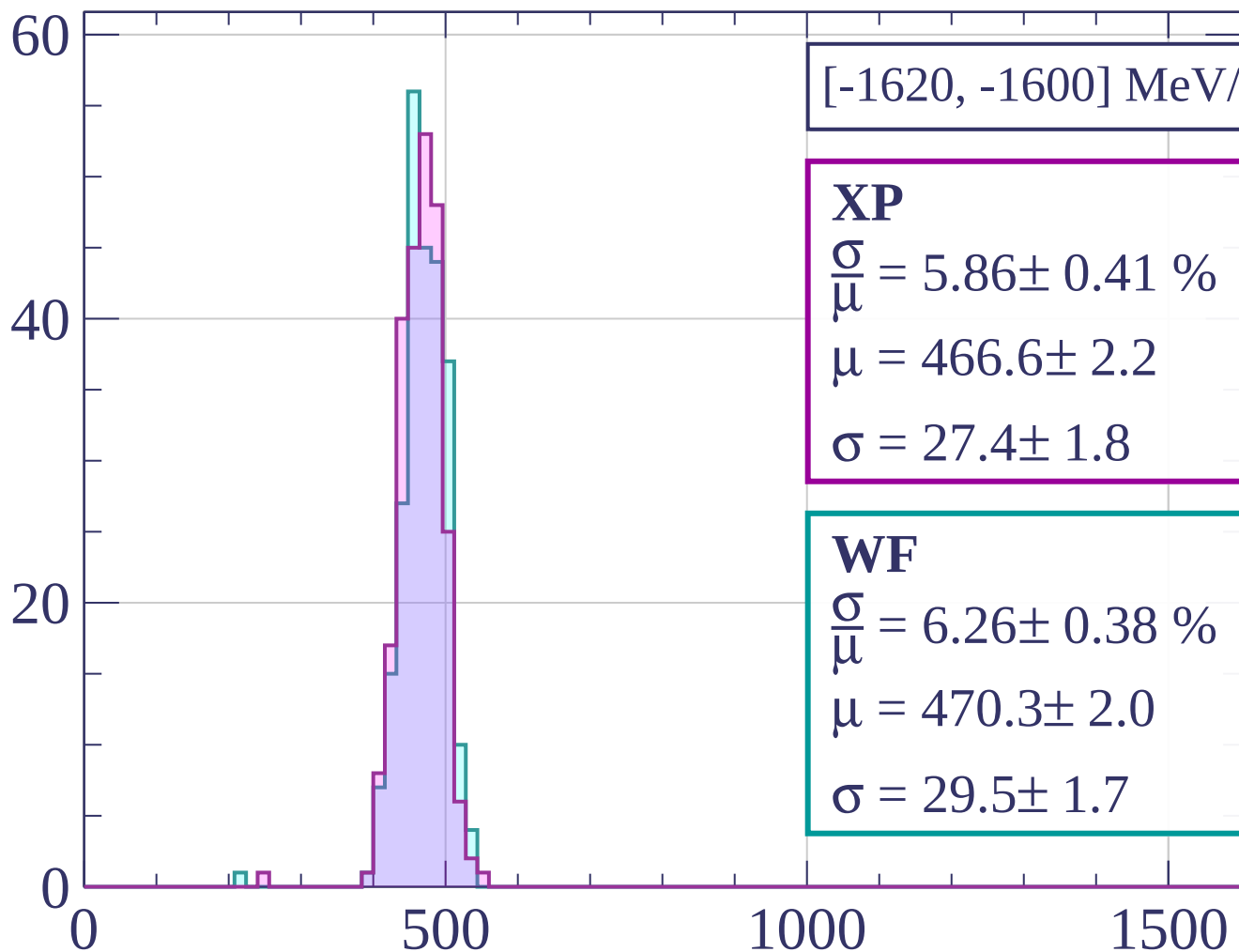
Count



Count



Count



[-1620, -1600] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.86 \pm 0.41 \%$$

$$\mu = 466.6 \pm 2.2$$

$$\sigma = 27.4 \pm 1.8$$

WF

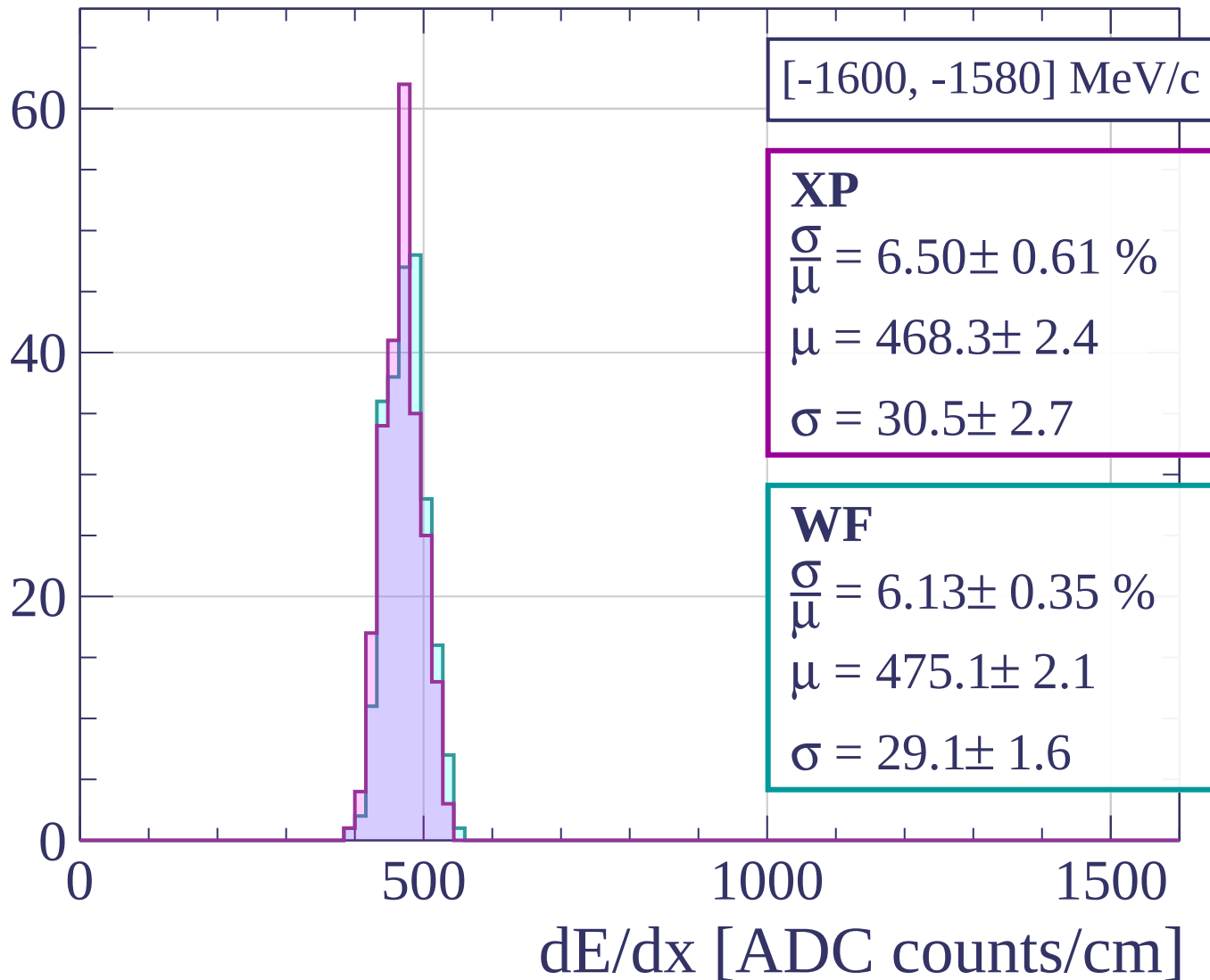
$$\frac{\sigma}{\mu} = 6.26 \pm 0.38 \%$$

$$\mu = 470.3 \pm 2.0$$

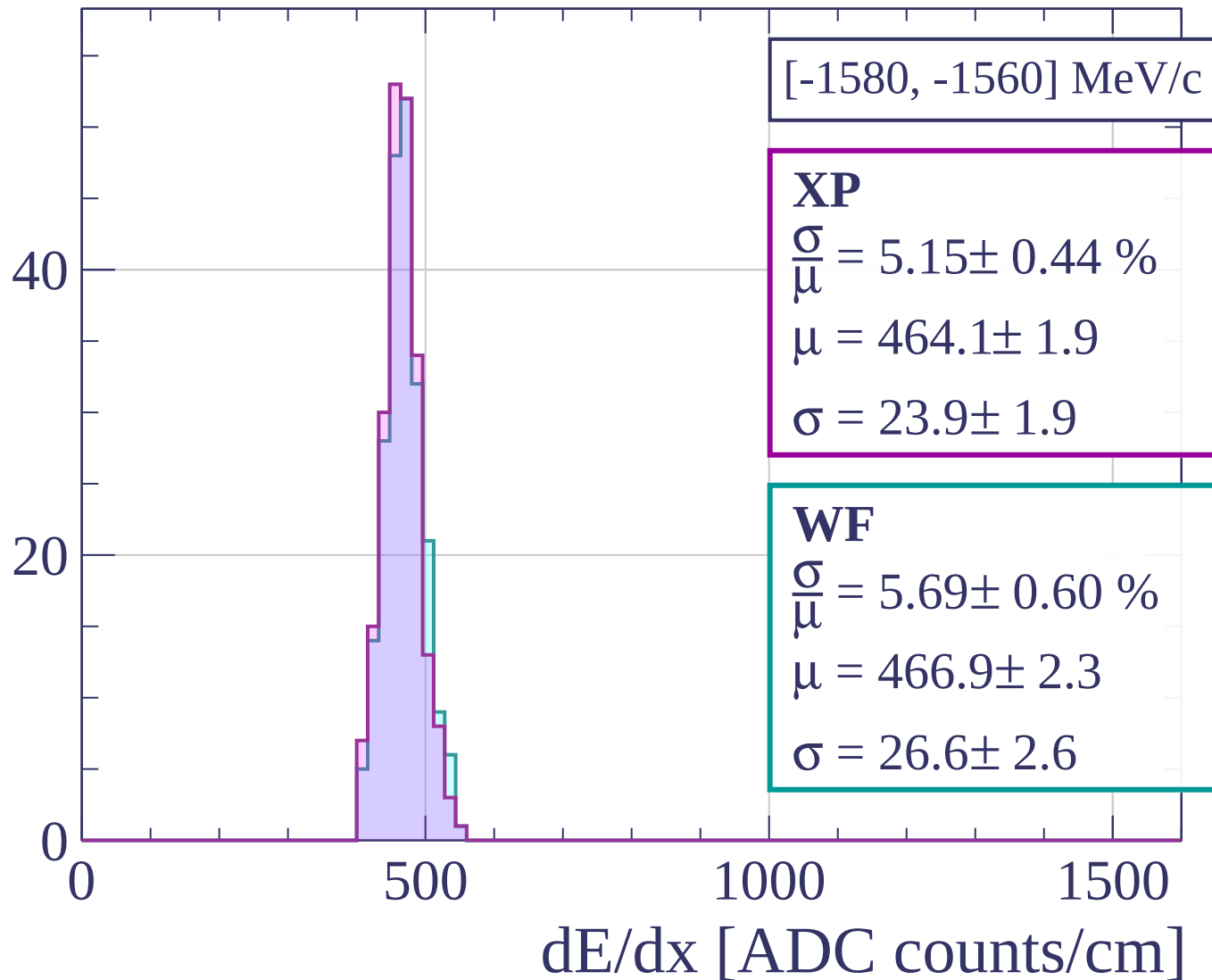
$$\sigma = 29.5 \pm 1.7$$

dE/dx [ADC counts/cm]

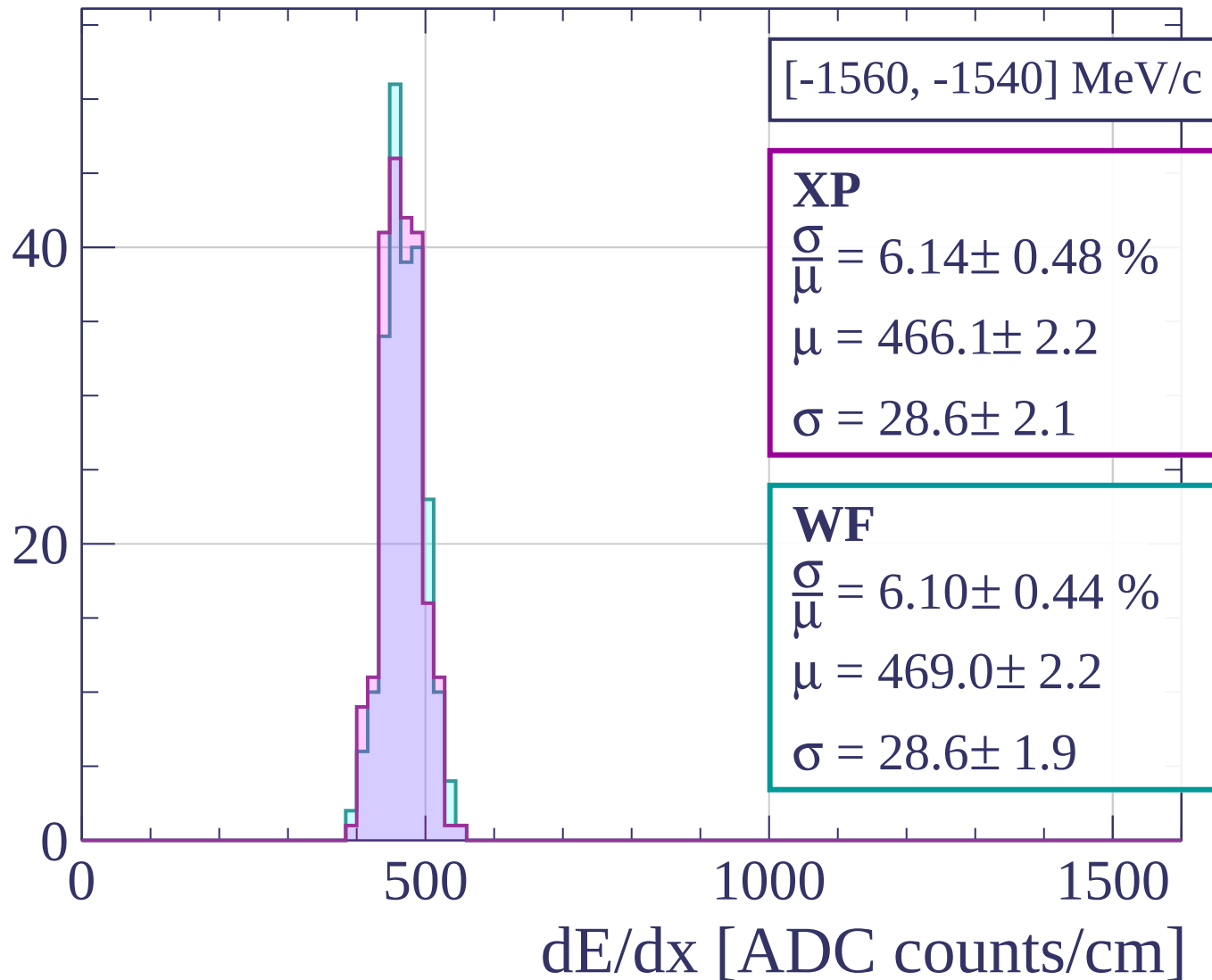
Count



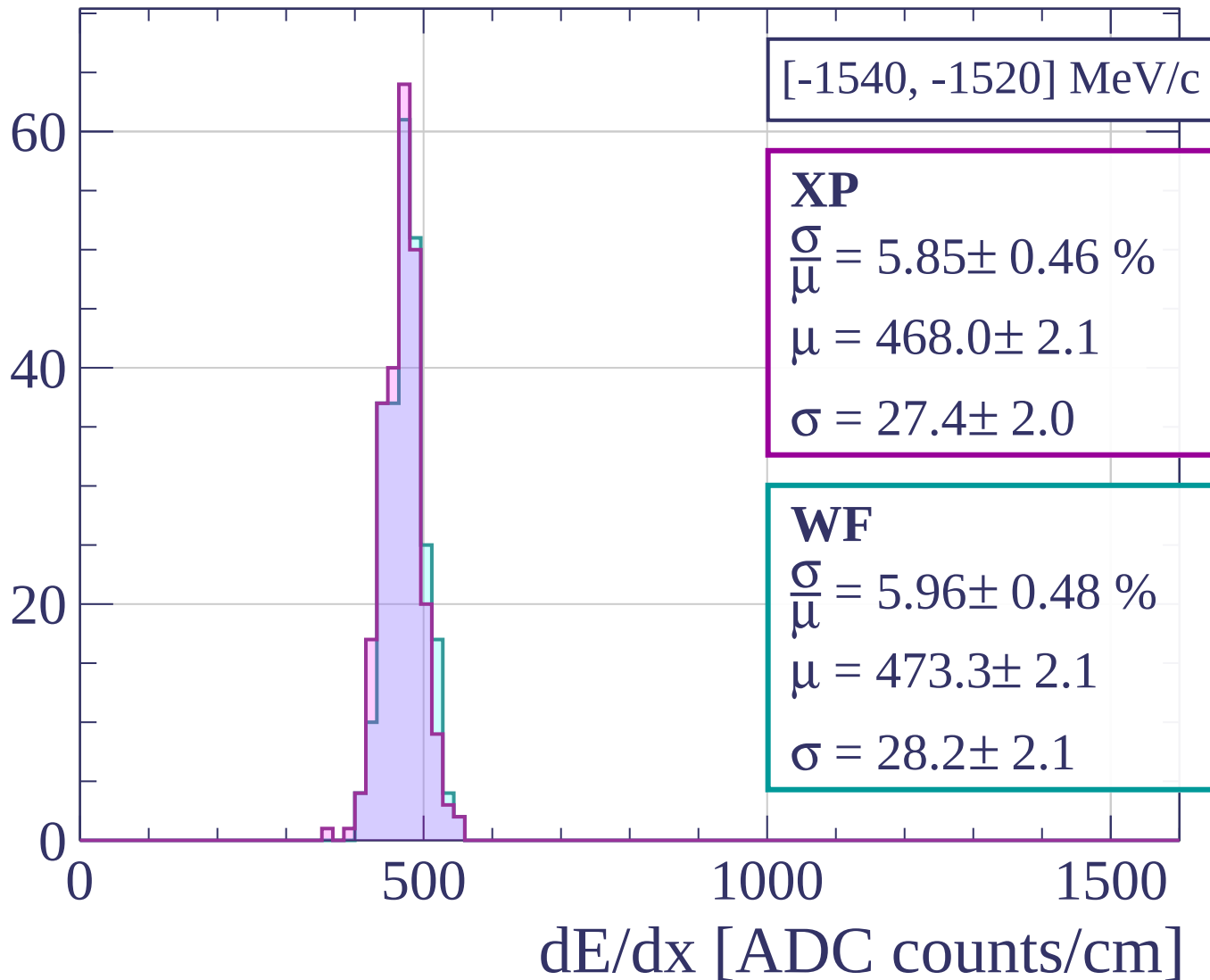
Count



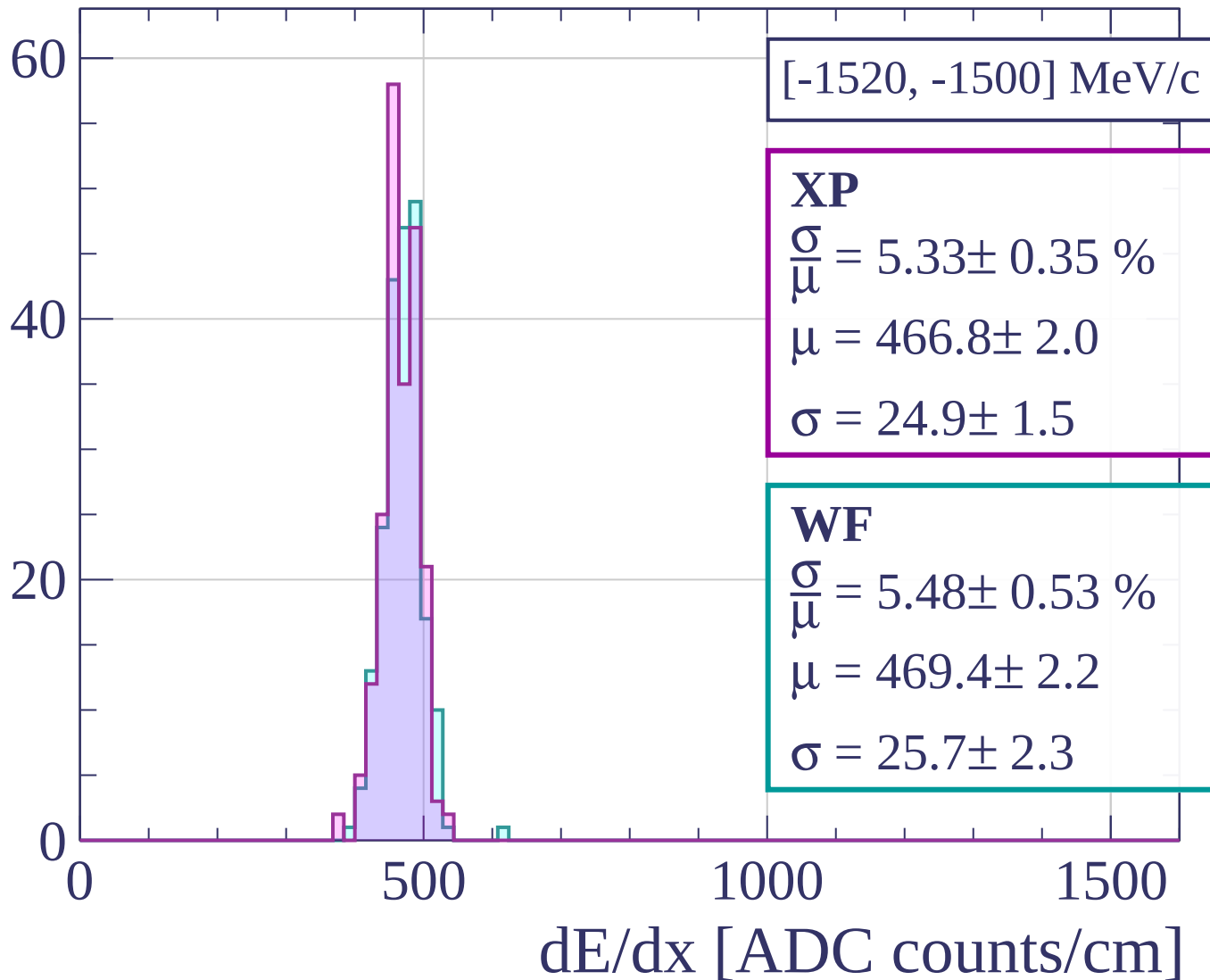
Count



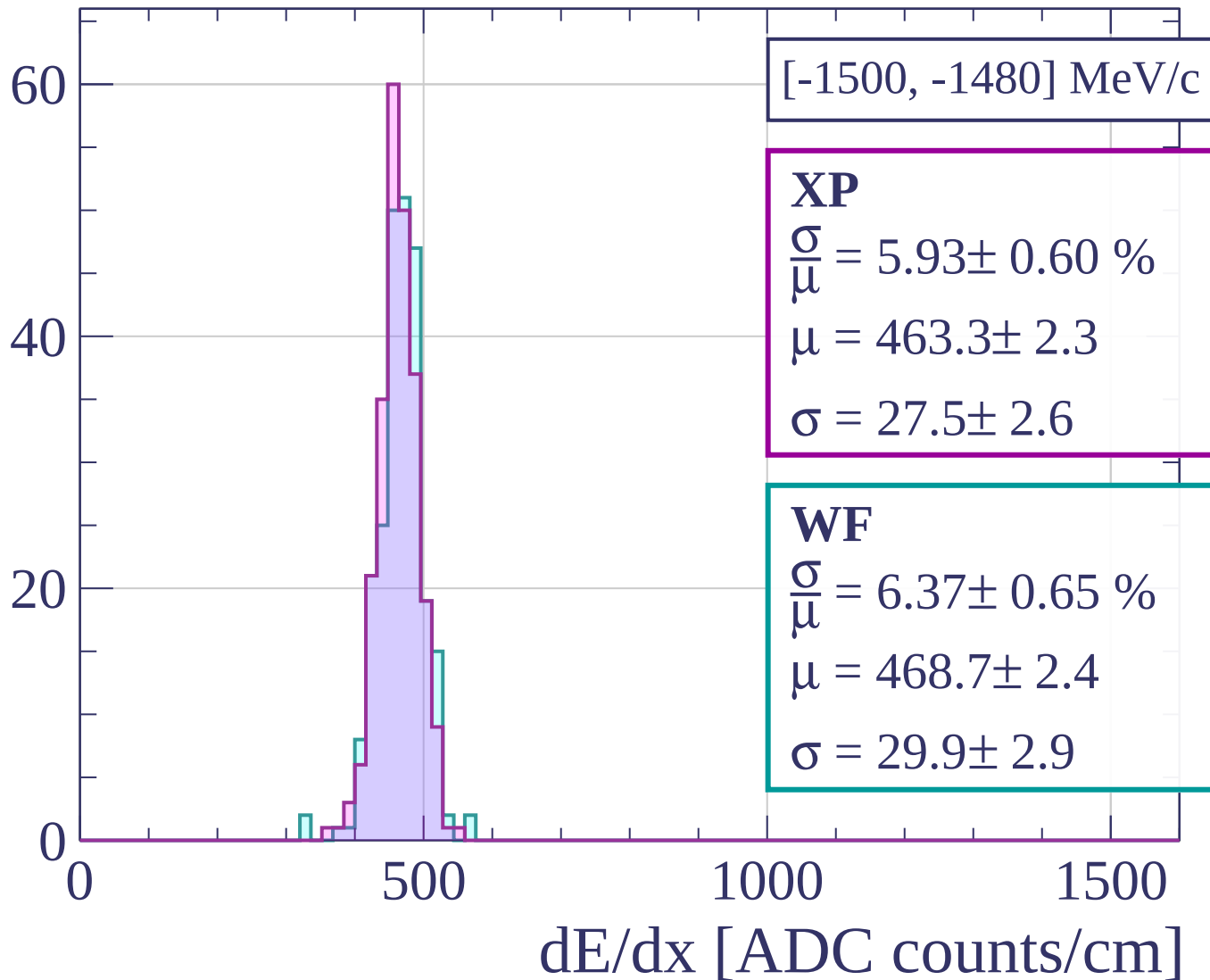
Count



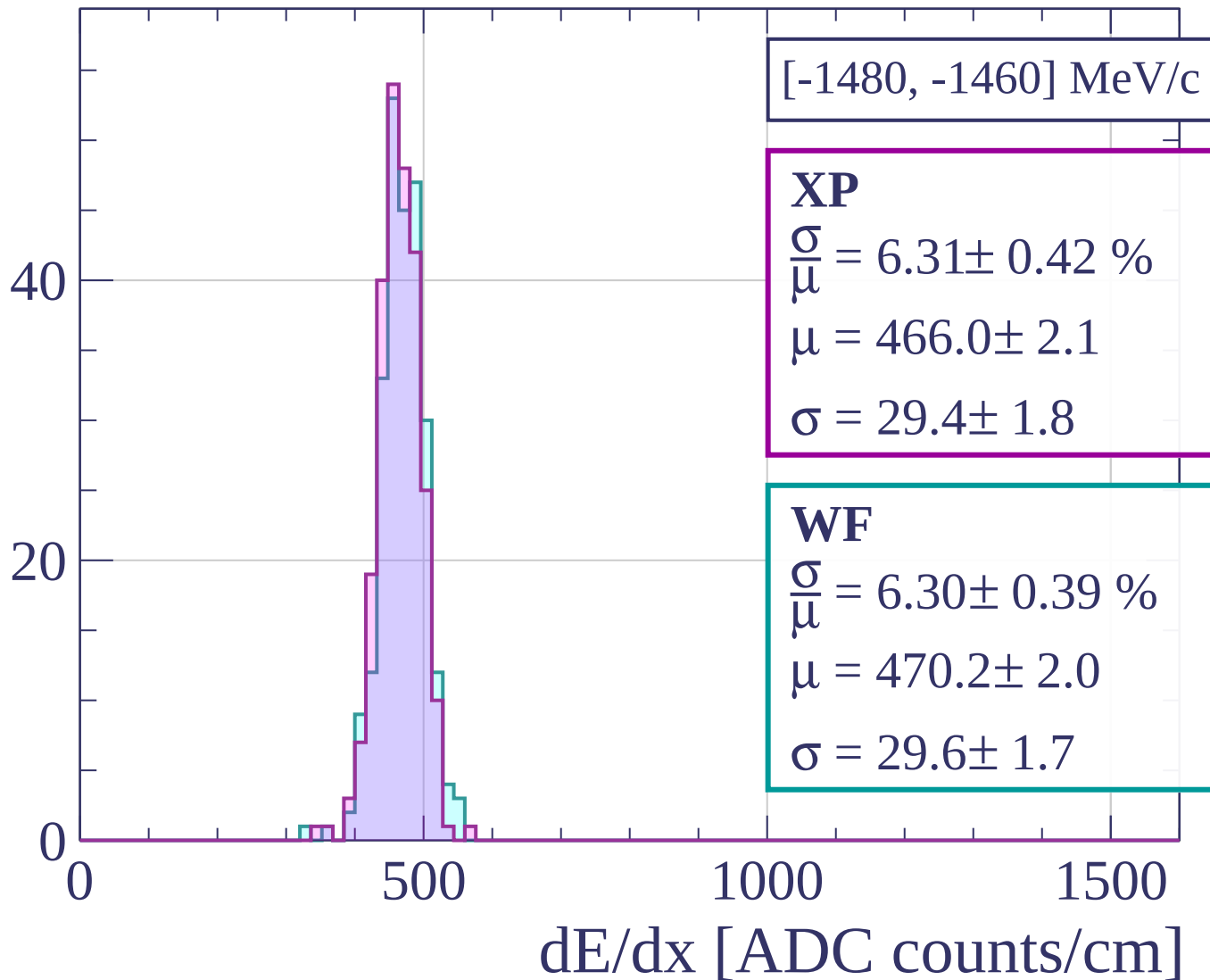
Count



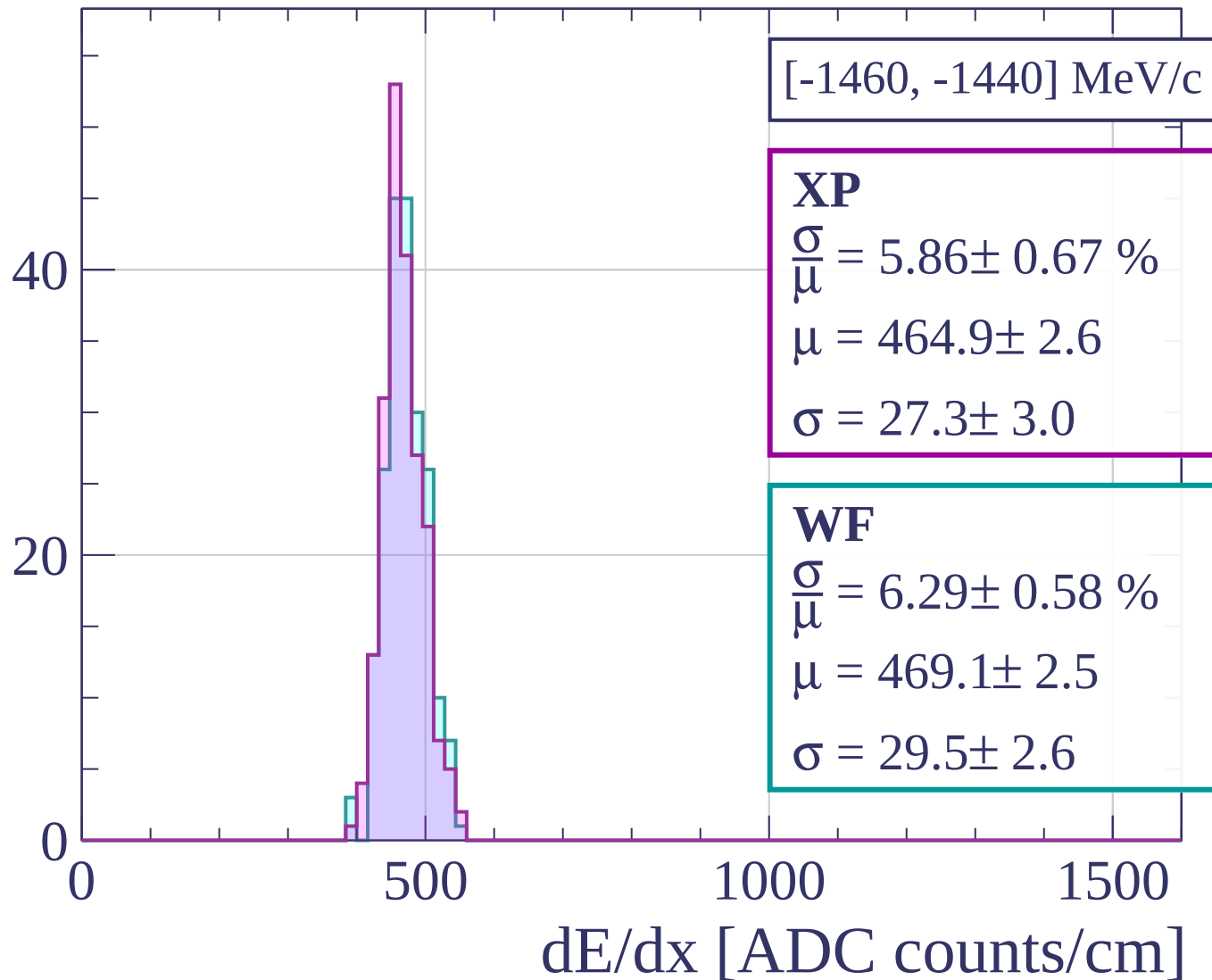
Count



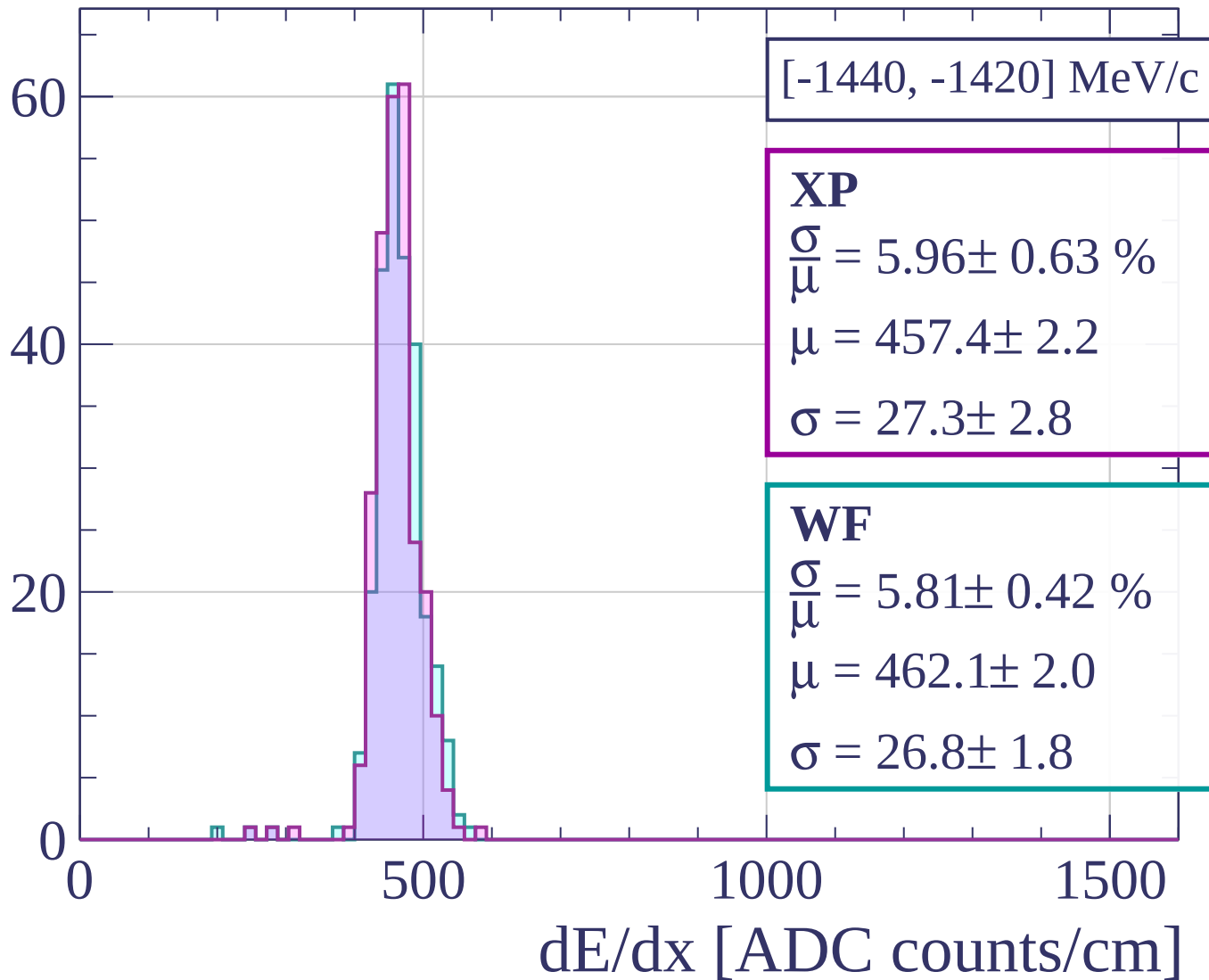
Count



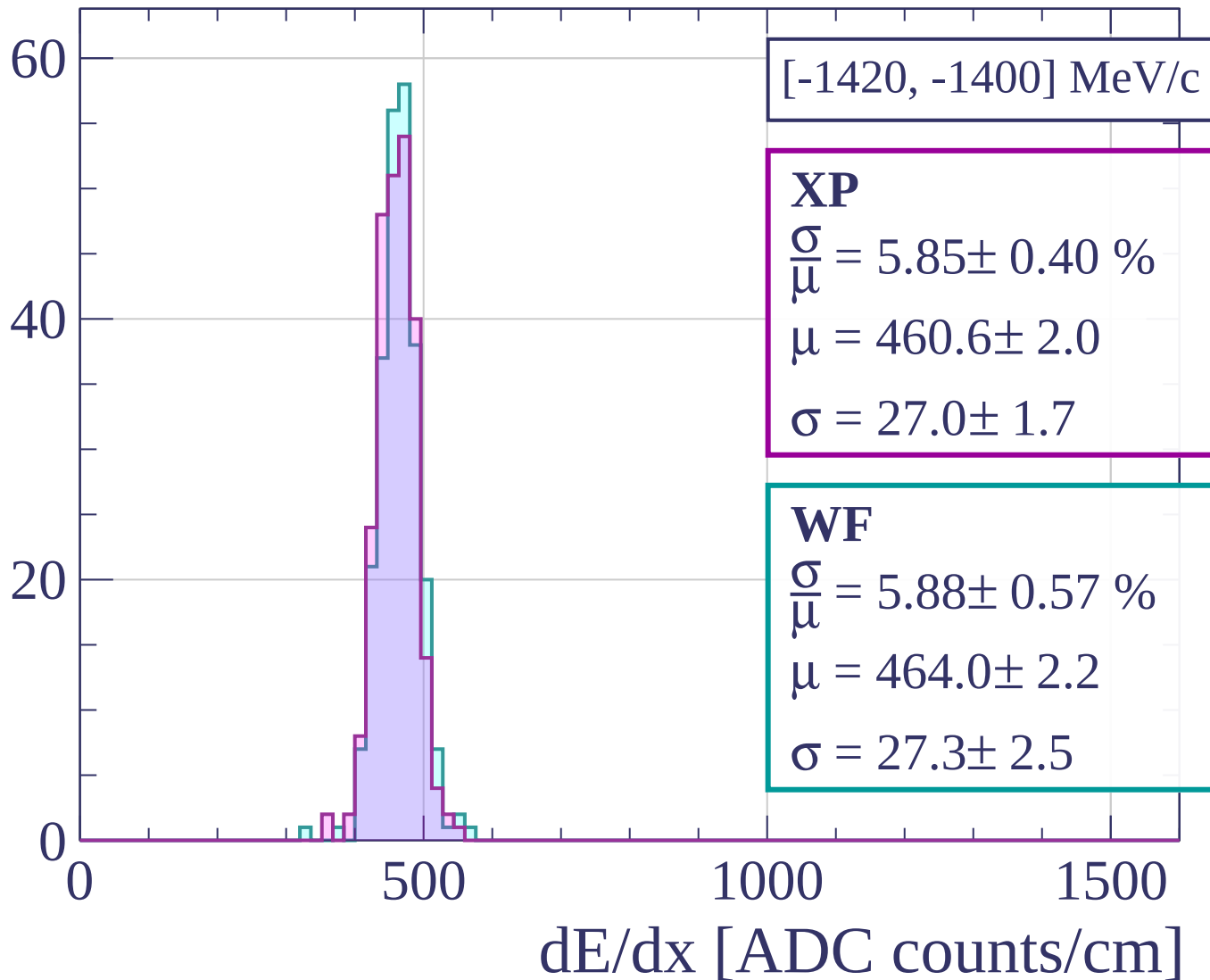
Count



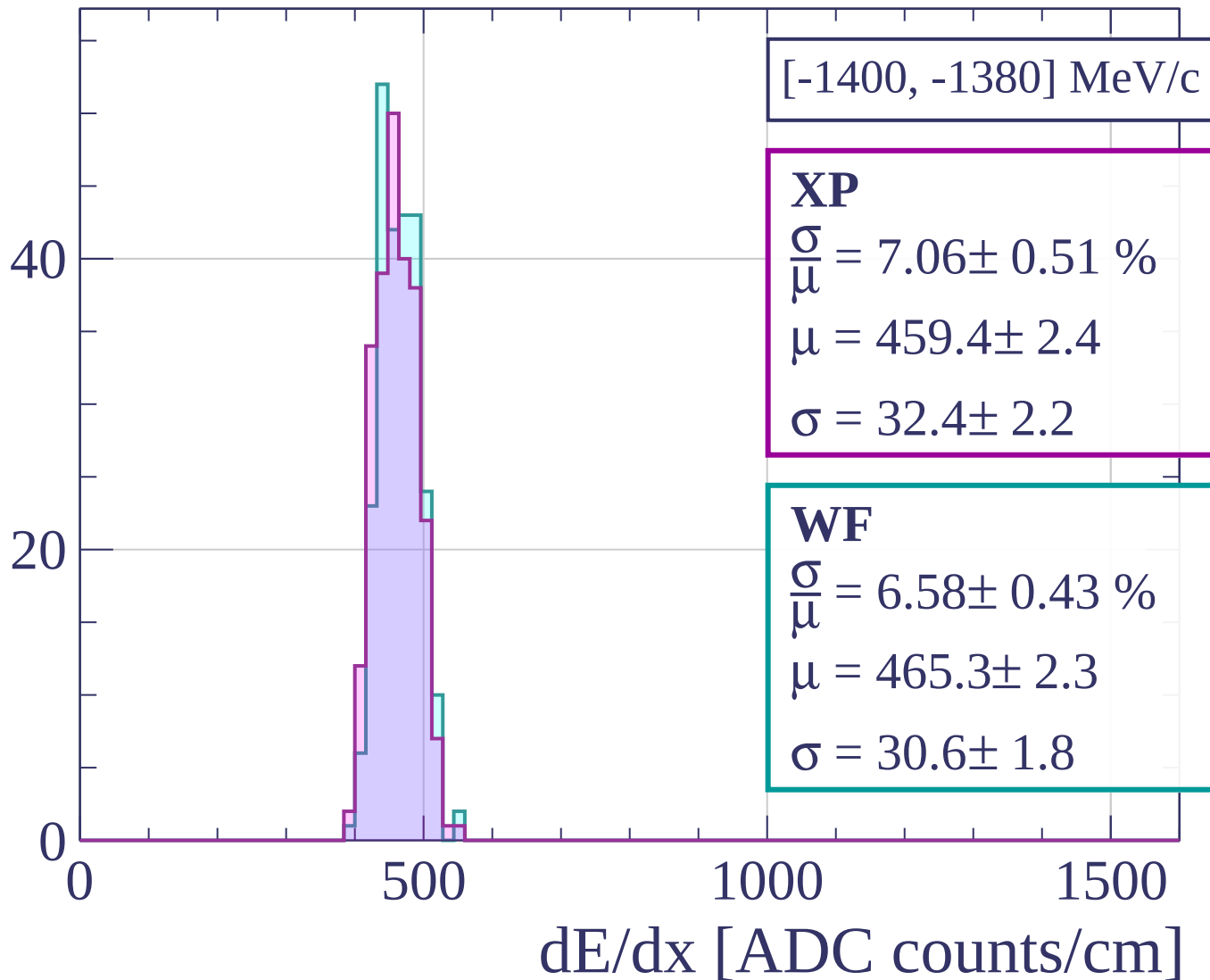
Count



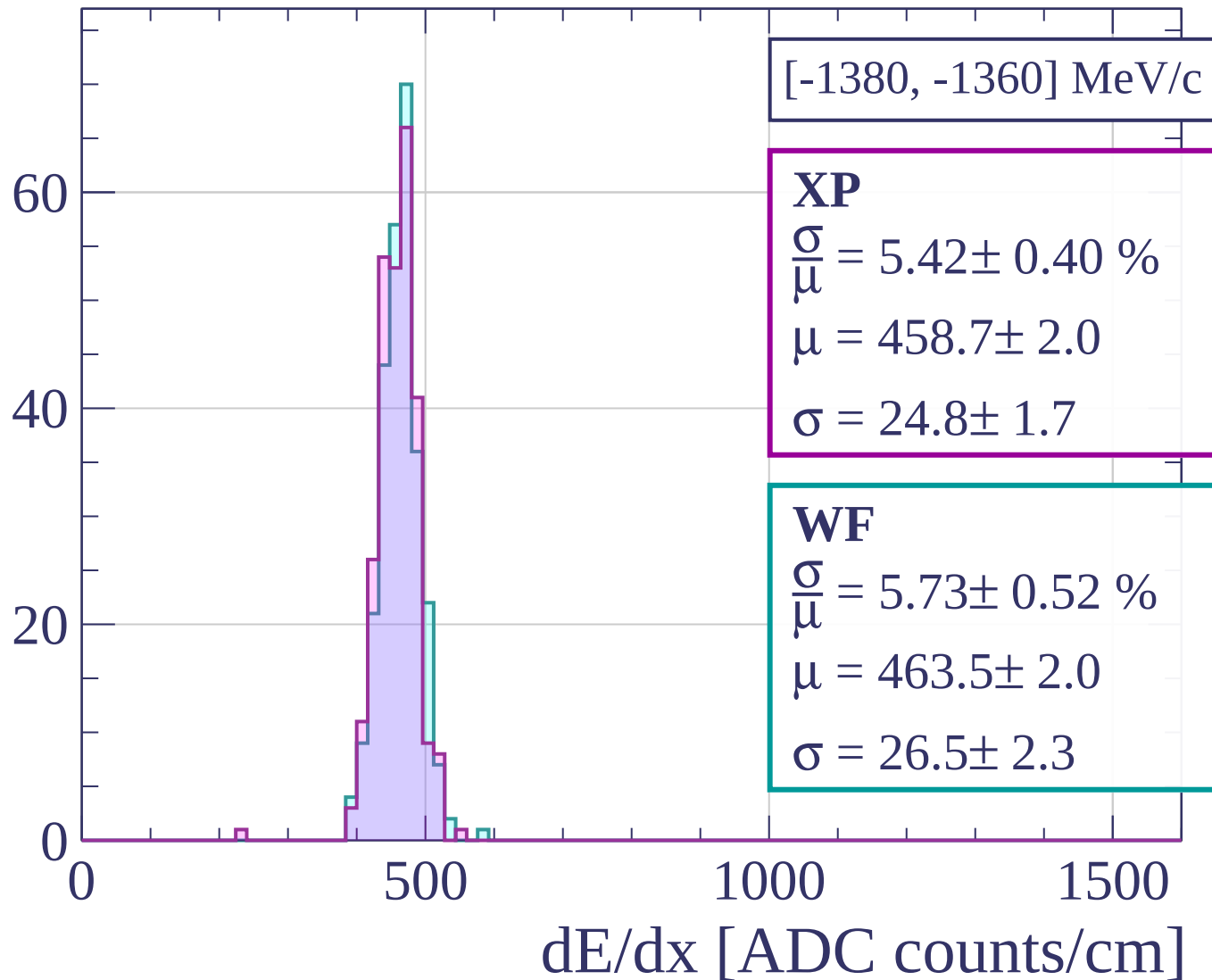
Count



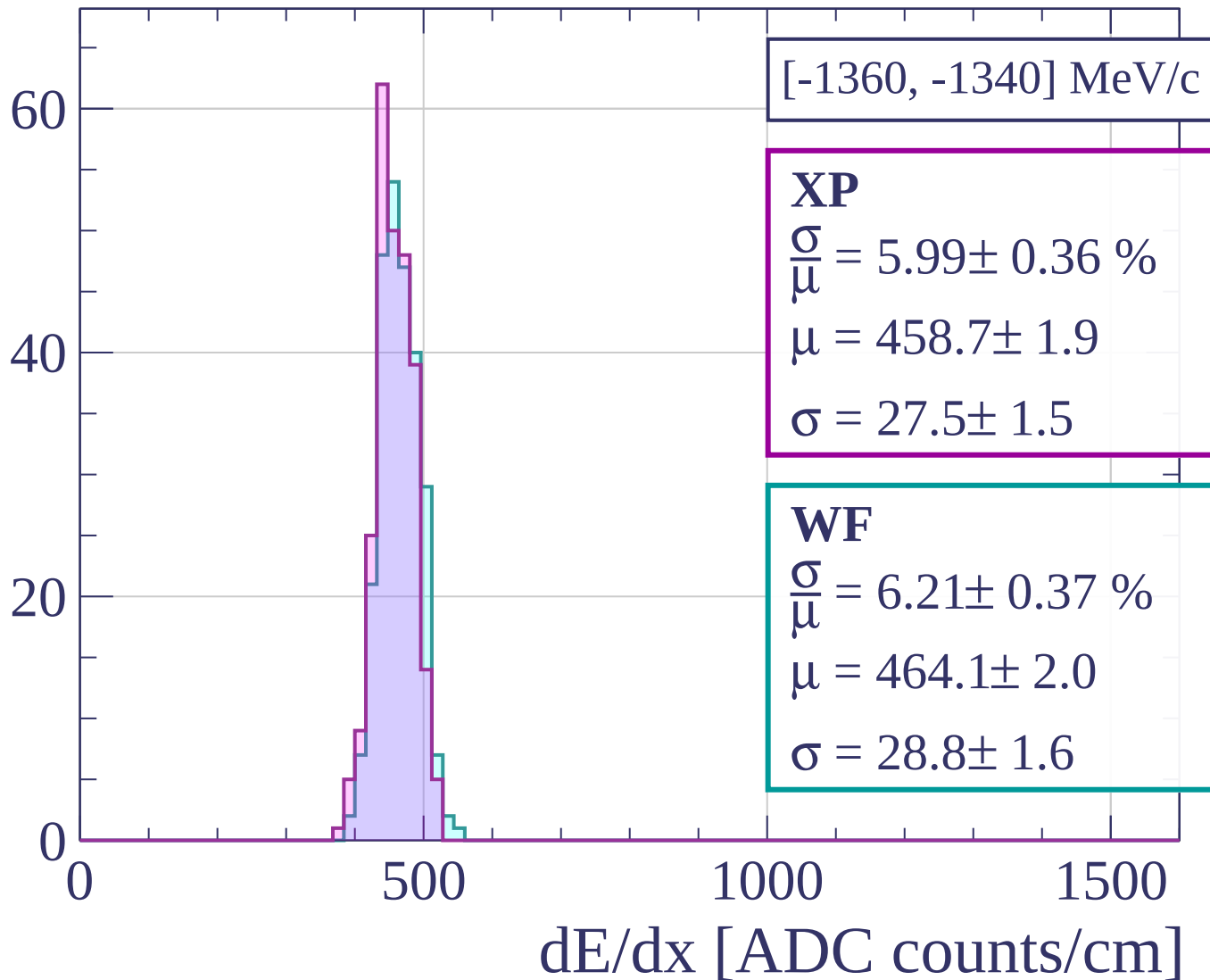
Count



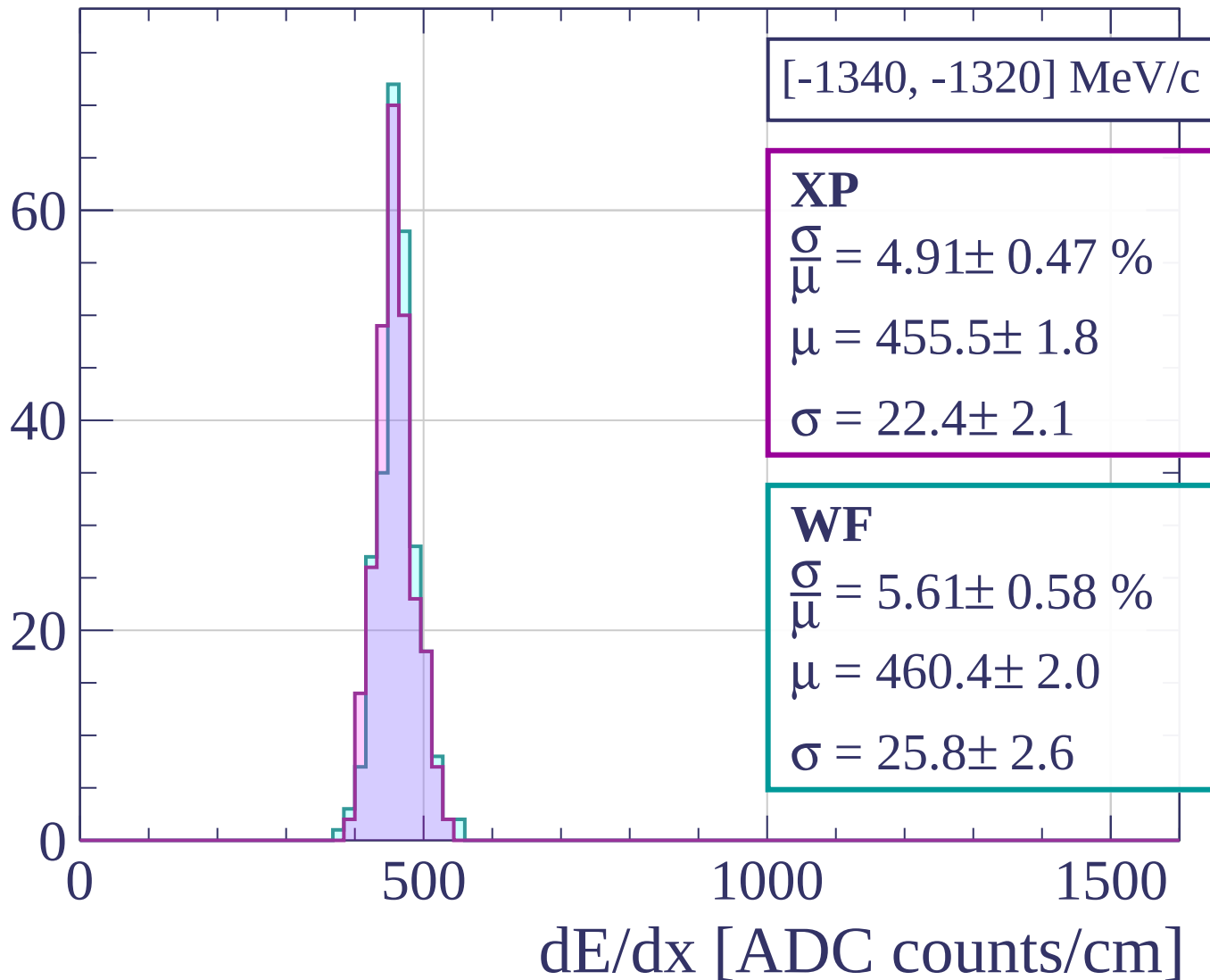
Count



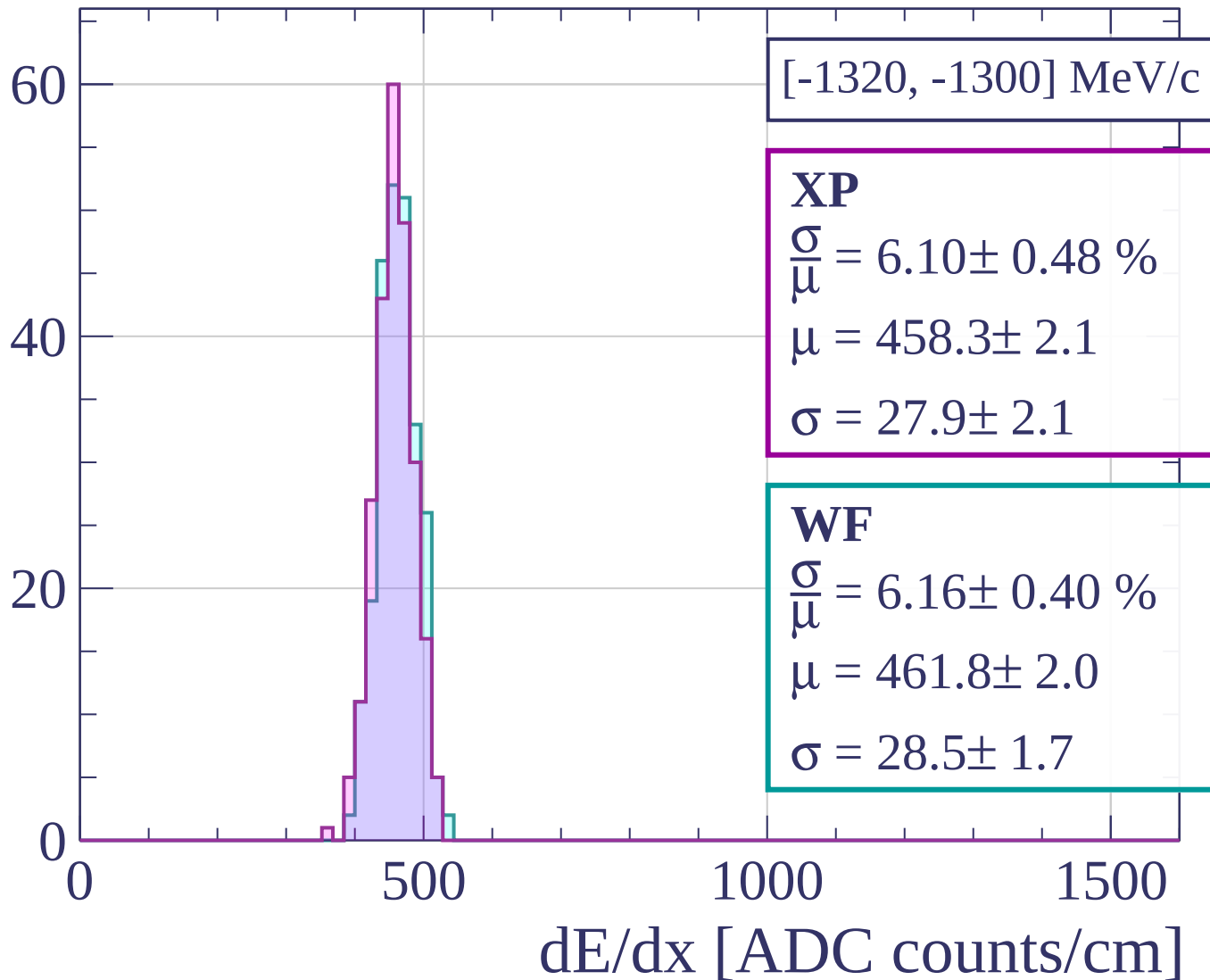
Count



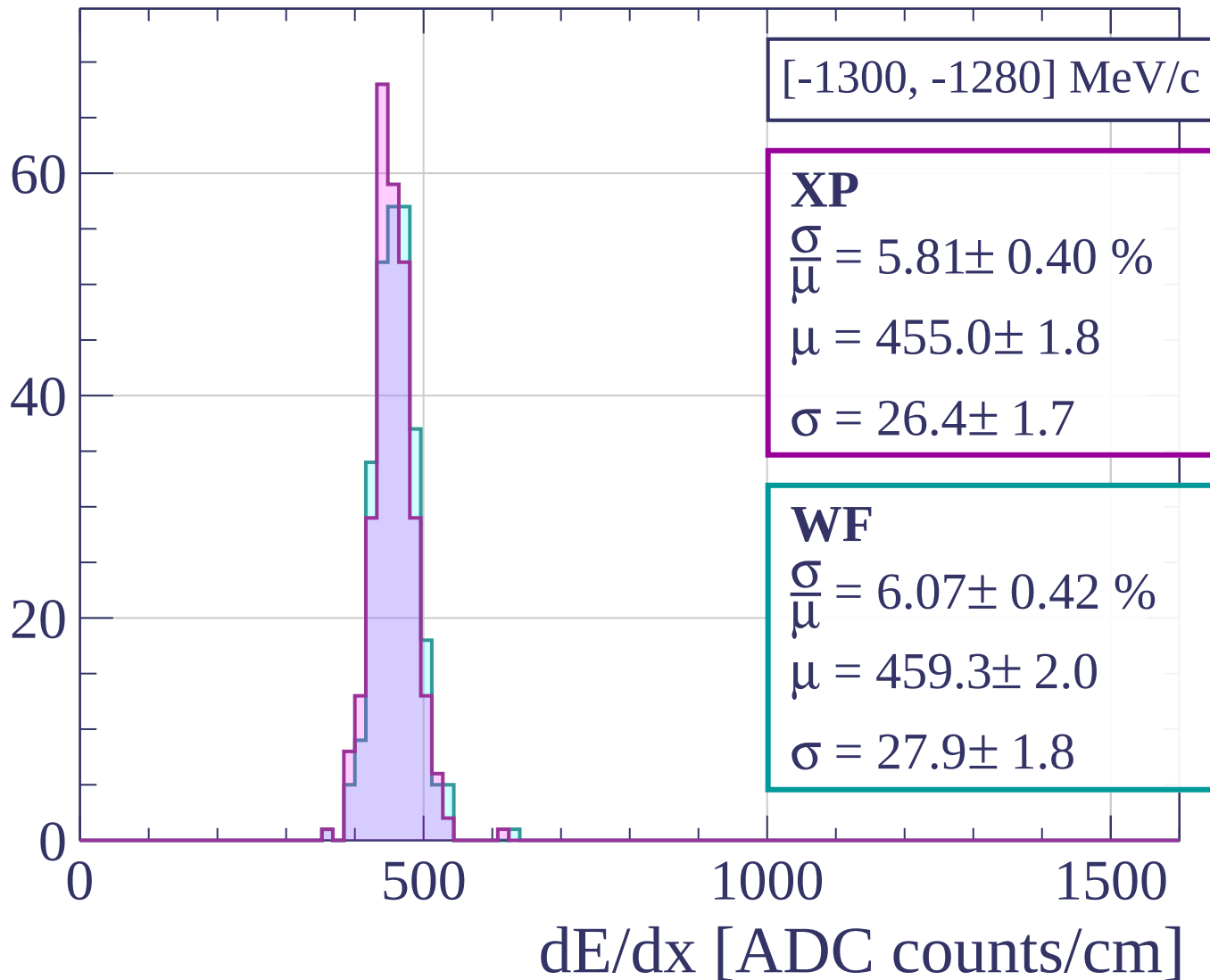
Count



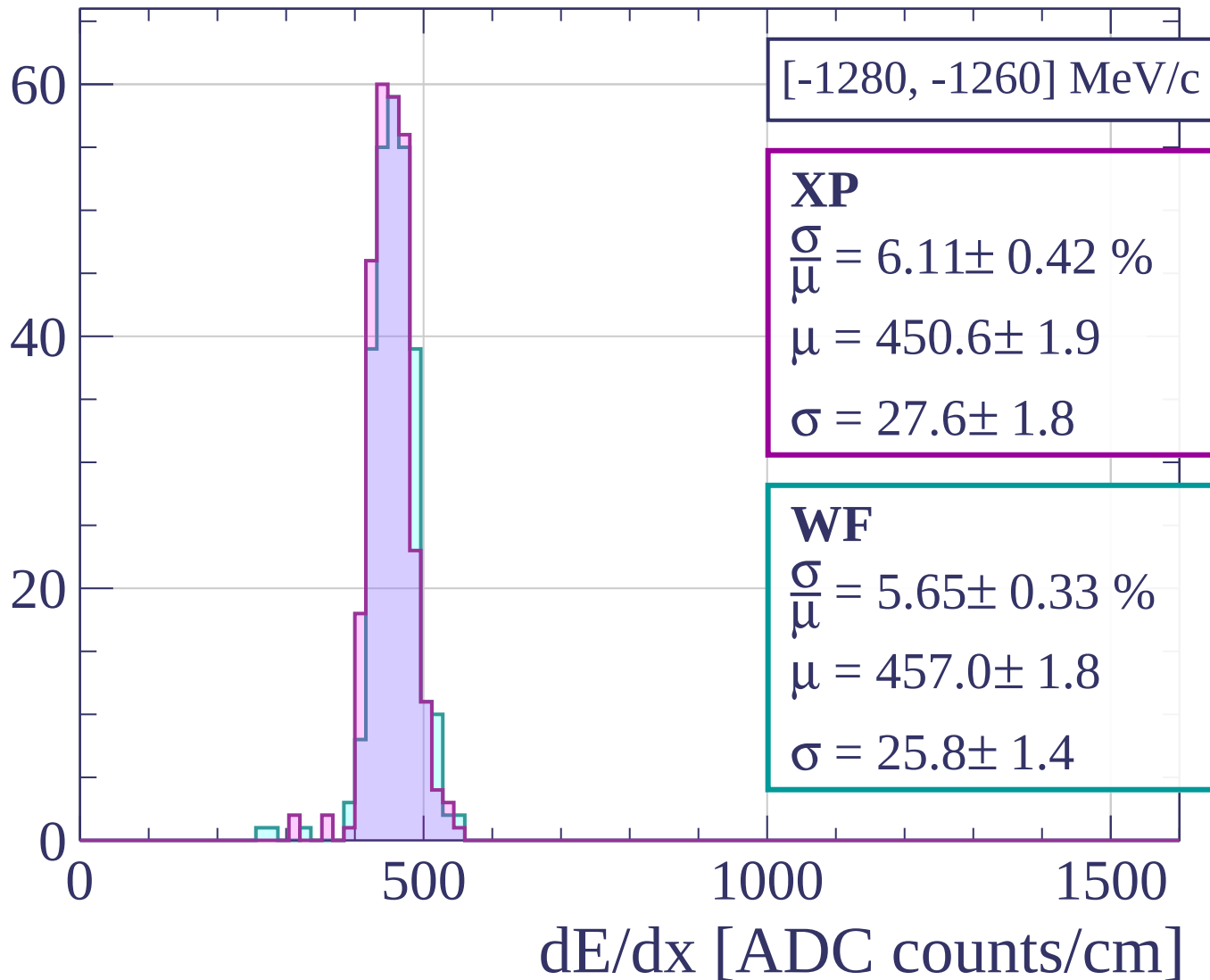
Count



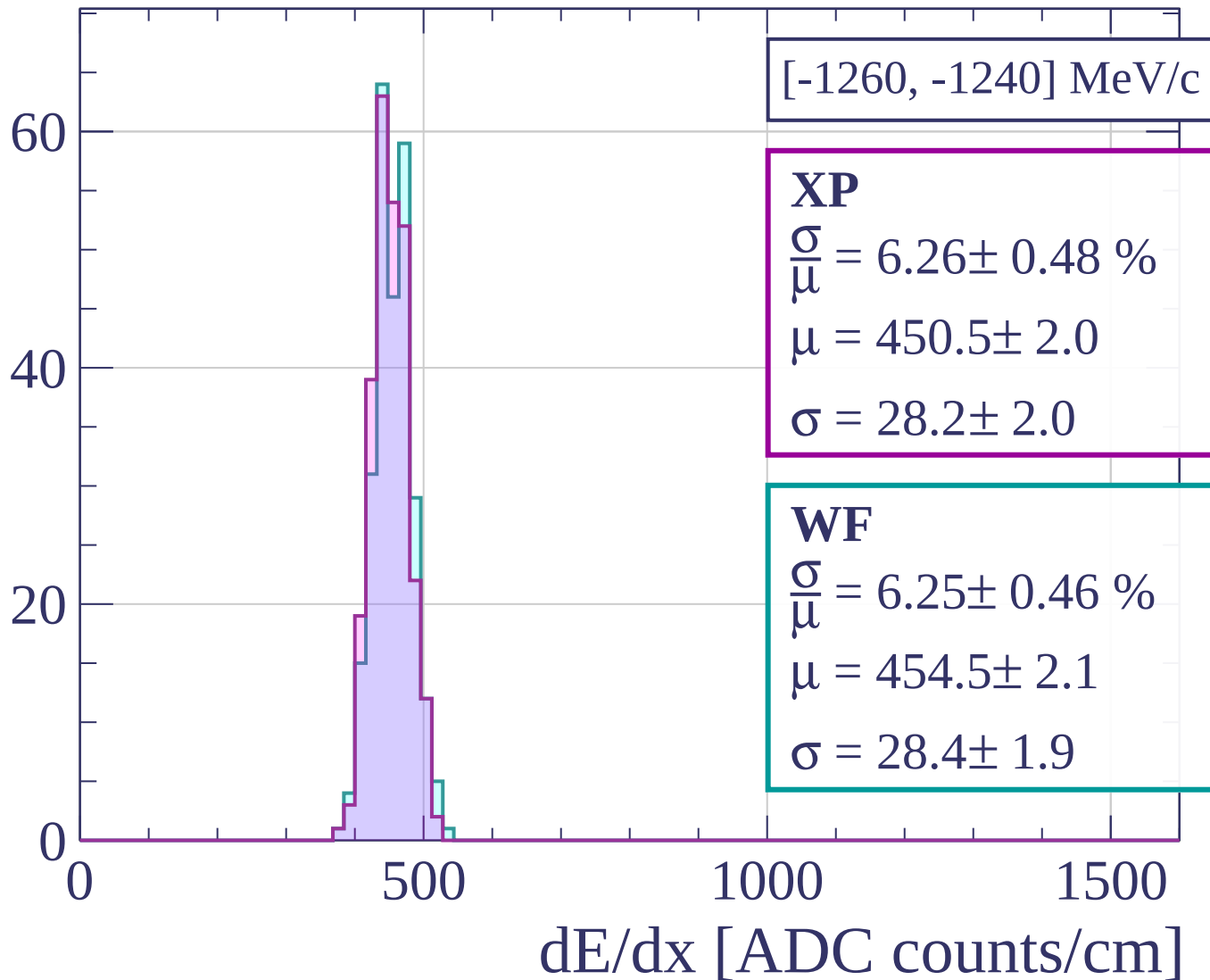
Count



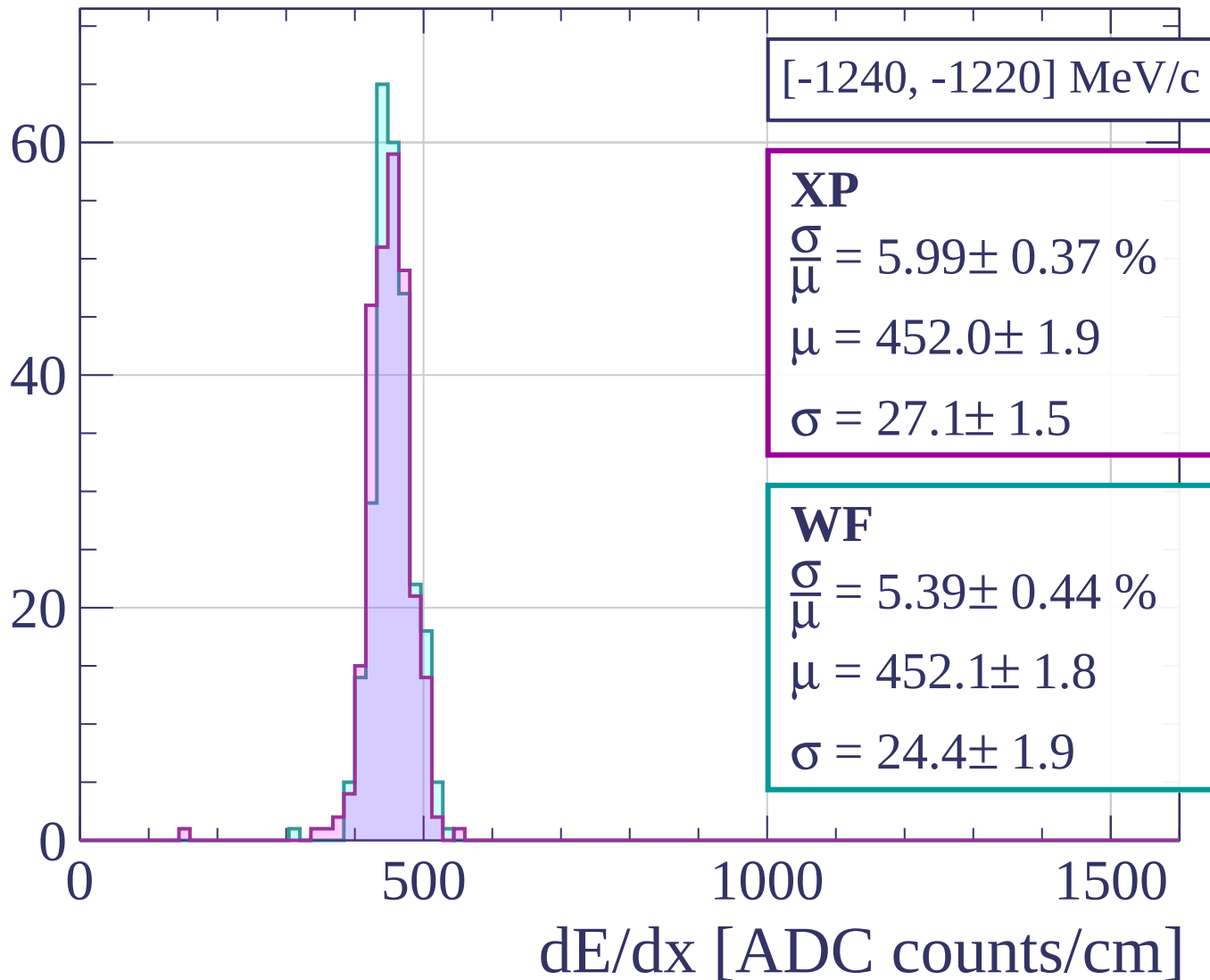
Count

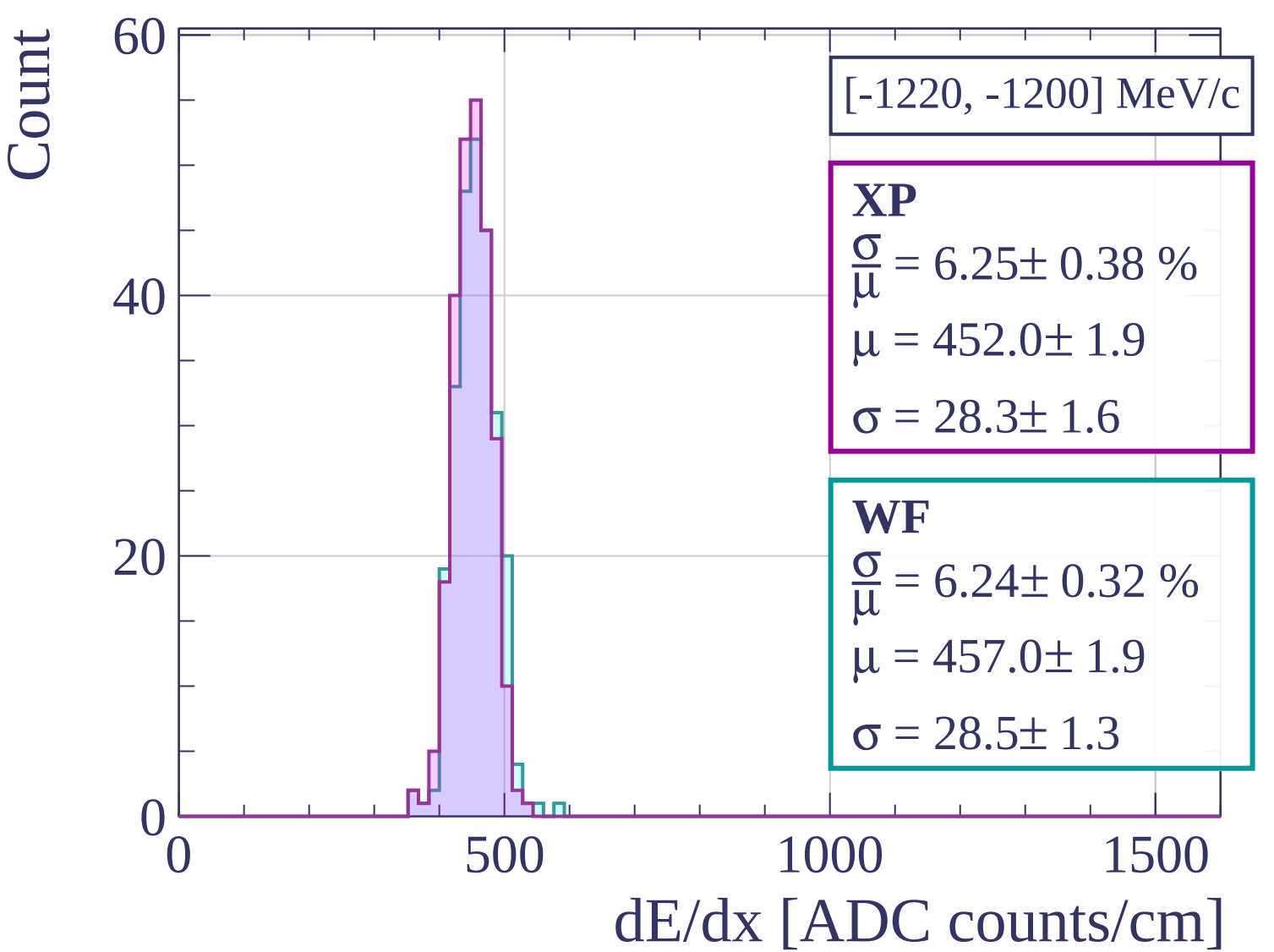


Count

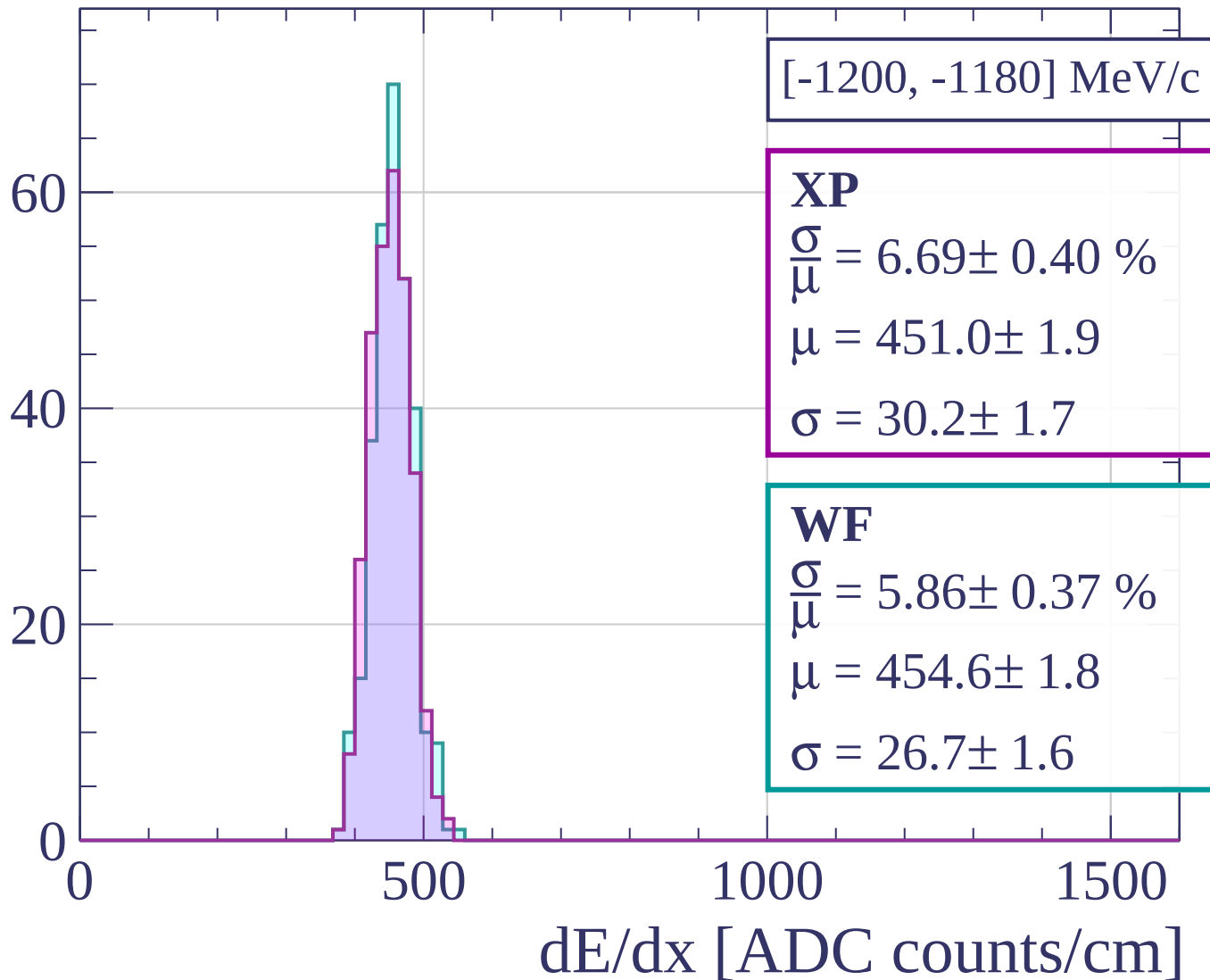


Count

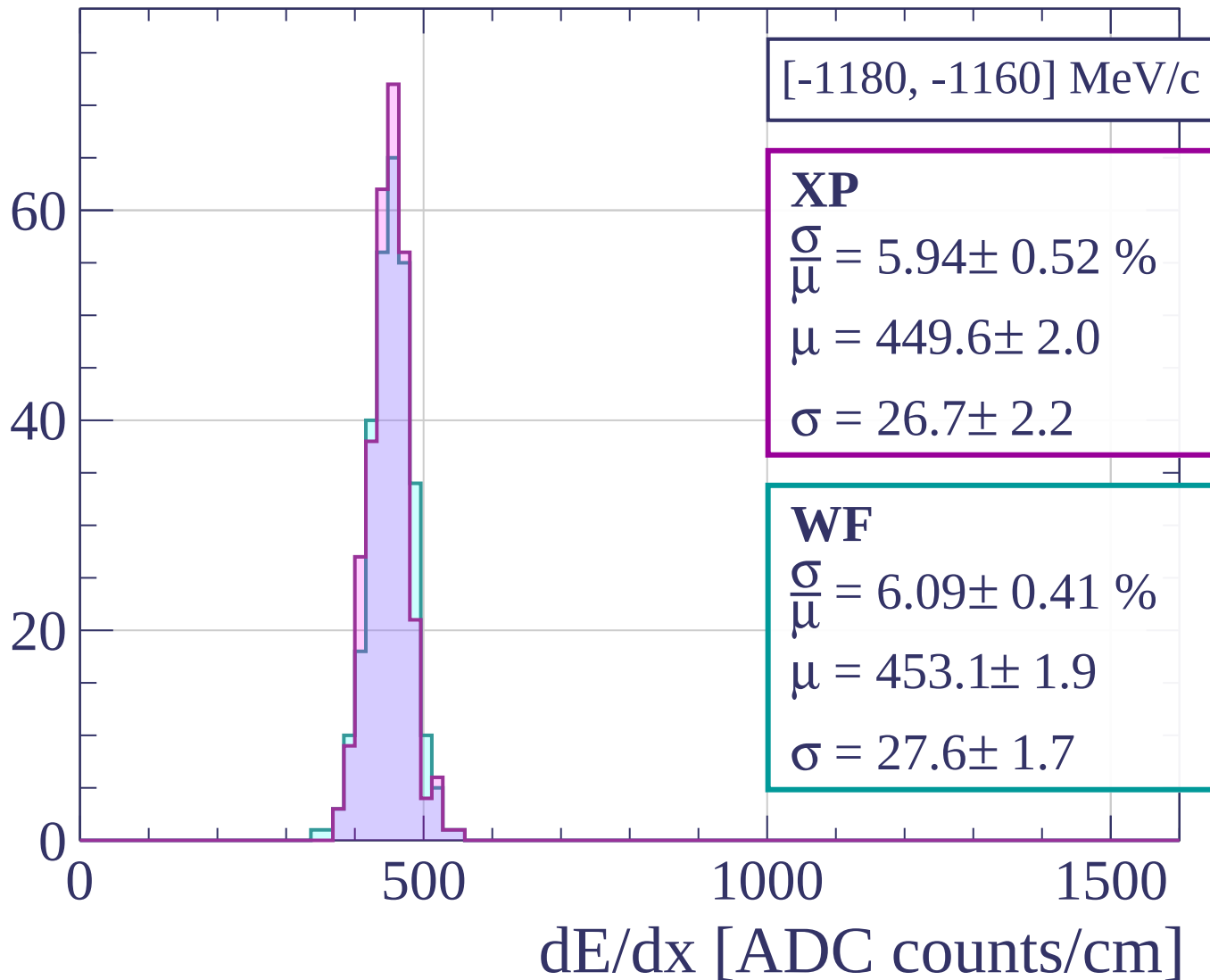




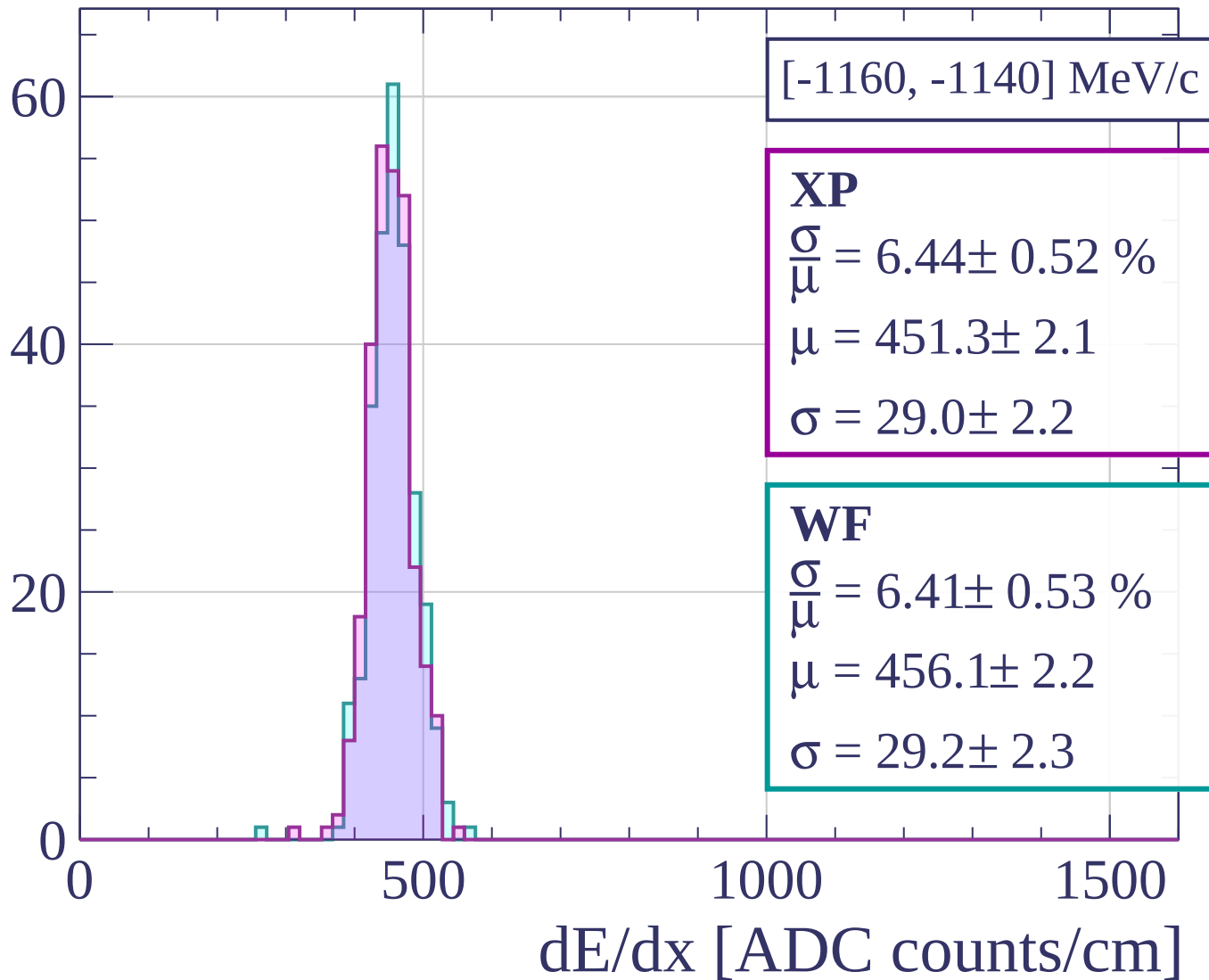
Count



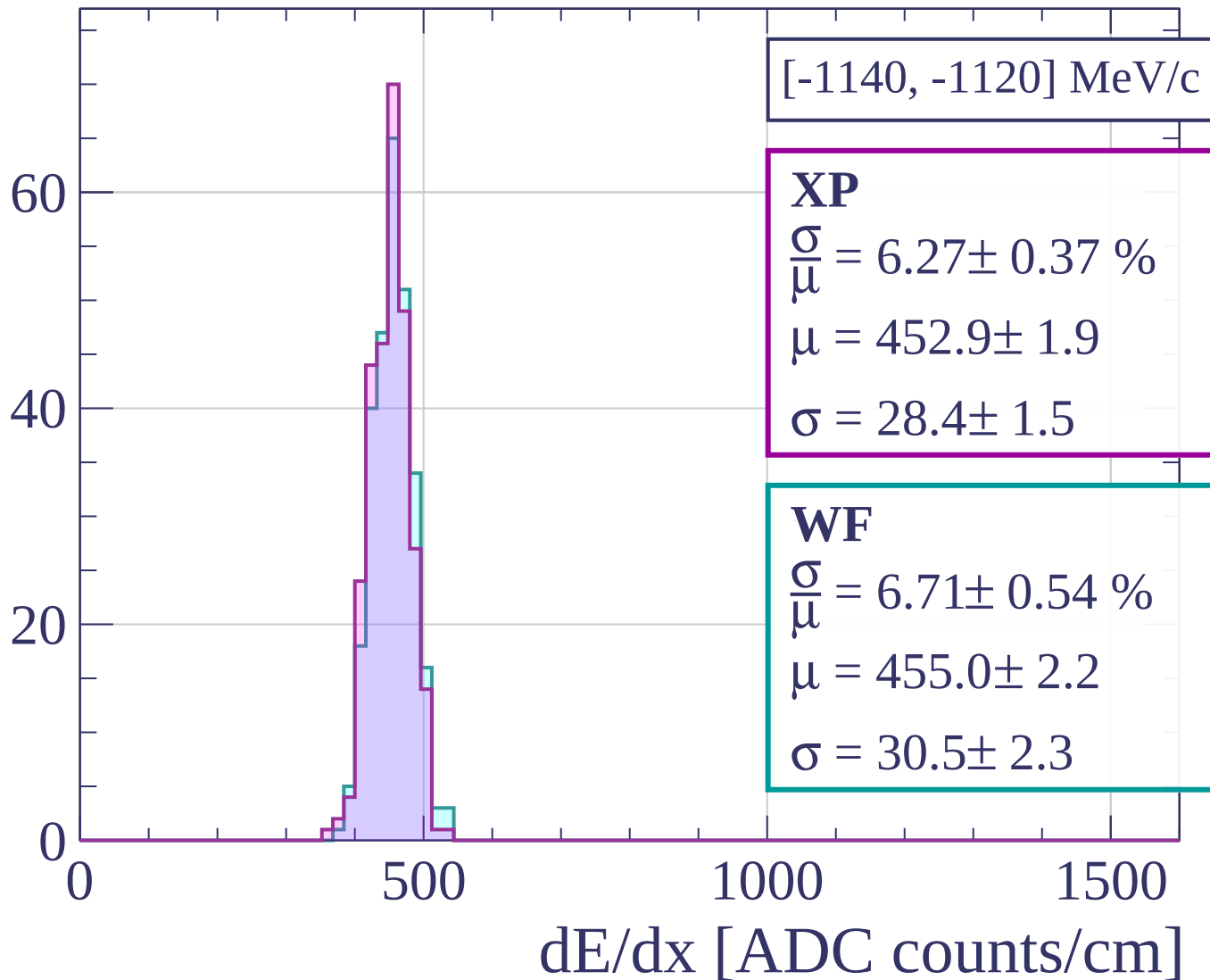
Count



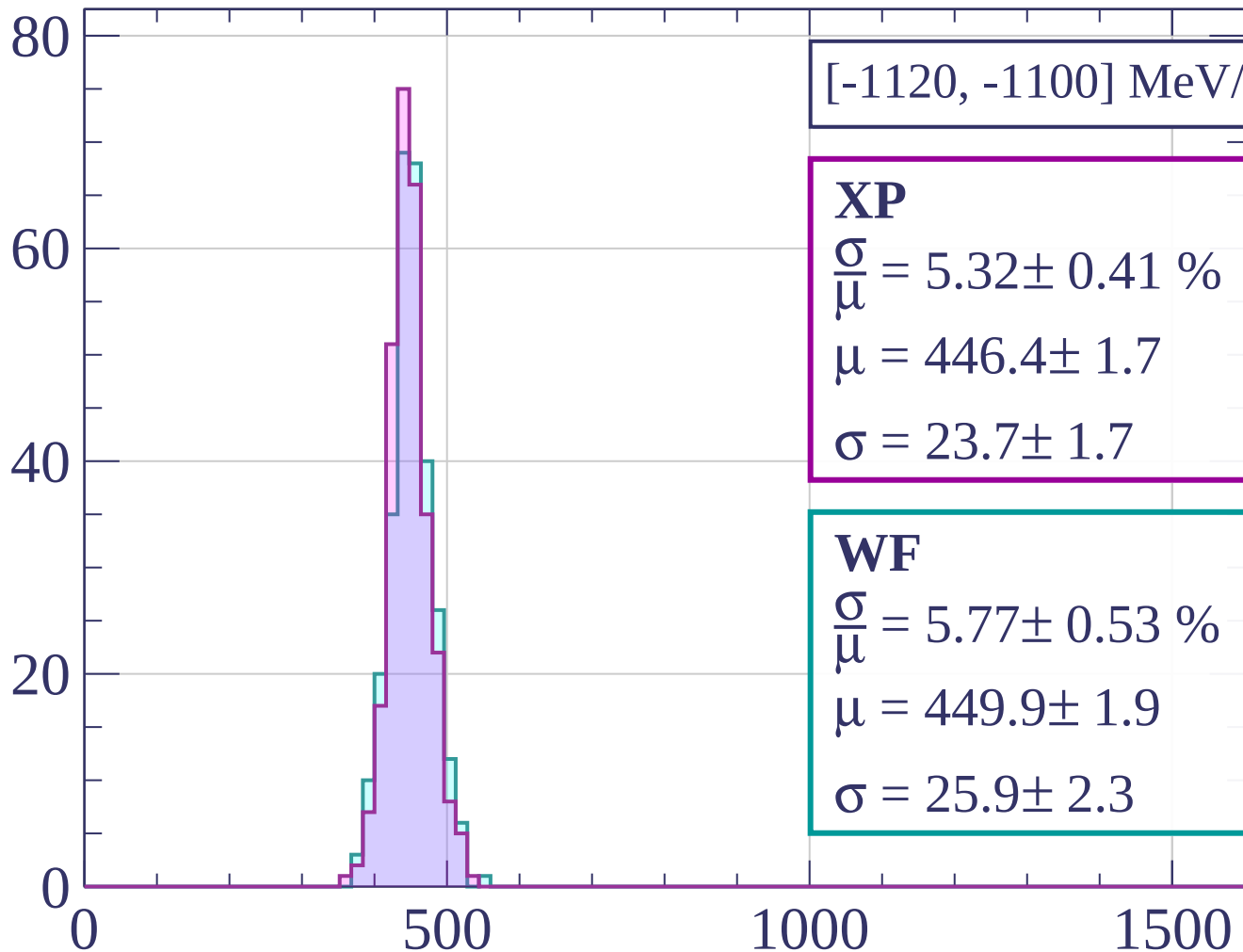
Count



Count



Count



[-1120, -1100] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.32 \pm 0.41 \%$$

$$\mu = 446.4 \pm 1.7$$

$$\sigma = 23.7 \pm 1.7$$

WF

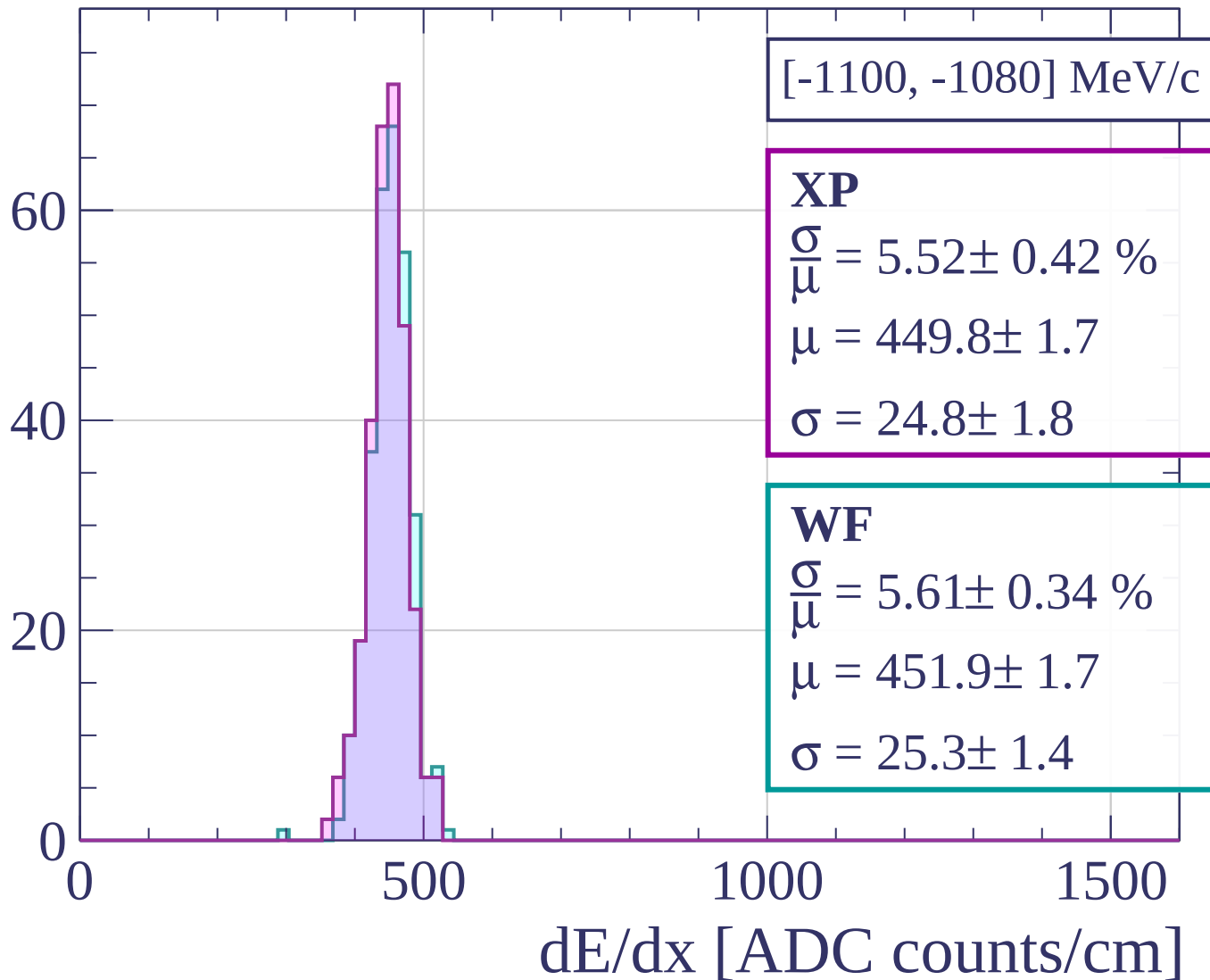
$$\frac{\sigma}{\mu} = 5.77 \pm 0.53 \%$$

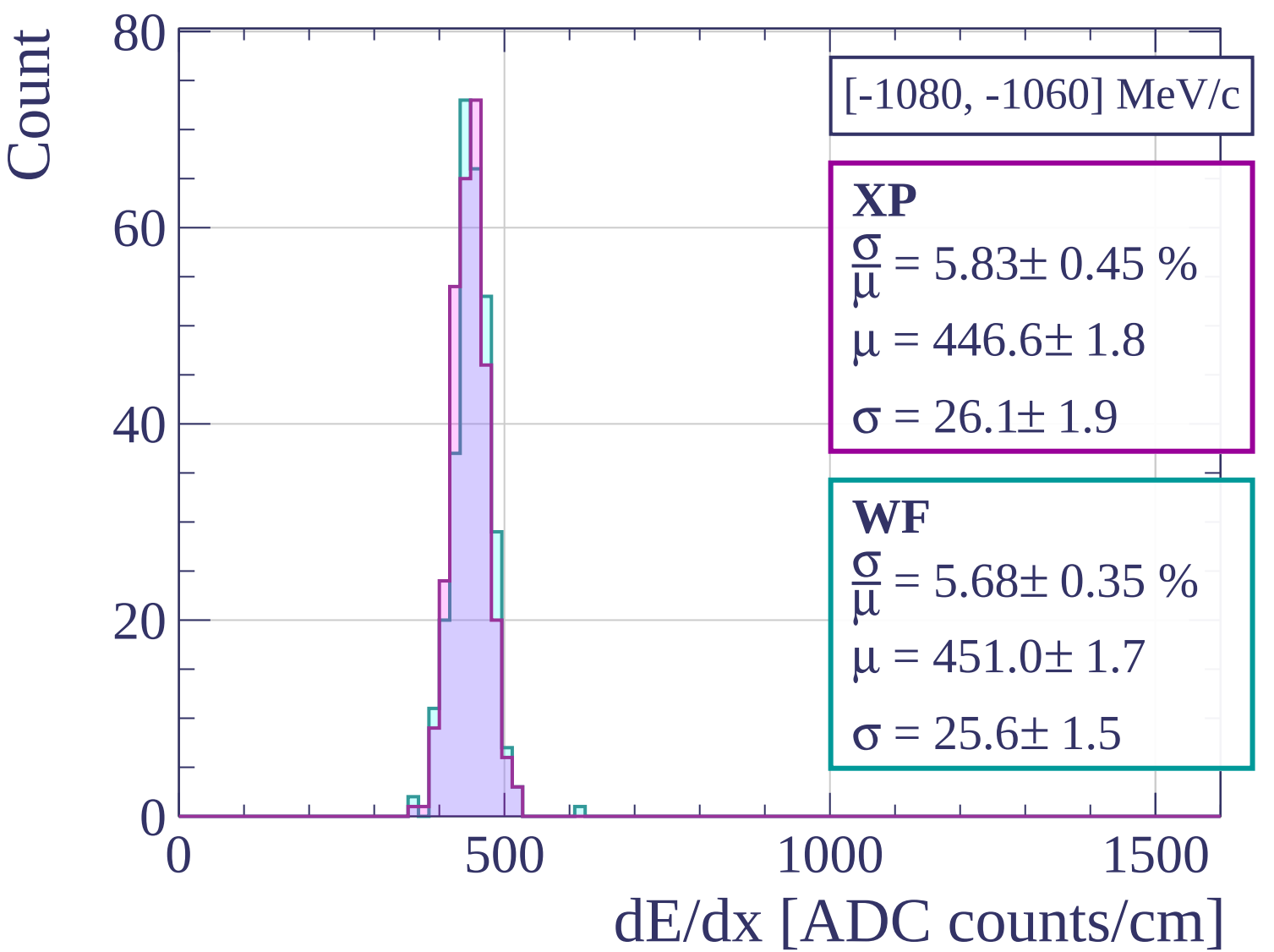
$$\mu = 449.9 \pm 1.9$$

$$\sigma = 25.9 \pm 2.3$$

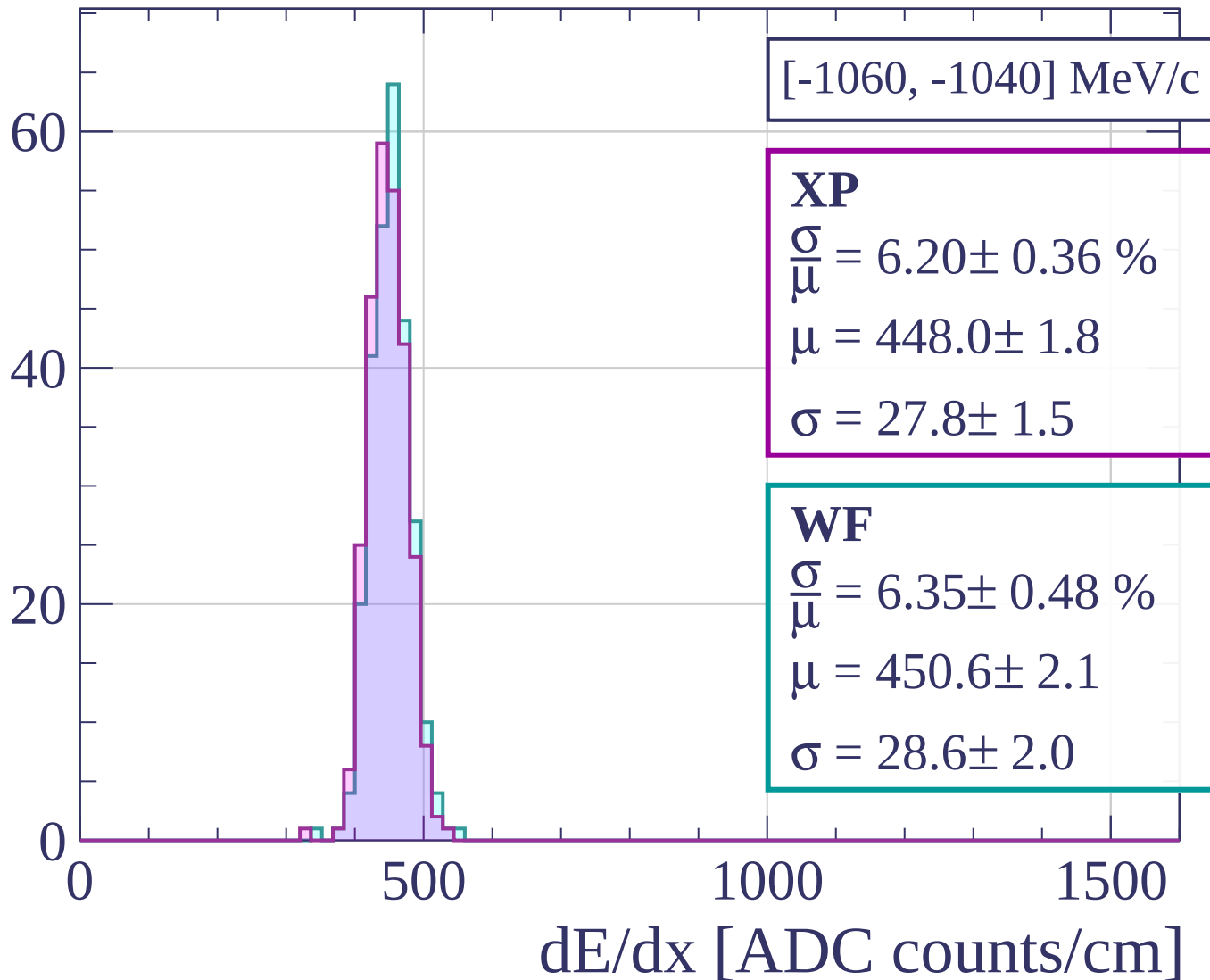
dE/dx [ADC counts/cm]

Count

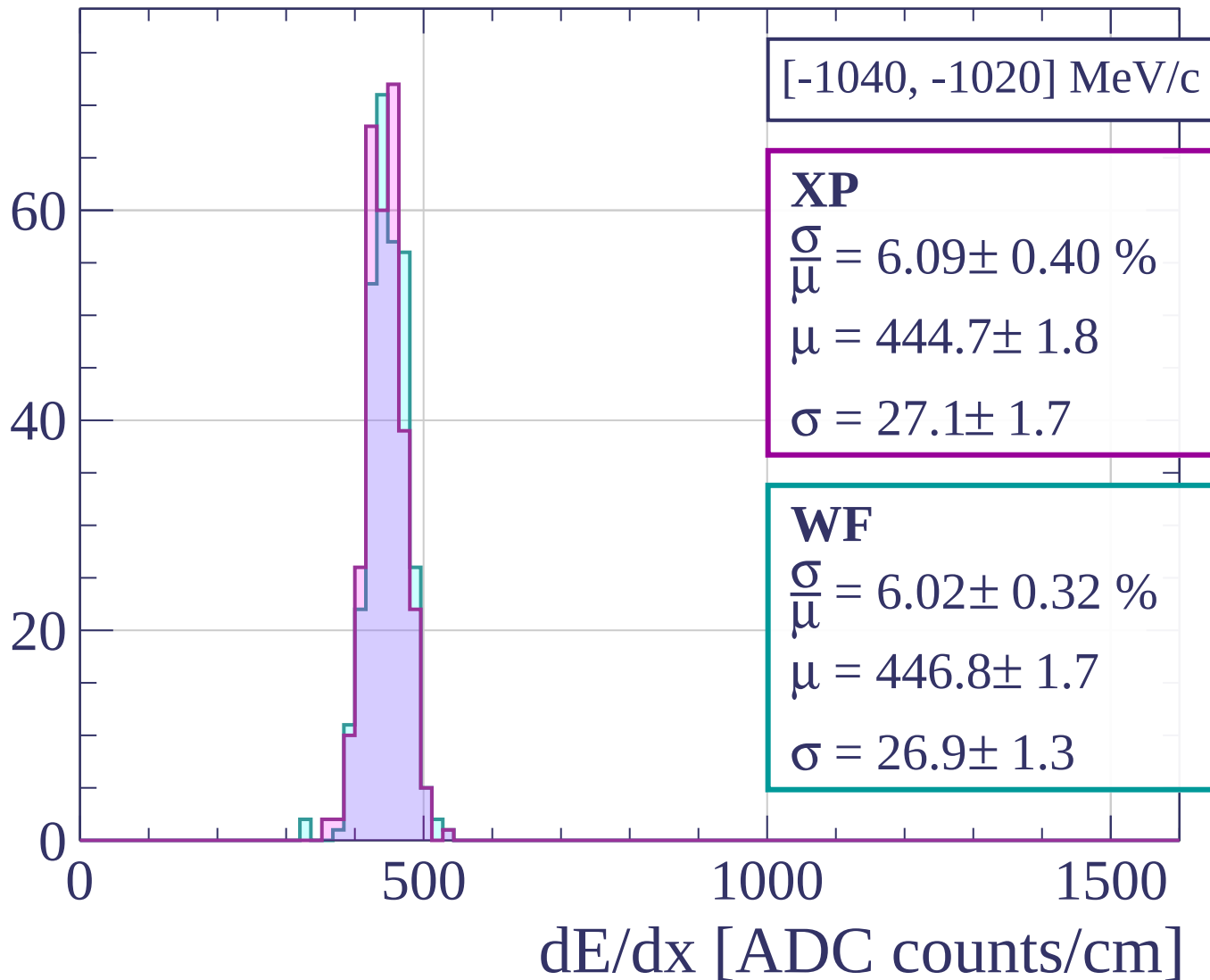




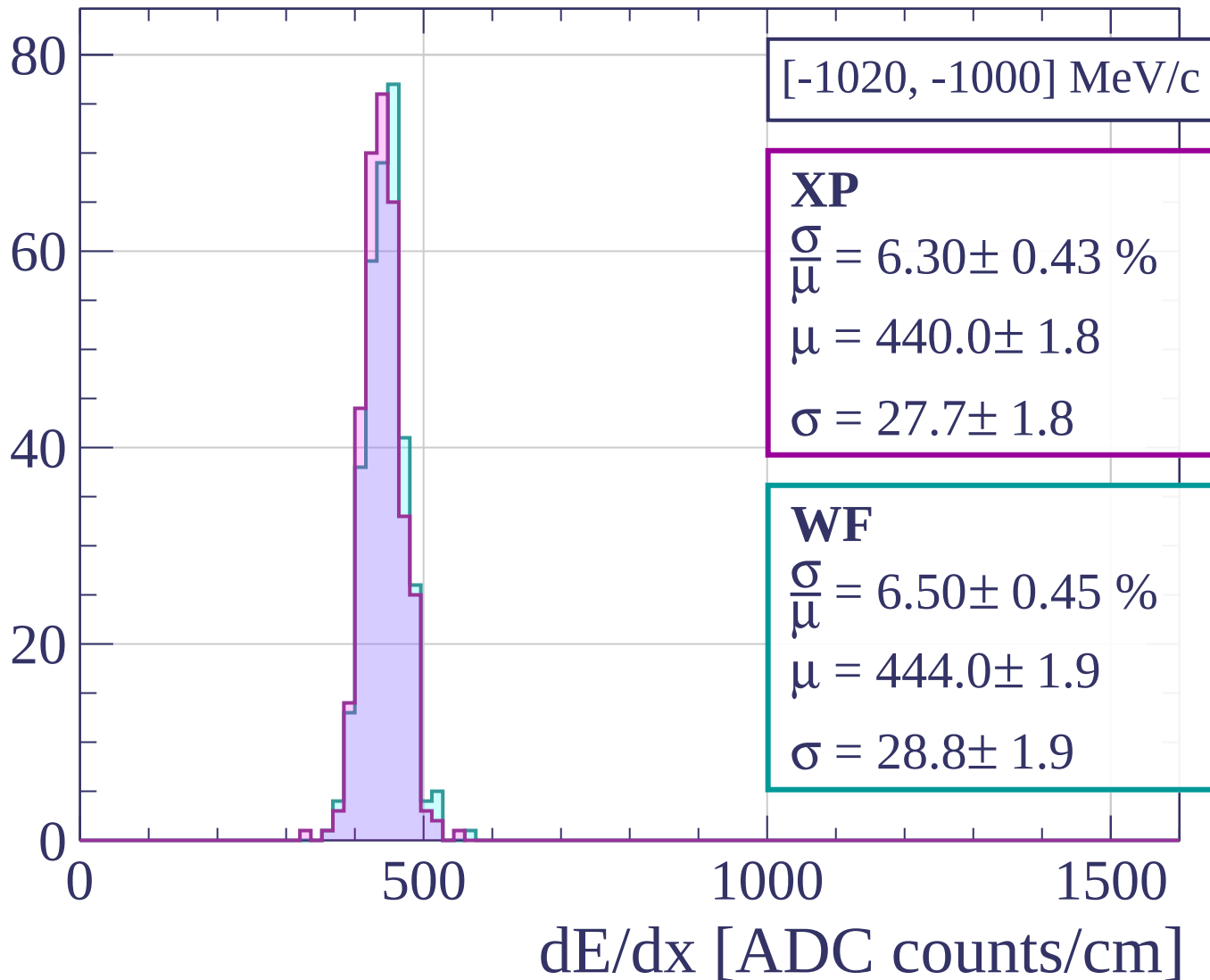
Count



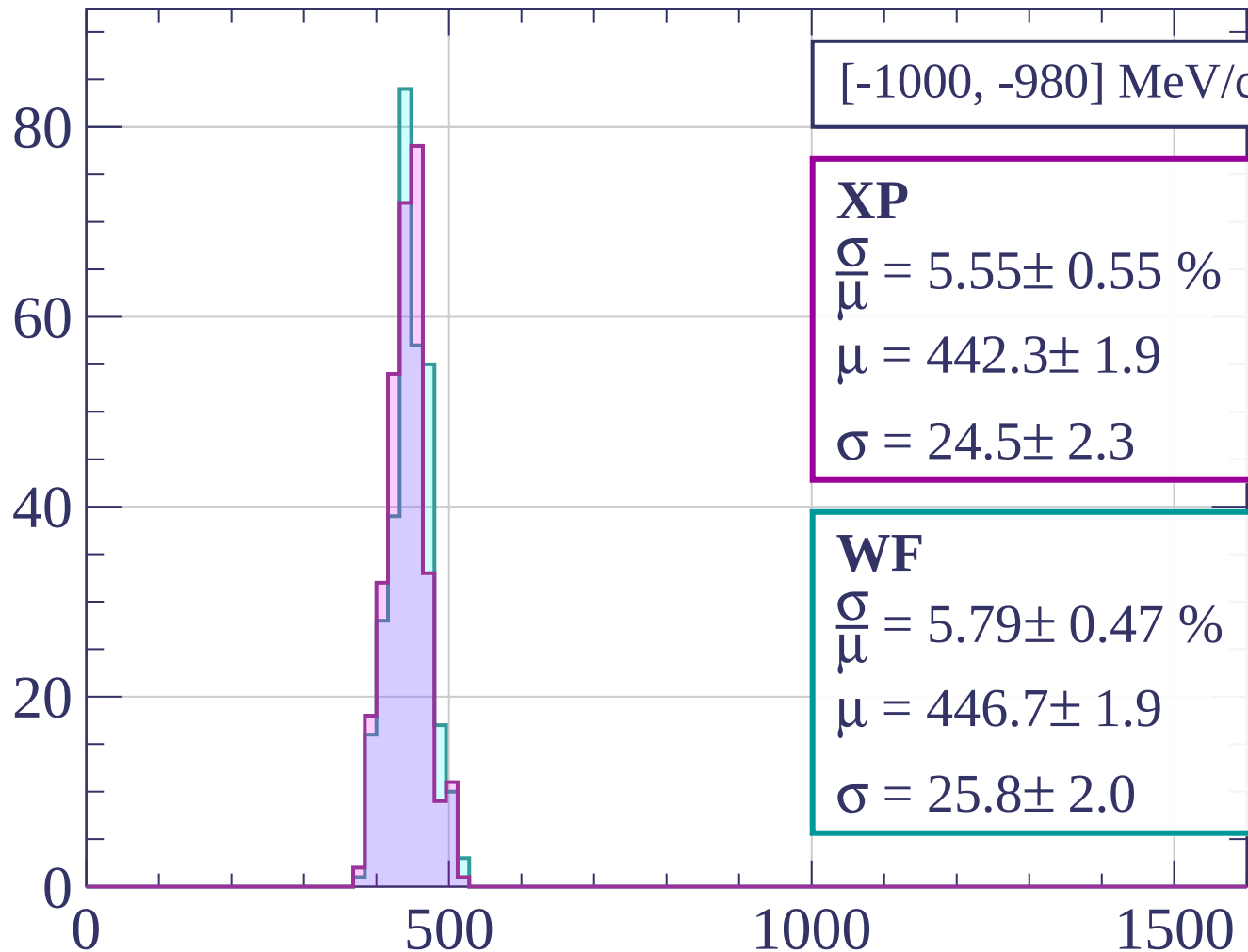
Count



Count

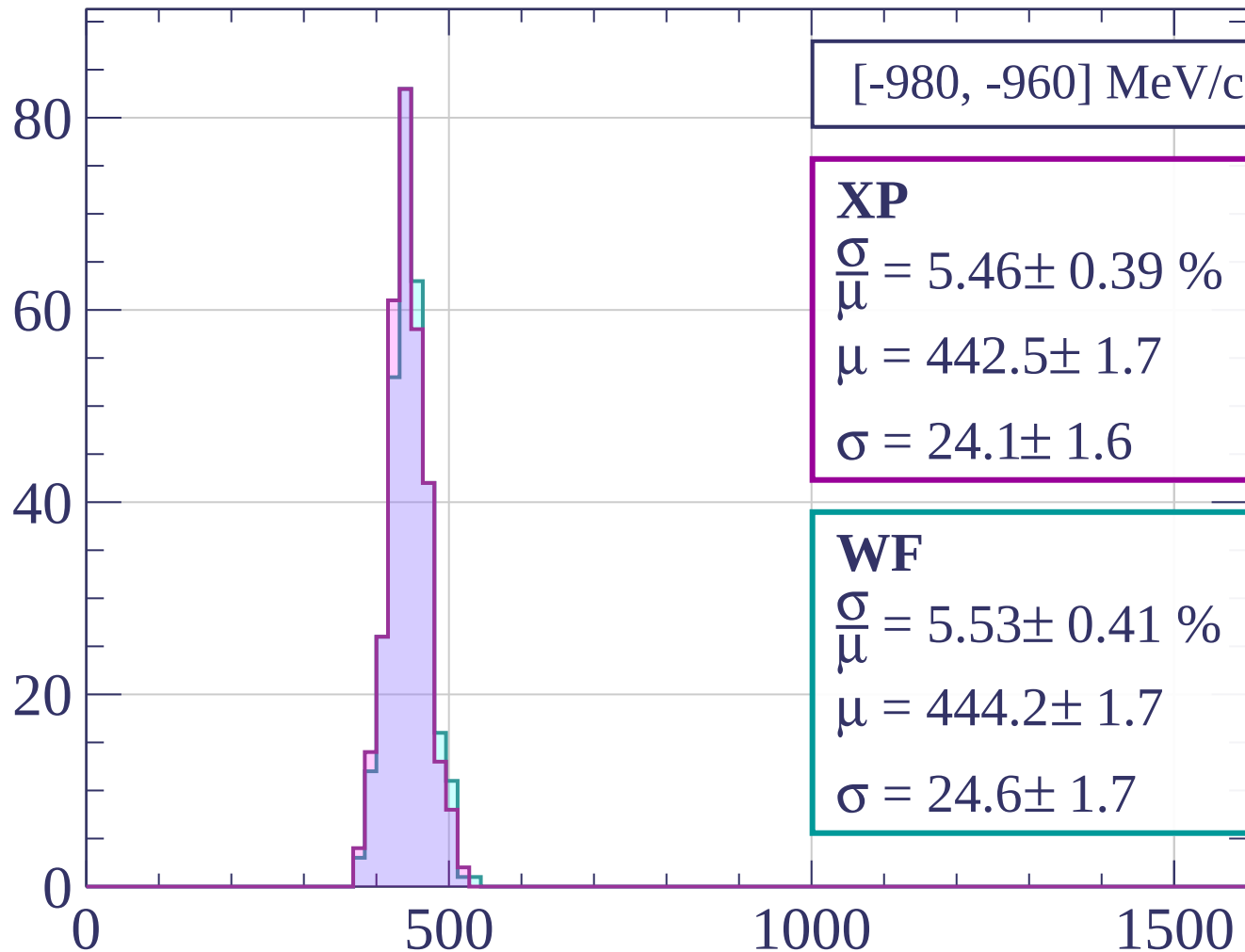


Count



dE/dx [ADC counts/cm]

Count



$[-980, -960]$ MeV/c

XP

$$\frac{\sigma}{\mu} = 5.46 \pm 0.39 \%$$

$$\mu = 442.5 \pm 1.7$$

$$\sigma = 24.1 \pm 1.6$$

WF

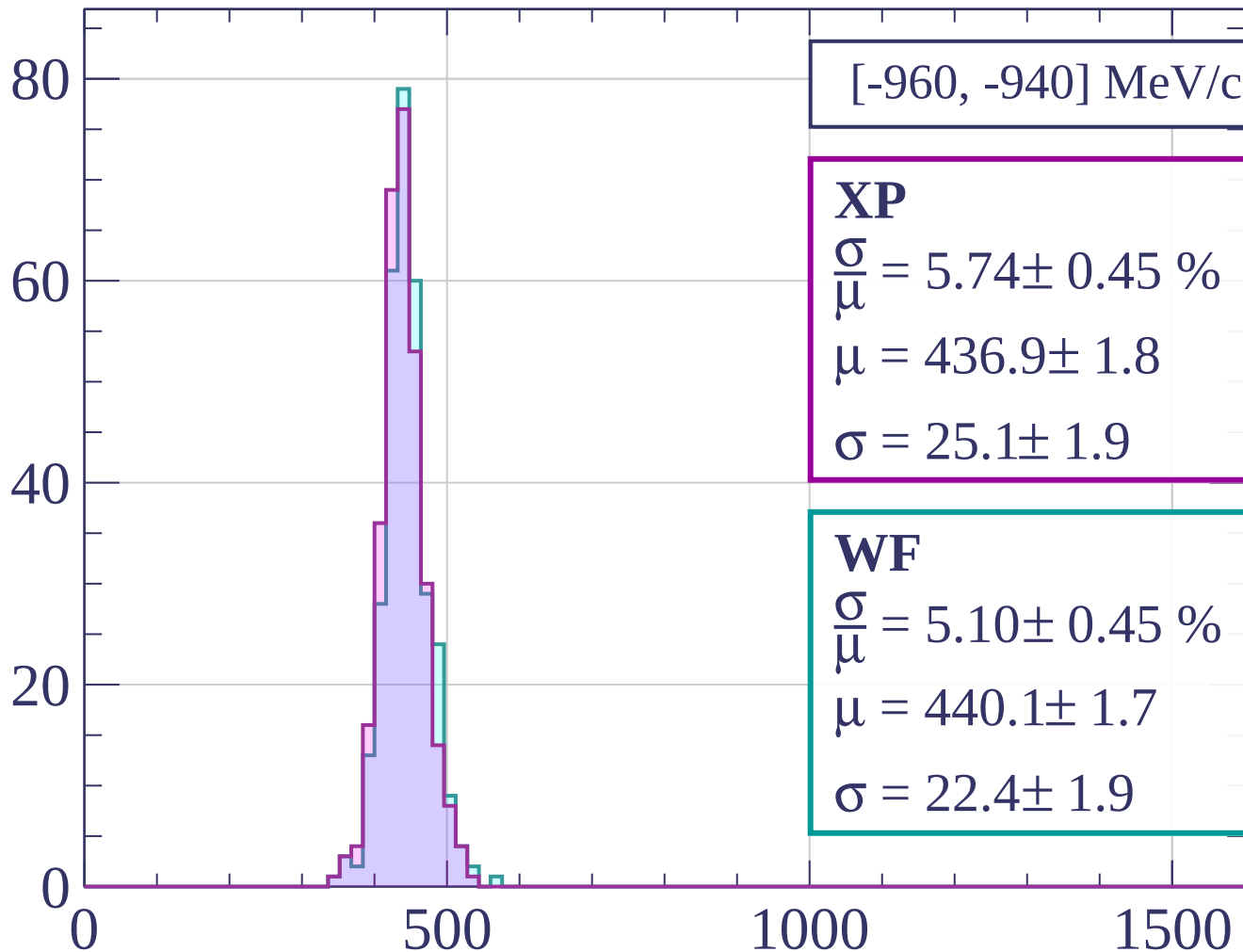
$$\frac{\sigma}{\mu} = 5.53 \pm 0.41 \%$$

$$\mu = 444.2 \pm 1.7$$

$$\sigma = 24.6 \pm 1.7$$

dE/dx [ADC counts/cm]

Count



[-960, -940] MeV/c

XP

$$\frac{\sigma}{\bar{\mu}} = 5.74 \pm 0.45 \%$$

$$\mu = 436.9 \pm 1.8$$

$$\sigma = 25.1 \pm 1.9$$

WF

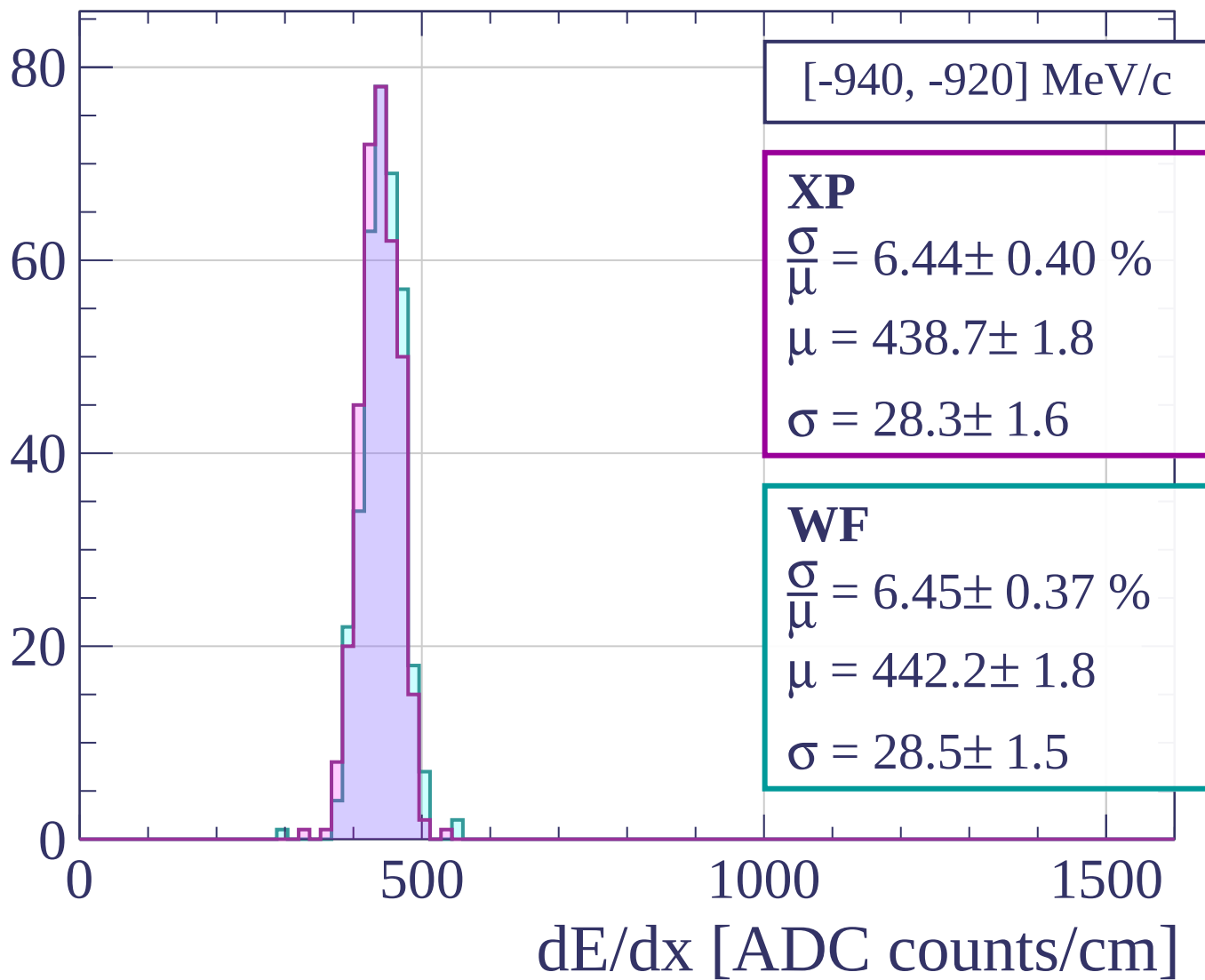
$$\frac{\sigma}{\bar{\mu}} = 5.10 \pm 0.45 \%$$

$$\mu = 440.1 \pm 1.7$$

$$\sigma = 22.4 \pm 1.9$$

dE/dx [ADC counts/cm]

Count



Count

[-920, -900] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.56 \pm 0.34 \%$$

$$\mu = 435.2 \pm 1.6$$

$$\sigma = 28.6 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 6.91 \pm 0.49 \%$$

$$\mu = 437.4 \pm 1.9$$

$$\sigma = 30.2 \pm 2.0$$

80

60

40

20

0

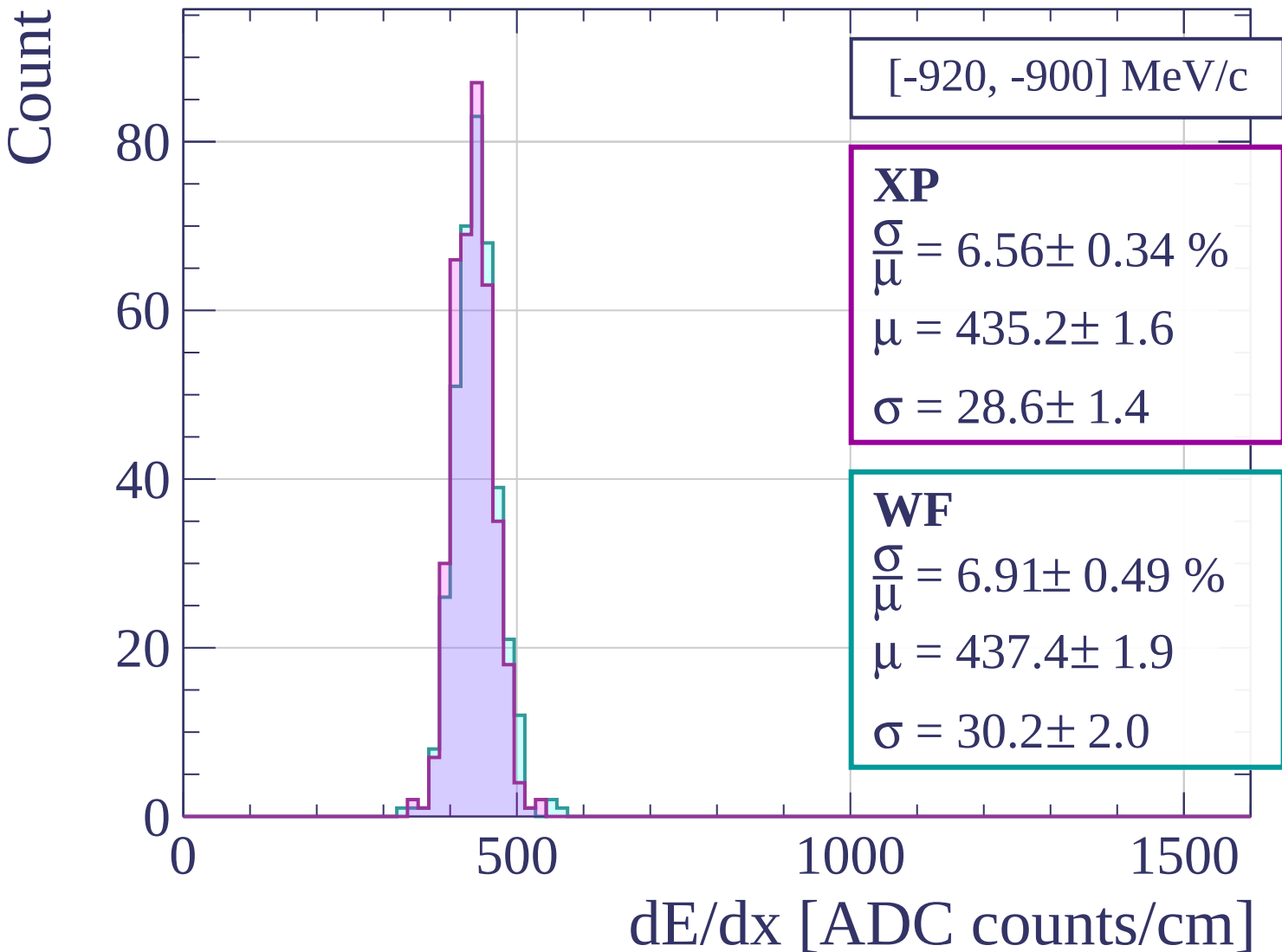
0

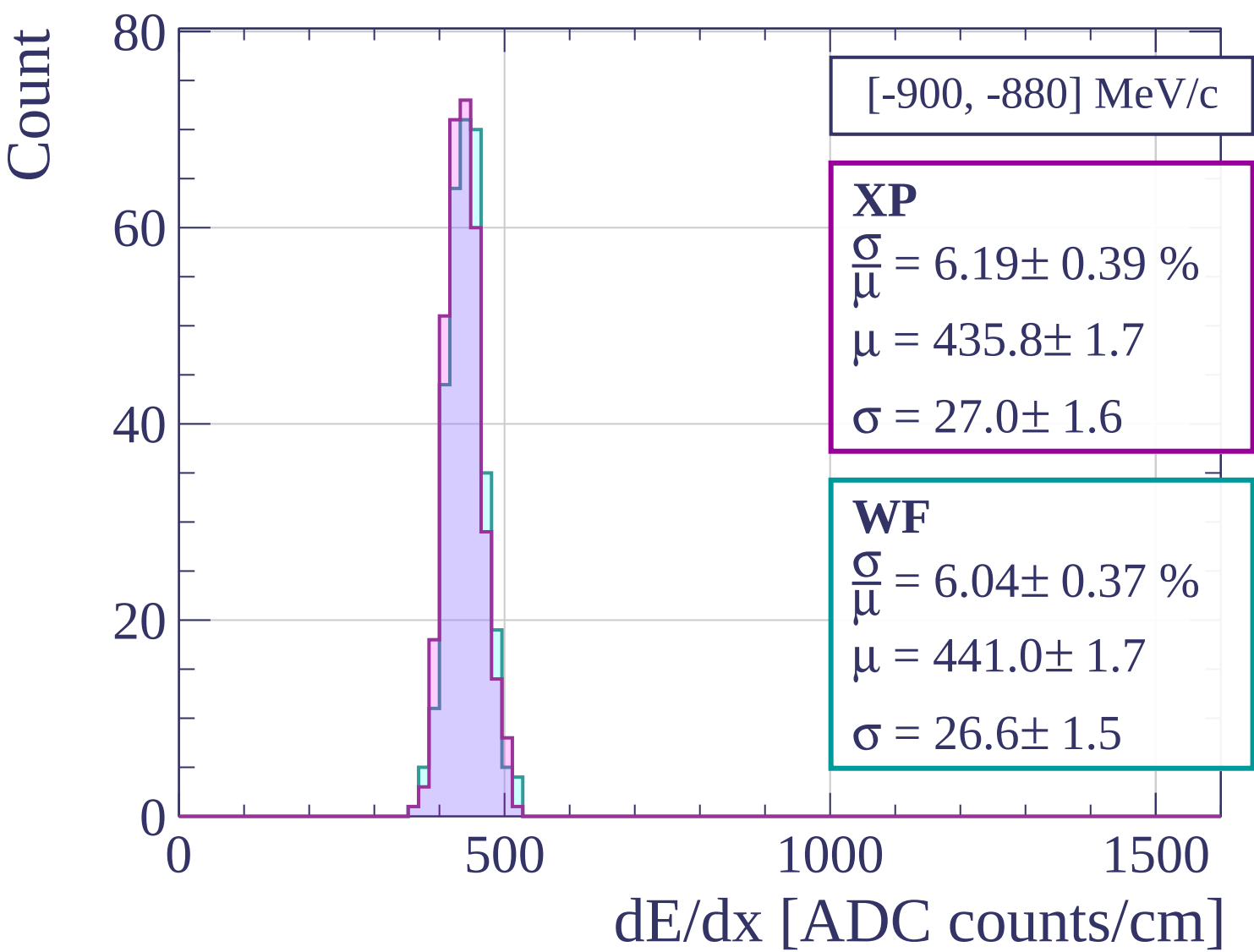
500

1000

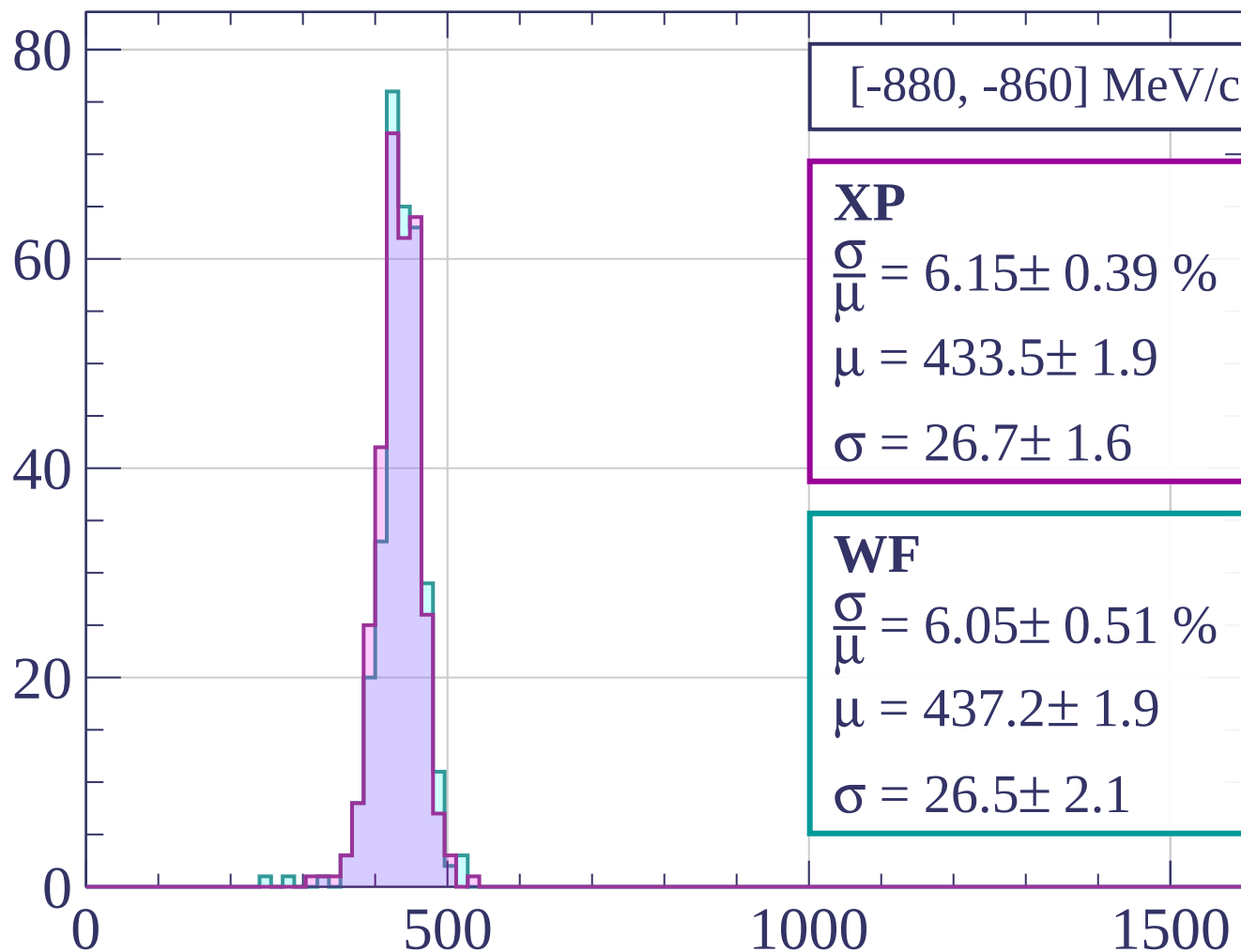
1500

dE/dx [ADC counts/cm]





Count



$[-880, -860]$ MeV/c

XP

$$\frac{\sigma}{\mu} = 6.15 \pm 0.39 \%$$

$$\mu = 433.5 \pm 1.9$$

$$\sigma = 26.7 \pm 1.6$$

WF

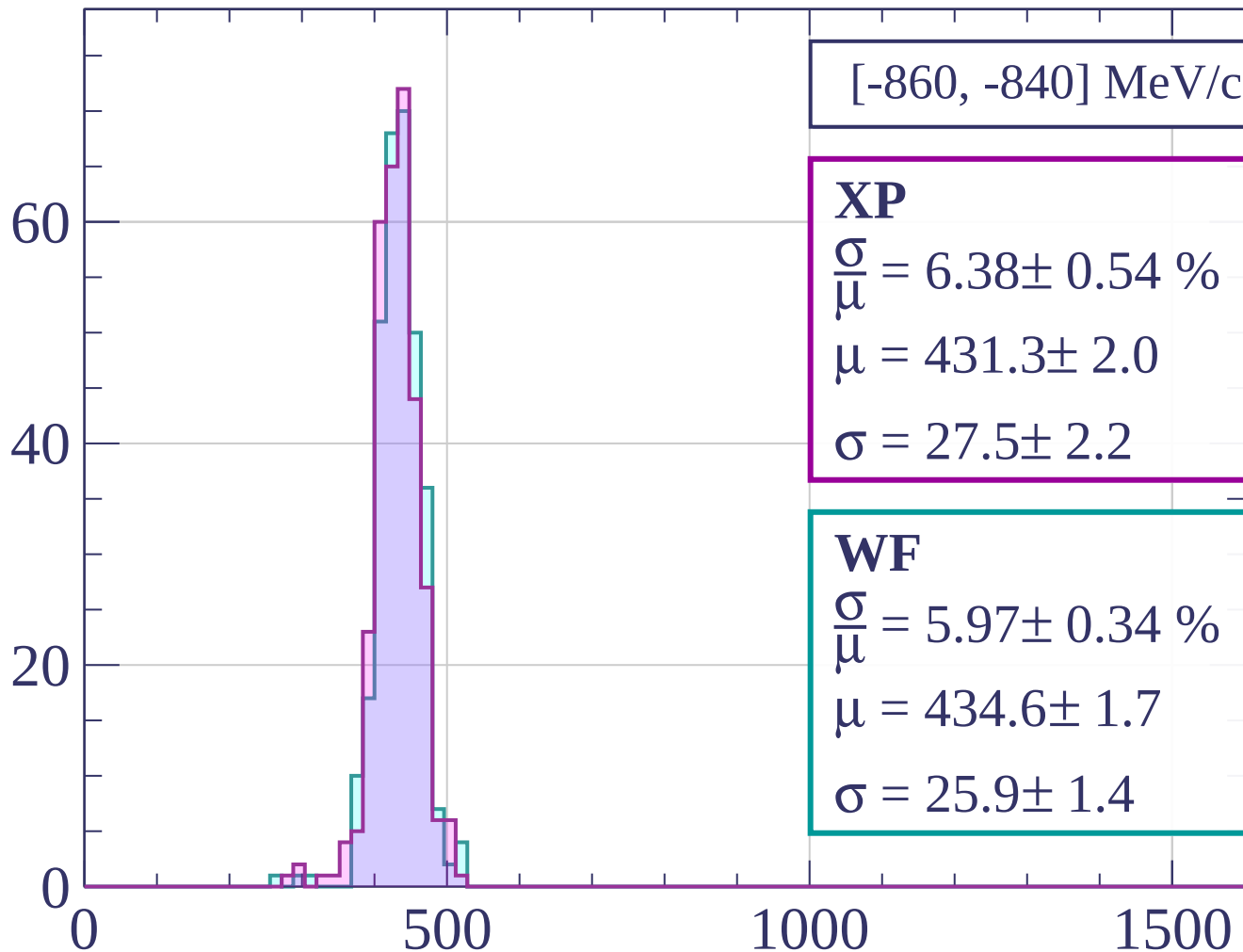
$$\frac{\sigma}{\mu} = 6.05 \pm 0.51 \%$$

$$\mu = 437.2 \pm 1.9$$

$$\sigma = 26.5 \pm 2.1$$

dE/dx [ADC counts/cm]

Count



[-860, -840] MeV/c

XP

$$\frac{\sigma}{\bar{\mu}} = 6.38 \pm 0.54 \%$$

$$\mu = 431.3 \pm 2.0$$

$$\sigma = 27.5 \pm 2.2$$

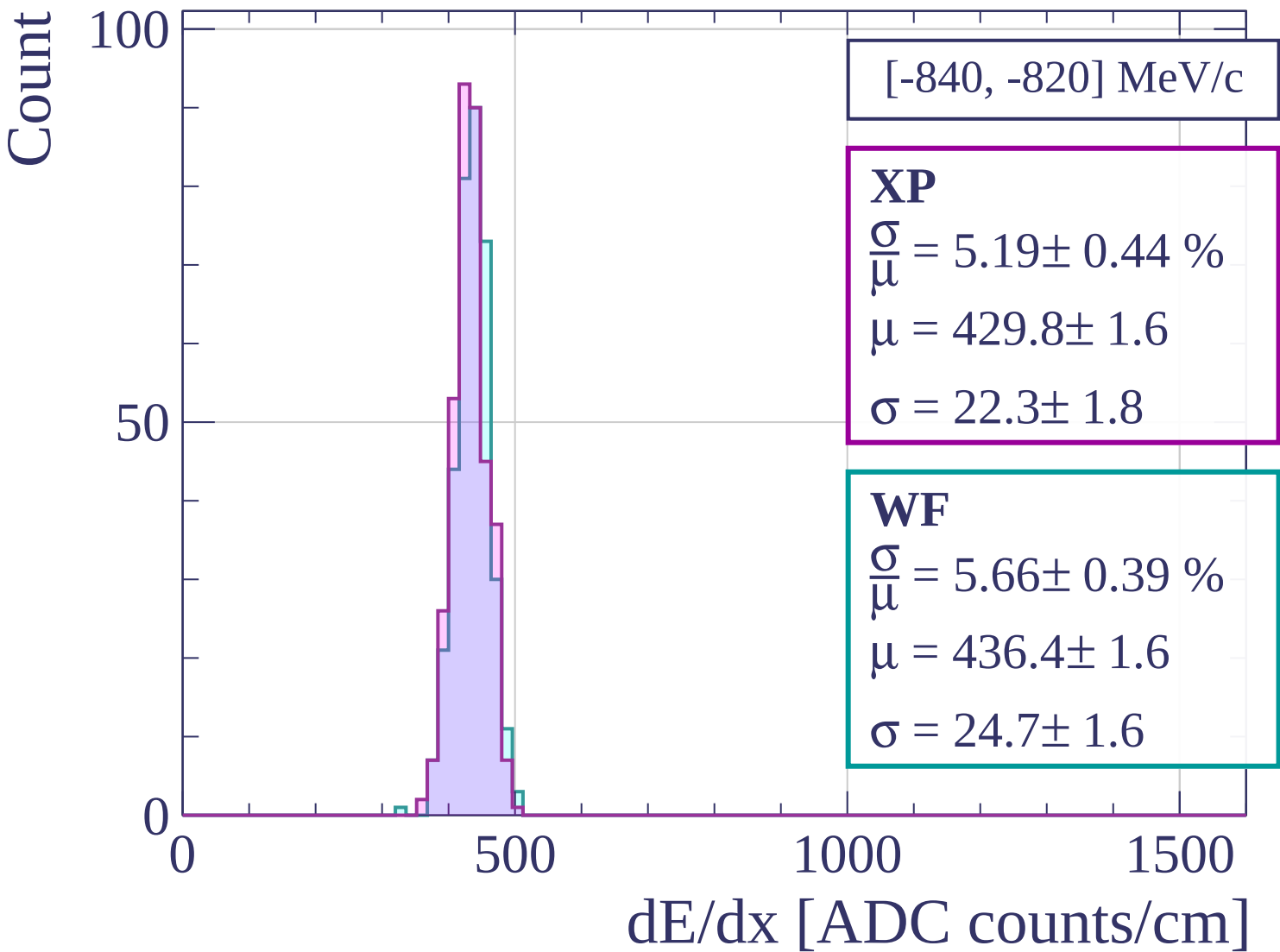
WF

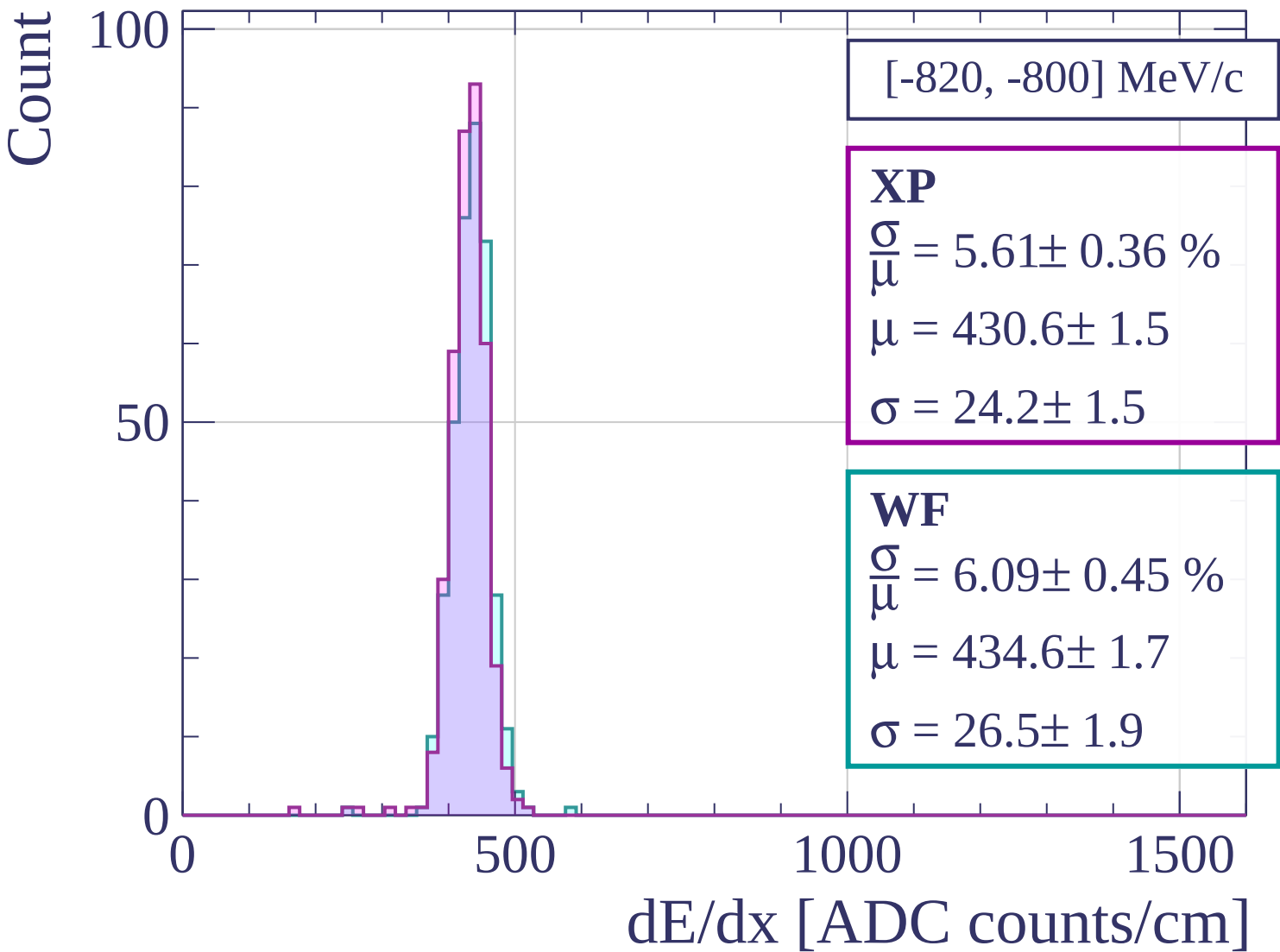
$$\frac{\sigma}{\bar{\mu}} = 5.97 \pm 0.34 \%$$

$$\mu = 434.6 \pm 1.7$$

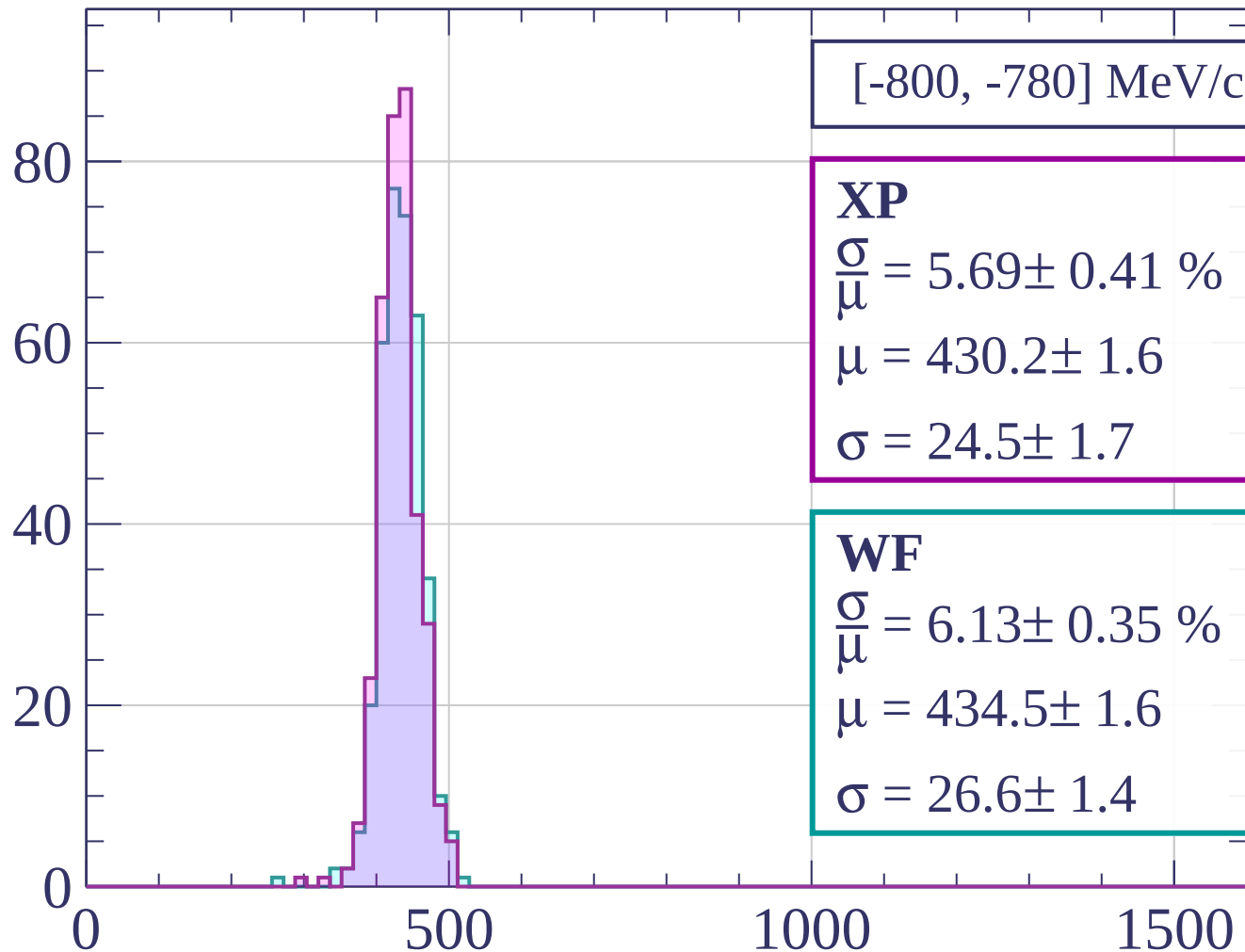
$$\sigma = 25.9 \pm 1.4$$

dE/dx [ADC counts/cm]

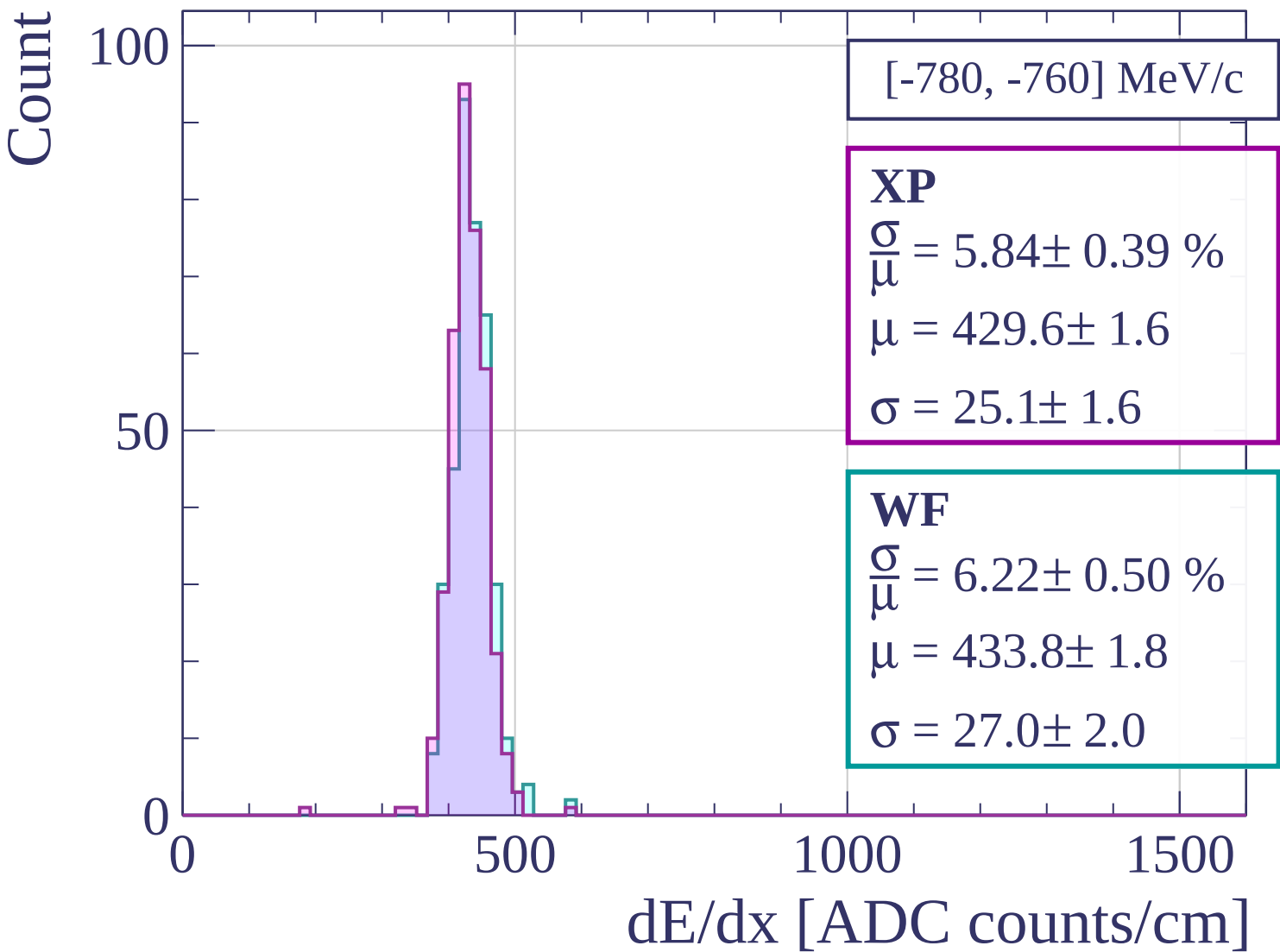


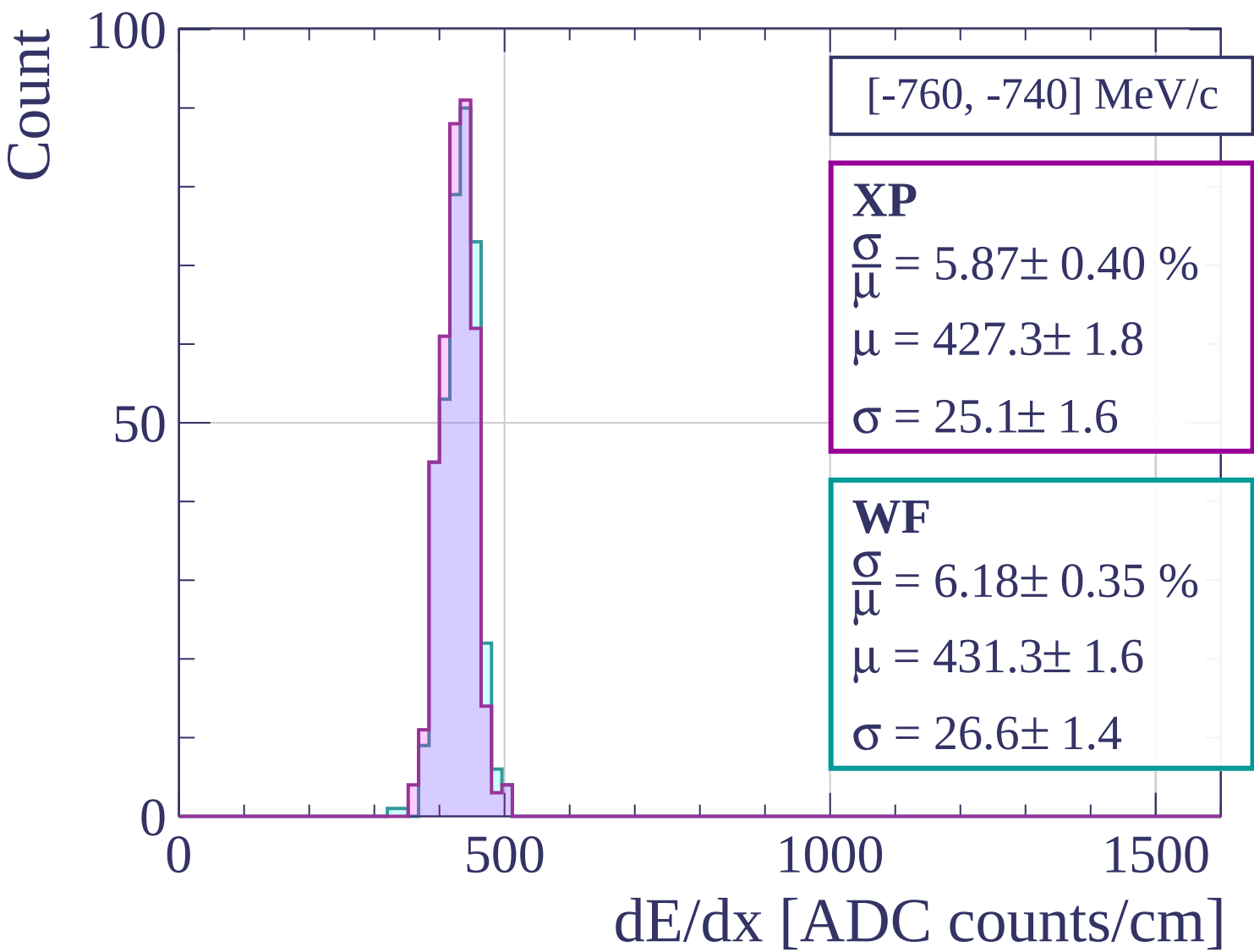


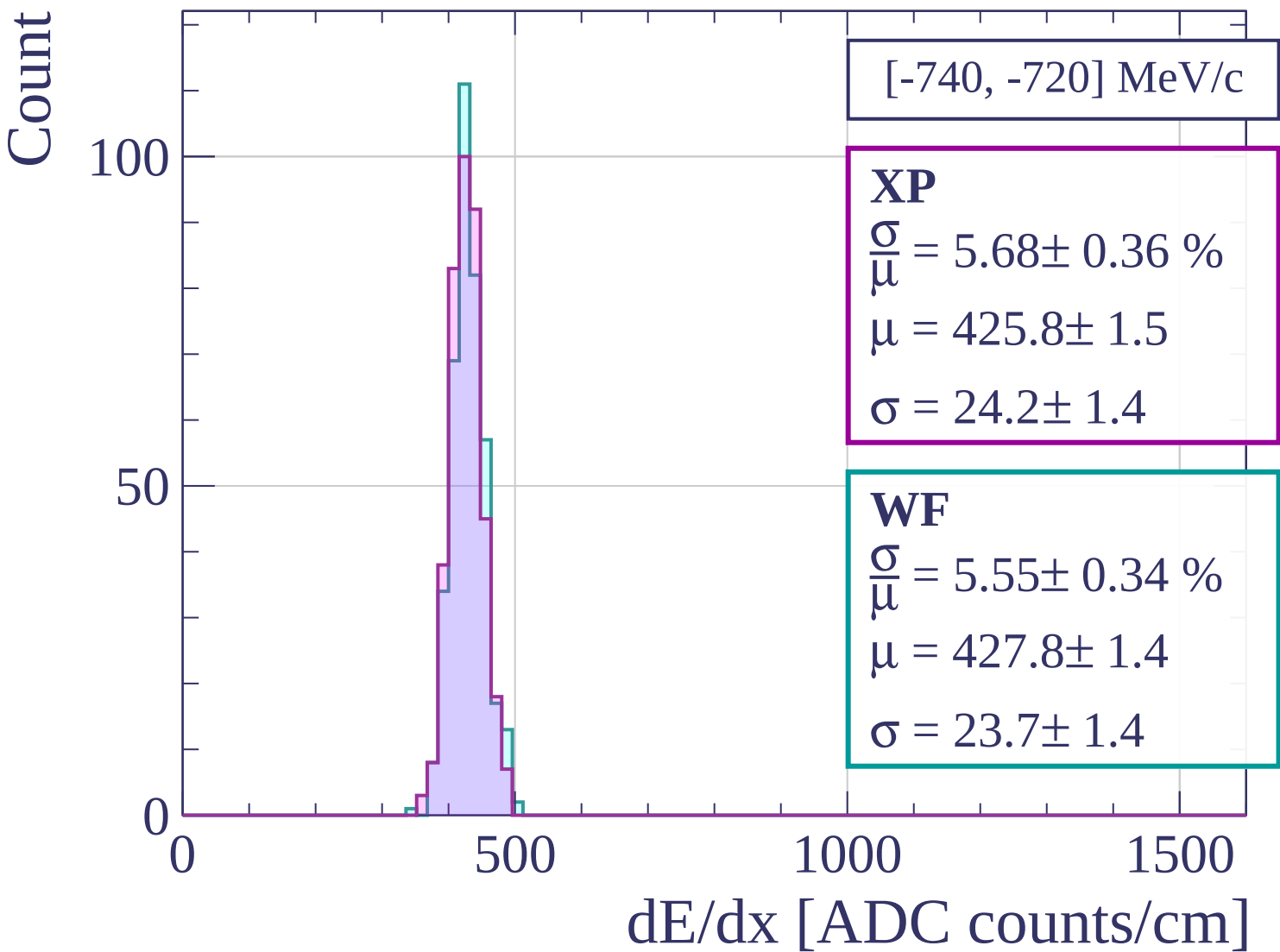
Count



dE/dx [ADC counts/cm]







Count

[-720, -700] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.81 \pm 0.29 \%$$

$$\mu = 426.0 \pm 1.4$$

$$\sigma = 24.8 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 5.25 \pm 0.42 \%$$

$$\mu = 428.1 \pm 1.5$$

$$\sigma = 22.5 \pm 1.7$$

100

50

0

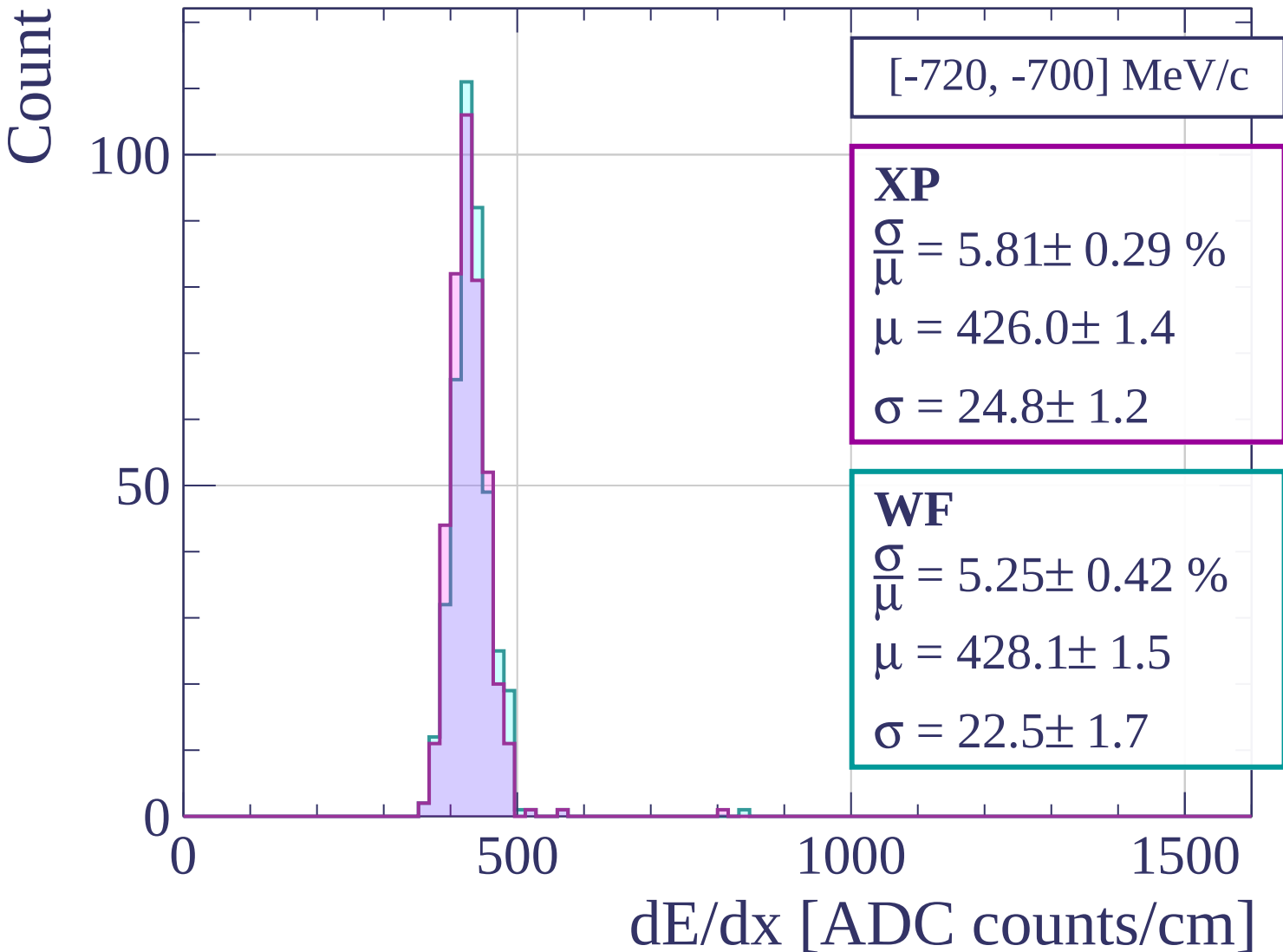
0

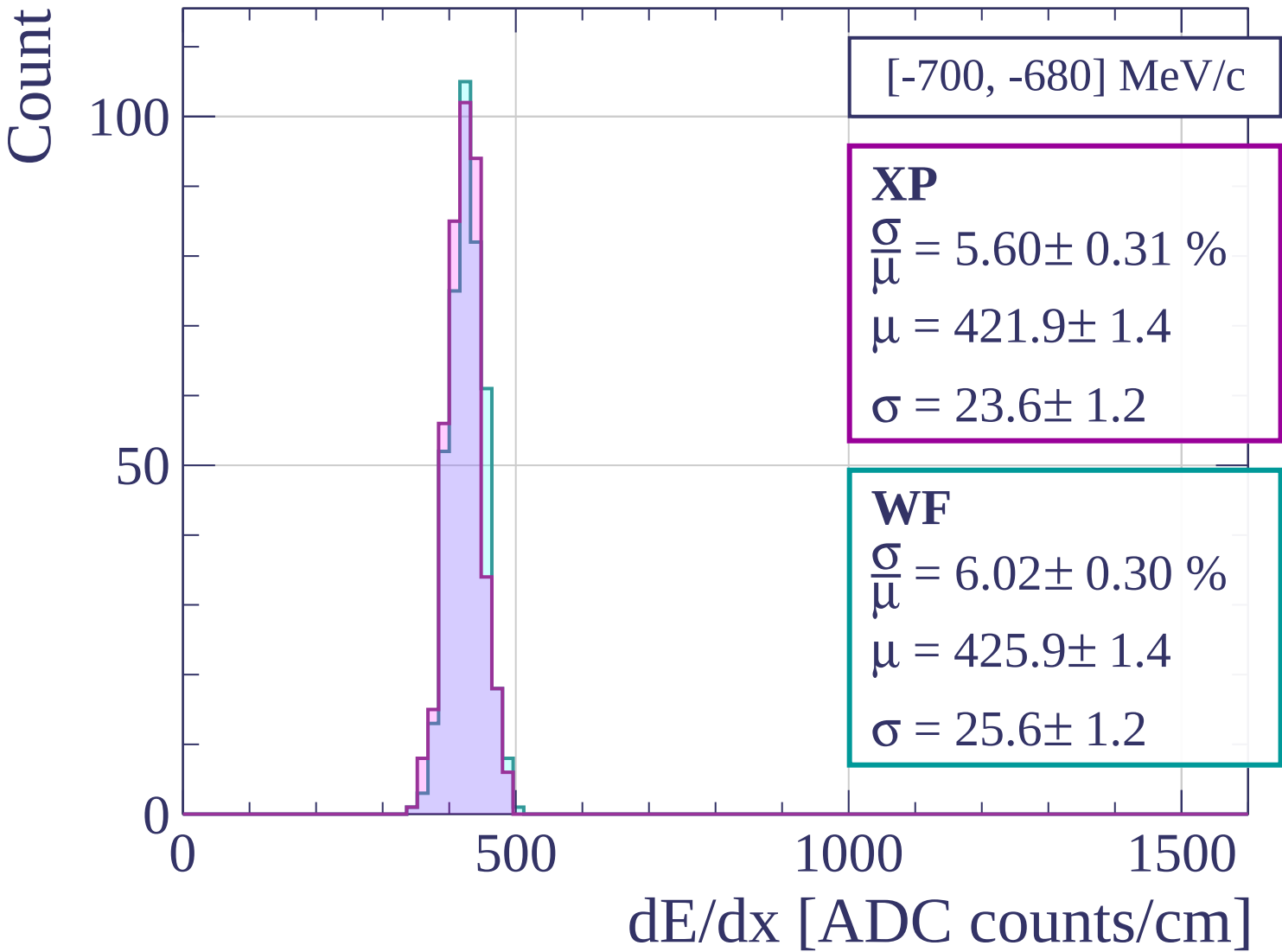
500

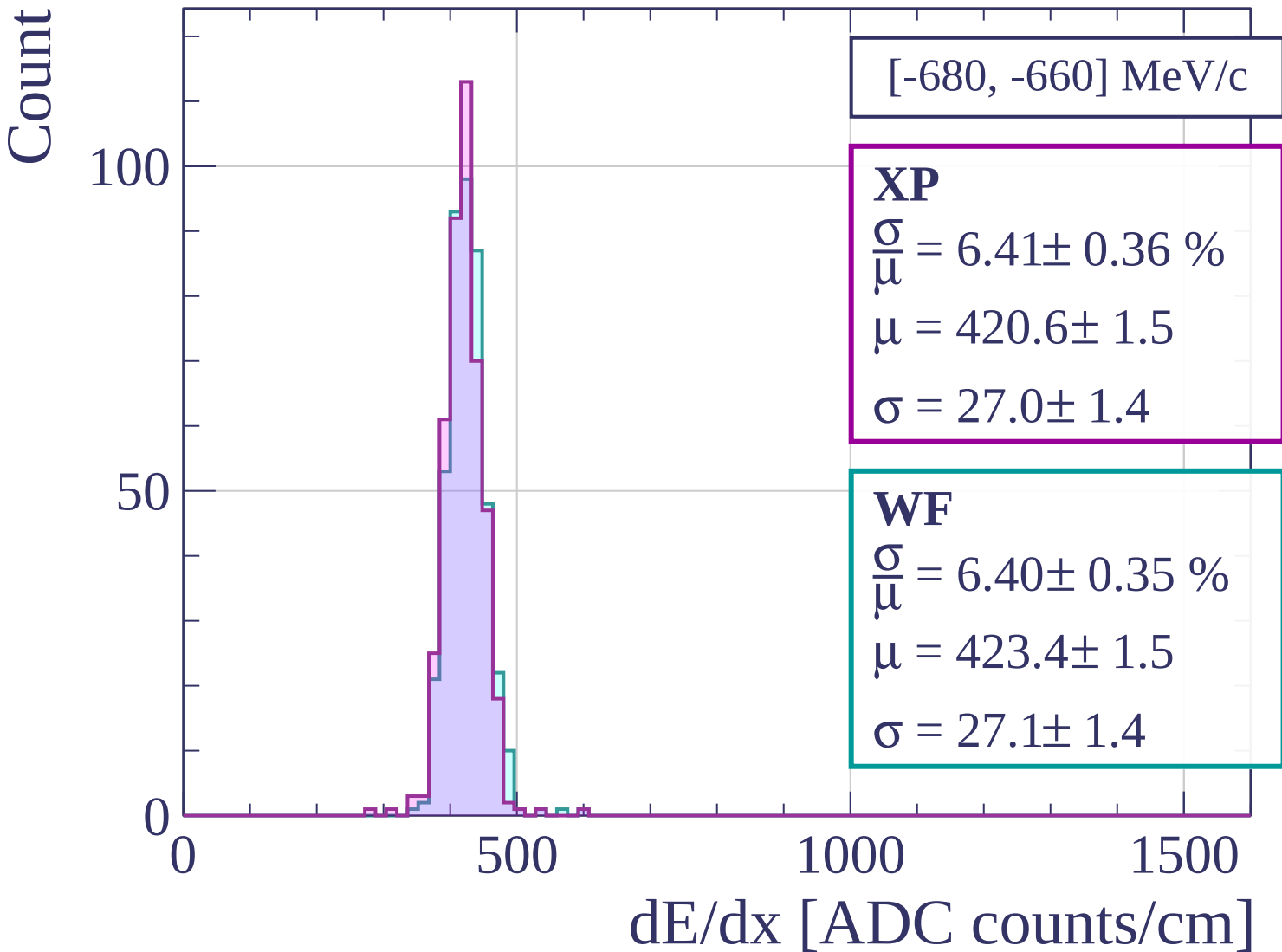
1000

1500

dE/dx [ADC counts/cm]







Count

[-660, -640] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.98 \pm 0.31 \%$$

$$\mu = 424.0 \pm 1.4$$

$$\sigma = 25.4 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 6.22 \pm 0.32 \%$$

$$\mu = 427.5 \pm 1.4$$

$$\sigma = 26.6 \pm 1.3$$

100

50

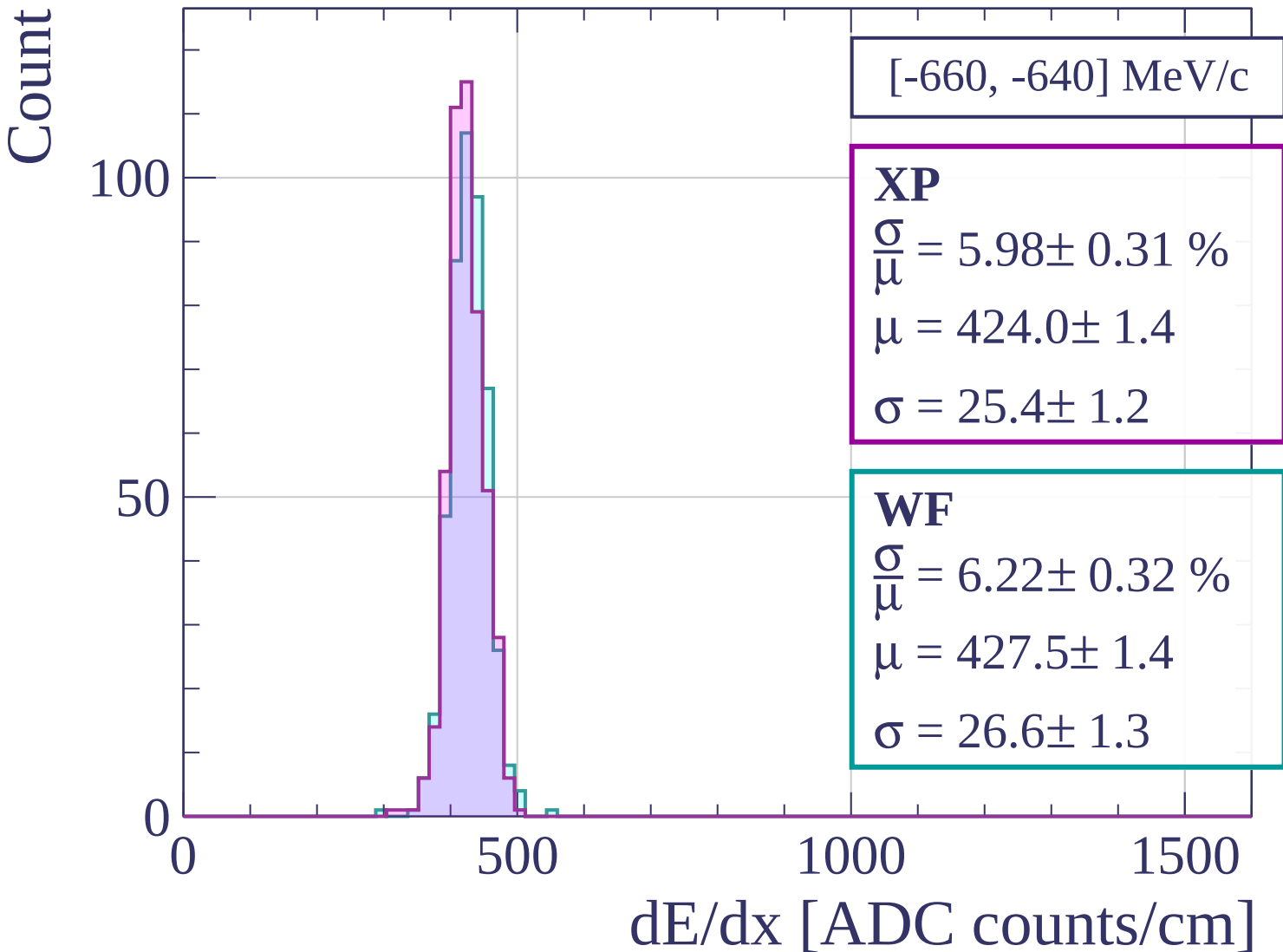
0

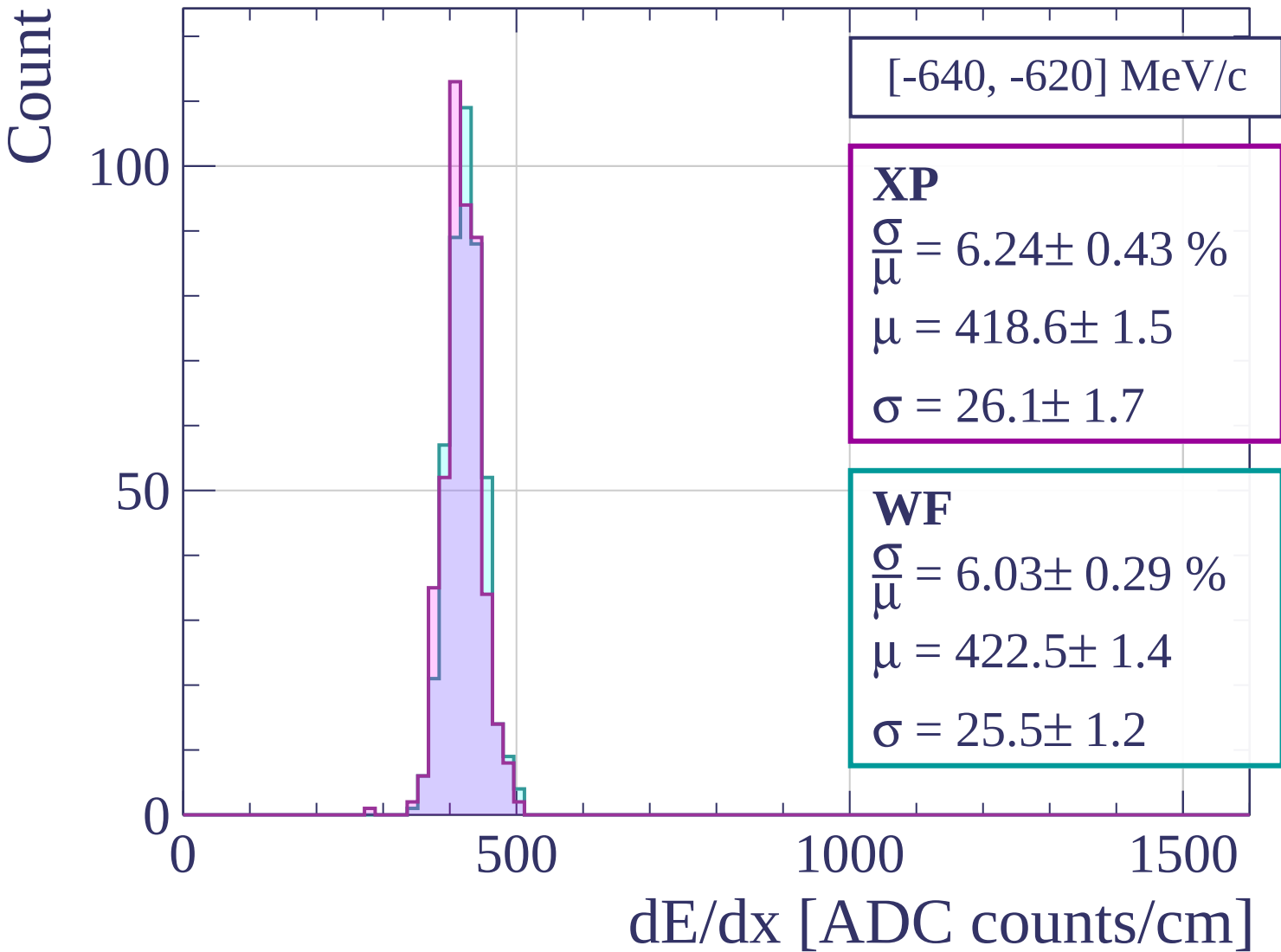
500

1000

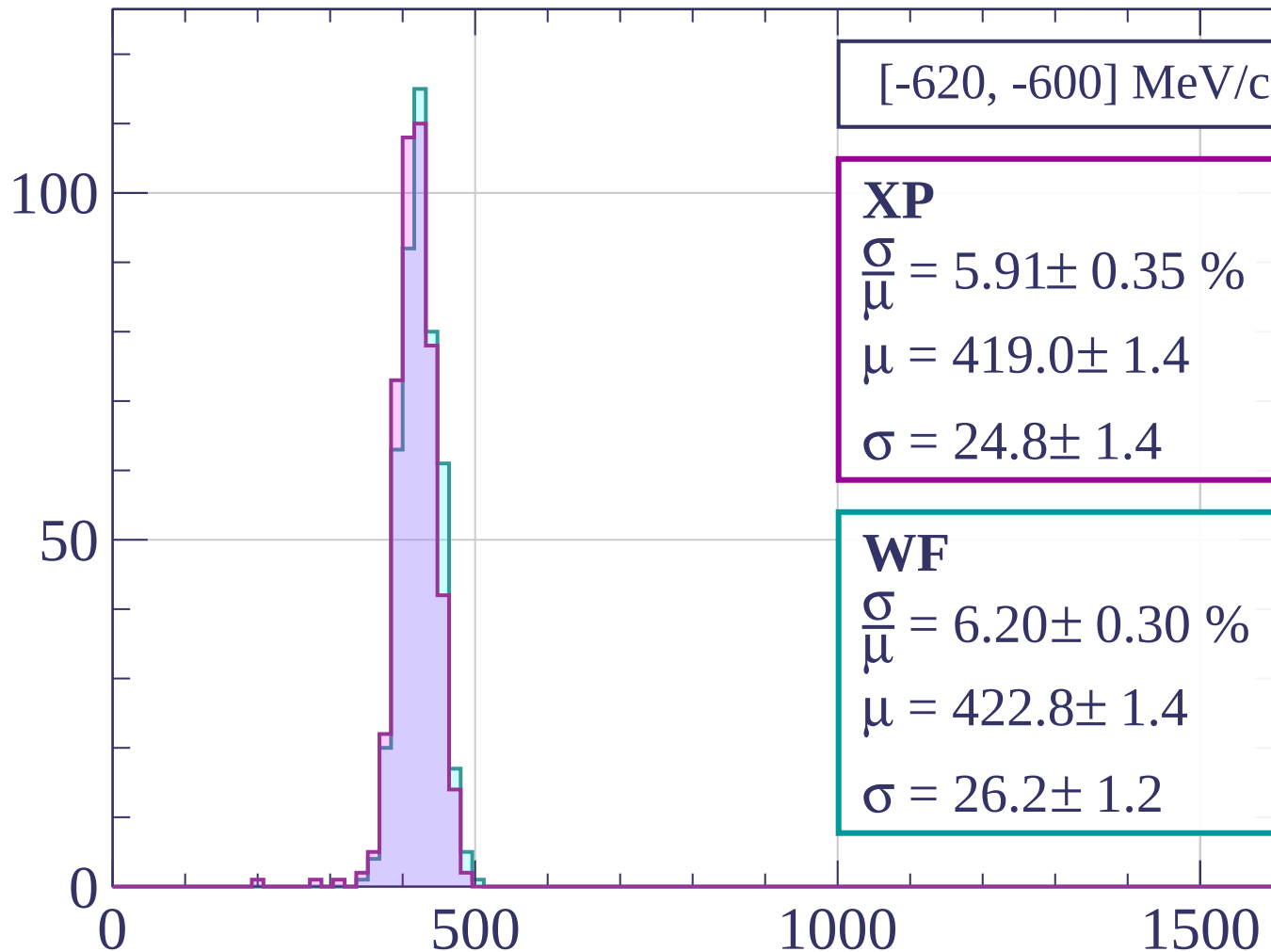
1500

dE/dx [ADC counts/cm]





Count



[-620, -600] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.91 \pm 0.35 \%$$

$$\mu = 419.0 \pm 1.4$$

$$\sigma = 24.8 \pm 1.4$$

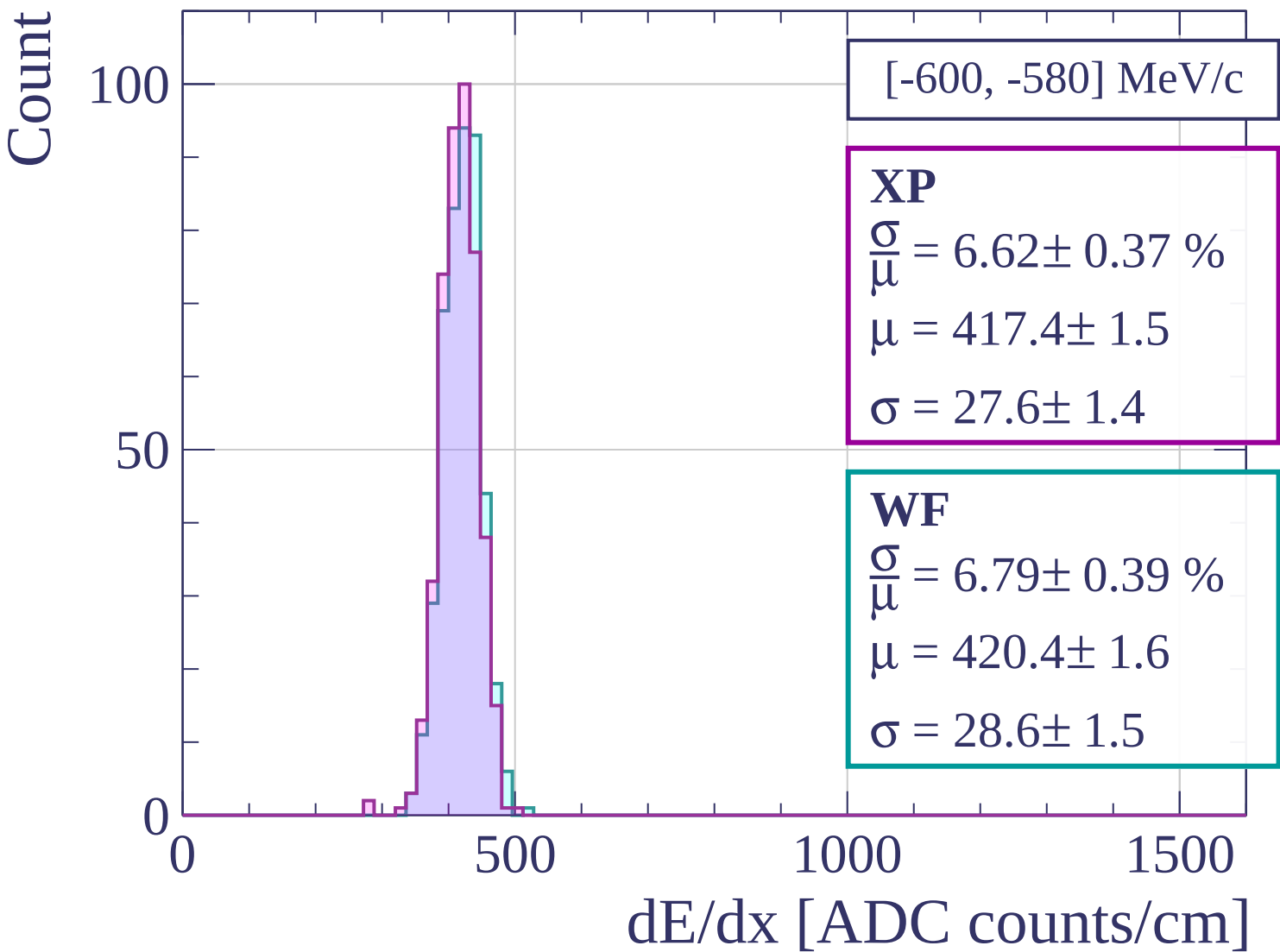
WF

$$\frac{\sigma}{\mu} = 6.20 \pm 0.30 \%$$

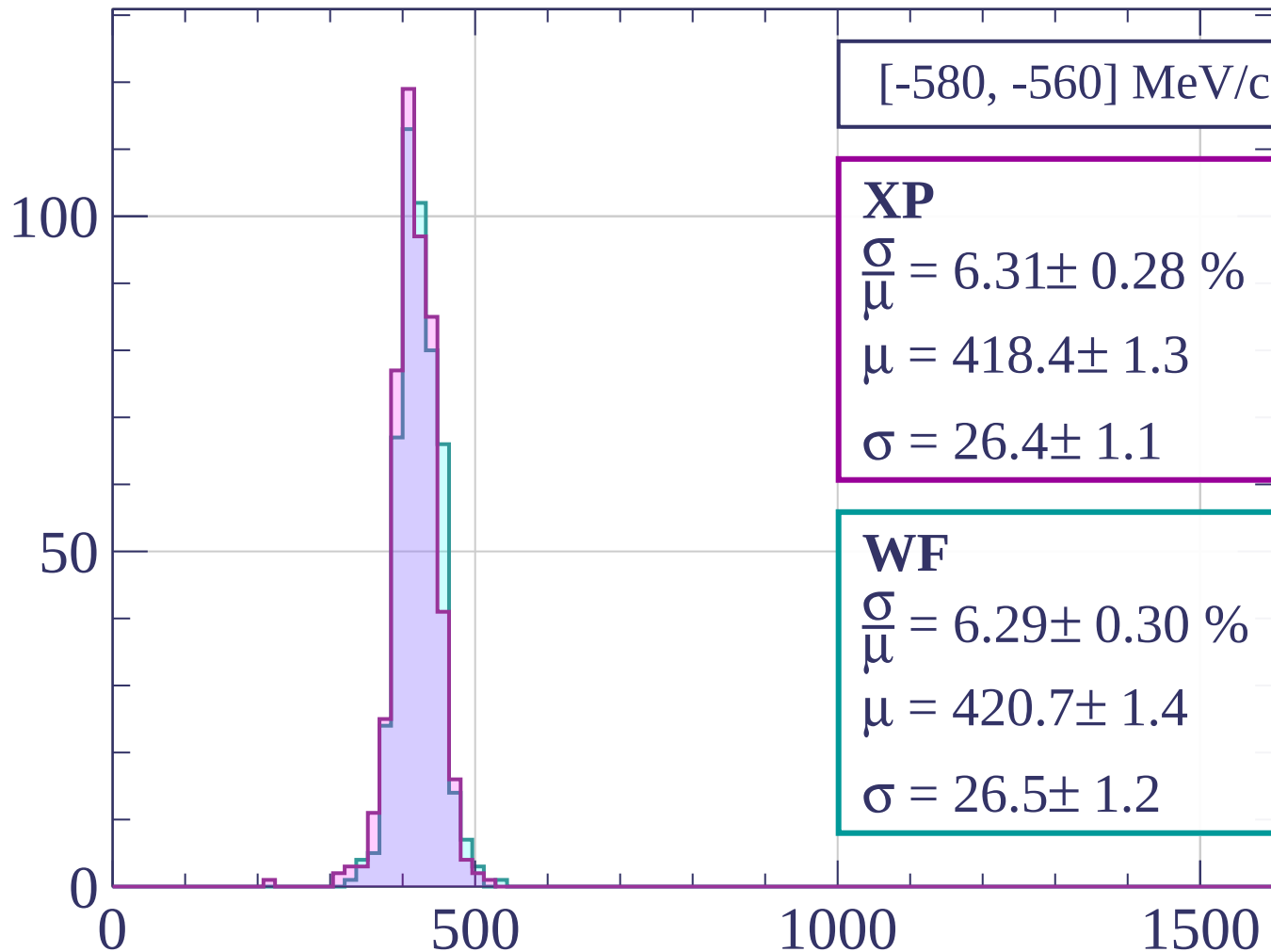
$$\mu = 422.8 \pm 1.4$$

$$\sigma = 26.2 \pm 1.2$$

dE/dx [ADC counts/cm]



Count



[-580, -560] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.31 \pm 0.28 \%$$

$$\mu = 418.4 \pm 1.3$$

$$\sigma = 26.4 \pm 1.1$$

WF

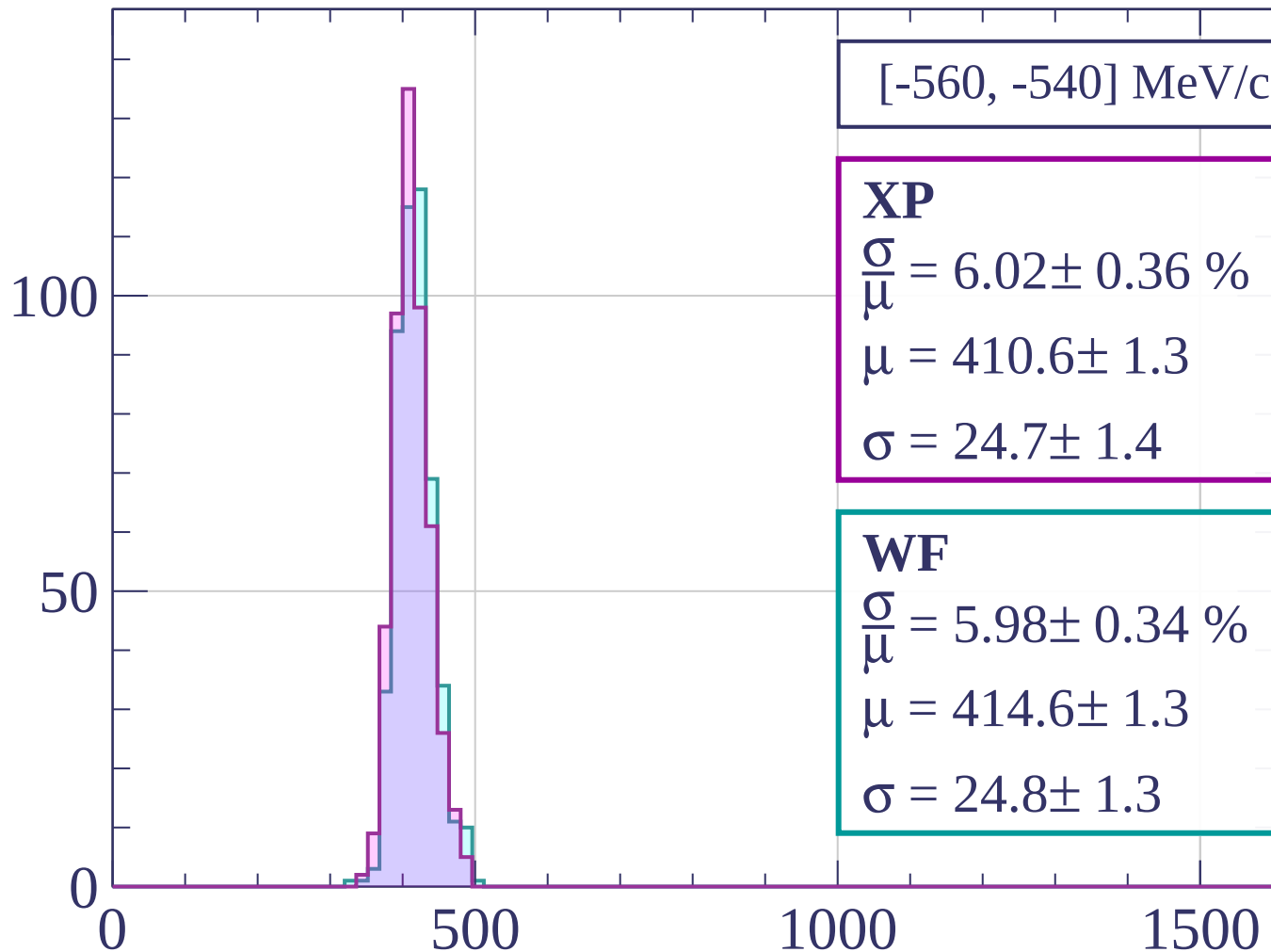
$$\frac{\sigma}{\mu} = 6.29 \pm 0.30 \%$$

$$\mu = 420.7 \pm 1.4$$

$$\sigma = 26.5 \pm 1.2$$

dE/dx [ADC counts/cm]

Count



$[-560, -540] \text{ MeV/c}$

XP

$$\frac{\sigma}{\mu} = 6.02 \pm 0.36 \%$$

$$\mu = 410.6 \pm 1.3$$

$$\sigma = 24.7 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 5.98 \pm 0.34 \%$$

$$\mu = 414.6 \pm 1.3$$

$$\sigma = 24.8 \pm 1.3$$

dE/dx [ADC counts/cm]

Count

[-540, -520] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.07 \pm 0.28 \%$$

$$\mu = 414.5 \pm 1.4$$

$$\sigma = 25.2 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 6.68 \pm 0.50 \%$$

$$\mu = 416.3 \pm 1.6$$

$$\sigma = 27.8 \pm 2.0$$

0

50

100

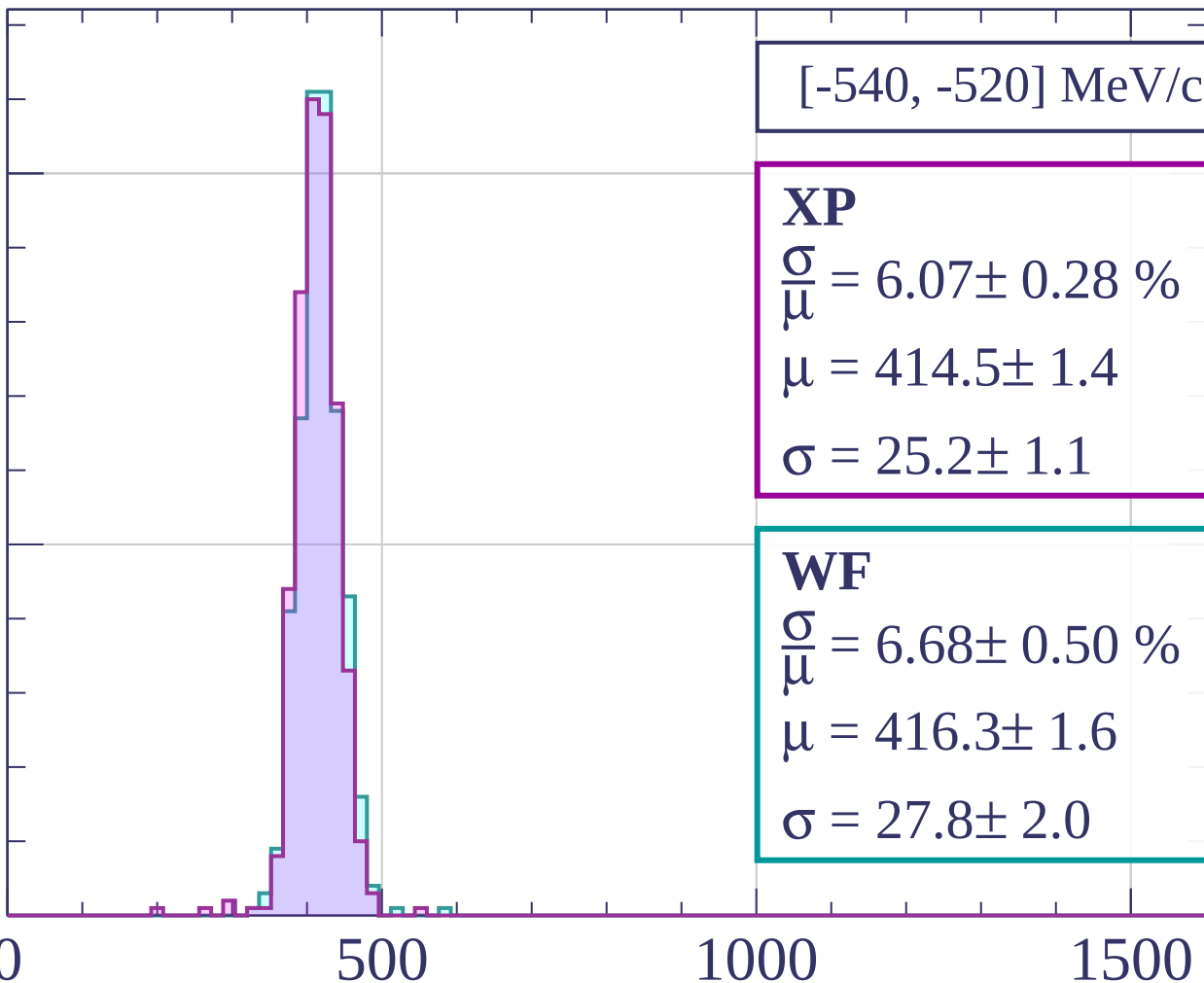
0

500

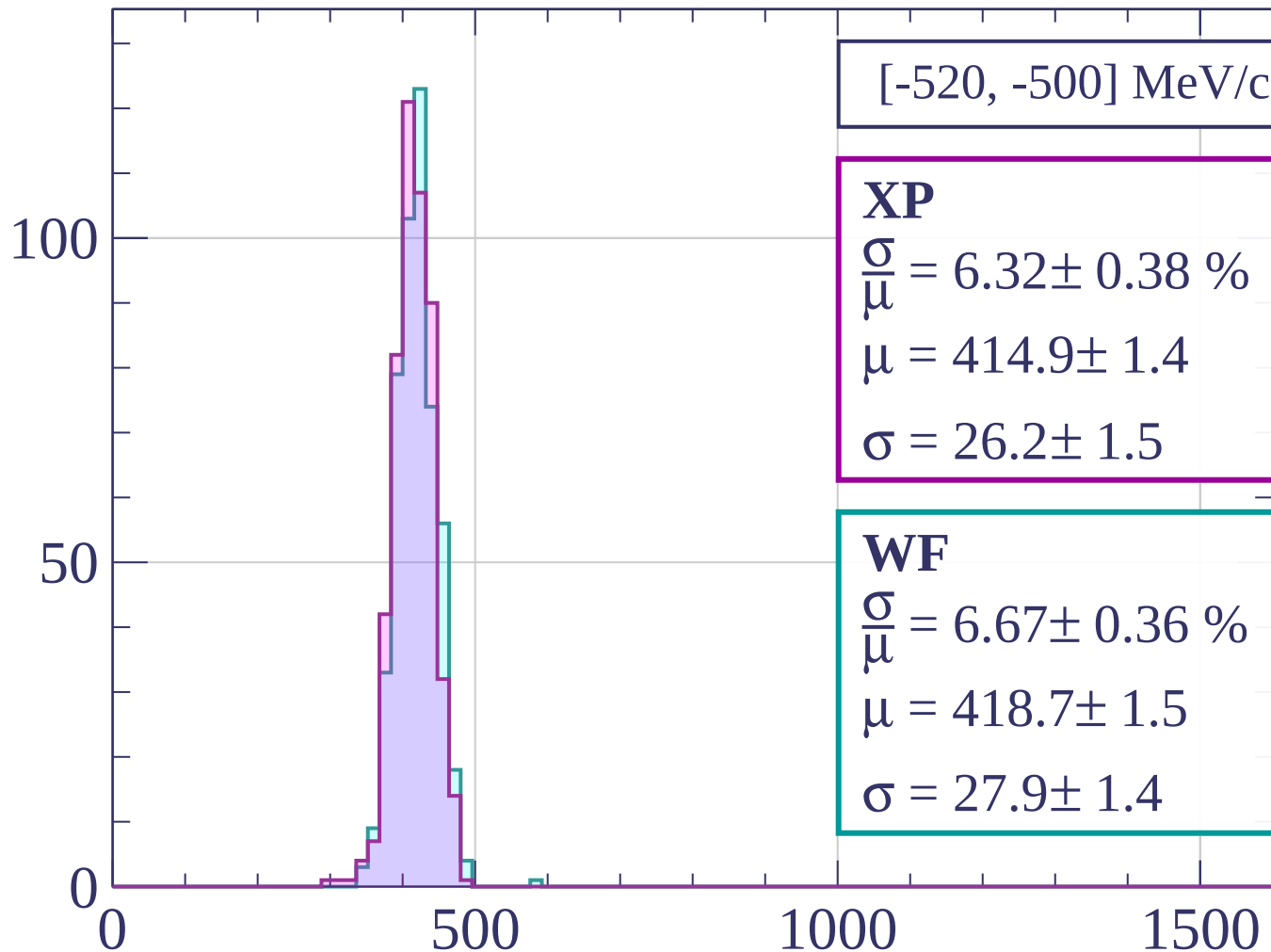
1000

1500

dE/dx [ADC counts/cm]



Count



[-520, -500] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.32 \pm 0.38 \%$$

$$\mu = 414.9 \pm 1.4$$

$$\sigma = 26.2 \pm 1.5$$

WF

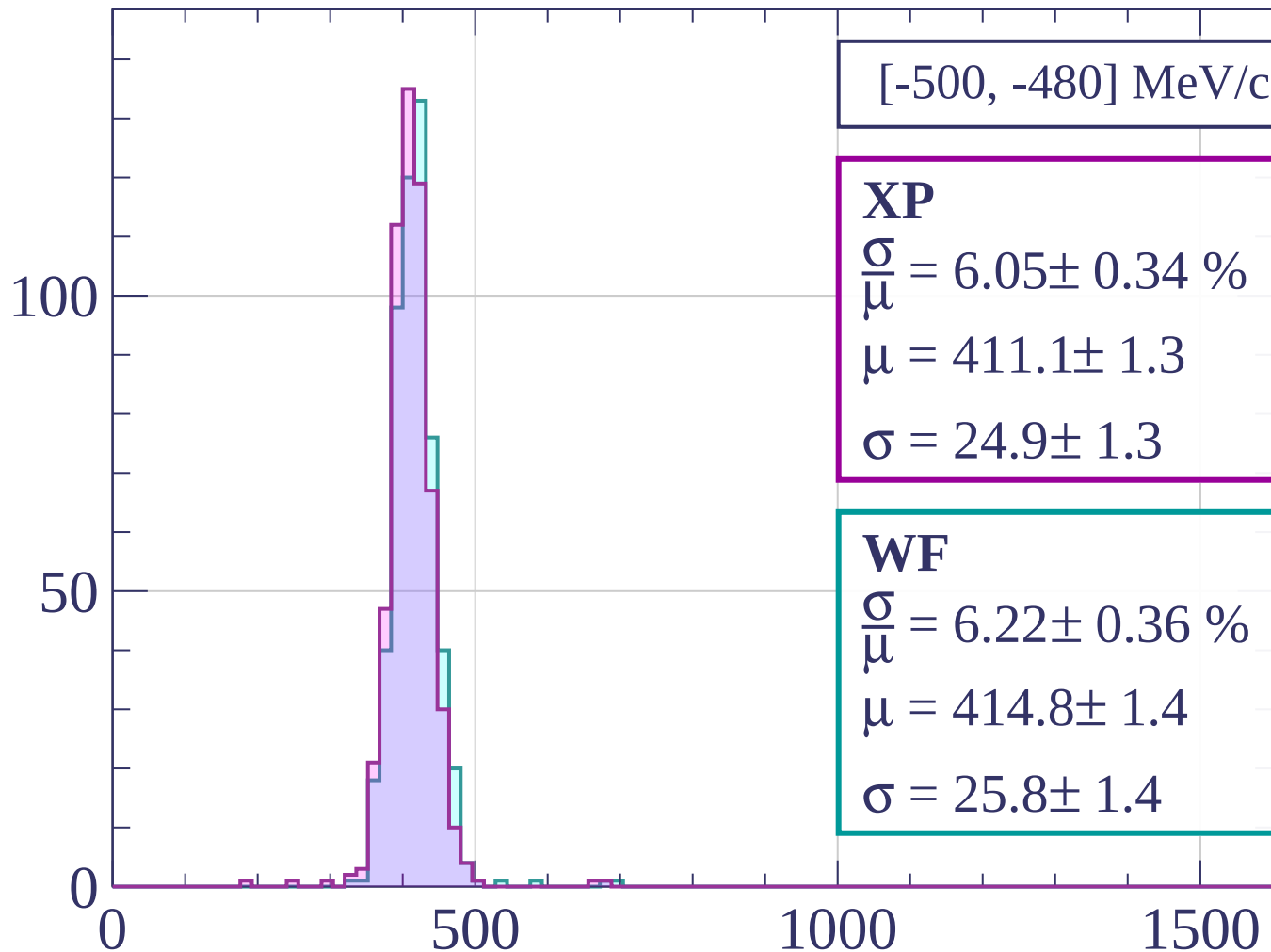
$$\frac{\sigma}{\mu} = 6.67 \pm 0.36 \%$$

$$\mu = 418.7 \pm 1.5$$

$$\sigma = 27.9 \pm 1.4$$

dE/dx [ADC counts/cm]

Count



[-500, -480] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.05 \pm 0.34 \%$$

$$\mu = 411.1 \pm 1.3$$

$$\sigma = 24.9 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 6.22 \pm 0.36 \%$$

$$\mu = 414.8 \pm 1.4$$

$$\sigma = 25.8 \pm 1.4$$

dE/dx [ADC counts/cm]

Count

[-480, -460] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.17 \pm 0.36 \%$$

$$\mu = 412.5 \pm 1.4$$

$$\sigma = 25.5 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 5.95 \pm 0.29 \%$$

$$\mu = 415.9 \pm 1.3$$

$$\sigma = 24.7 \pm 1.1$$

100

50

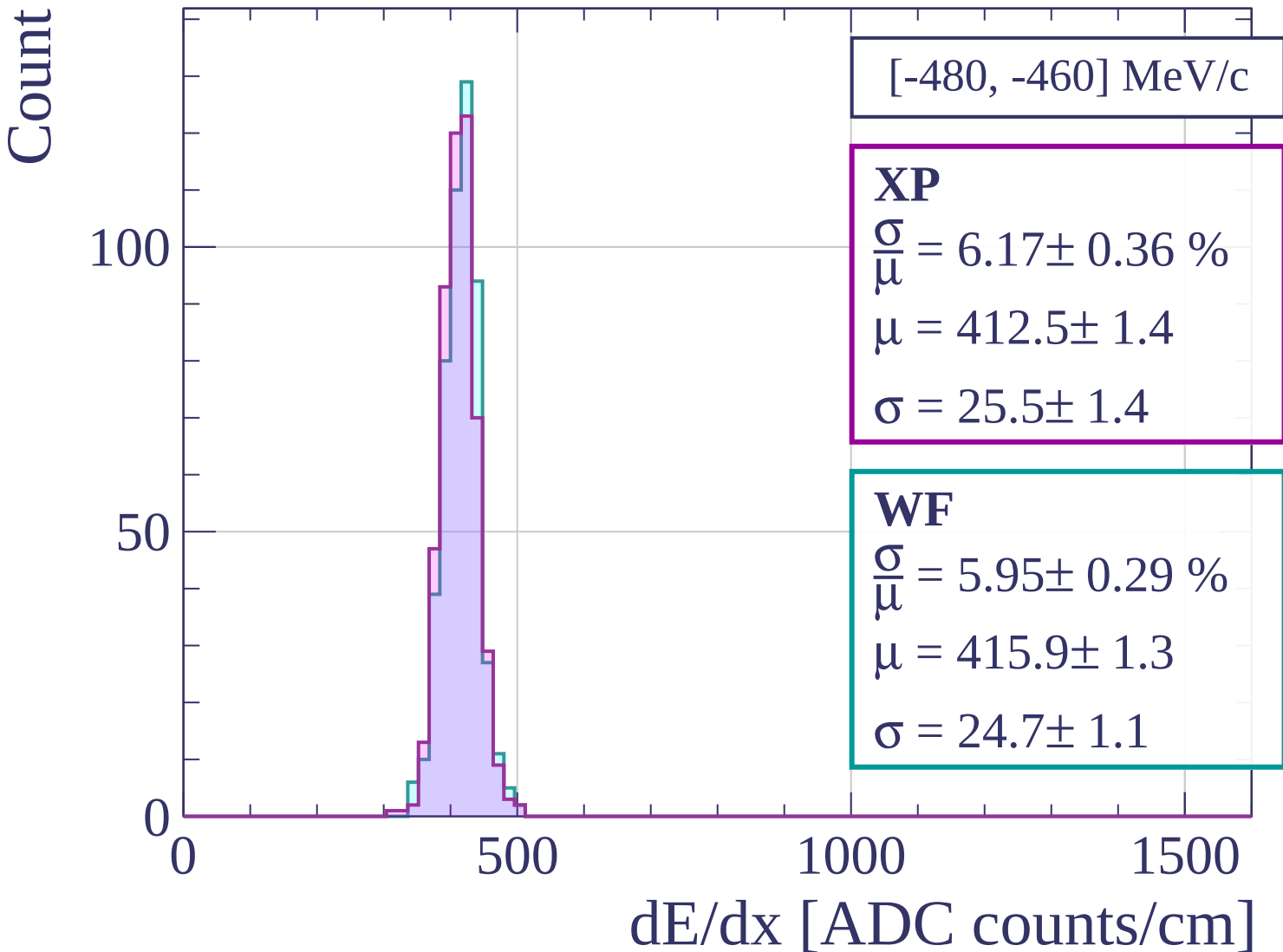
0

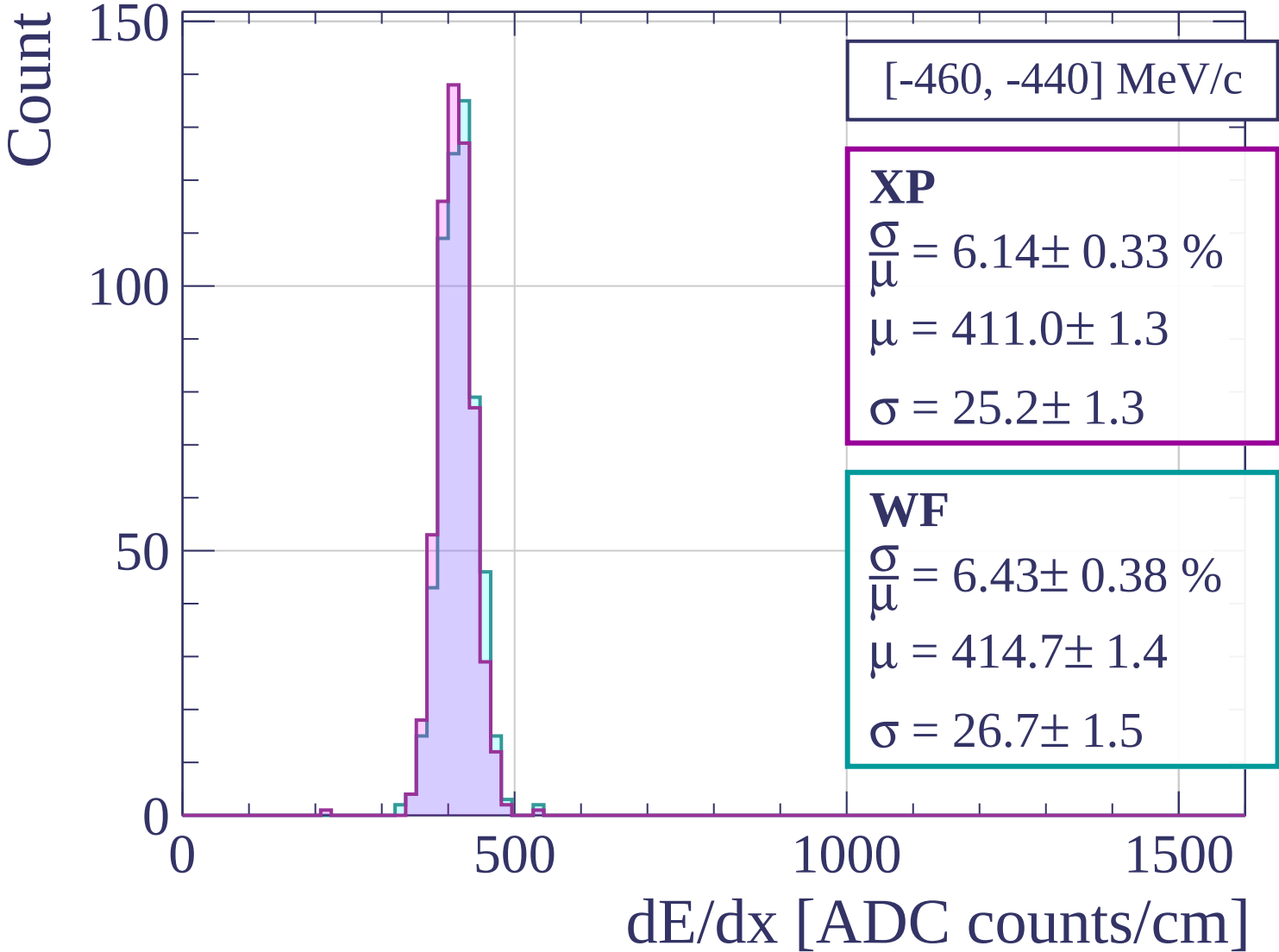
500

1000

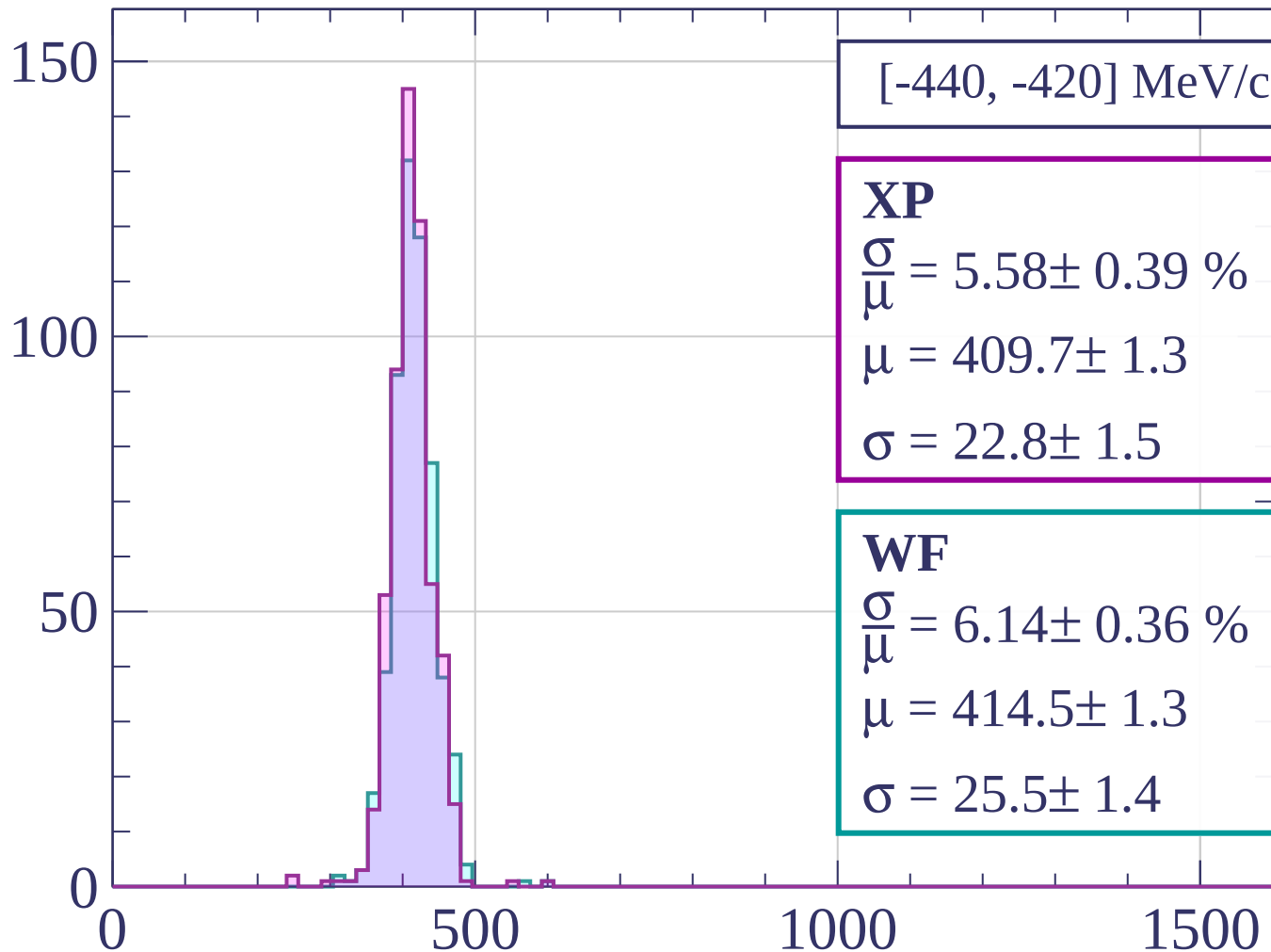
1500

dE/dx [ADC counts/cm]



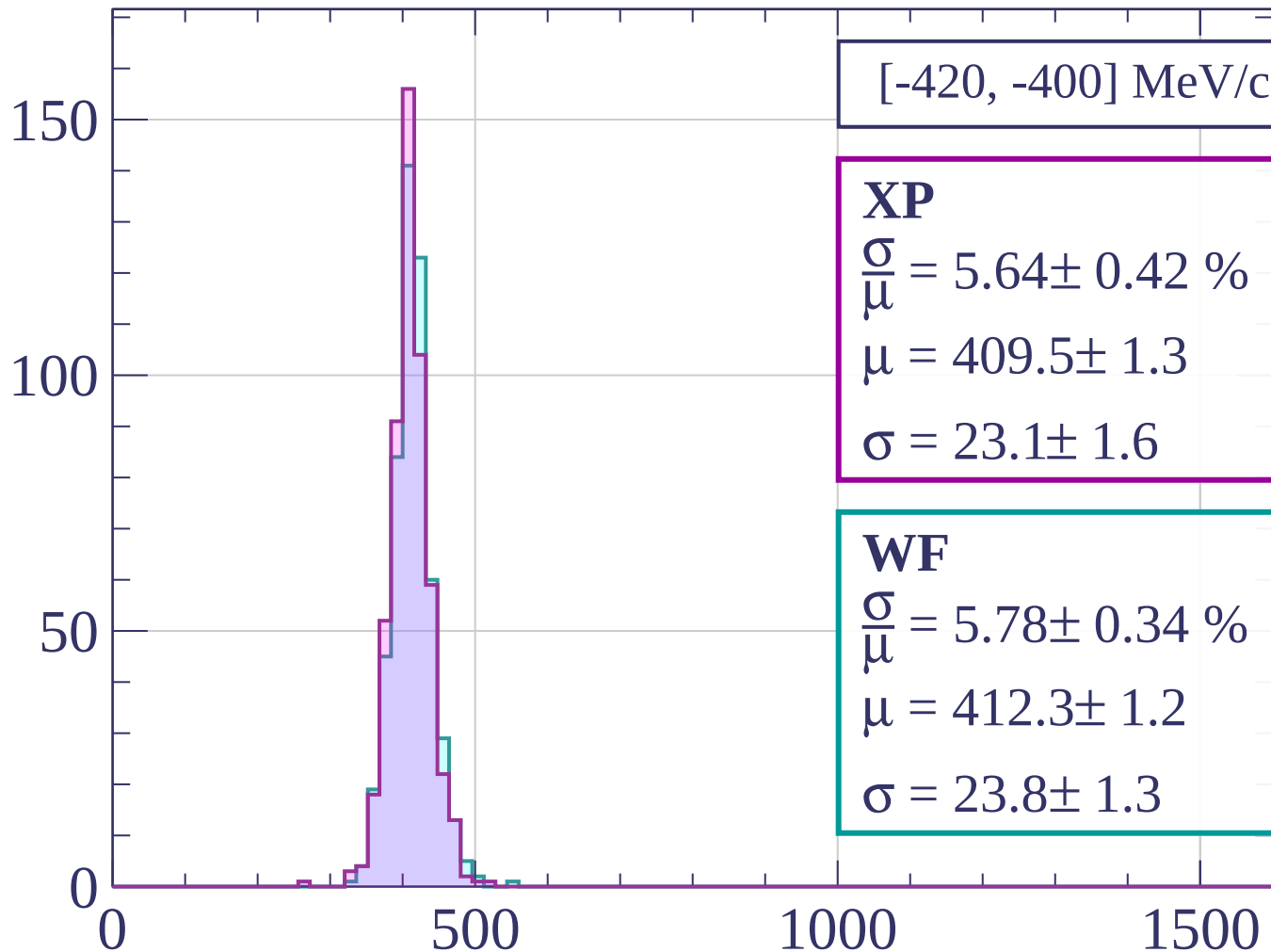


Count



dE/dx [ADC counts/cm]

Count



[-420, -400] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.64 \pm 0.42 \%$$

$$\mu = 409.5 \pm 1.3$$

$$\sigma = 23.1 \pm 1.6$$

WF

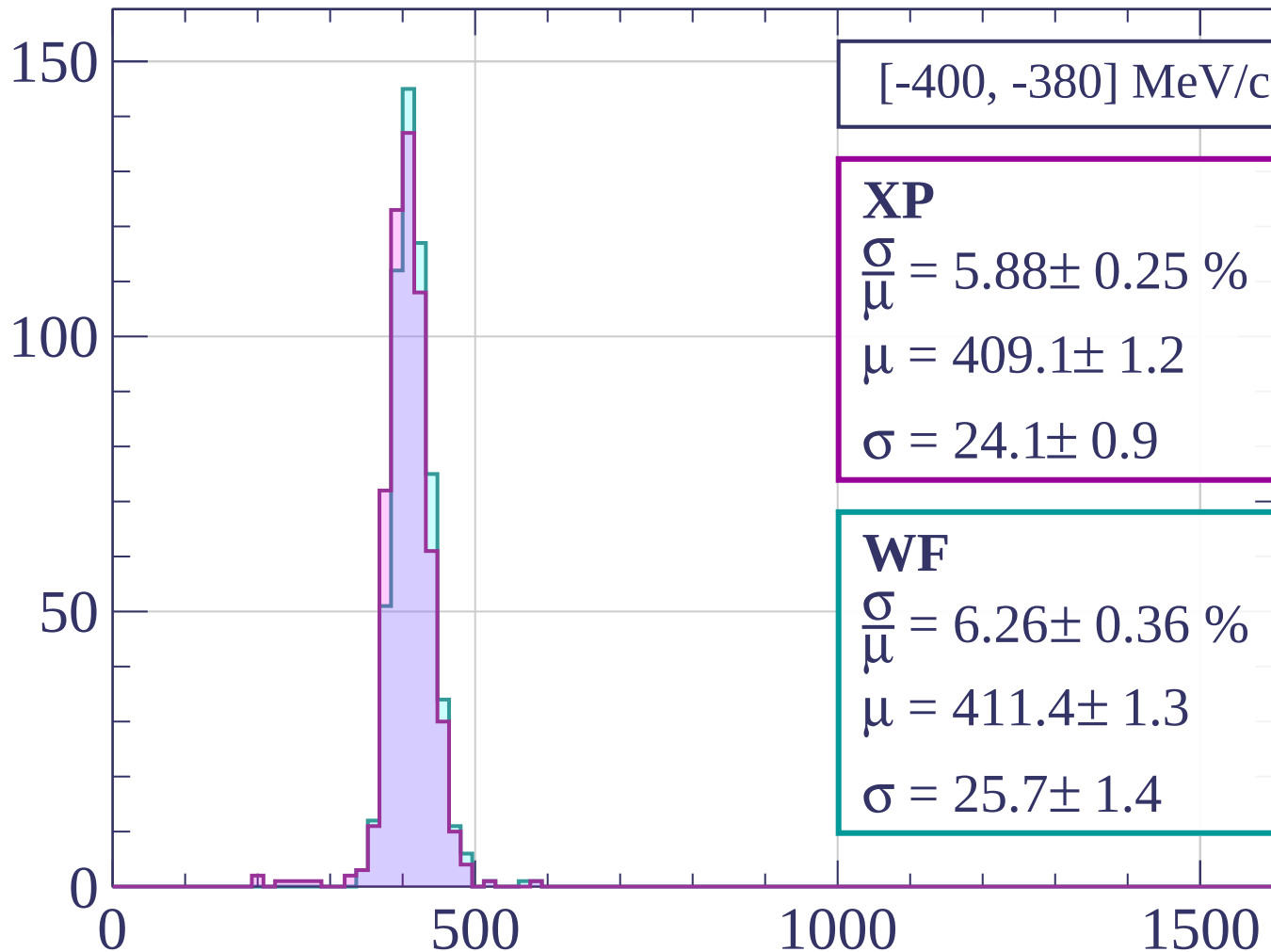
$$\frac{\sigma}{\mu} = 5.78 \pm 0.34 \%$$

$$\mu = 412.3 \pm 1.2$$

$$\sigma = 23.8 \pm 1.3$$

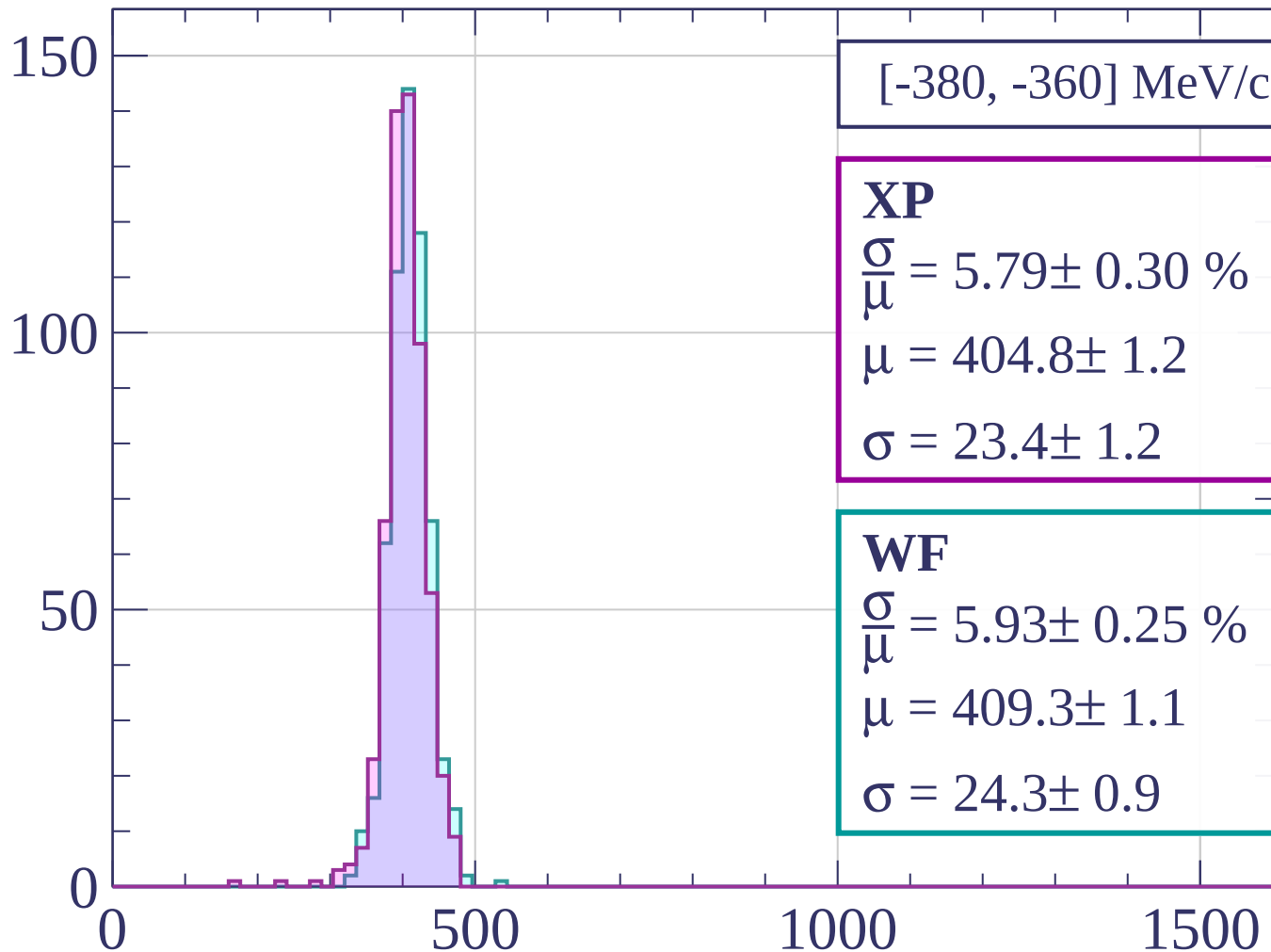
dE/dx [ADC counts/cm]

Count



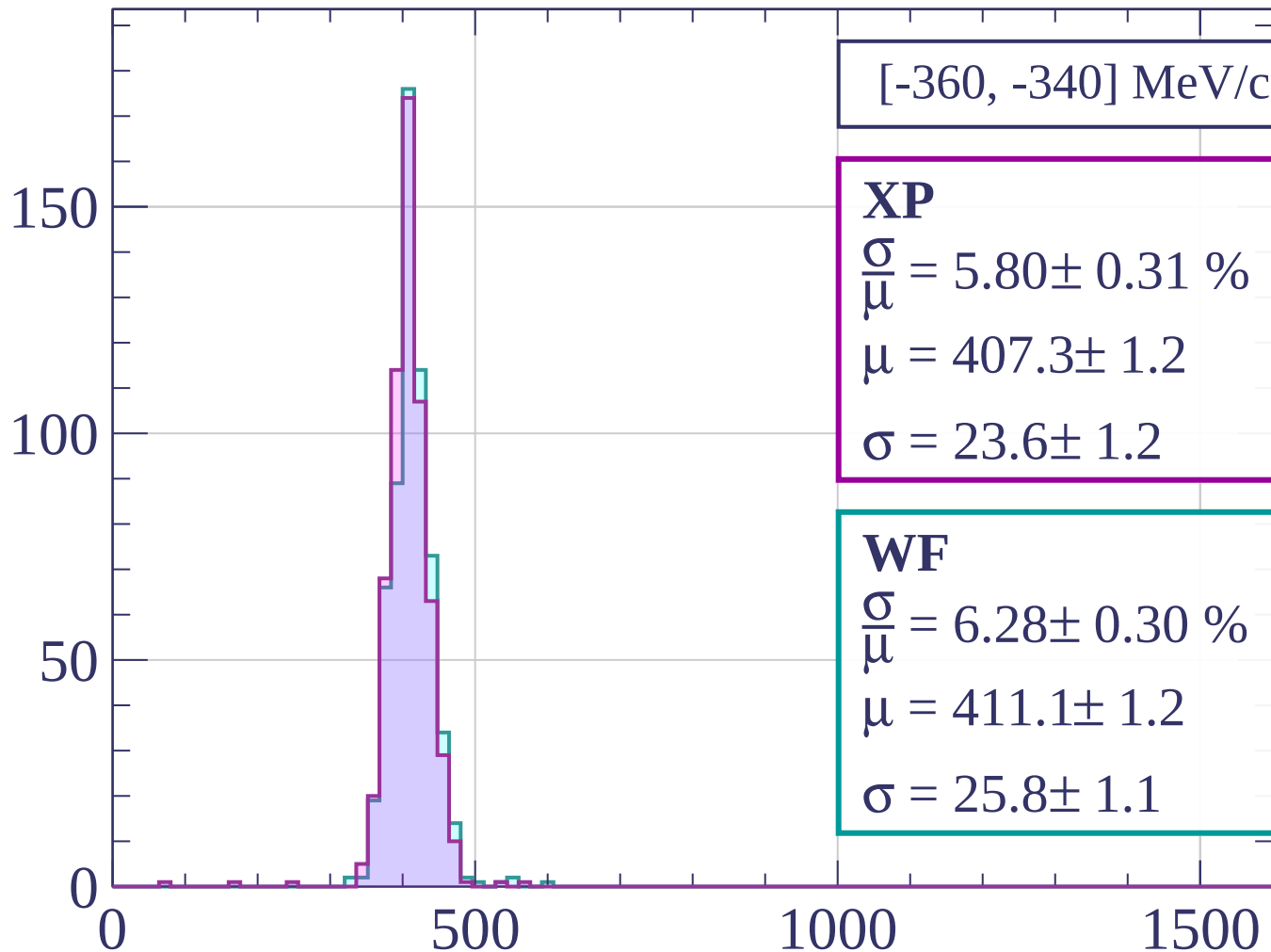
dE/dx [ADC counts/cm]

Count



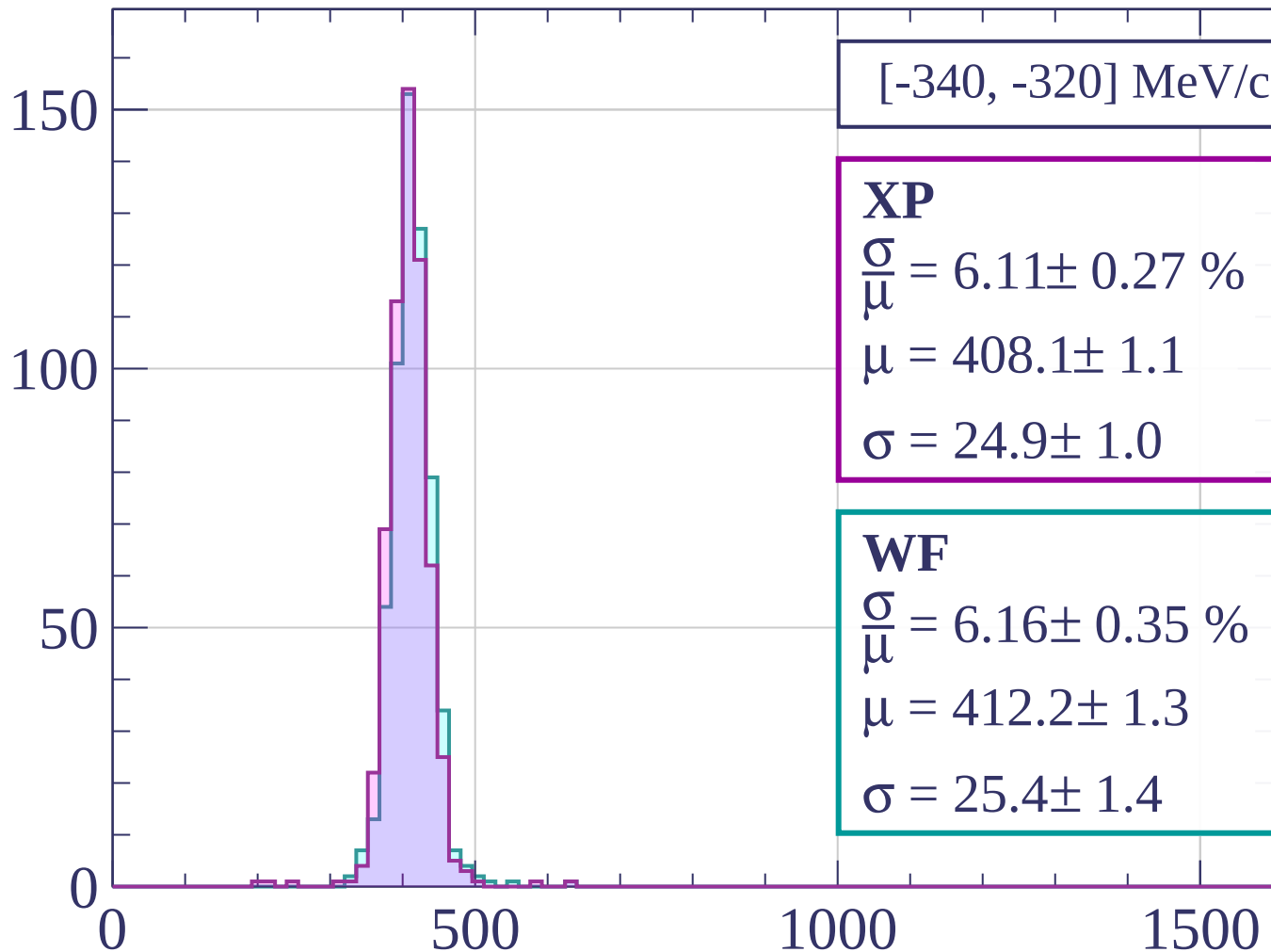
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



$[-340, -320]$ MeV/c

XP

$$\frac{\sigma}{\mu} = 6.11 \pm 0.27 \%$$

$$\mu = 408.1 \pm 1.1$$

$$\sigma = 24.9 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 6.16 \pm 0.35 \%$$

$$\mu = 412.2 \pm 1.3$$

$$\sigma = 25.4 \pm 1.4$$

dE/dx [ADC counts/cm]

Count

[-320, -300] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.35 \pm 0.29 \%$$

$$\mu = 412.3 \pm 1.3$$

$$\sigma = 26.2 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 6.75 \pm 0.28 \%$$

$$\mu = 417.4 \pm 1.3$$

$$\sigma = 28.2 \pm 1.1$$

100

50

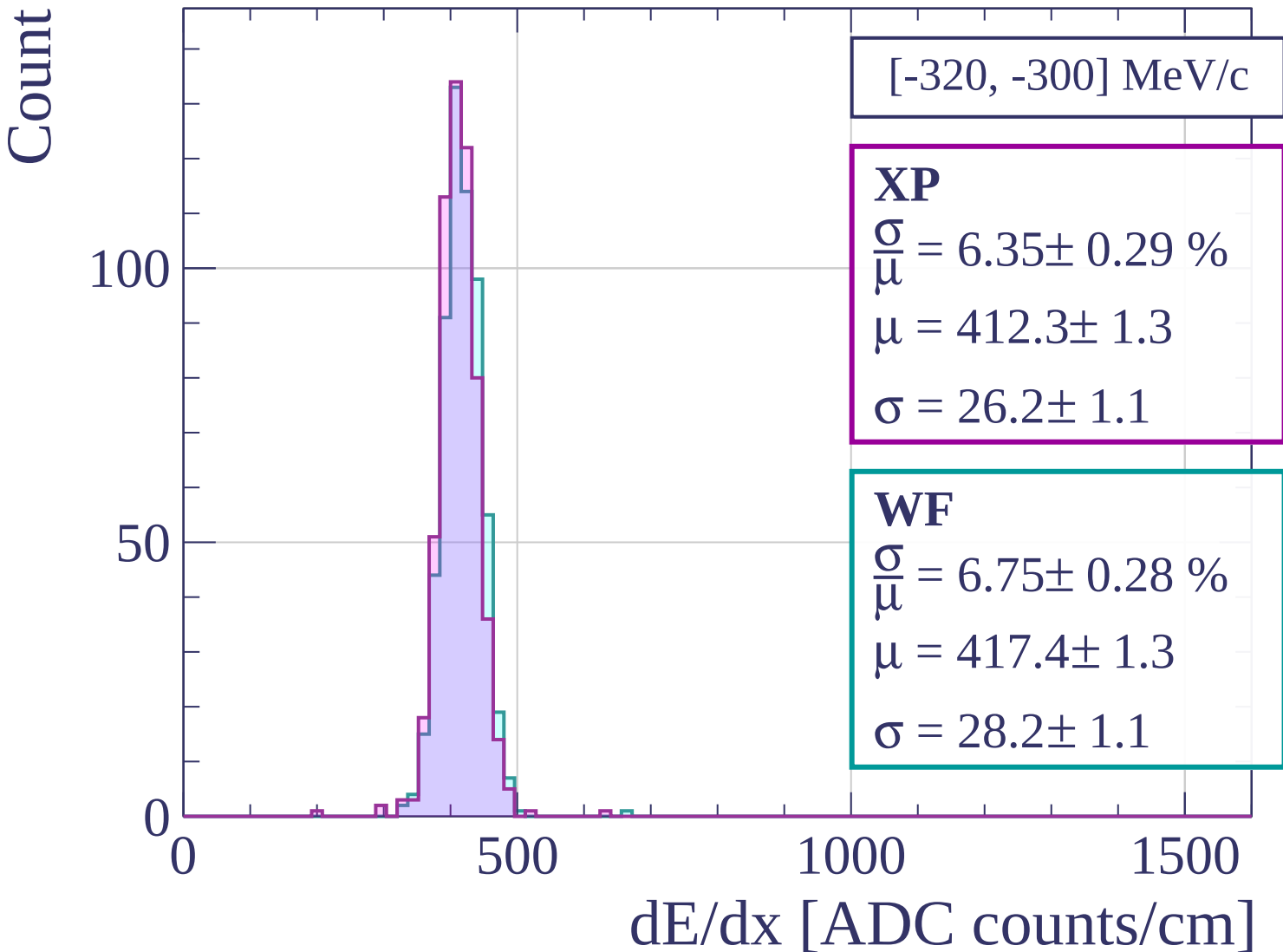
0

500

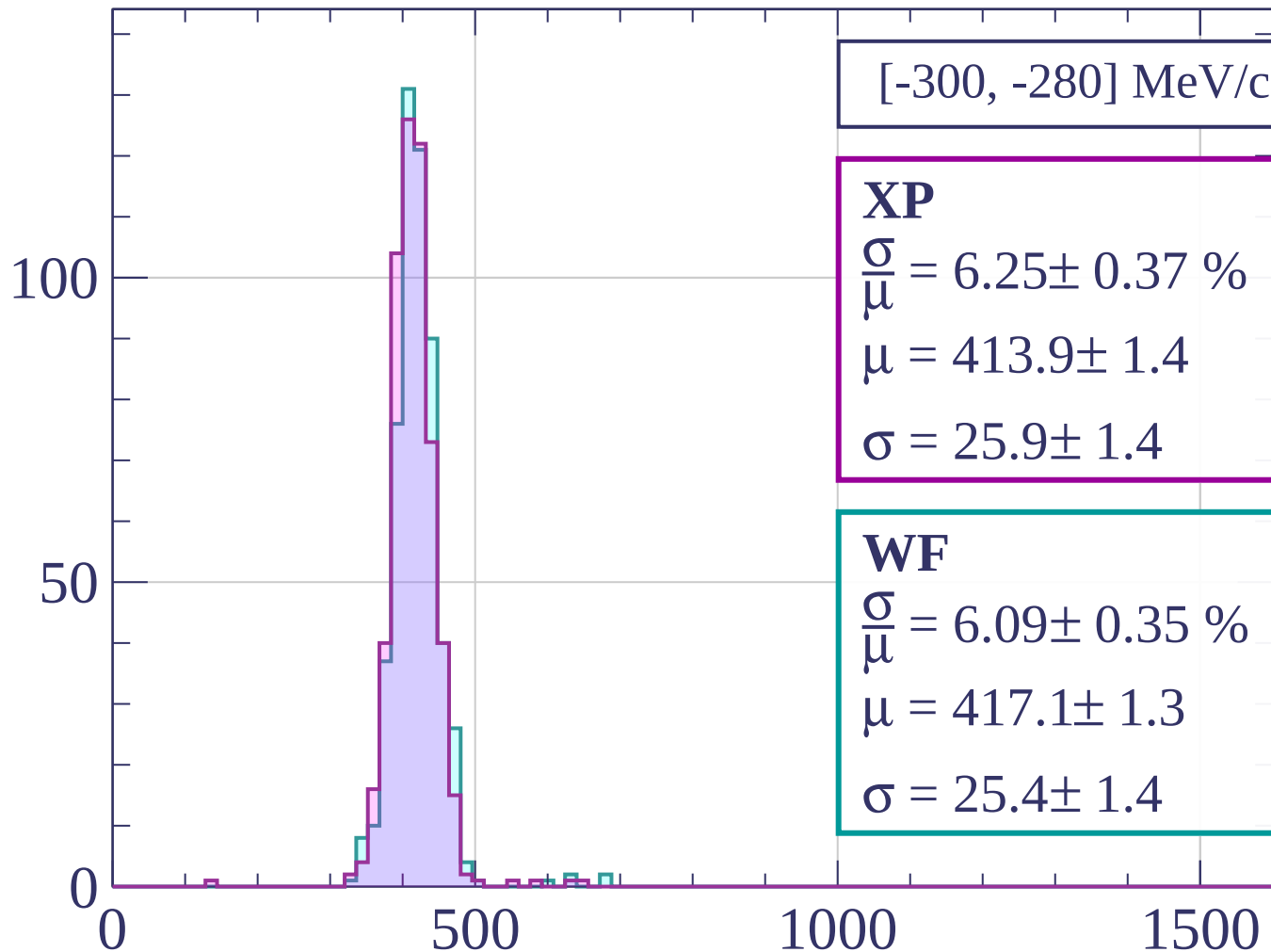
1000

1500

dE/dx [ADC counts/cm]



Count



[-300, -280] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.25 \pm 0.37 \%$$

$$\mu = 413.9 \pm 1.4$$

$$\sigma = 25.9 \pm 1.4$$

WF

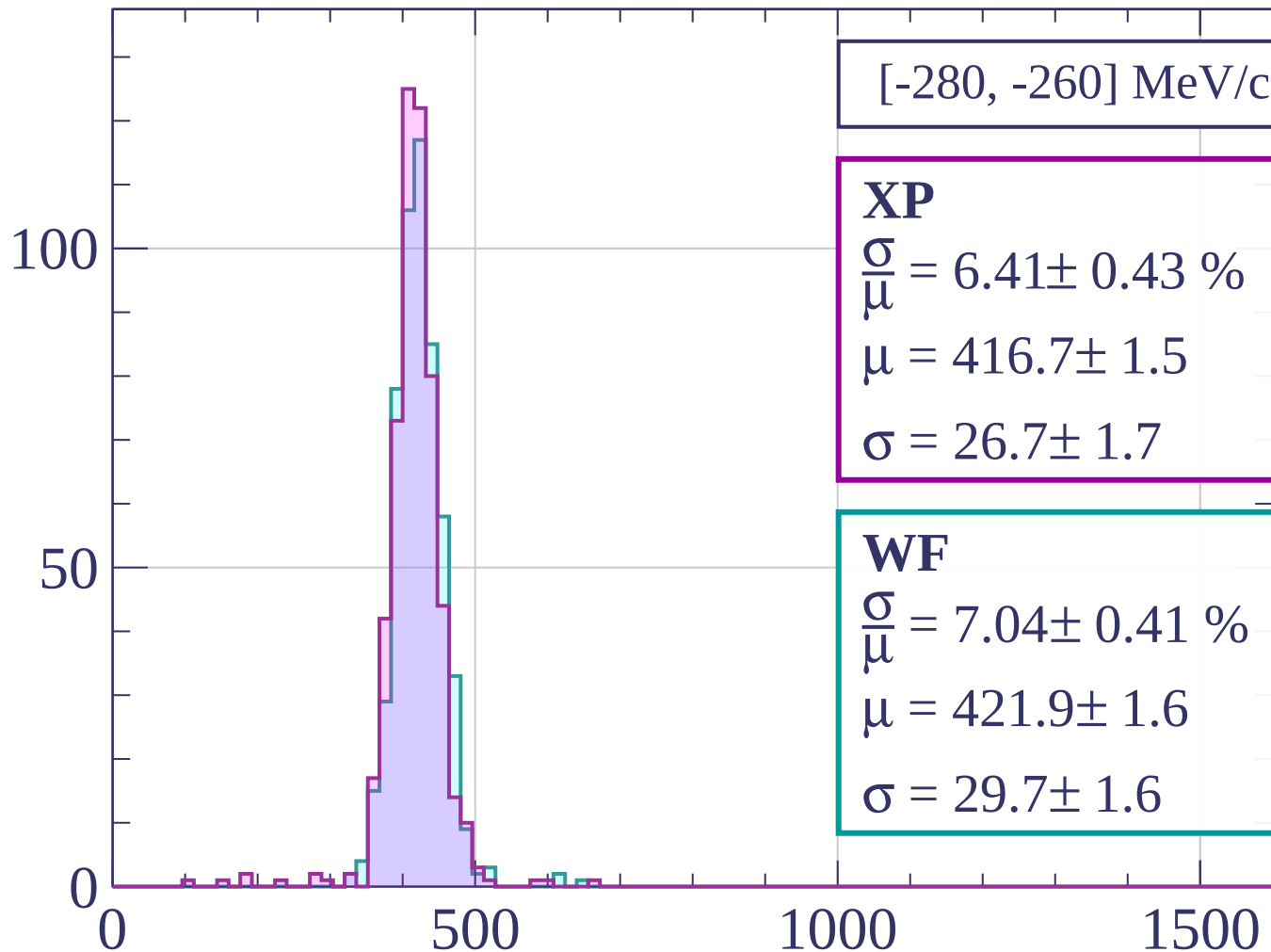
$$\frac{\sigma}{\mu} = 6.09 \pm 0.35 \%$$

$$\mu = 417.1 \pm 1.3$$

$$\sigma = 25.4 \pm 1.4$$

dE/dx [ADC counts/cm]

Count



$[-280, -260] \text{ MeV/c}$

XP

$$\frac{\sigma}{\mu} = 6.41 \pm 0.43 \%$$

$$\mu = 416.7 \pm 1.5$$

$$\sigma = 26.7 \pm 1.7$$

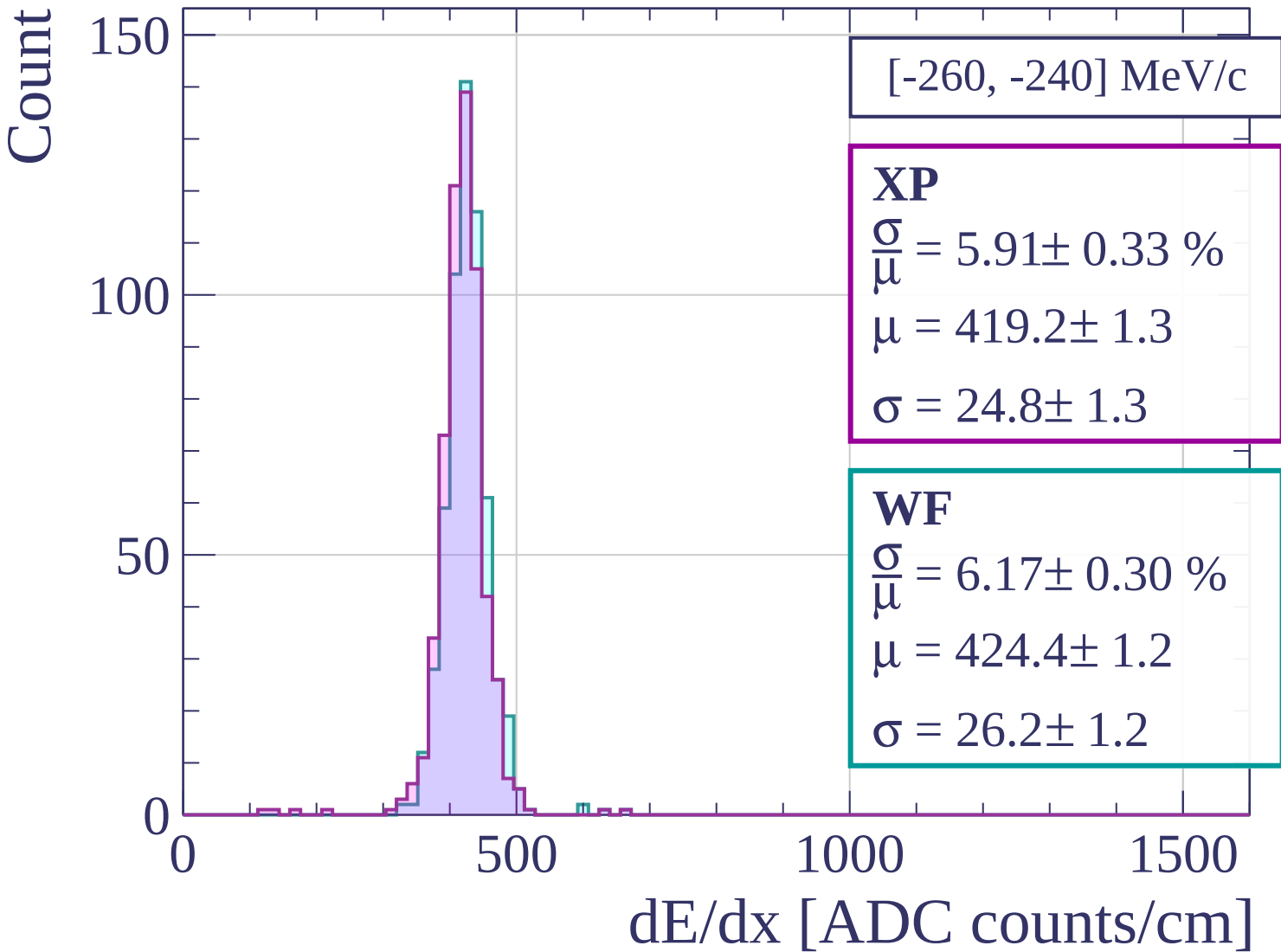
WF

$$\frac{\sigma}{\mu} = 7.04 \pm 0.41 \%$$

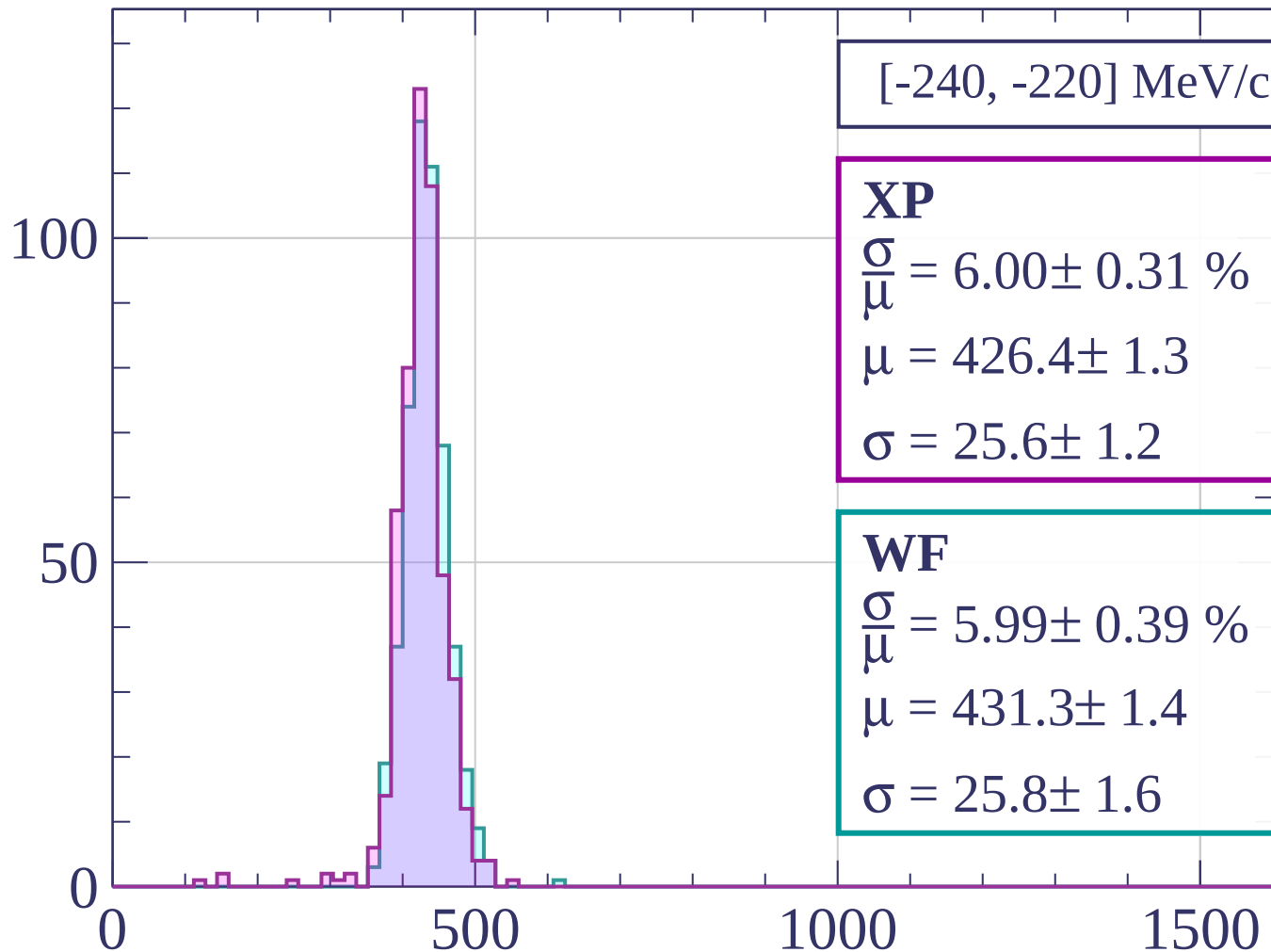
$$\mu = 421.9 \pm 1.6$$

$$\sigma = 29.7 \pm 1.6$$

dE/dx [ADC counts/cm]



Count



$[-240, -220] \text{ MeV/c}$

XP

$$\frac{\sigma}{\bar{\mu}} = 6.00 \pm 0.31 \%$$

$$\mu = 426.4 \pm 1.3$$

$$\sigma = 25.6 \pm 1.2$$

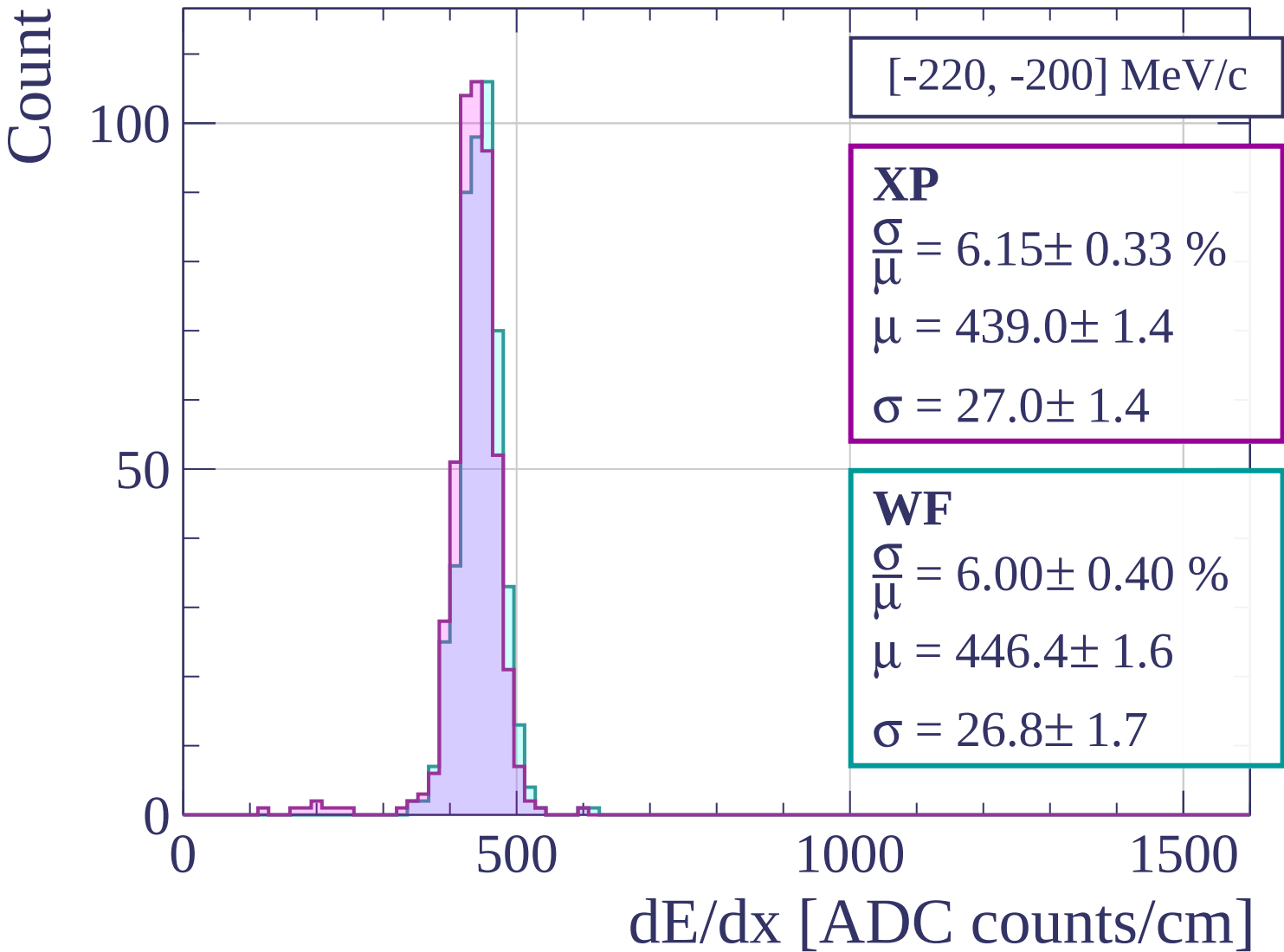
WF

$$\frac{\sigma}{\bar{\mu}} = 5.99 \pm 0.39 \%$$

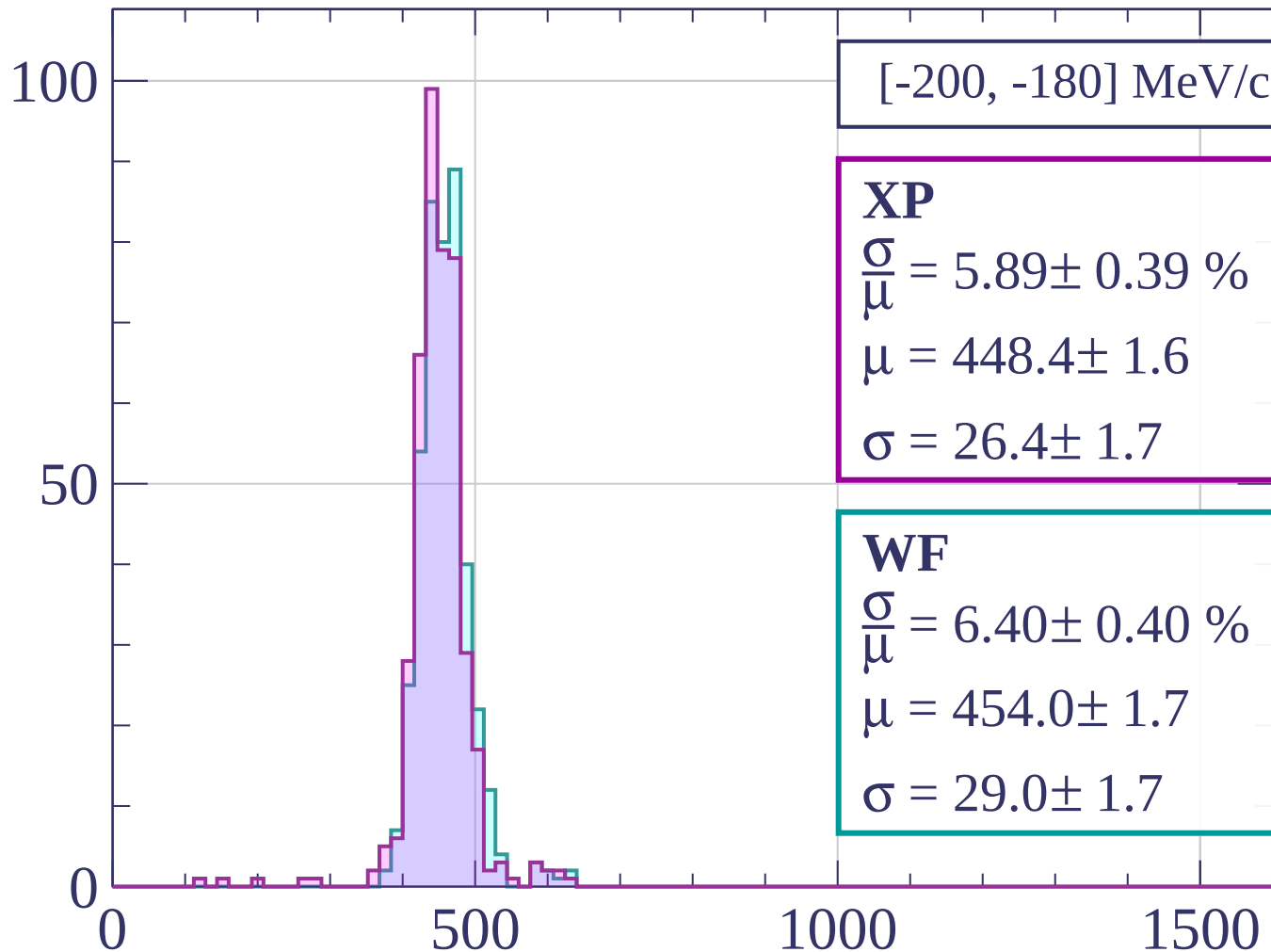
$$\mu = 431.3 \pm 1.4$$

$$\sigma = 25.8 \pm 1.6$$

dE/dx [ADC counts/cm]



Count



[-200, -180] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.89 \pm 0.39 \%$$

$$\mu = 448.4 \pm 1.6$$

$$\sigma = 26.4 \pm 1.7$$

WF

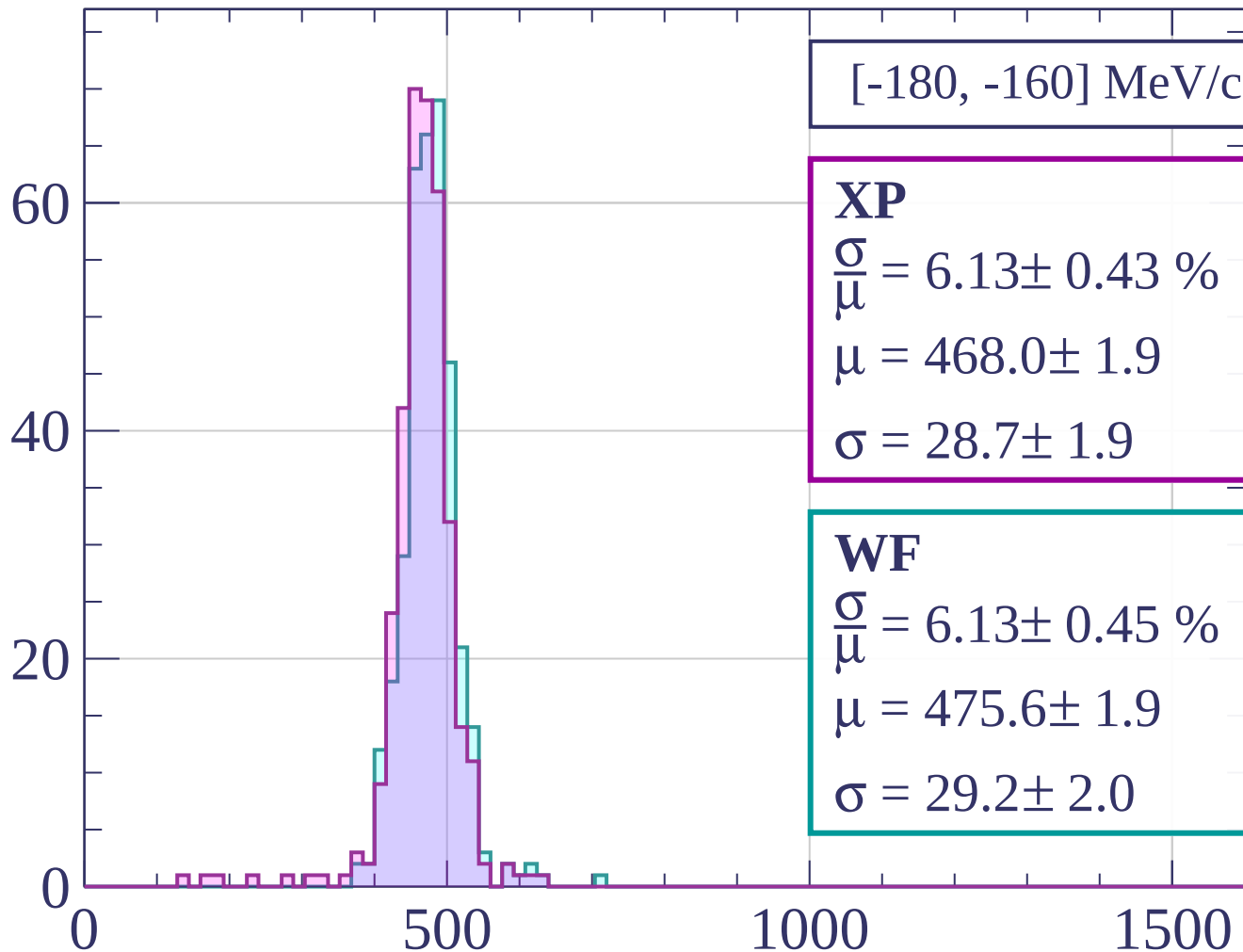
$$\frac{\sigma}{\mu} = 6.40 \pm 0.40 \%$$

$$\mu = 454.0 \pm 1.7$$

$$\sigma = 29.0 \pm 1.7$$

dE/dx [ADC counts/cm]

Count



[-180, -160] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.13 \pm 0.43 \%$$

$$\mu = 468.0 \pm 1.9$$

$$\sigma = 28.7 \pm 1.9$$

WF

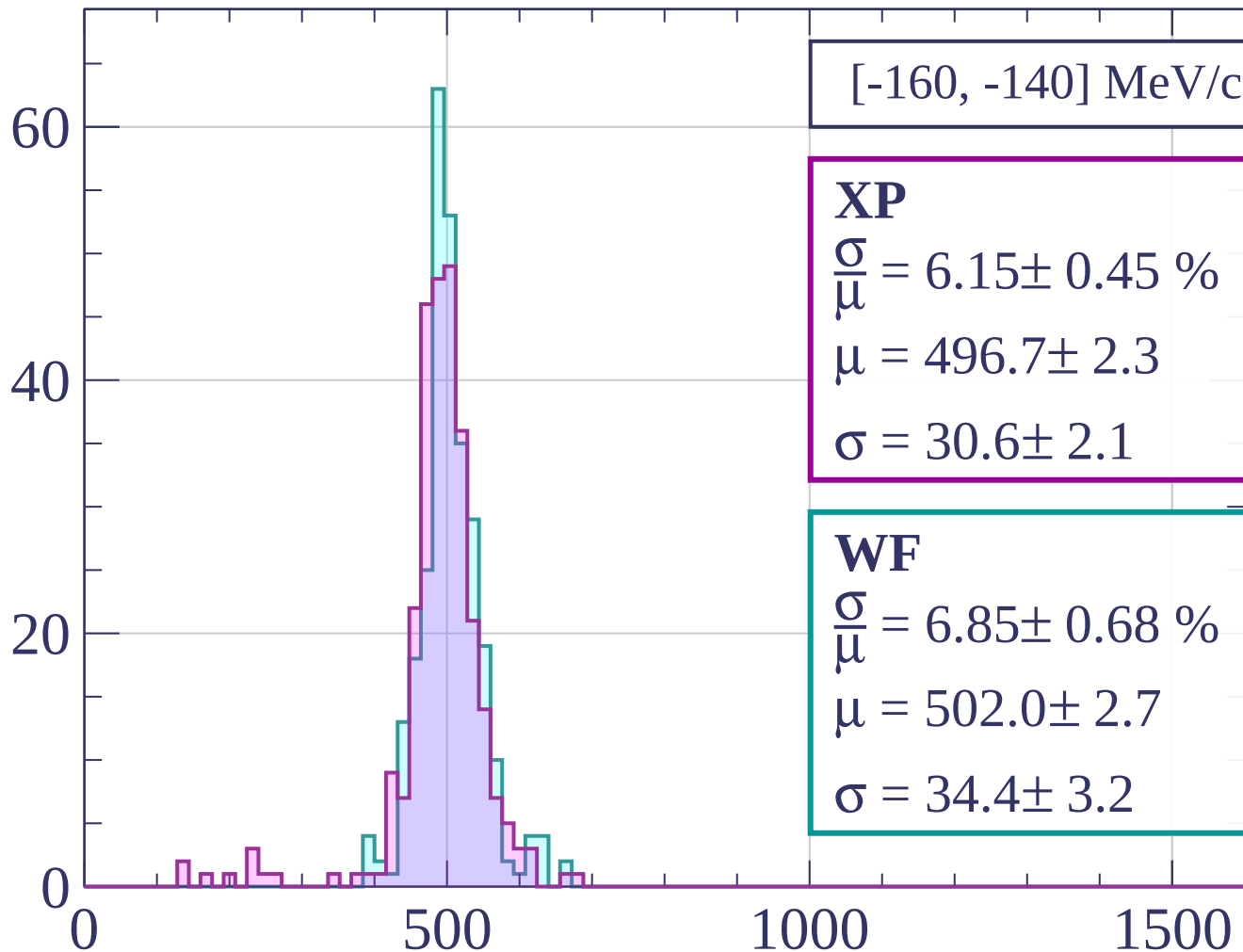
$$\frac{\sigma}{\mu} = 6.13 \pm 0.45 \%$$

$$\mu = 475.6 \pm 1.9$$

$$\sigma = 29.2 \pm 2.0$$

dE/dx [ADC counts/cm]

Count



[-160, -140] MeV/c

XP

$$\frac{\sigma}{\bar{\mu}} = 6.15 \pm 0.45 \%$$

$$\mu = 496.7 \pm 2.3$$

$$\sigma = 30.6 \pm 2.1$$

WF

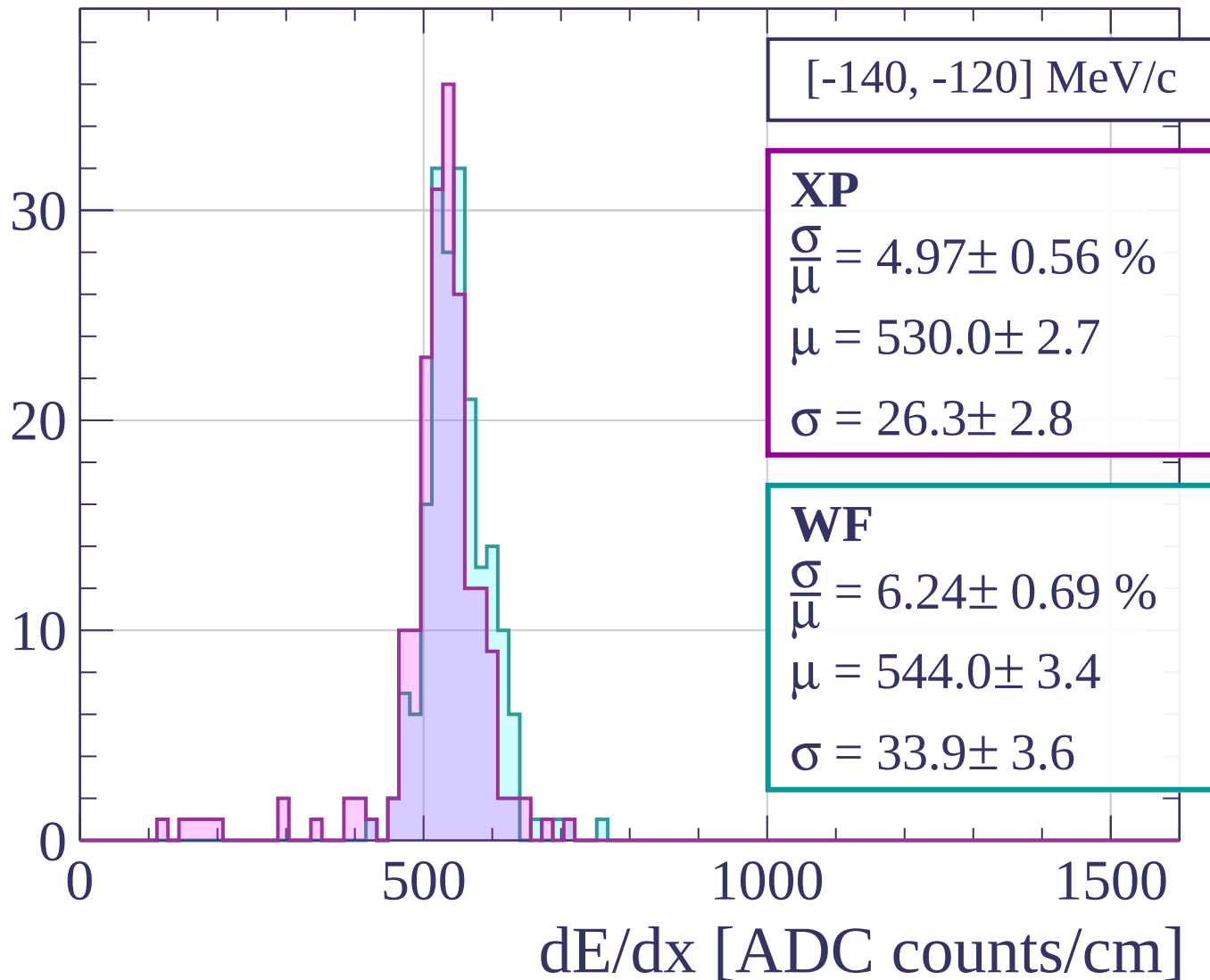
$$\frac{\sigma}{\bar{\mu}} = 6.85 \pm 0.68 \%$$

$$\mu = 502.0 \pm 2.7$$

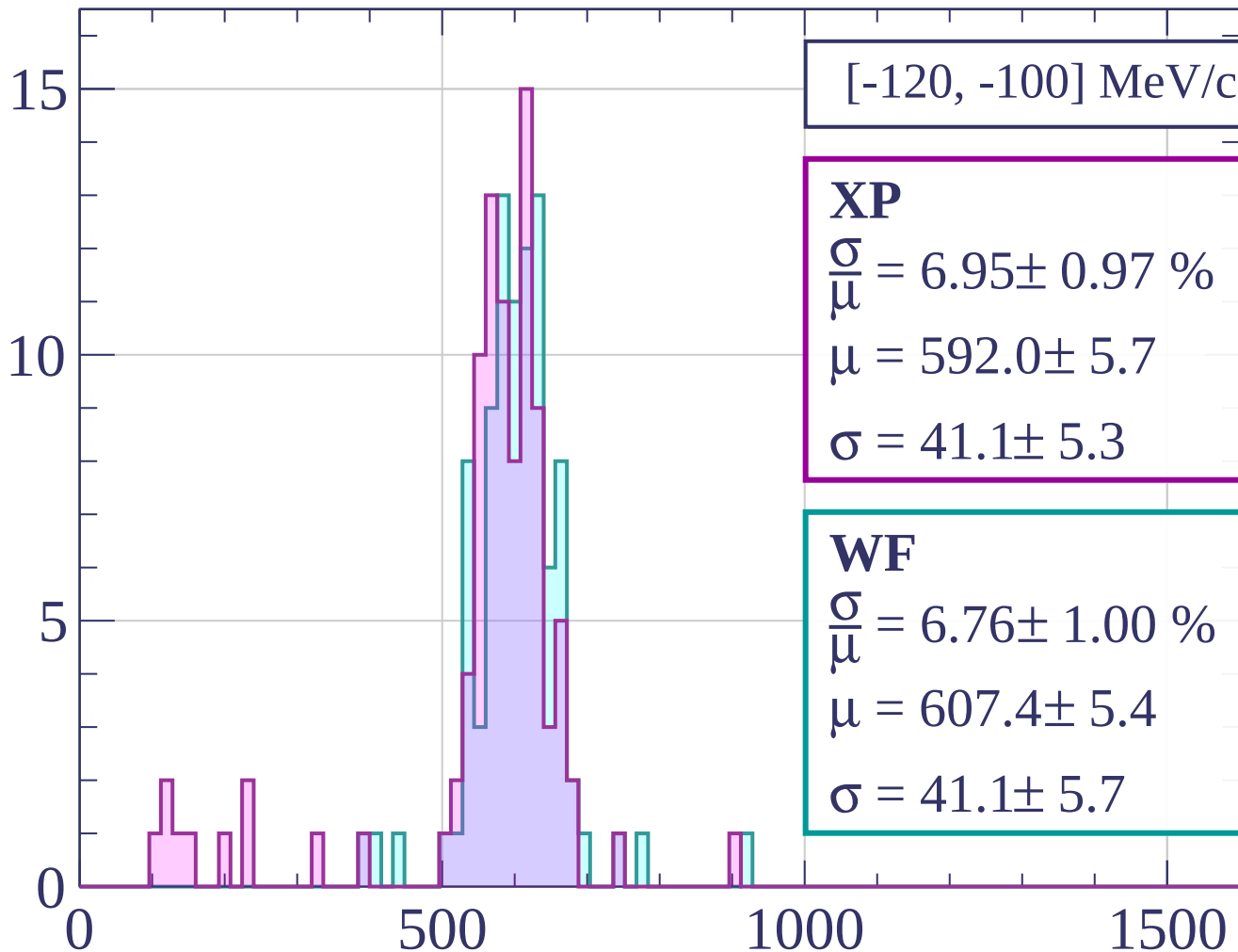
$$\sigma = 34.4 \pm 3.2$$

dE/dx [ADC counts/cm]

Count



Count



[-120, -100] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.95 \pm 0.97 \%$$

$$\mu = 592.0 \pm 5.7$$

$$\sigma = 41.1 \pm 5.3$$

WF

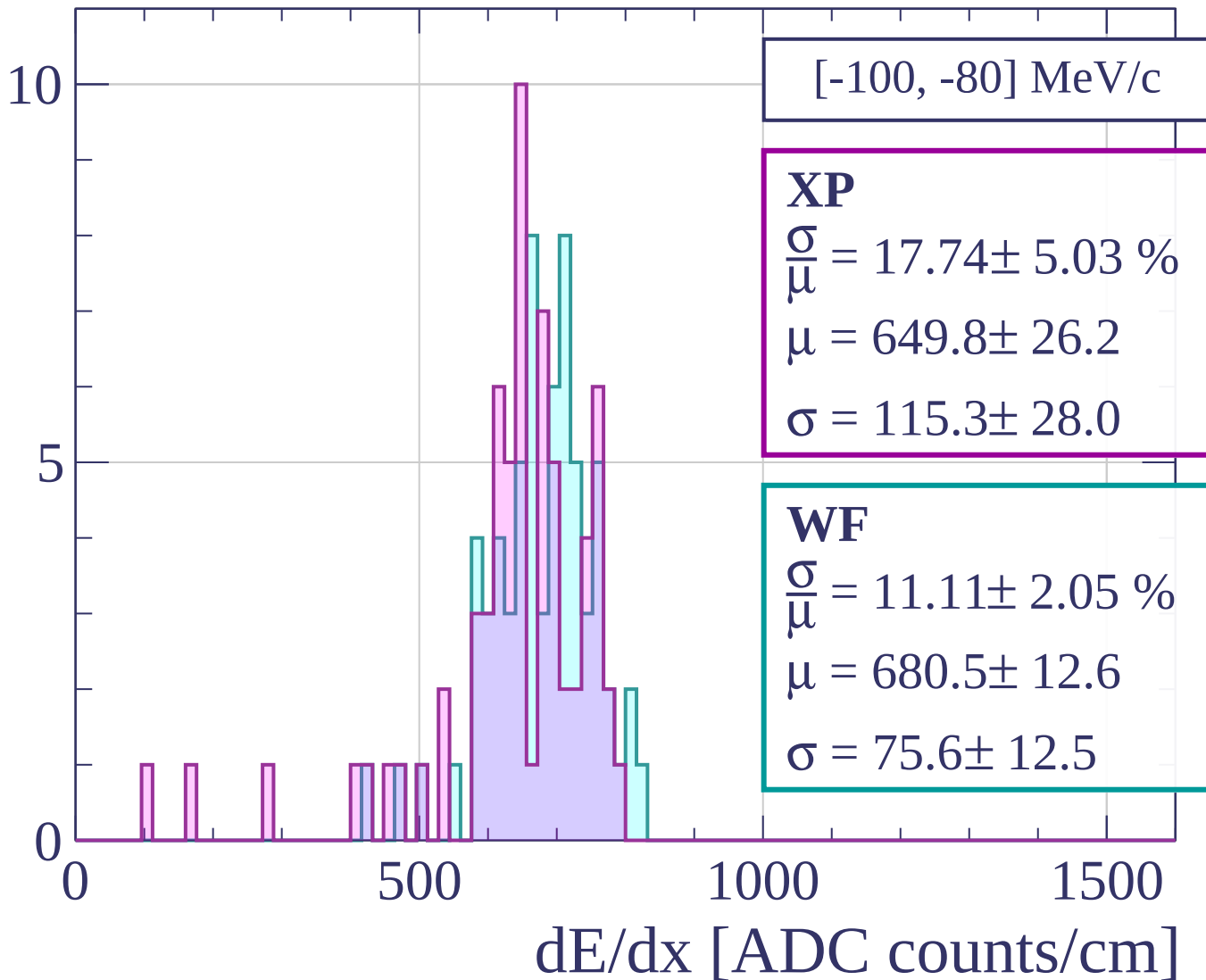
$$\frac{\sigma}{\mu} = 6.76 \pm 1.00 \%$$

$$\mu = 607.4 \pm 5.4$$

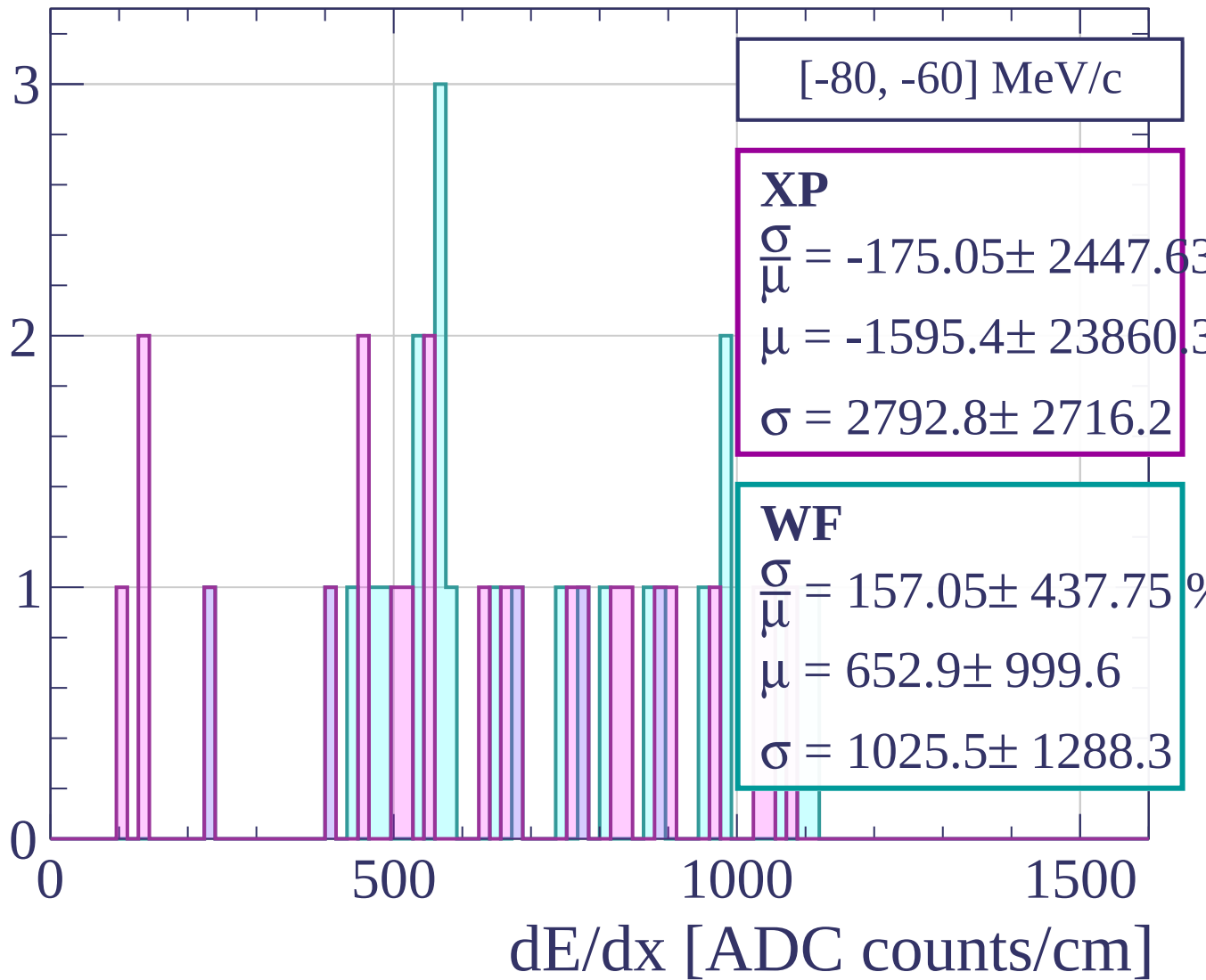
$$\sigma = 41.1 \pm 5.7$$

dE/dx [ADC counts/cm]

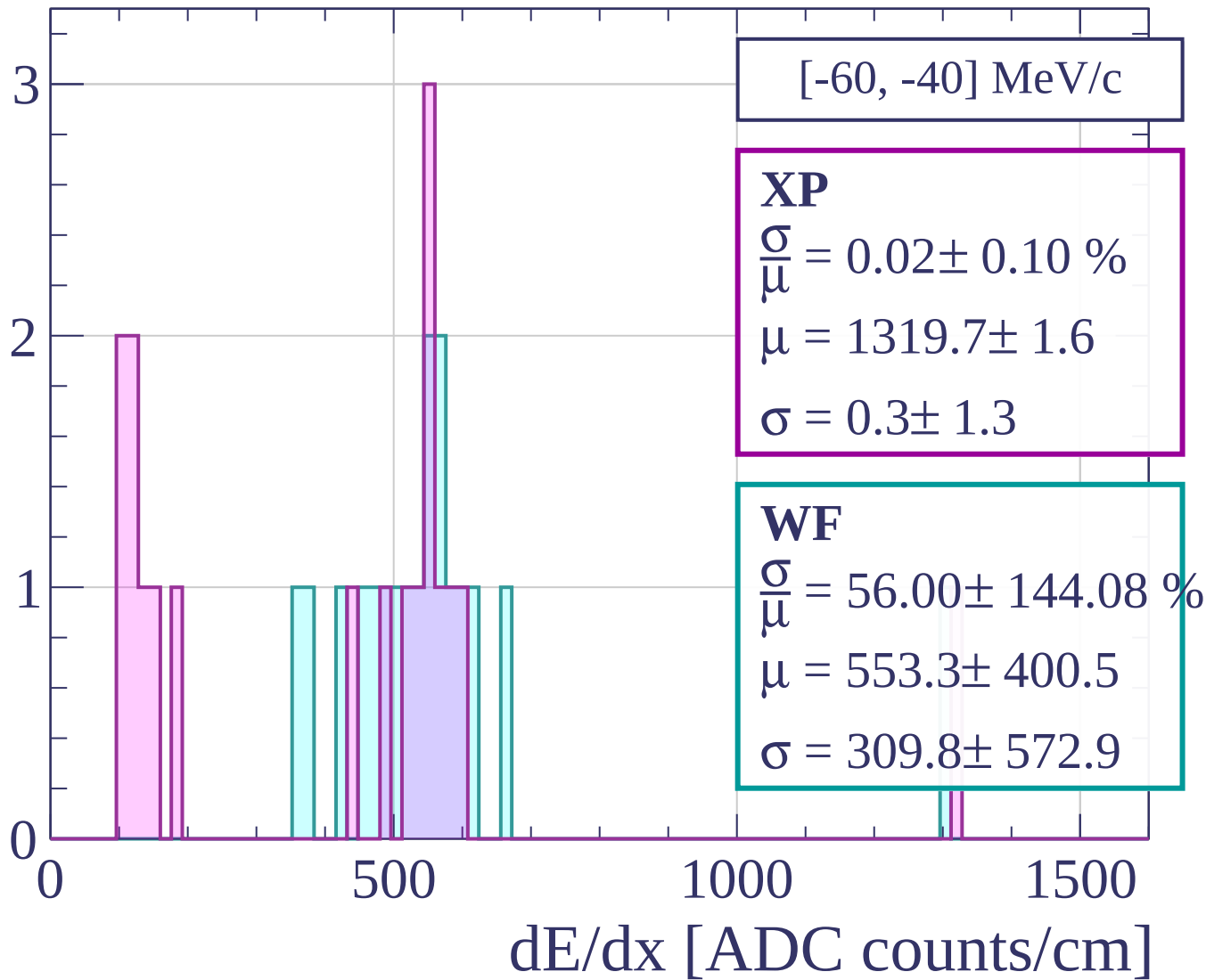
Count



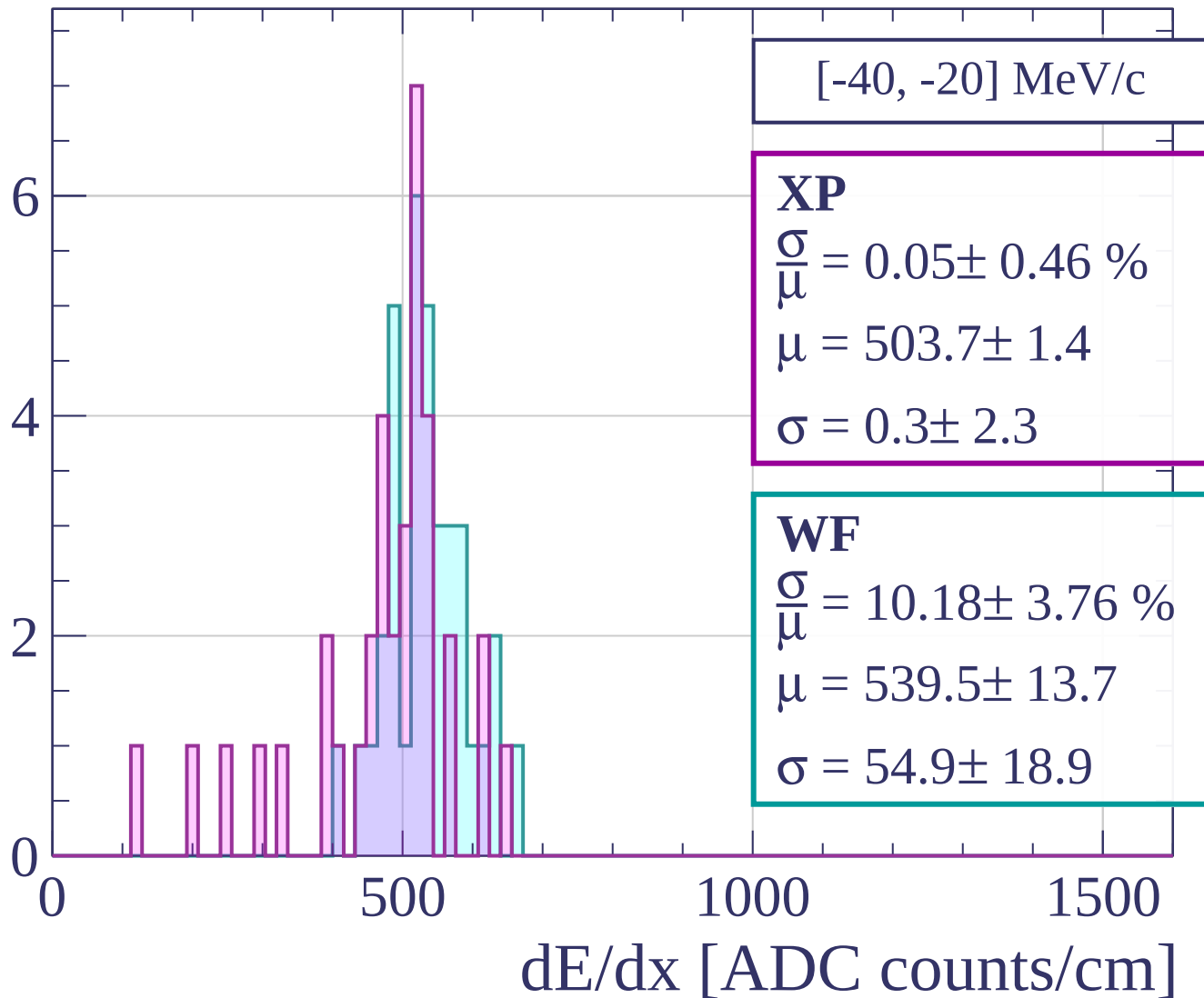
Count



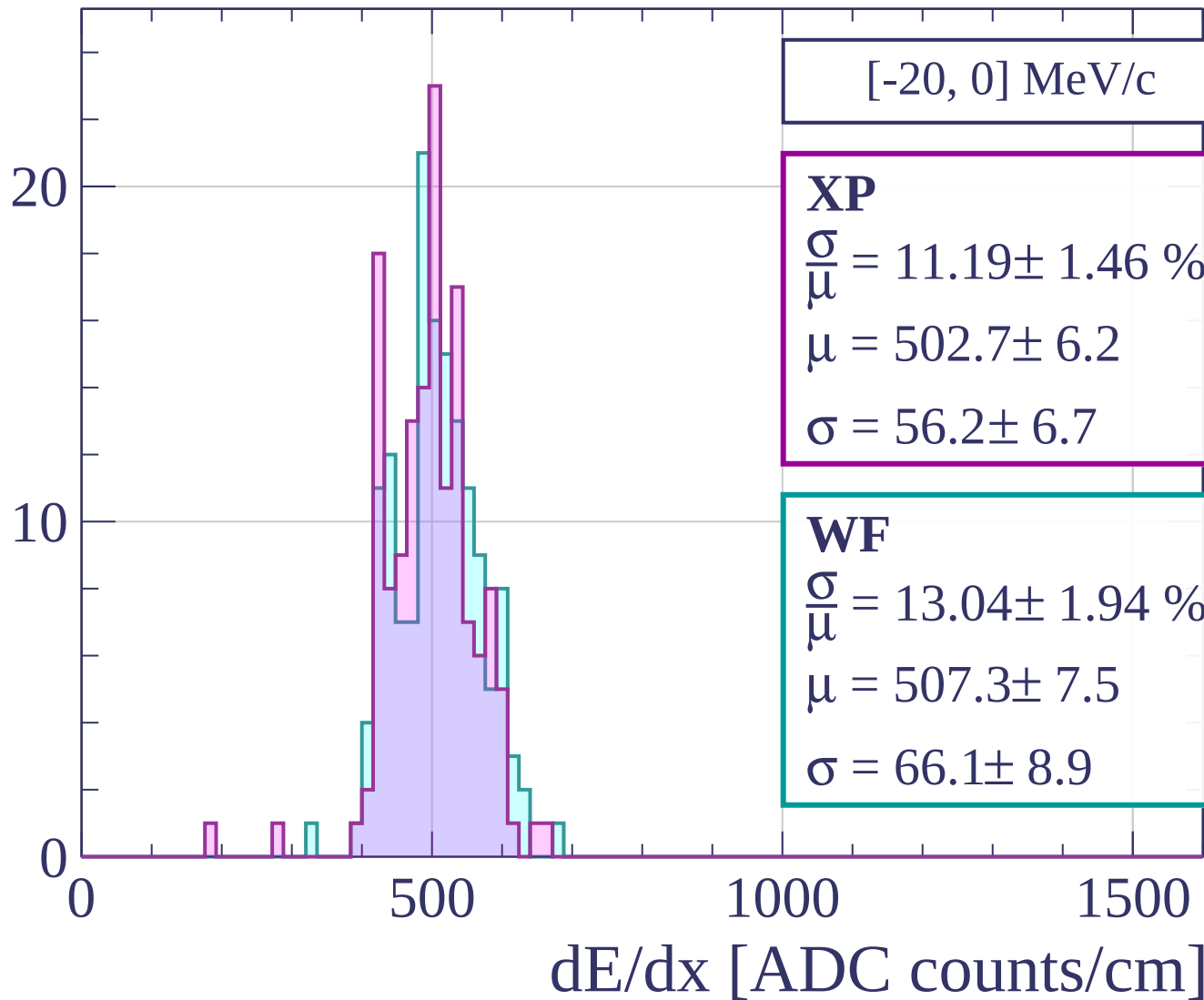
Count



Count



Count



Count

[0, 20] MeV/c

XP

$$\frac{\sigma}{\mu} = 12.51 \pm 0.55 \%$$

$$\mu = 471.0 \pm 2.7$$

$$\sigma = 58.9 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 12.14 \pm 0.49 \%$$

$$\mu = 481.4 \pm 2.7$$

$$\sigma = 58.5 \pm 2.1$$

80

60

40

20

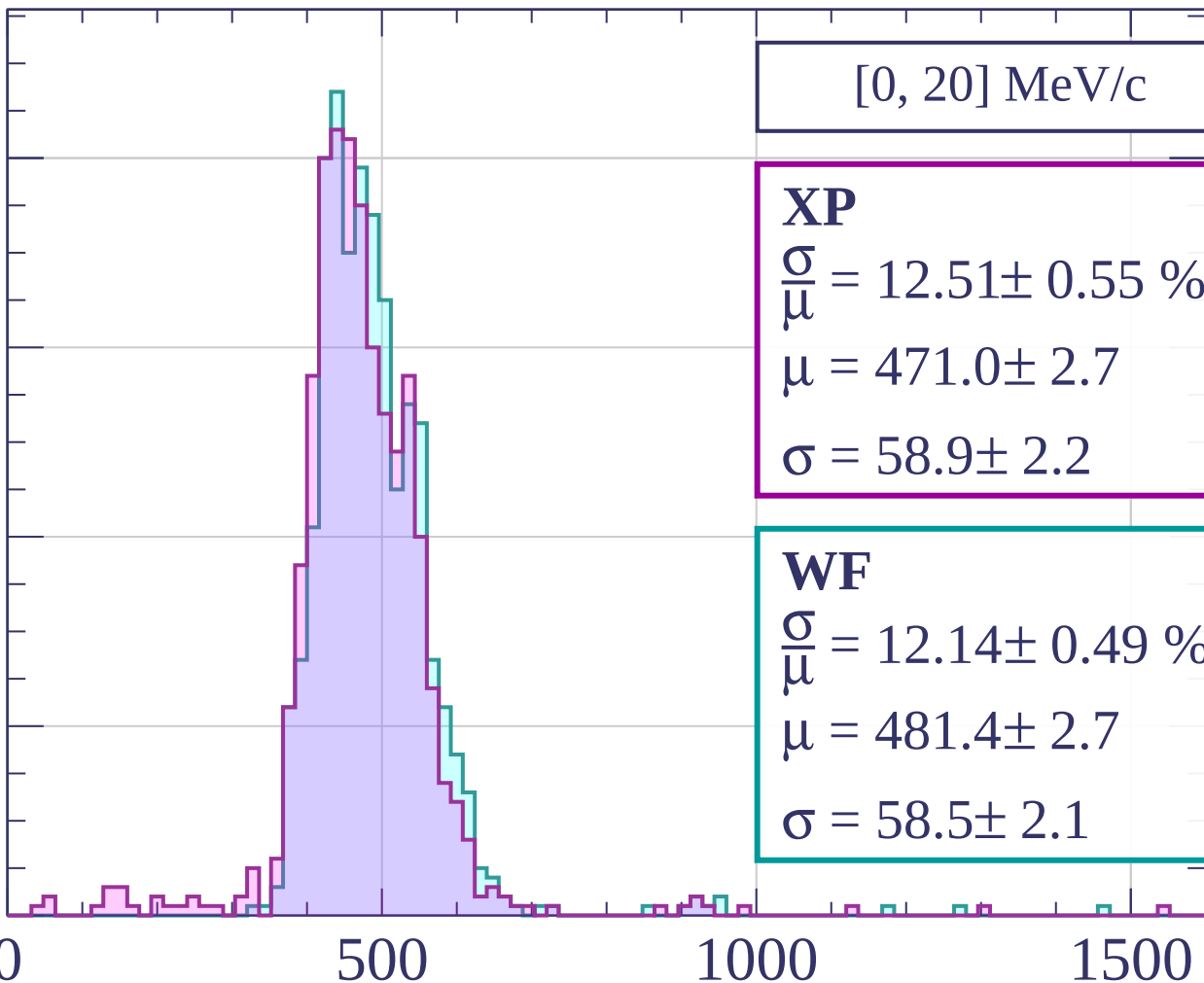
0

500

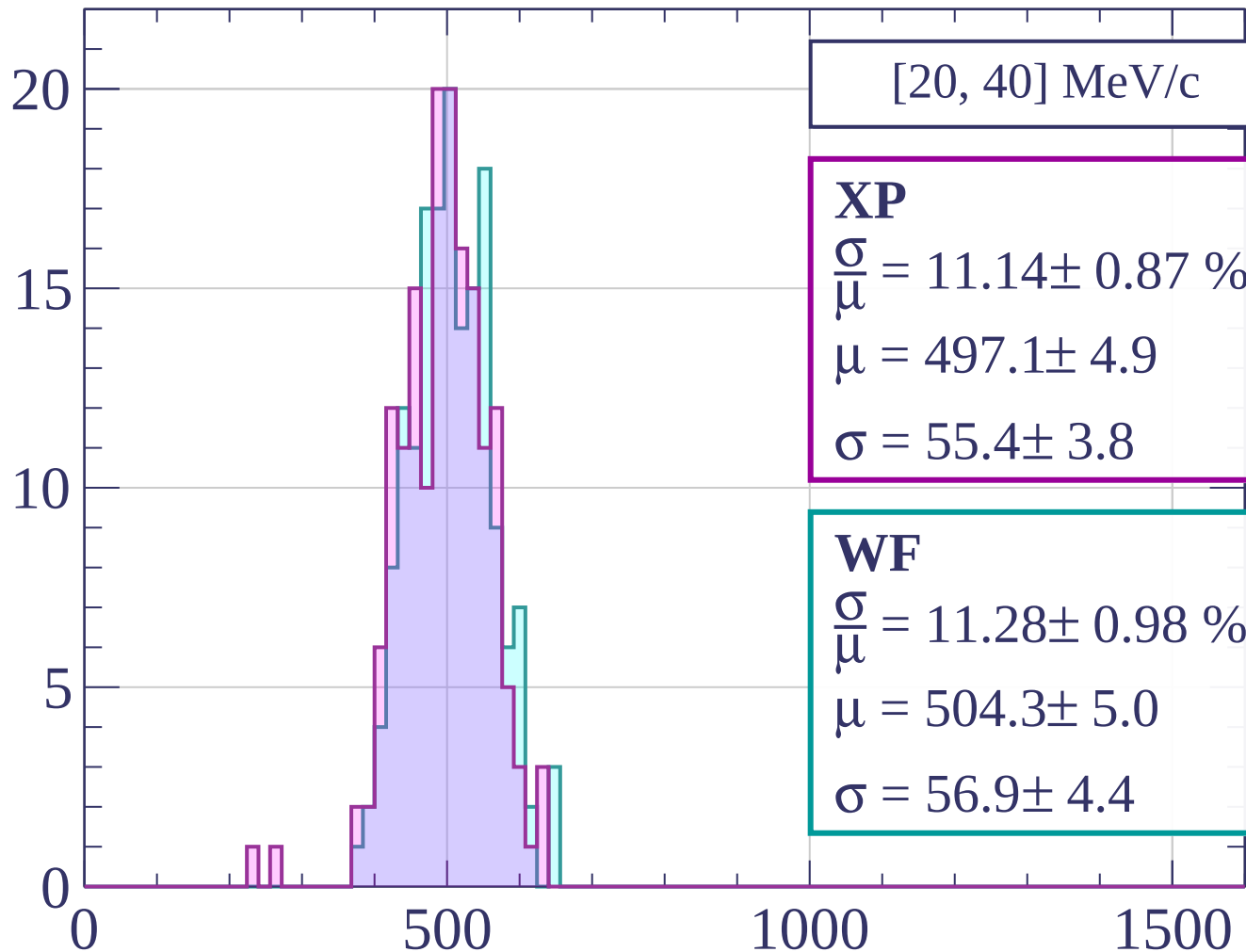
1000

1500

dE/dx [ADC counts/cm]

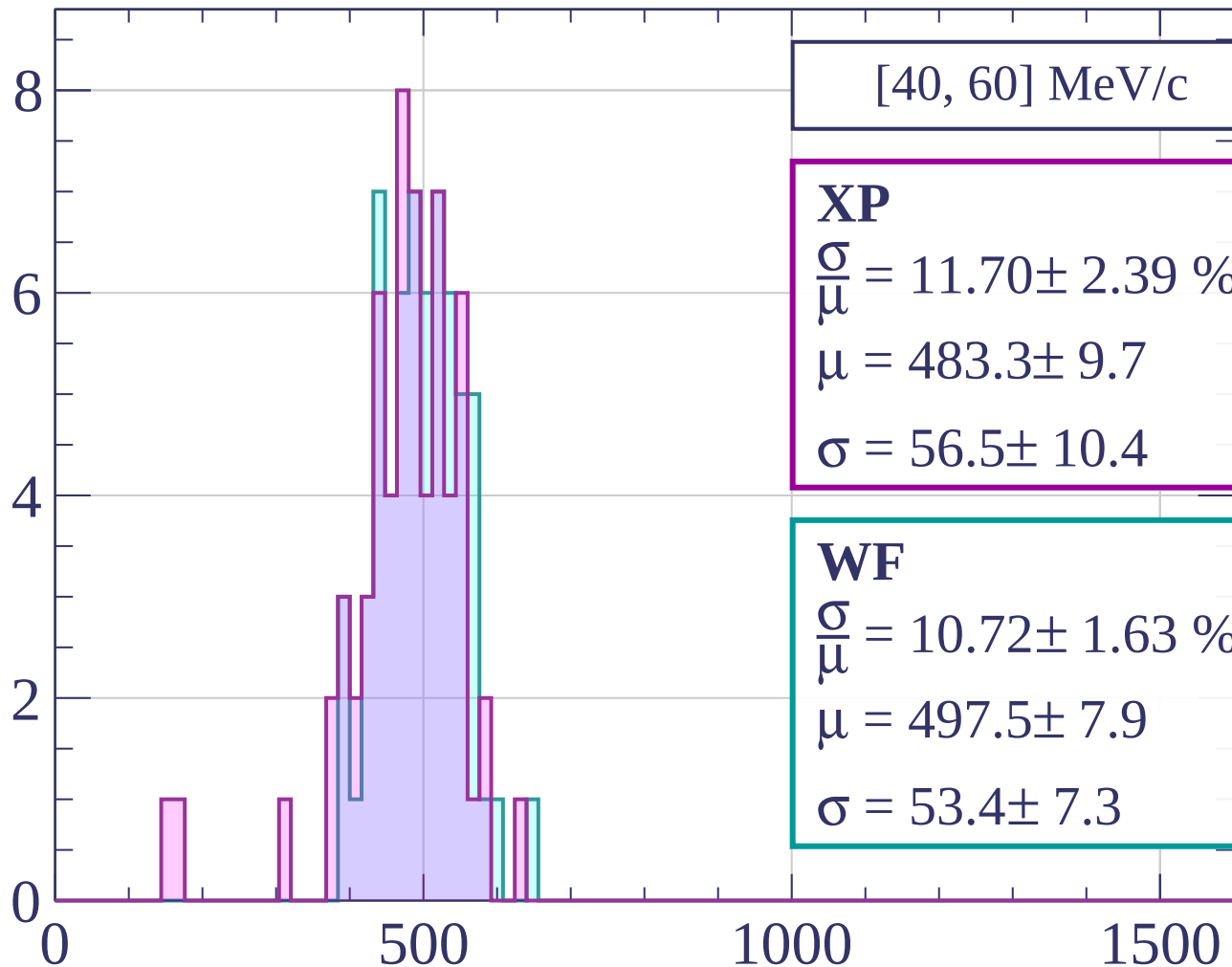


Count



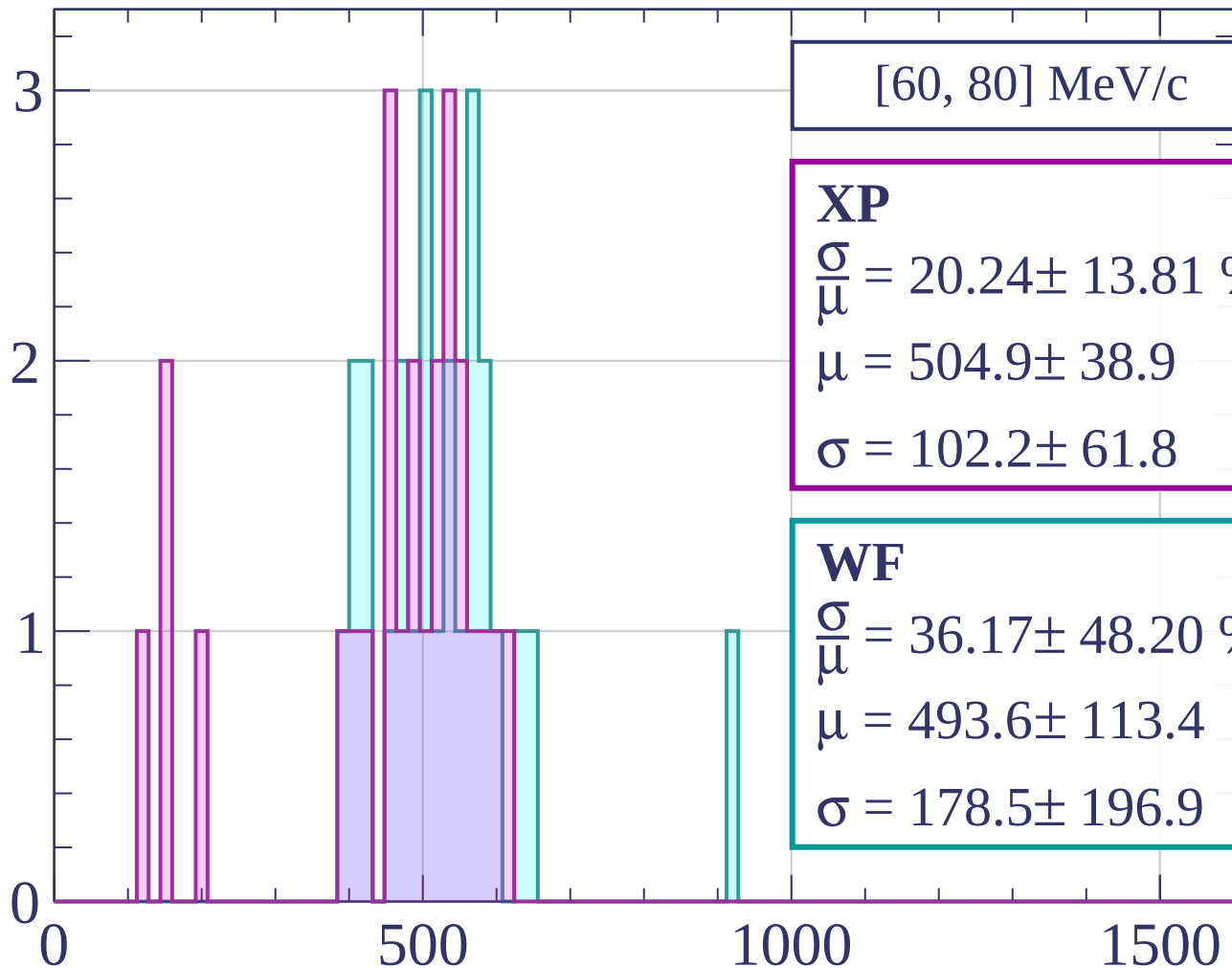
dE/dx [ADC counts/cm]

Count



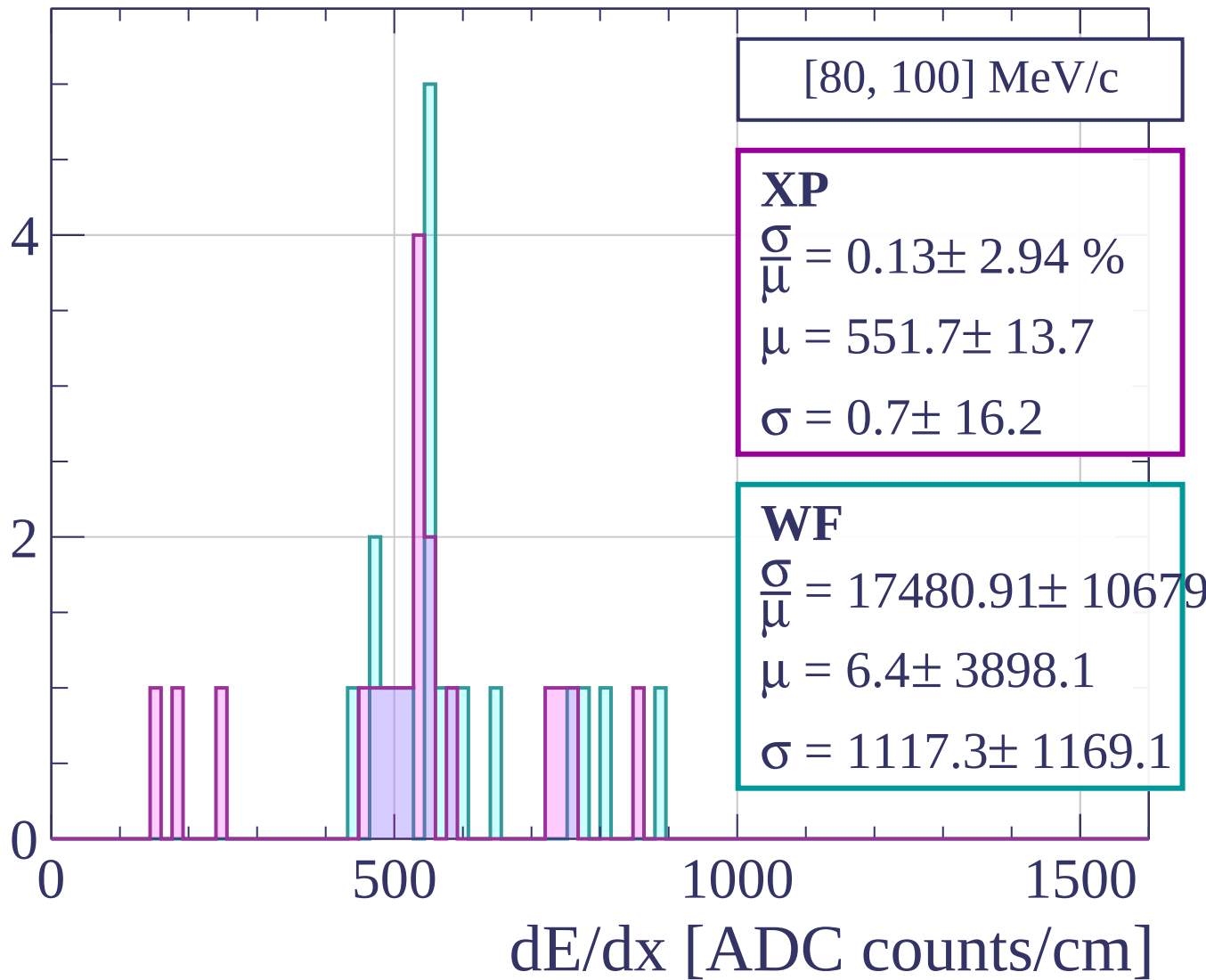
dE/dx [ADC counts/cm]

Count

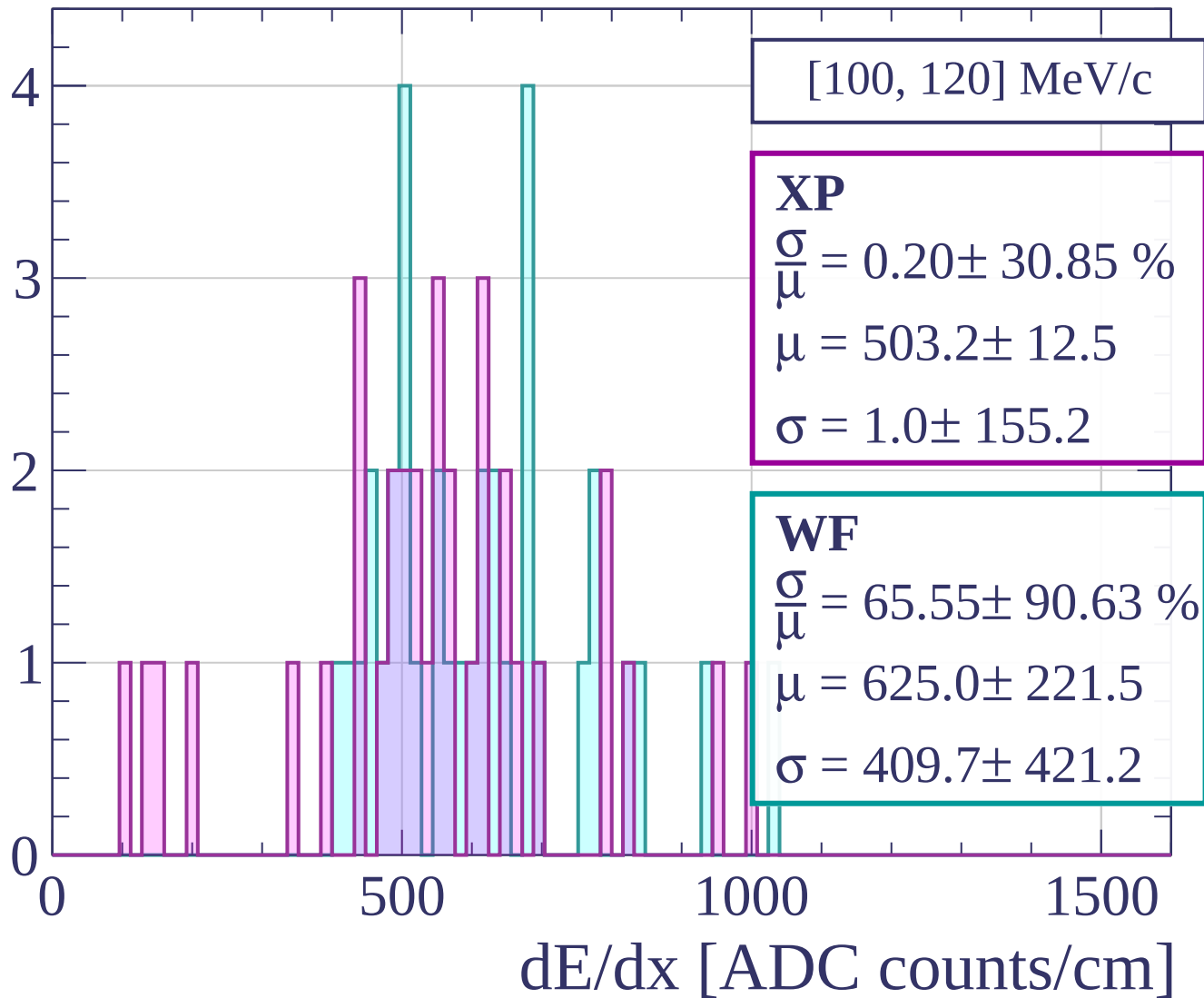


dE/dx [ADC counts/cm]

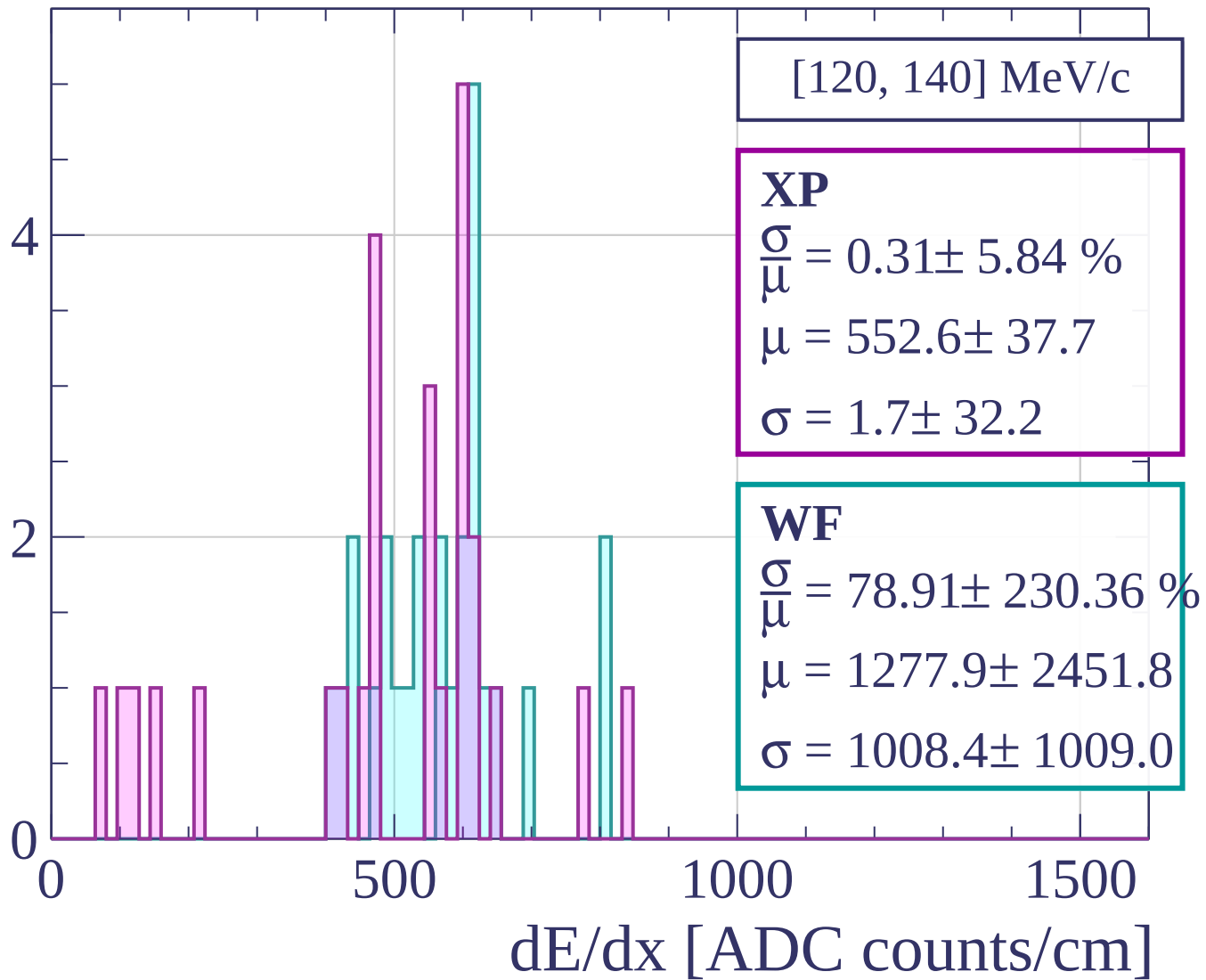
Count



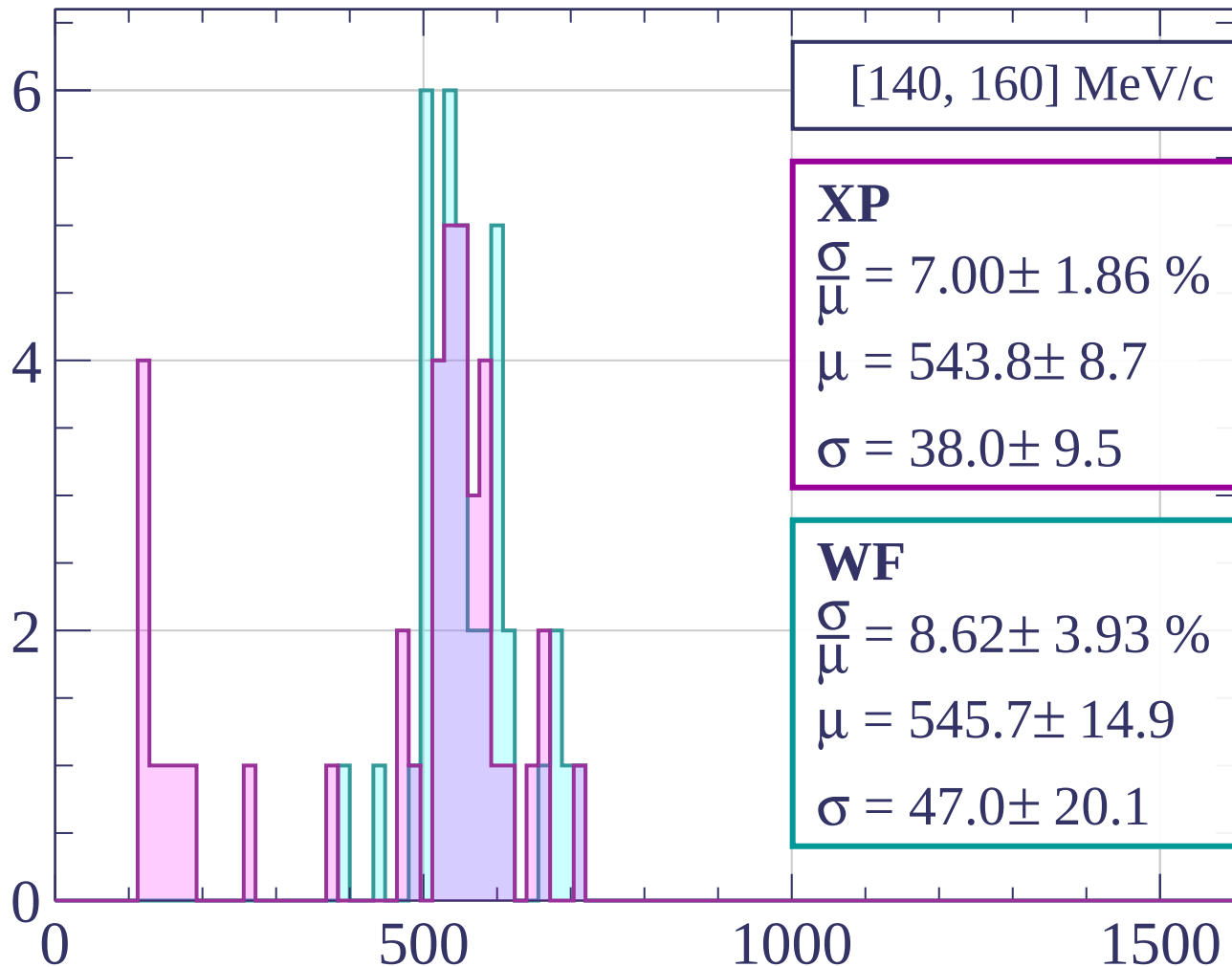
Count



Count



Count



[140, 160] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.00 \pm 1.86 \%$$

$$\mu = 543.8 \pm 8.7$$

$$\sigma = 38.0 \pm 9.5$$

WF

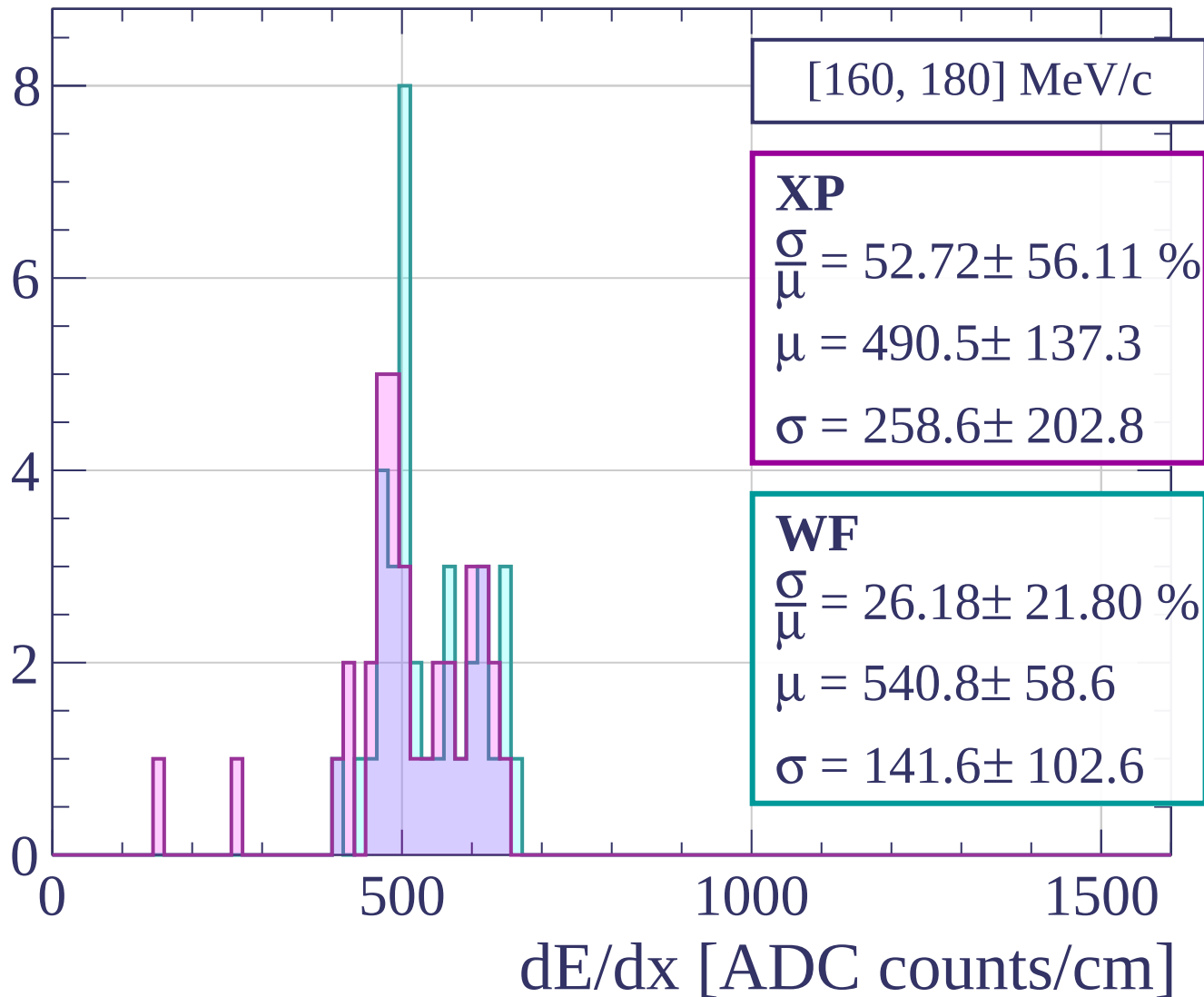
$$\frac{\sigma}{\mu} = 8.62 \pm 3.93 \%$$

$$\mu = 545.7 \pm 14.9$$

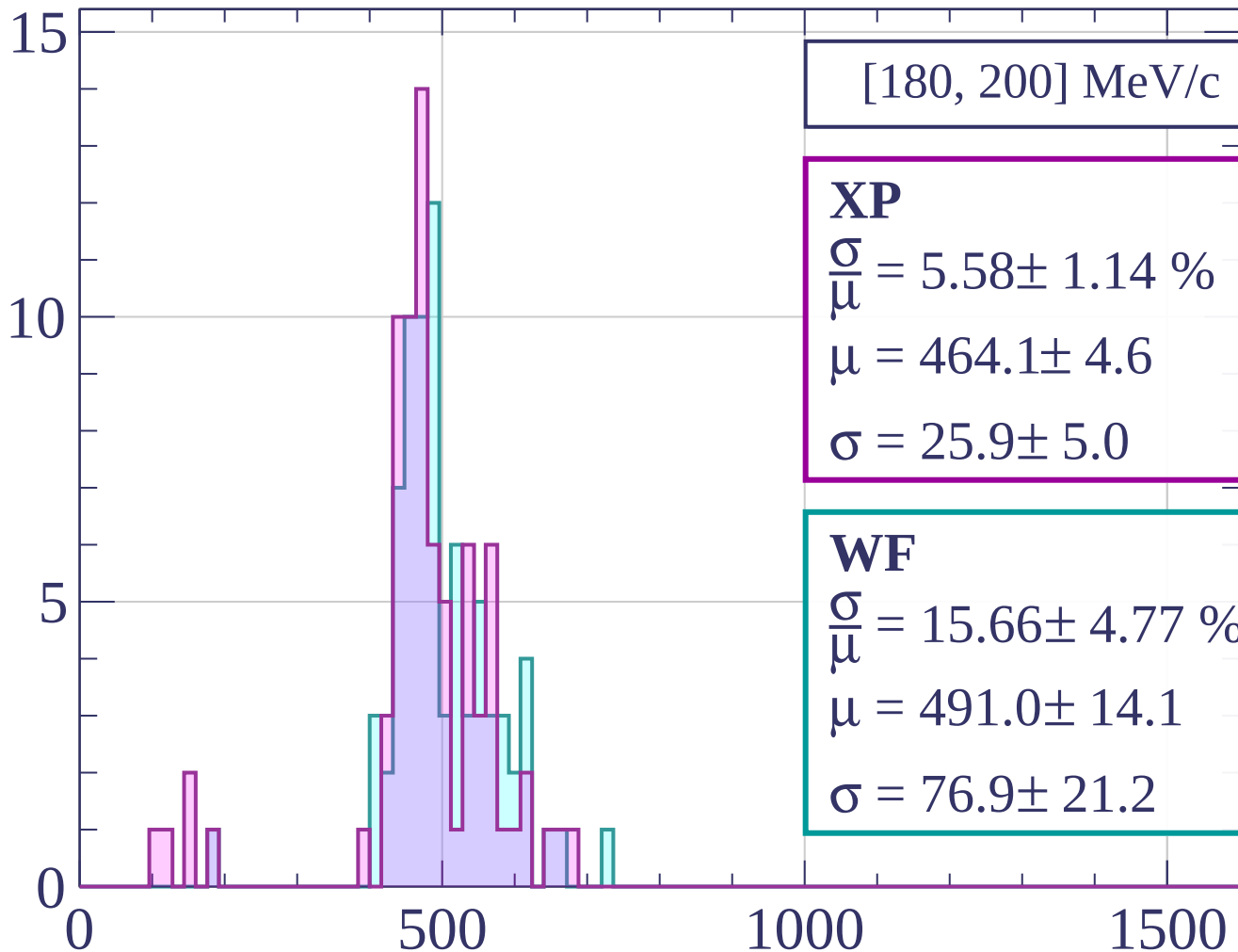
$$\sigma = 47.0 \pm 20.1$$

dE/dx [ADC counts/cm]

Count



Count



[180, 200] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.58 \pm 1.14 \%$$

$$\mu = 464.1 \pm 4.6$$

$$\sigma = 25.9 \pm 5.0$$

WF

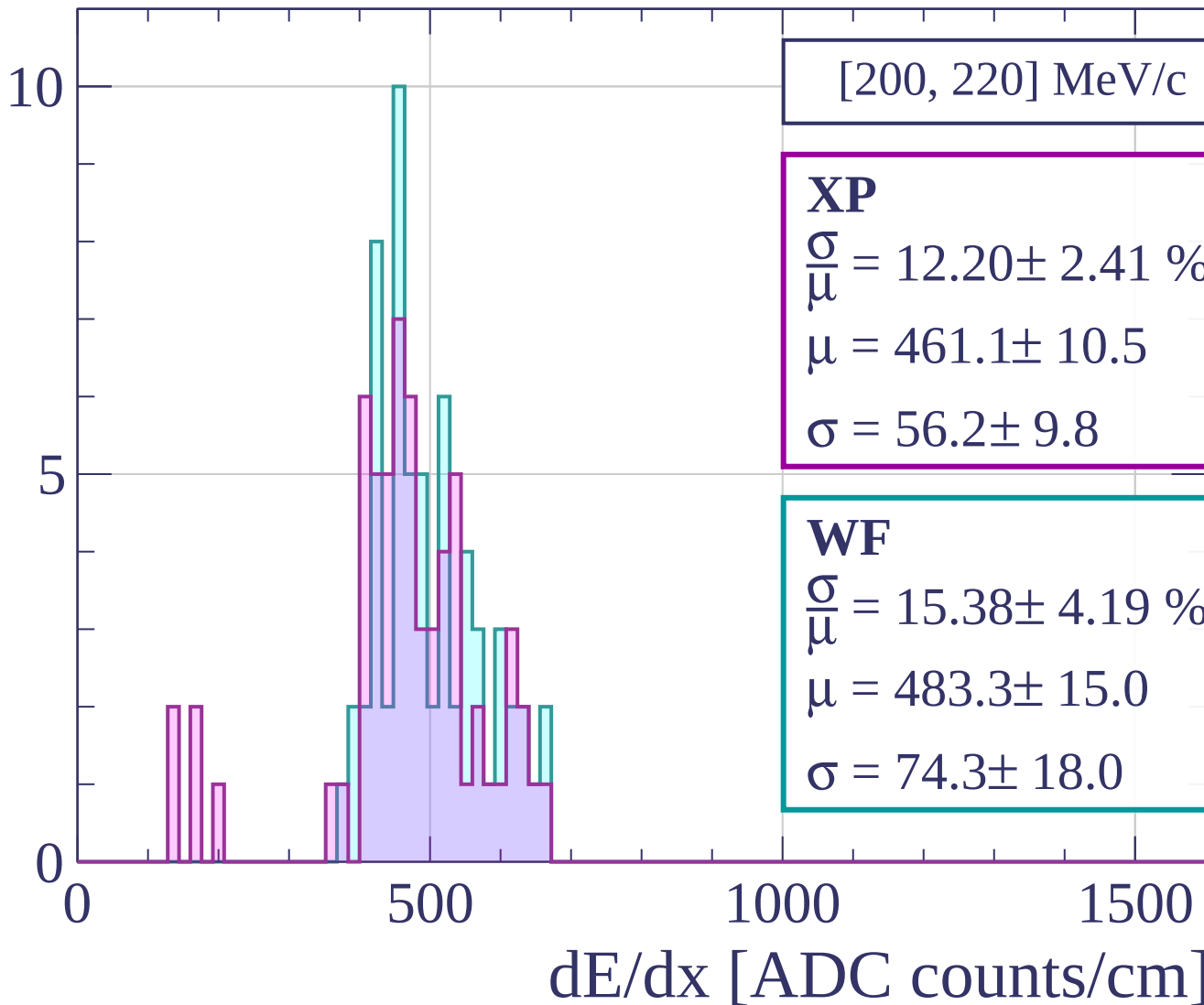
$$\frac{\sigma}{\mu} = 15.66 \pm 4.77 \%$$

$$\mu = 491.0 \pm 14.1$$

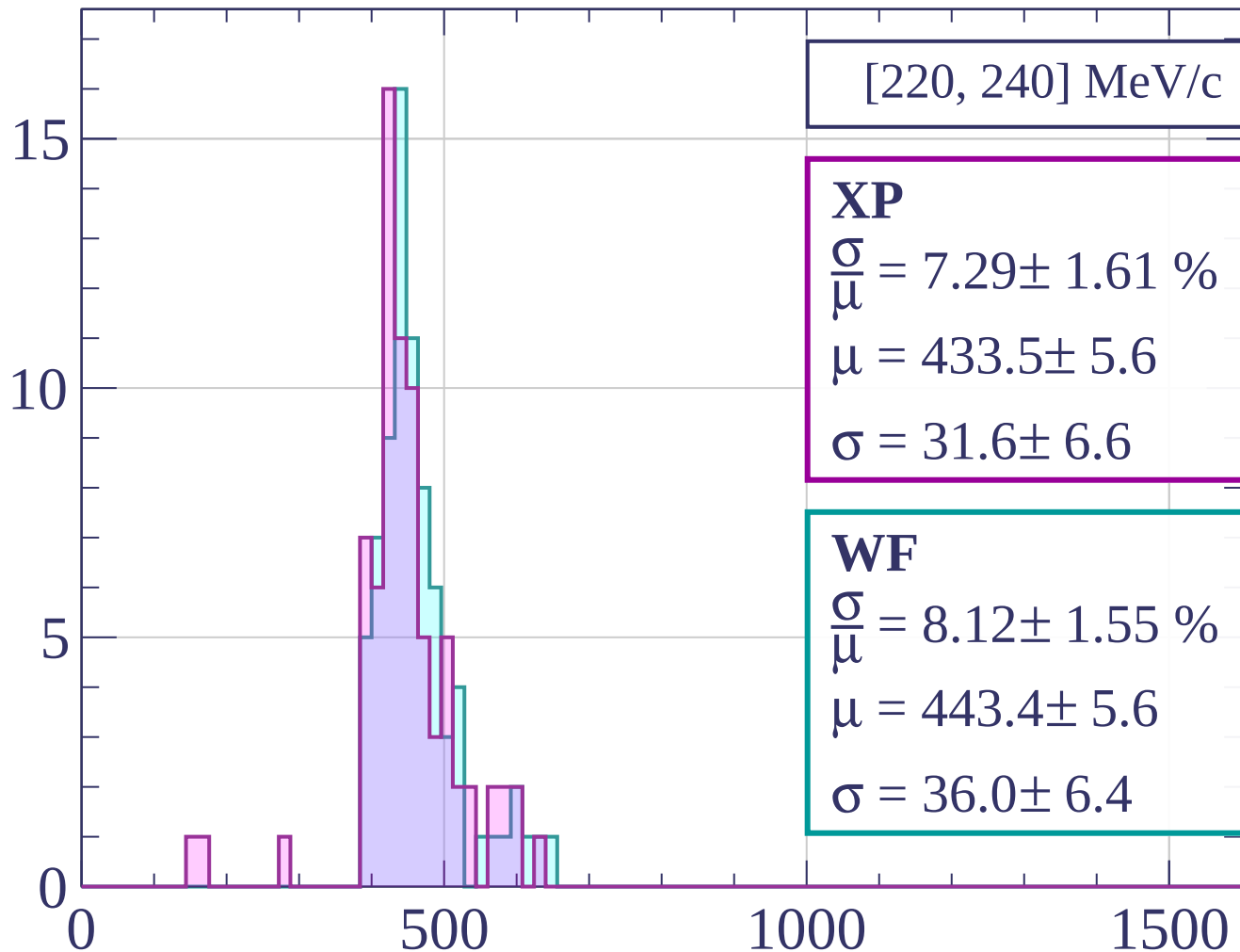
$$\sigma = 76.9 \pm 21.2$$

dE/dx [ADC counts/cm]

Count

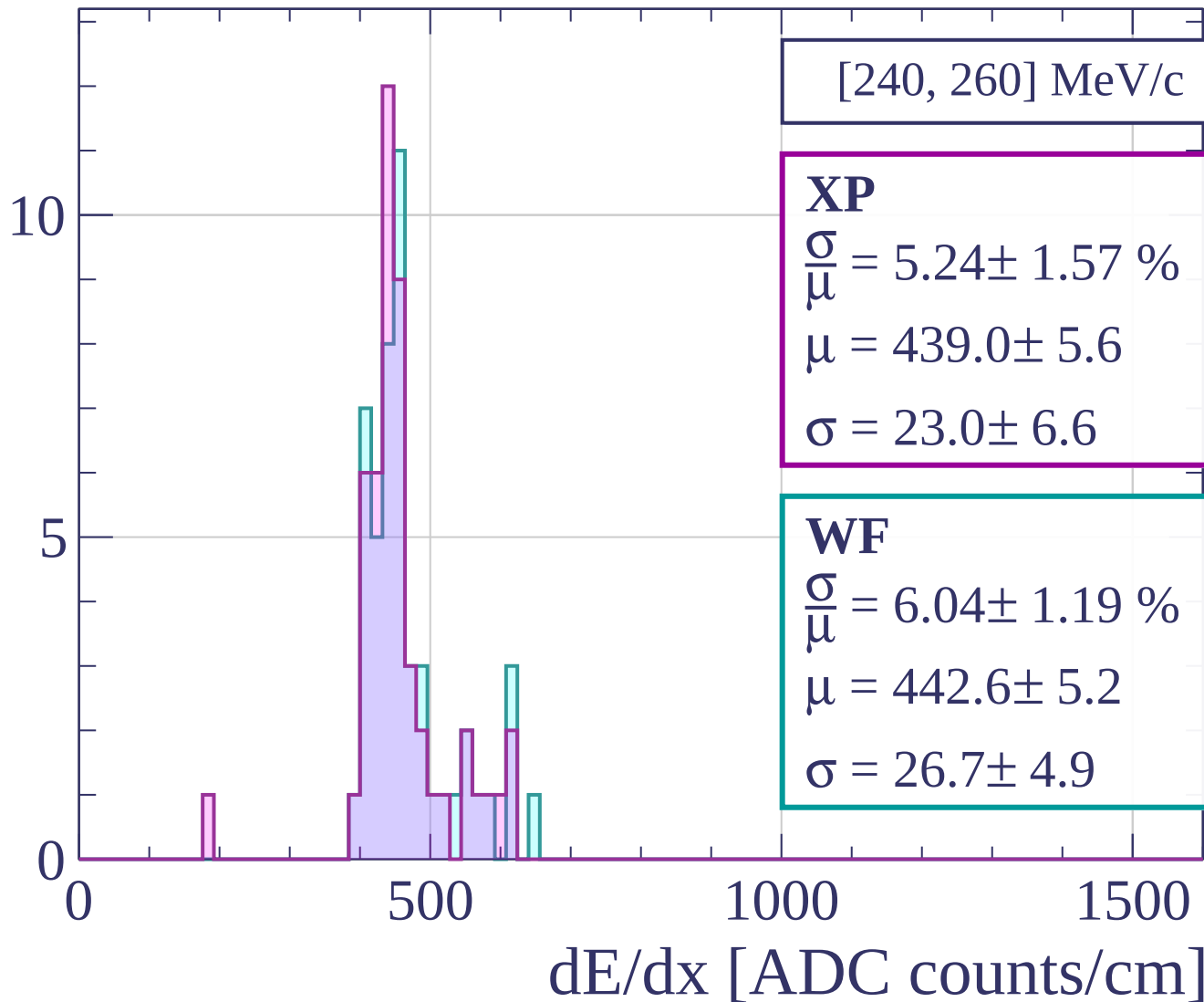


Count

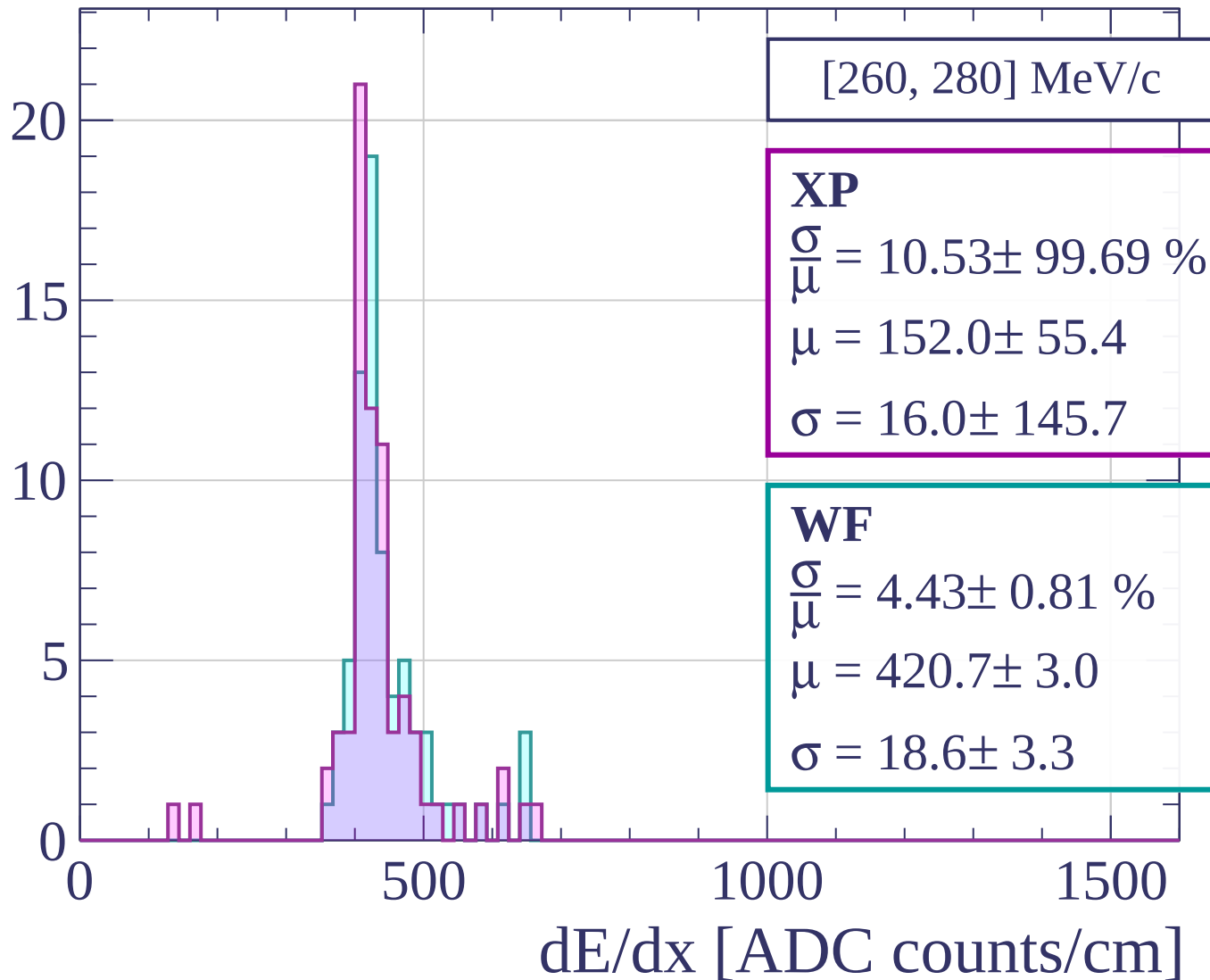


dE/dx [ADC counts/cm]

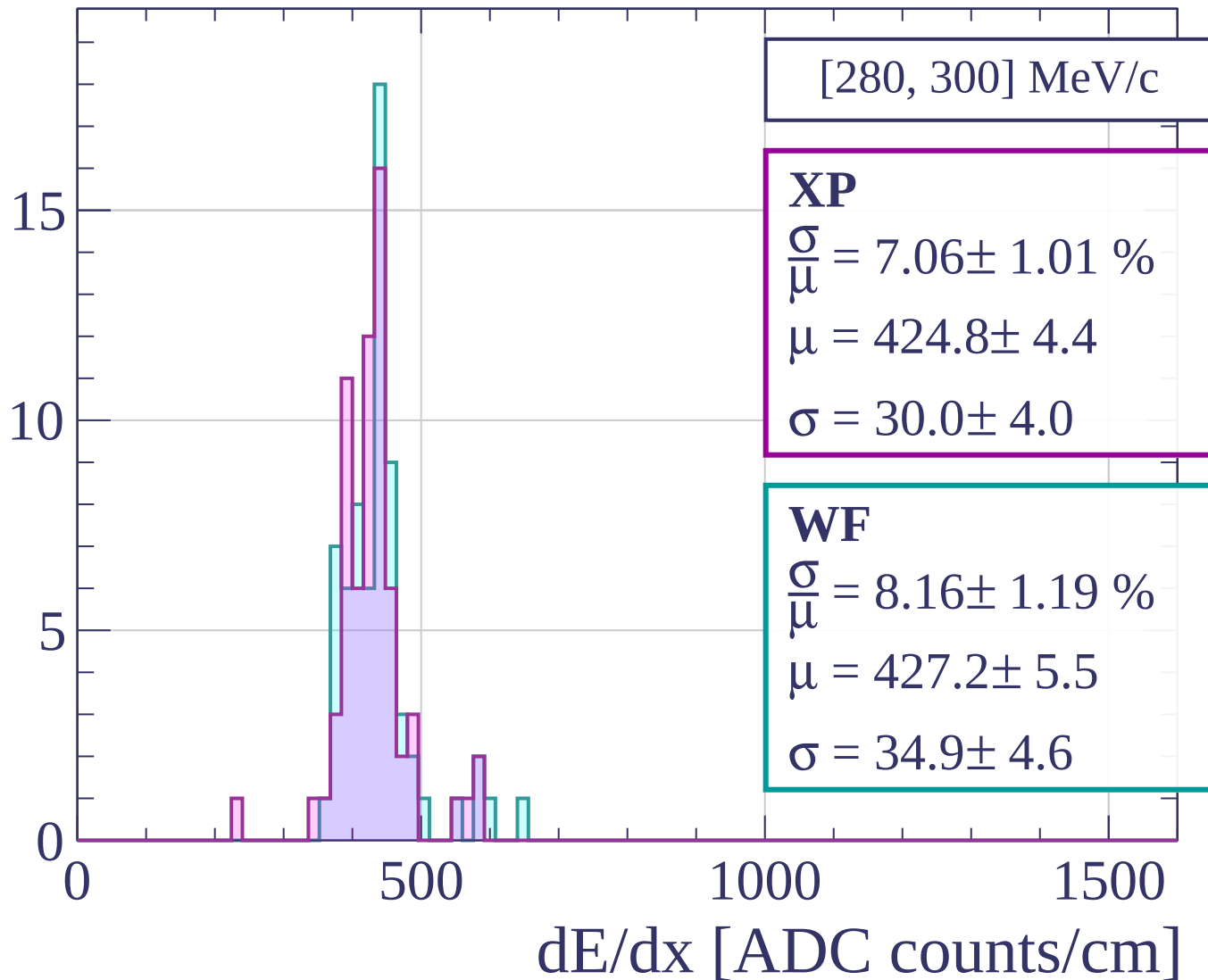
Count



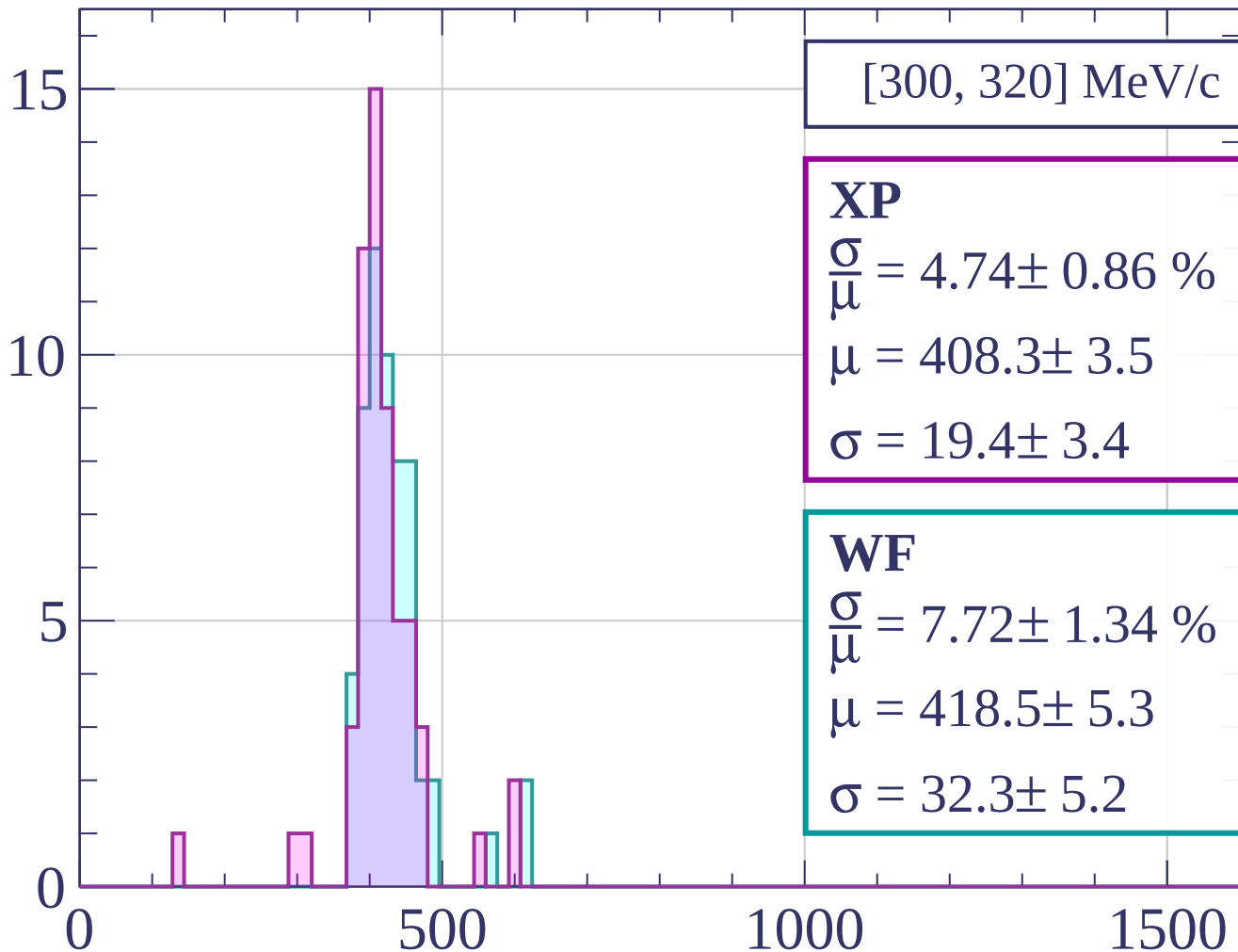
Count



Count

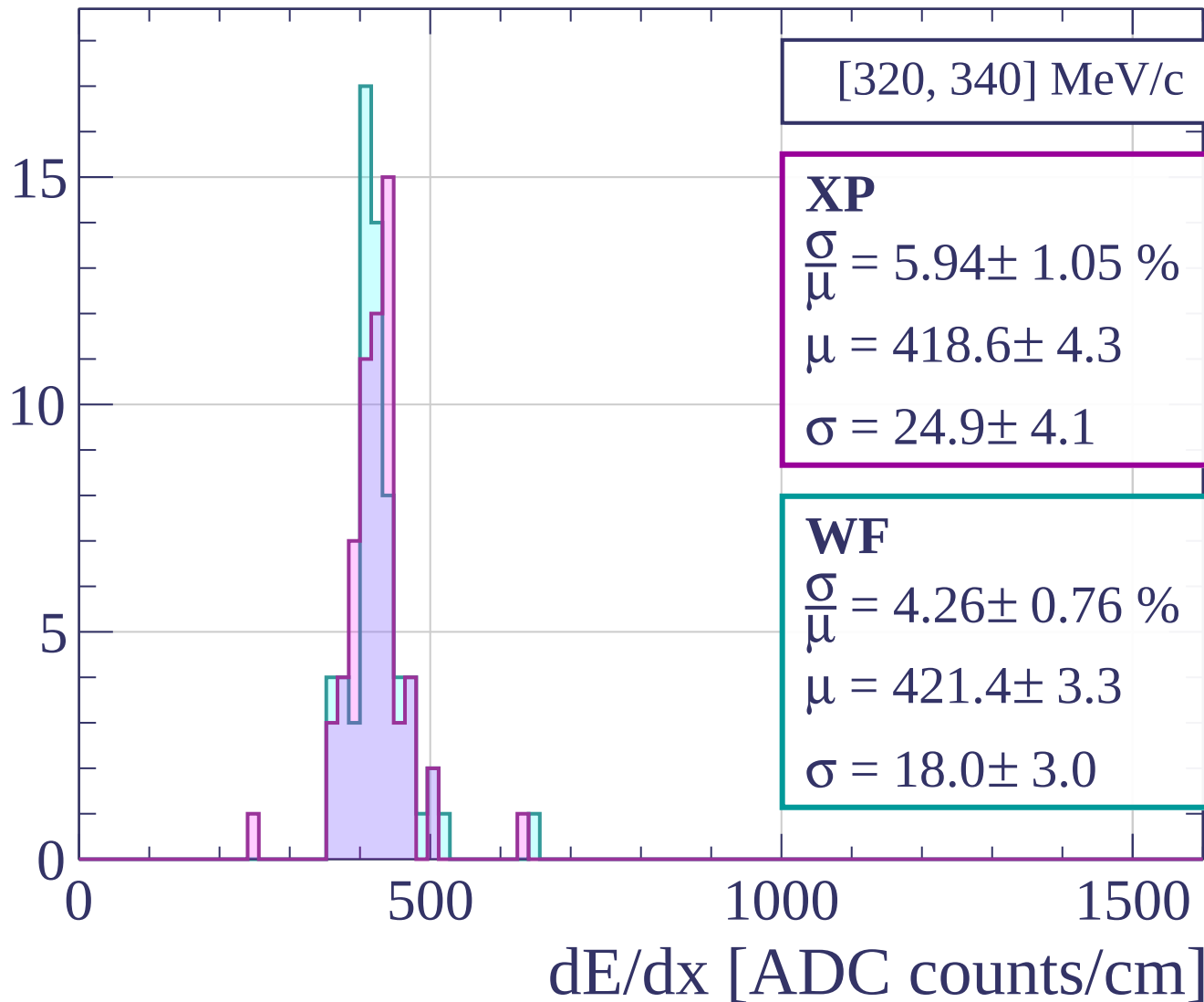


Count

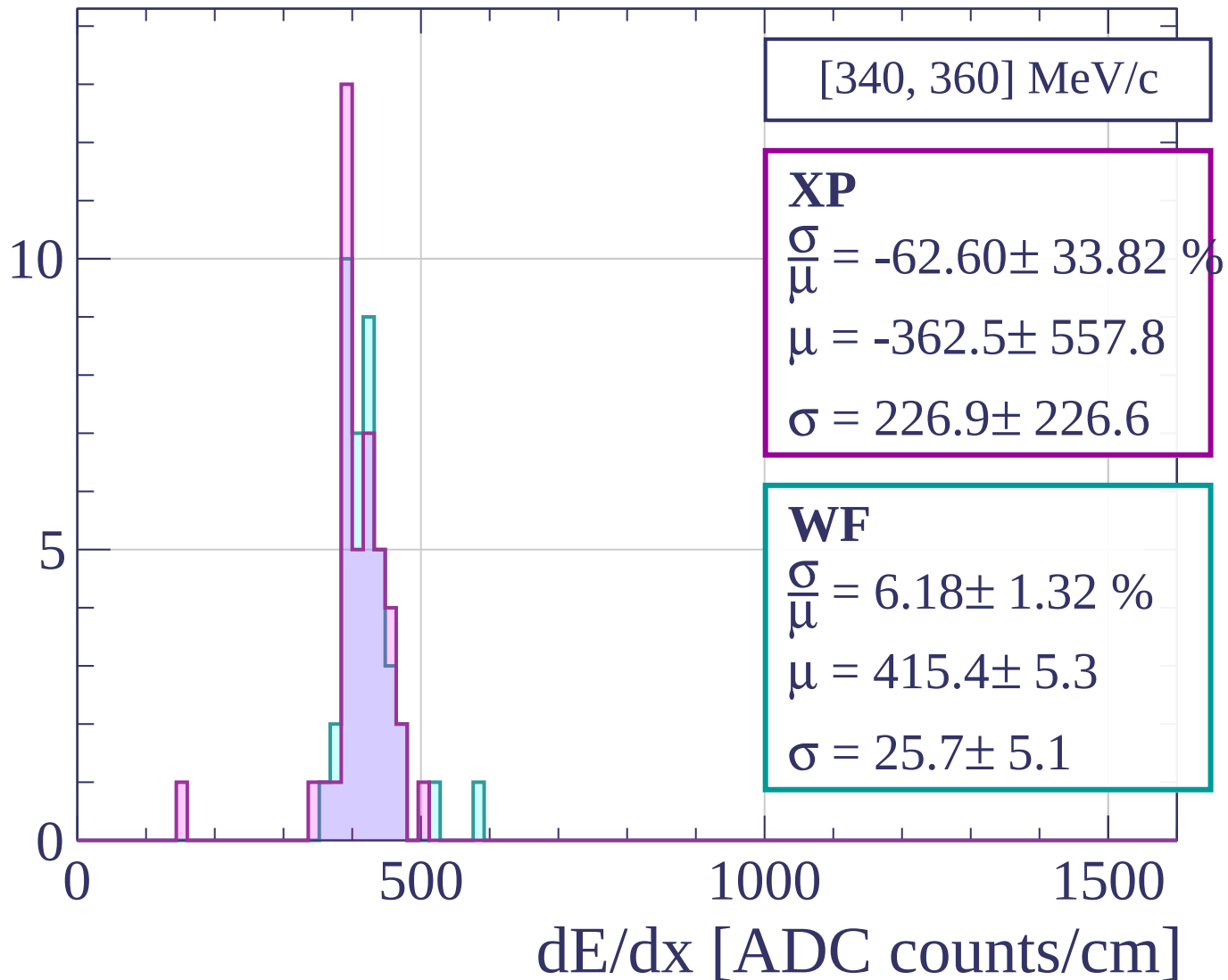


dE/dx [ADC counts/cm]

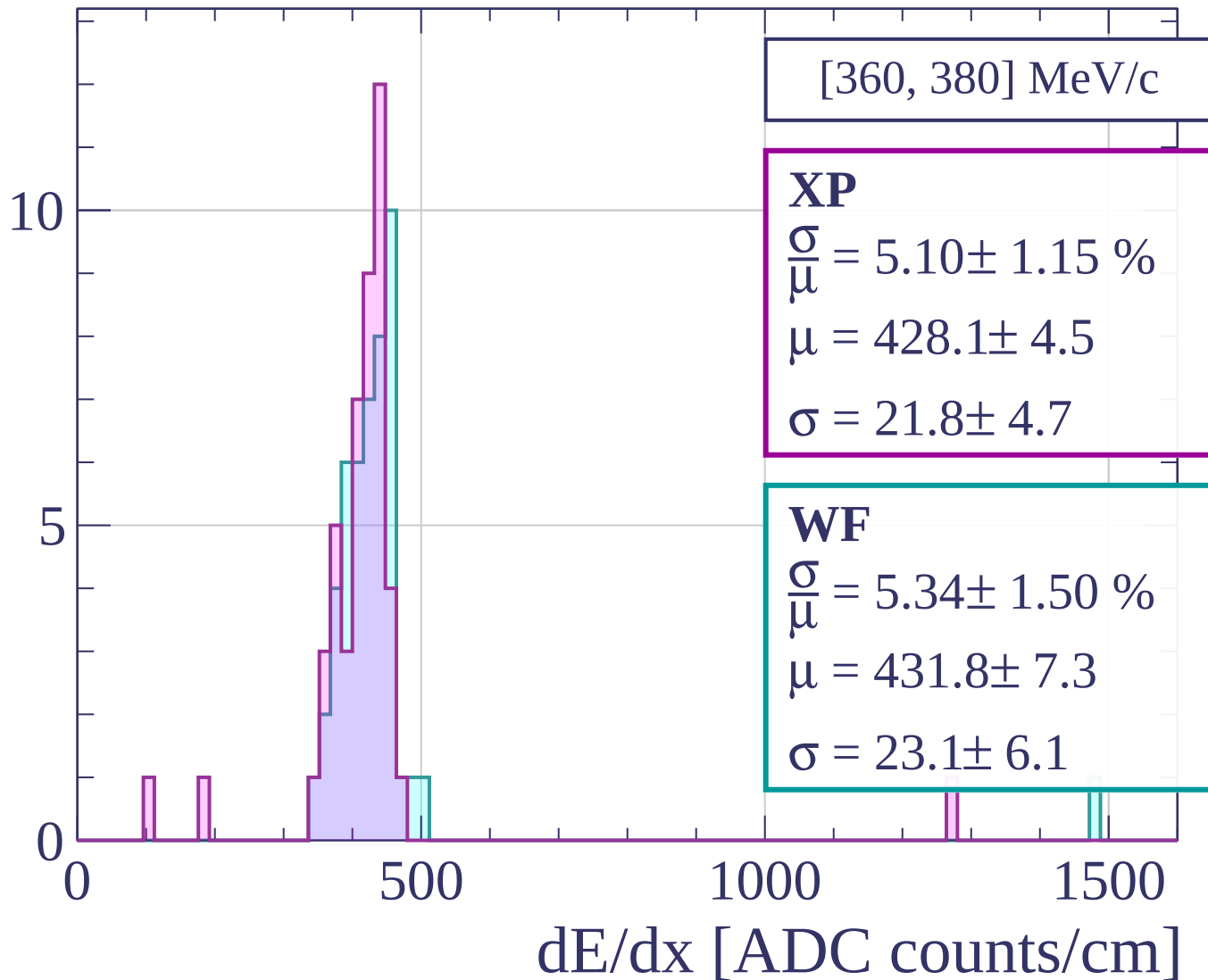
Count



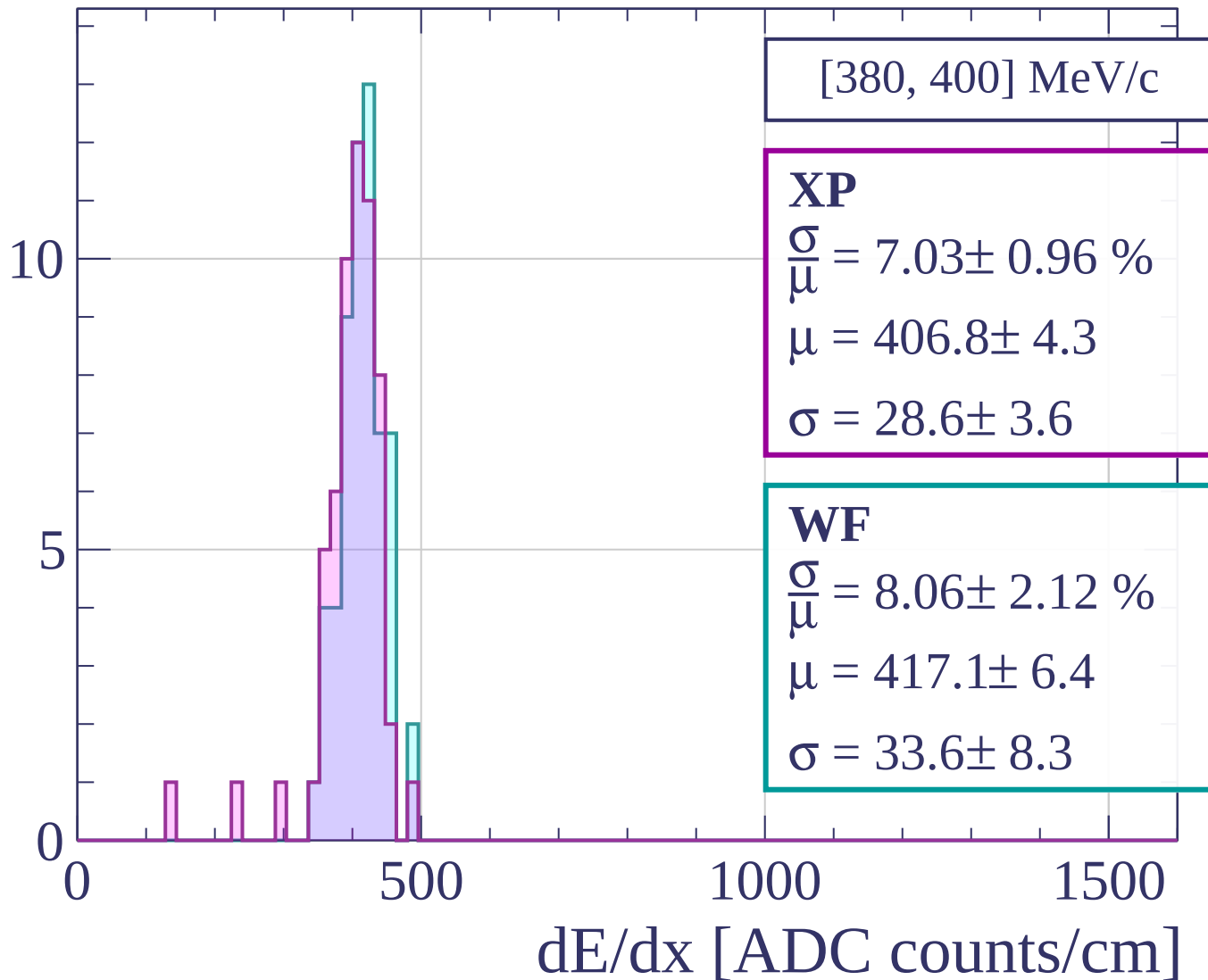
Count



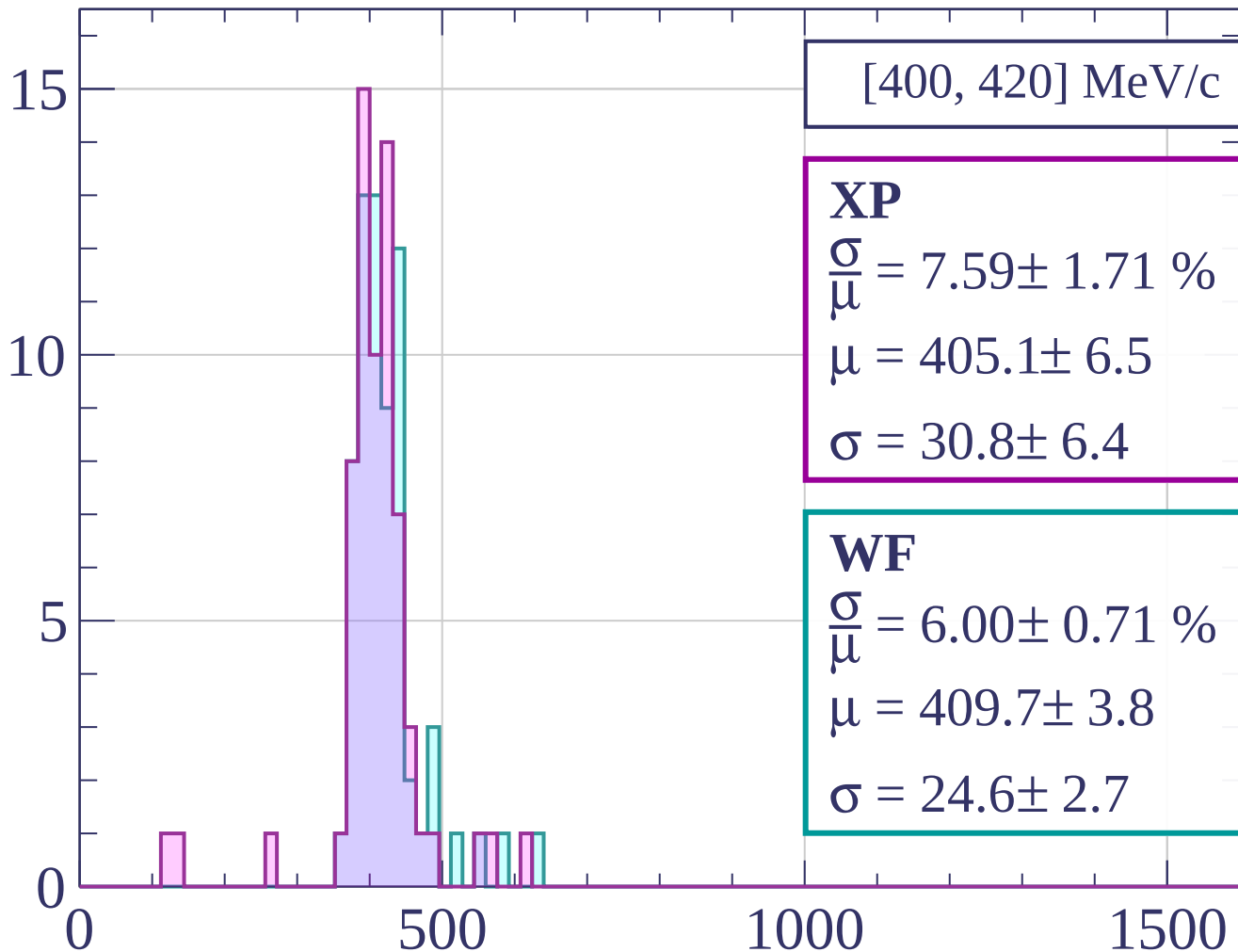
Count



Count

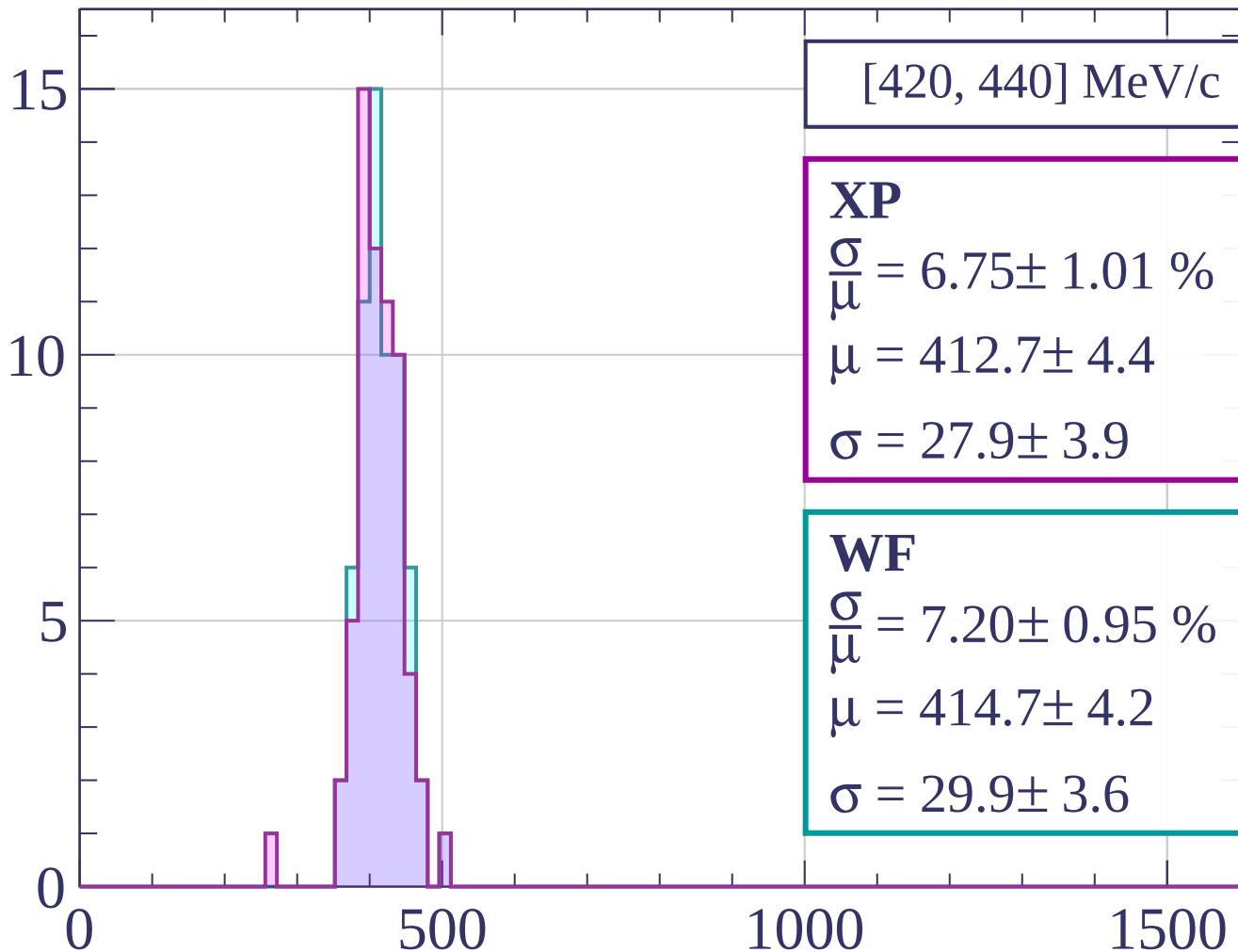


Count



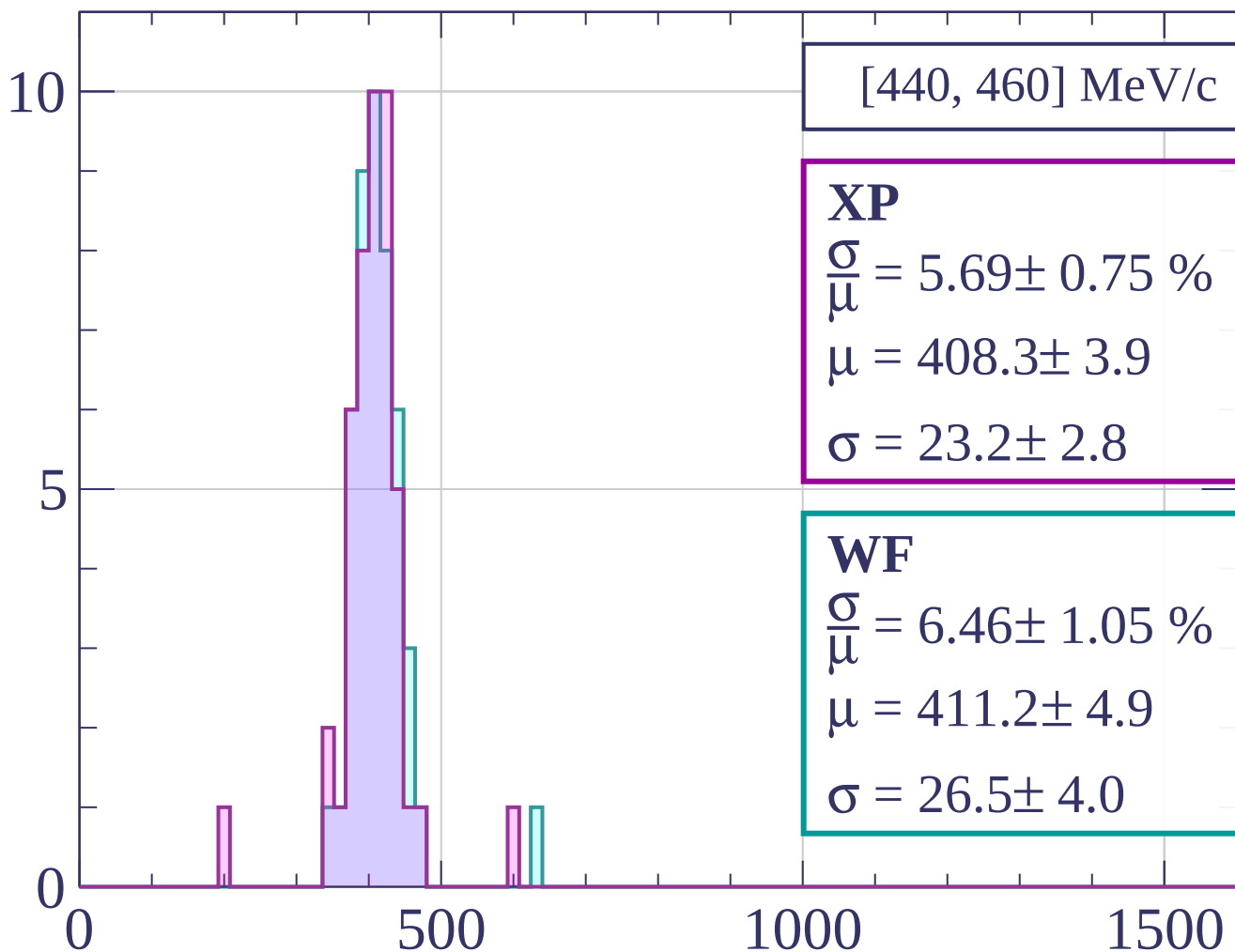
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



[440, 460] MeV/c

XP

$$\frac{\sigma}{\bar{\mu}} = 5.69 \pm 0.75 \%$$

$$\mu = 408.3 \pm 3.9$$

$$\sigma = 23.2 \pm 2.8$$

WF

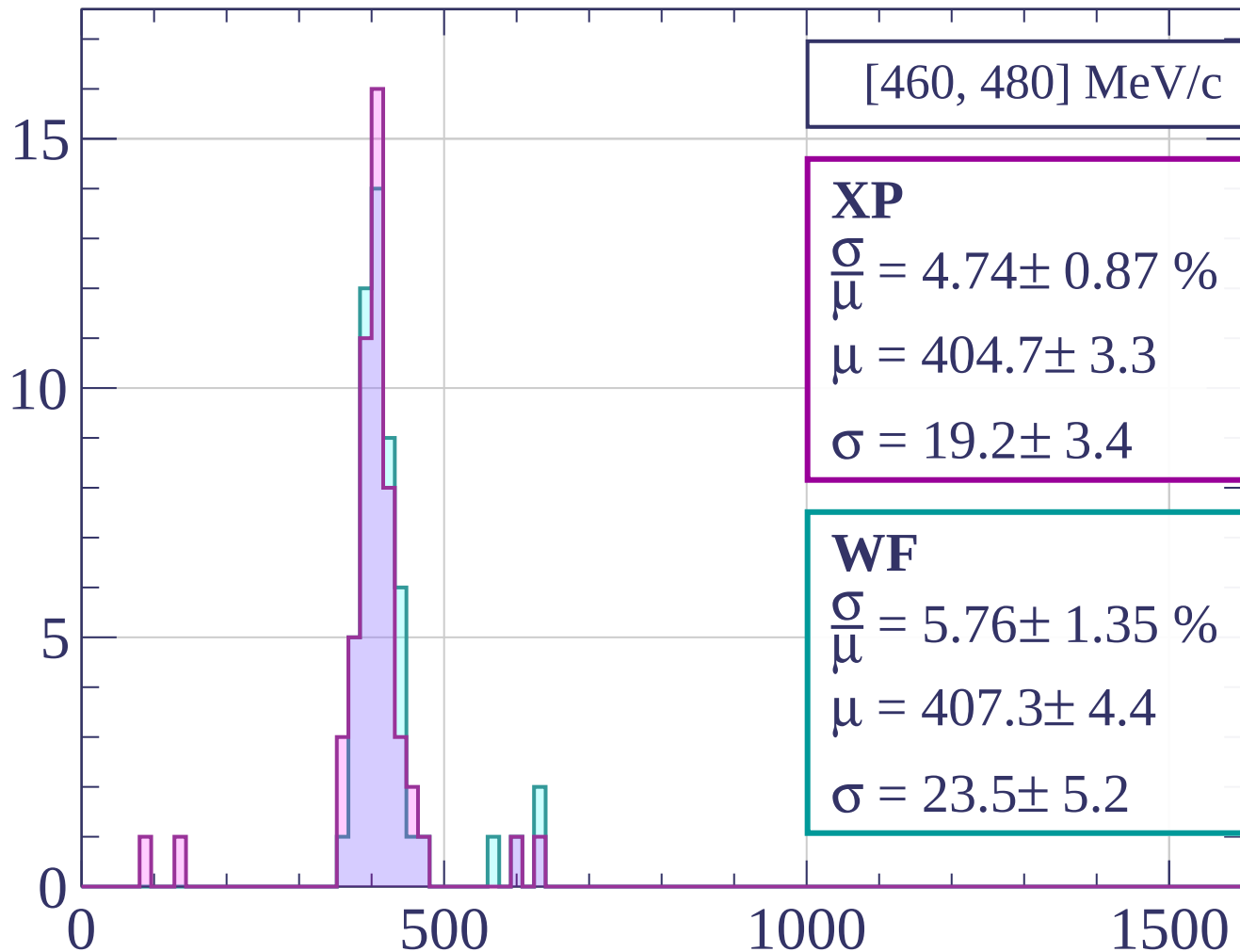
$$\frac{\sigma}{\bar{\mu}} = 6.46 \pm 1.05 \%$$

$$\mu = 411.2 \pm 4.9$$

$$\sigma = 26.5 \pm 4.0$$

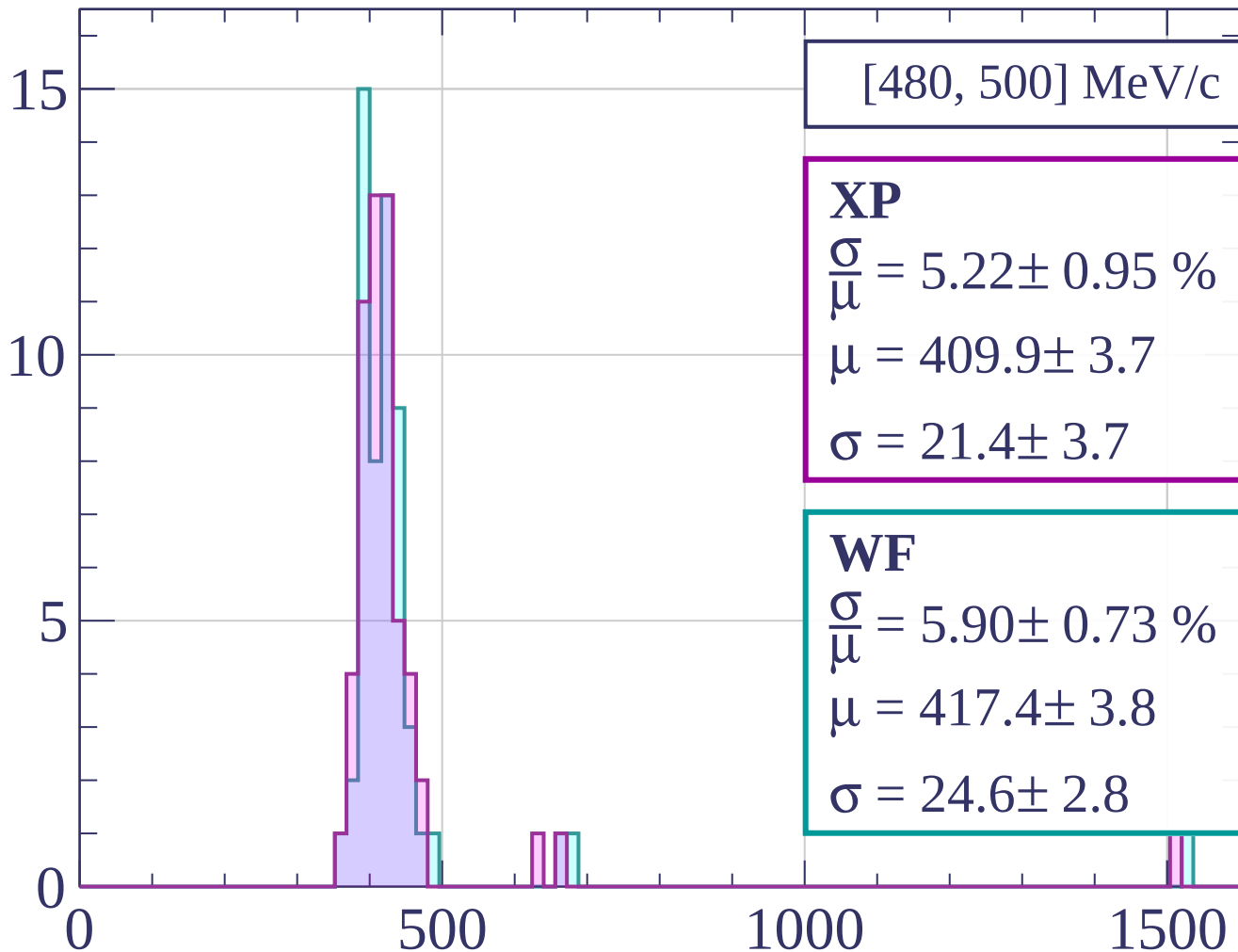
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



[480, 500] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.22 \pm 0.95 \%$$

$$\mu = 409.9 \pm 3.7$$

$$\sigma = 21.4 \pm 3.7$$

WF

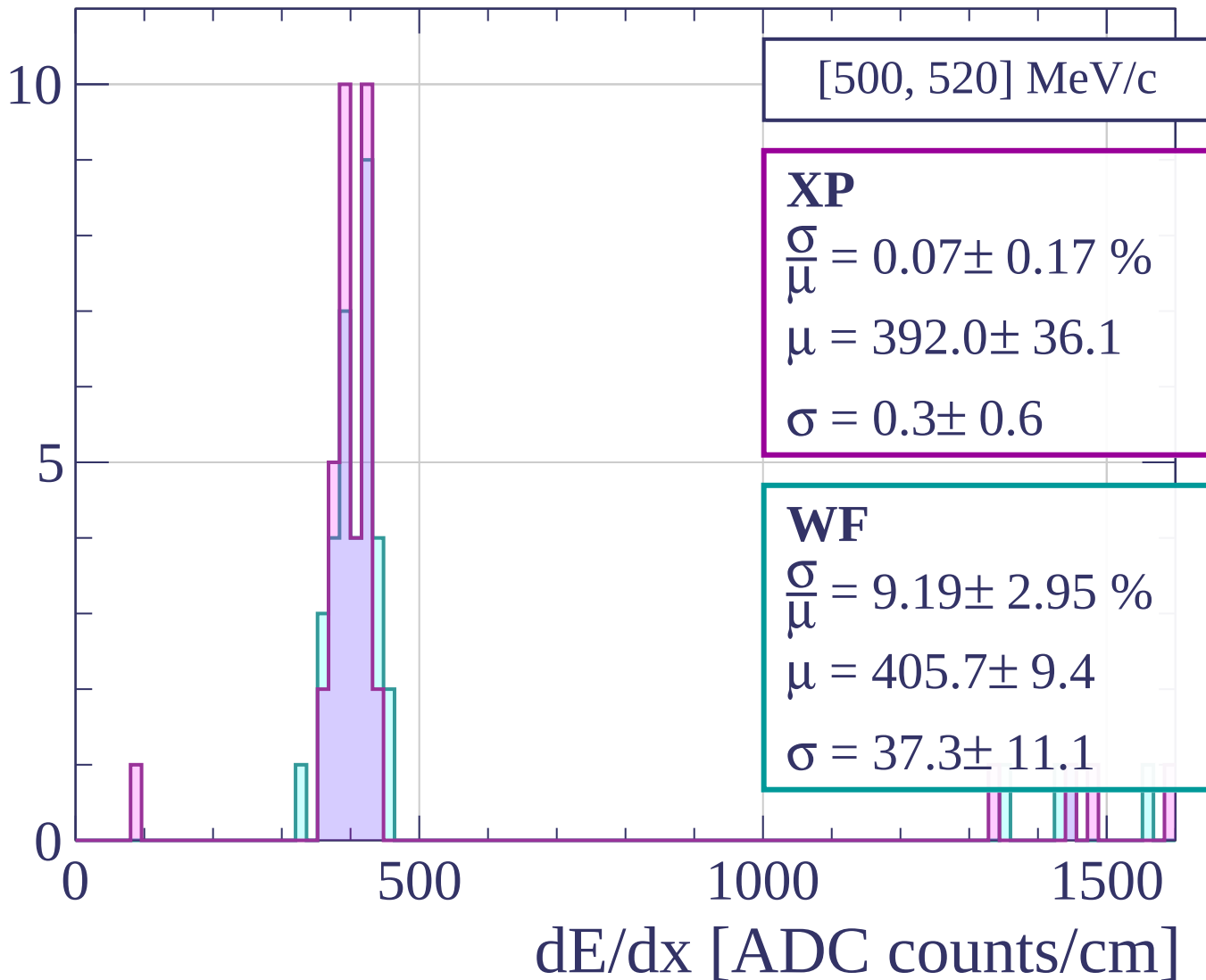
$$\frac{\sigma}{\mu} = 5.90 \pm 0.73 \%$$

$$\mu = 417.4 \pm 3.8$$

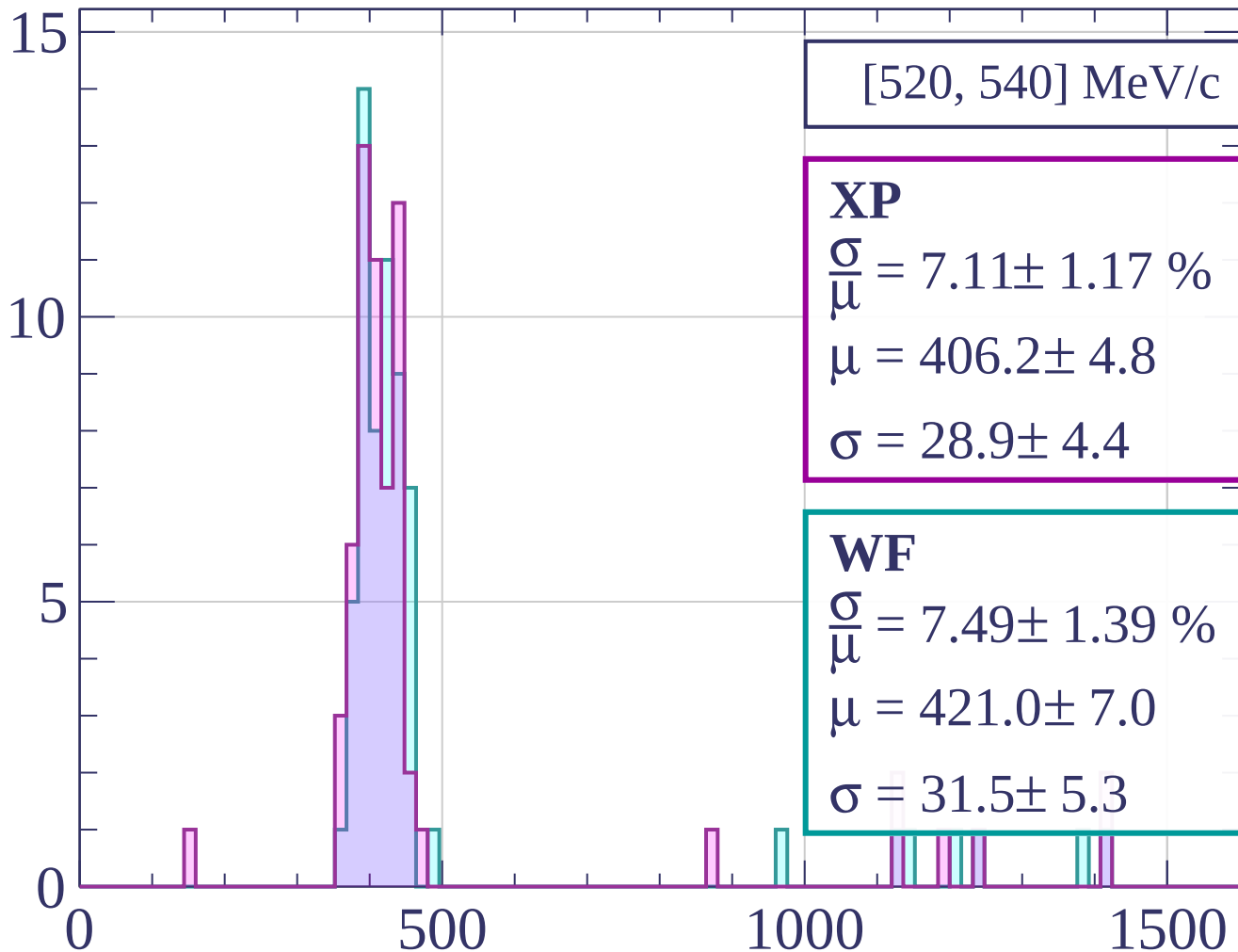
$$\sigma = 24.6 \pm 2.8$$

dE/dx [ADC counts/cm]

Count

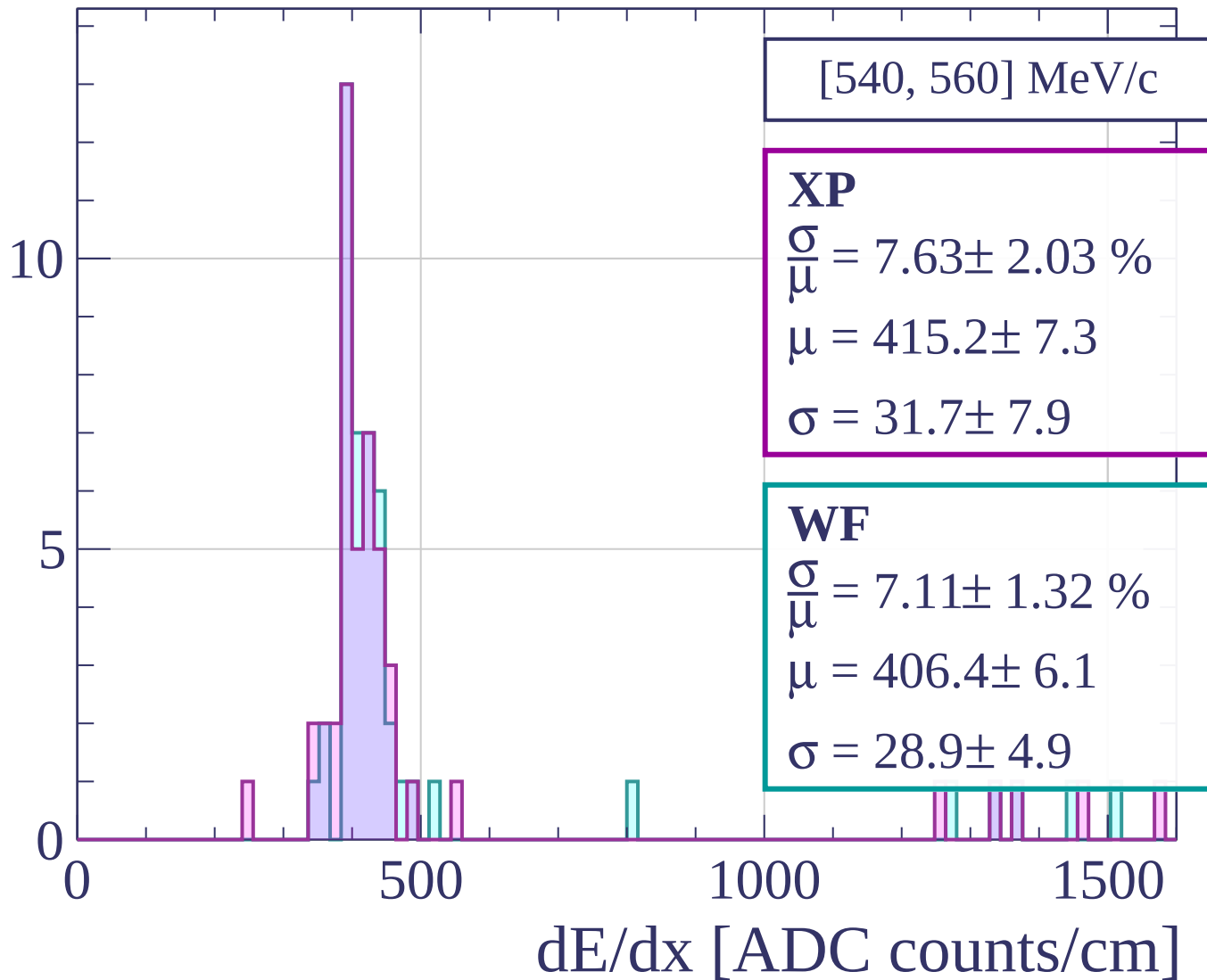


Count

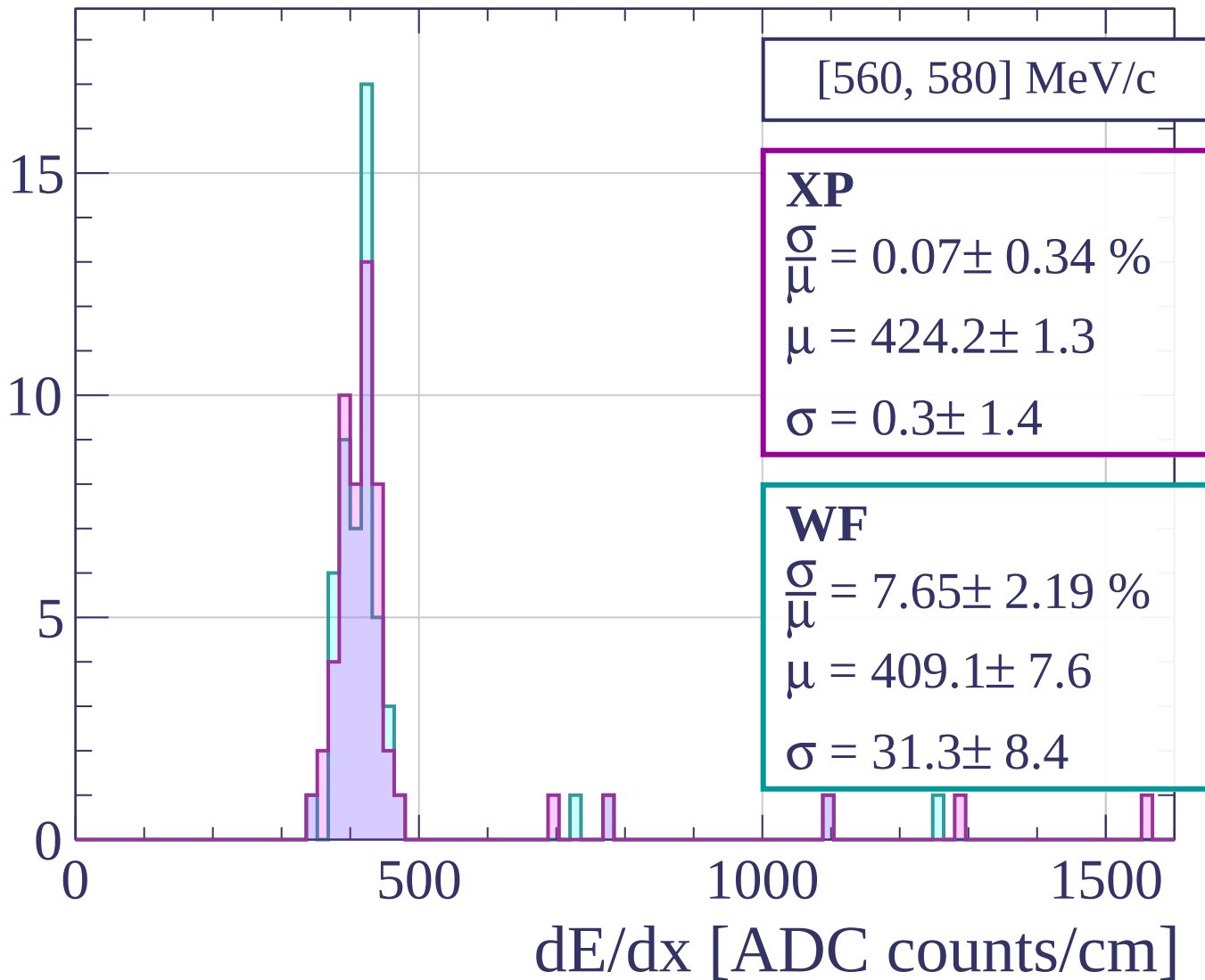


dE/dx [ADC counts/cm]

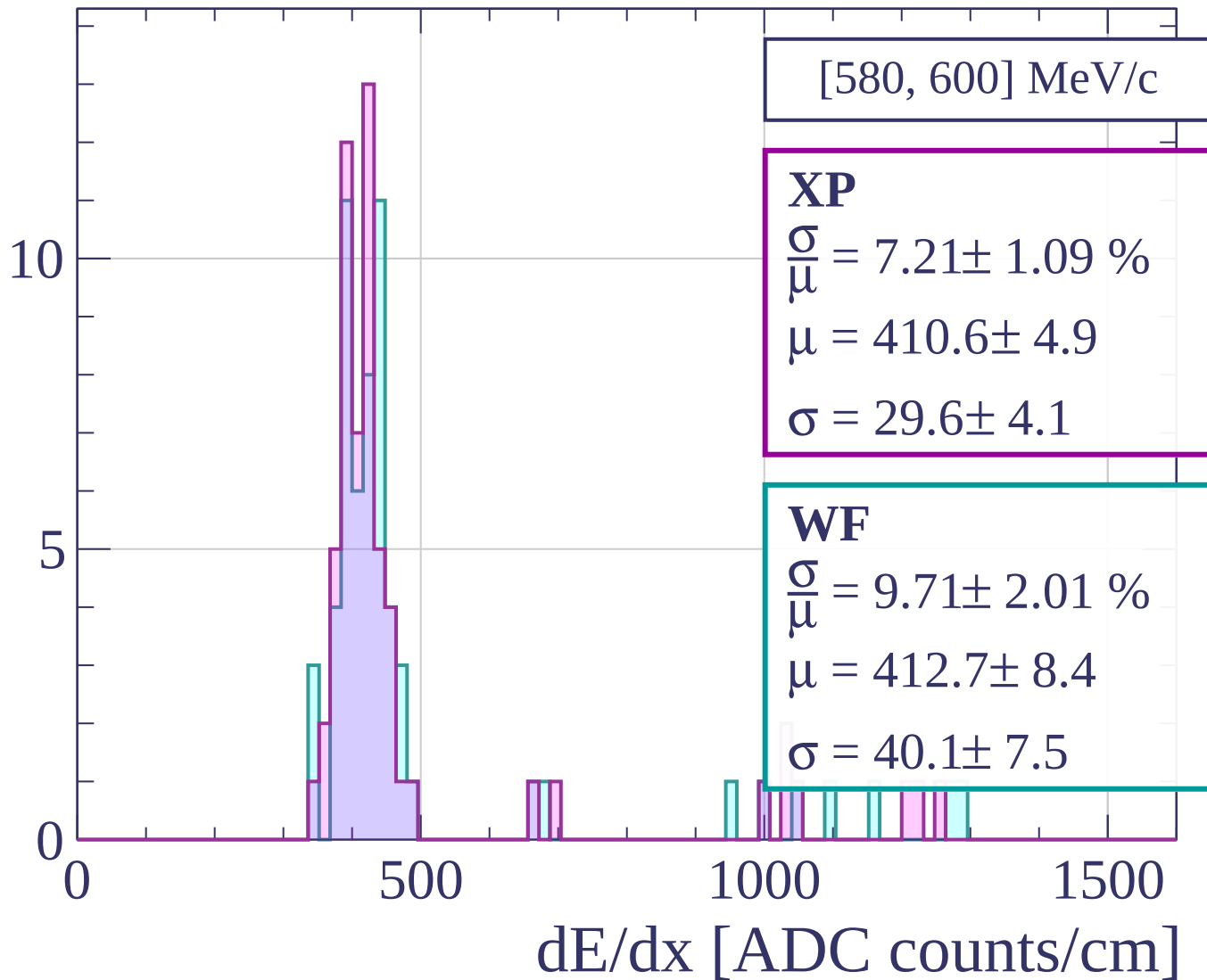
Count



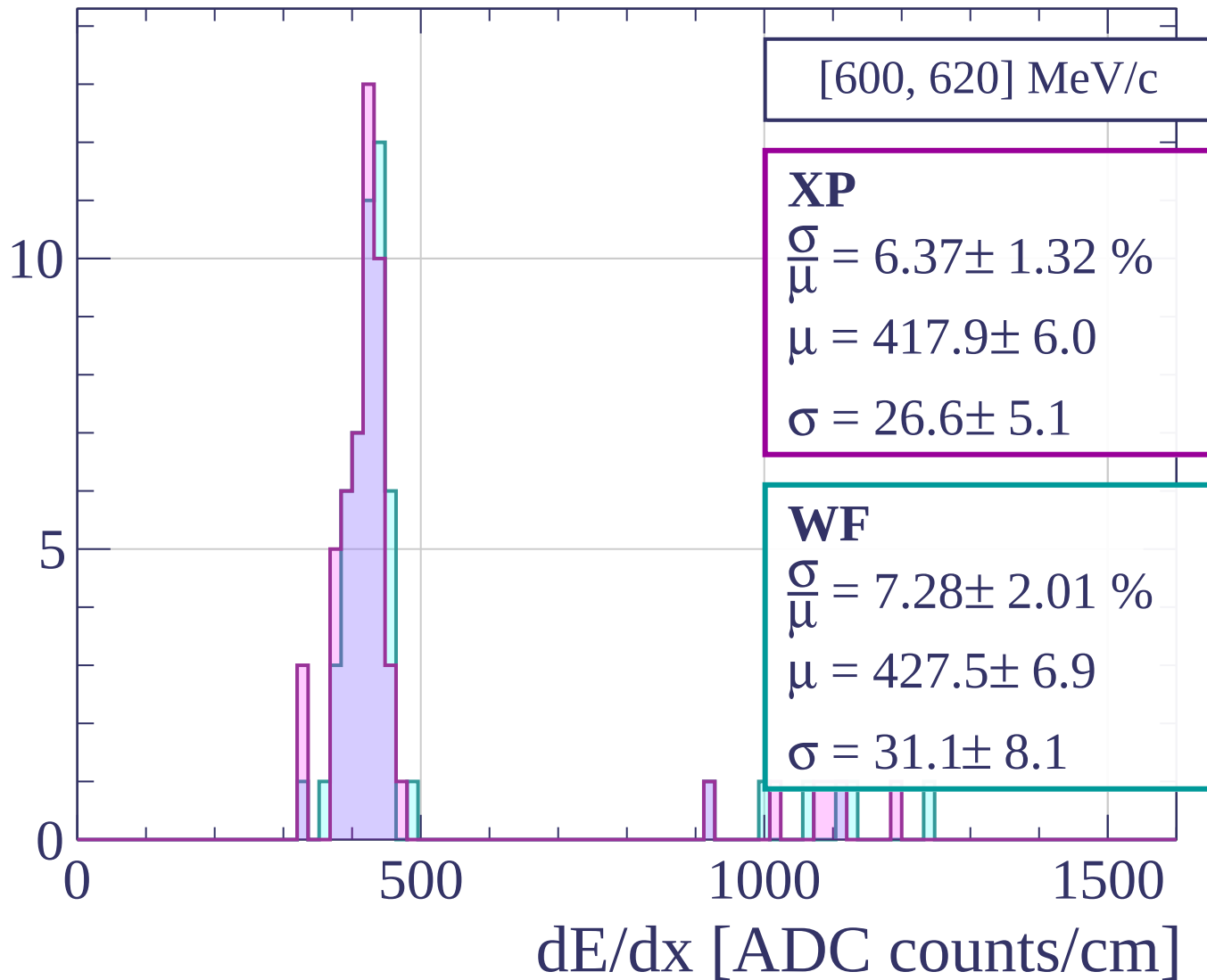
Count



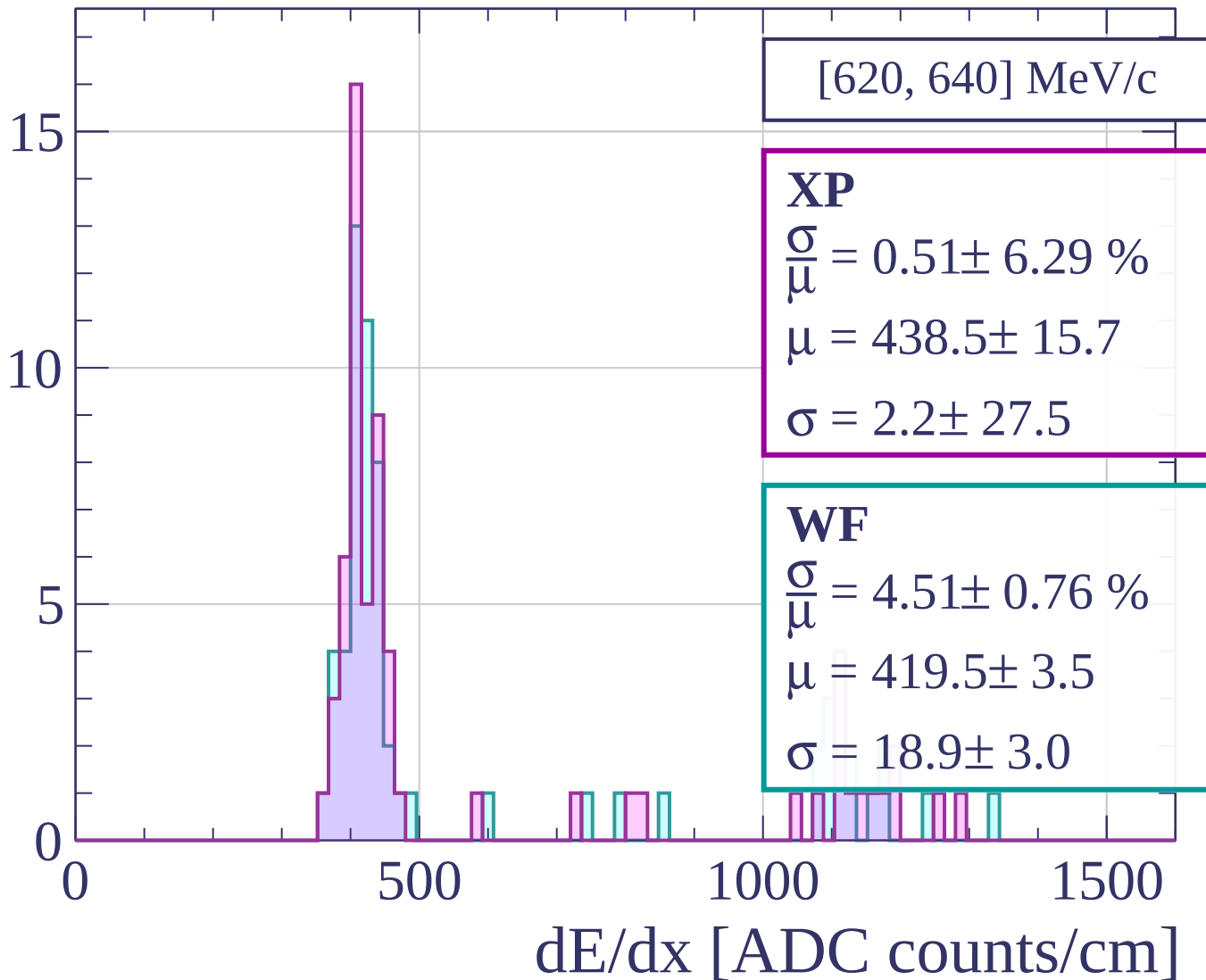
Count



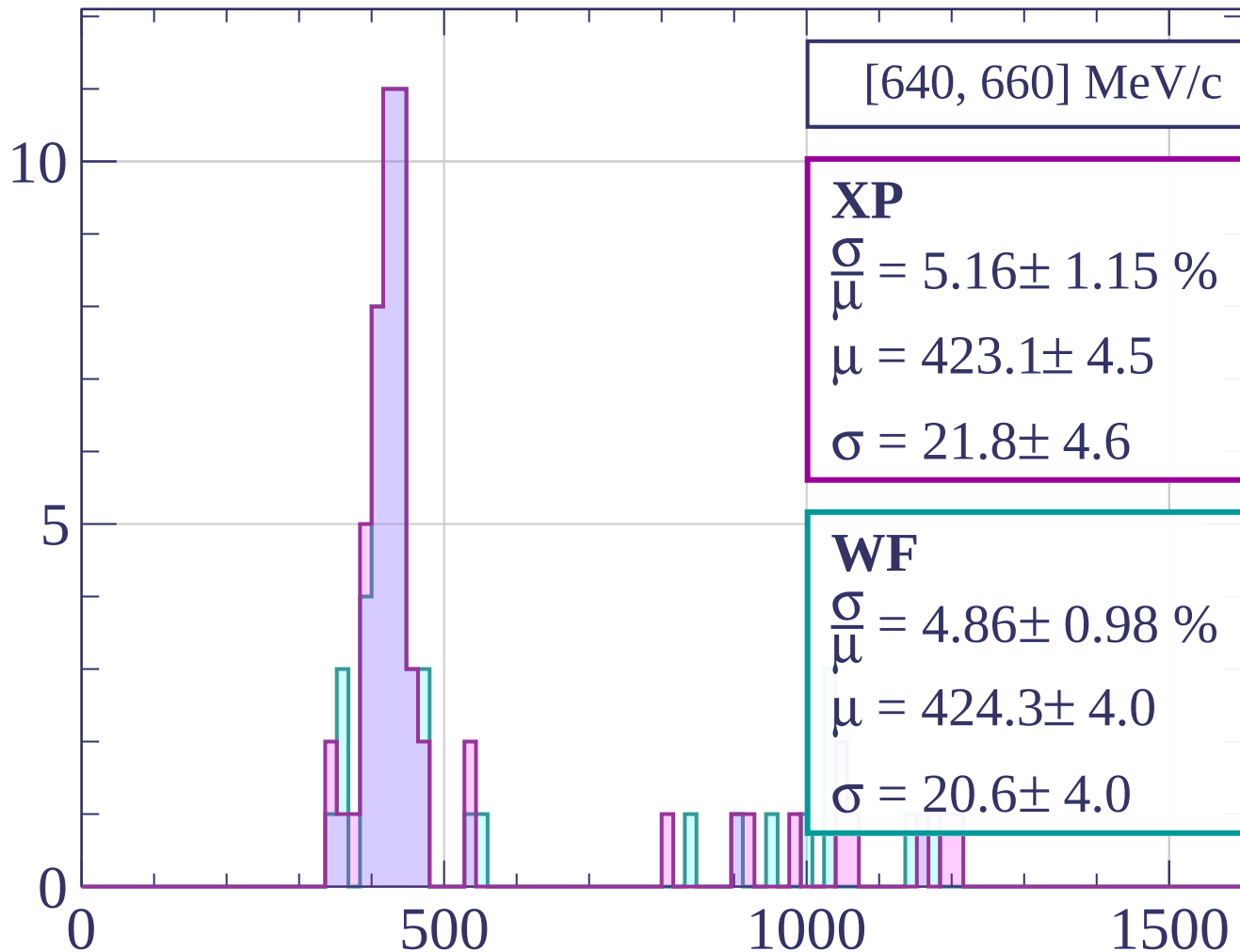
Count



Count

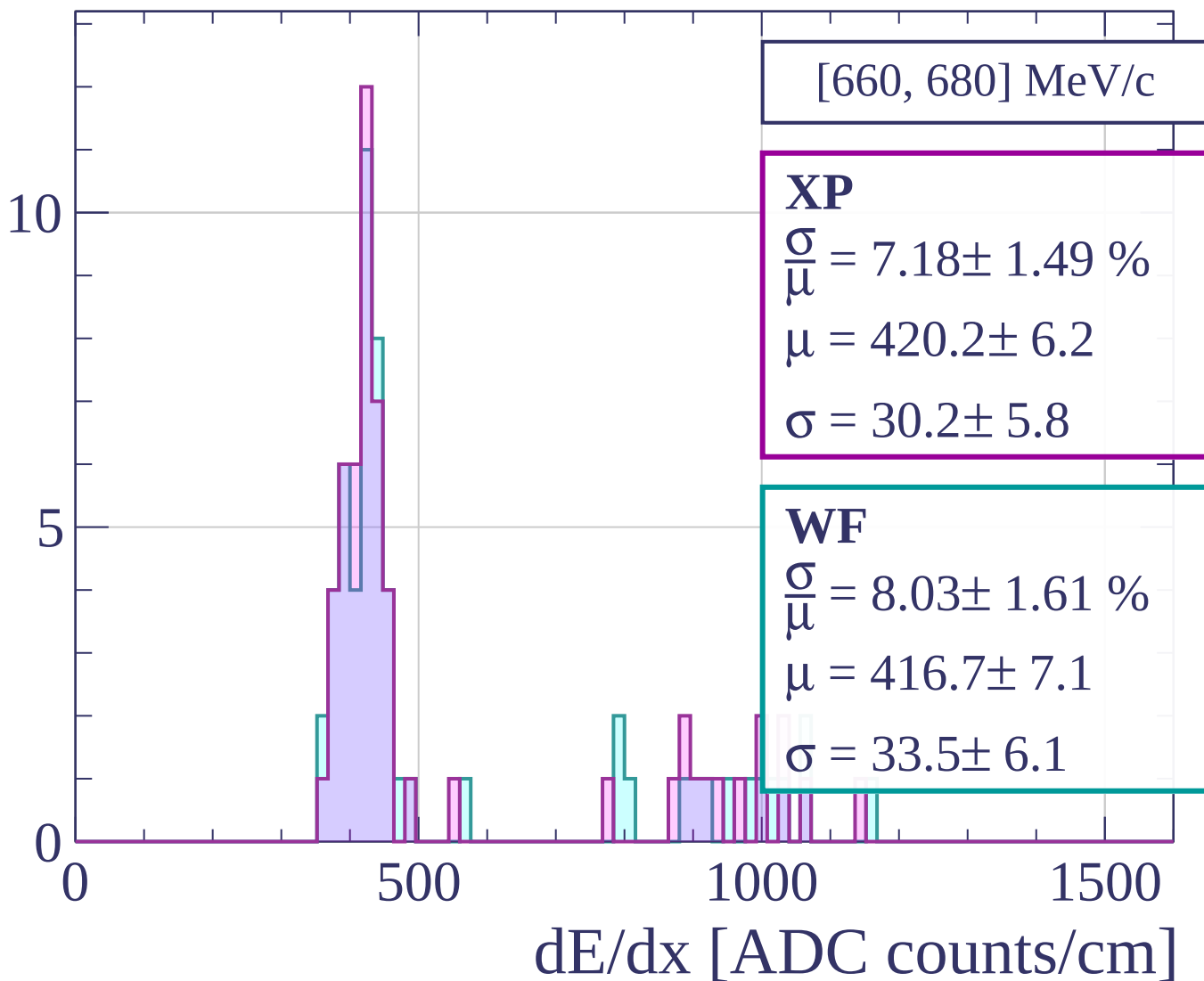


Count

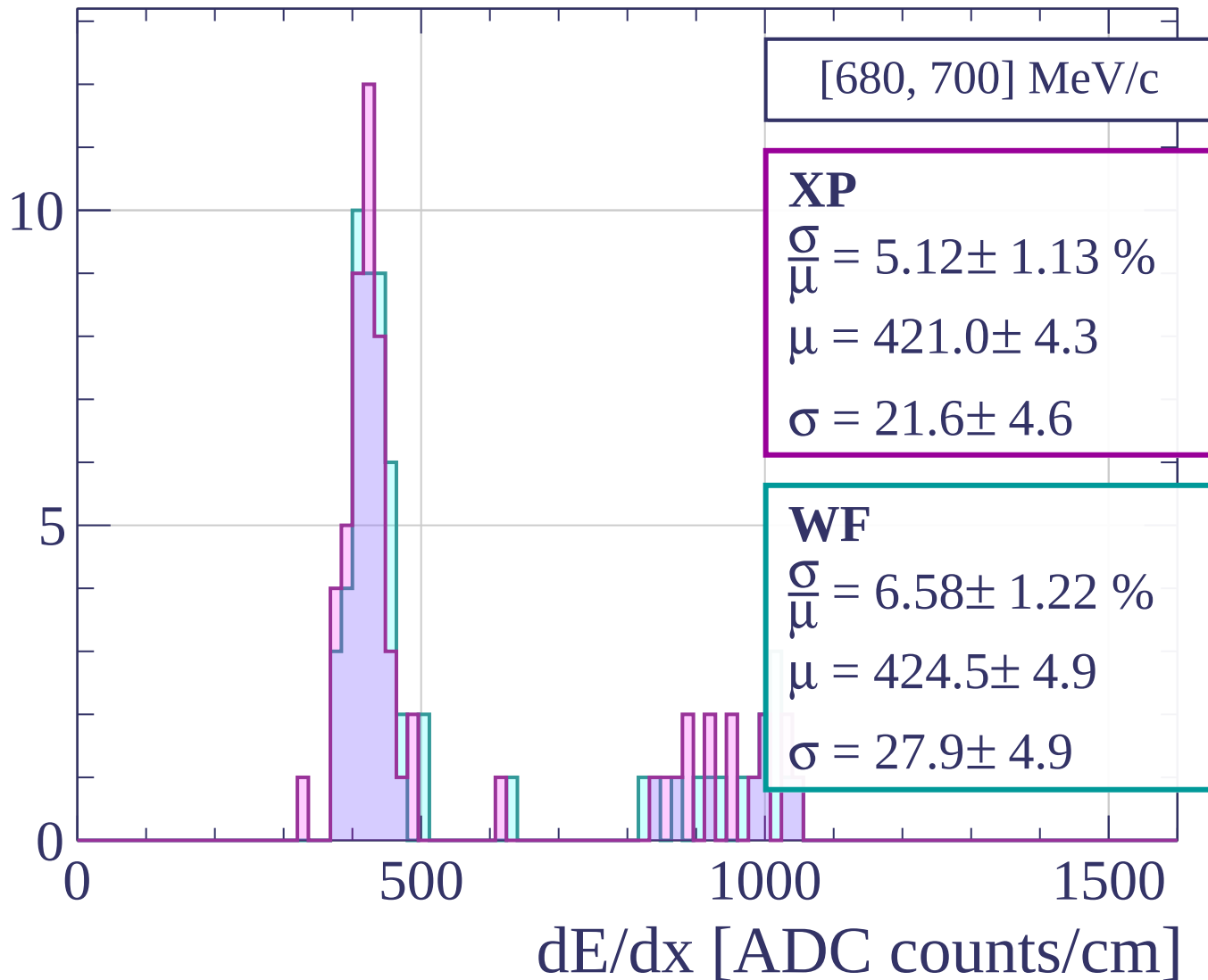


dE/dx [ADC counts/cm]

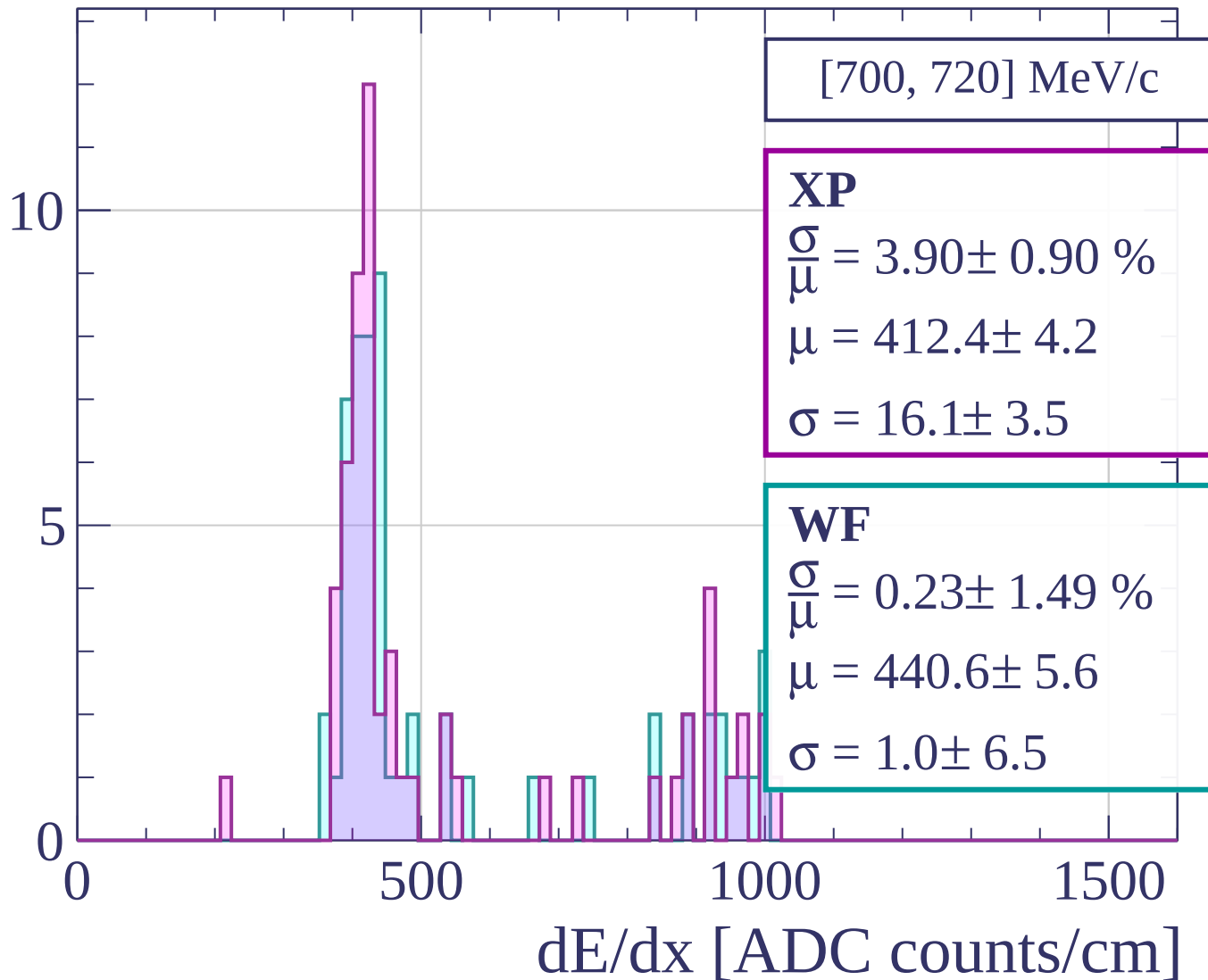
Count



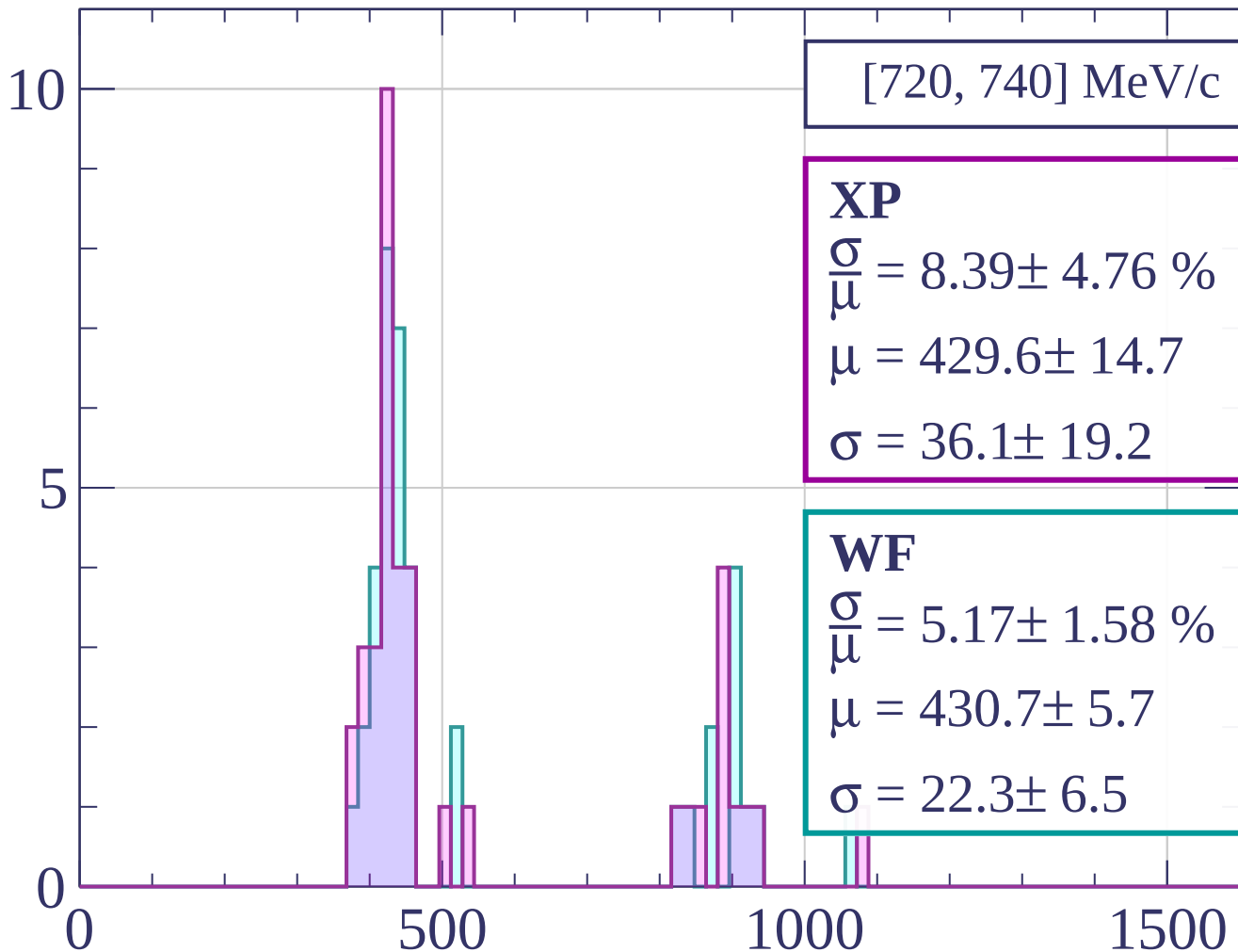
Count



Count



Count



[720, 740] MeV/c

XP

$$\frac{\sigma}{\mu} = 8.39 \pm 4.76 \%$$

$$\mu = 429.6 \pm 14.7$$

$$\sigma = 36.1 \pm 19.2$$

WF

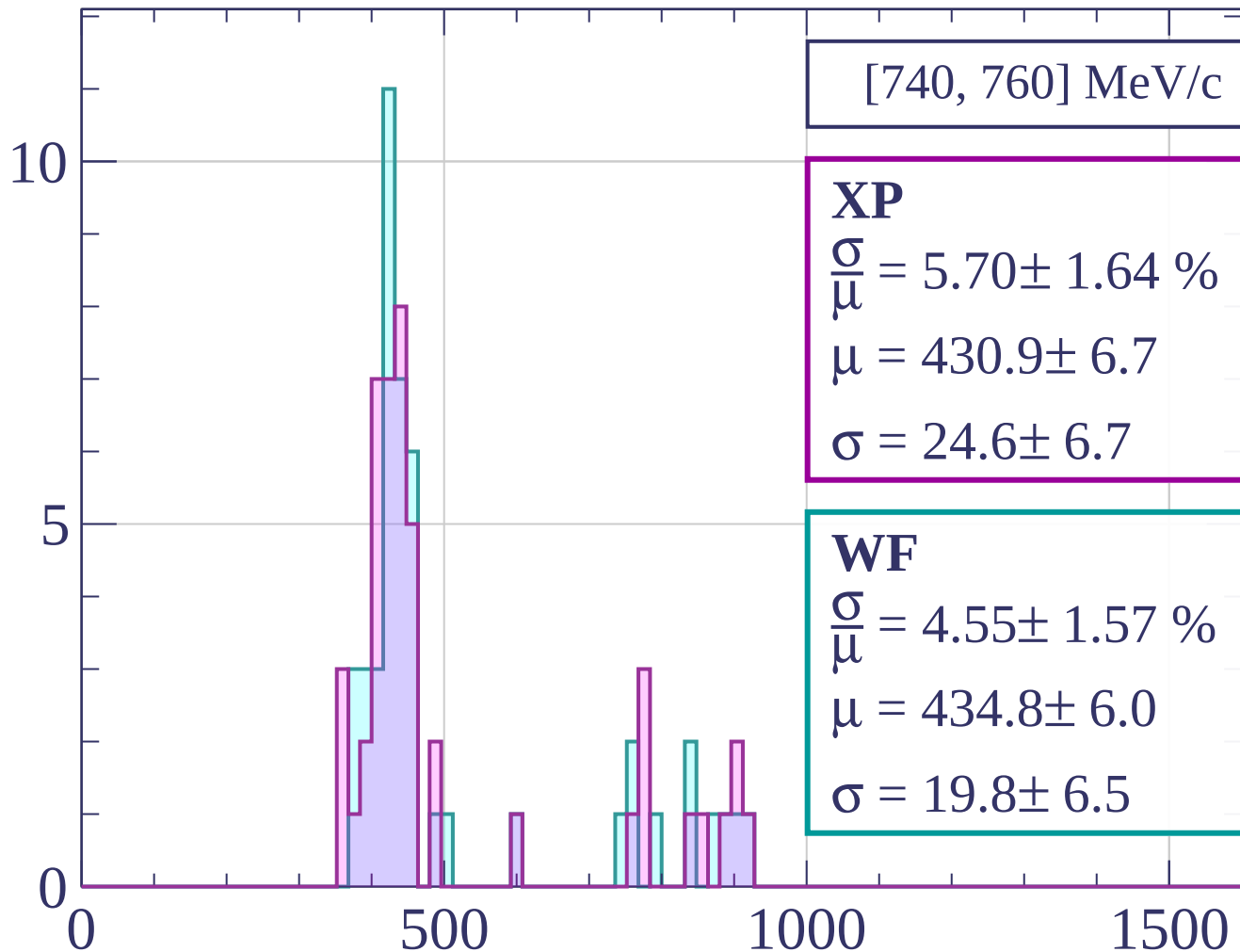
$$\frac{\sigma}{\mu} = 5.17 \pm 1.58 \%$$

$$\mu = 430.7 \pm 5.7$$

$$\sigma = 22.3 \pm 6.5$$

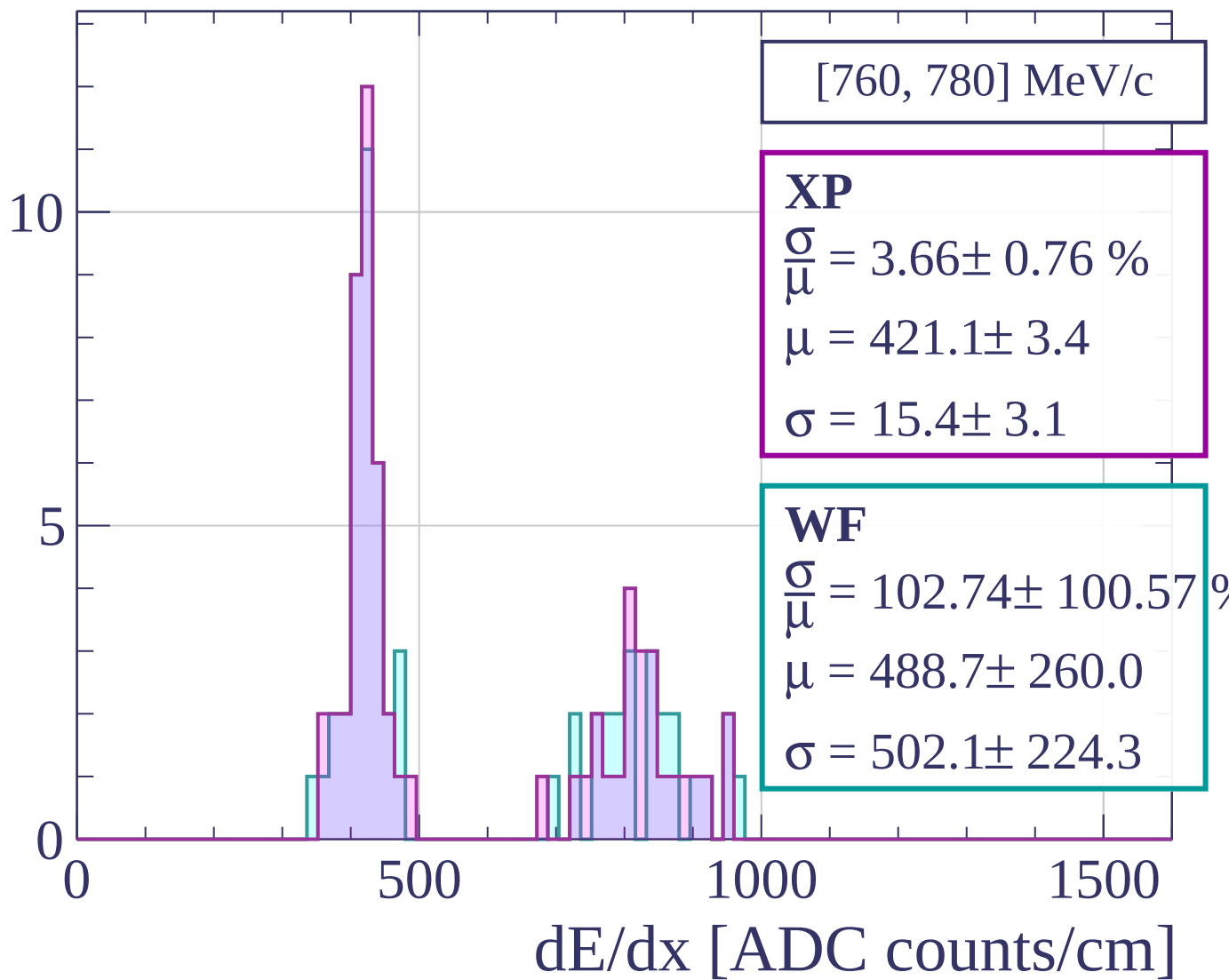
dE/dx [ADC counts/cm]

Count

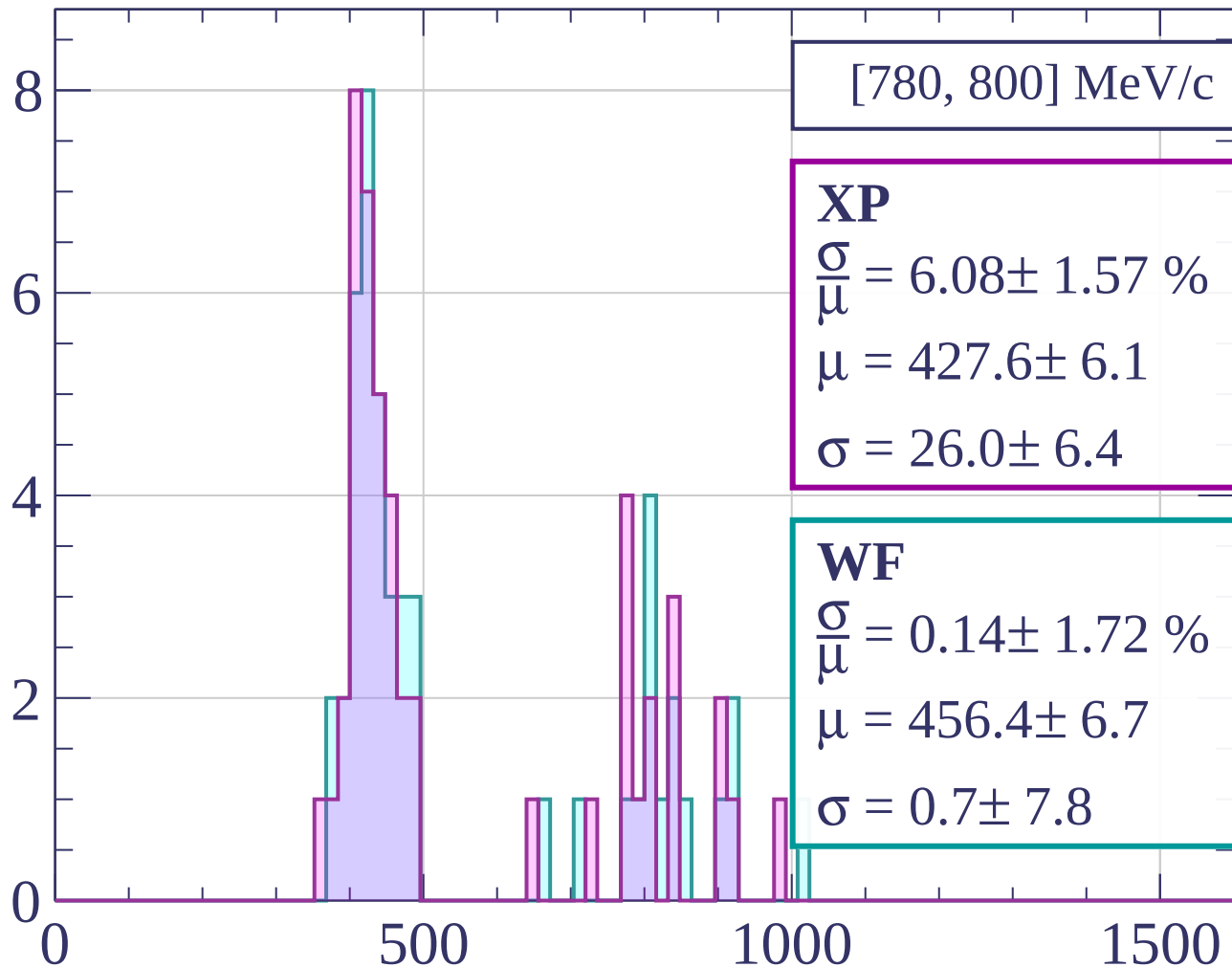


dE/dx [ADC counts/cm]

Count

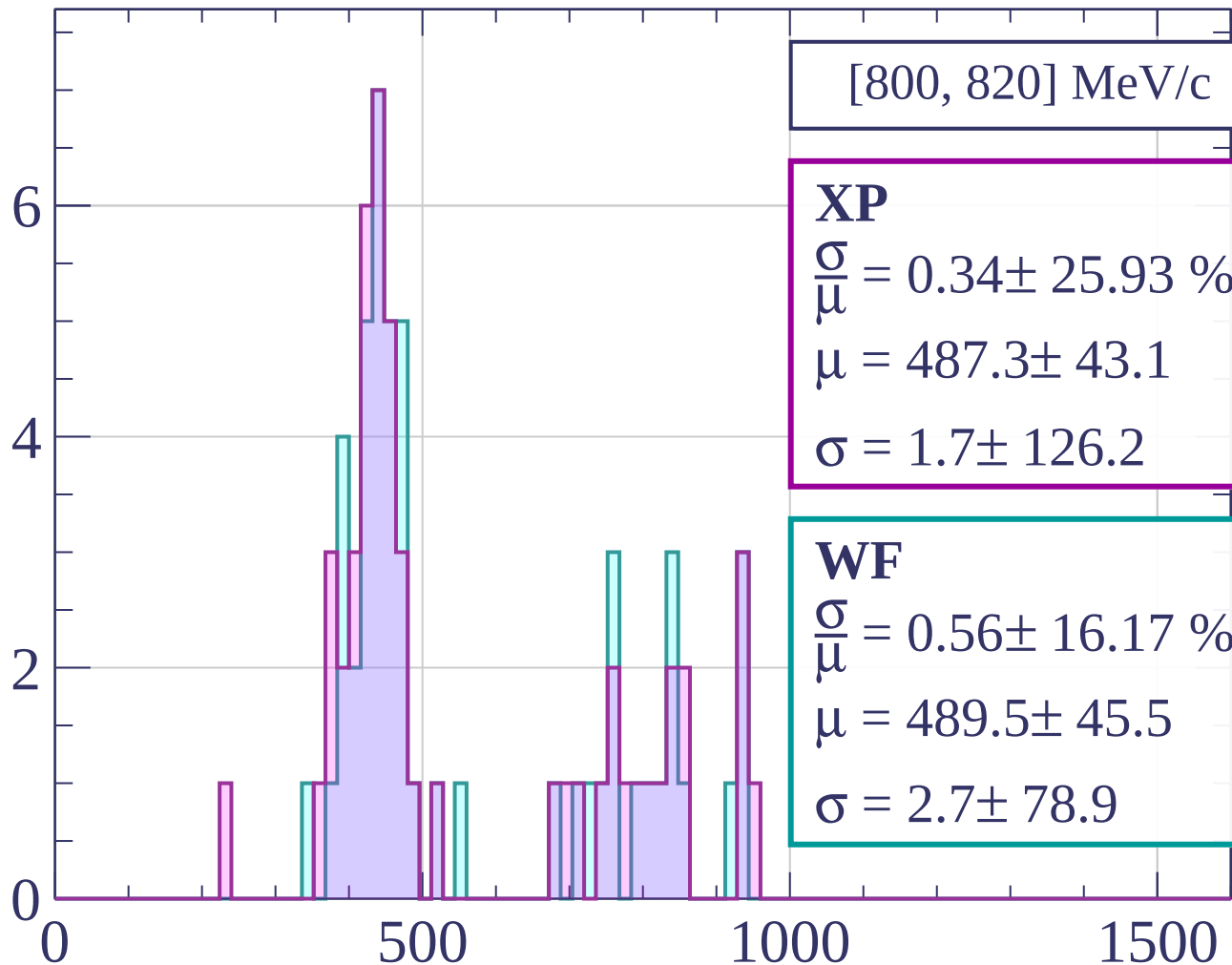


Count



dE/dx [ADC counts/cm]

Count



[800, 820] MeV/c

XP

$$\frac{\sigma}{\mu} = 0.34 \pm 25.93 \%$$

$$\mu = 487.3 \pm 43.1$$

$$\sigma = 1.7 \pm 126.2$$

WF

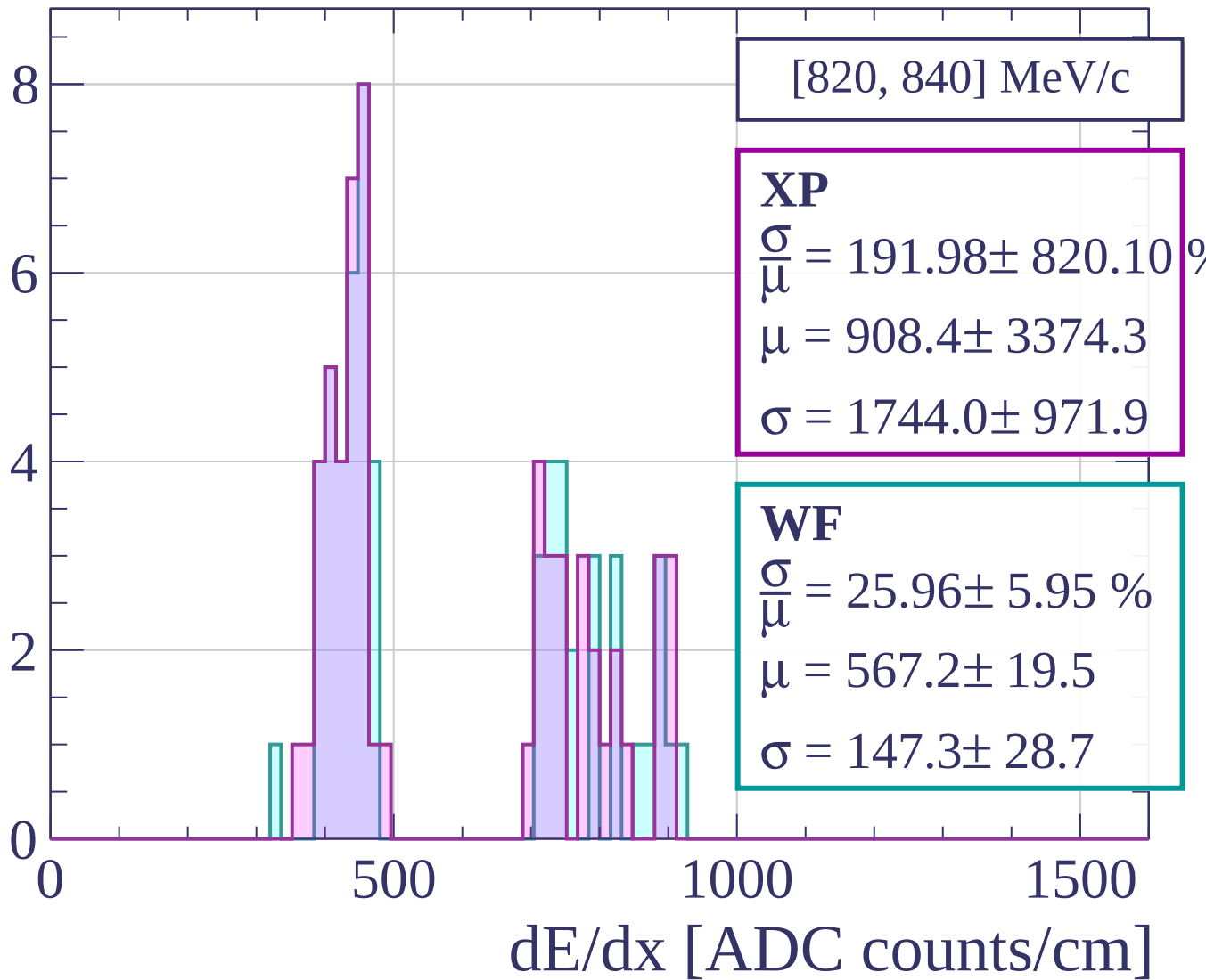
$$\frac{\sigma}{\mu} = 0.56 \pm 16.17 \%$$

$$\mu = 489.5 \pm 45.5$$

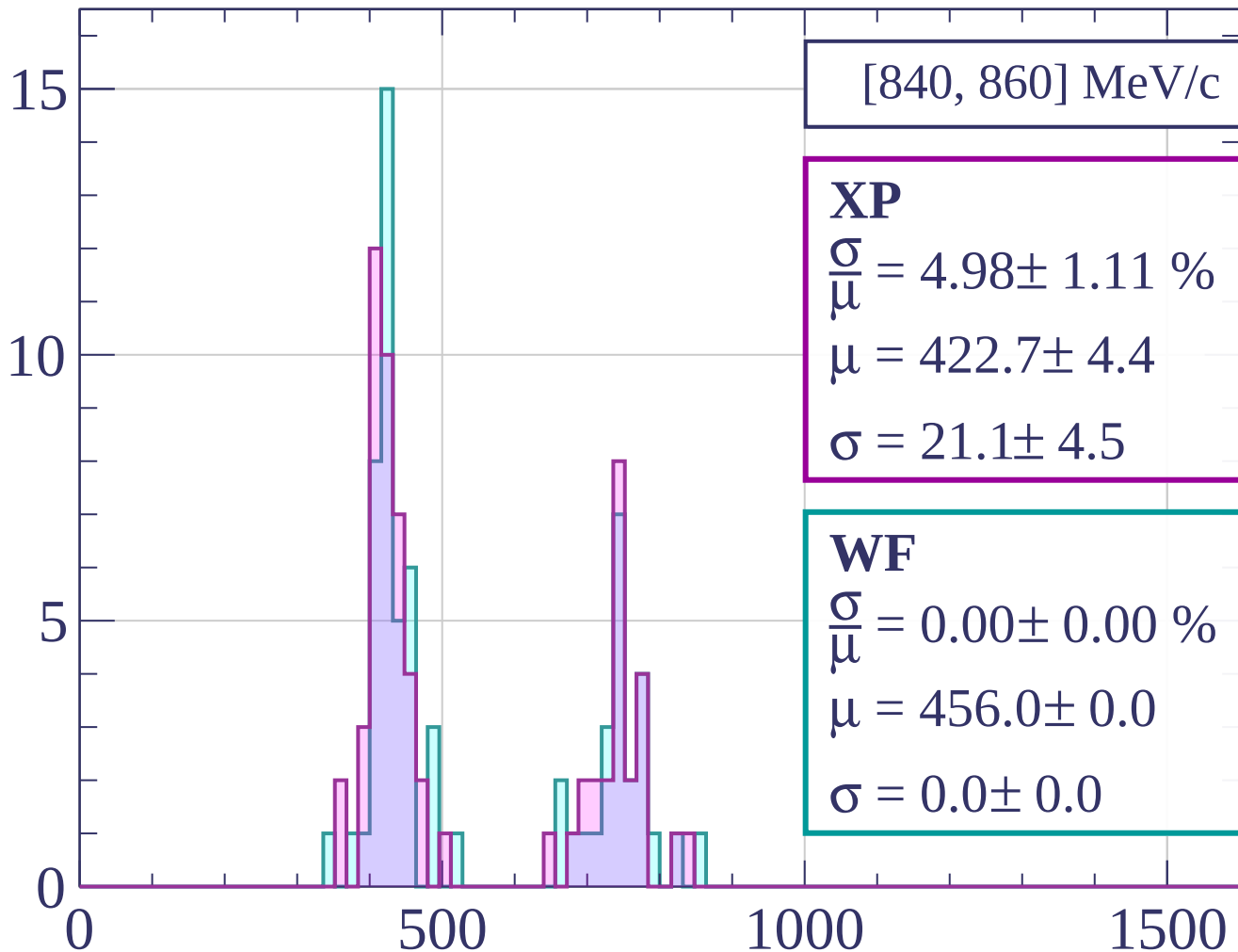
$$\sigma = 2.7 \pm 78.9$$

dE/dx [ADC counts/cm]

Count

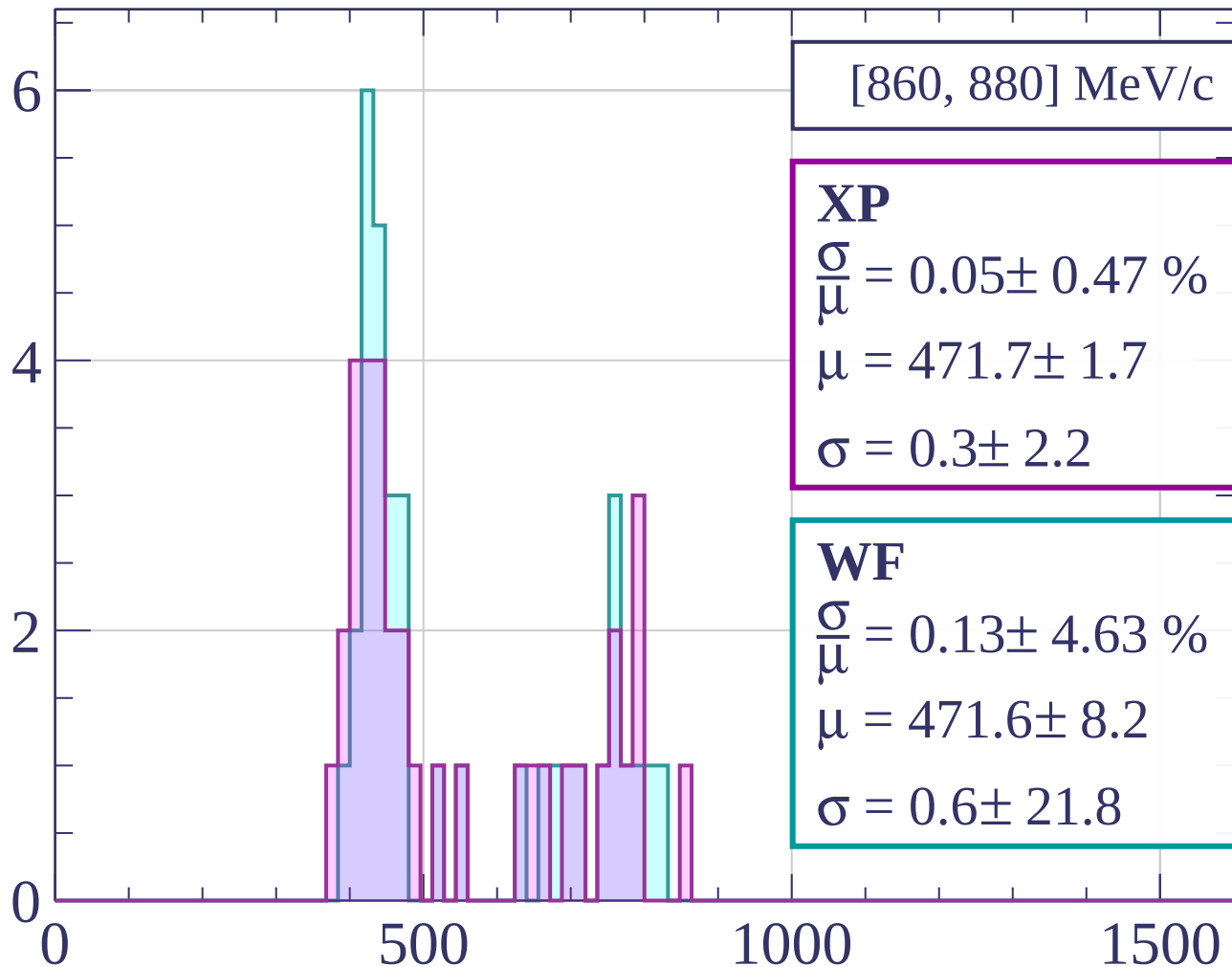


Count



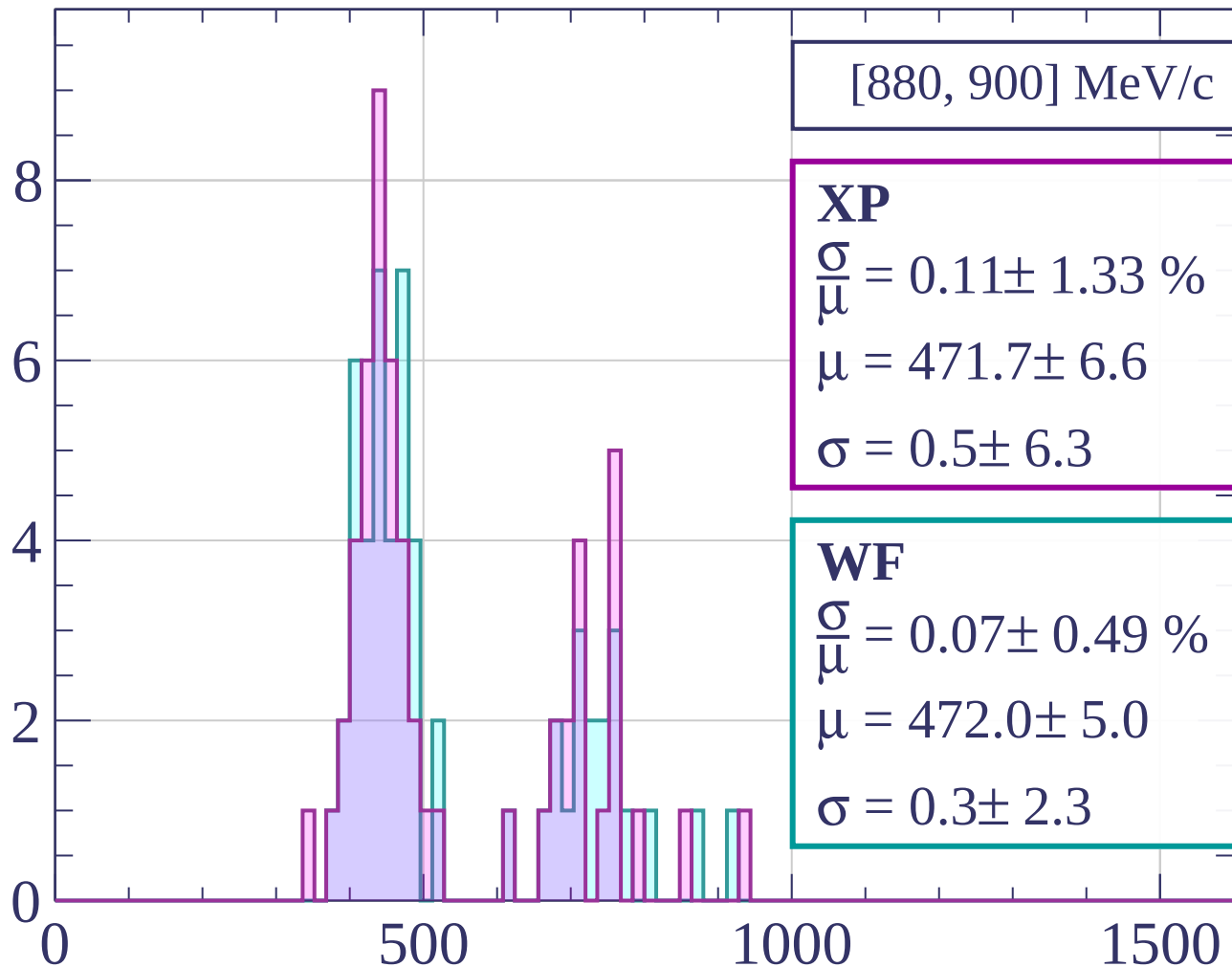
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



[880, 900] MeV/c

XP

$$\frac{\sigma}{\bar{\mu}} = 0.11 \pm 1.33 \%$$

$$\mu = 471.7 \pm 6.6$$

$$\sigma = 0.5 \pm 6.3$$

WF

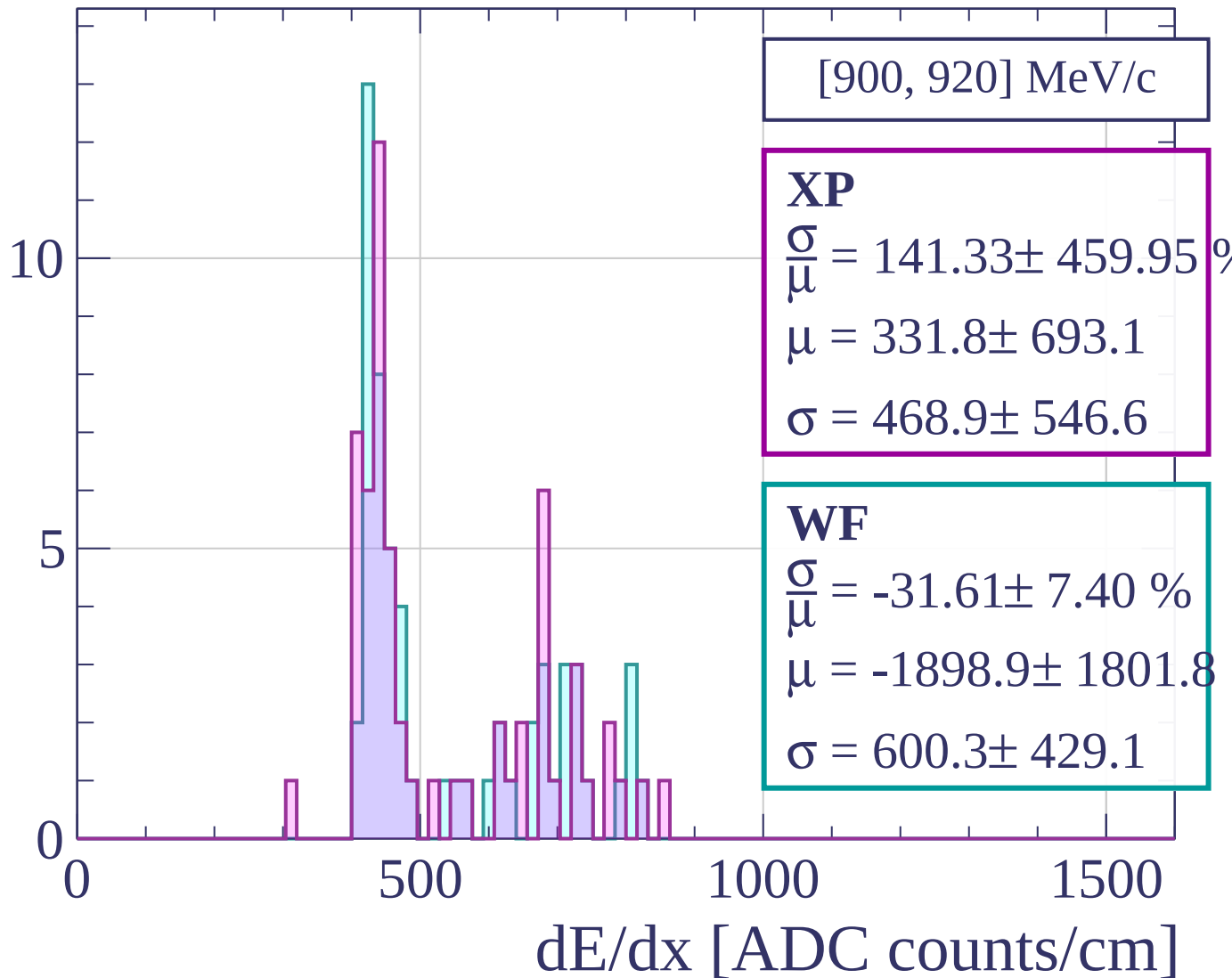
$$\frac{\sigma}{\bar{\mu}} = 0.07 \pm 0.49 \%$$

$$\mu = 472.0 \pm 5.0$$

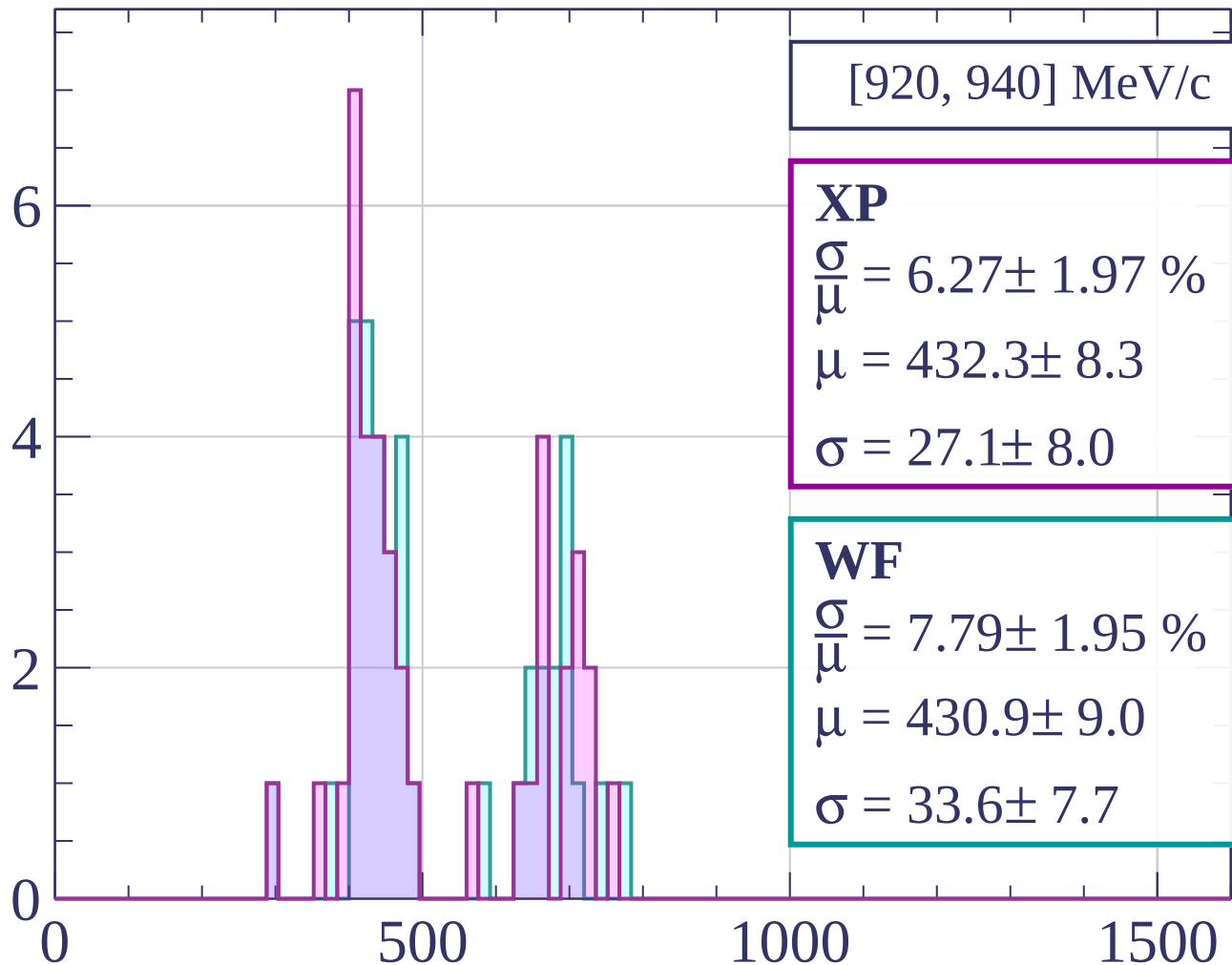
$$\sigma = 0.3 \pm 2.3$$

dE/dx [ADC counts/cm]

Count



Count



[920, 940] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.27 \pm 1.97 \%$$

$$\mu = 432.3 \pm 8.3$$

$$\sigma = 27.1 \pm 8.0$$

WF

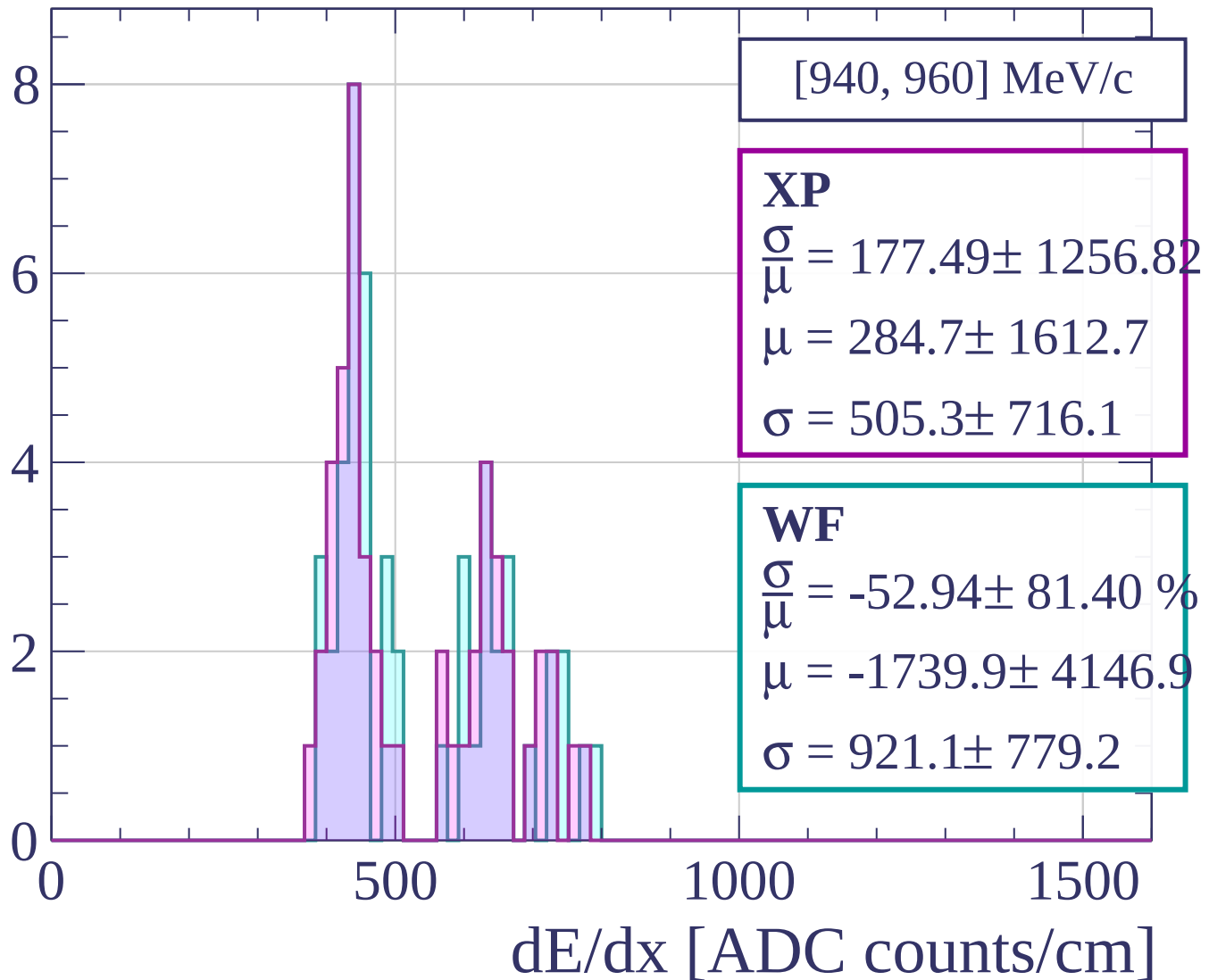
$$\frac{\sigma}{\mu} = 7.79 \pm 1.95 \%$$

$$\mu = 430.9 \pm 9.0$$

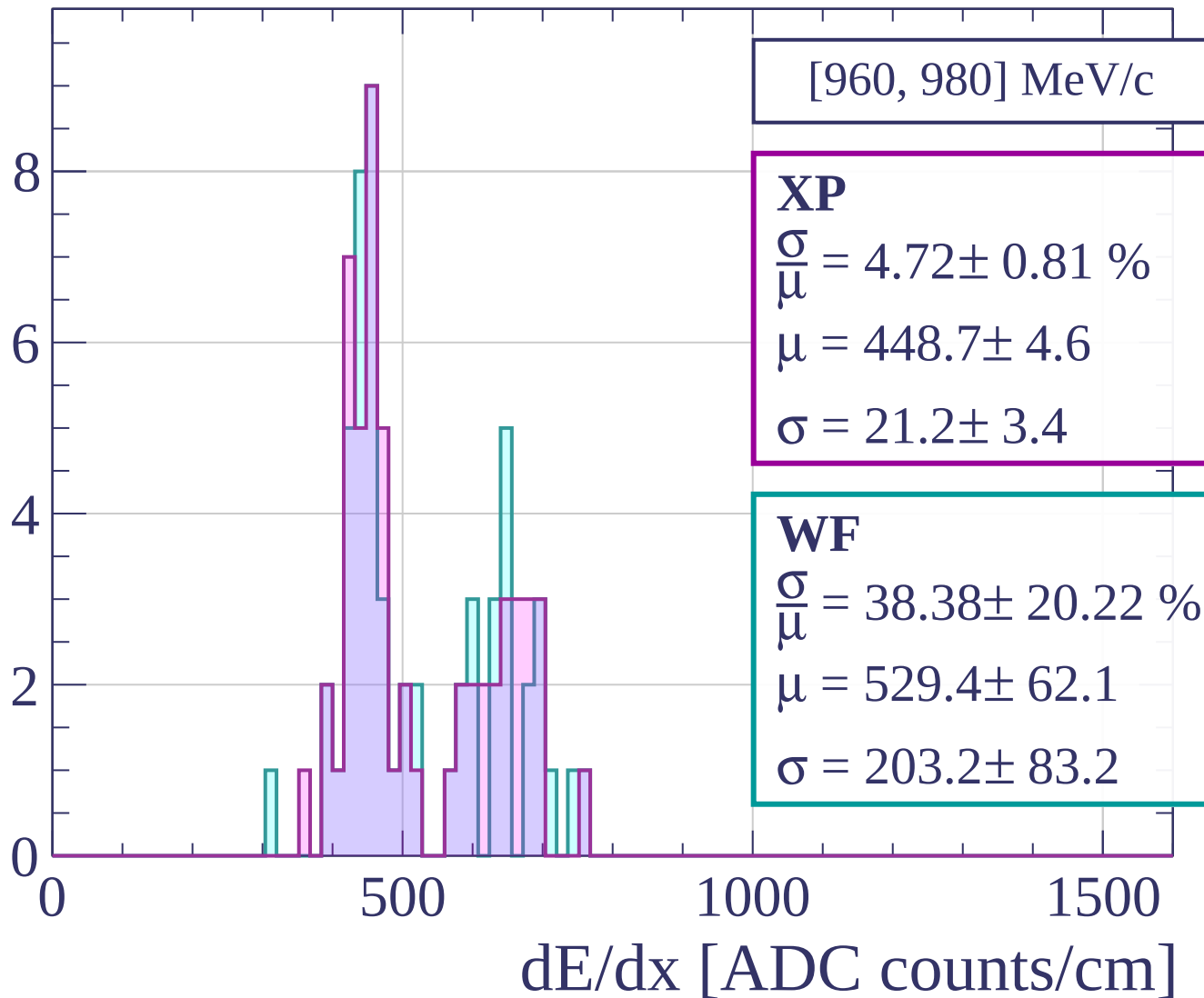
$$\sigma = 33.6 \pm 7.7$$

dE/dx [ADC counts/cm]

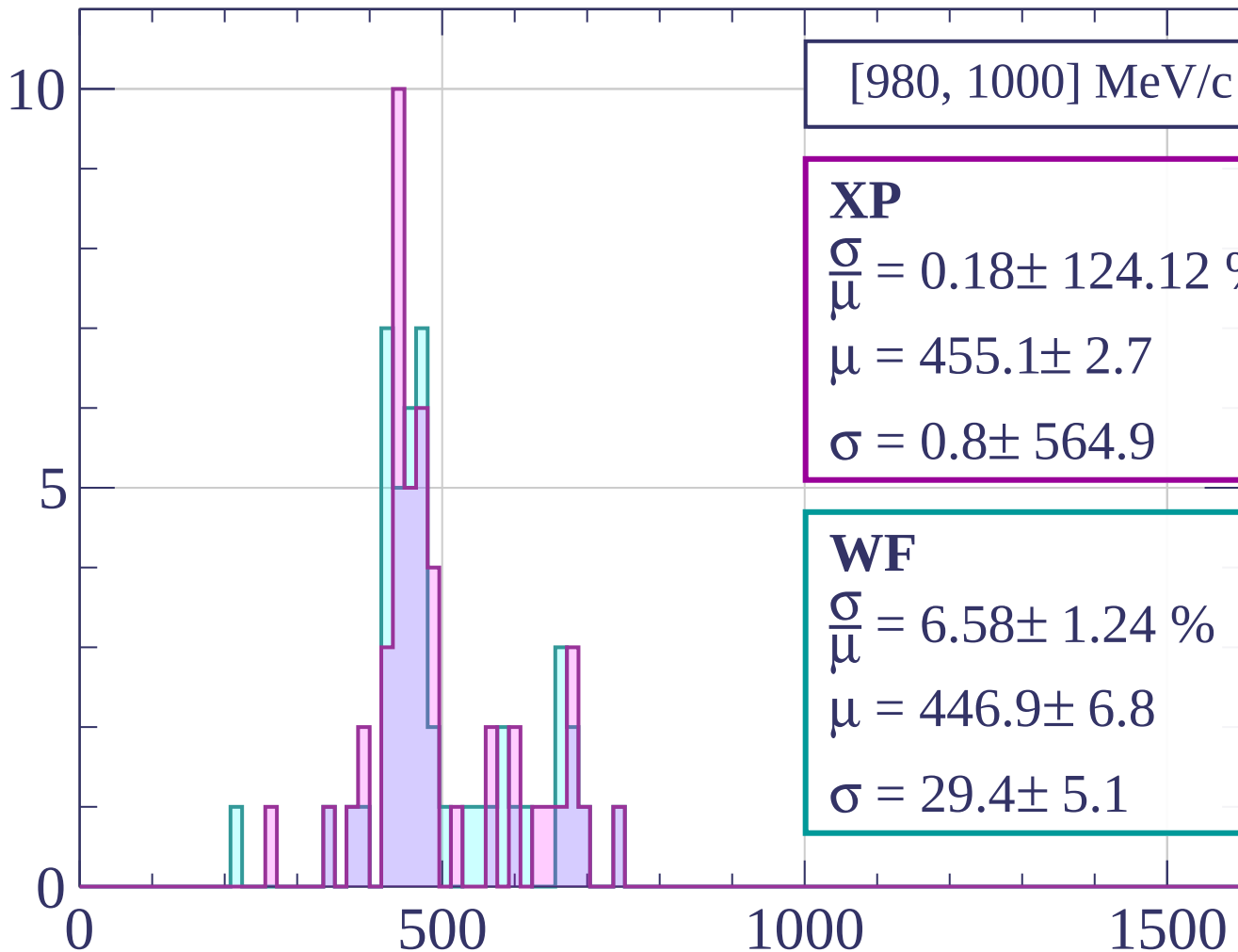
Count



Count

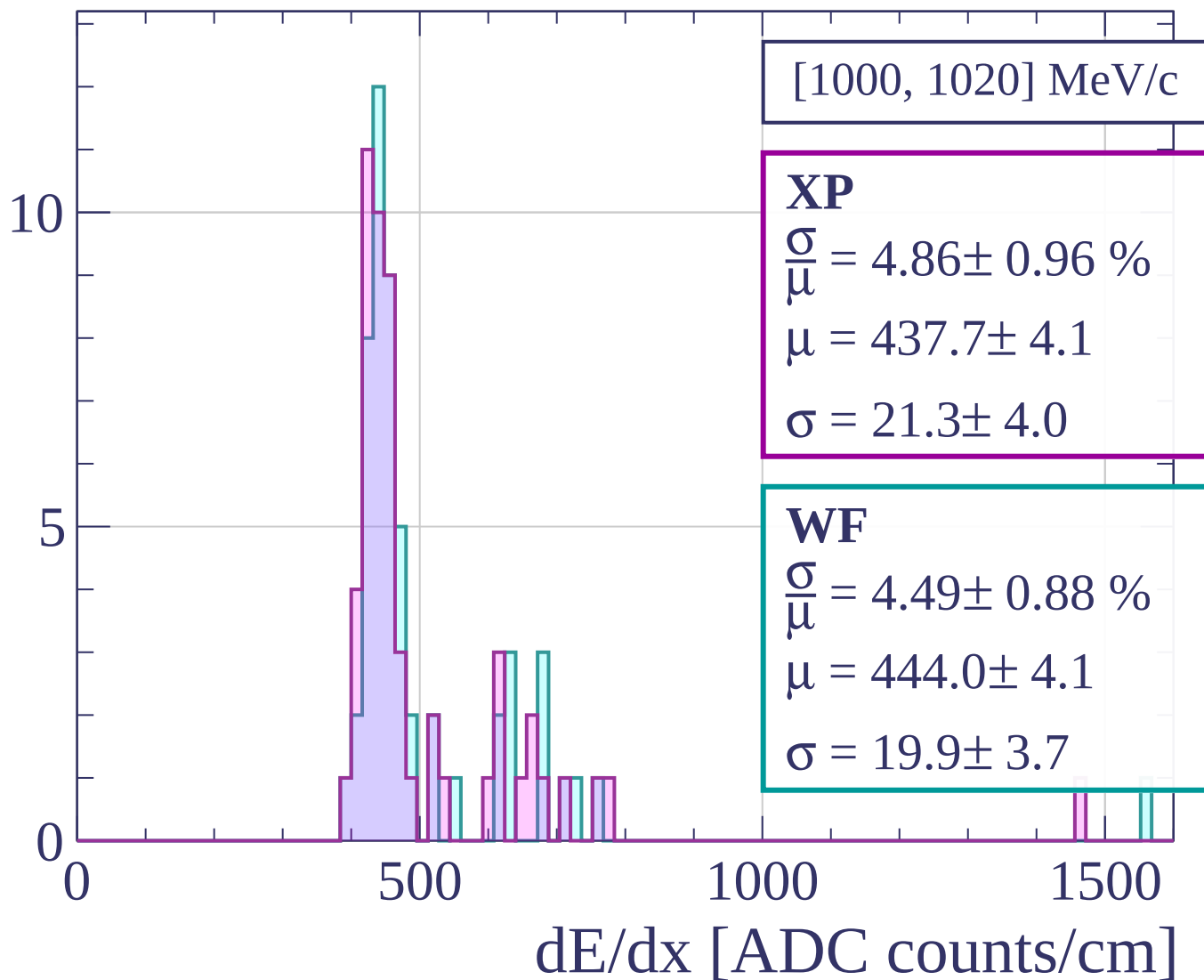


Count

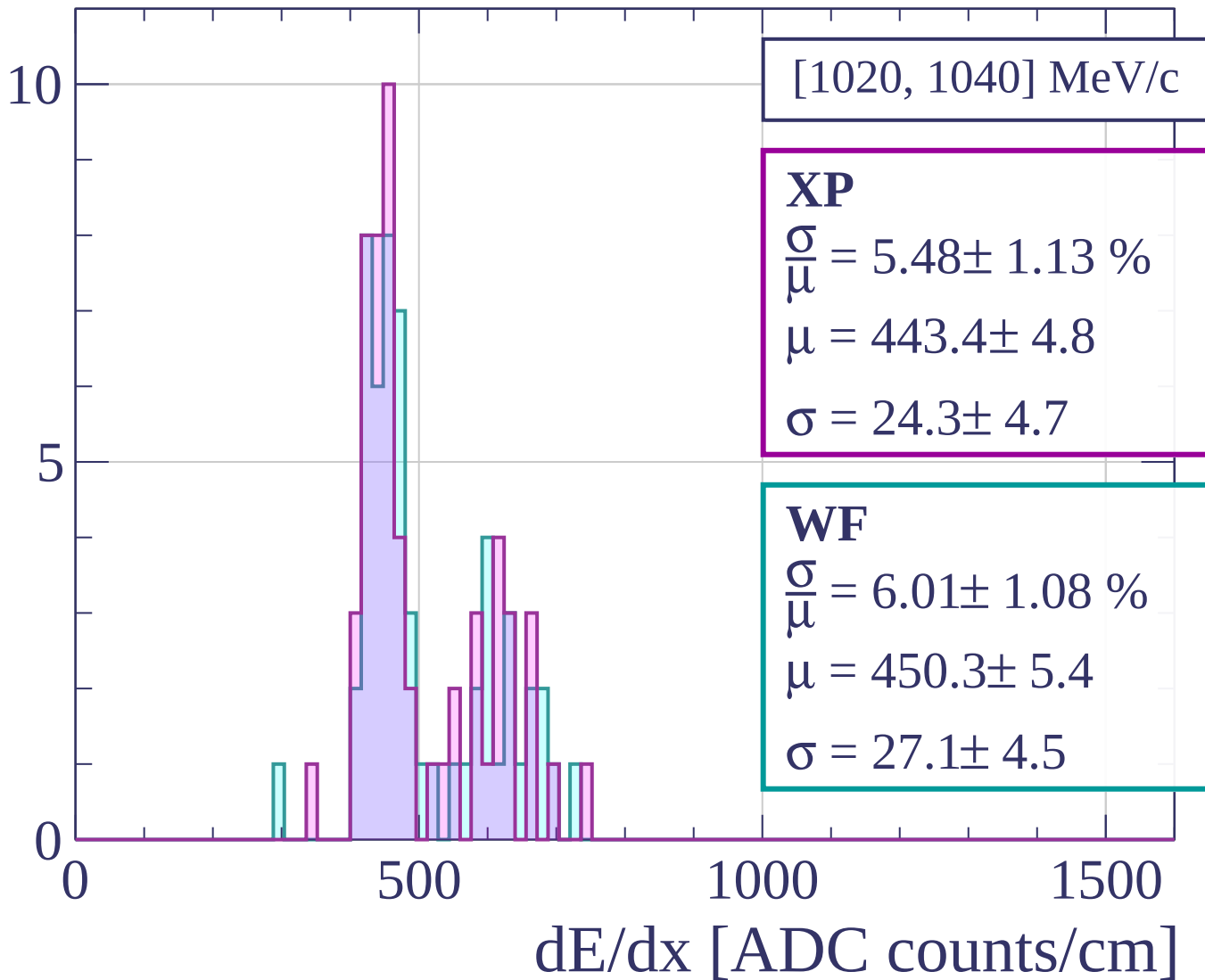


dE/dx [ADC counts/cm]

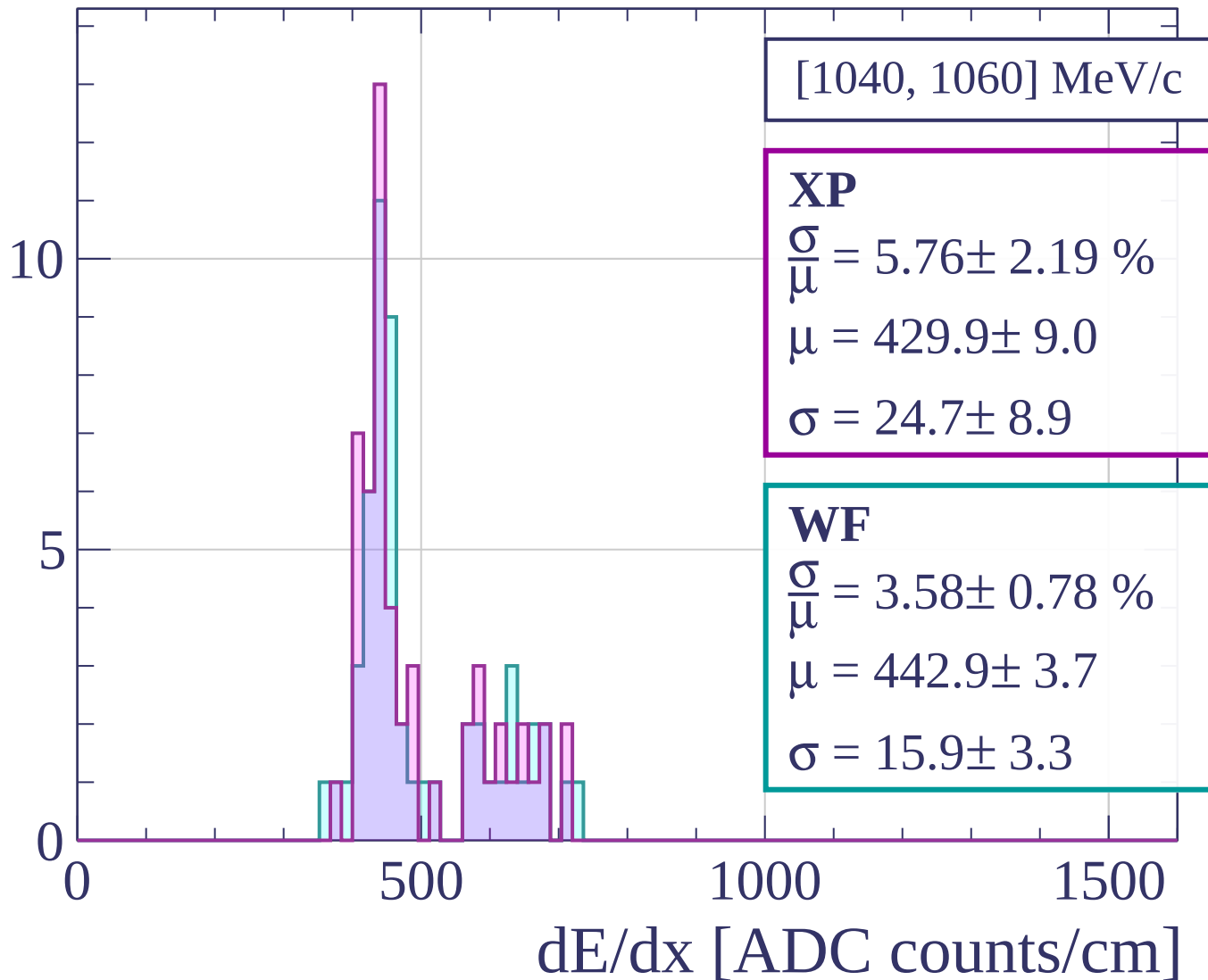
Count



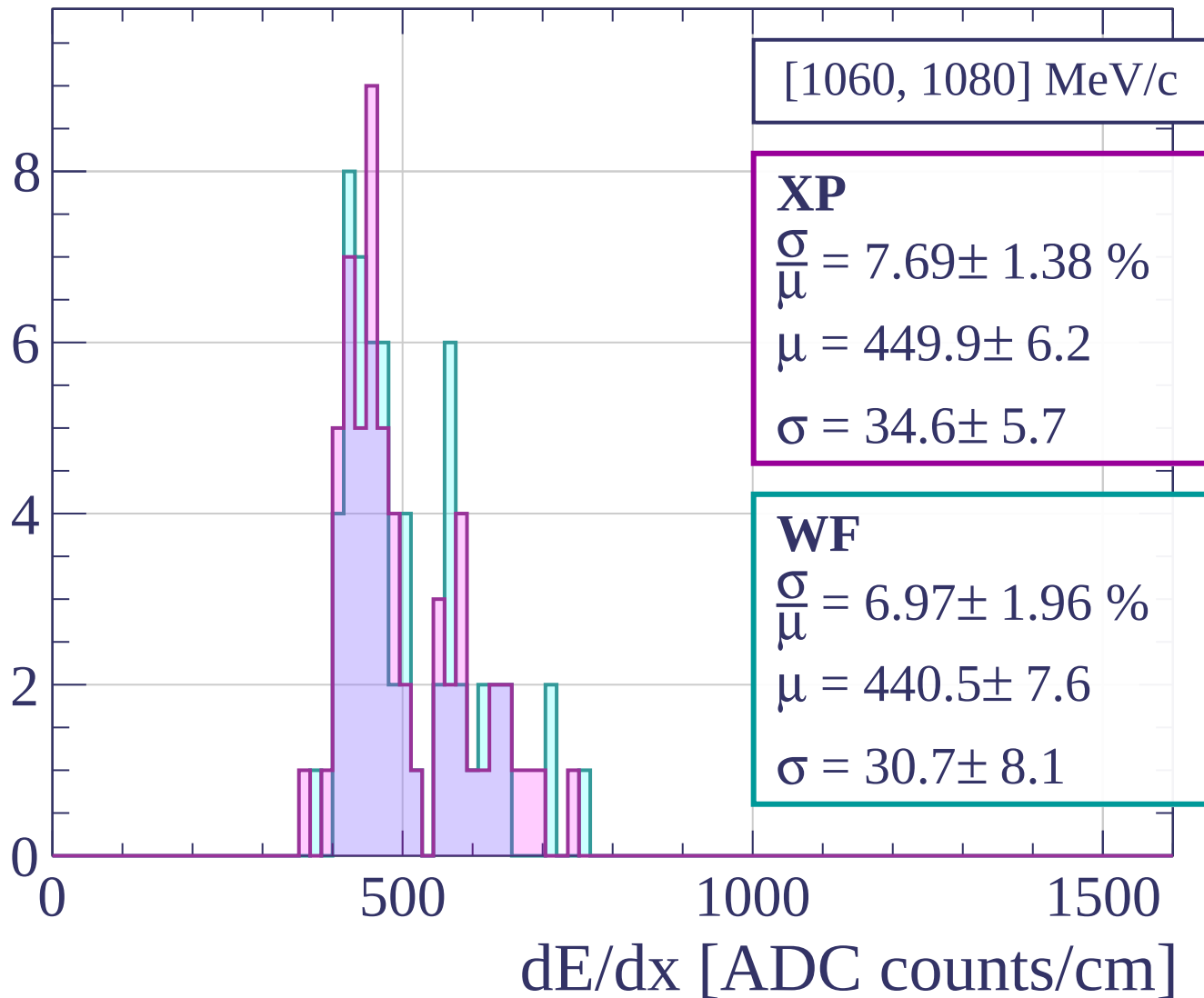
Count



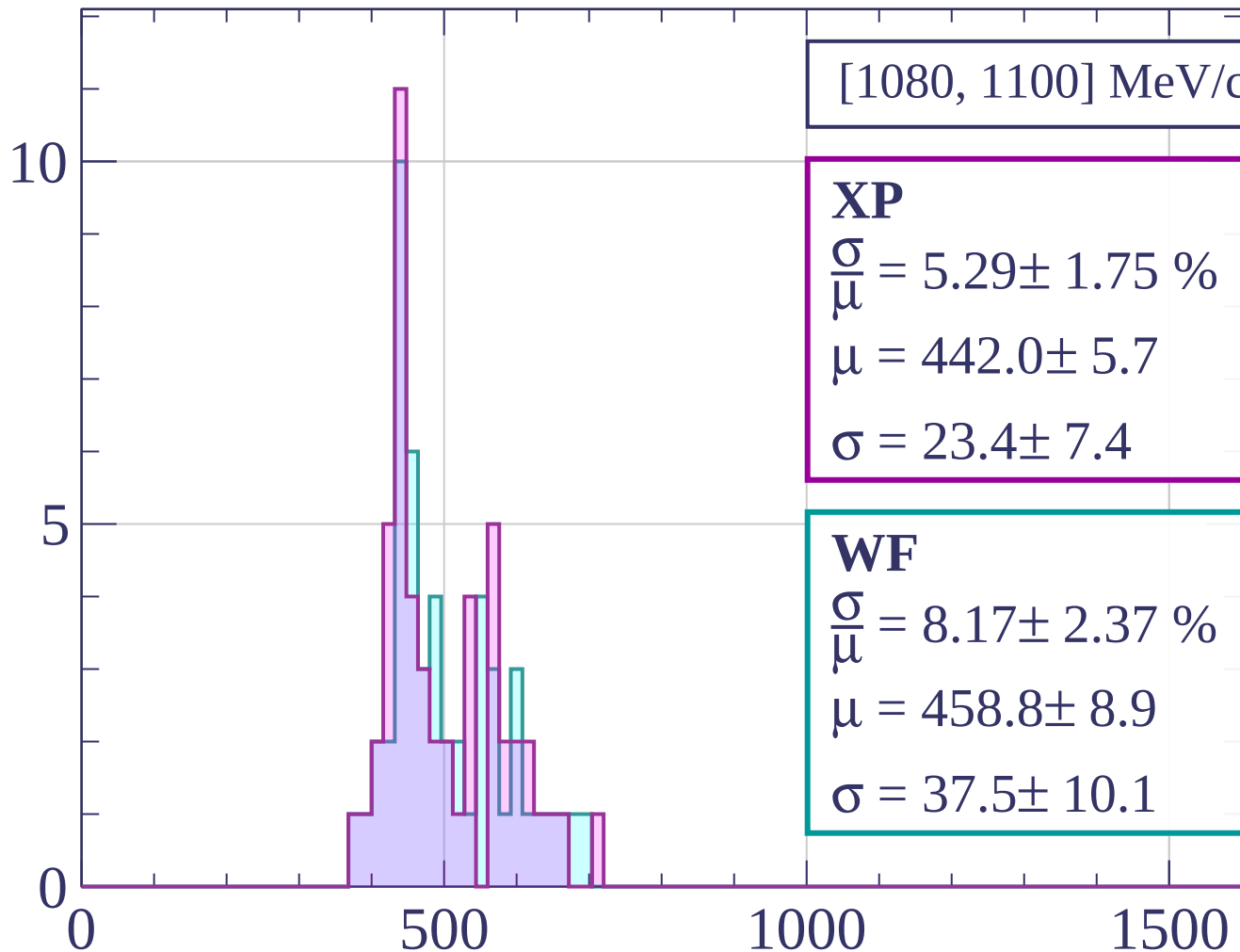
Count



Count

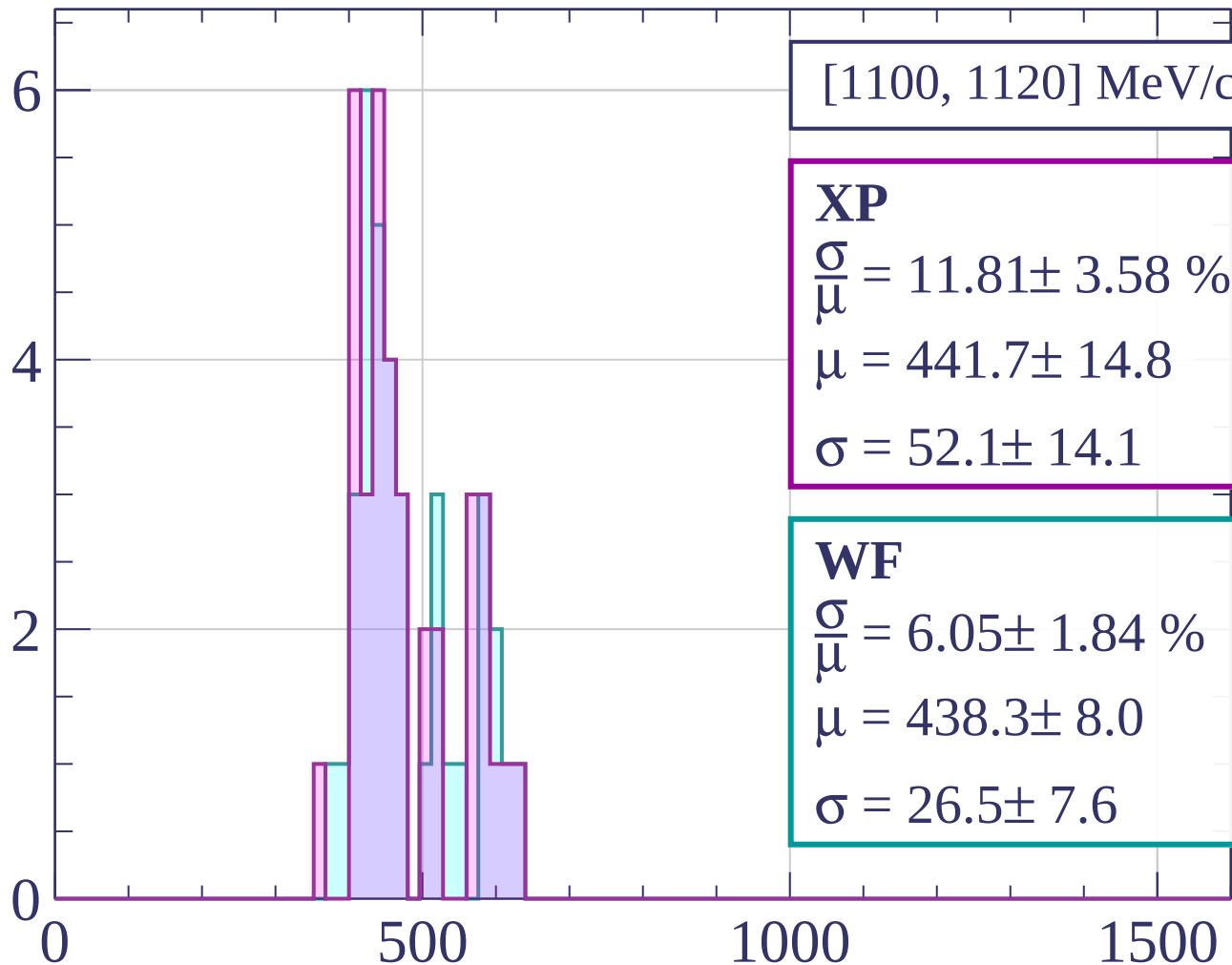


Count



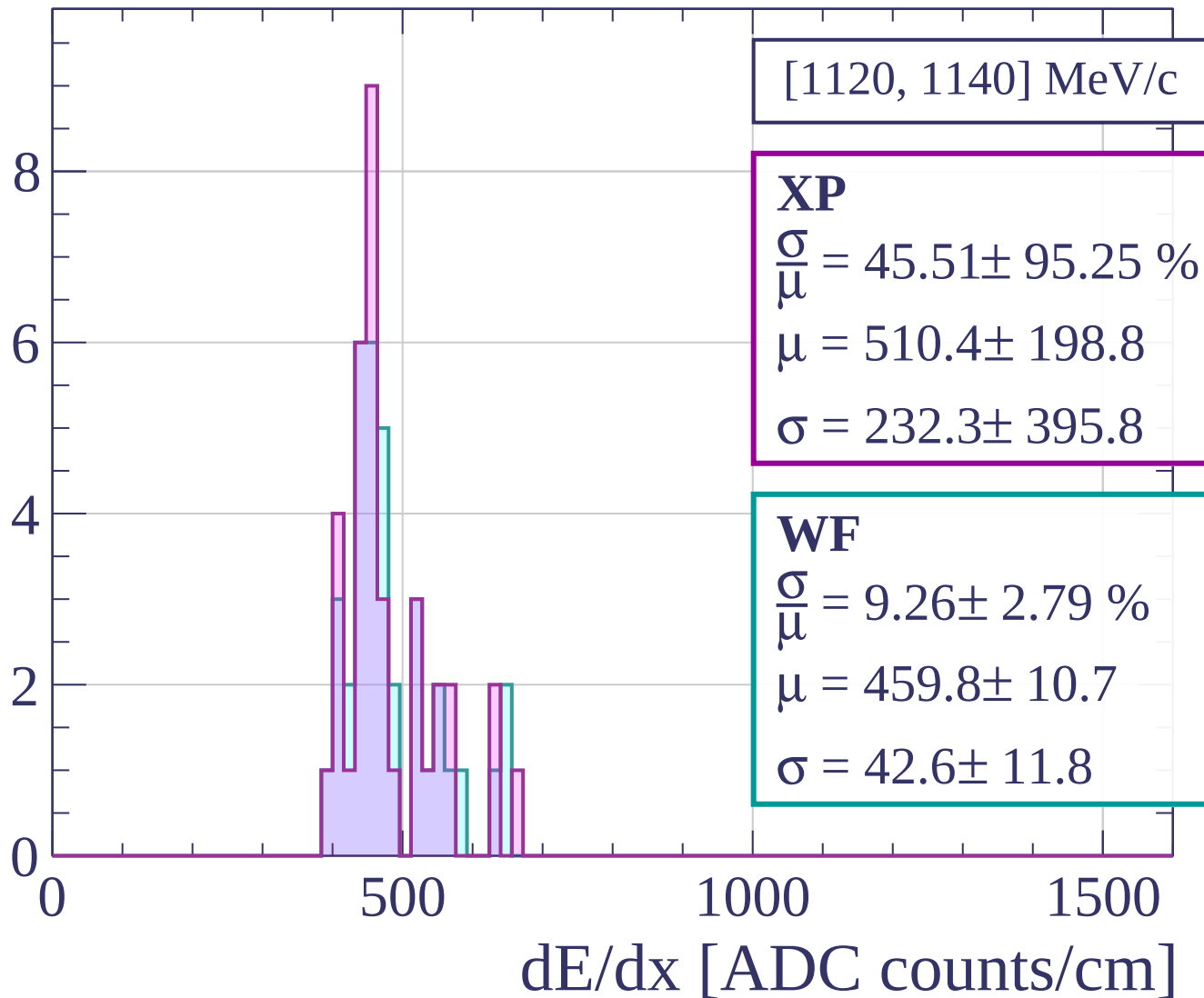
dE/dx [ADC counts/cm]

Count

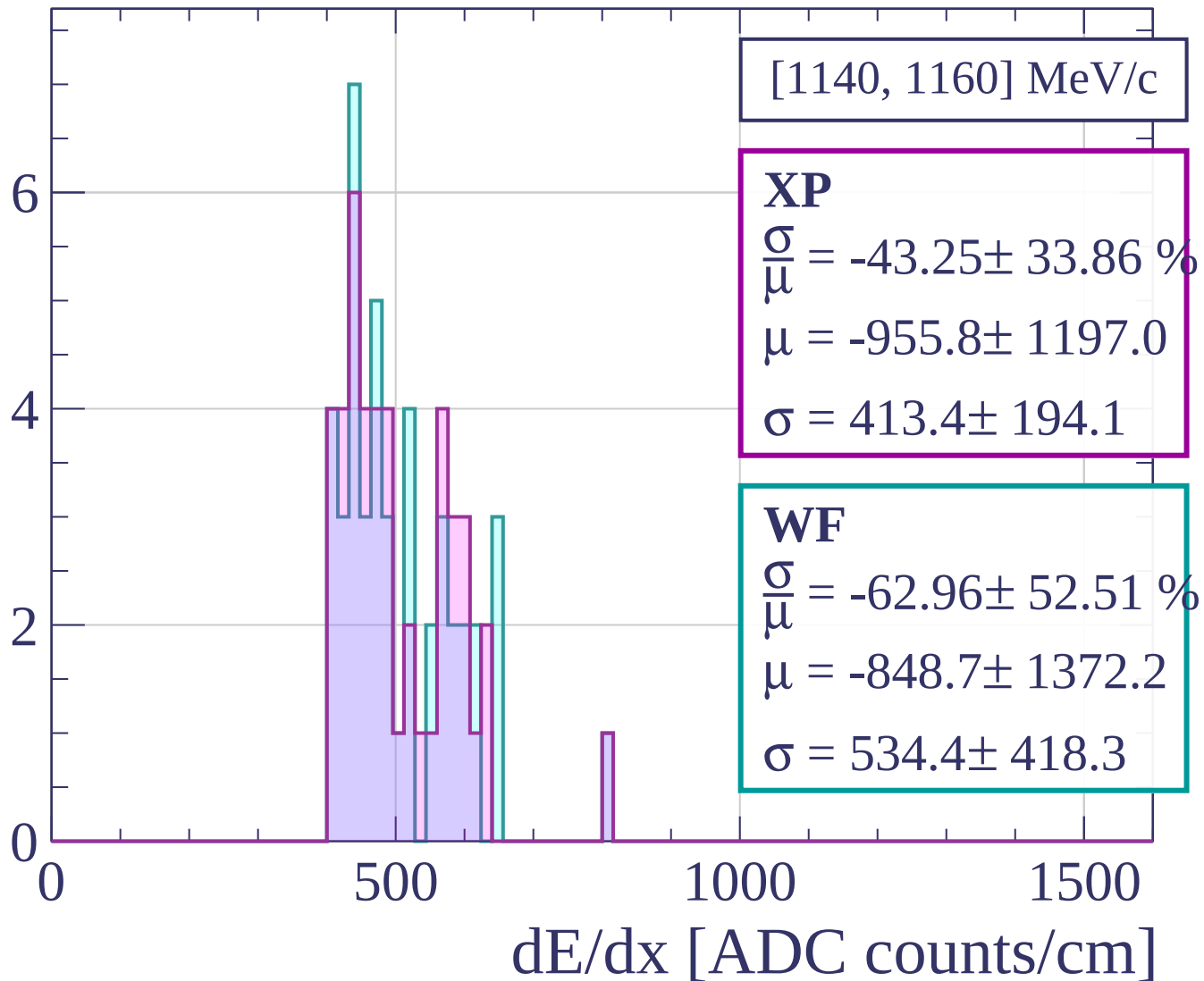


dE/dx [ADC counts/cm]

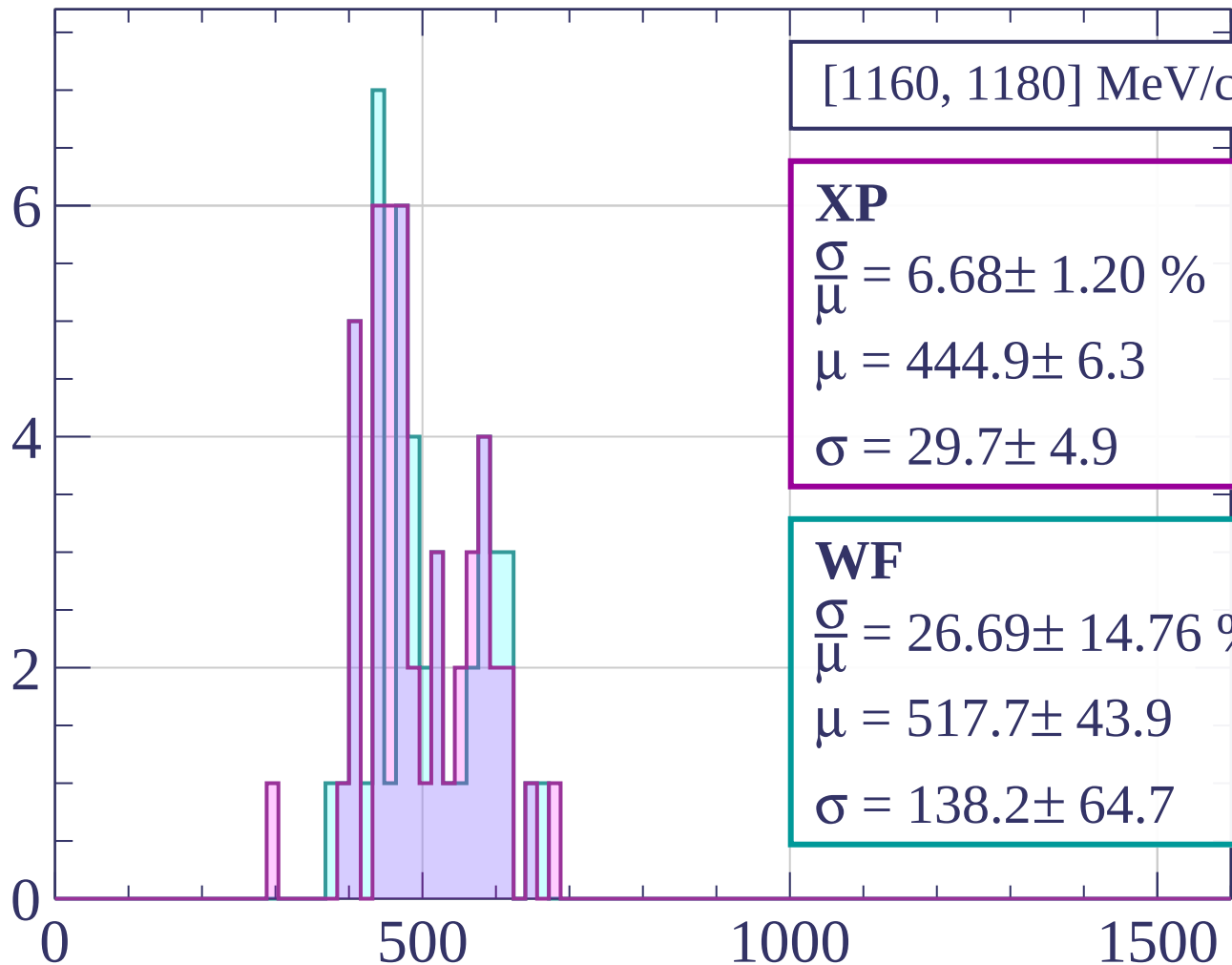
Count



Count

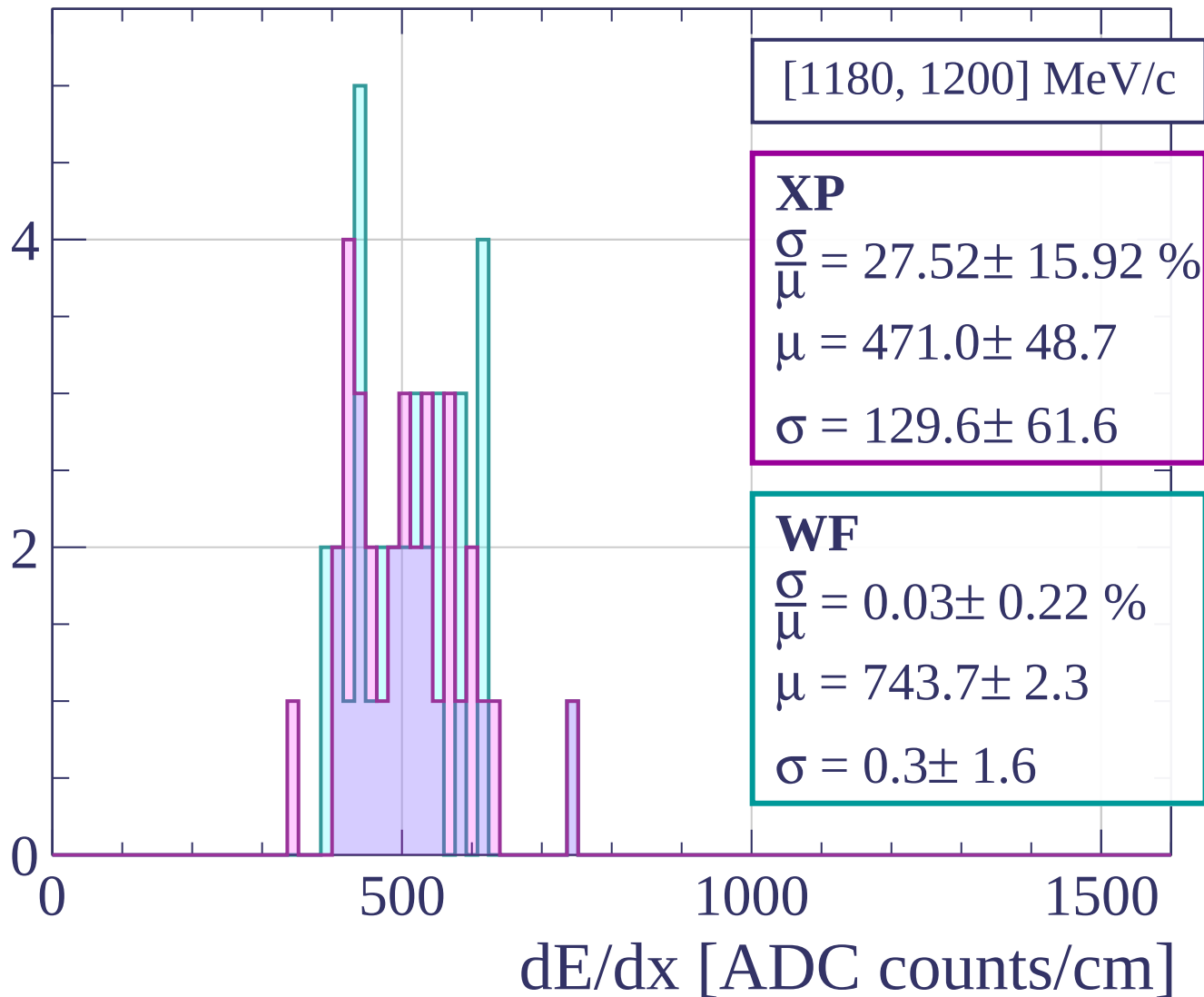


Count

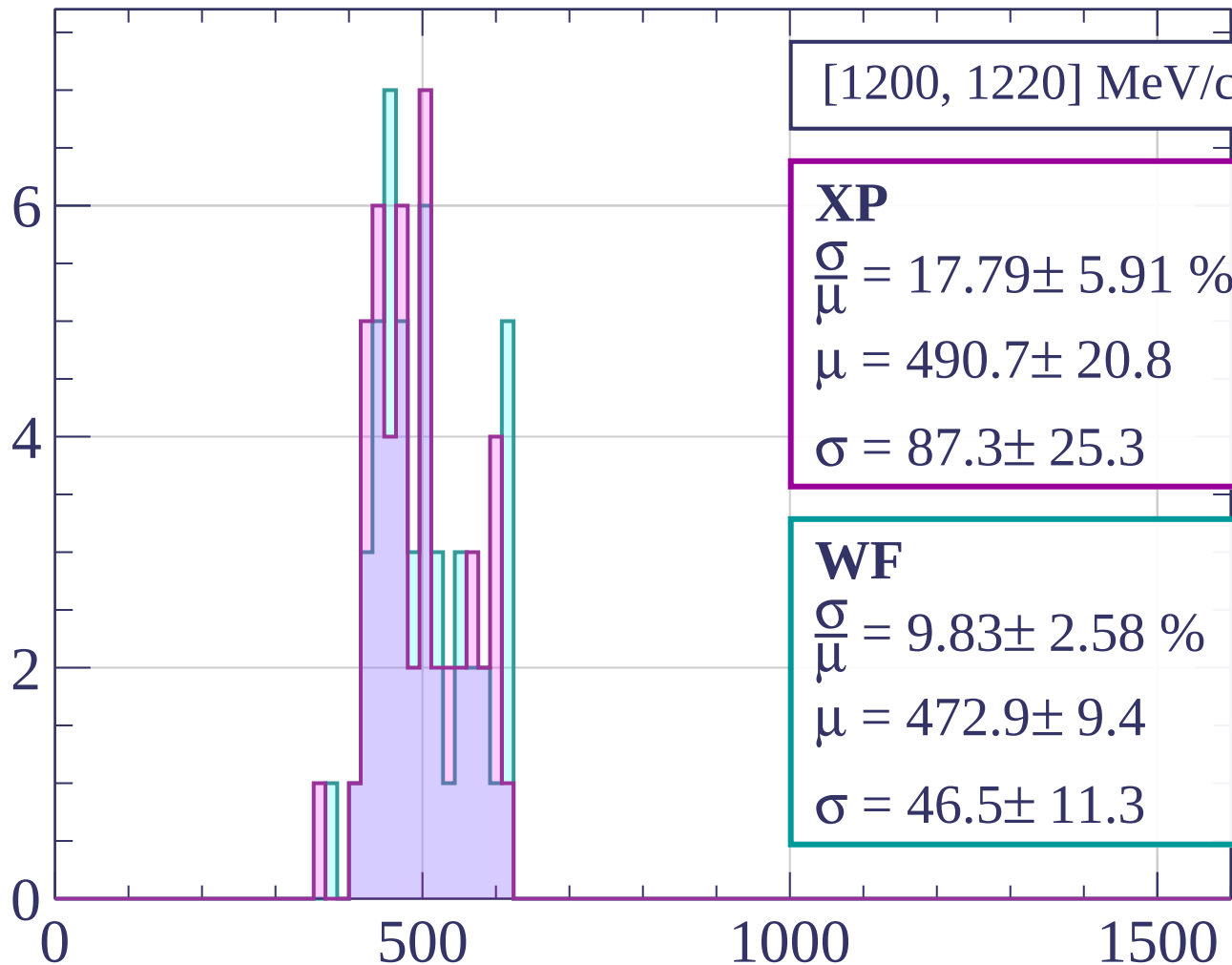


dE/dx [ADC counts/cm]

Count



Count



[1200, 1220] MeV/c

XP

$$\frac{\sigma}{\mu} = 17.79 \pm 5.91 \%$$

$$\mu = 490.7 \pm 20.8$$

$$\sigma = 87.3 \pm 25.3$$

WF

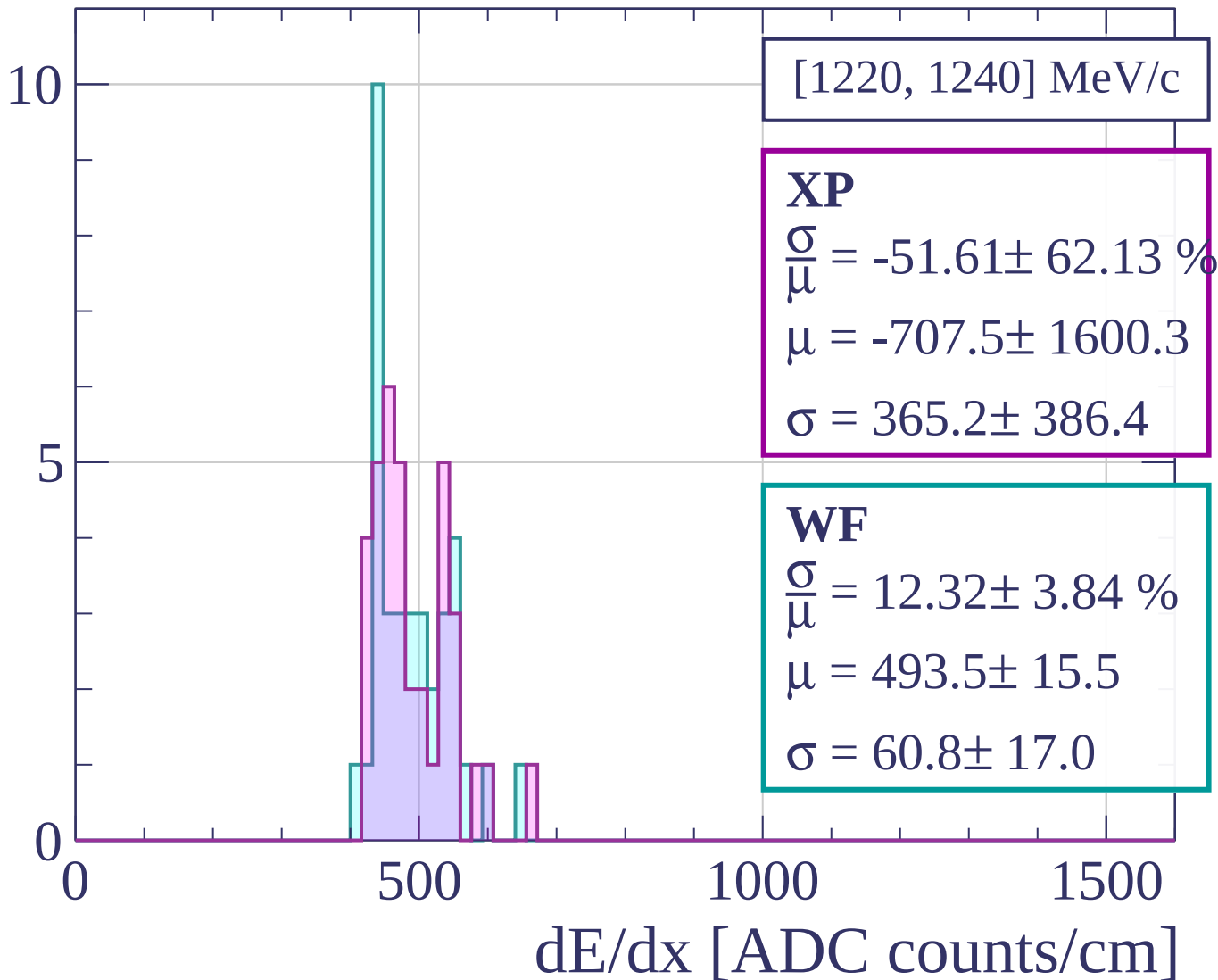
$$\frac{\sigma}{\mu} = 9.83 \pm 2.58 \%$$

$$\mu = 472.9 \pm 9.4$$

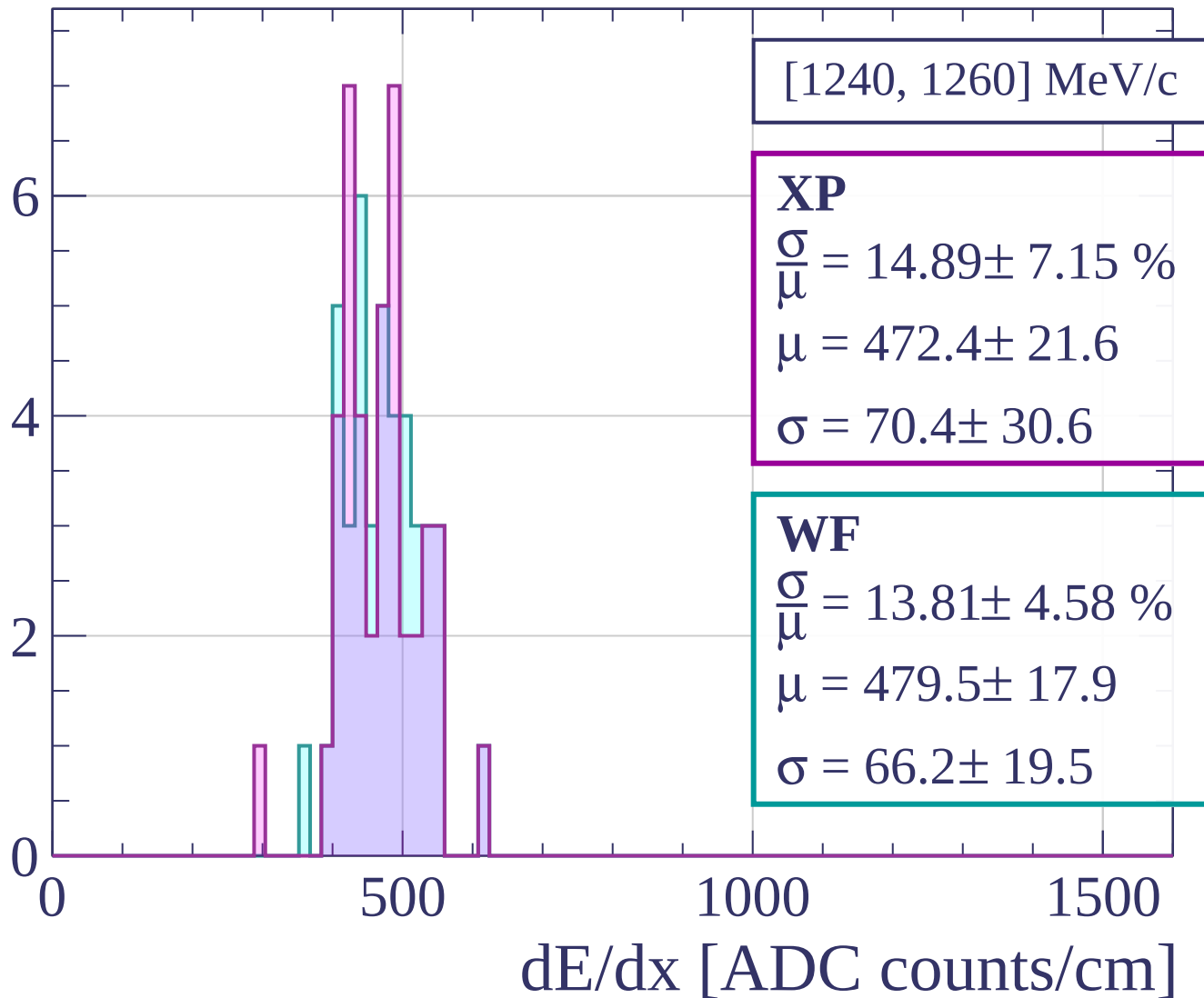
$$\sigma = 46.5 \pm 11.3$$

dE/dx [ADC counts/cm]

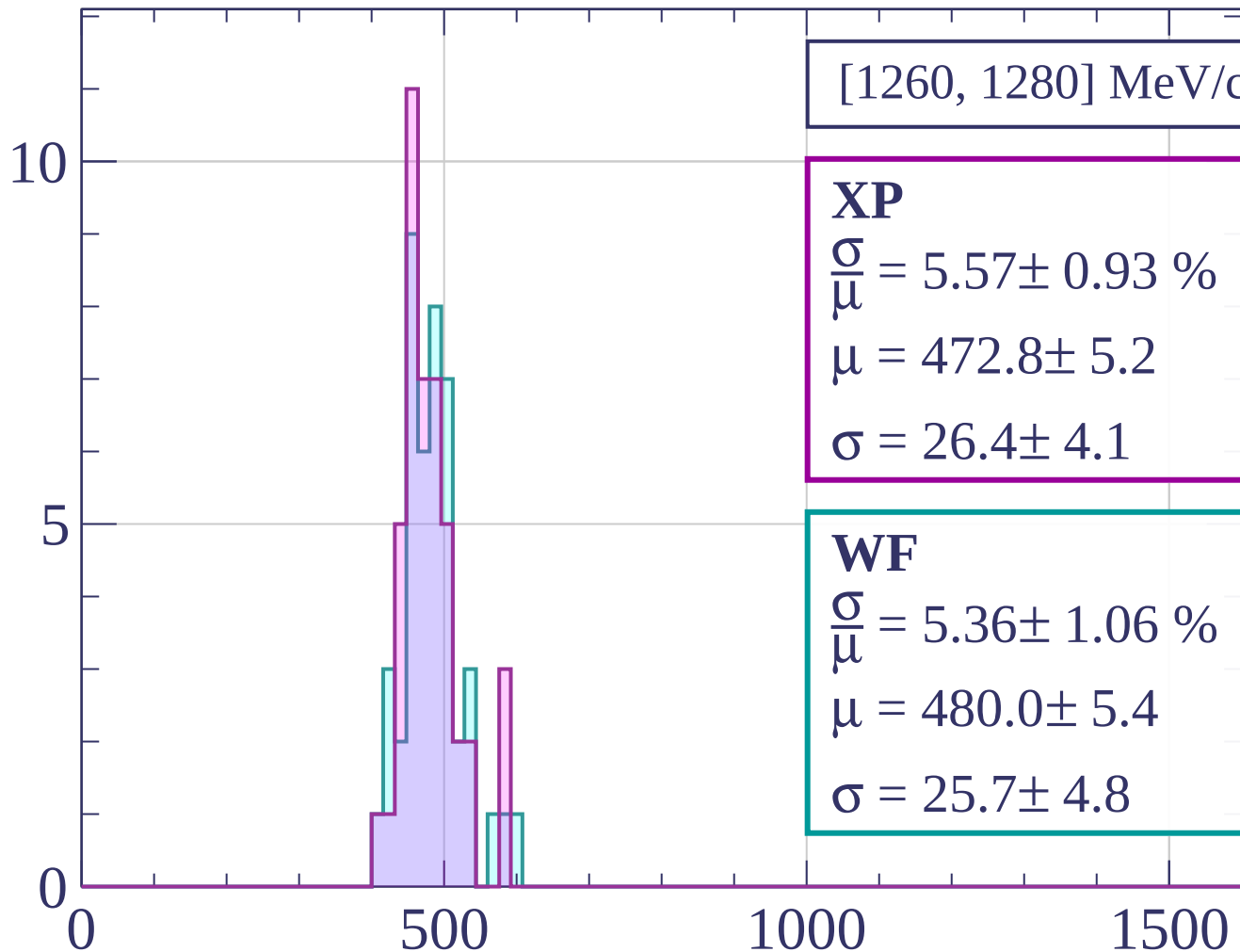
Count



Count

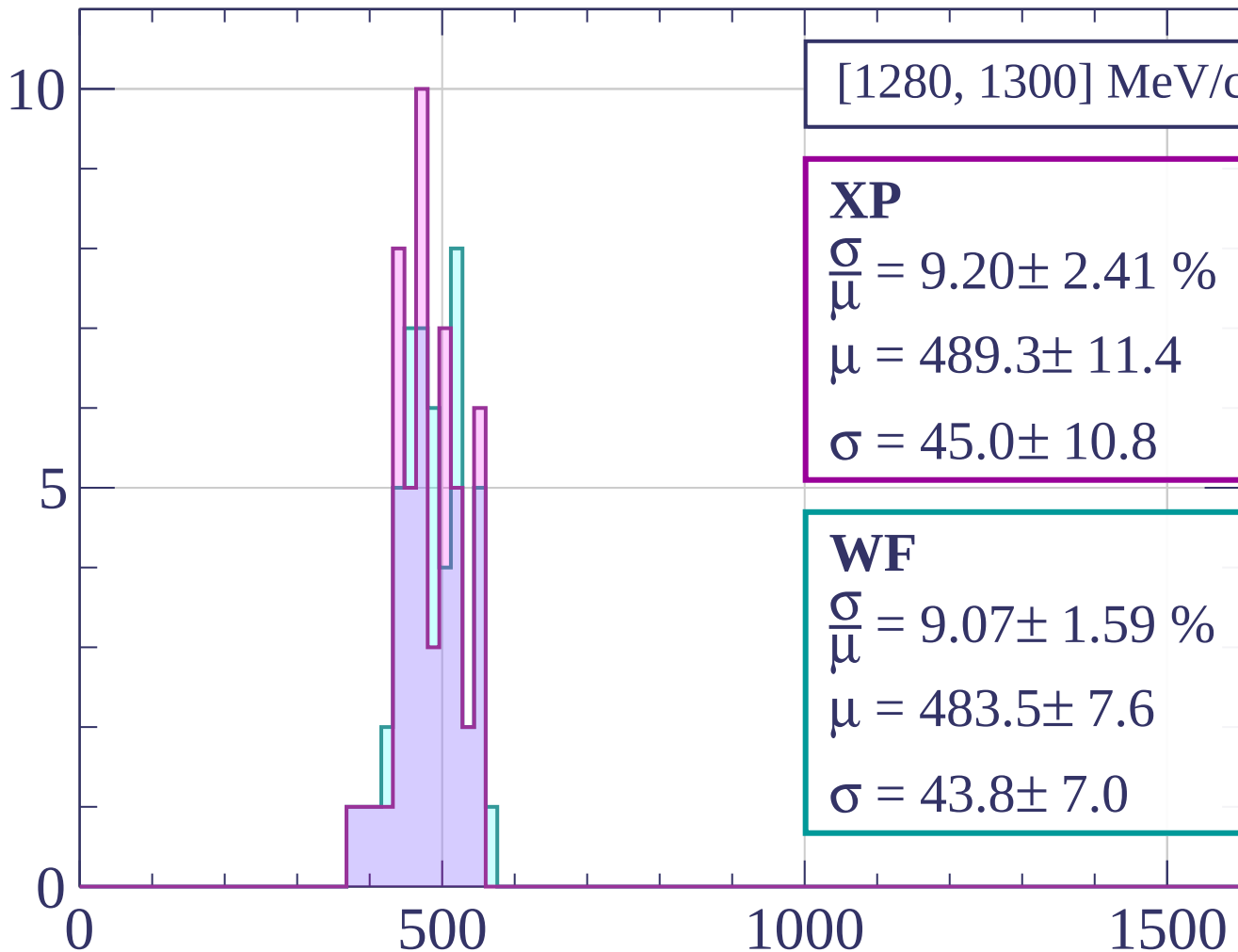


Count



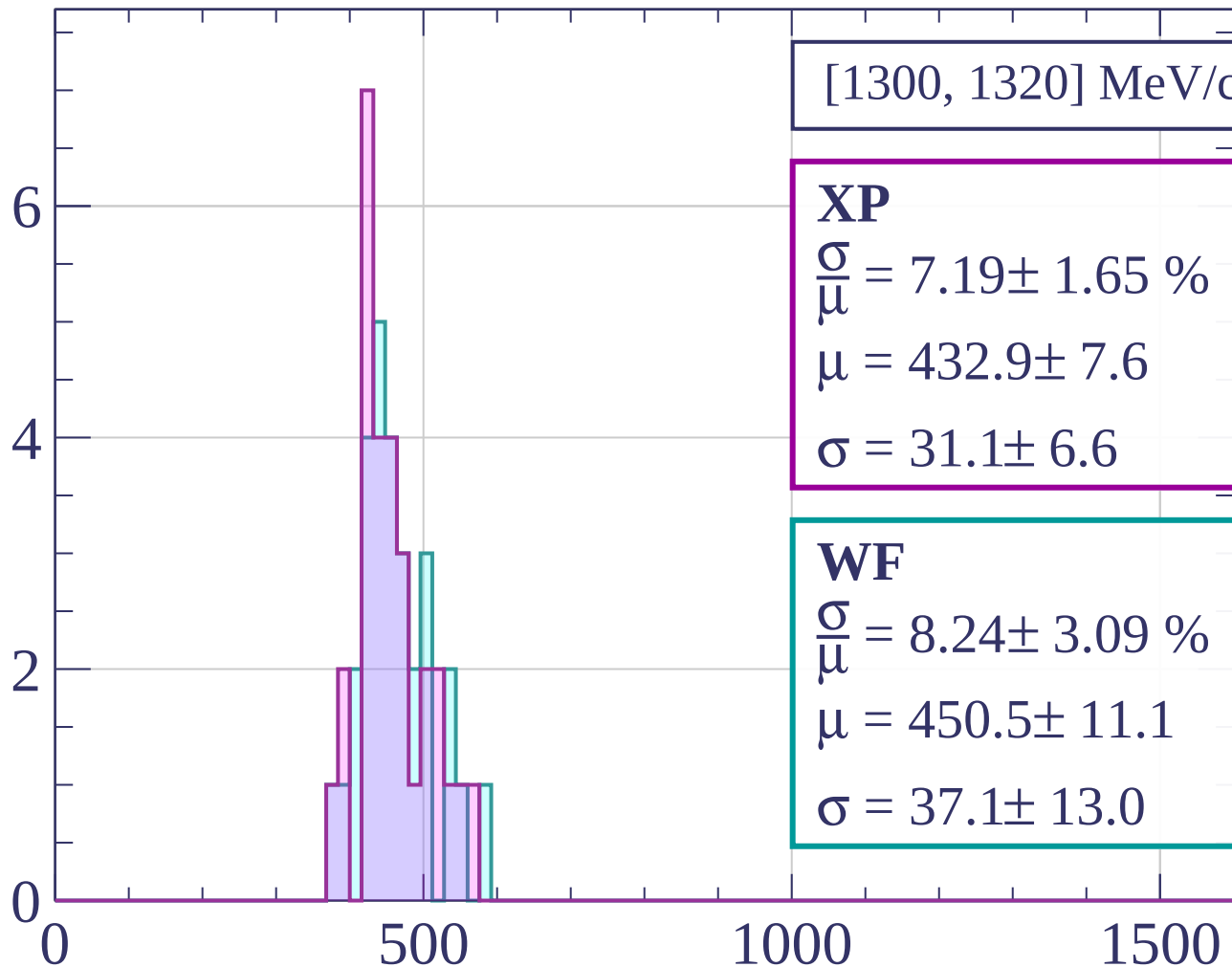
dE/dx [ADC counts/cm]

Count



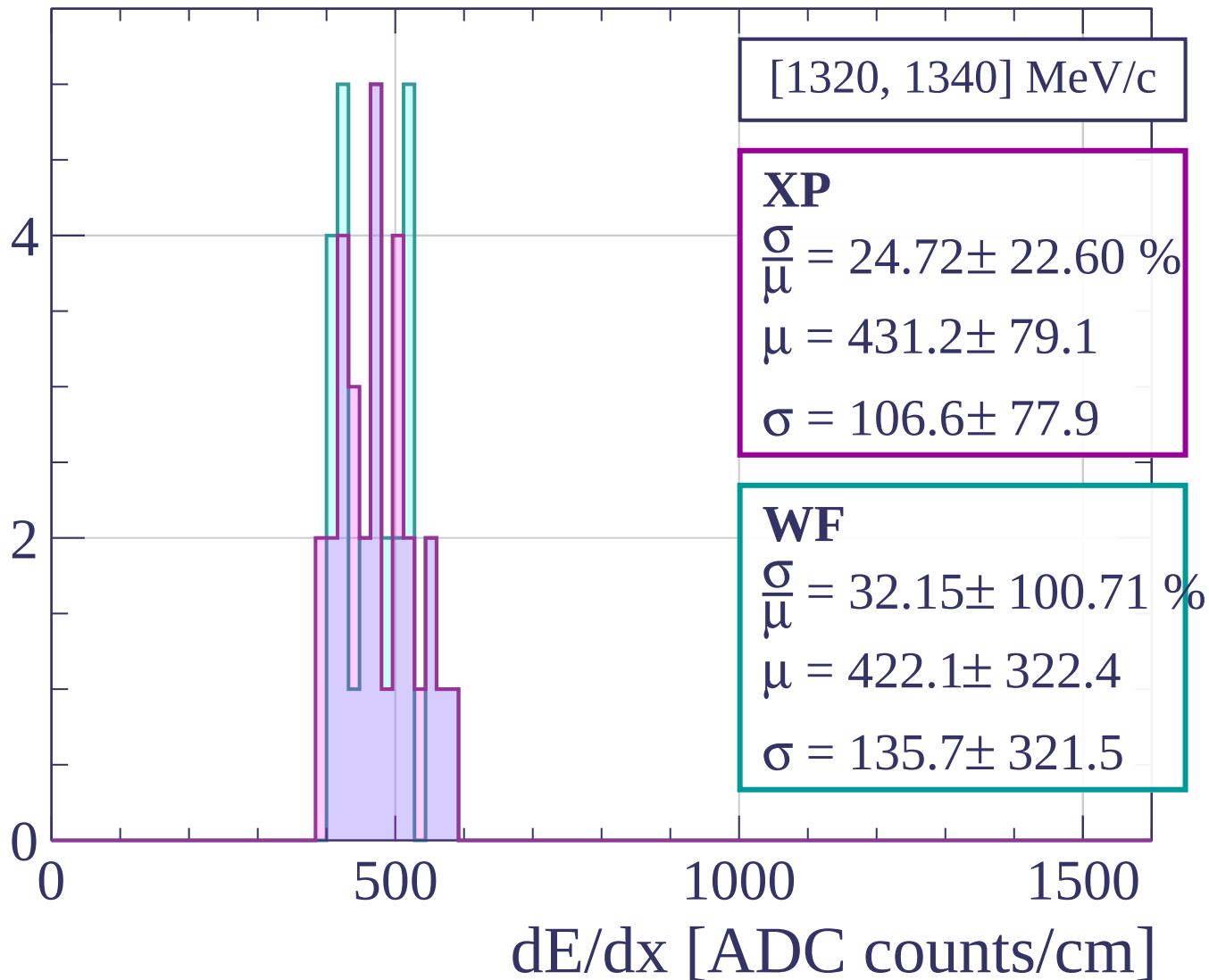
dE/dx [ADC counts/cm]

Count

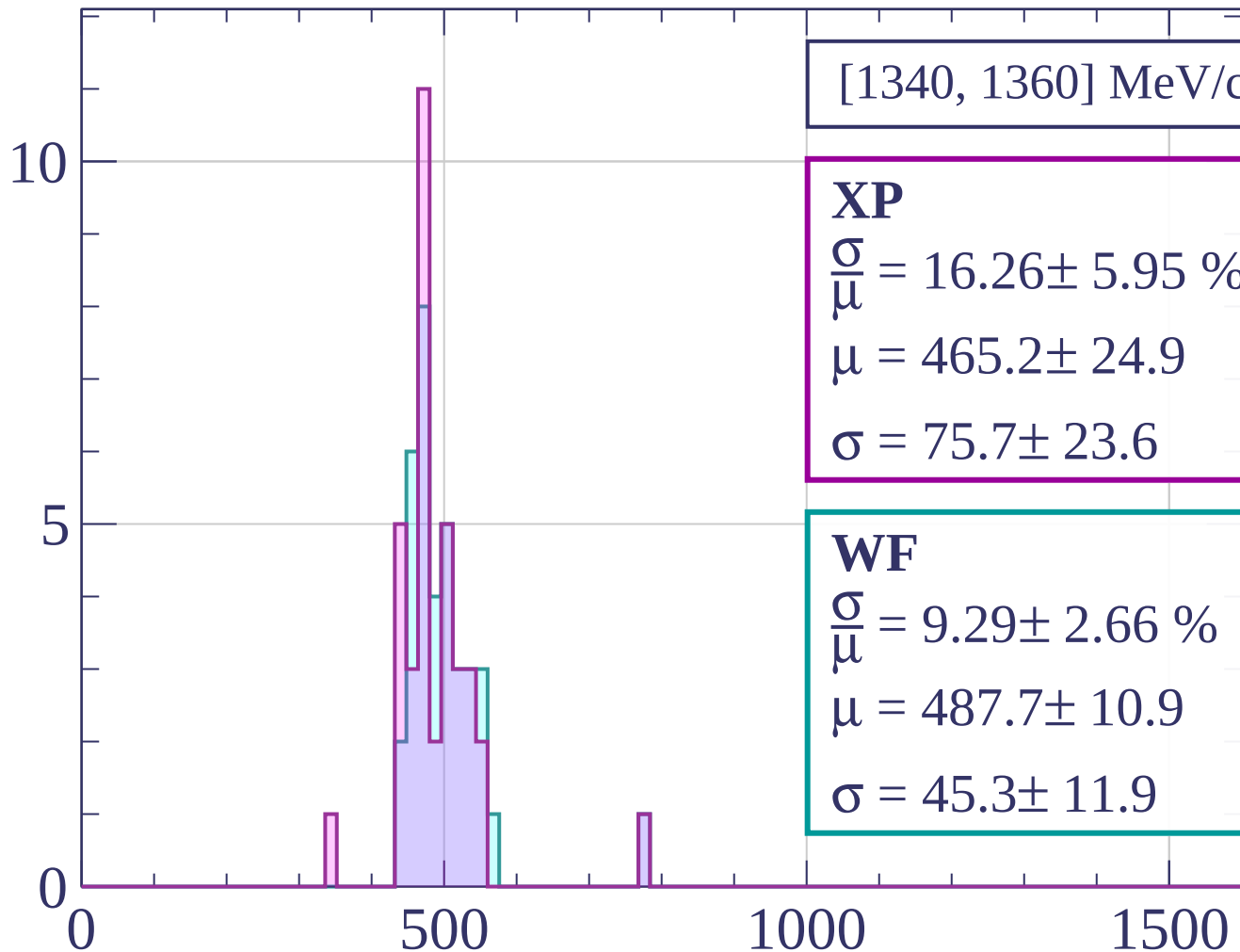


dE/dx [ADC counts/cm]

Count

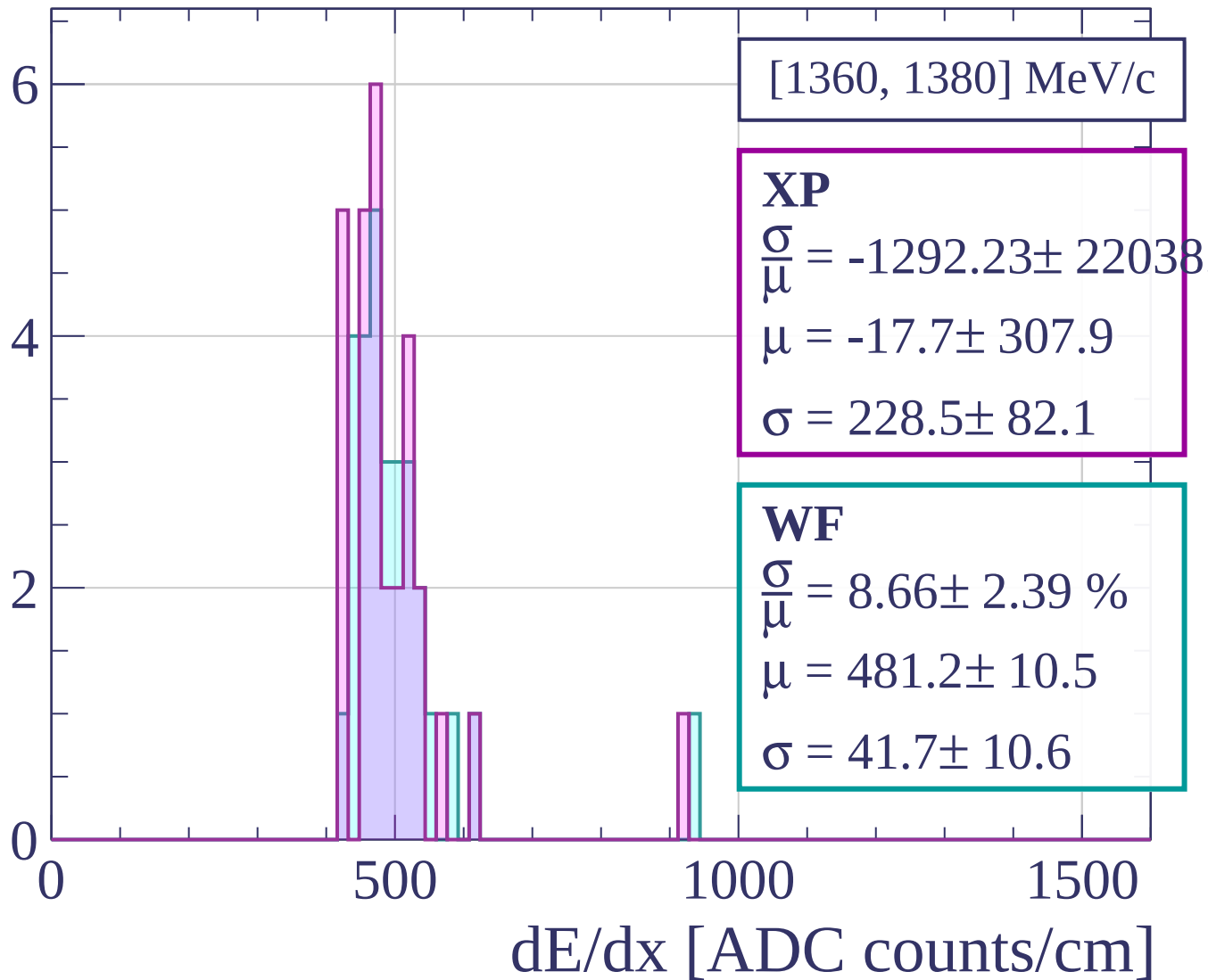


Count

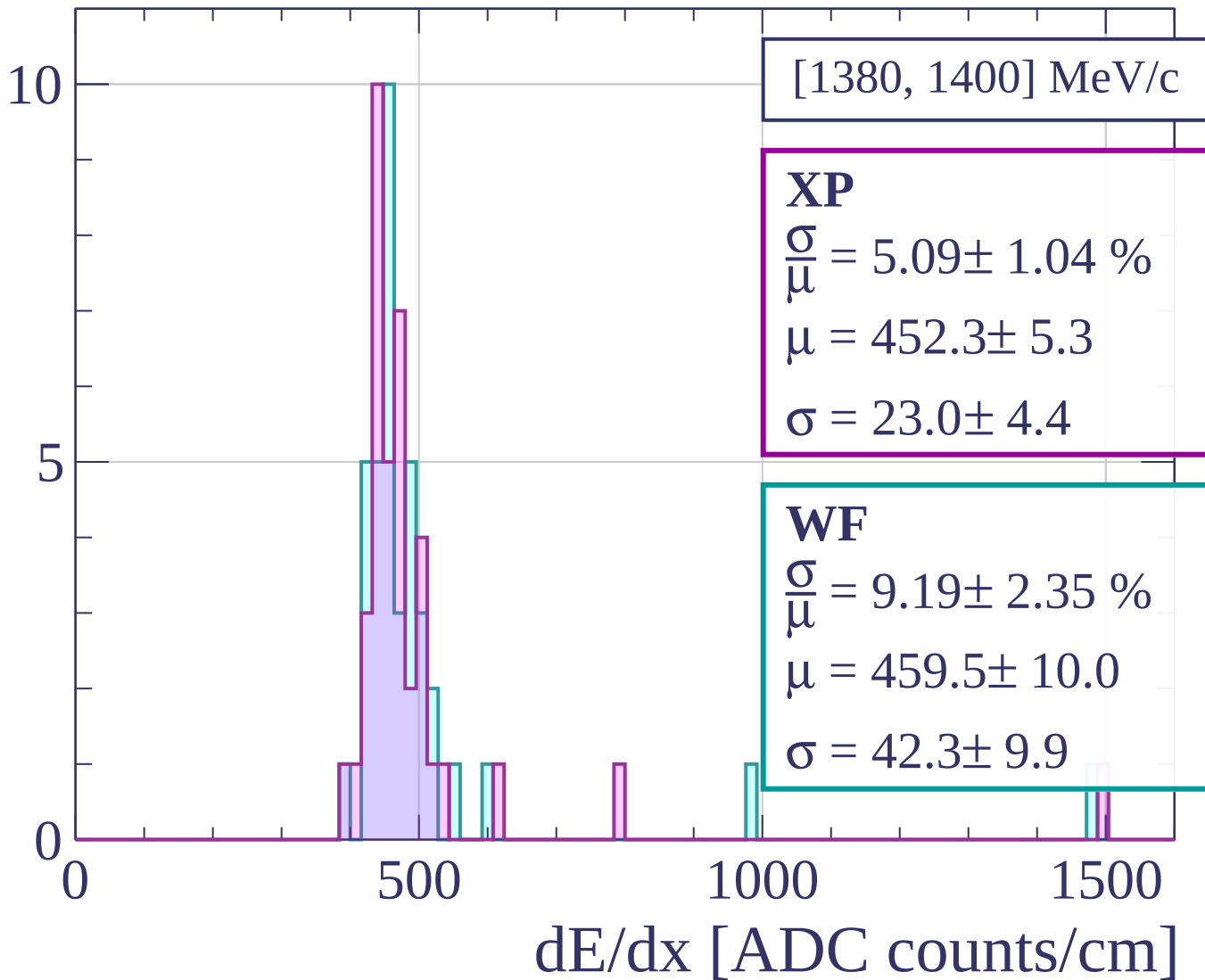


dE/dx [ADC counts/cm]

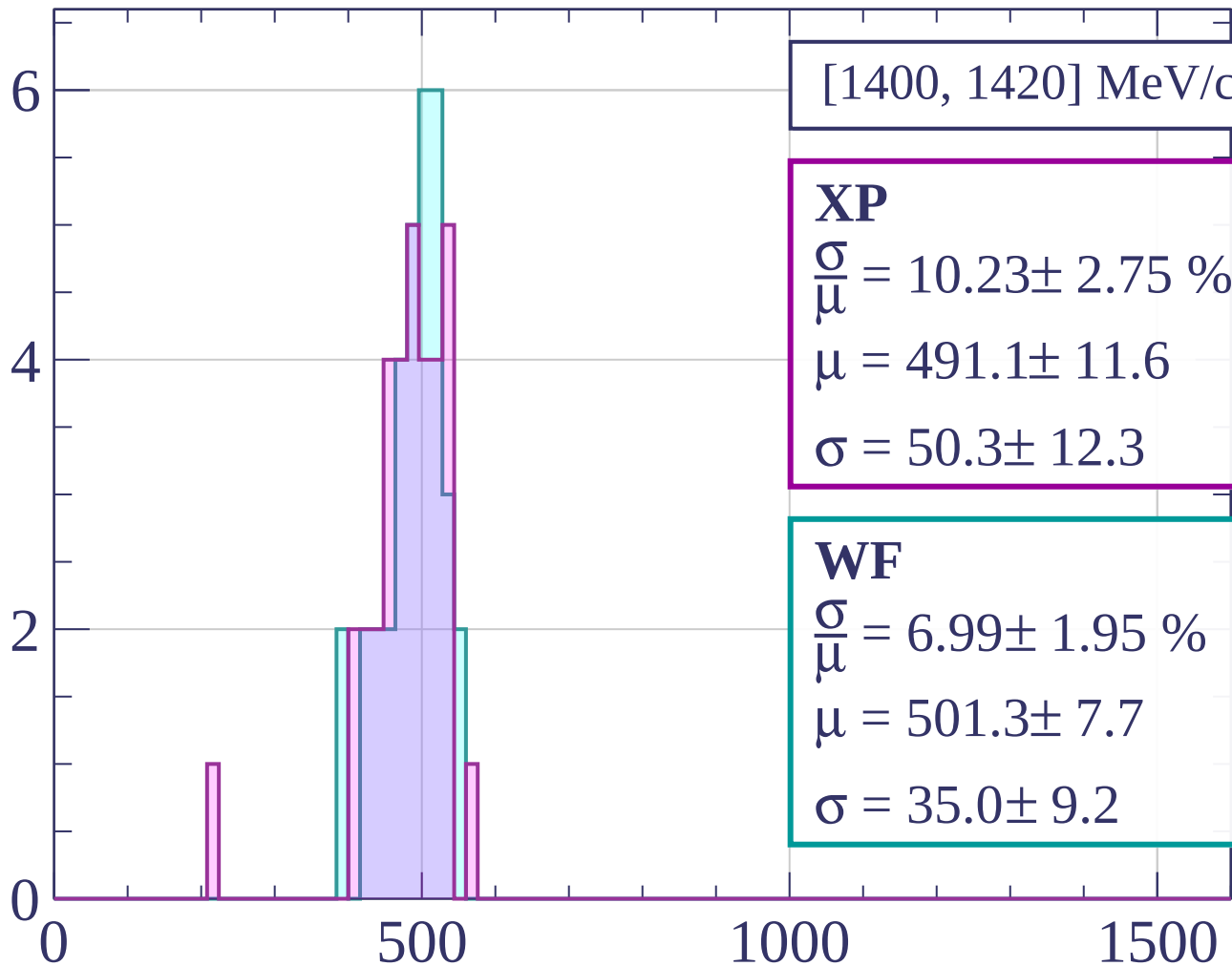
Count



Count



Count



[1400, 1420] MeV/c

XP

$$\frac{\sigma}{\mu} = 10.23 \pm 2.75 \%$$

$$\mu = 491.1 \pm 11.6$$

$$\sigma = 50.3 \pm 12.3$$

WF

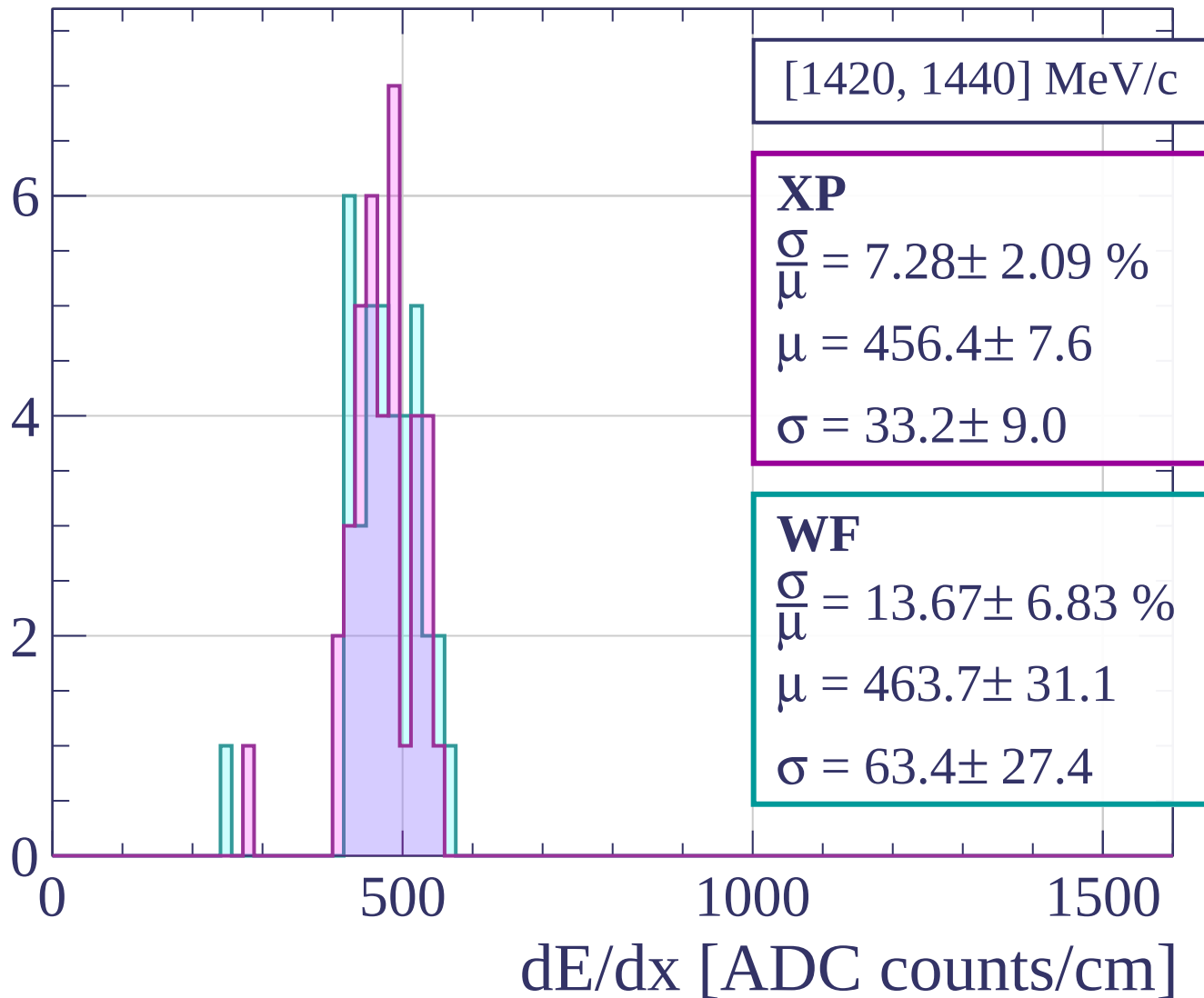
$$\frac{\sigma}{\mu} = 6.99 \pm 1.95 \%$$

$$\mu = 501.3 \pm 7.7$$

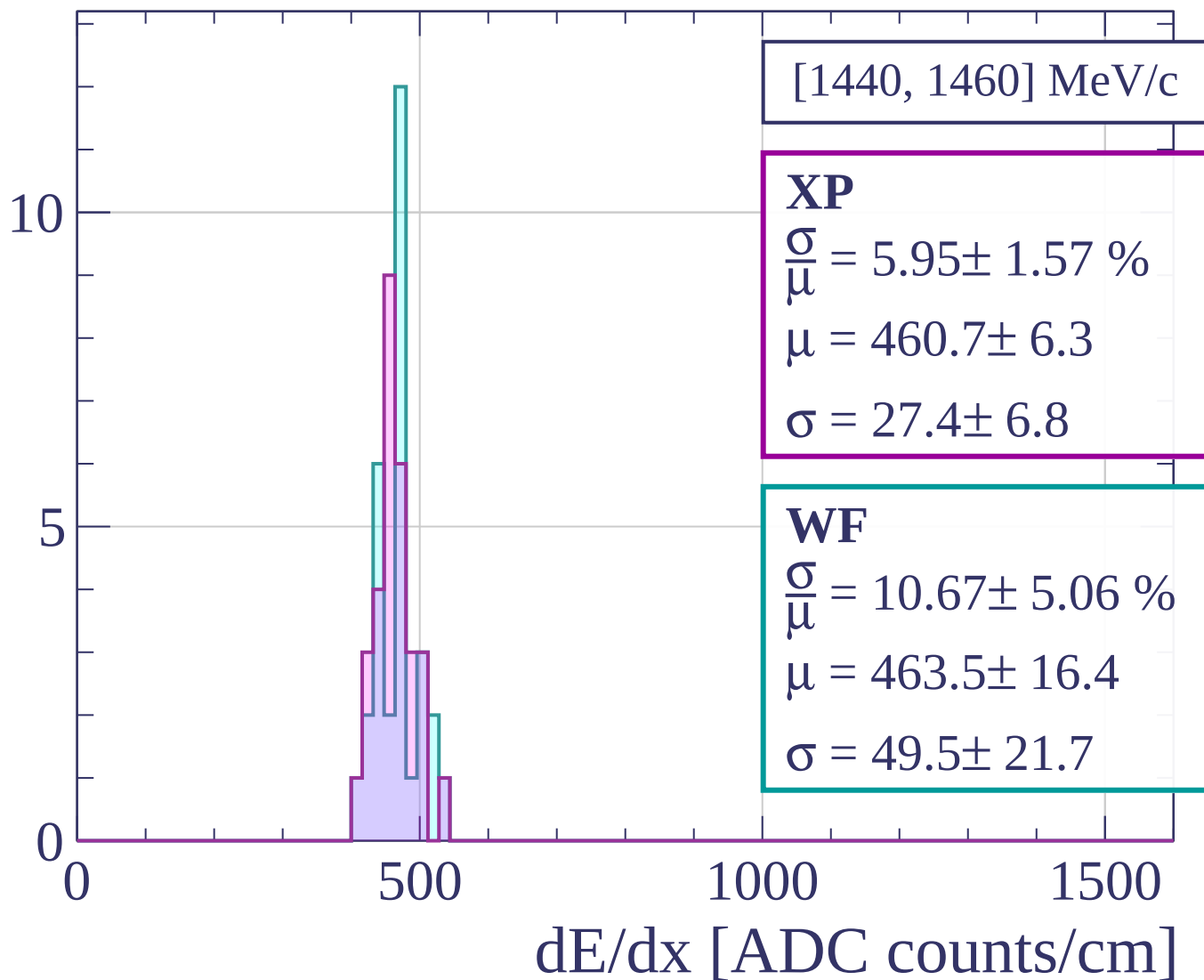
$$\sigma = 35.0 \pm 9.2$$

dE/dx [ADC counts/cm]

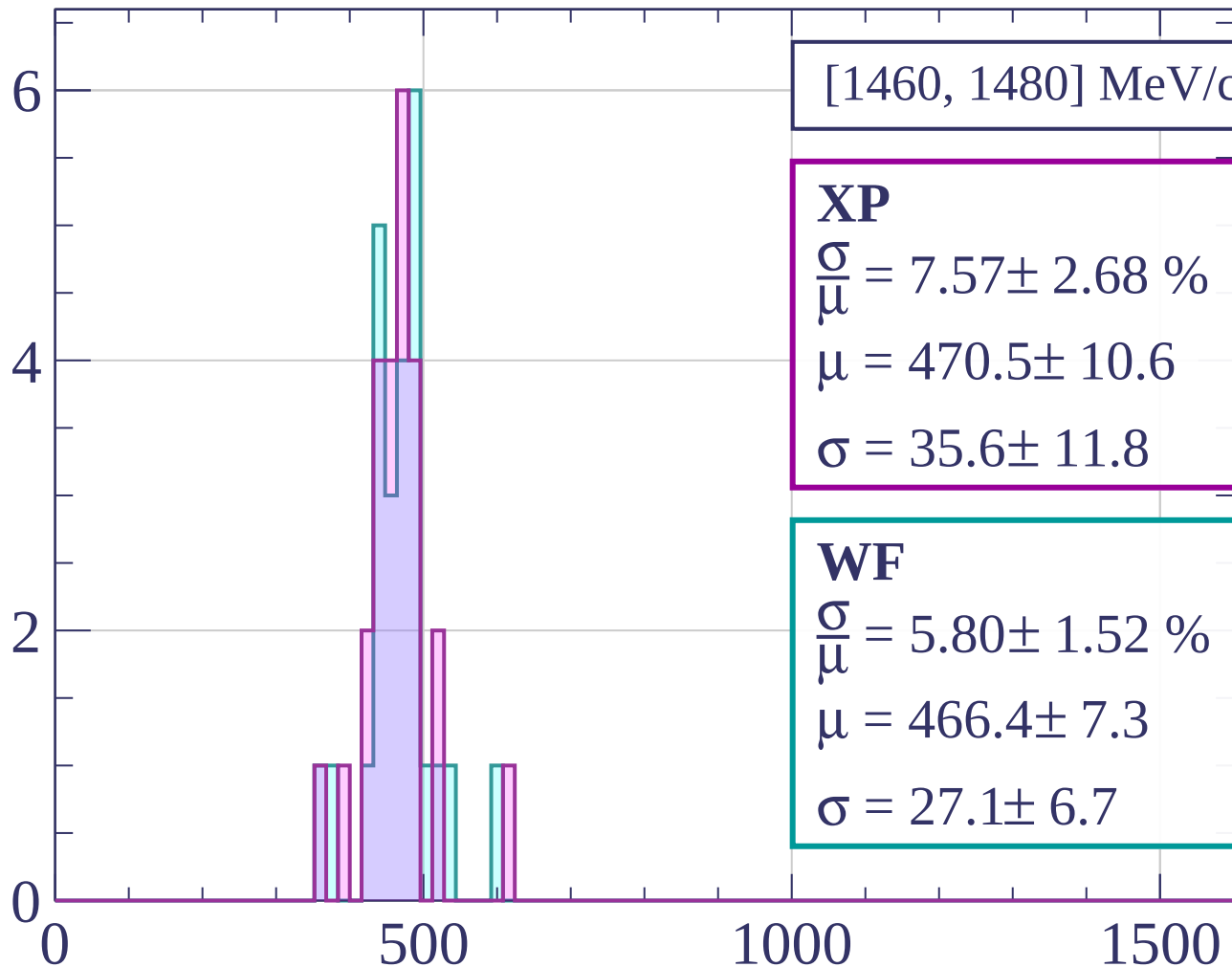
Count



Count



Count



[1460, 1480] MeV/c

XP

$$\frac{\sigma}{\mu} = 7.57 \pm 2.68 \%$$

$$\mu = 470.5 \pm 10.6$$

$$\sigma = 35.6 \pm 11.8$$

WF

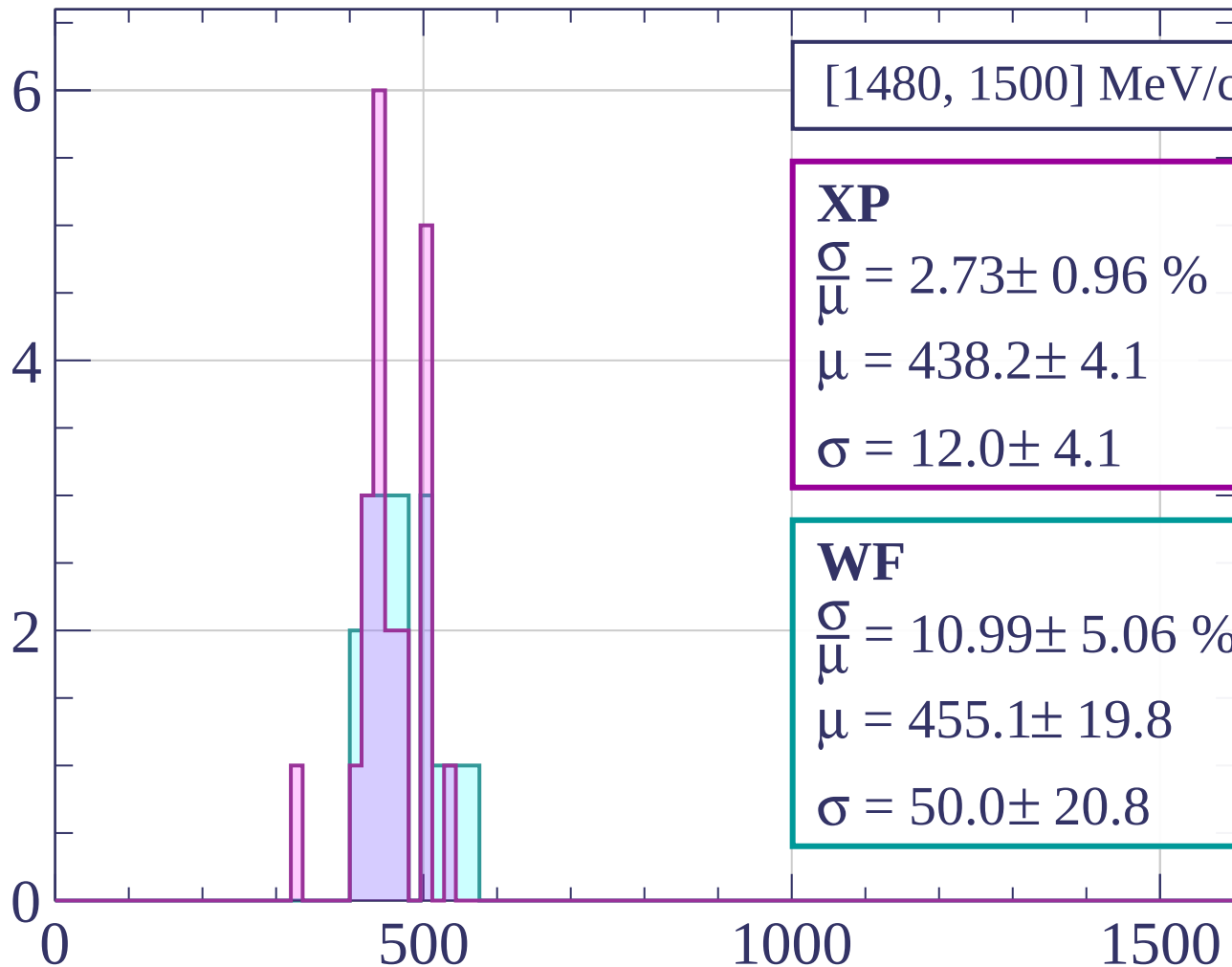
$$\frac{\sigma}{\mu} = 5.80 \pm 1.52 \%$$

$$\mu = 466.4 \pm 7.3$$

$$\sigma = 27.1 \pm 6.7$$

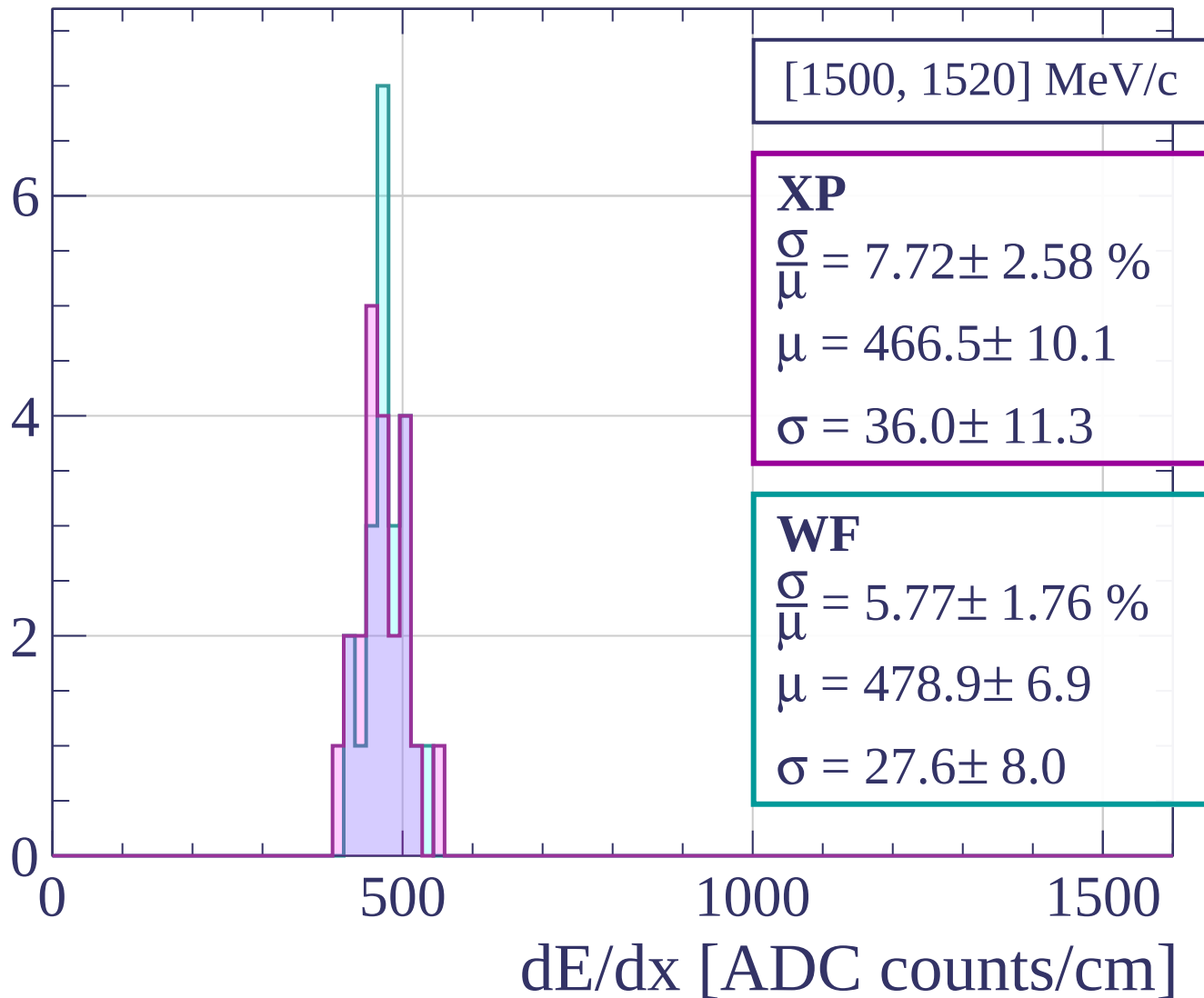
dE/dx [ADC counts/cm]

Count

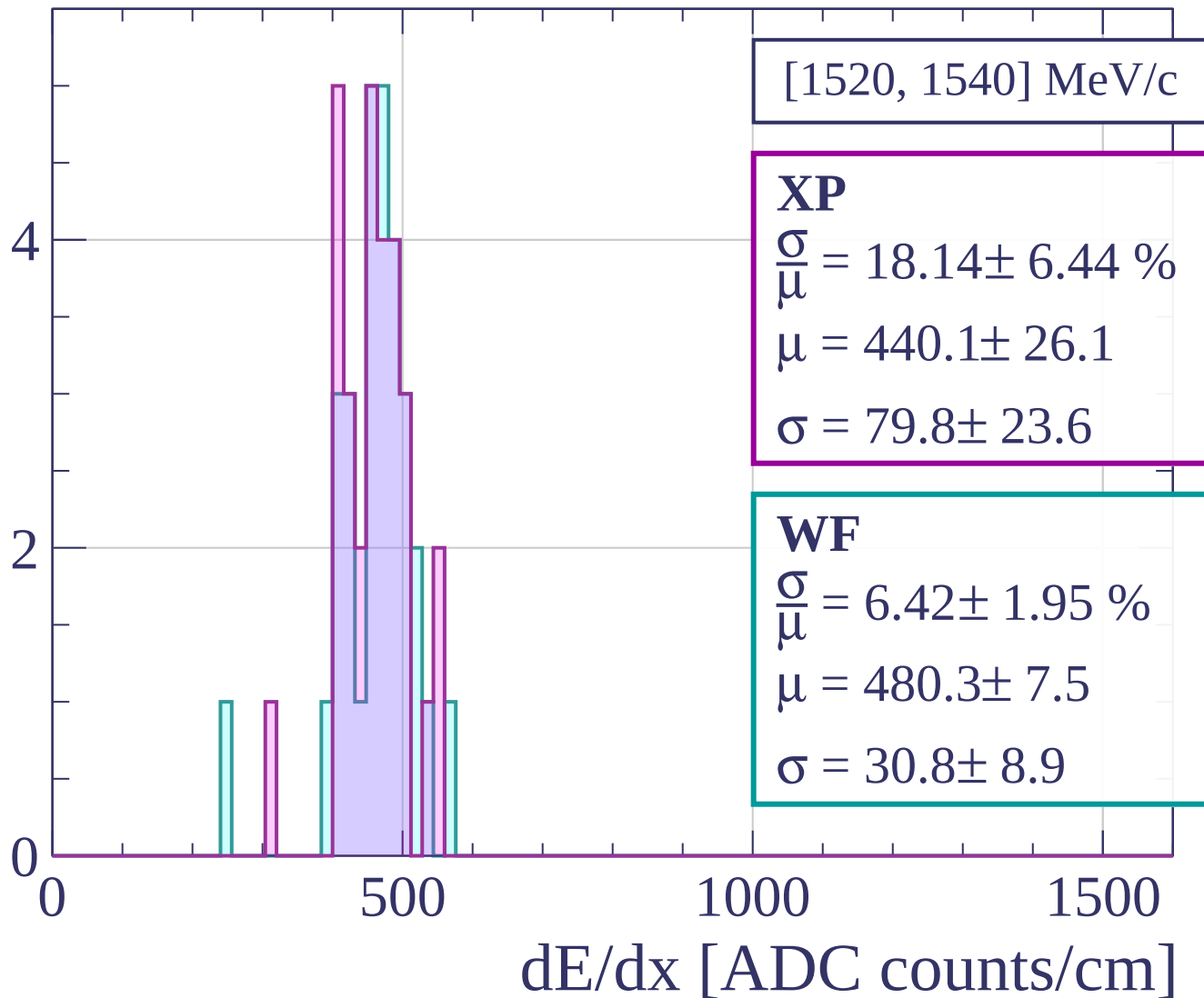


dE/dx [ADC counts/cm]

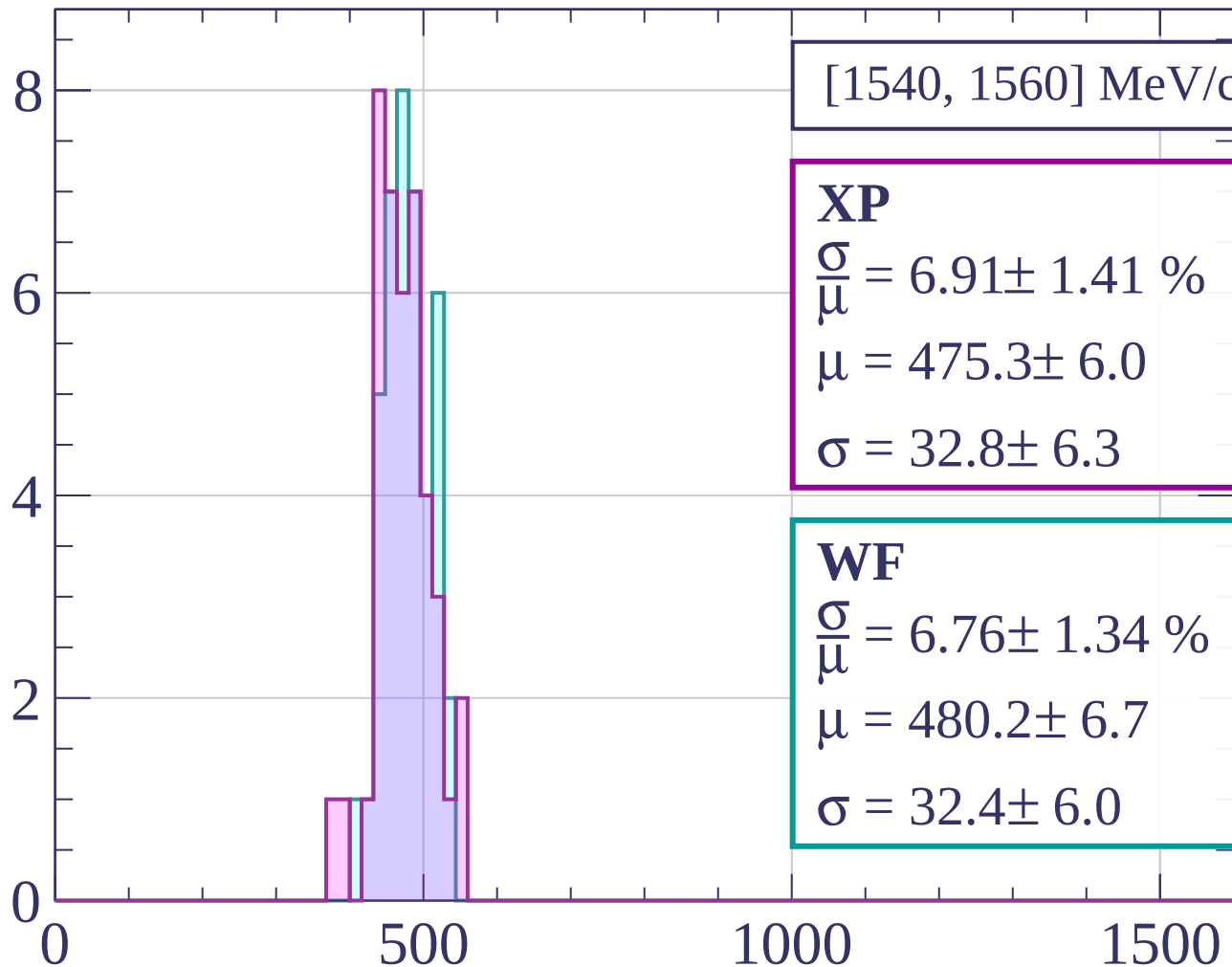
Count



Count

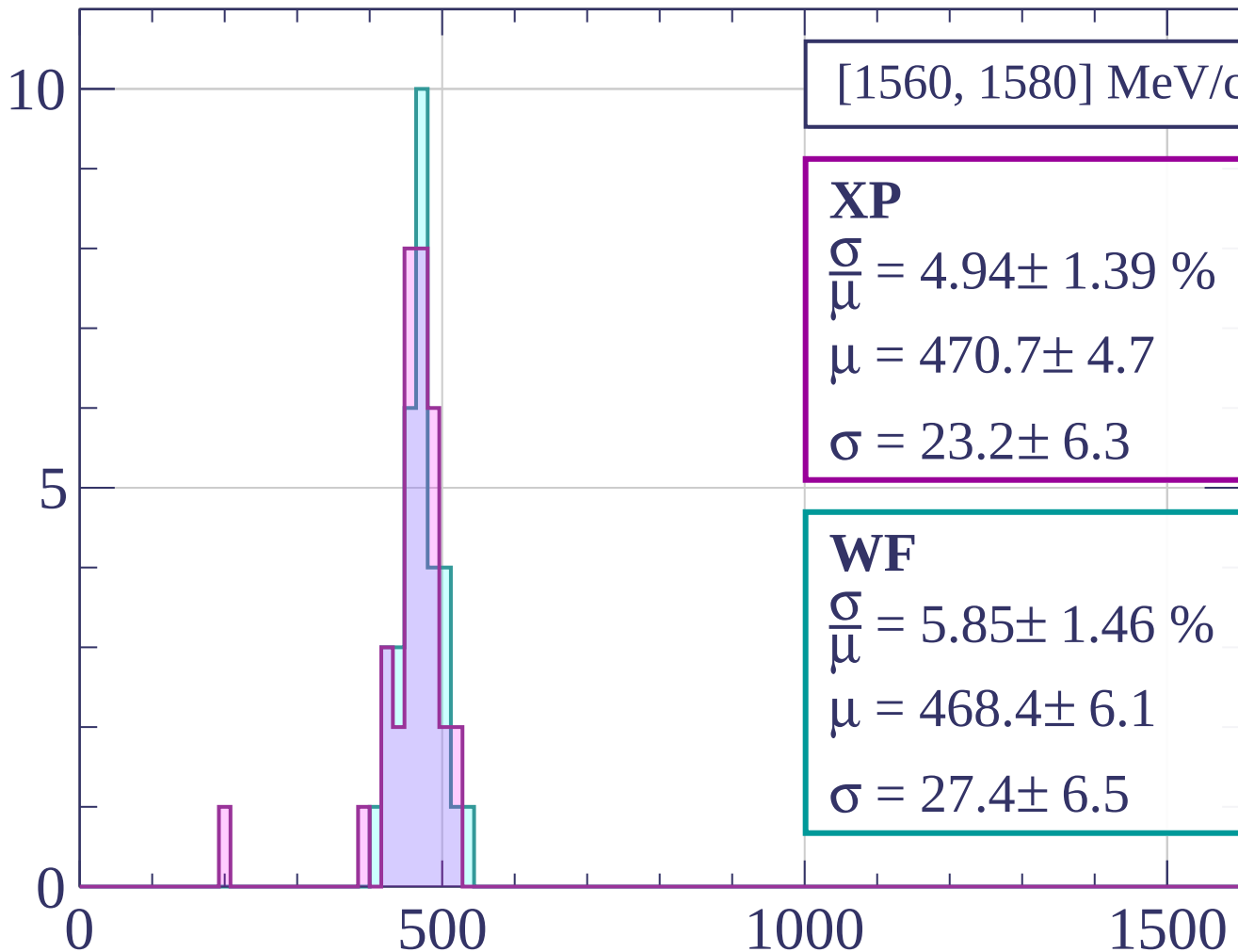


Count



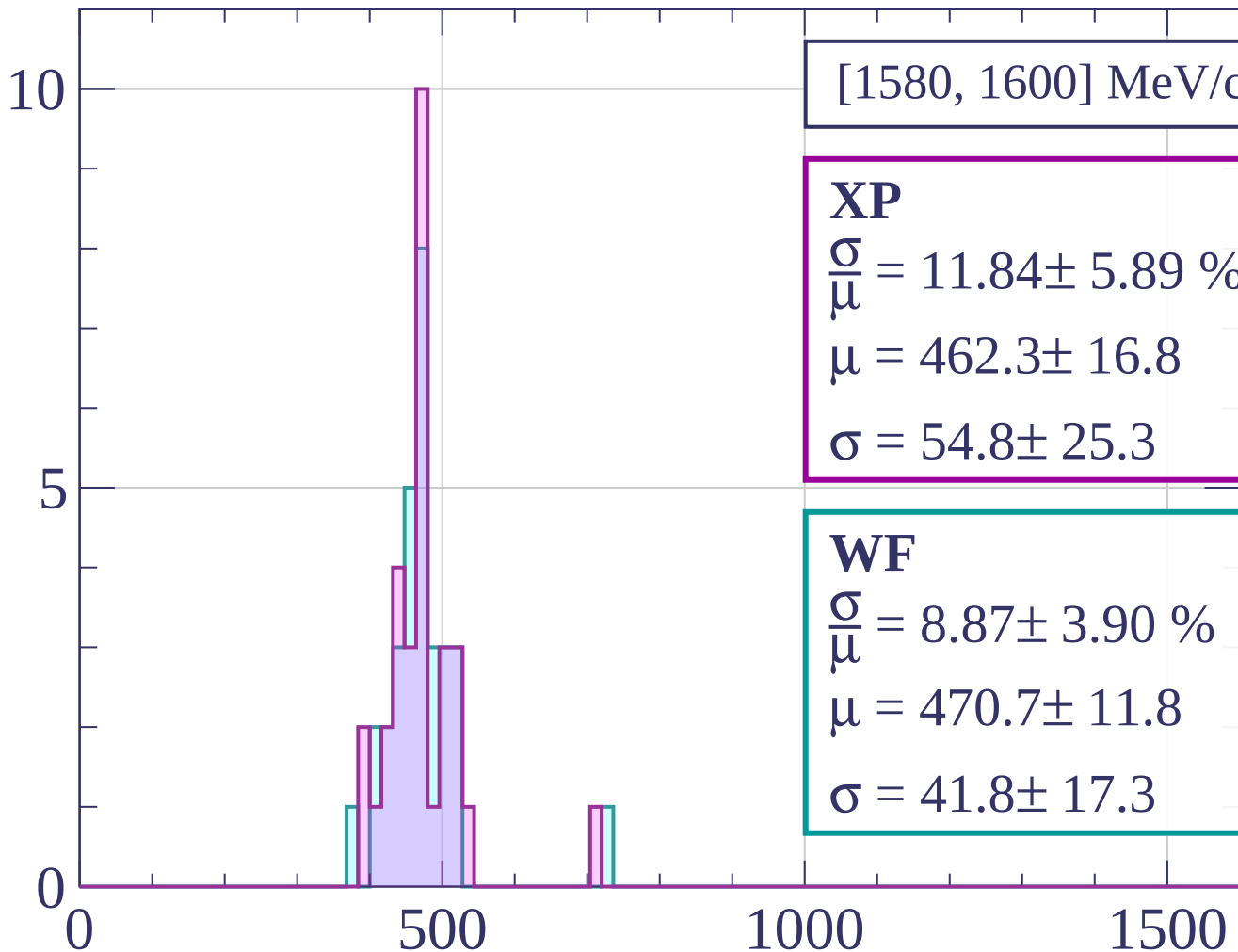
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



[1580, 1600] MeV/c

XP

$$\frac{\sigma}{\mu} = 11.84 \pm 5.89 \%$$

$$\mu = 462.3 \pm 16.8$$

$$\sigma = 54.8 \pm 25.3$$

WF

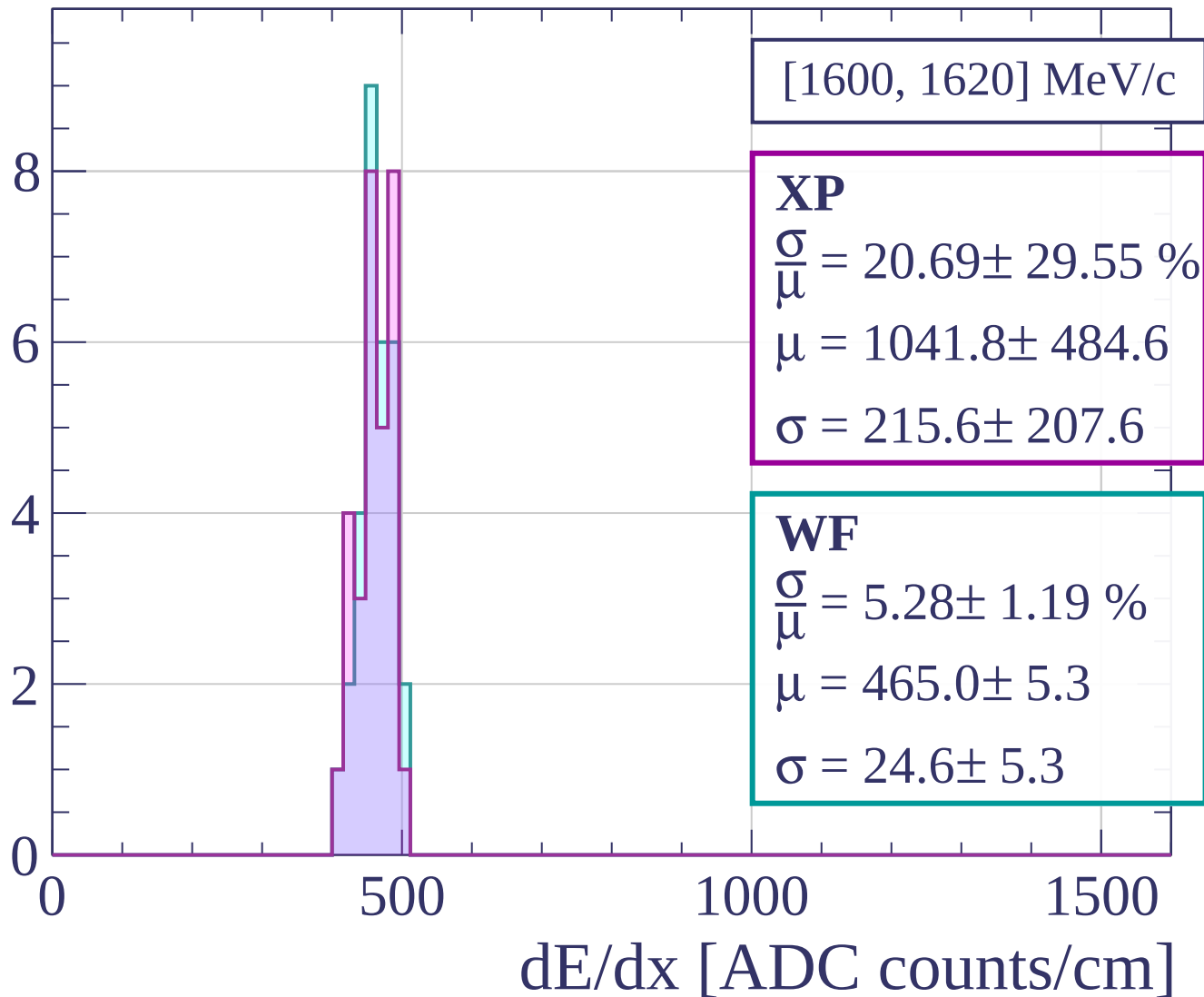
$$\frac{\sigma}{\mu} = 8.87 \pm 3.90 \%$$

$$\mu = 470.7 \pm 11.8$$

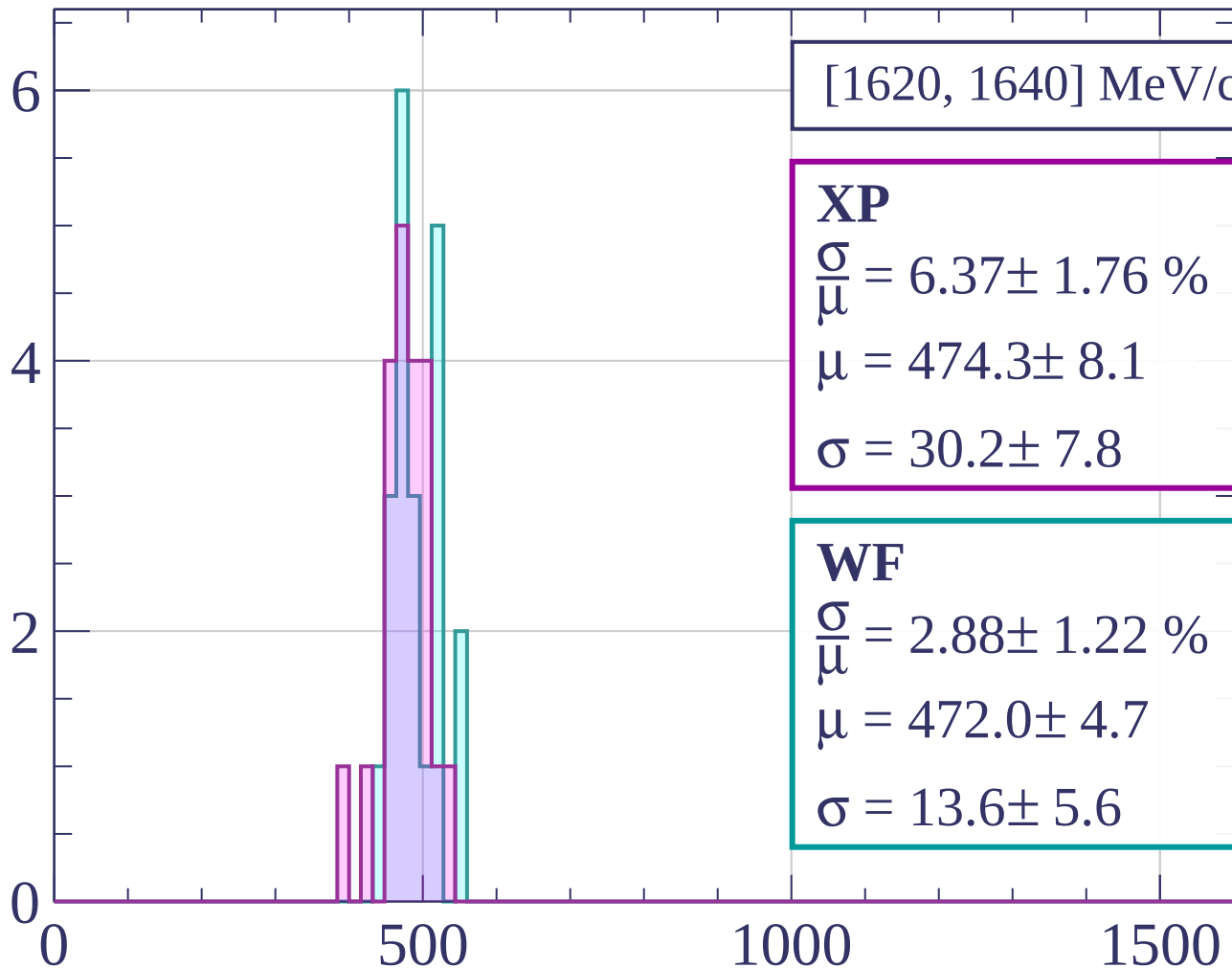
$$\sigma = 41.8 \pm 17.3$$

dE/dx [ADC counts/cm]

Count



Count



[1620, 1640] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.37 \pm 1.76 \%$$

$$\mu = 474.3 \pm 8.1$$

$$\sigma = 30.2 \pm 7.8$$

WF

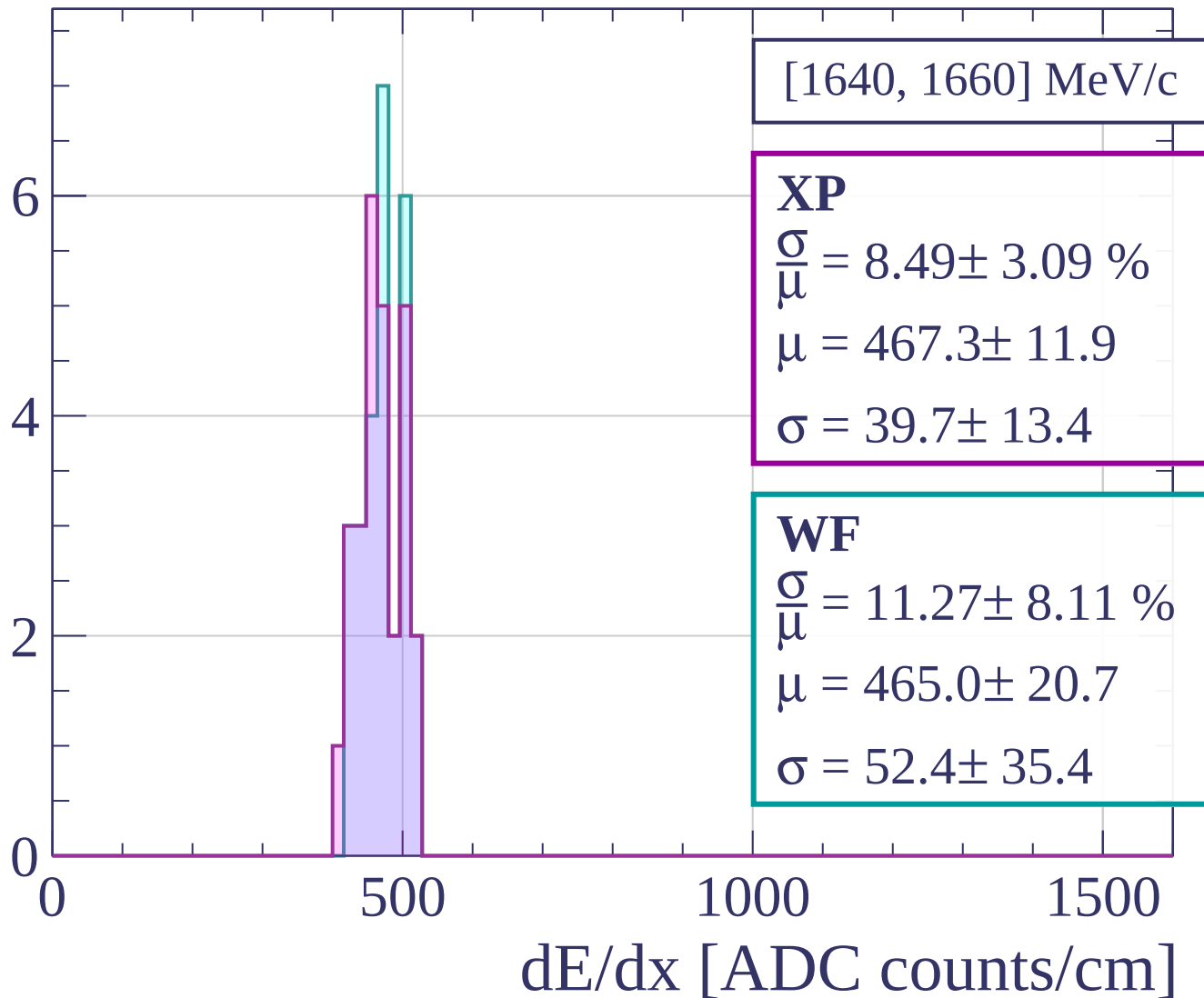
$$\frac{\sigma}{\mu} = 2.88 \pm 1.22 \%$$

$$\mu = 472.0 \pm 4.7$$

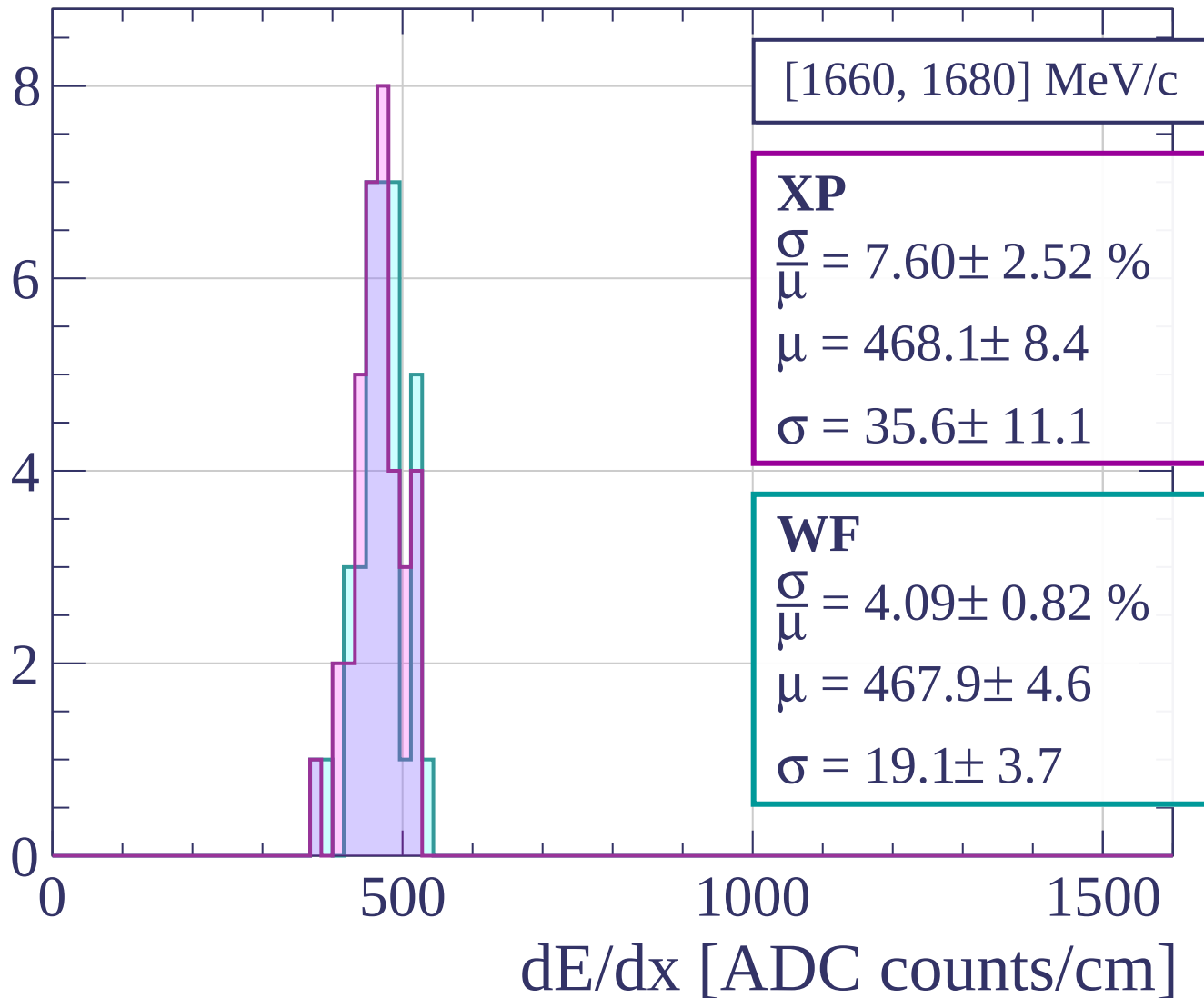
$$\sigma = 13.6 \pm 5.6$$

dE/dx [ADC counts/cm]

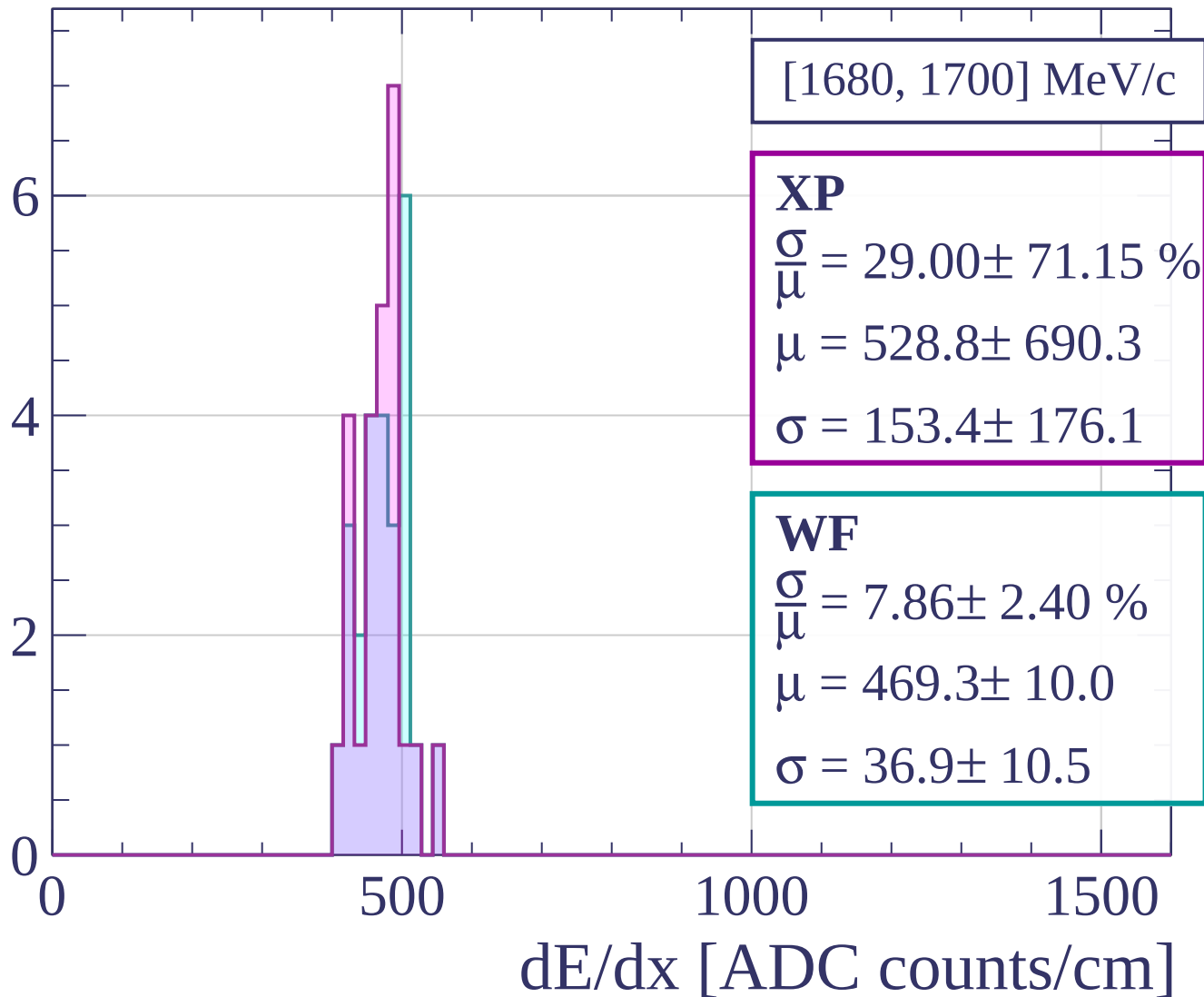
Count



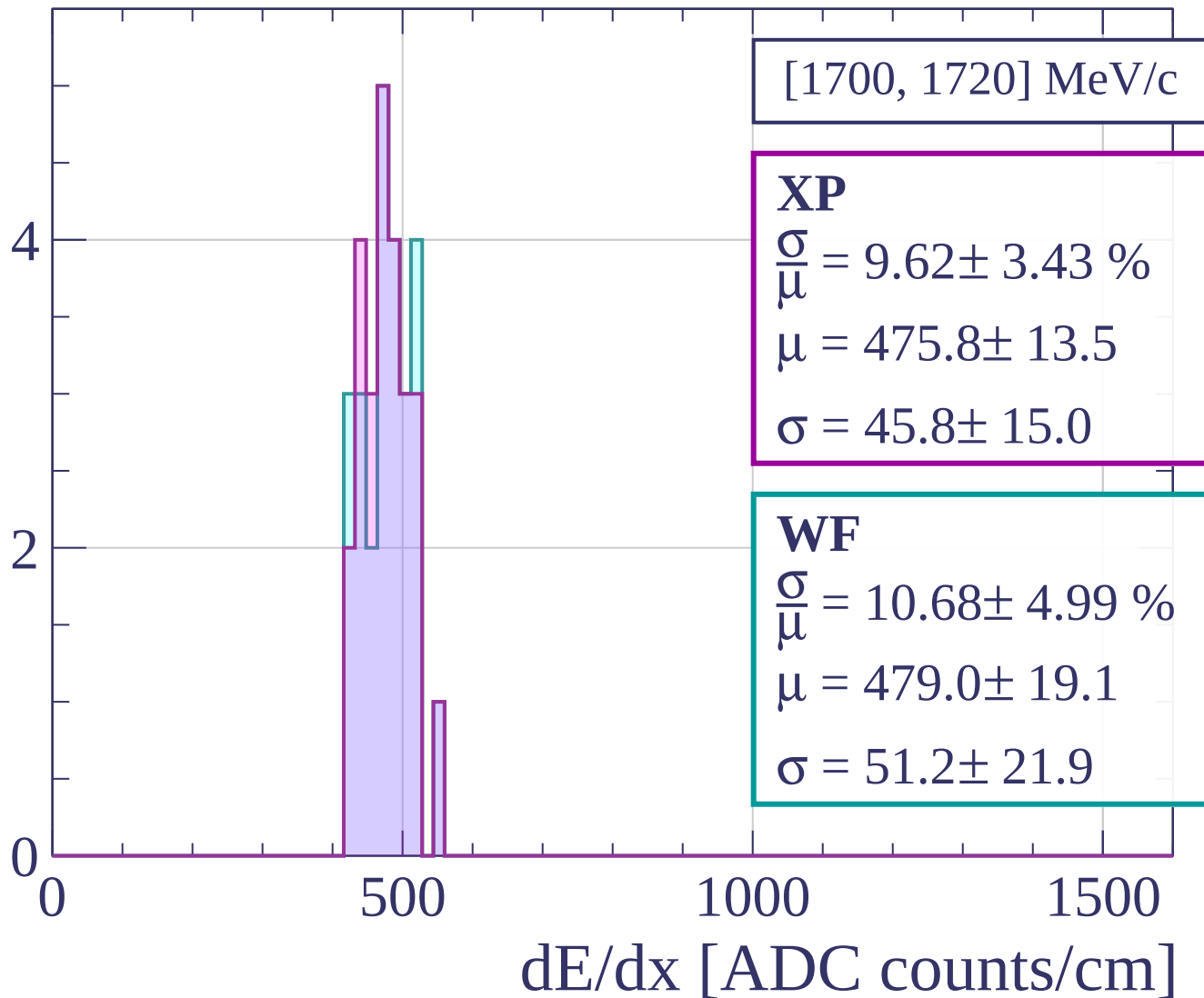
Count



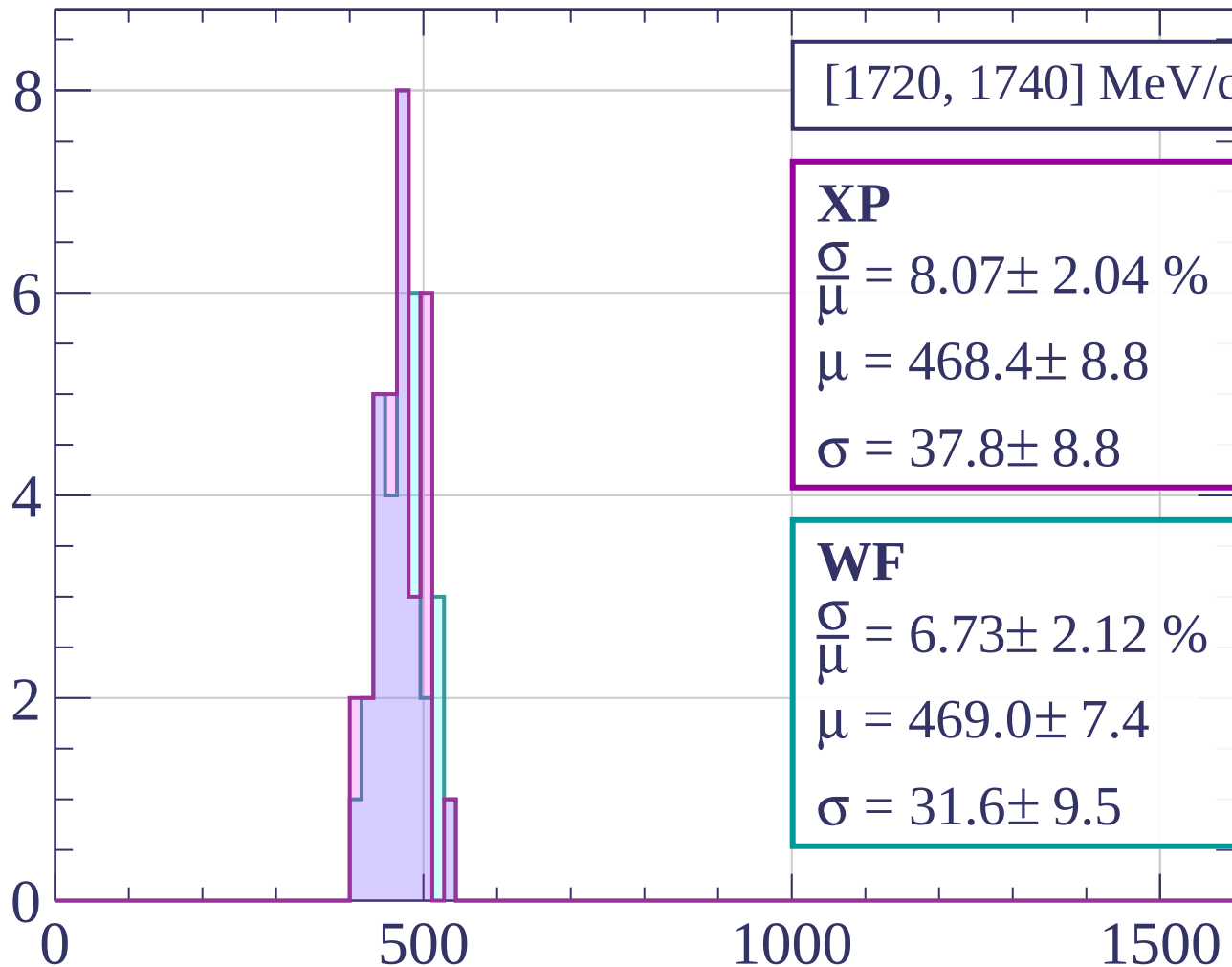
Count



Count

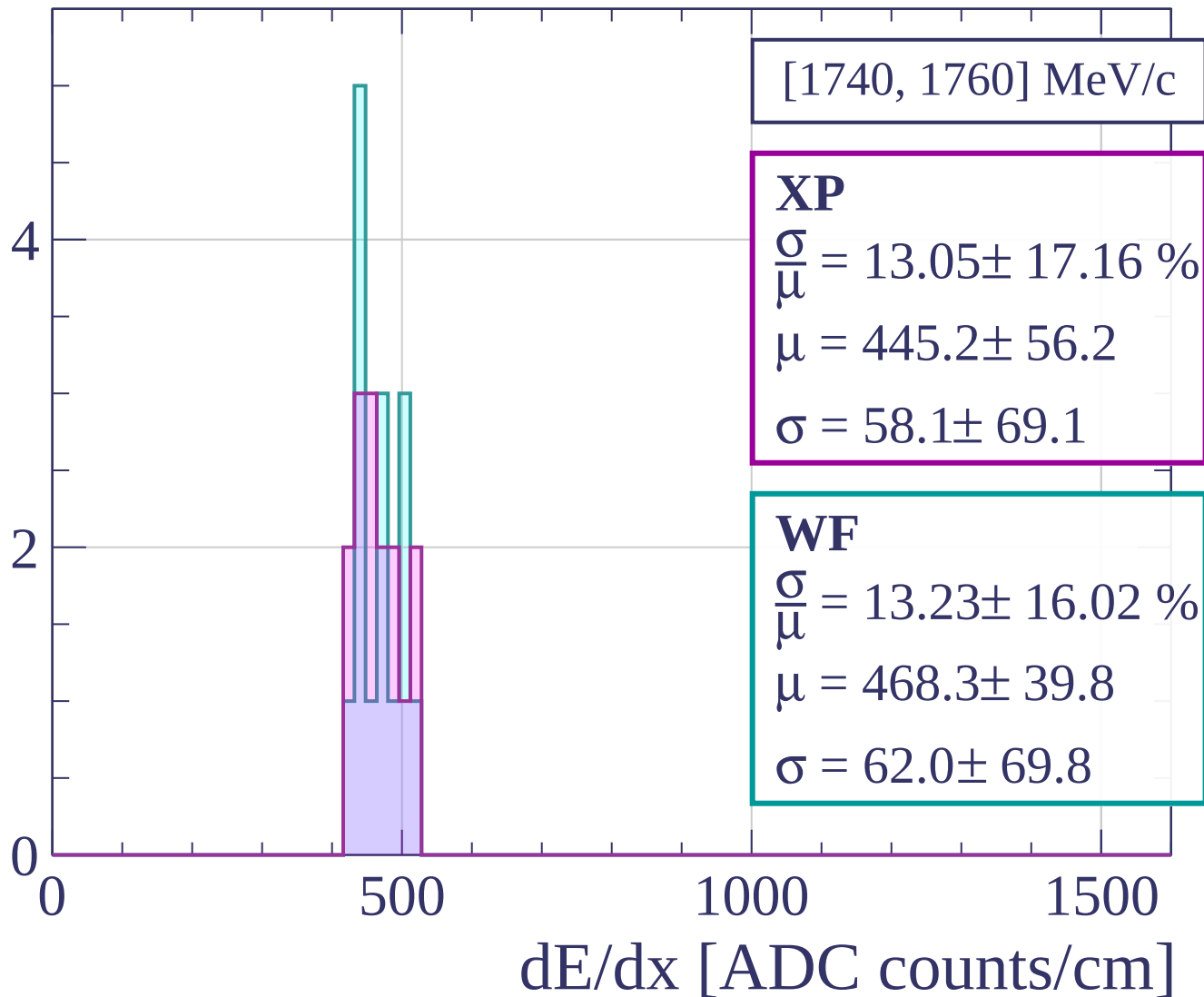


Count

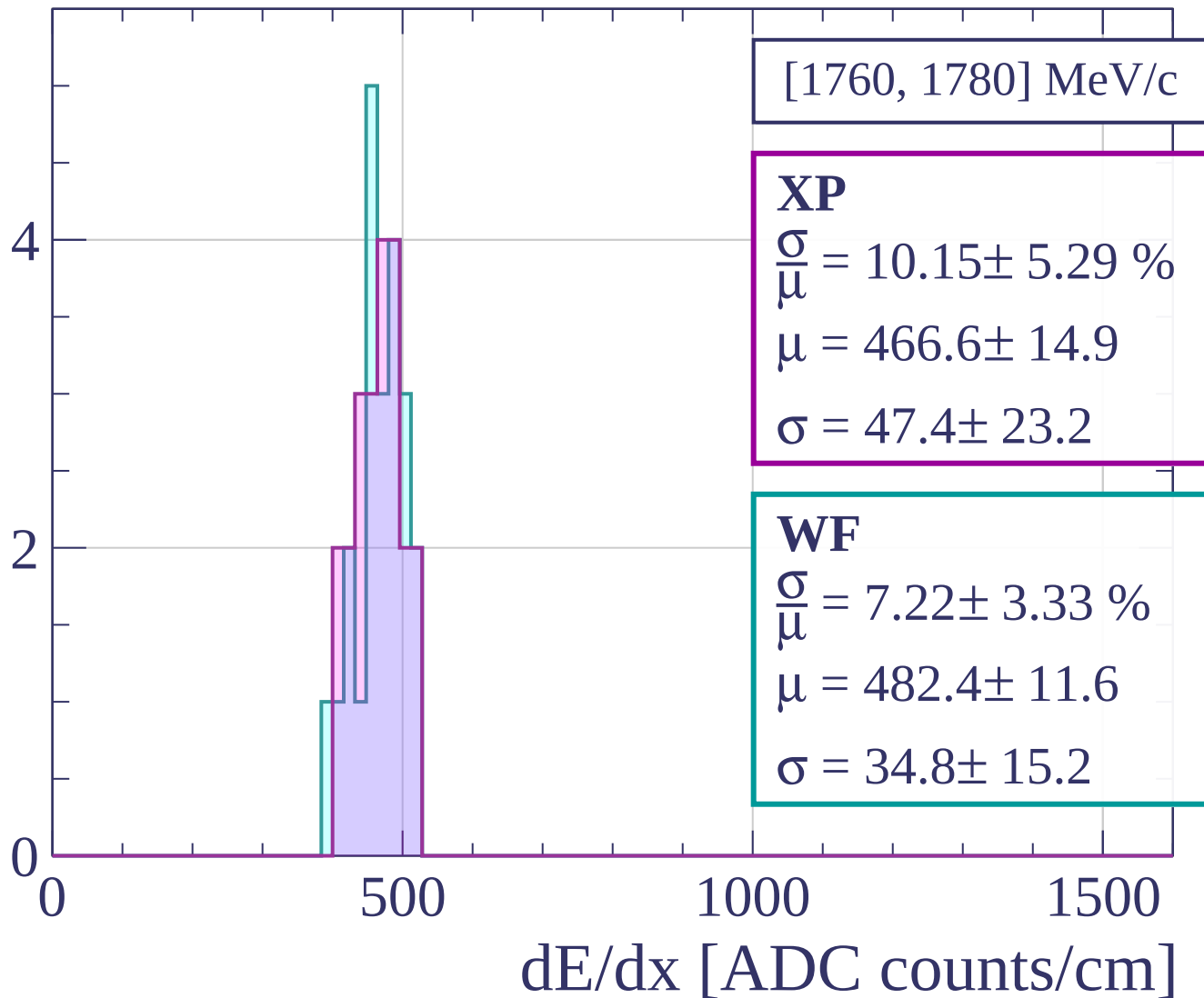


dE/dx [ADC counts/cm]

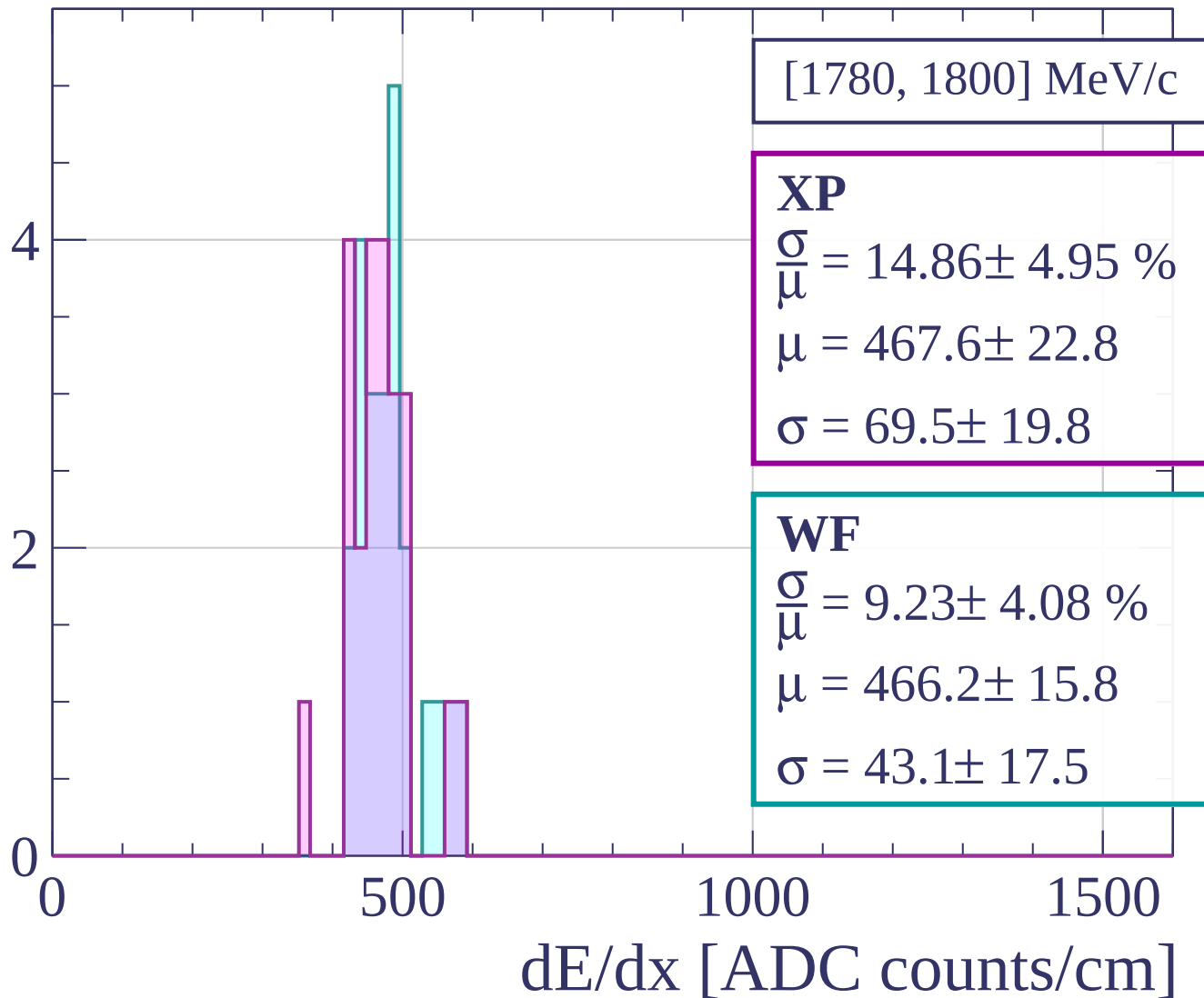
Count



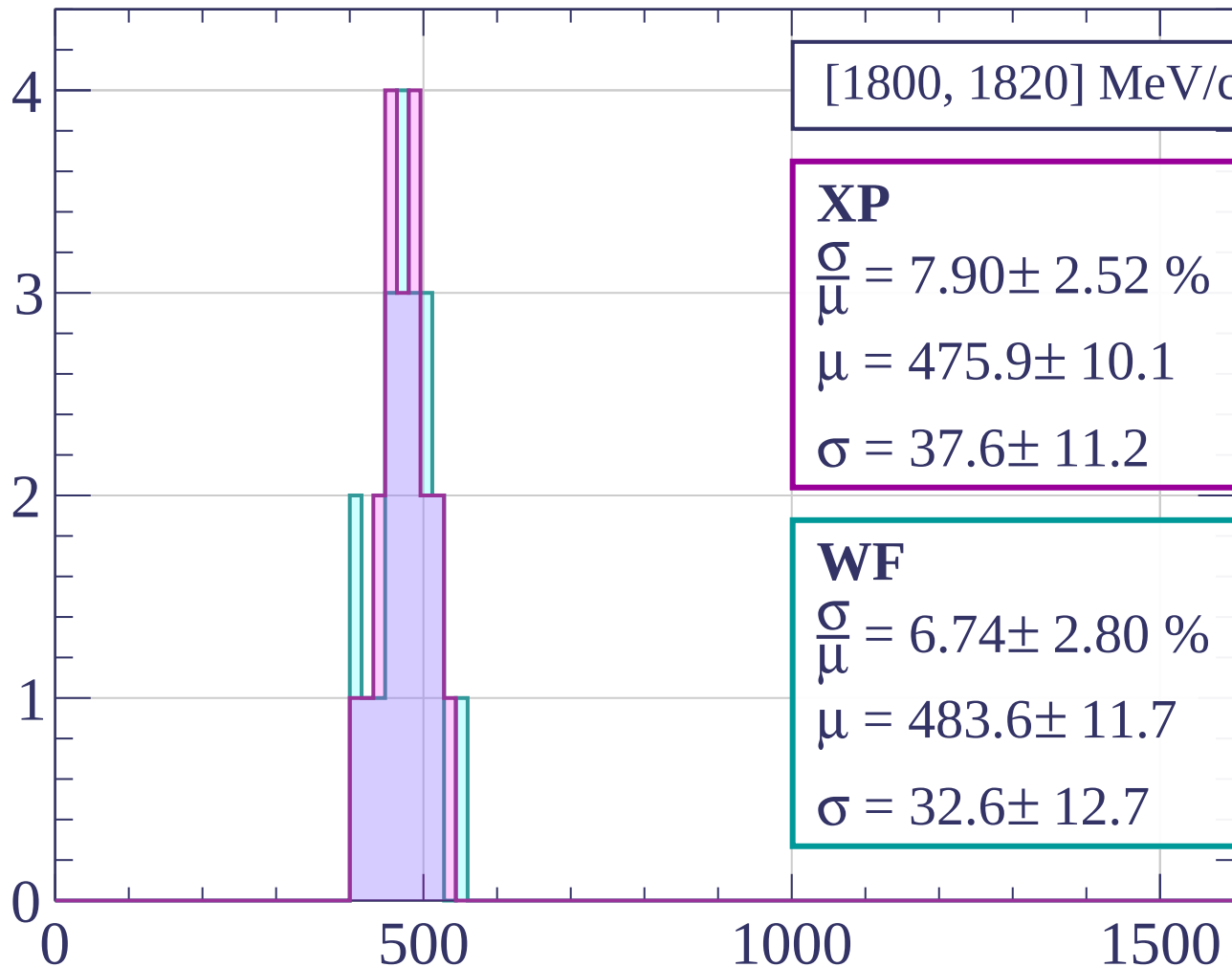
Count



Count

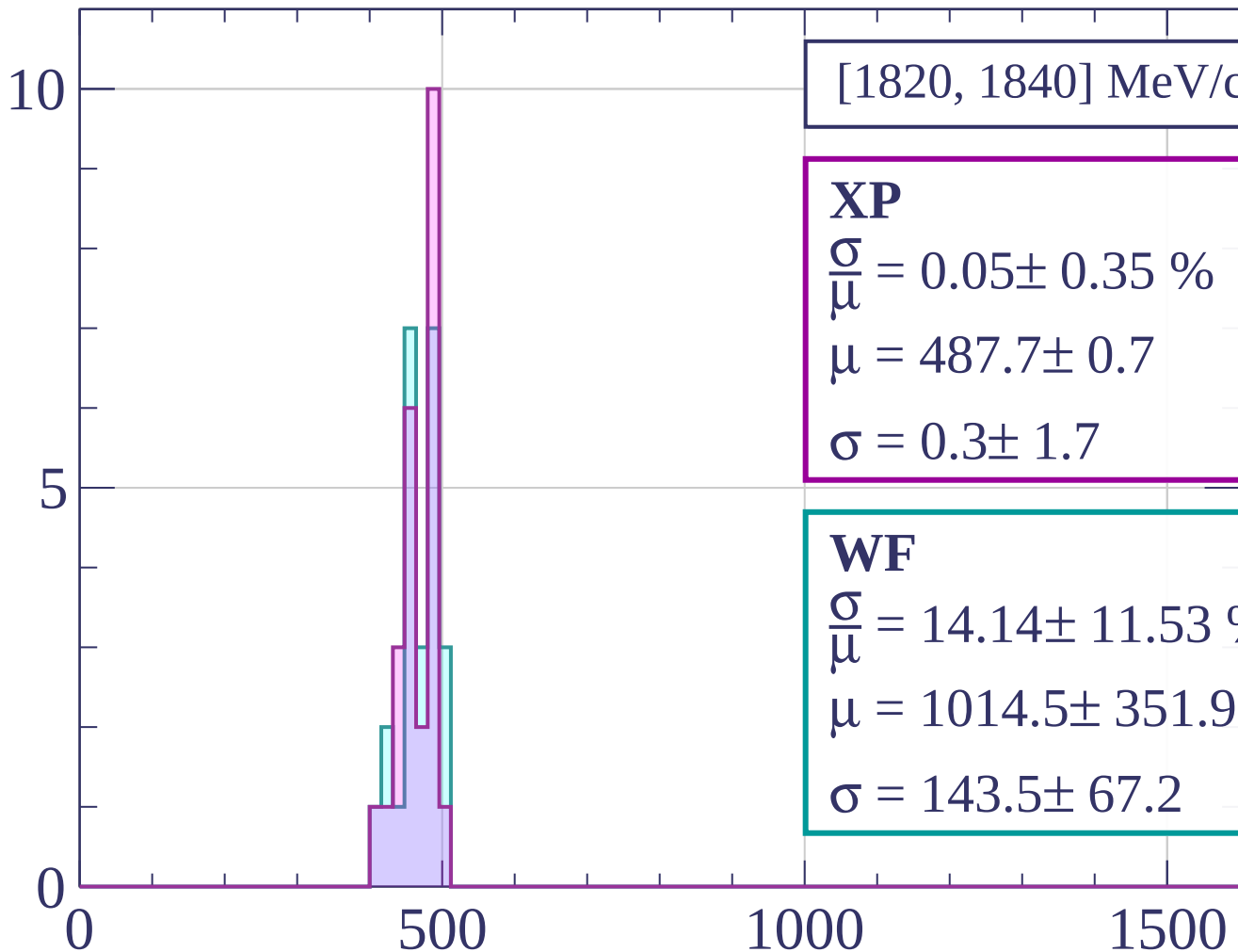


Count



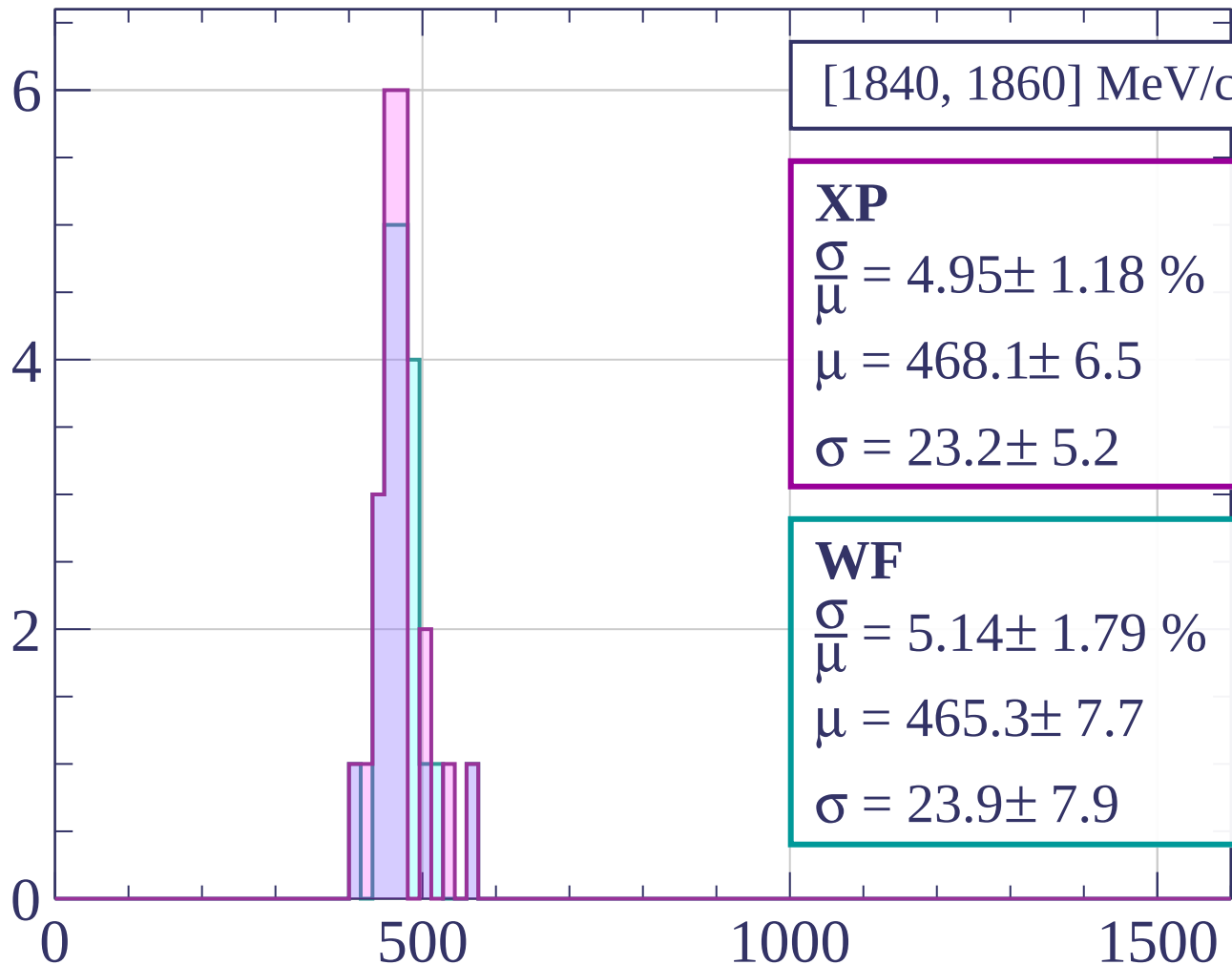
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



[1840, 1860] MeV/c

XP

$$\frac{\sigma}{\mu} = 4.95 \pm 1.18 \%$$

$$\mu = 468.1 \pm 6.5$$

$$\sigma = 23.2 \pm 5.2$$

WF

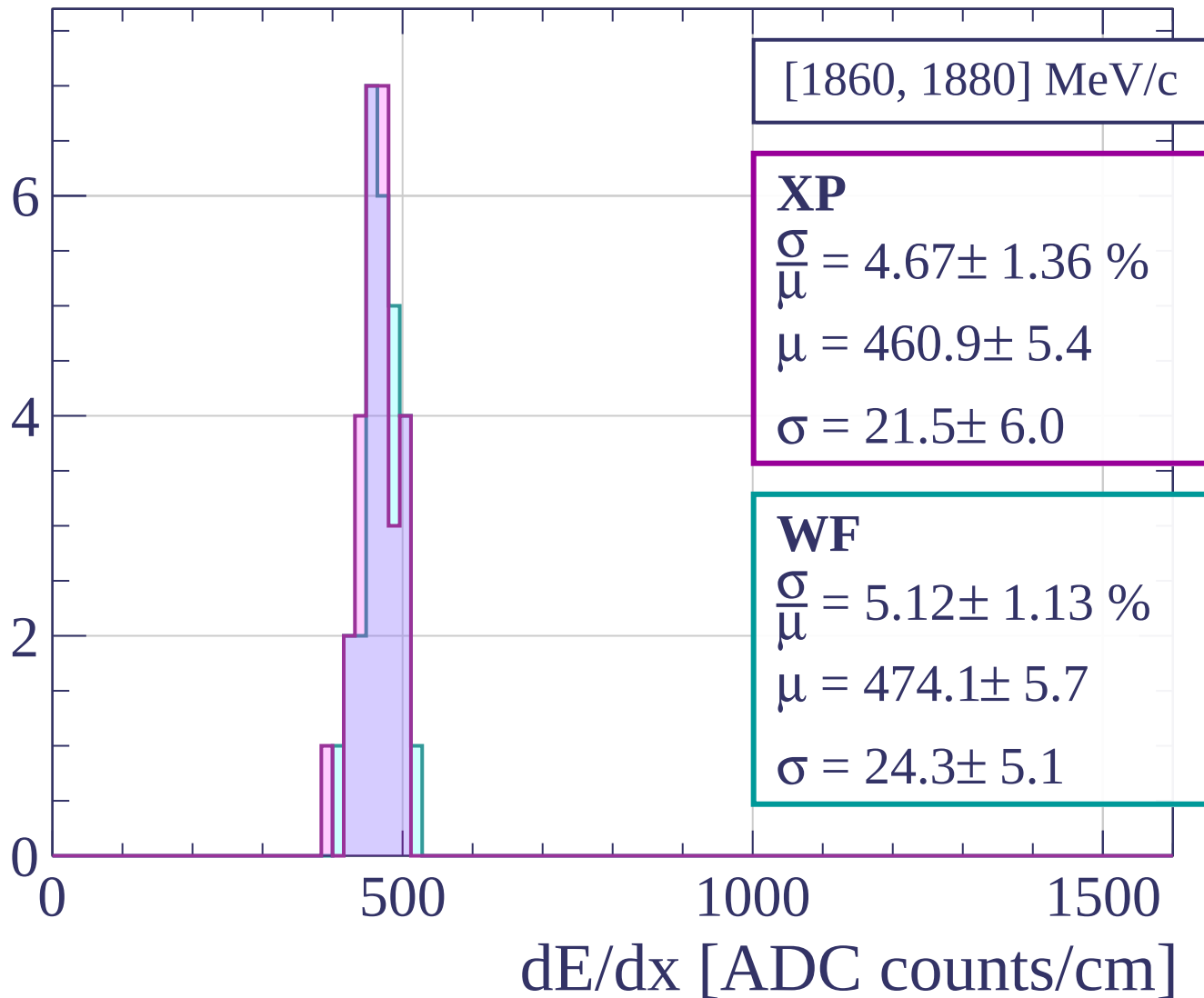
$$\frac{\sigma}{\mu} = 5.14 \pm 1.79 \%$$

$$\mu = 465.3 \pm 7.7$$

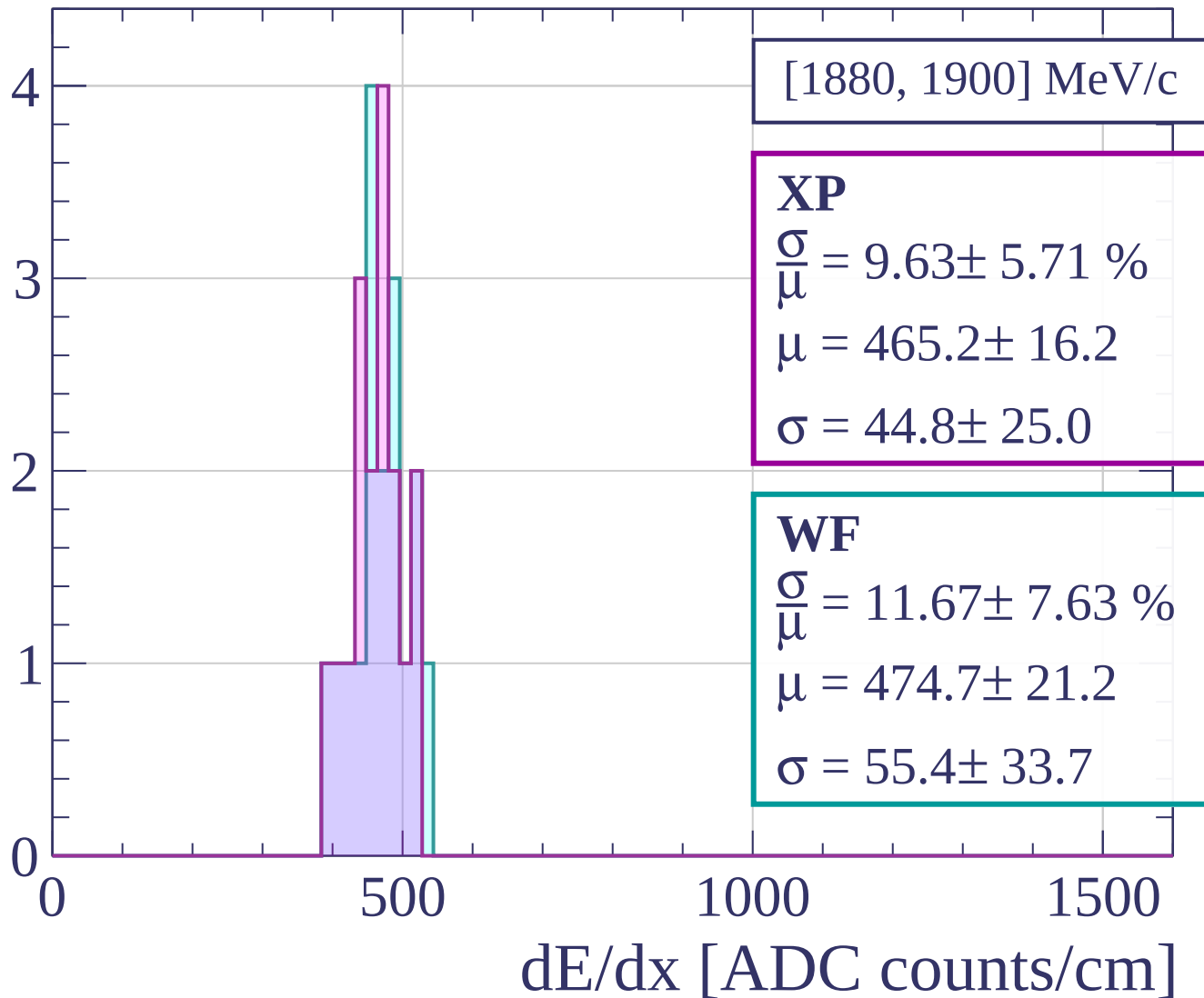
$$\sigma = 23.9 \pm 7.9$$

dE/dx [ADC counts/cm]

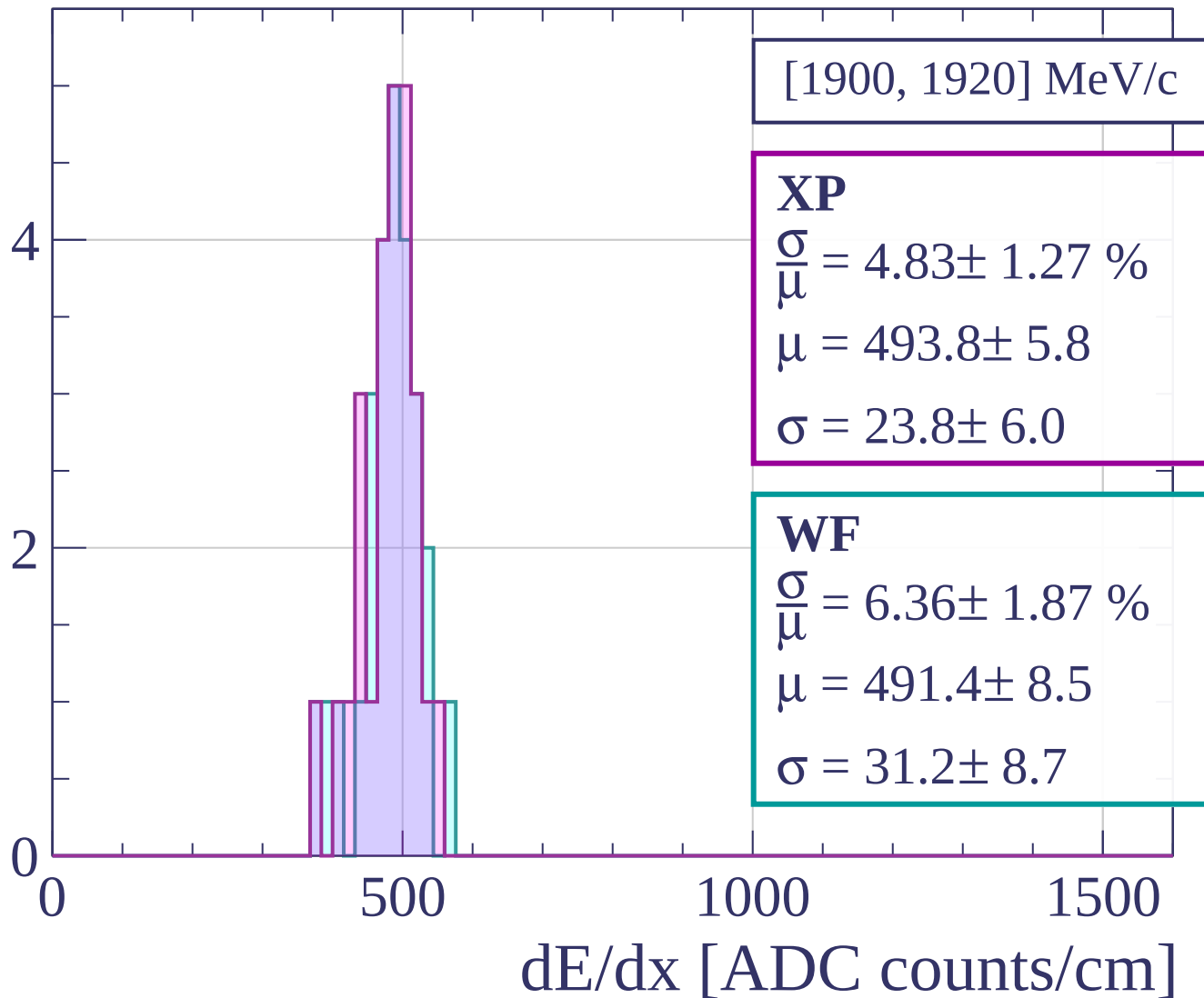
Count



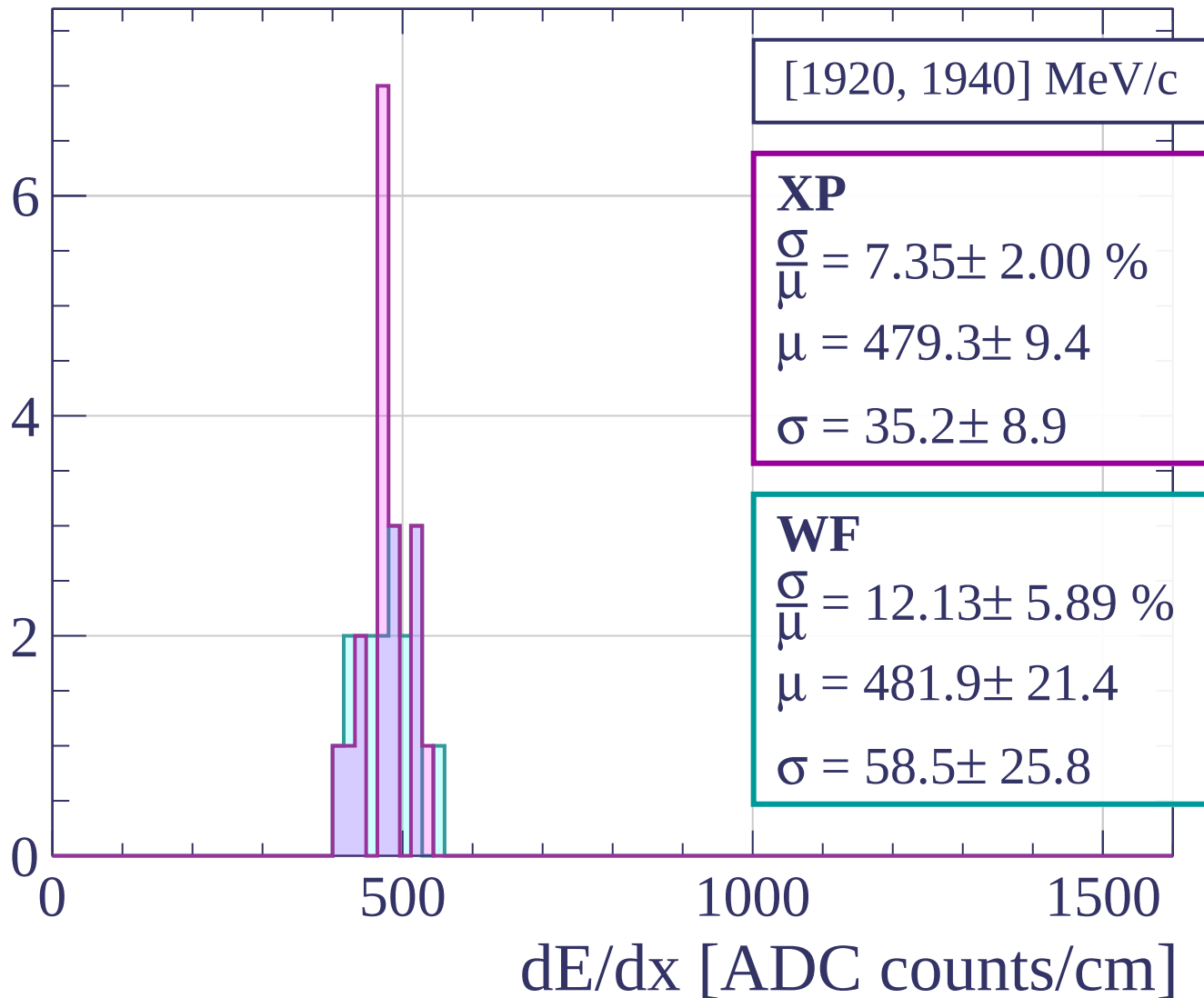
Count



Count



Count



Count

[1940, 1960] MeV/c

XP

$$\frac{\sigma}{\mu} = 19.99 \pm 21.64 \%$$

$$\mu = 461.7 \pm 54.9$$

$$\sigma = 92.3 \pm 89.0$$

WF

$$\frac{\sigma}{\mu} = 16.40 \pm 13.12 \%$$

$$\mu = 462.7 \pm 34.6$$

$$\sigma = 75.9 \pm 55.0$$

2.0
1.5
1.0
0.5
0.0

0

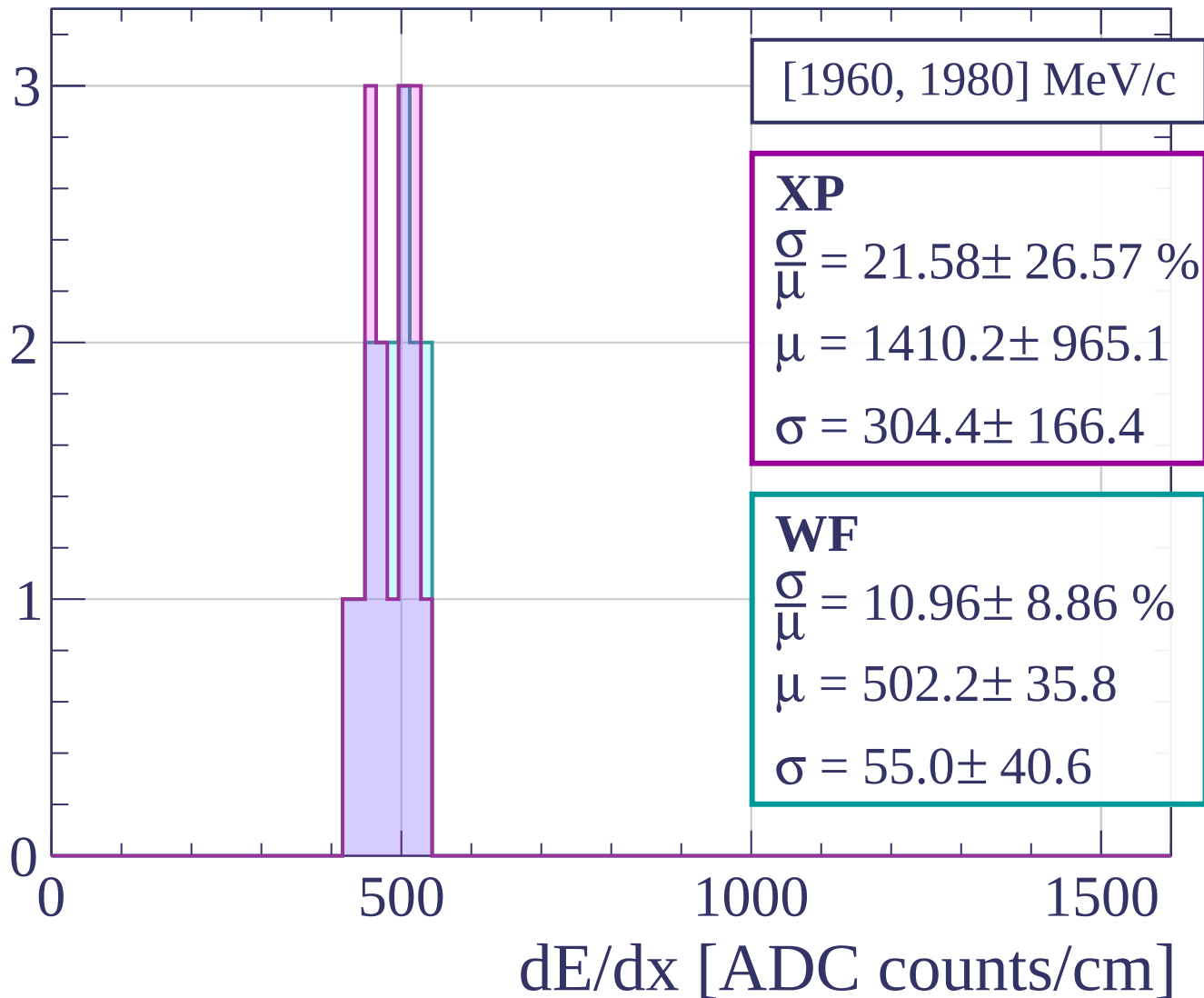
500

1000

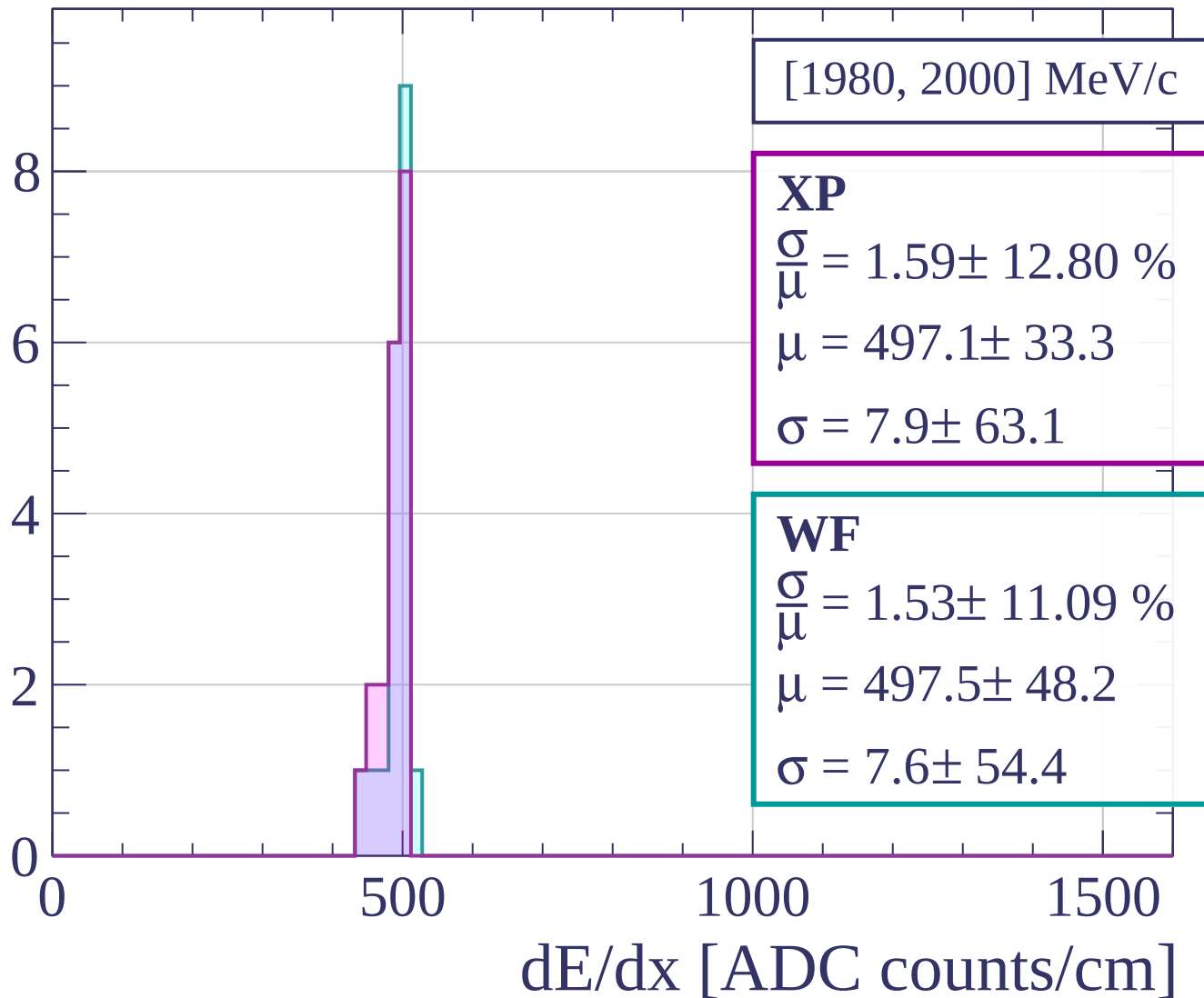
1500

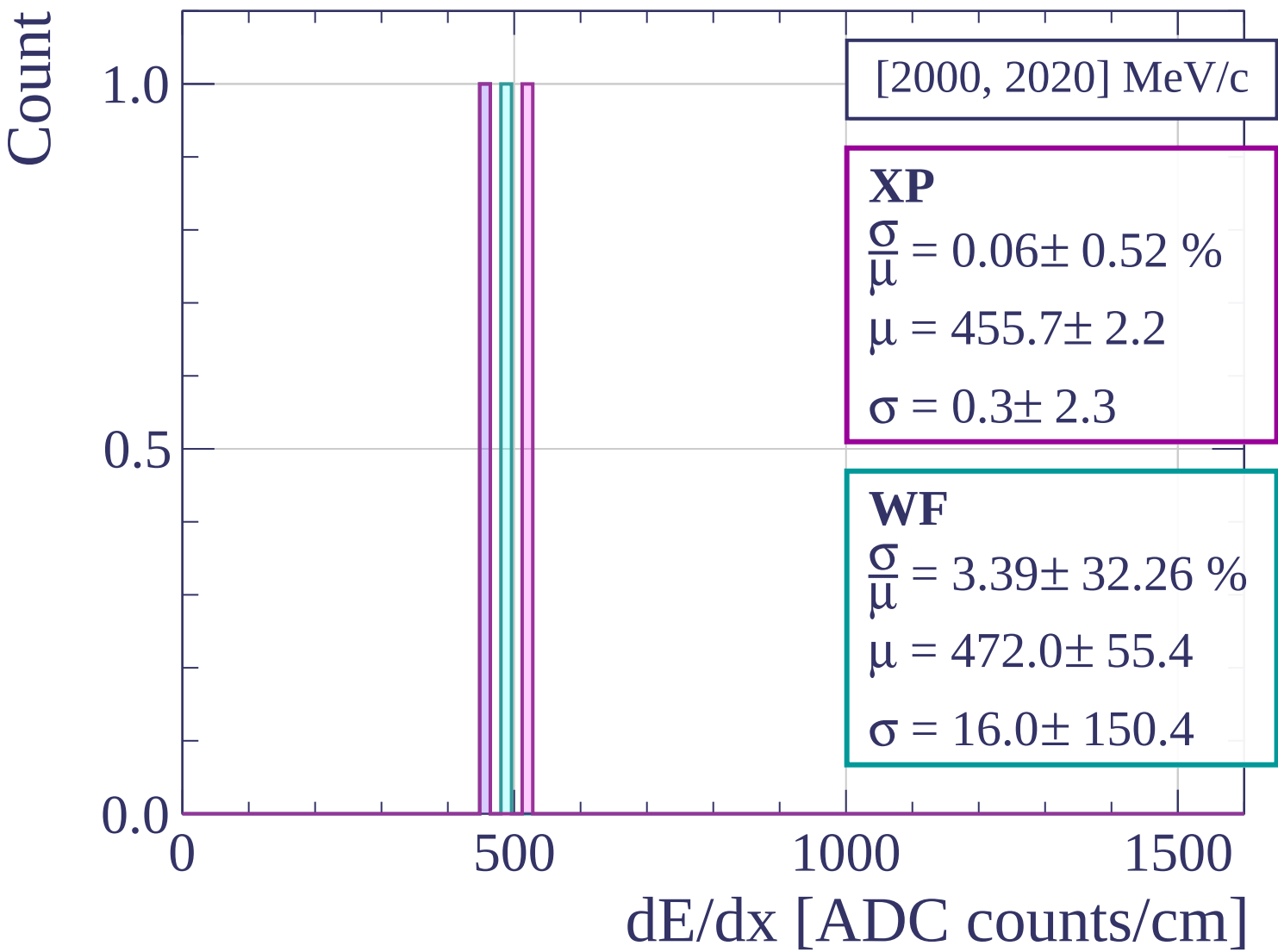
dE/dx [ADC counts/cm]

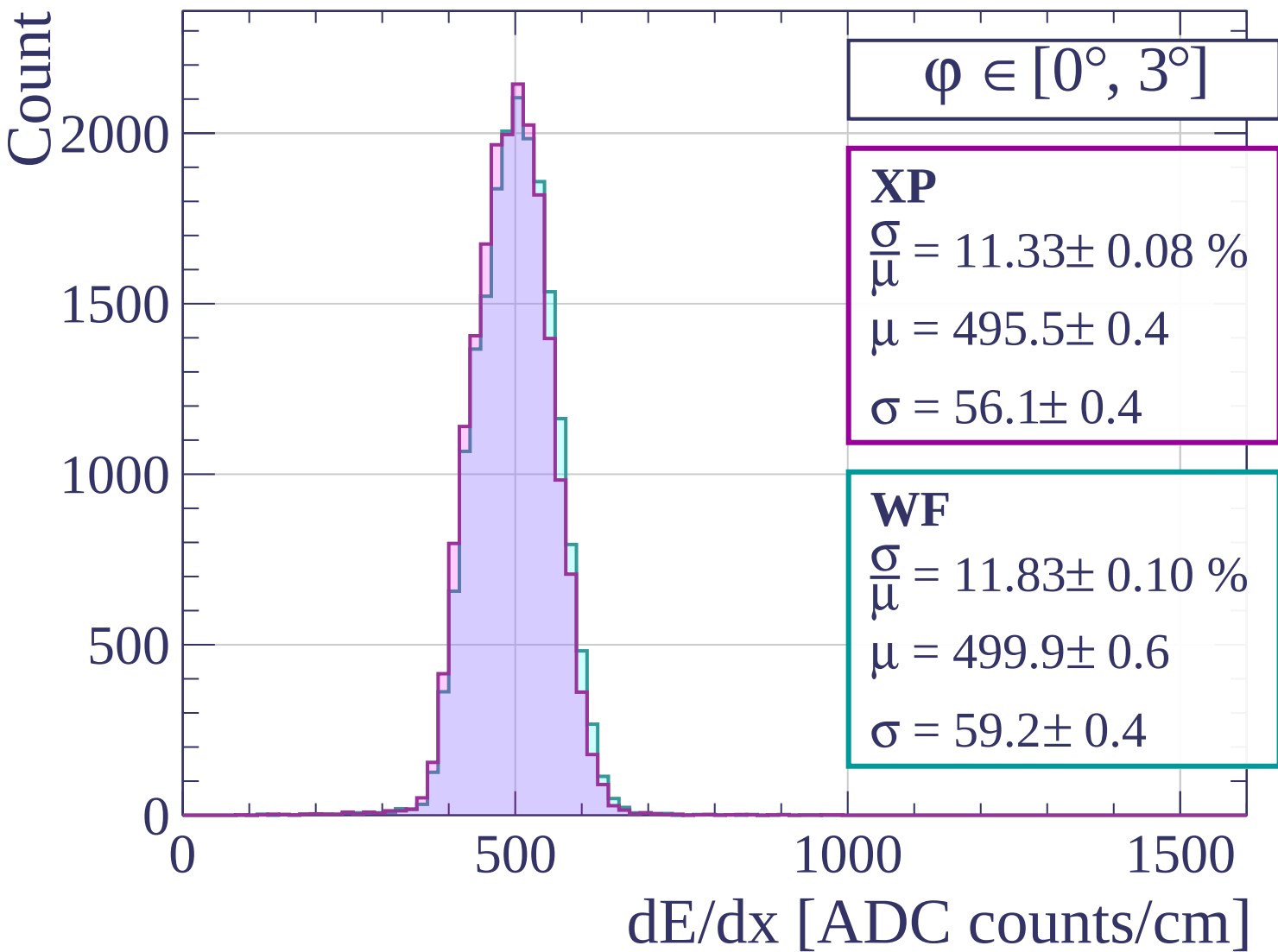
Count

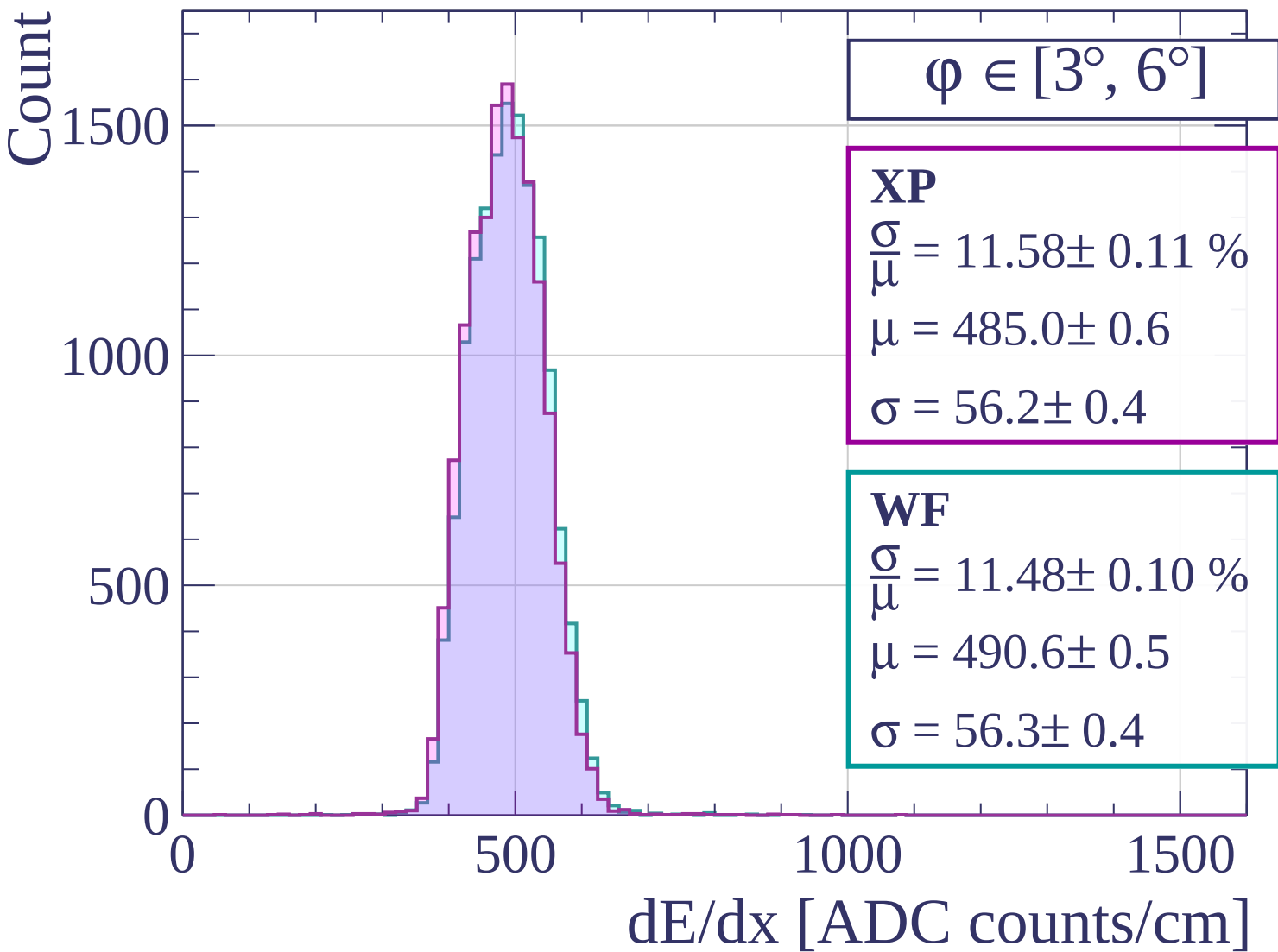


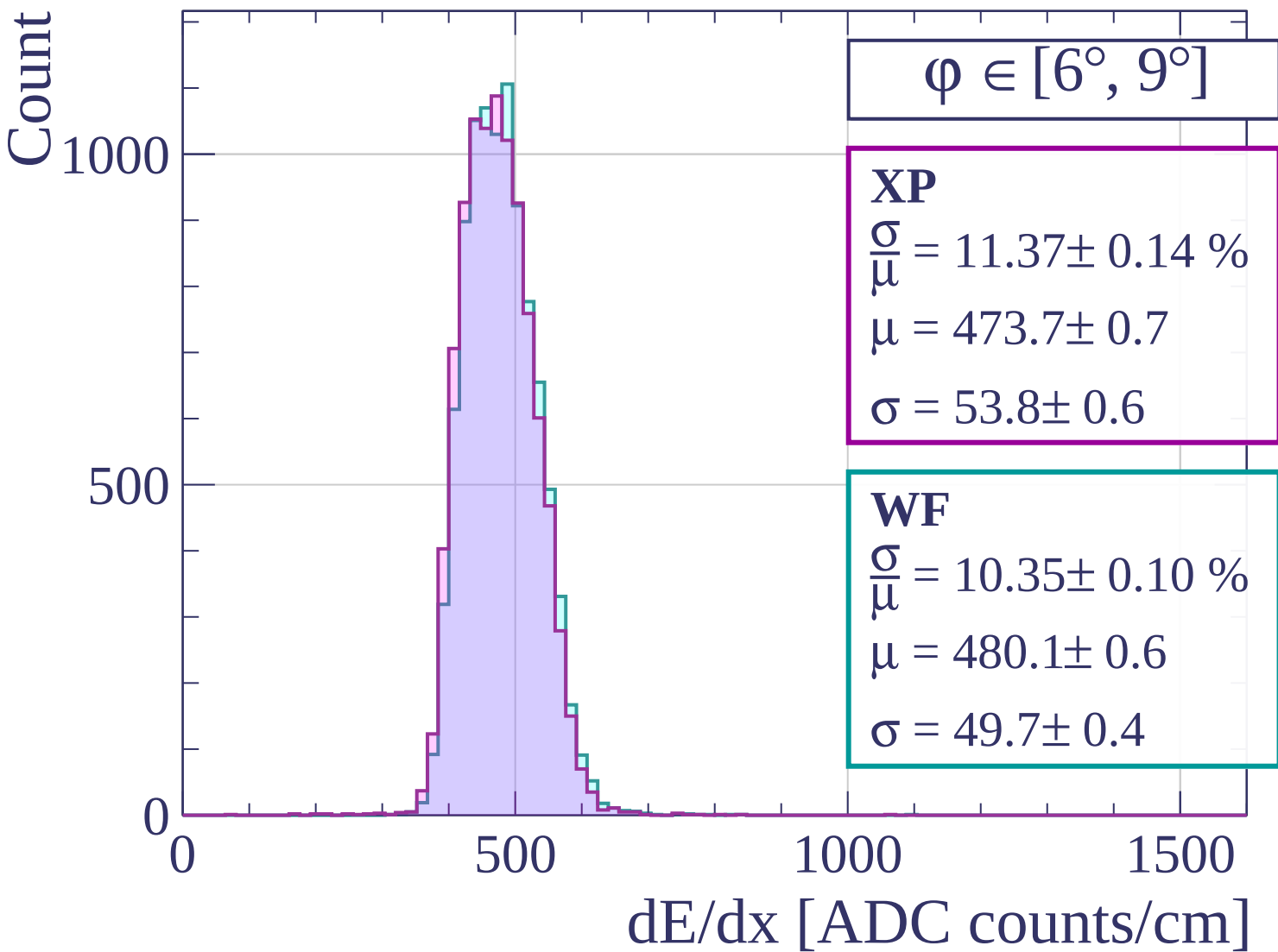
Count











Count

$$\varphi \in [9^\circ, 12^\circ]$$

XP

$$\frac{\sigma}{\mu} = 10.40 \pm 0.18 \%$$

$$\mu = 456.7 \pm 0.8$$

$$\sigma = 47.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.48 \pm 0.12 \%$$

$$\mu = 463.9 \pm 0.7$$

$$\sigma = 44.0 \pm 0.5$$

800

600

400

200

0

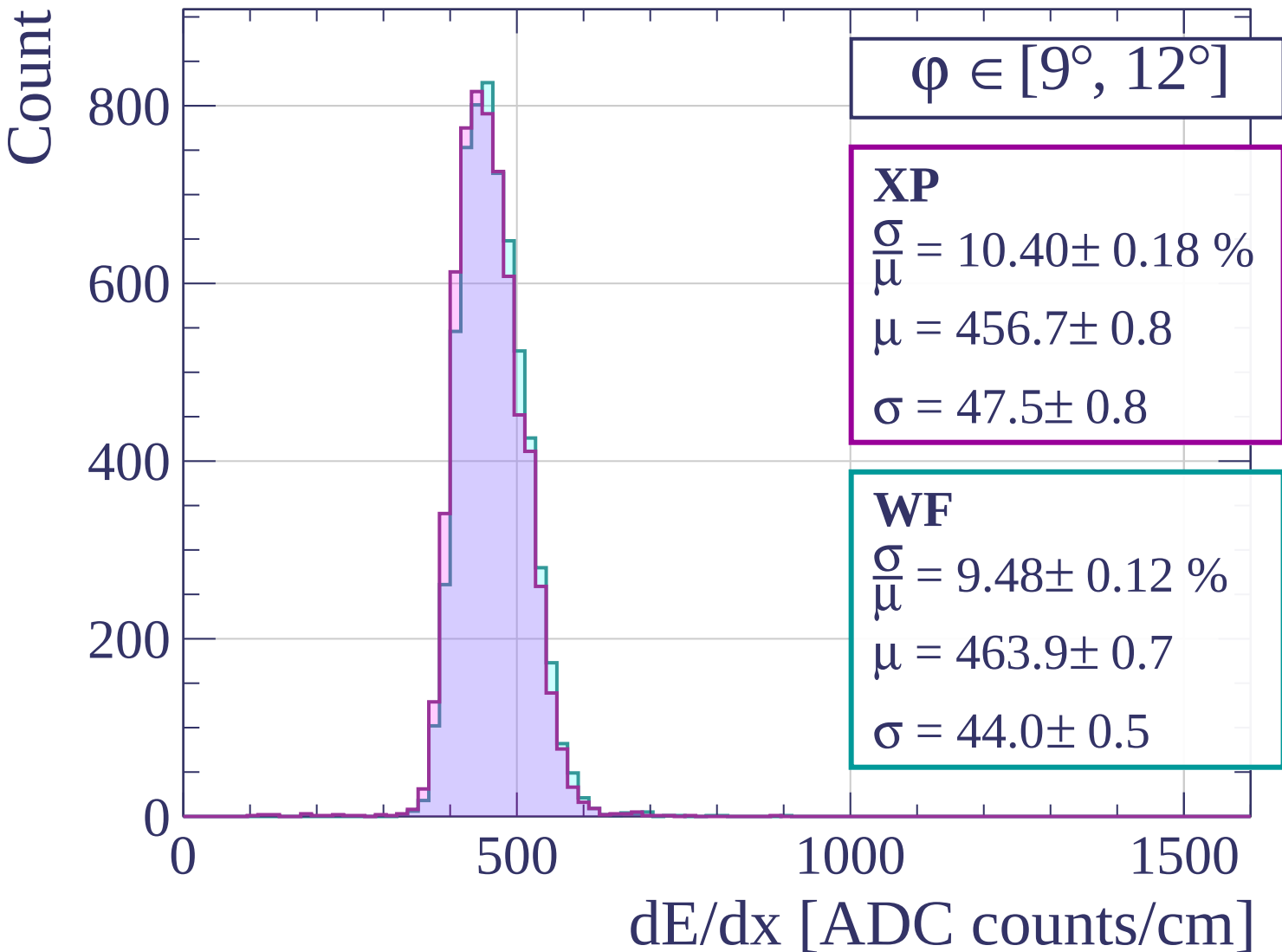
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\phi \in [12^\circ, 15^\circ]$

XP

$$\frac{\sigma}{\mu} = 9.50 \pm 0.22 \%$$

$$\mu = 445.9 \pm 0.9$$

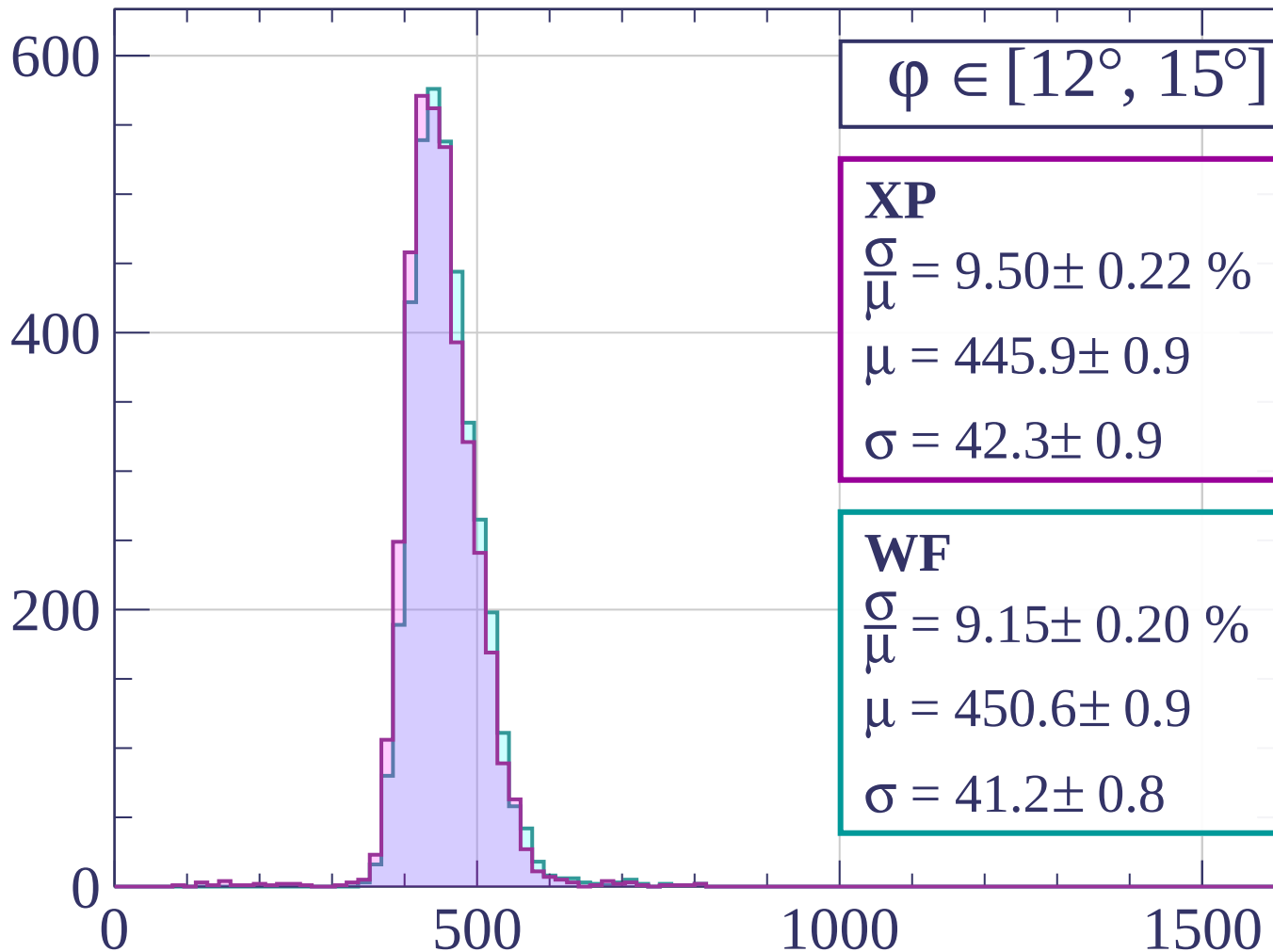
$$\sigma = 42.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.15 \pm 0.20 \%$$

$$\mu = 450.6 \pm 0.9$$

$$\sigma = 41.2 \pm 0.8$$



dE/dx [ADC counts/cm]

Count

$\varphi \in [15^\circ, 18^\circ]$

XP

$$\frac{\sigma}{\mu} = 8.76 \pm 0.25 \%$$

$$\mu = 435.5 \pm 1.0$$

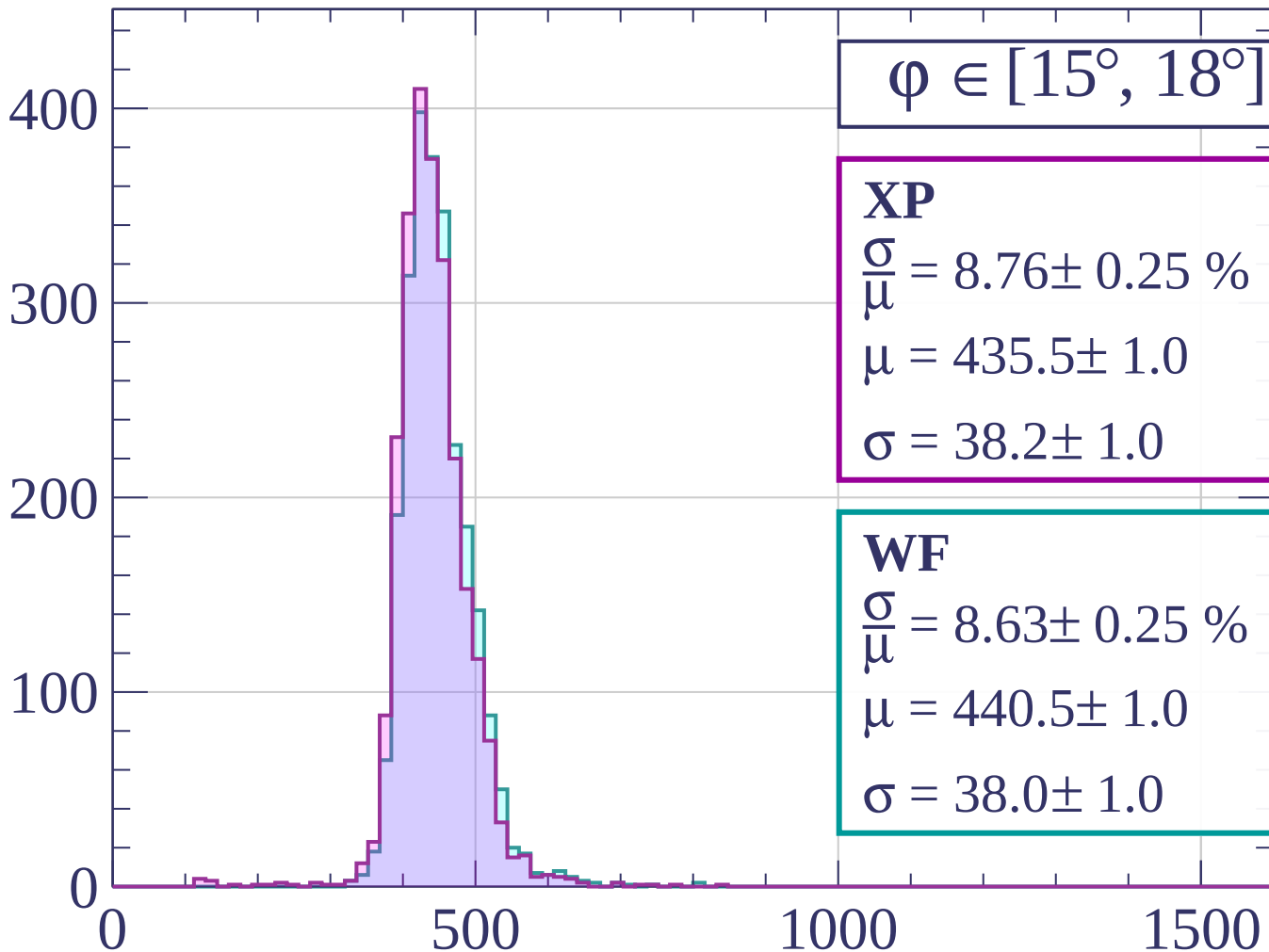
$$\sigma = 38.2 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 8.63 \pm 0.25 \%$$

$$\mu = 440.5 \pm 1.0$$

$$\sigma = 38.0 \pm 1.0$$



dE/dx [ADC counts/cm]

Count

$\varphi \in [18^\circ, 21^\circ]$

XP

$$\frac{\sigma}{\mu} = 7.58 \pm 0.22 \%$$

$$\mu = 429.2 \pm 1.0$$

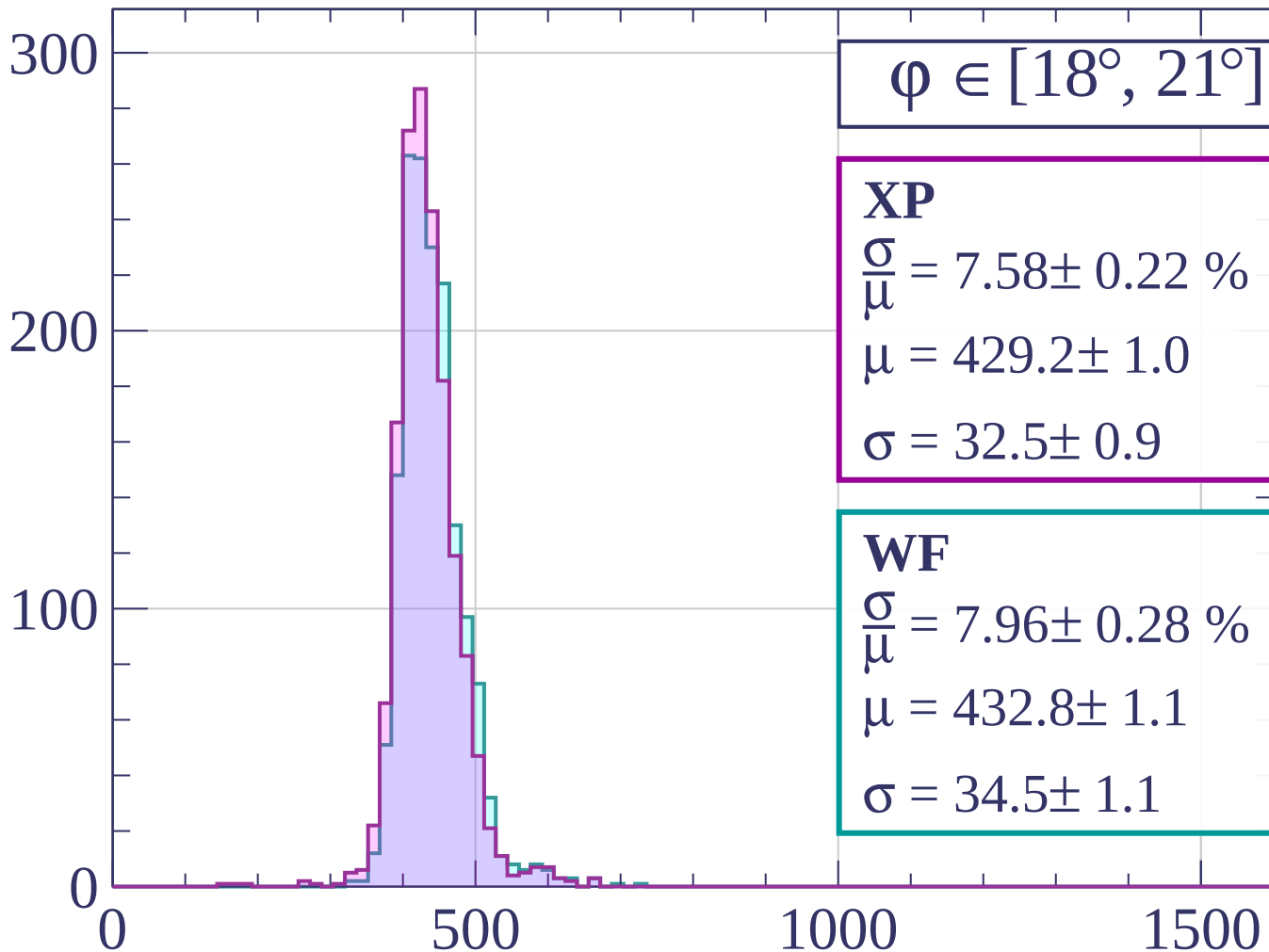
$$\sigma = 32.5 \pm 0.9$$

WF

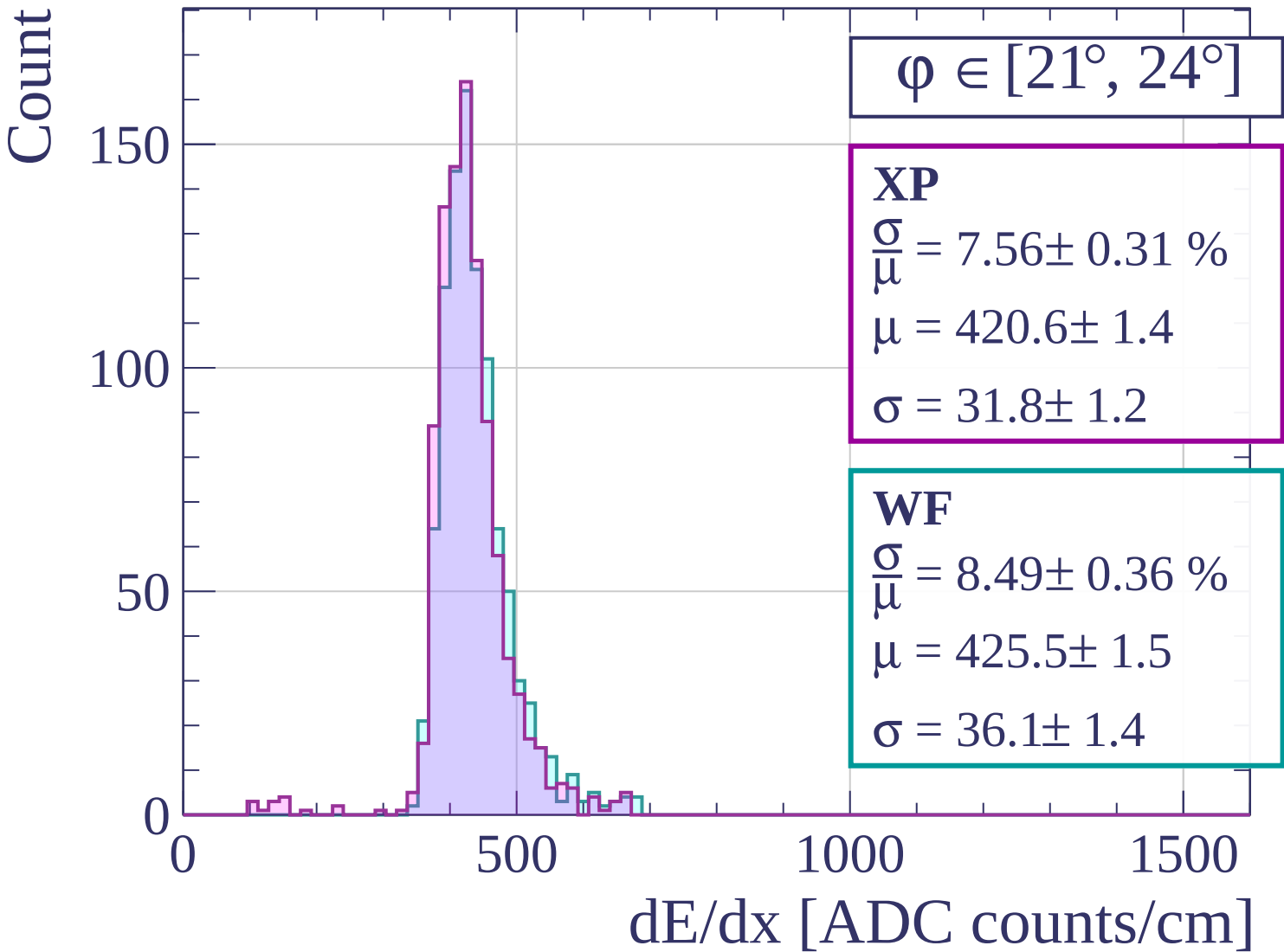
$$\frac{\sigma}{\mu} = 7.96 \pm 0.28 \%$$

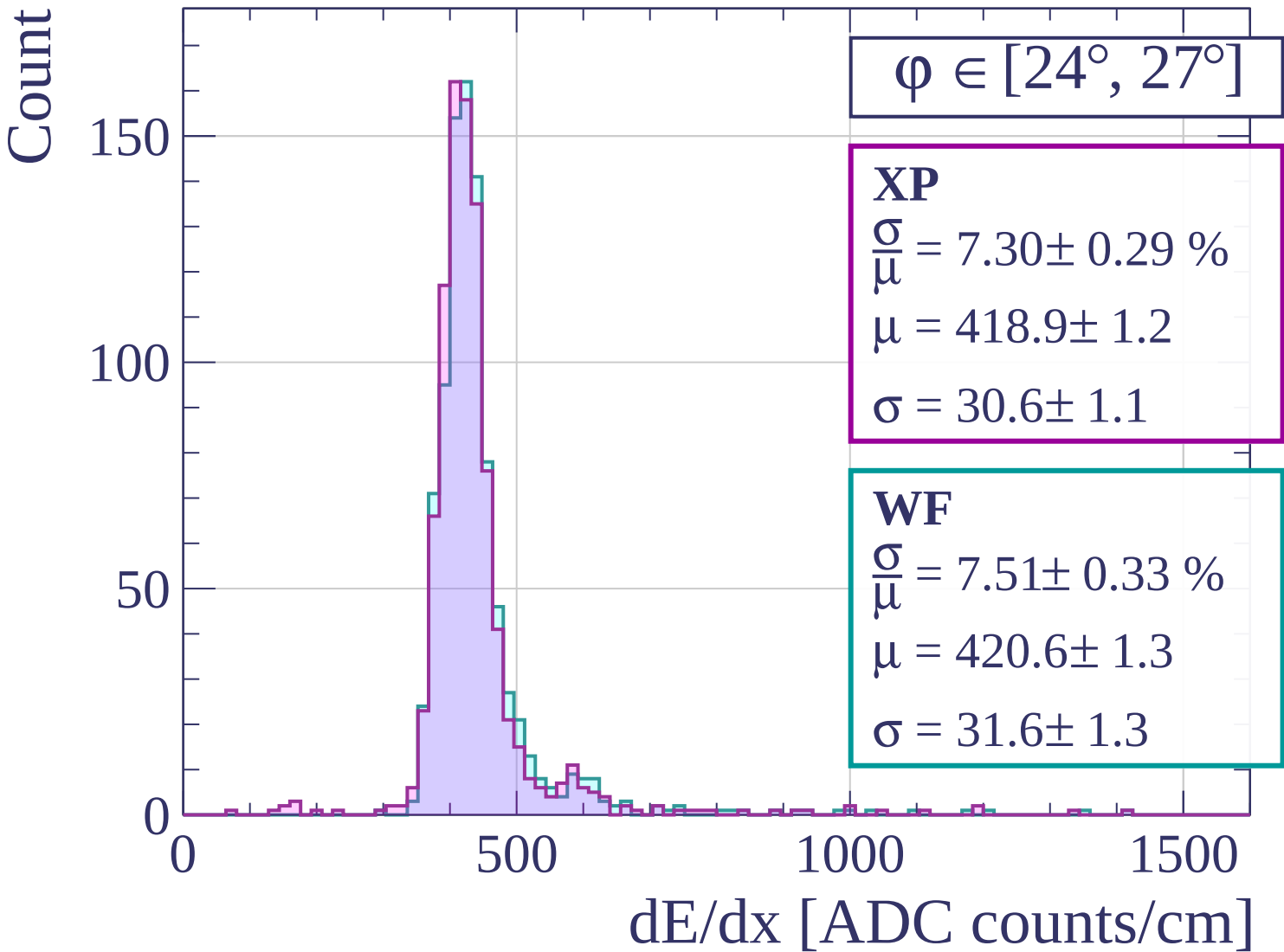
$$\mu = 432.8 \pm 1.1$$

$$\sigma = 34.5 \pm 1.1$$



dE/dx [ADC counts/cm]





Count

$\phi \in [27^\circ, 30^\circ]$

XP

$$\frac{\sigma}{\mu} = 7.92 \pm 0.36 \%$$

$$\mu = 417.8 \pm 1.4$$

$$\sigma = 33.1 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 7.88 \pm 0.34 \%$$

$$\mu = 419.1 \pm 1.4$$

$$\sigma = 33.0 \pm 1.3$$

150

100

50

0

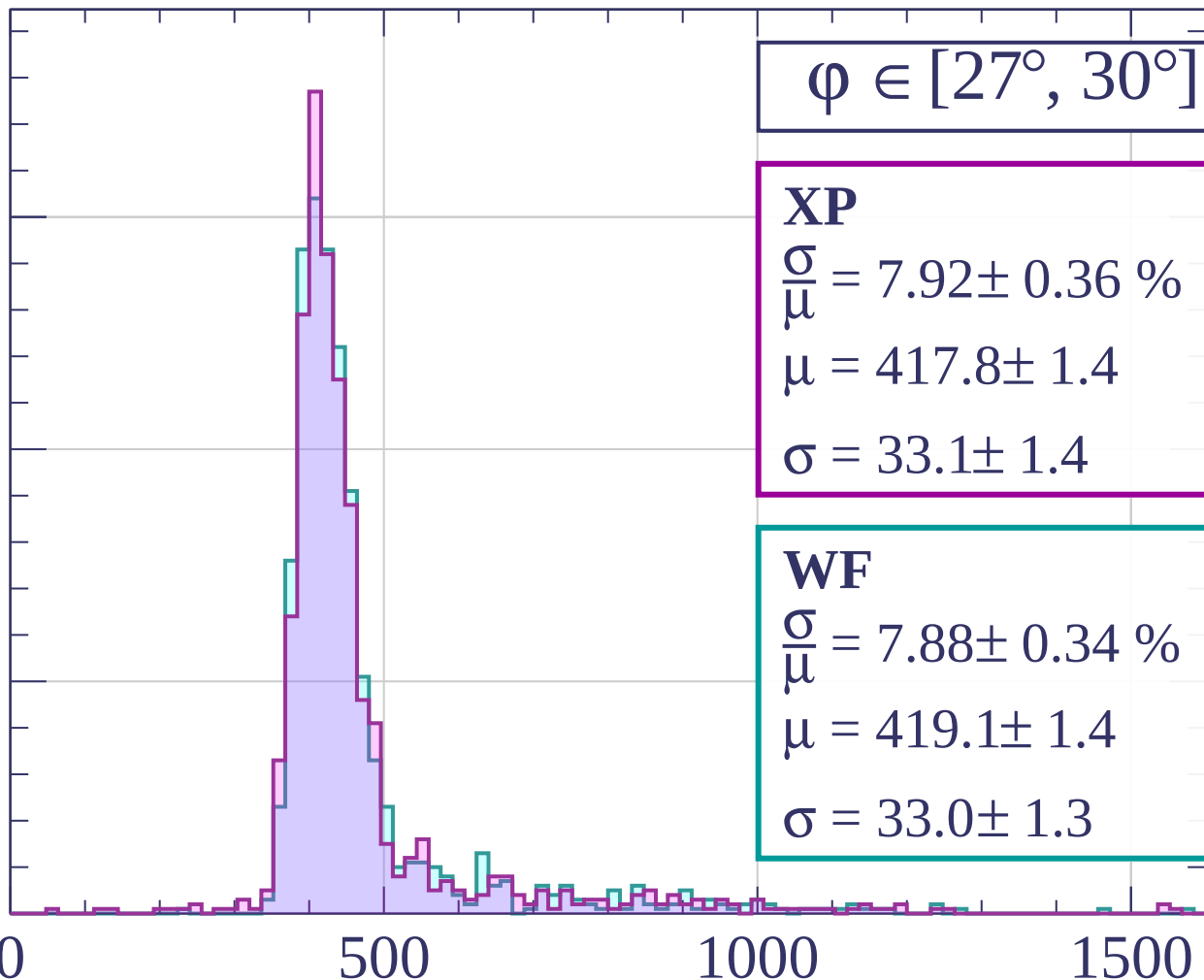
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\phi \in [30^\circ, 33^\circ]$

XP

$$\frac{\sigma}{\mu} = 7.46 \pm 0.28 \%$$

$$\mu = 419.0 \pm 1.1$$

$$\sigma = 31.3 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 7.59 \pm 0.29 \%$$

$$\mu = 420.1 \pm 1.2$$

$$\sigma = 31.9 \pm 1.1$$

200

150

100

50

0

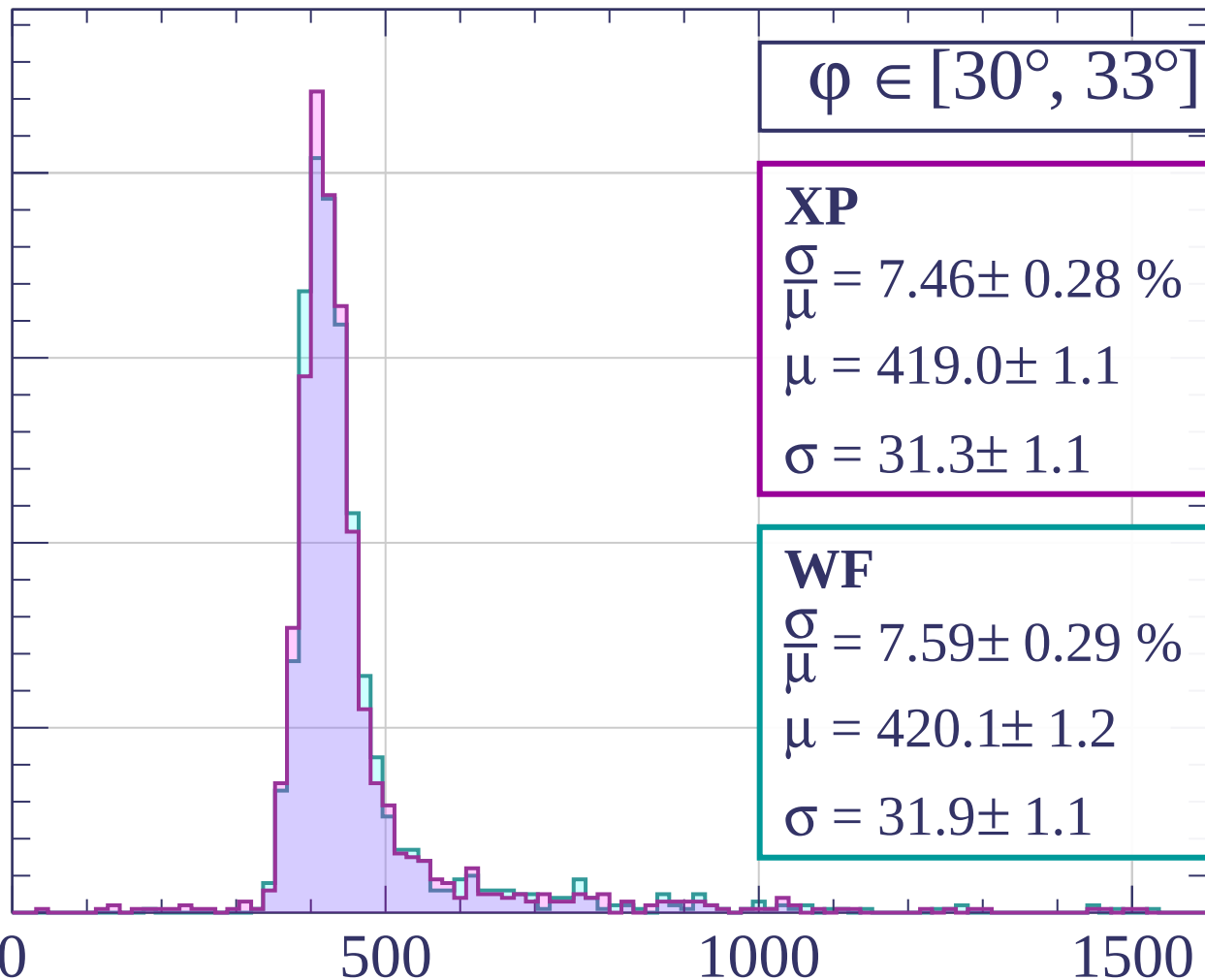
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\varphi \in [33^\circ, 36^\circ]$

XP

$$\frac{\sigma}{\mu} = 7.53 \pm 0.29 \%$$

$$\mu = 419.8 \pm 1.2$$

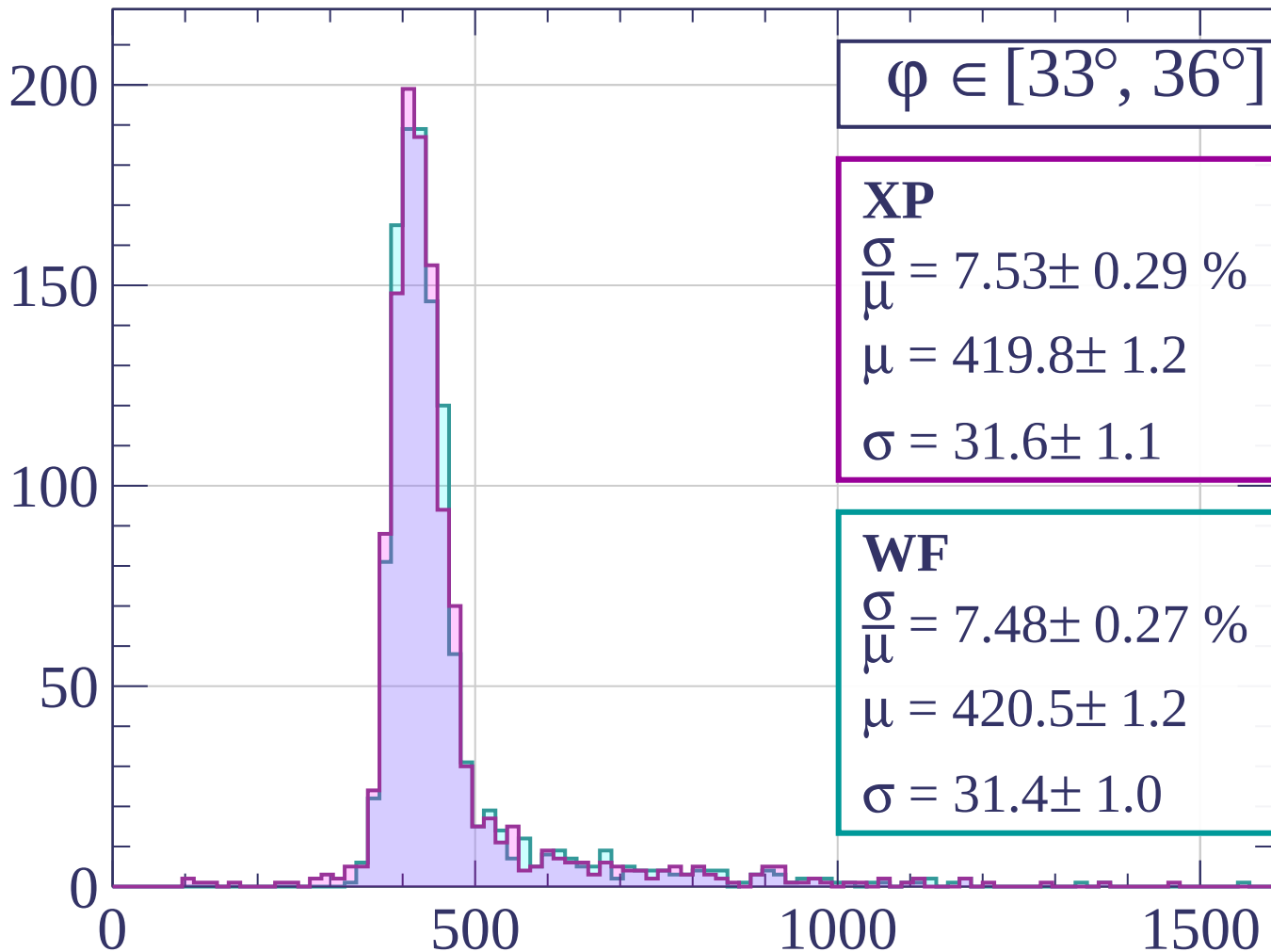
$$\sigma = 31.6 \pm 1.1$$

WF

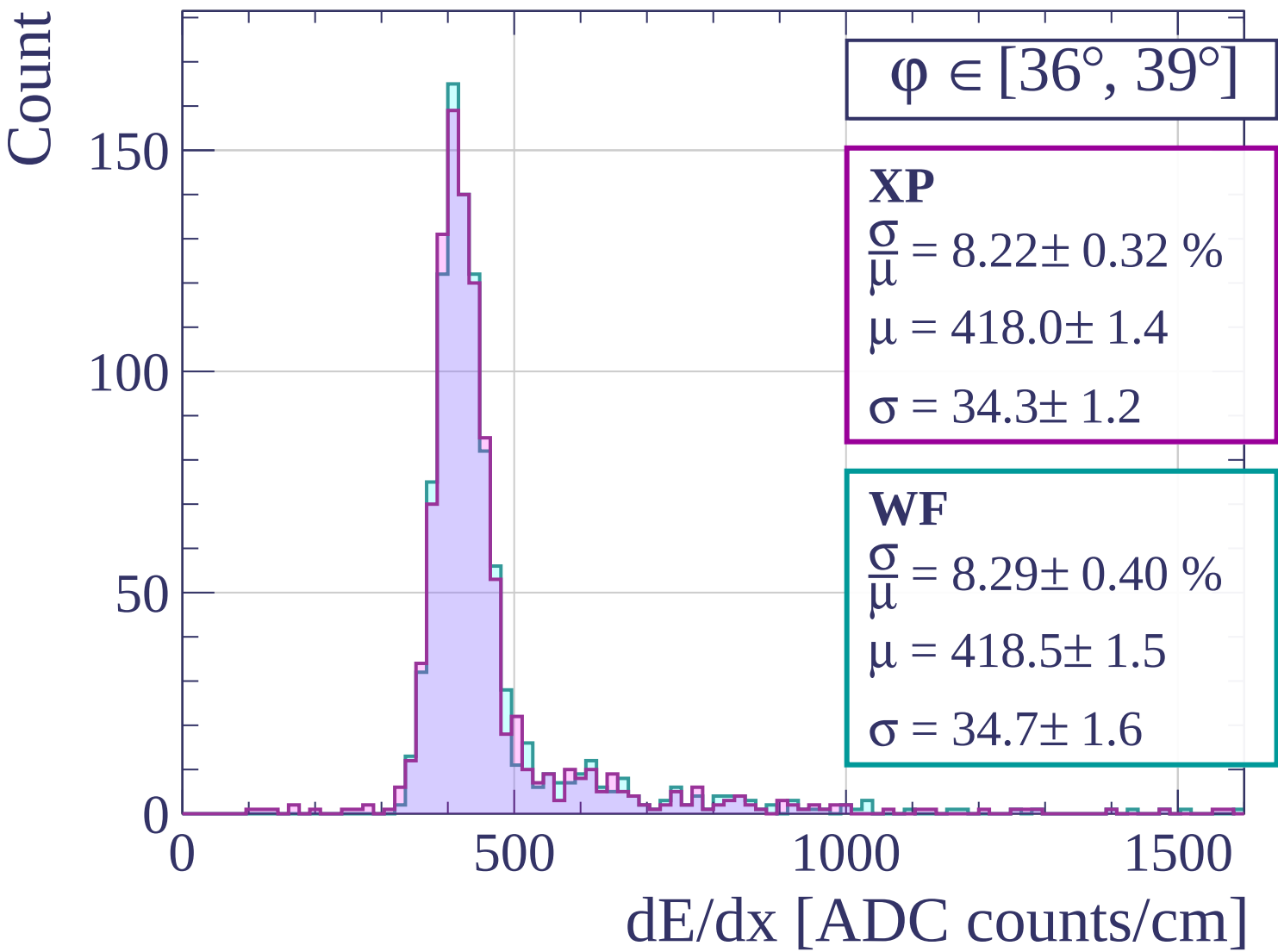
$$\frac{\sigma}{\mu} = 7.48 \pm 0.27 \%$$

$$\mu = 420.5 \pm 1.2$$

$$\sigma = 31.4 \pm 1.0$$



dE/dx [ADC counts/cm]



Count

$\phi \in [39^\circ, 42^\circ]$

XP

$$\frac{\sigma}{\mu} = 8.41 \pm 0.44 \%$$

$$\mu = 414.0 \pm 1.7$$

$$\sigma = 34.8 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 8.55 \pm 0.48 \%$$

$$\mu = 414.6 \pm 1.7$$

$$\sigma = 35.5 \pm 1.8$$

100

50

0

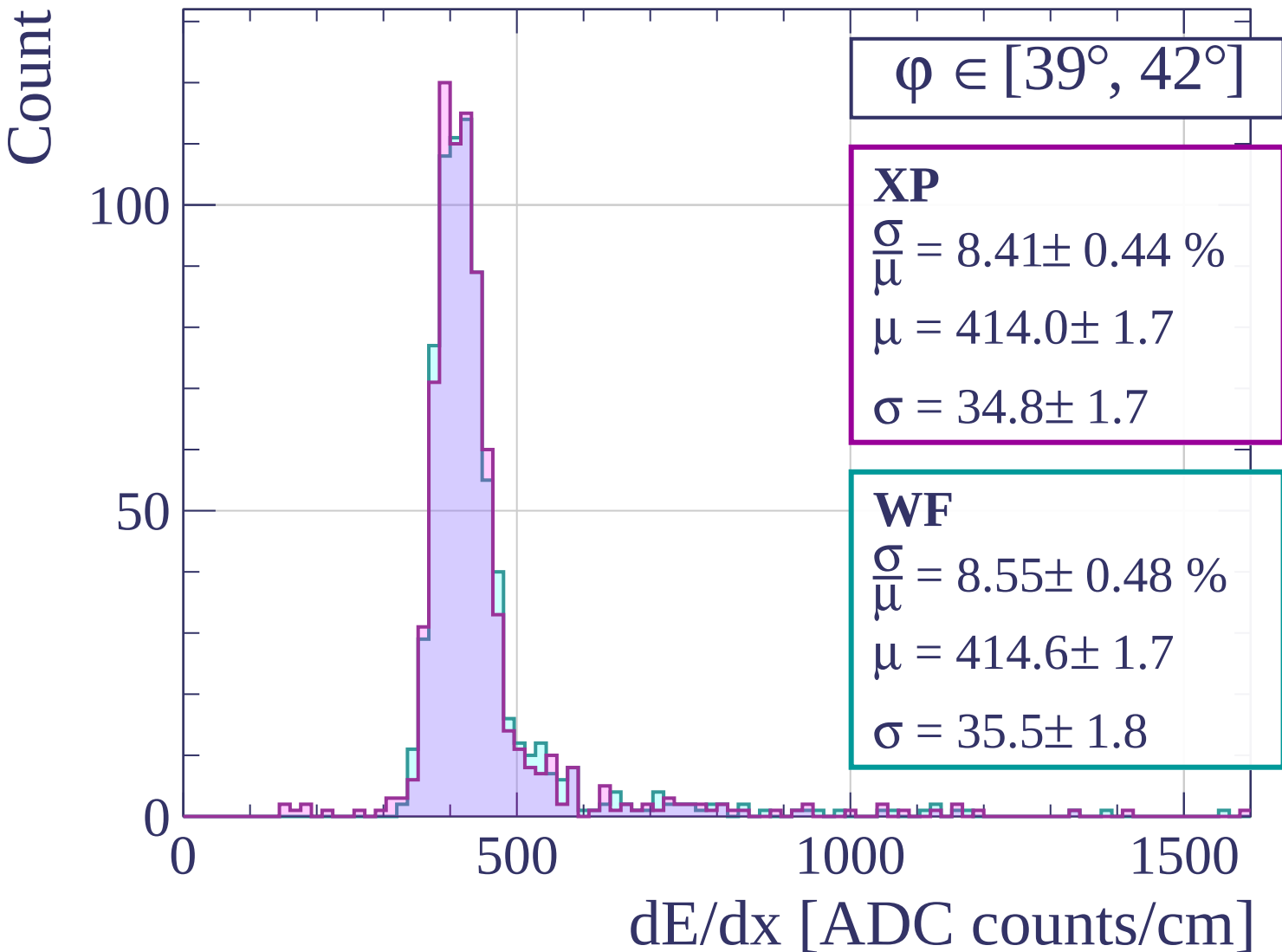
0

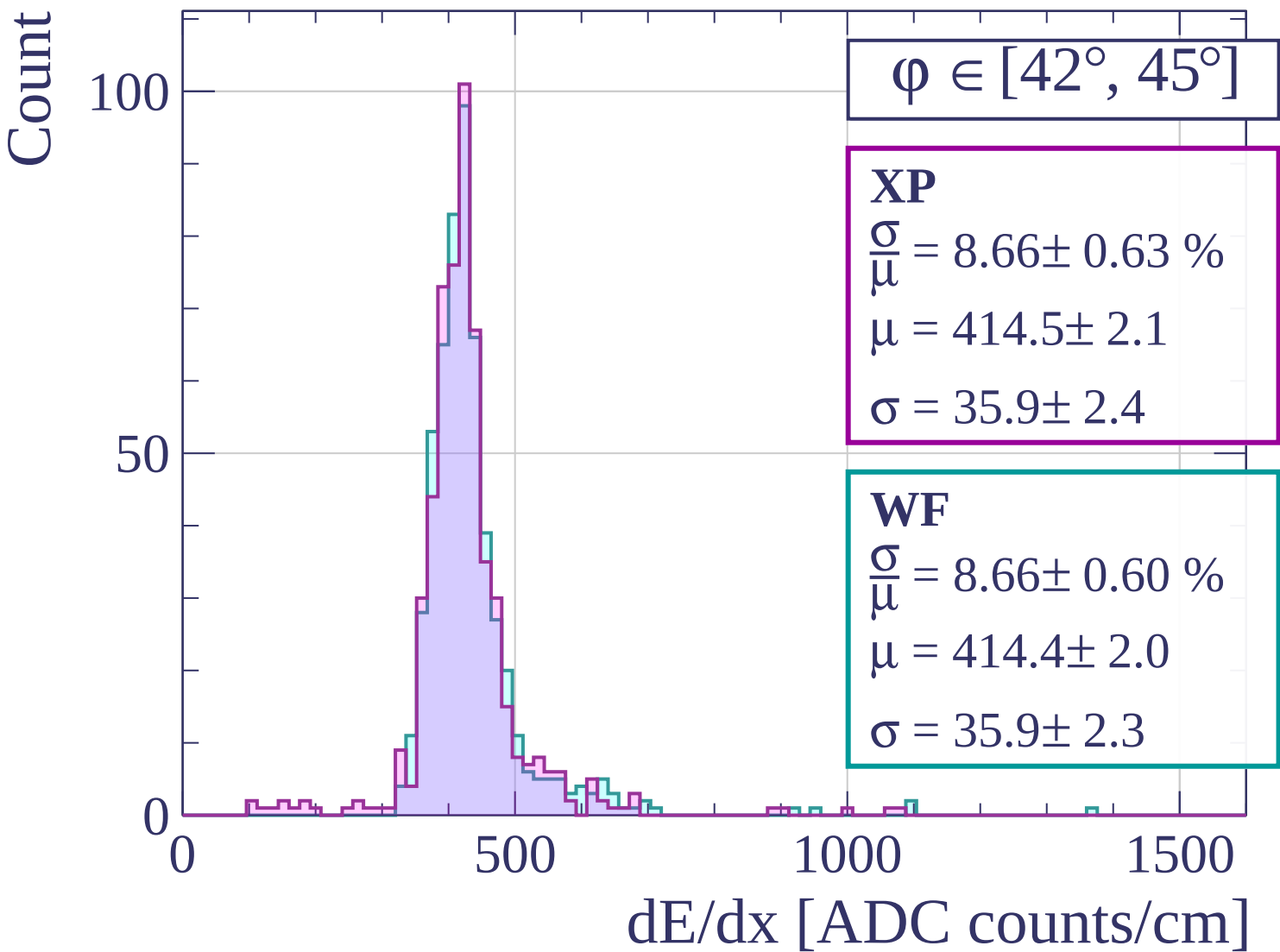
500

1000

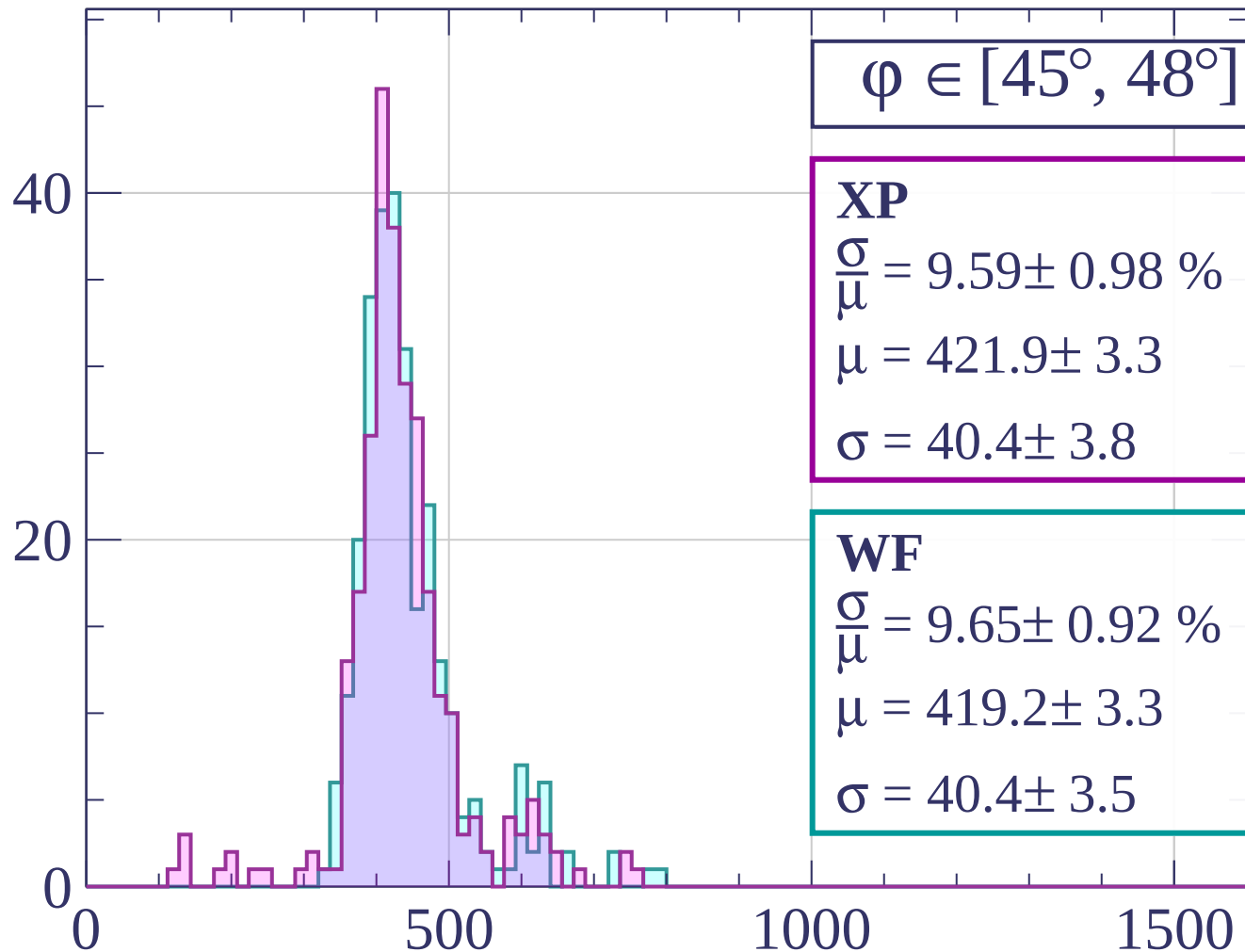
1500

dE/dx [ADC counts/cm]



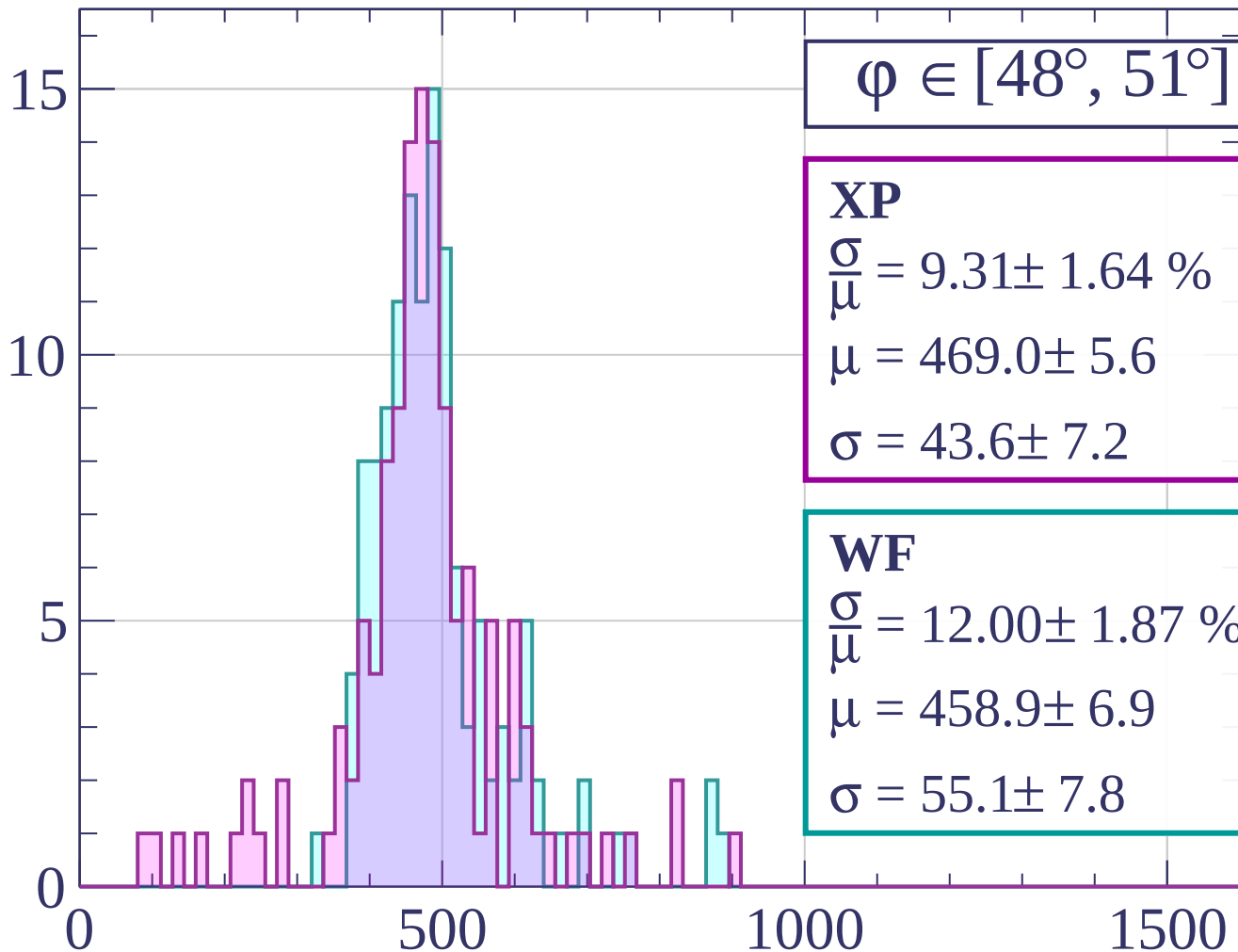


Count



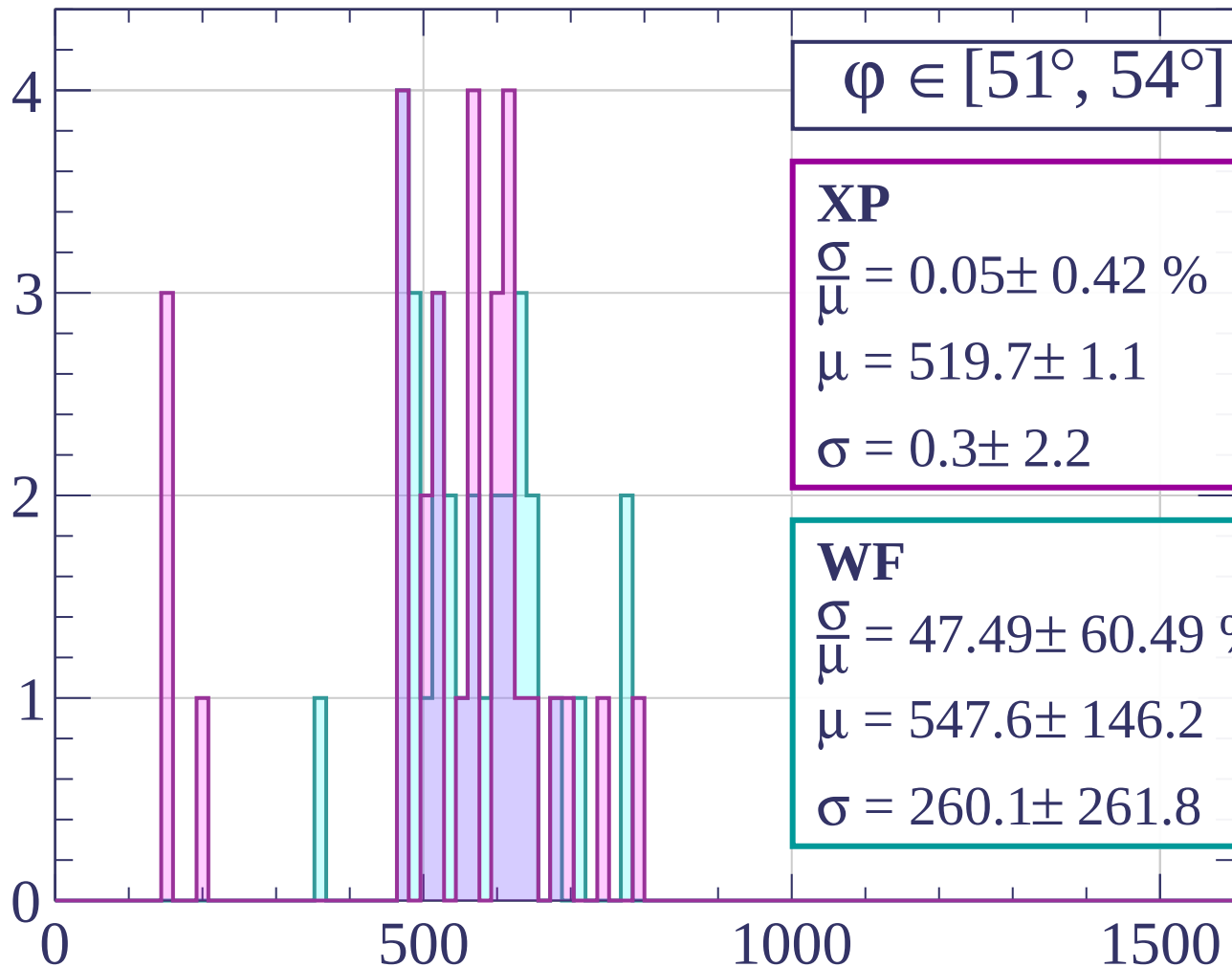
dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count

$\phi \in [54^\circ, 57^\circ]$

XP

$$\frac{\sigma}{\mu} = 142.42 \pm 455.00 \%$$

$$\mu = 520.0 \pm 811.7$$

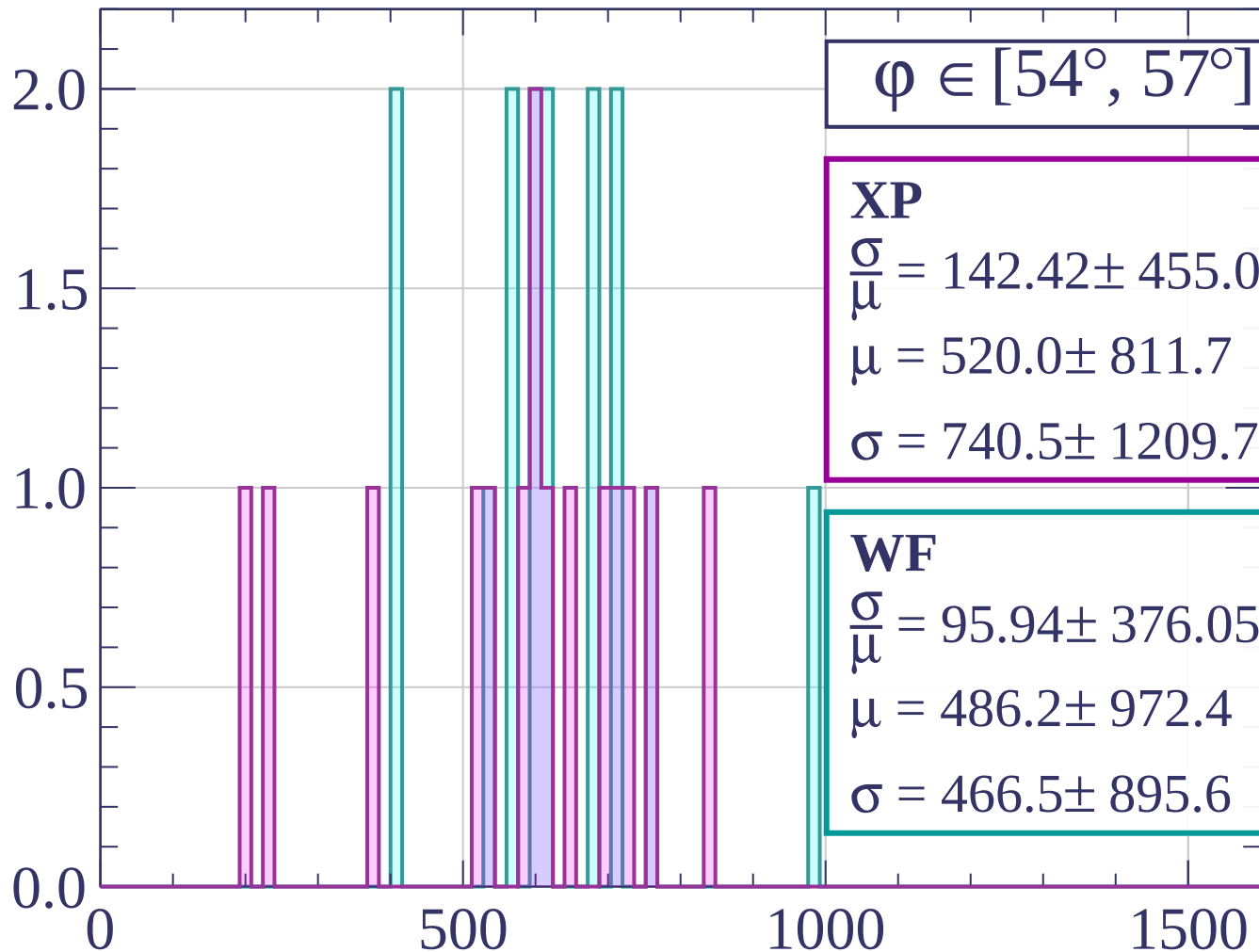
$$\sigma = 740.5 \pm 1209.7$$

WF

$$\frac{\sigma}{\mu} = 95.94 \pm 376.05 \%$$

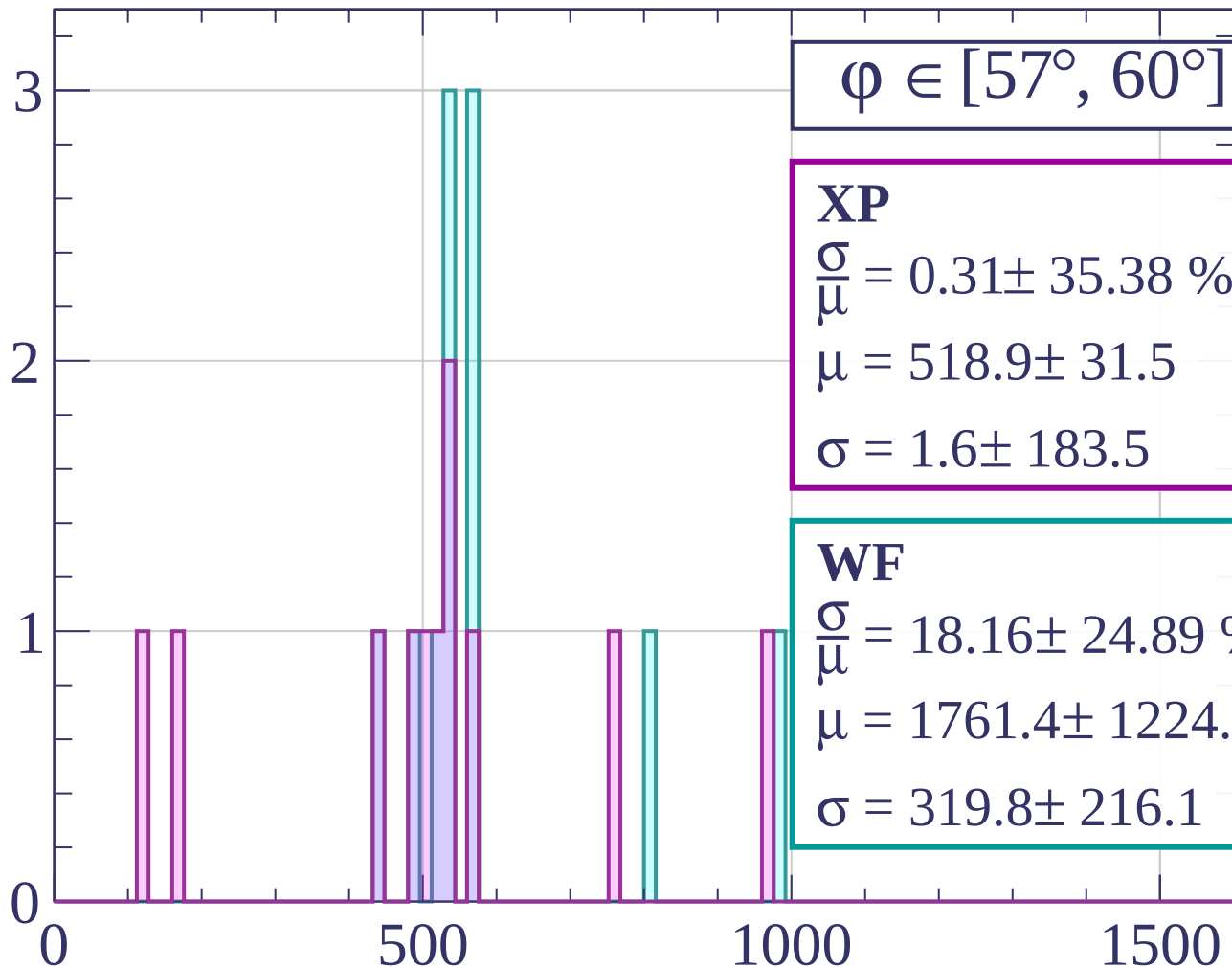
$$\mu = 486.2 \pm 972.4$$

$$\sigma = 466.5 \pm 895.6$$



dE/dx [ADC counts/cm]

Count



$\phi \in [57^\circ, 60^\circ]$

XP

$$\frac{\sigma}{\mu} = 0.31 \pm 35.38 \%$$

$$\mu = 518.9 \pm 31.5$$

$$\sigma = 1.6 \pm 183.5$$

WF

$$\frac{\sigma}{\mu} = 18.16 \pm 24.89 \%$$

$$\mu = 1761.4 \pm 1224.7$$

$$\sigma = 319.8 \pm 216.1$$

dE/dx [ADC counts/cm]

Count

$\phi \in [60^\circ, 63^\circ]$

XP

$$\frac{\sigma}{\mu} = 6.37 \pm 159.17 \%$$

$$\mu = 1099.2 \pm 261.5$$

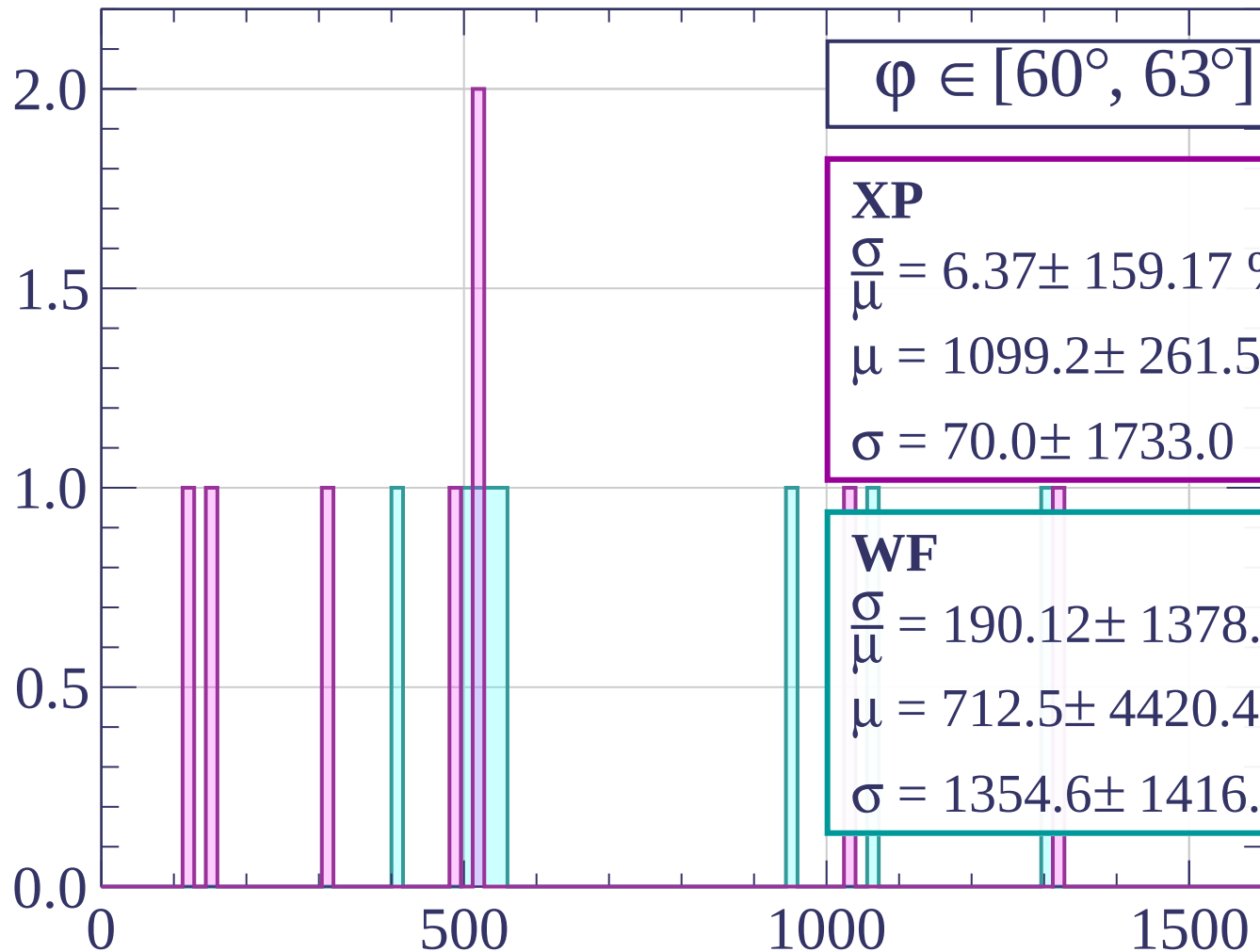
$$\sigma = 70.0 \pm 1733.0$$

WF

$$\frac{\sigma}{\mu} = 190.12 \pm 1378.22$$

$$\mu = 712.5 \pm 4420.4$$

$$\sigma = 1354.6 \pm 1416.1$$



dE/dx [ADC counts/cm]

Count

$\varphi \in [63^\circ, 66^\circ]$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0

0.5

0.0

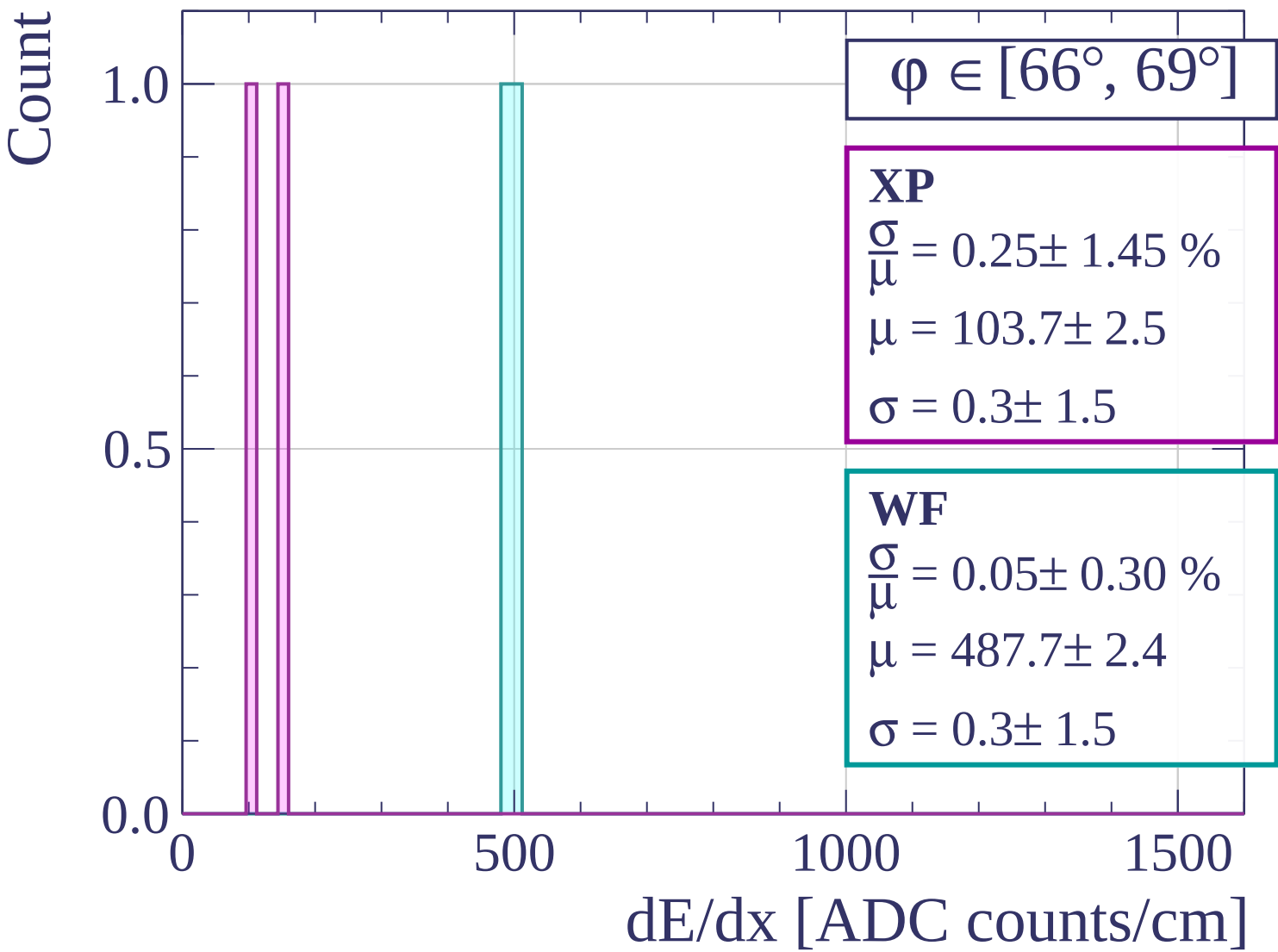
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\phi \in [69^\circ, 72^\circ]$

XP

$$\frac{\sigma}{\mu} = 11.76 \pm 112.74 \%$$

$$\mu = 136.0 \pm 55.4$$

$$\sigma = 16.0 \pm 146.8$$

WF

$$\frac{\sigma}{\mu} = 0.07 \pm 0.52 \%$$

$$\mu = 375.7 \pm 2.3$$

$$\sigma = 0.3 \pm 2.0$$

1.0

0.5

0.0

0

500

1000

1500

dE/dx [ADC counts/cm]

Count

$\phi \in [72^\circ, 75^\circ]$

XP

$$\frac{\sigma}{\mu} = 13.95 \pm 86.36 \%$$

$$\mu = 344.0 \pm 148.7$$

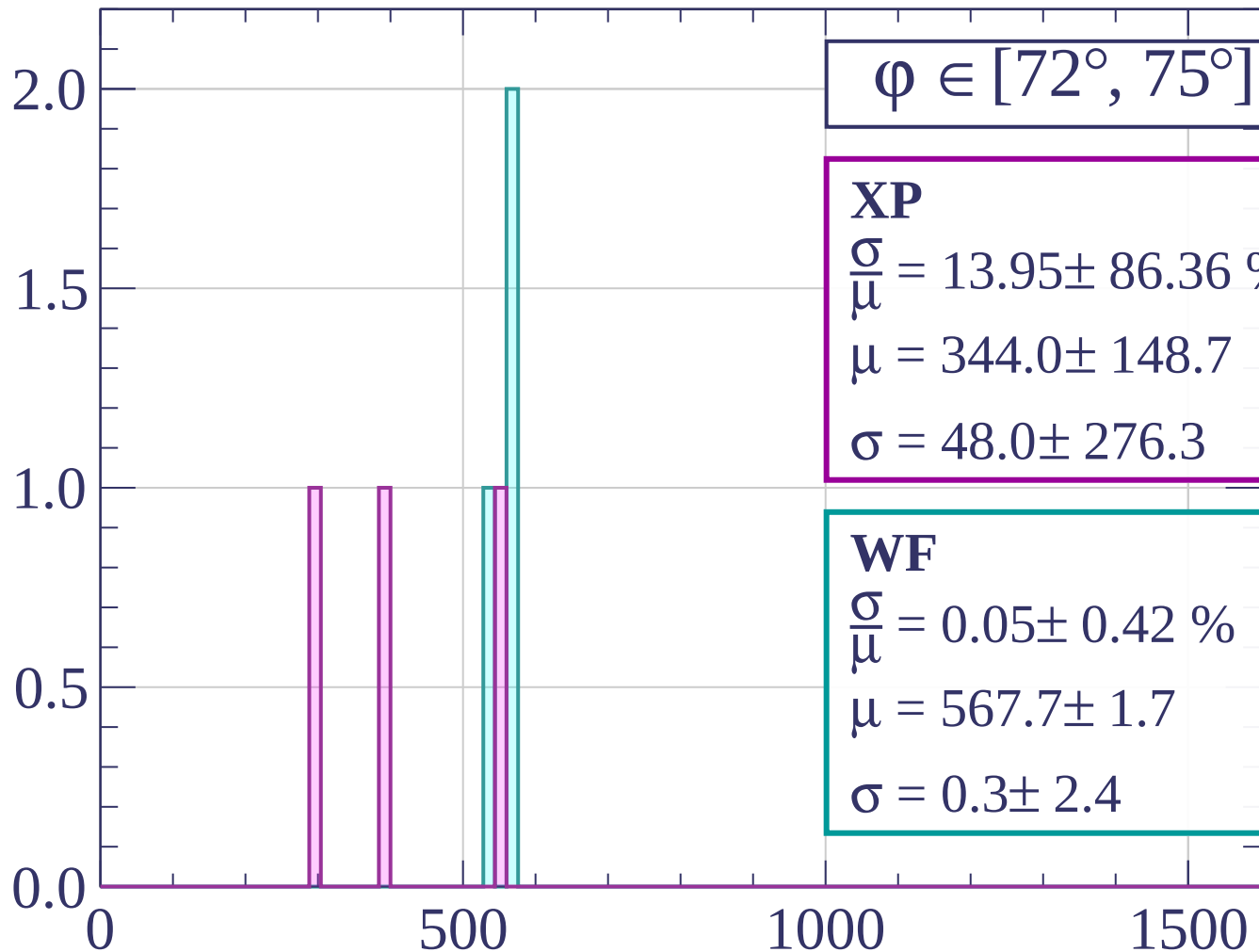
$$\sigma = 48.0 \pm 276.3$$

WF

$$\frac{\sigma}{\mu} = 0.05 \pm 0.42 \%$$

$$\mu = 567.7 \pm 1.7$$

$$\sigma = 0.3 \pm 2.4$$



dE/dx [ADC counts/cm]

Count

$\varphi \in [75^\circ, 78^\circ]$

XP

$$\frac{\sigma}{\mu} = 0.10 \pm 0.51 \%$$

$$\mu = 247.7 \pm 2.4$$

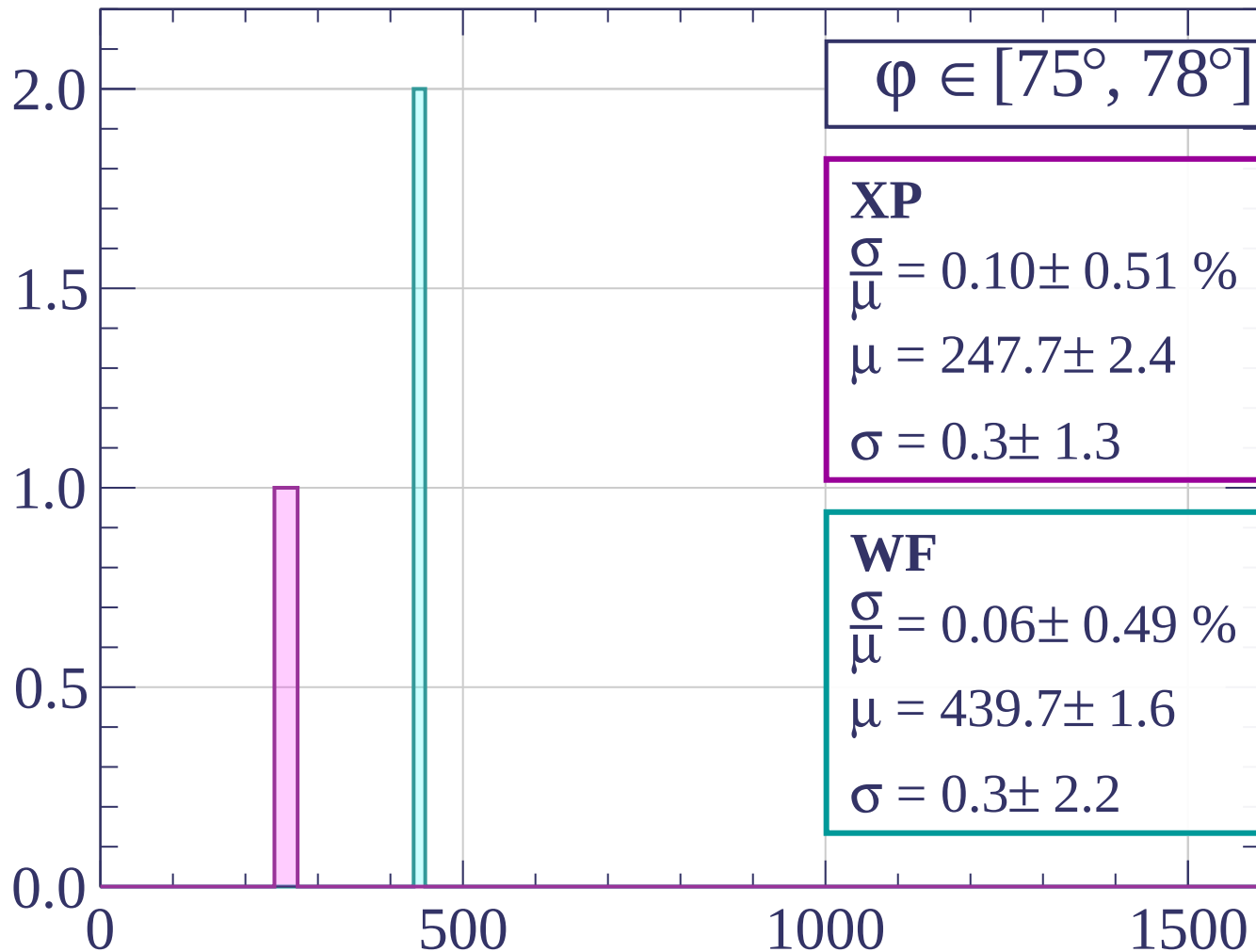
$$\sigma = 0.3 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 0.06 \pm 0.49 \%$$

$$\mu = 439.7 \pm 1.6$$

$$\sigma = 0.3 \pm 2.2$$



dE/dx [ADC counts/cm]

Count

$\varphi \in [78^\circ, 81^\circ]$

XP

$$\frac{\sigma}{\mu} = 50.00 \pm 613.70 \%$$

$$\mu = 304.0 \pm 847.0$$

$$\sigma = 152.0 \pm 1442.1$$

WF

$$\frac{\sigma}{\mu} = 10.61 \pm 56.72 \%$$

$$\mu = 528.0 \pm 180.1$$

$$\sigma = 56.0 \pm 280.4$$

1.0

0.5

0.0

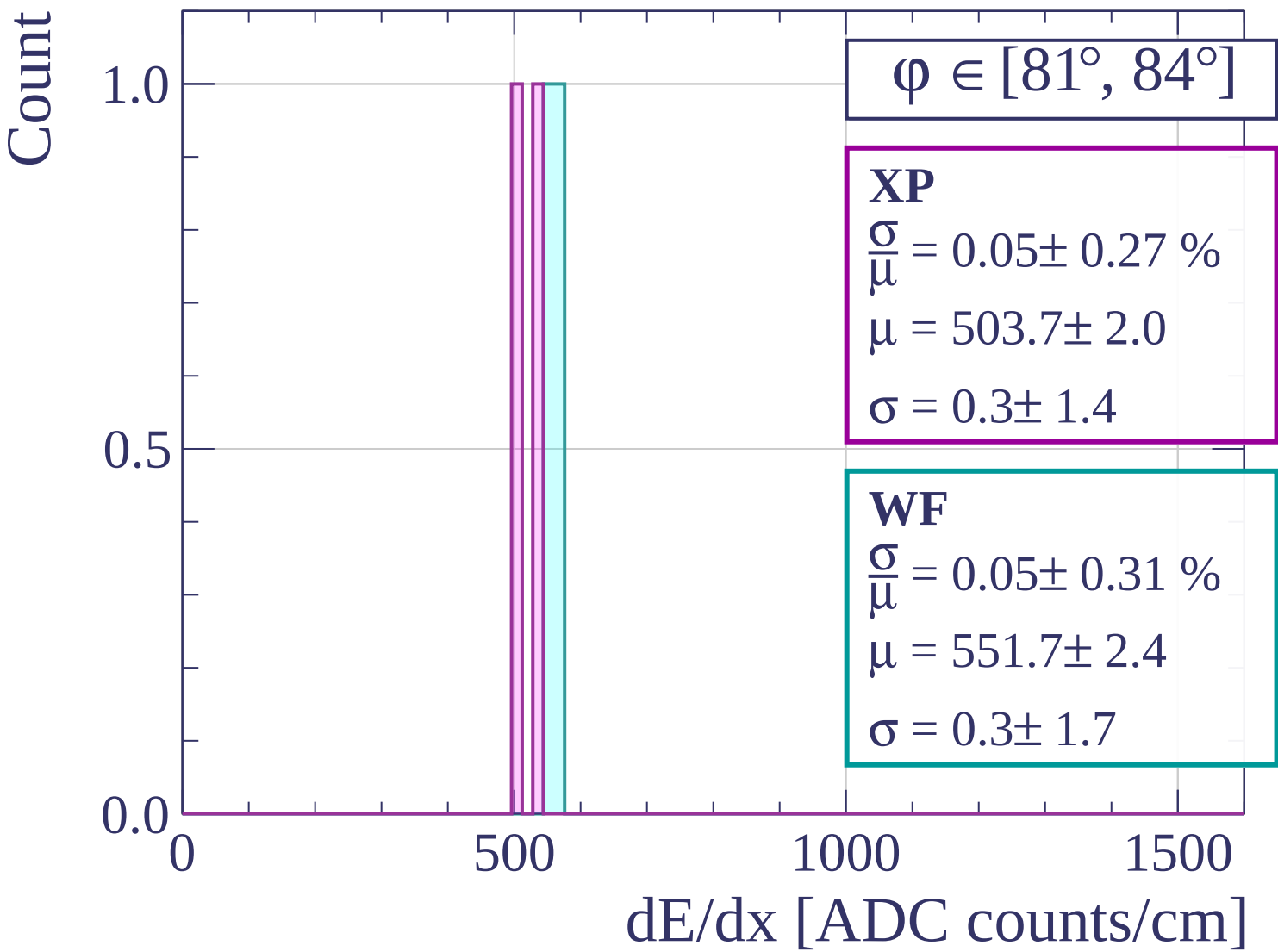
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\phi \in [84^\circ, 87^\circ]$

XP

$$\frac{\sigma}{\mu} = 12.77 \pm 78.48 \%$$

$$\mu = 376.0 \pm 148.7$$

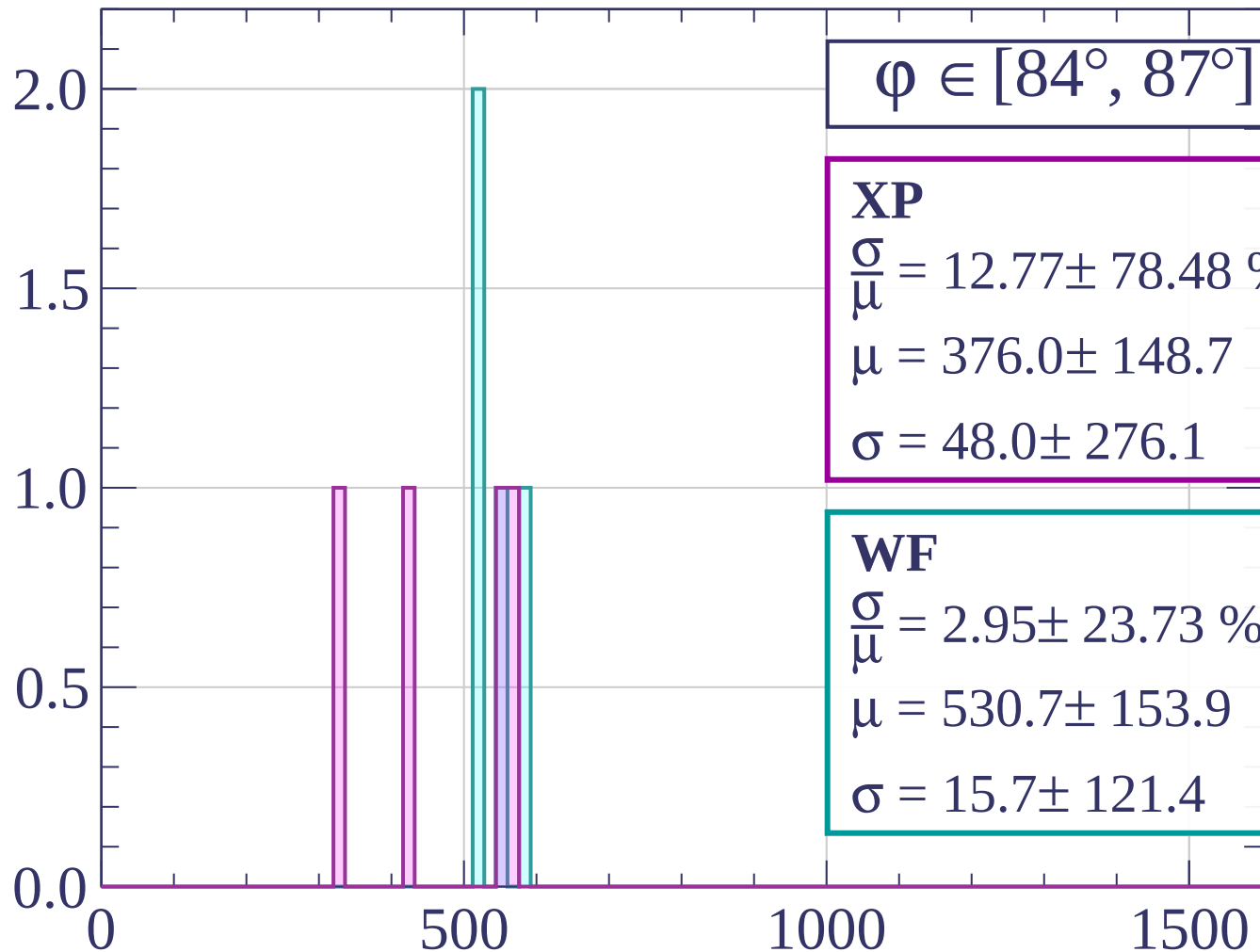
$$\sigma = 48.0 \pm 276.1$$

WF

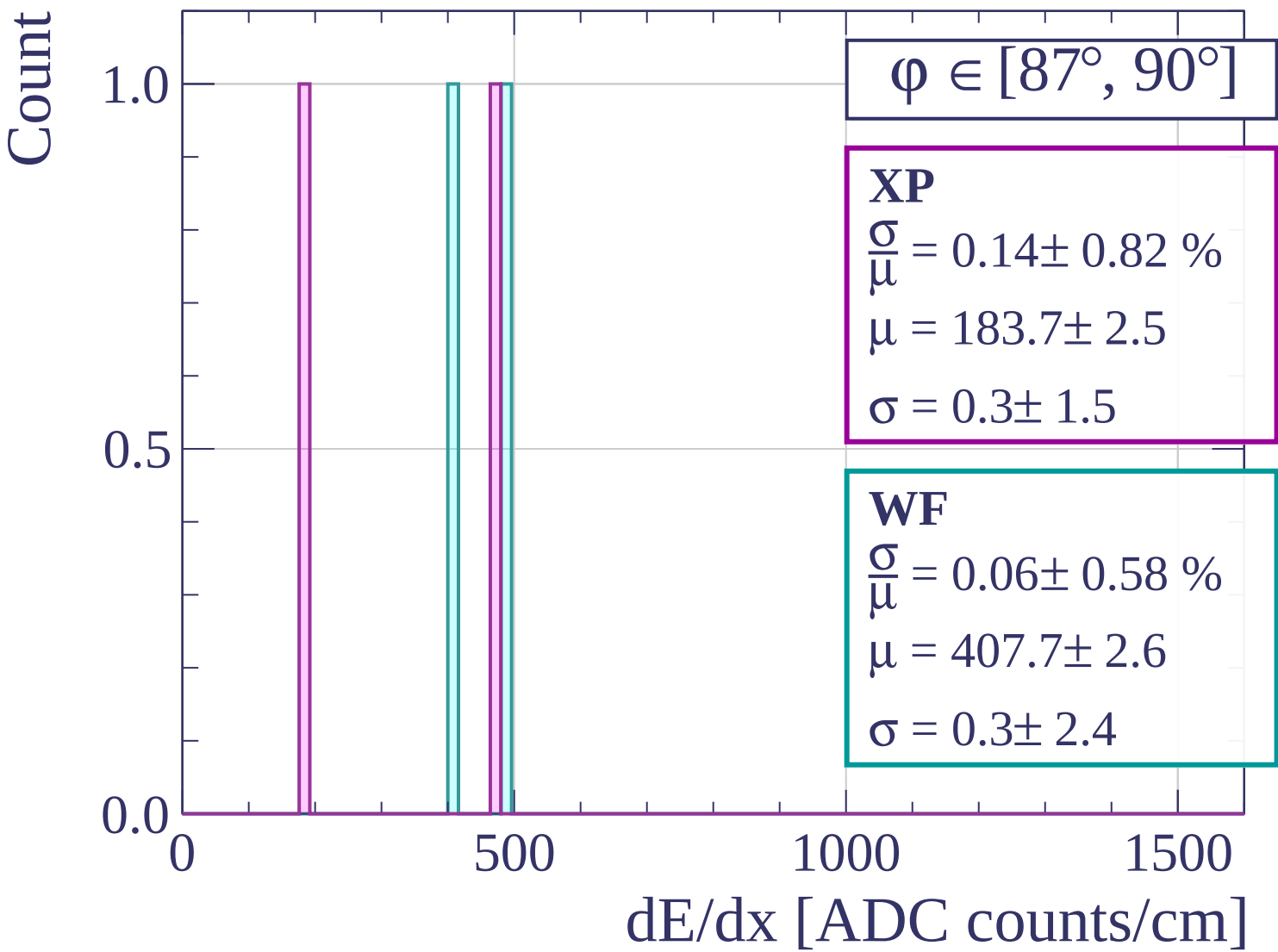
$$\frac{\sigma}{\mu} = 2.95 \pm 23.73 \%$$

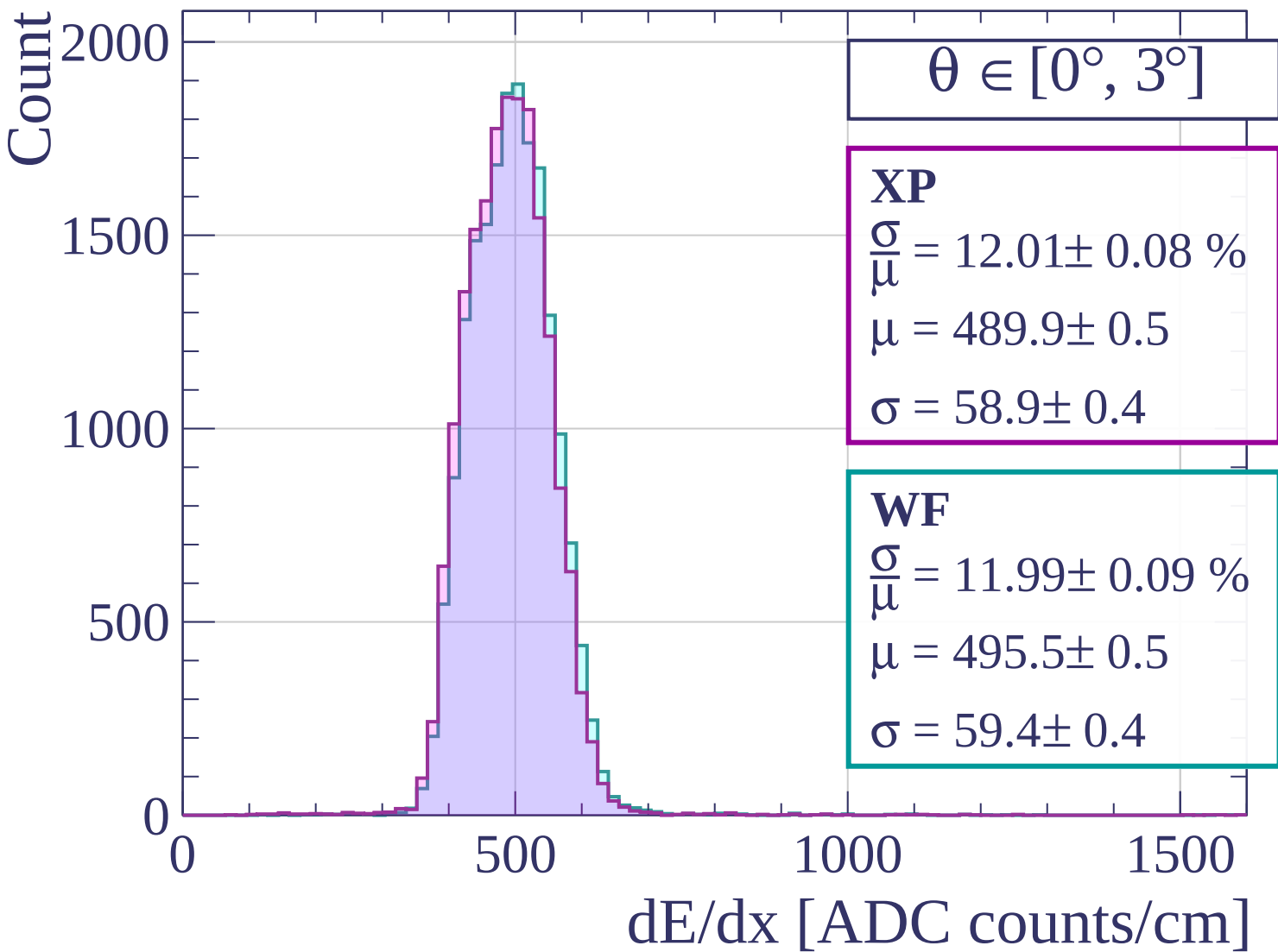
$$\mu = 530.7 \pm 153.9$$

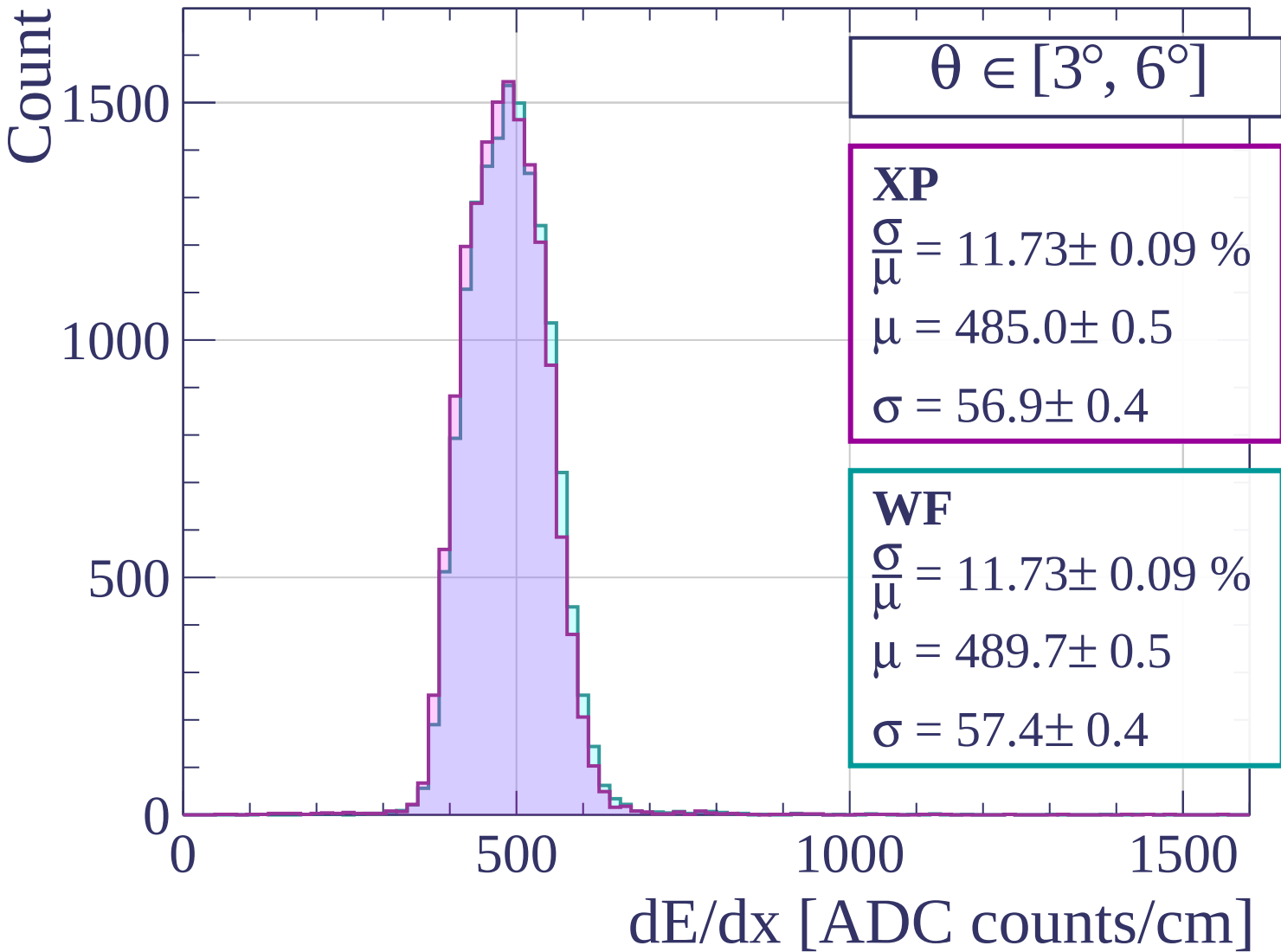
$$\sigma = 15.7 \pm 121.4$$

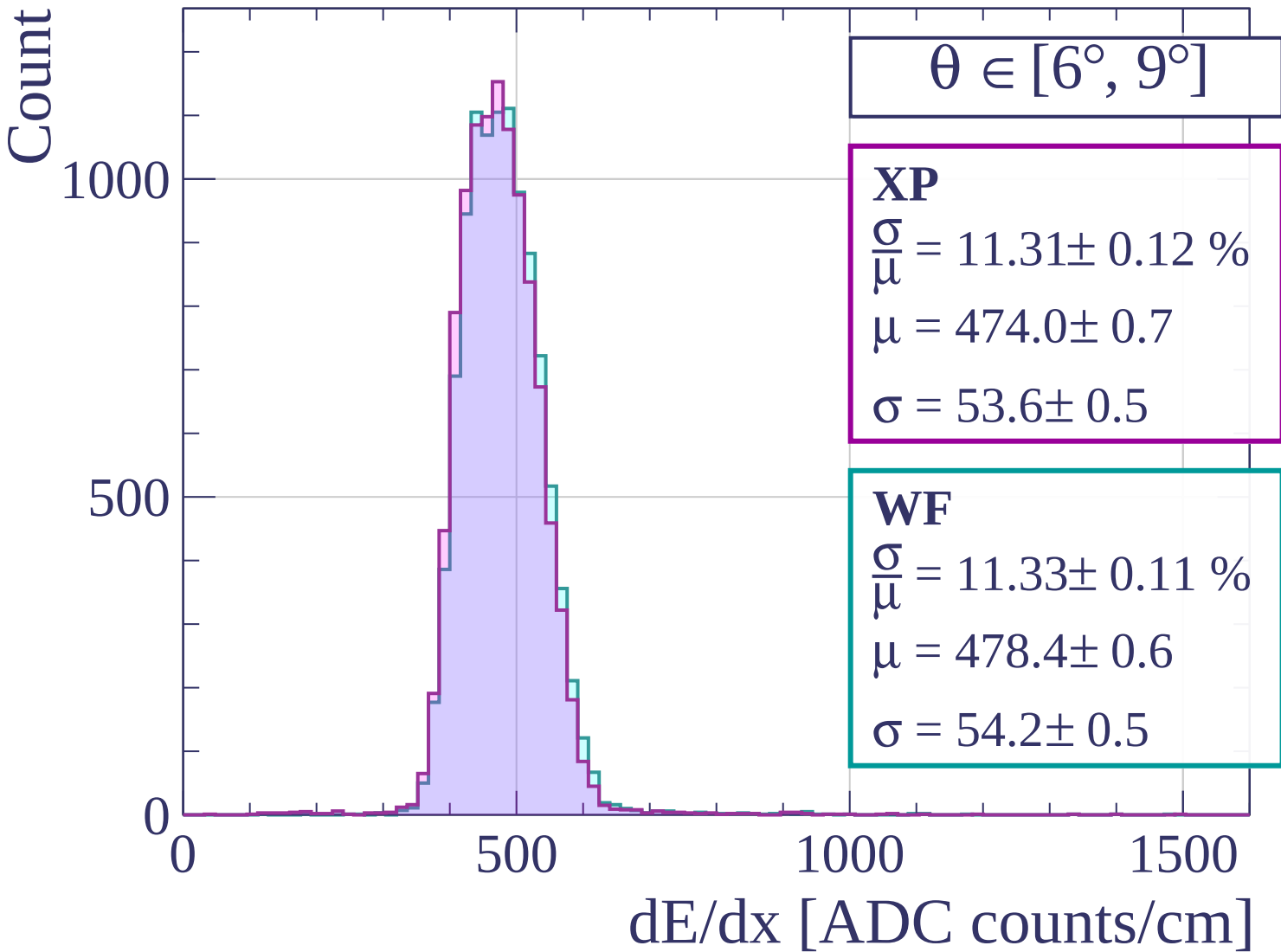


dE/dx [ADC counts/cm]









Count

$$\theta \in [9^\circ, 12^\circ]$$

XP

$$\frac{\sigma}{\mu} = 10.62 \pm 0.14 \%$$

$$\mu = 462.8 \pm 0.8$$

$$\sigma = 49.2 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 11.41 \pm 0.19 \%$$

$$\mu = 464.9 \pm 0.8$$

$$\sigma = 53.1 \pm 0.8$$

800

600

400

200

0

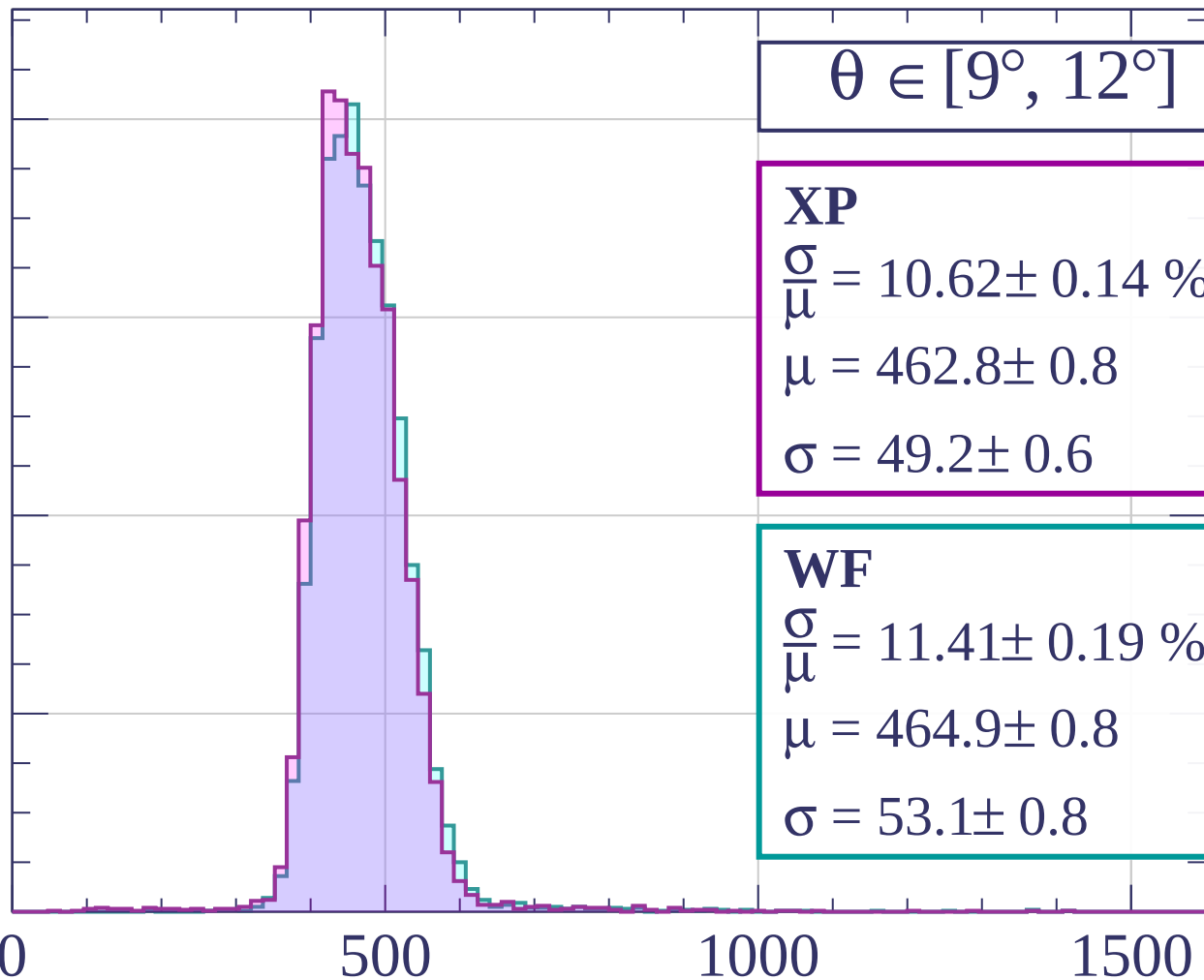
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [12^\circ, 15^\circ]$

XP

$$\frac{\sigma}{\mu} = 9.75 \pm 0.17 \%$$

$$\mu = 451.3 \pm 0.8$$

$$\sigma = 44.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.84 \pm 0.17 \%$$

$$\mu = 454.8 \pm 0.8$$

$$\sigma = 44.7 \pm 0.7$$

600

400

200

0

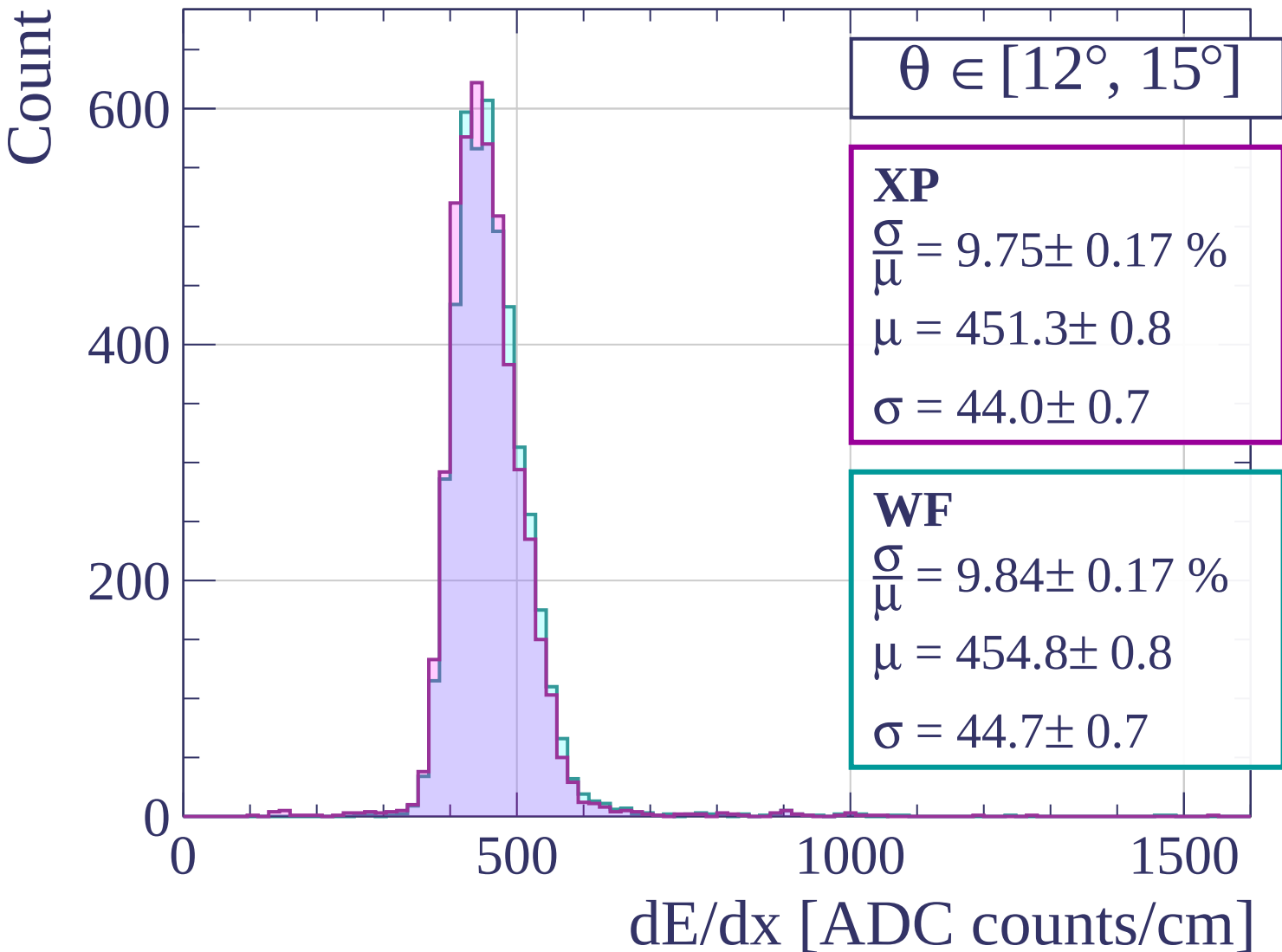
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [15^\circ, 18^\circ]$

XP

$$\frac{\sigma}{\mu} = 9.19 \pm 0.20 \%$$

$$\mu = 442.1 \pm 0.9$$

$$\sigma = 40.6 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.63 \pm 0.25 \%$$

$$\mu = 445.2 \pm 1.0$$

$$\sigma = 42.9 \pm 1.0$$

400

300

200

100

0

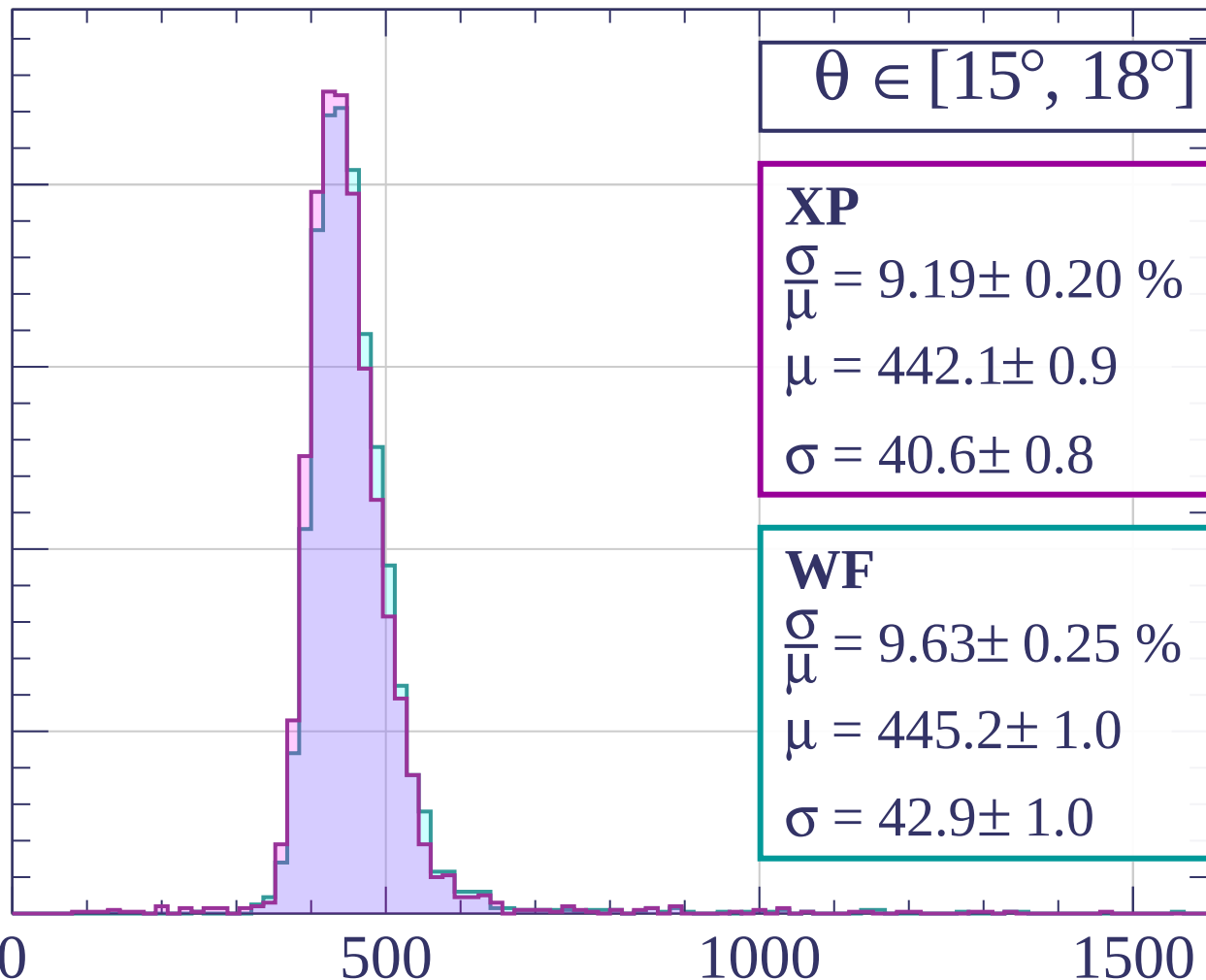
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [18^\circ, 21^\circ]$

XP

$$\frac{\sigma}{\mu} = 8.84 \pm 0.28 \%$$

$$\mu = 434.5 \pm 1.1$$

$$\sigma = 38.4 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 8.89 \pm 0.29 \%$$

$$\mu = 437.8 \pm 1.1$$

$$\sigma = 38.9 \pm 1.2$$

300

200

100

0

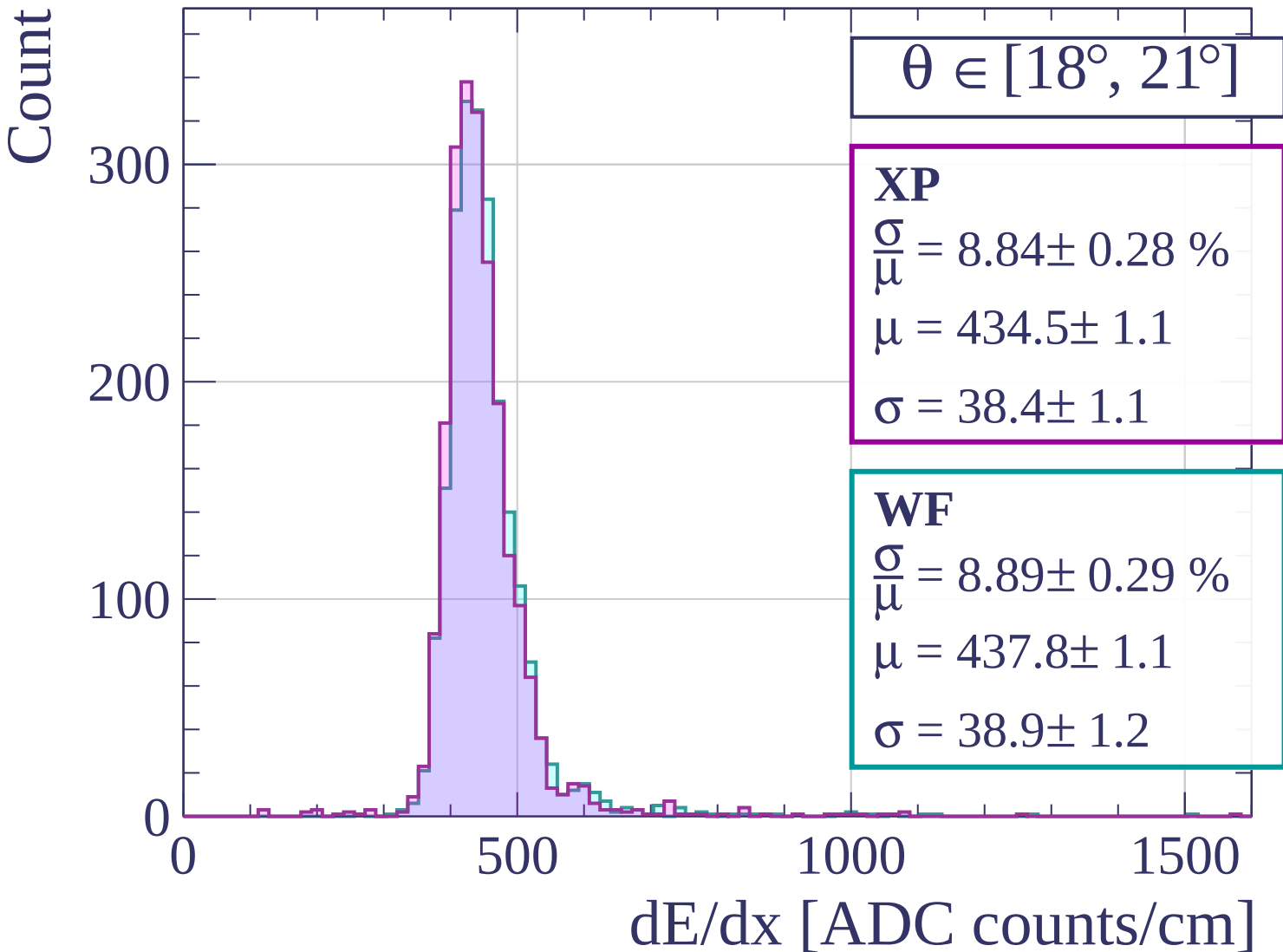
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [21^\circ, 24^\circ]$

XP

$$\frac{\sigma}{\mu} = 8.09 \pm 0.26 \%$$

$$\mu = 427.7 \pm 1.1$$

$$\sigma = 34.6 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 8.22 \pm 0.32 \%$$

$$\mu = 430.3 \pm 1.2$$

$$\sigma = 35.4 \pm 1.3$$

200

100

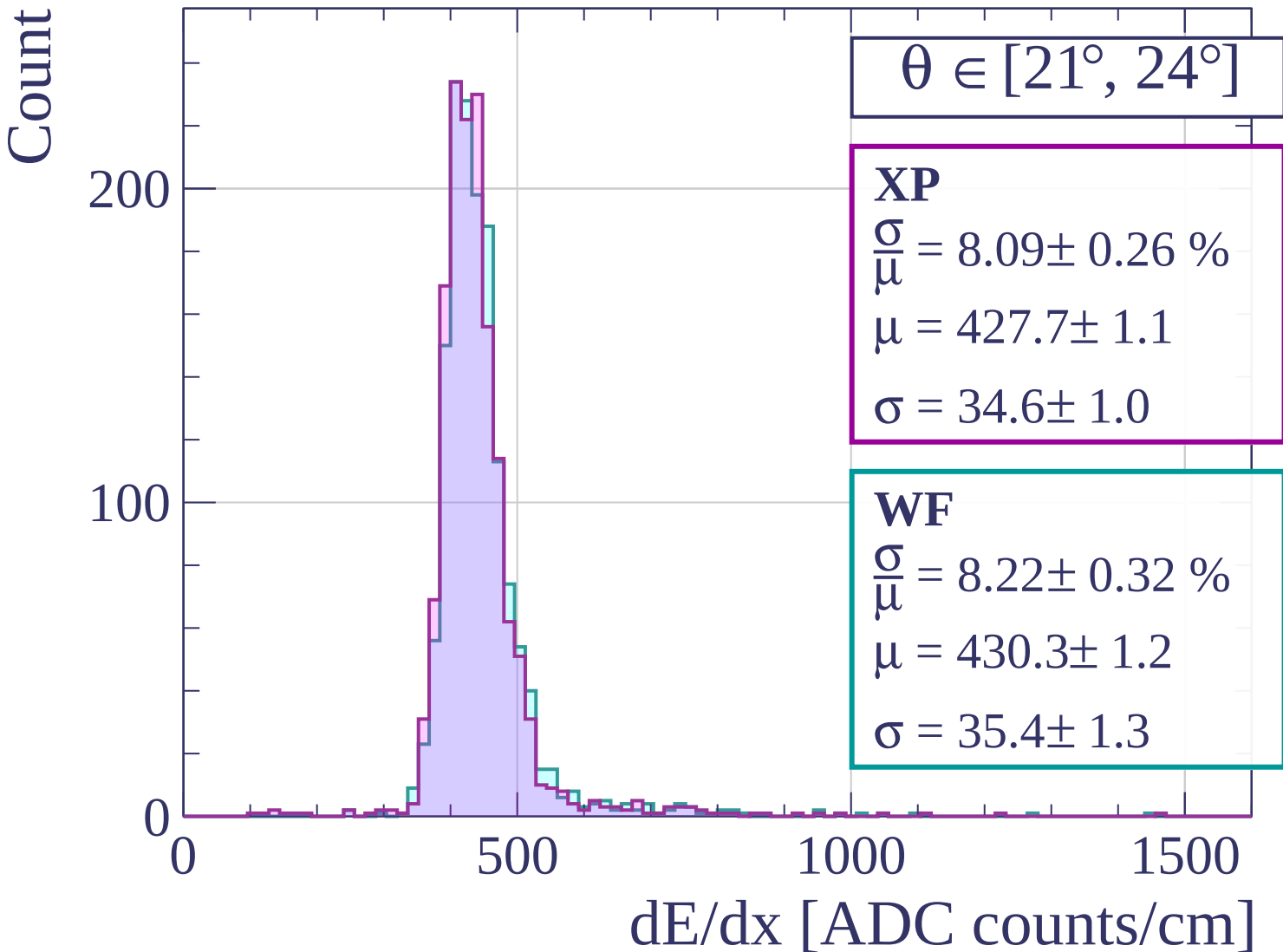
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\theta \in [24^\circ, 27^\circ]$

XP

$$\frac{\sigma}{\mu} = 8.04 \pm 0.32 \%$$

$$\mu = 428.6 \pm 1.4$$

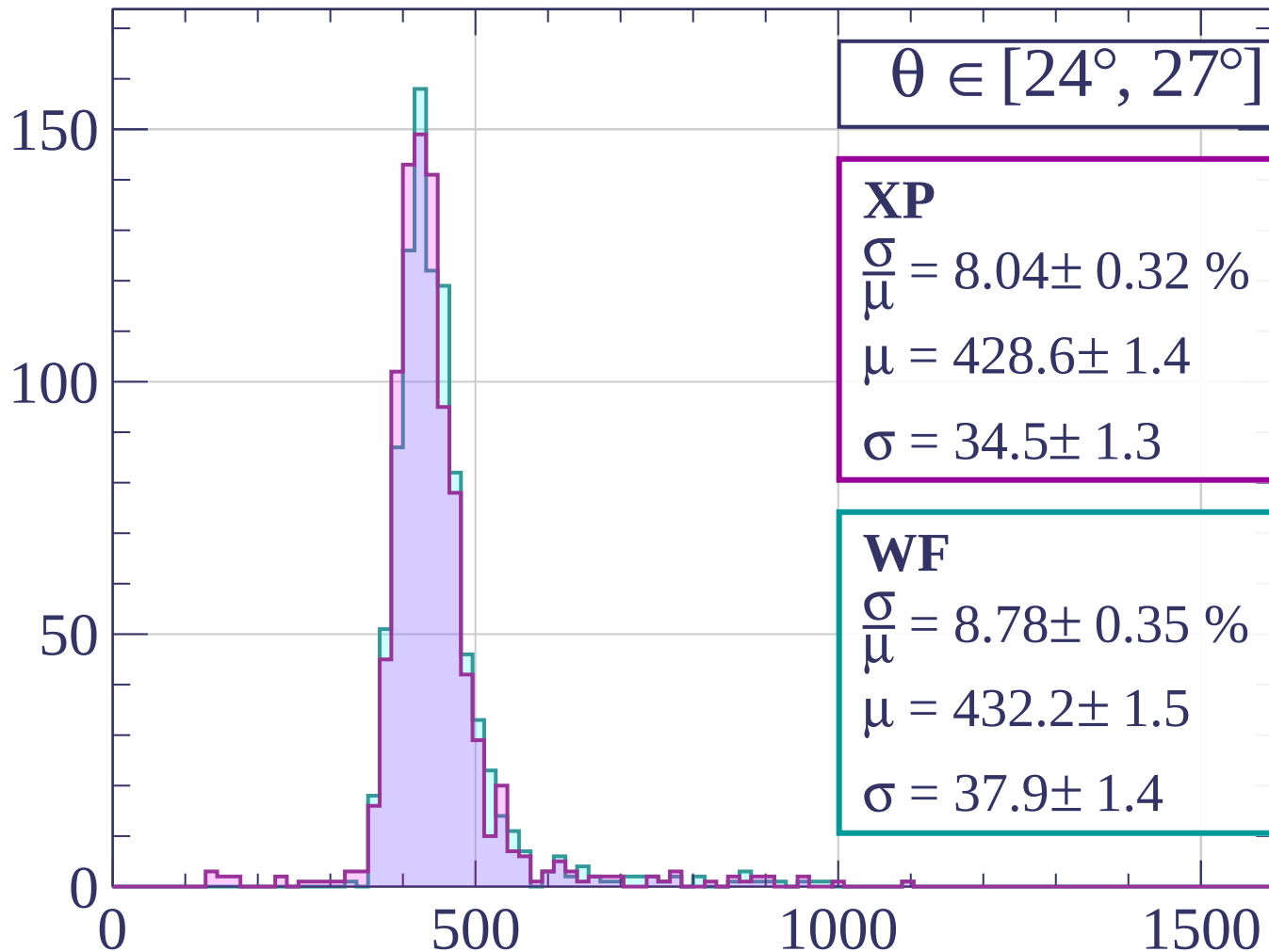
$$\sigma = 34.5 \pm 1.3$$

WF

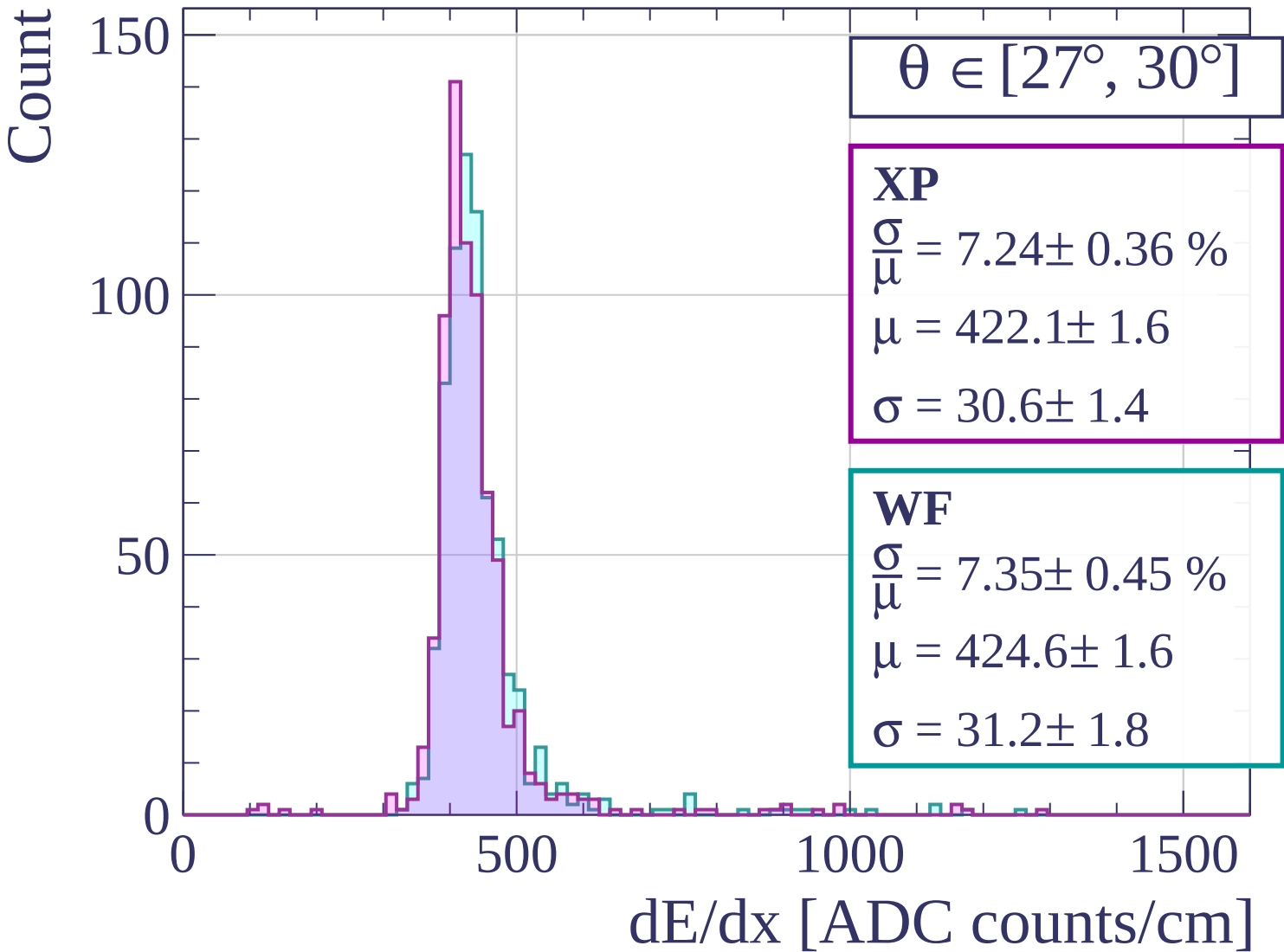
$$\frac{\sigma}{\mu} = 8.78 \pm 0.35 \%$$

$$\mu = 432.2 \pm 1.5$$

$$\sigma = 37.9 \pm 1.4$$



dE/dx [ADC counts/cm]



Count

$\theta \in [30^\circ, 33^\circ]$

XP

$$\frac{\sigma}{\mu} = 7.22 \pm 0.40 \%$$

$$\mu = 421.2 \pm 1.8$$

$$\sigma = 30.4 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 6.85 \pm 0.37 \%$$

$$\mu = 423.8 \pm 1.7$$

$$\sigma = 29.0 \pm 1.5$$

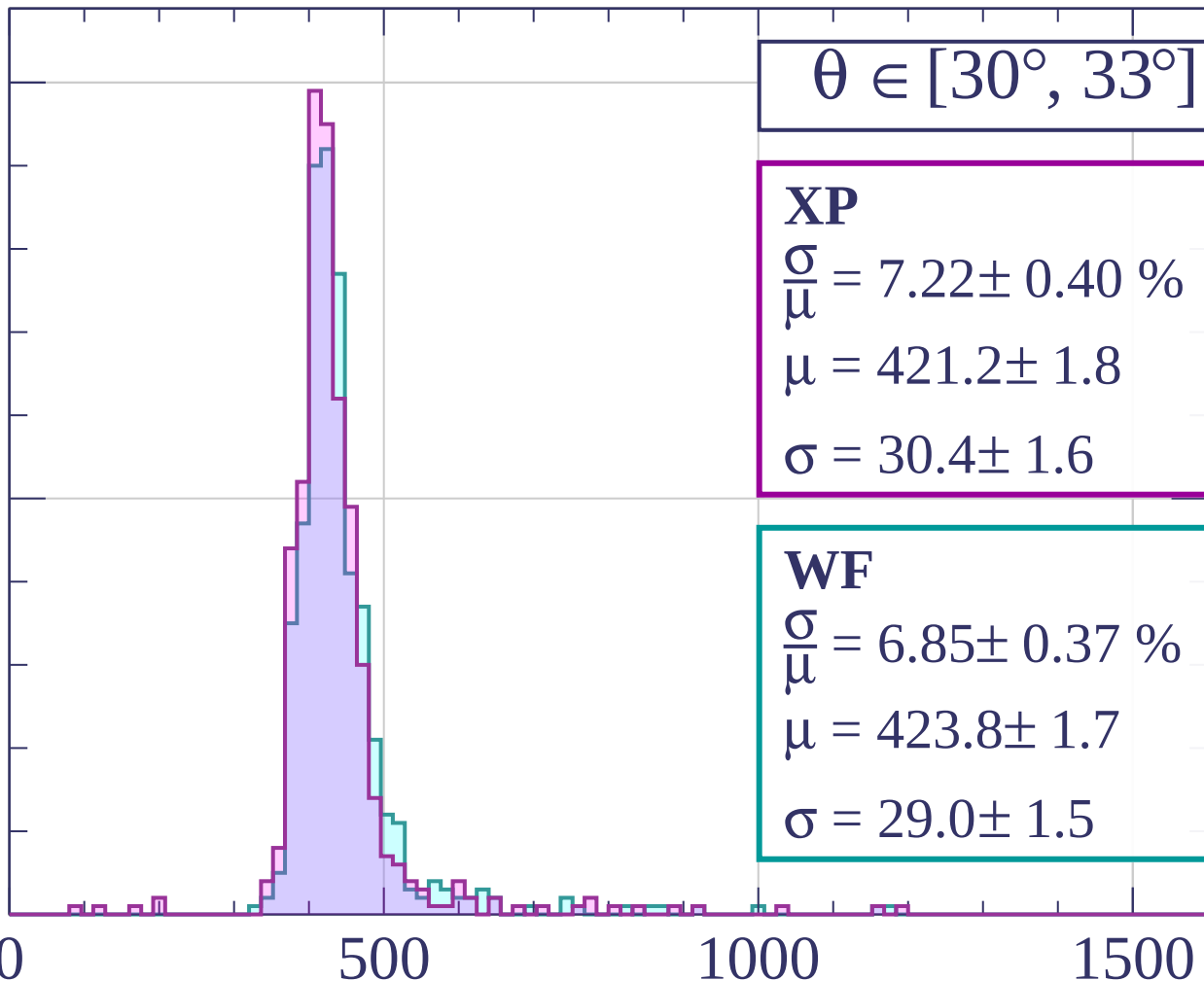
0

500

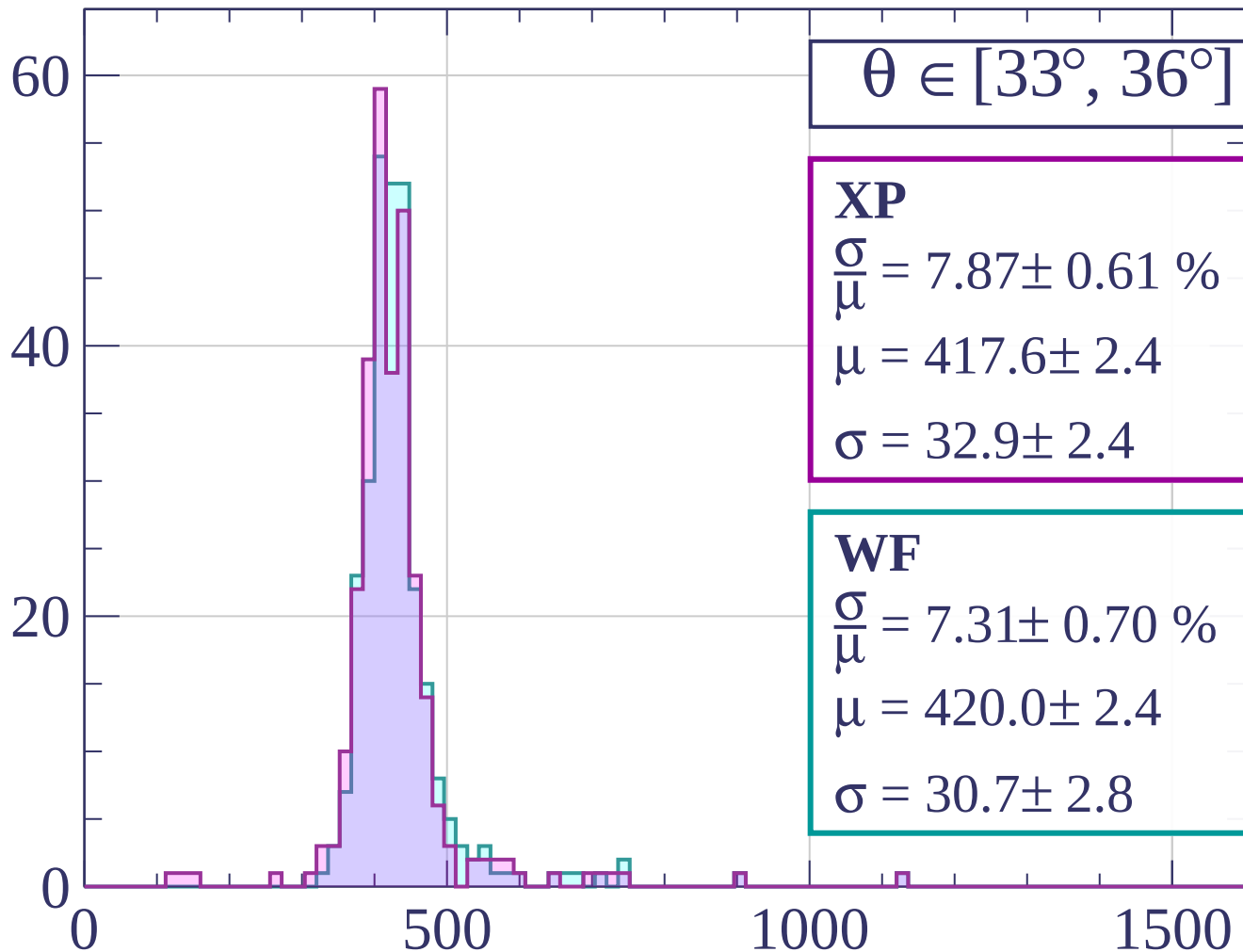
1000

1500

dE/dx [ADC counts/cm]



Count



$\theta \in [33^\circ, 36^\circ]$

XP

$$\frac{\sigma}{\bar{\mu}} = 7.87 \pm 0.61 \%$$

$$\mu = 417.6 \pm 2.4$$

$$\sigma = 32.9 \pm 2.4$$

WF

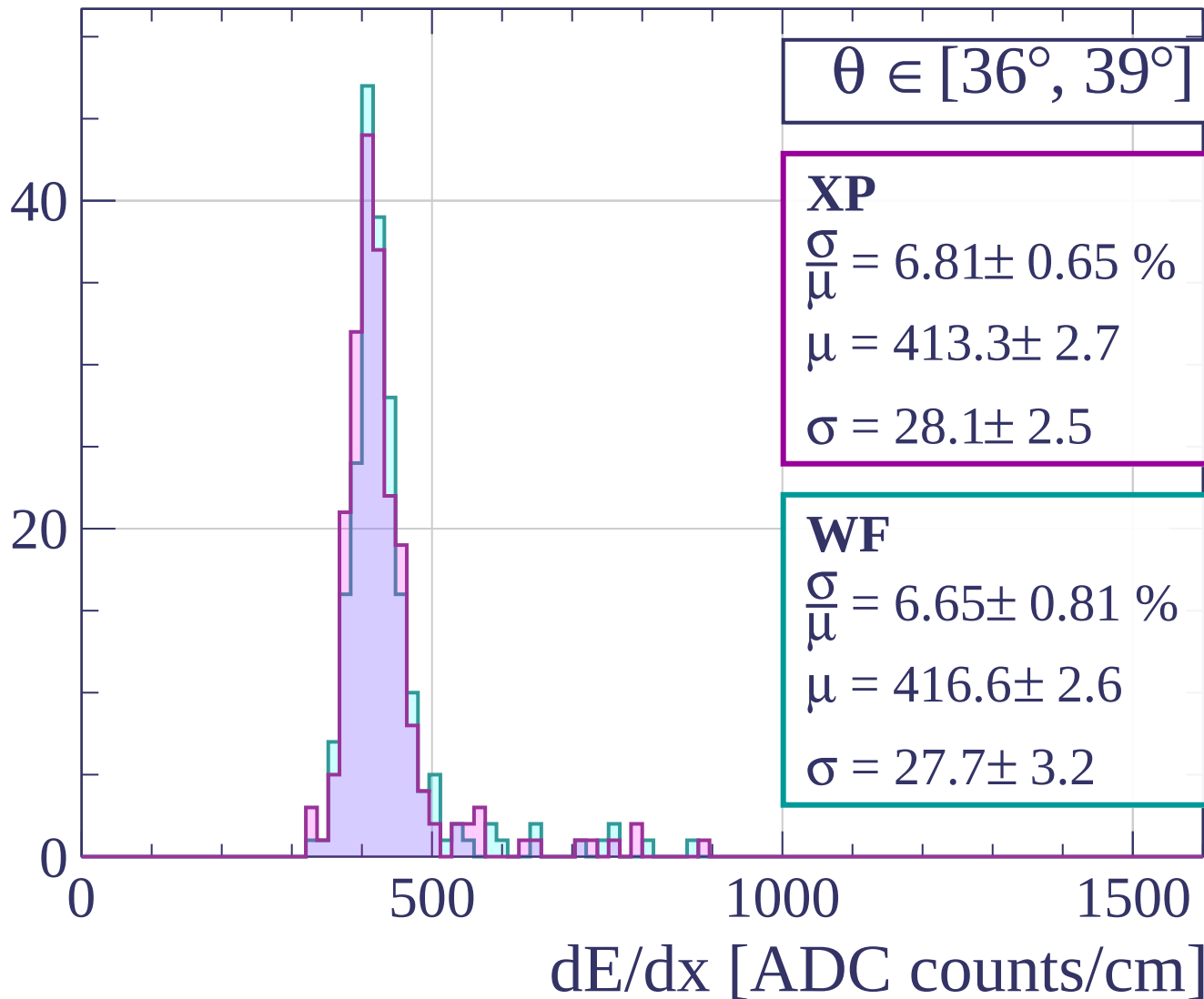
$$\frac{\sigma}{\bar{\mu}} = 7.31 \pm 0.70 \%$$

$$\mu = 420.0 \pm 2.4$$

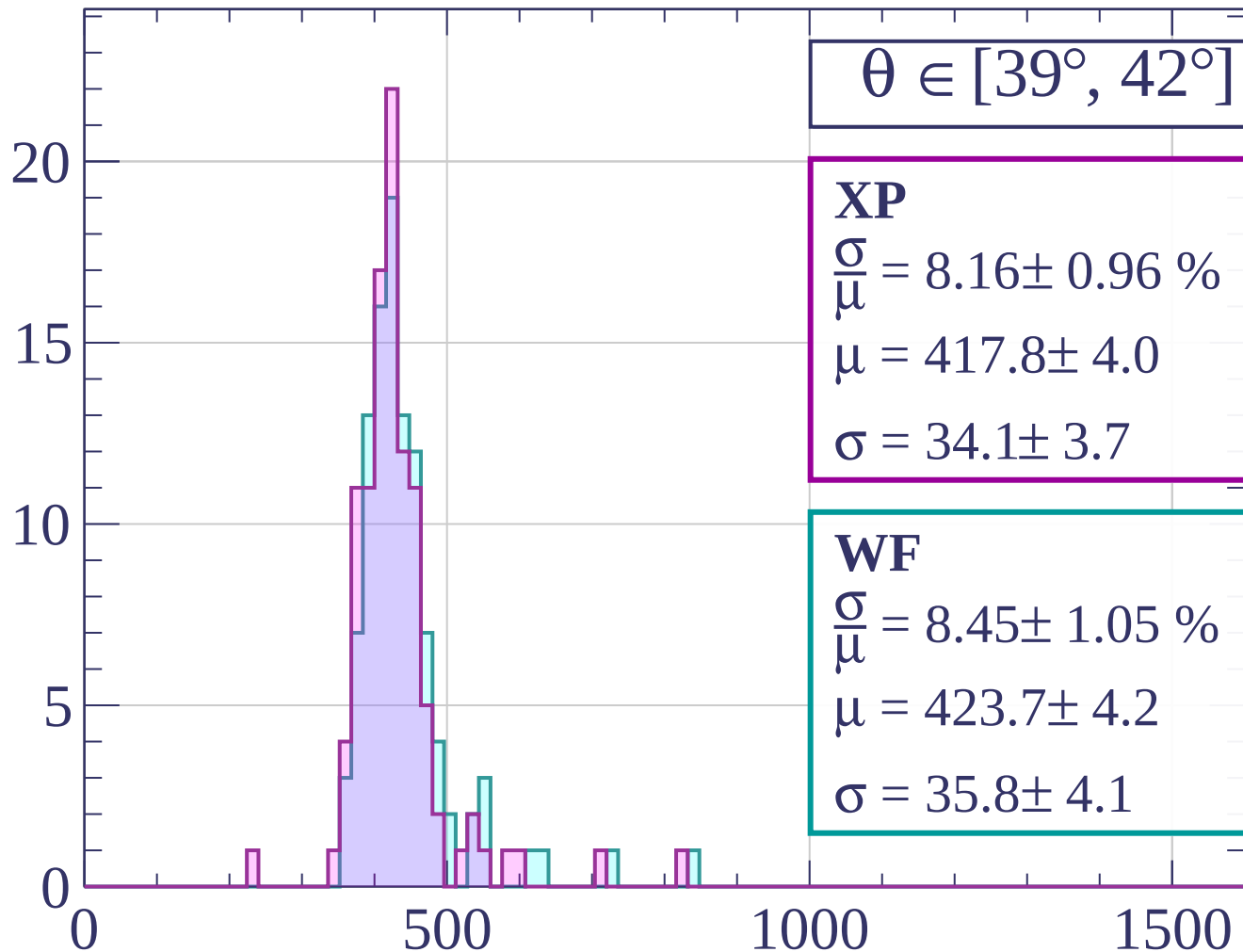
$$\sigma = 30.7 \pm 2.8$$

dE/dx [ADC counts/cm]

Count

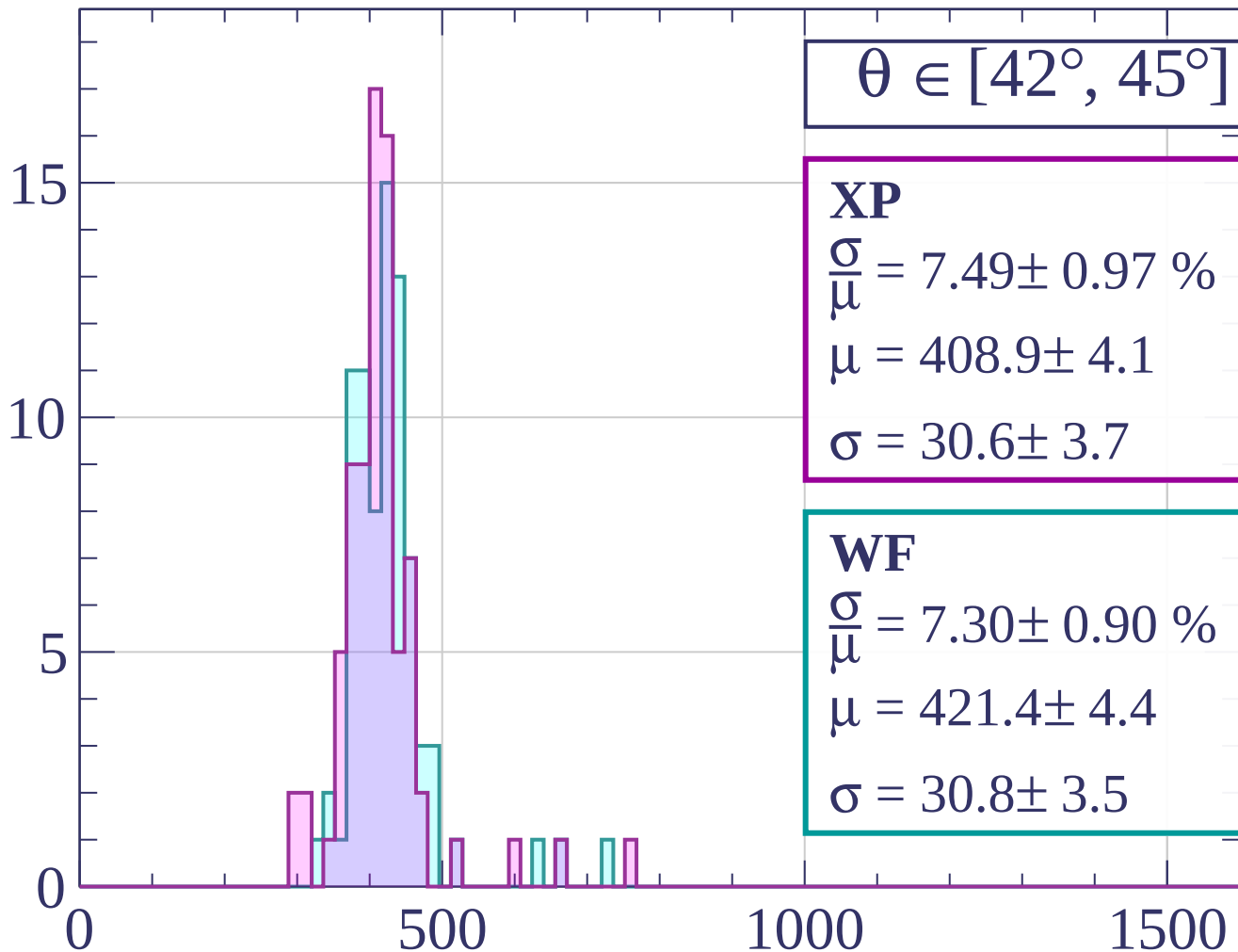


Count

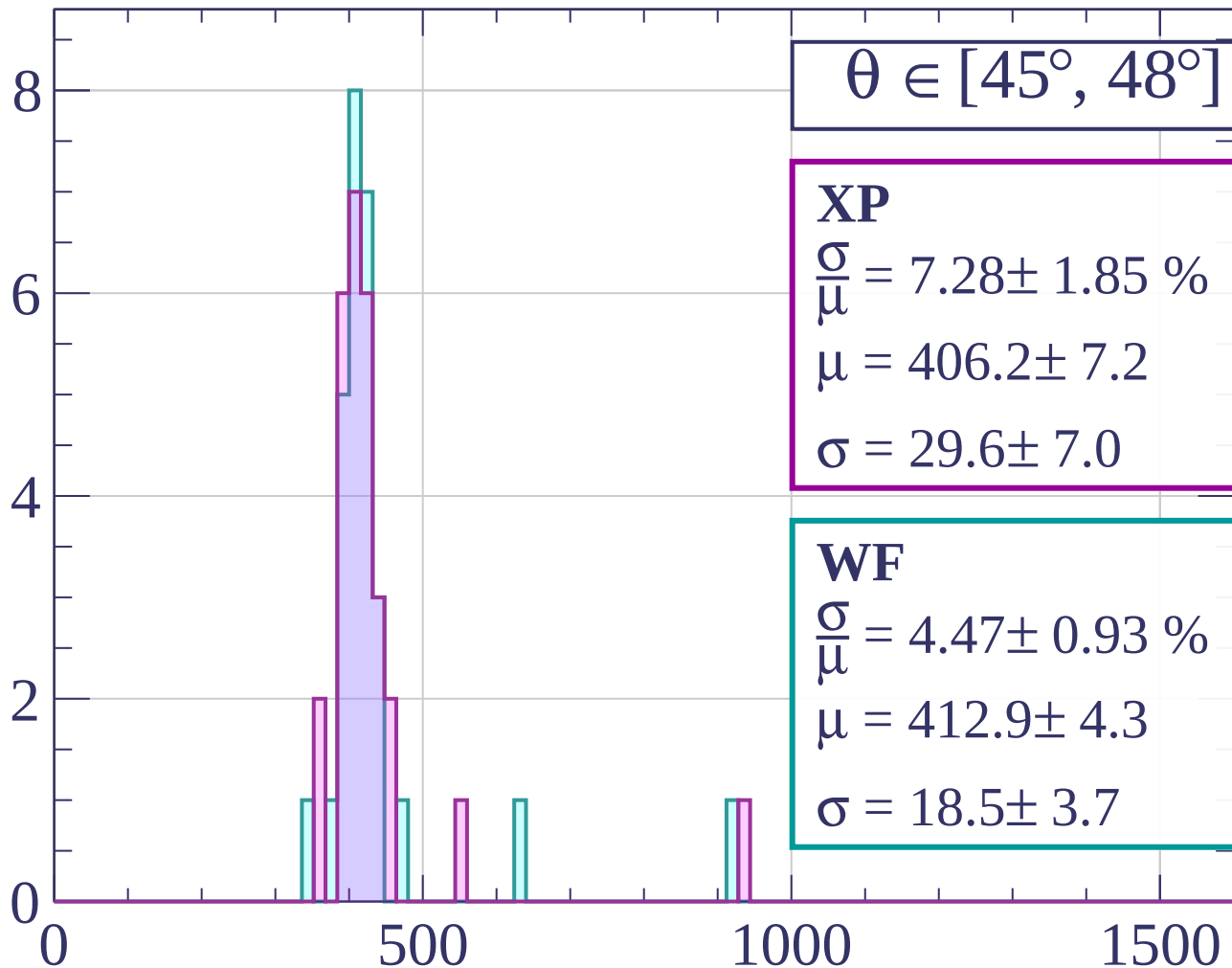


dE/dx [ADC counts/cm]

Count

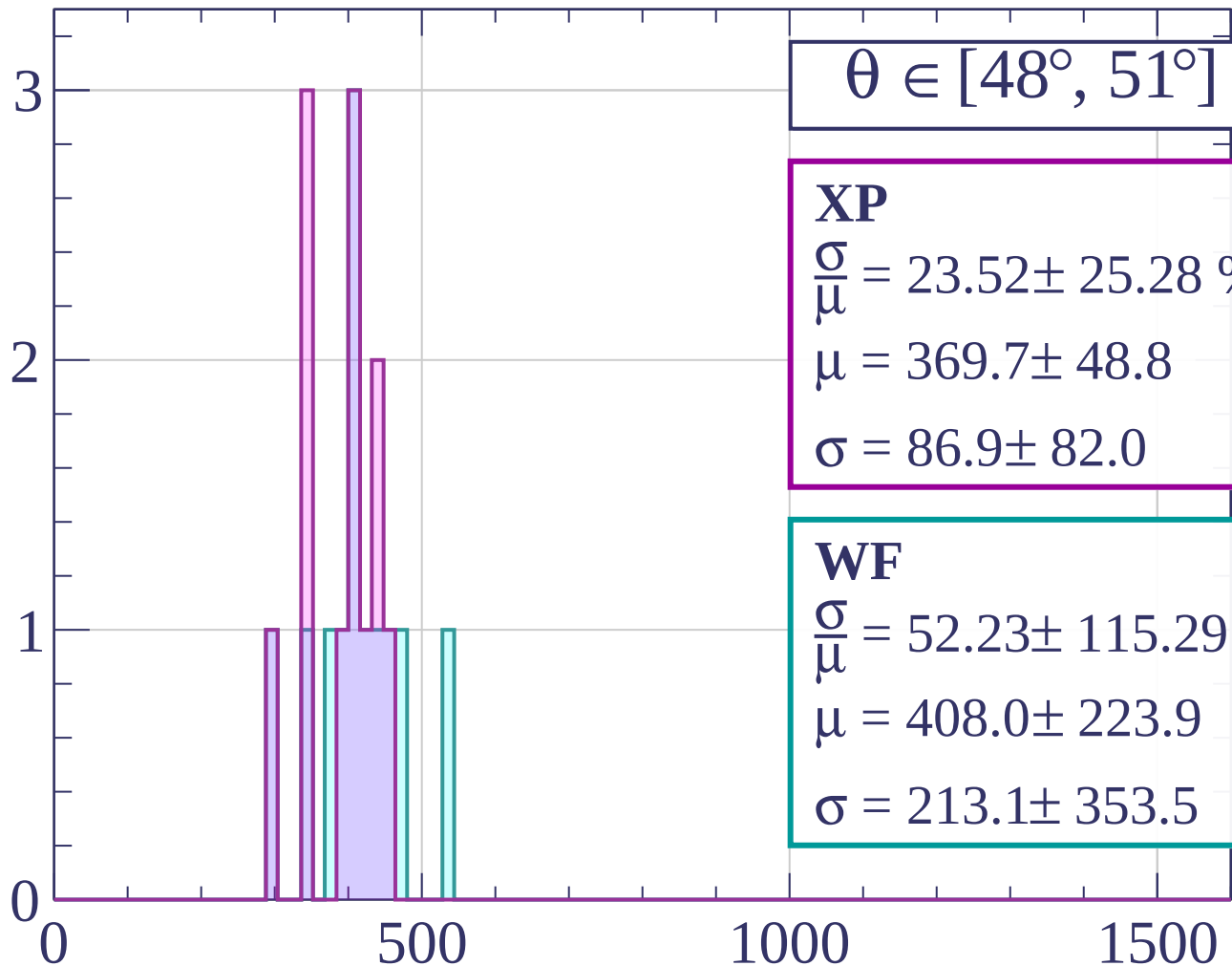


Count



dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count

$\theta \in [51^\circ, 54^\circ]$

XP

$$\frac{\sigma}{\mu} = -126.94 \pm 287.63 \%$$

$$\mu = -137.5 \pm 455.8$$

$$\sigma = 174.5 \pm 183.1$$

WF

$$\frac{\sigma}{\mu} = 27.79 \pm 52.52 \%$$

$$\mu = 995.4 \pm 1209.3$$

$$\sigma = 276.6 \pm 186.8$$

2.0
1.5
1.0
0.5
0.0

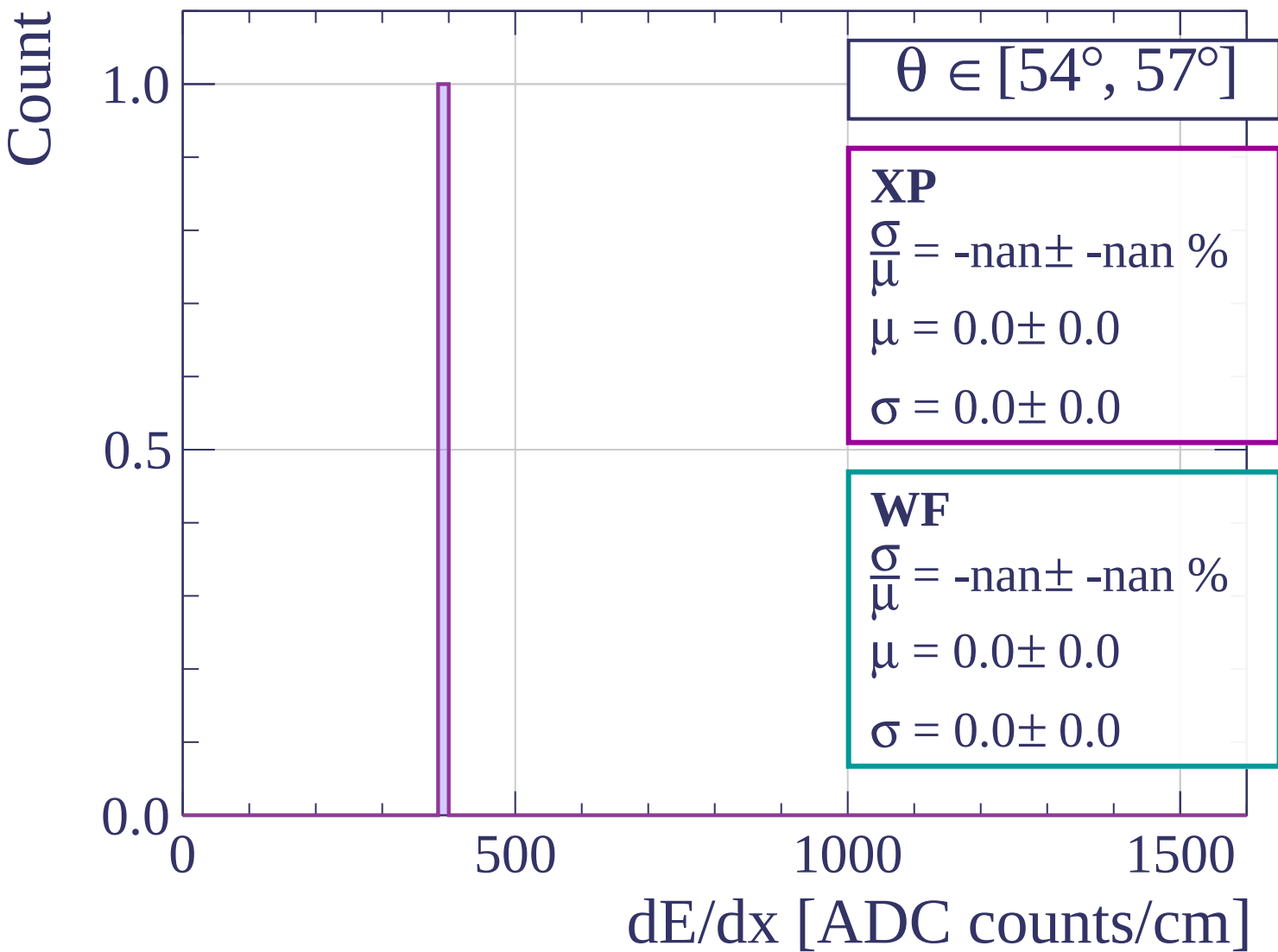
0

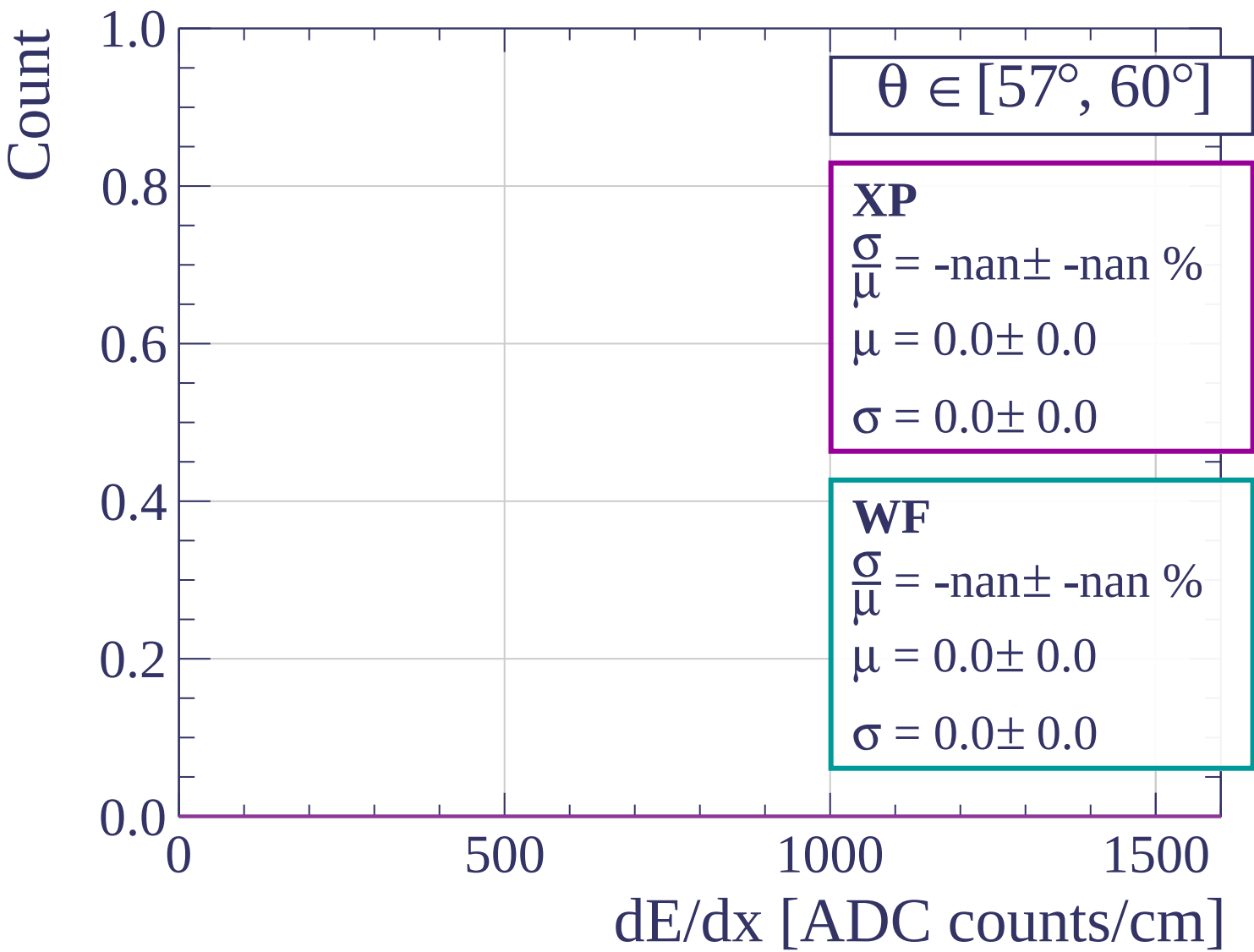
500

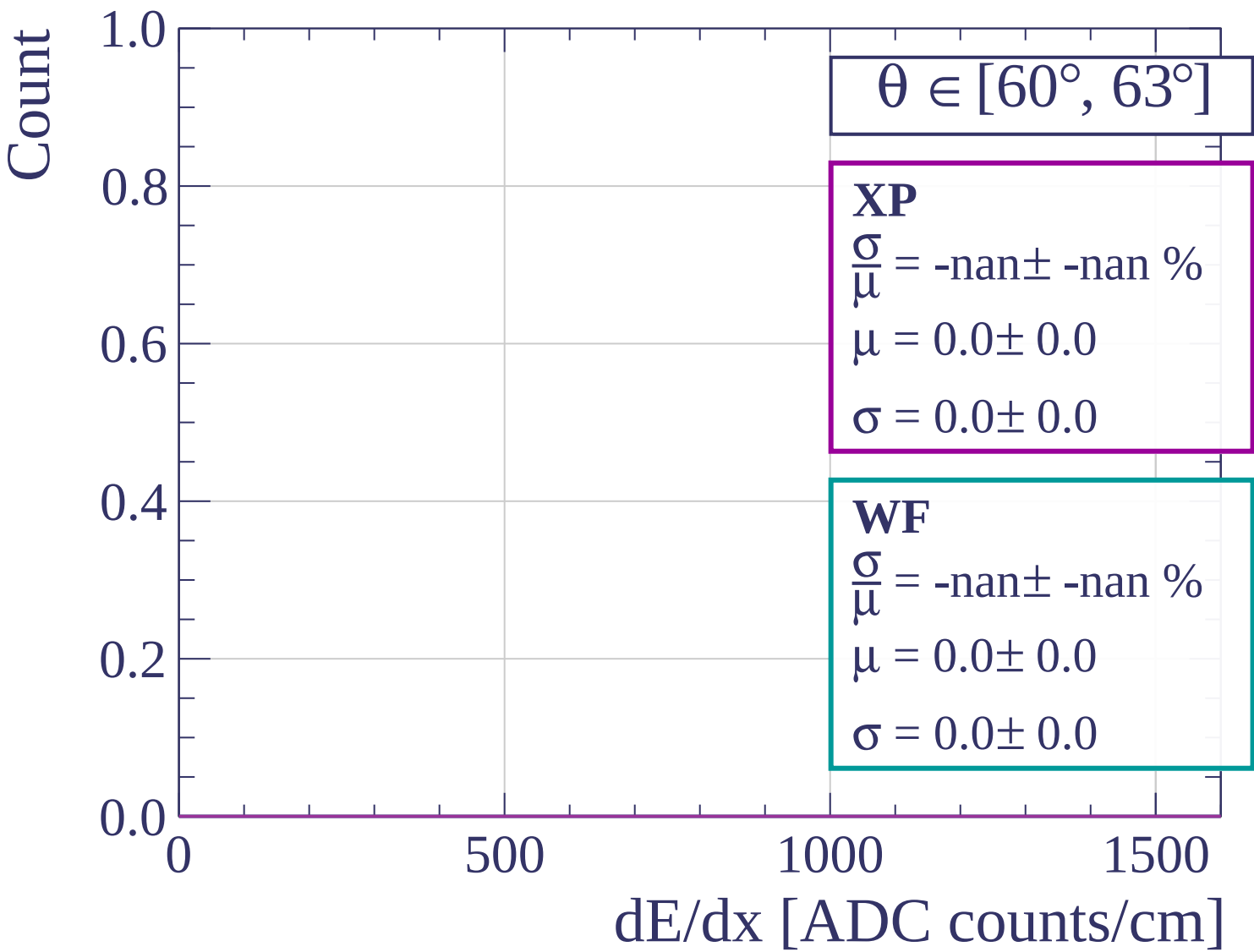
1000

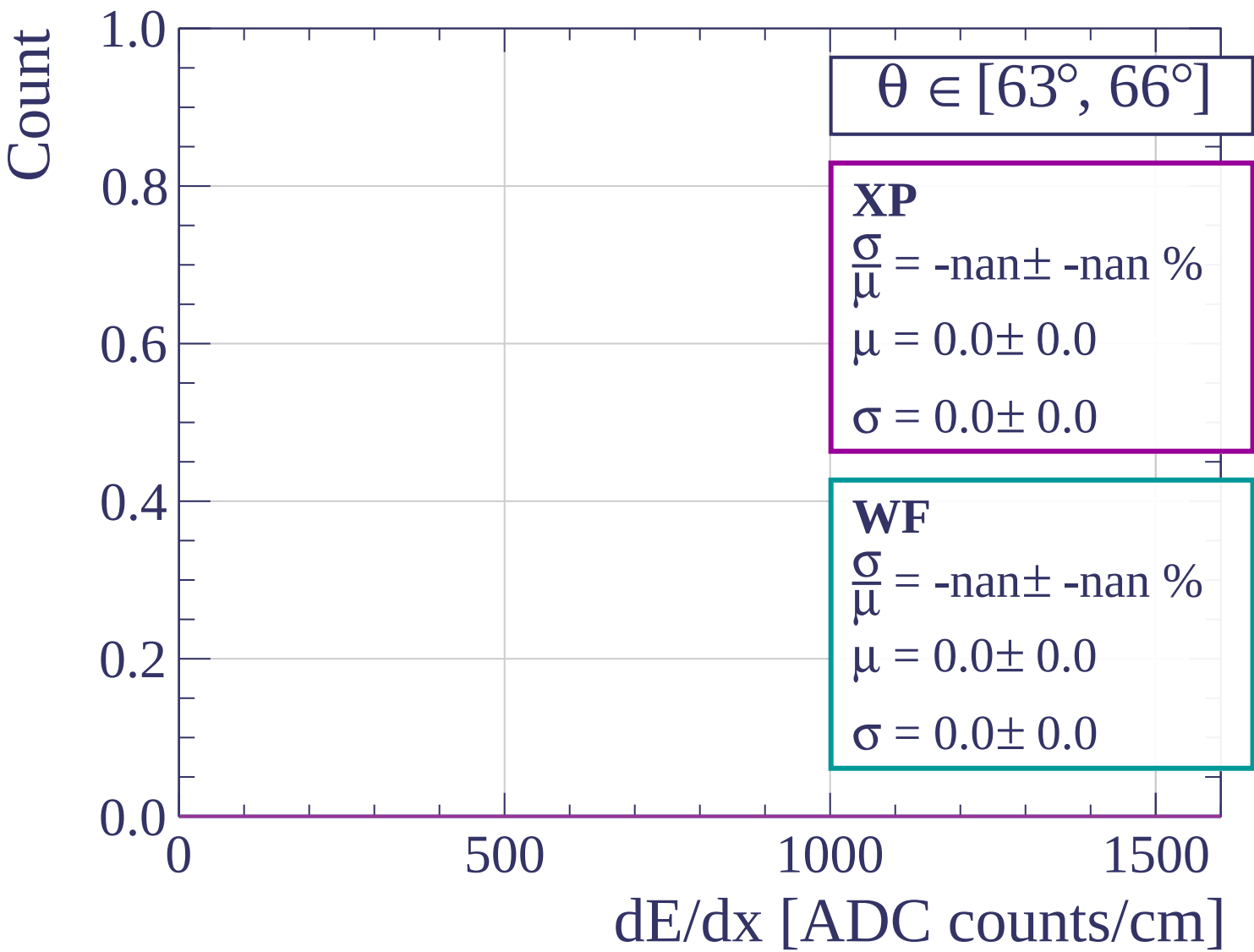
1500

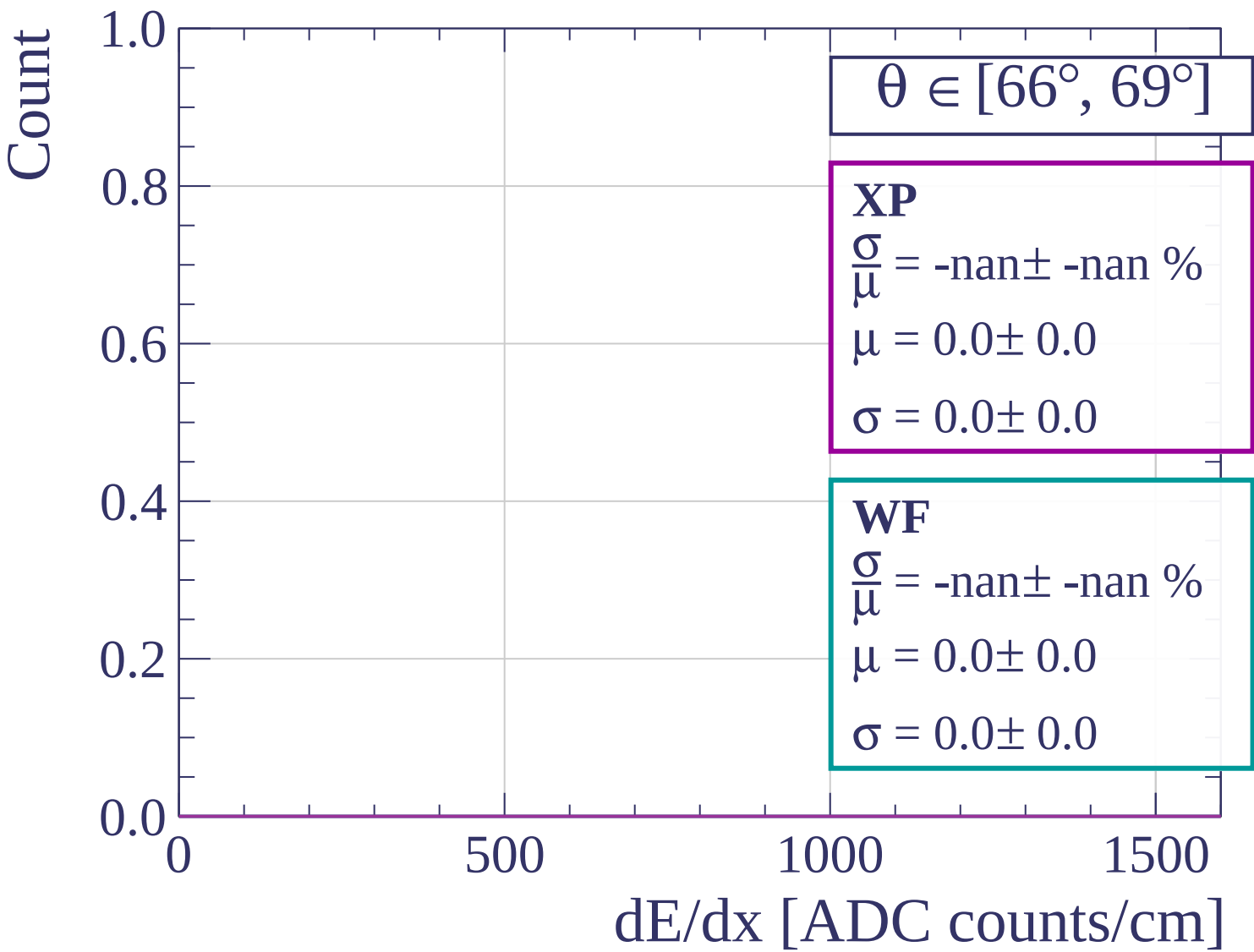
dE/dx [ADC counts/cm]

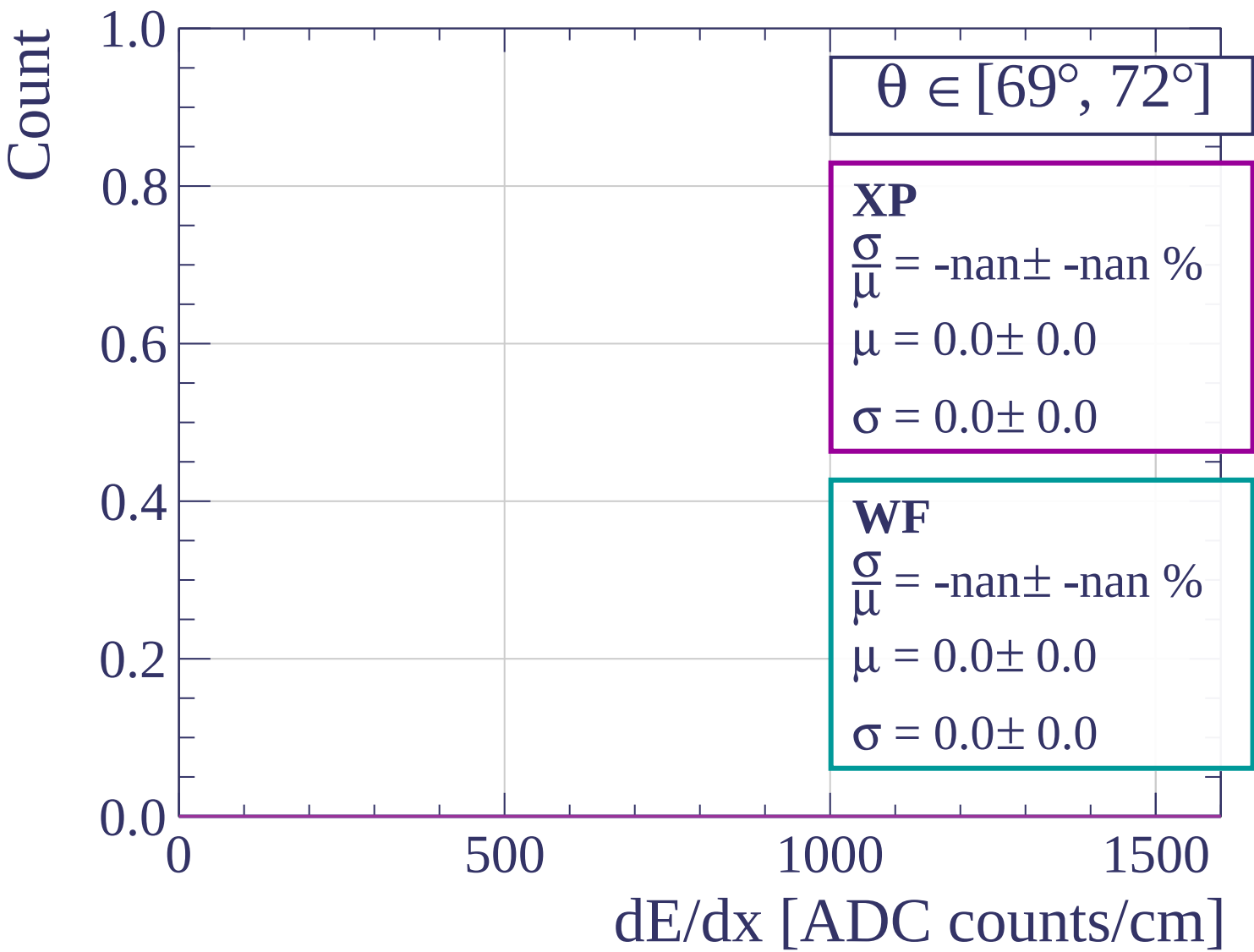


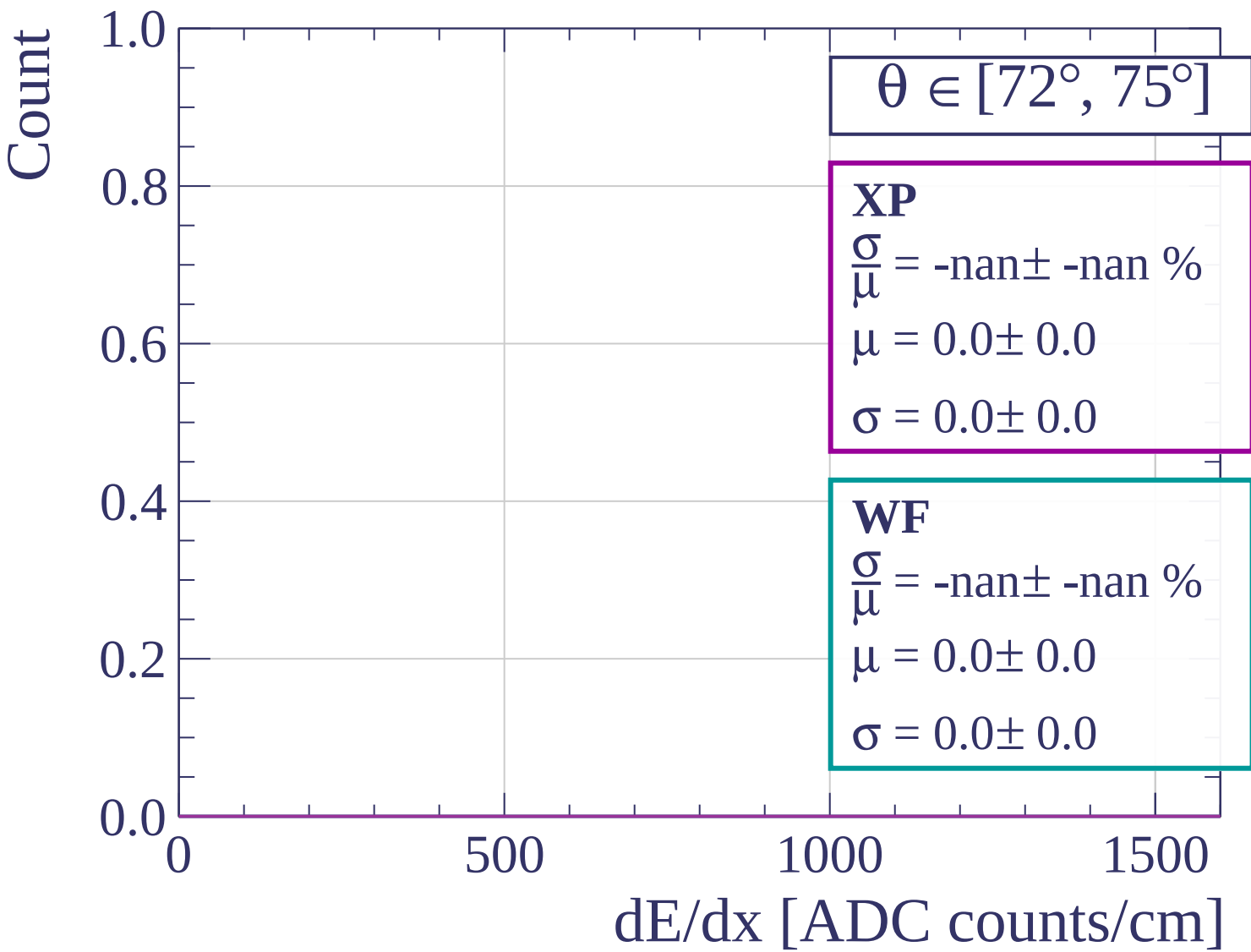


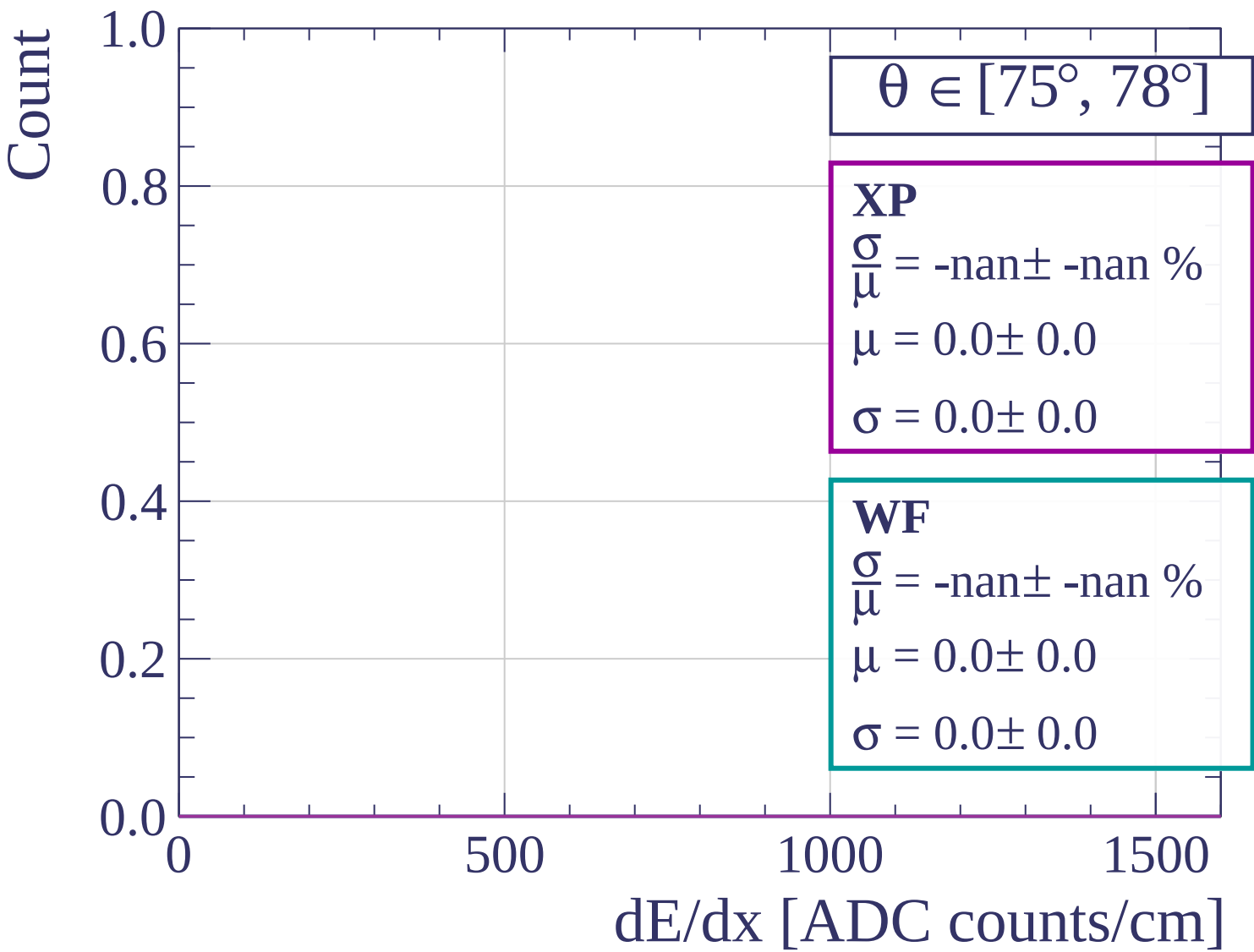


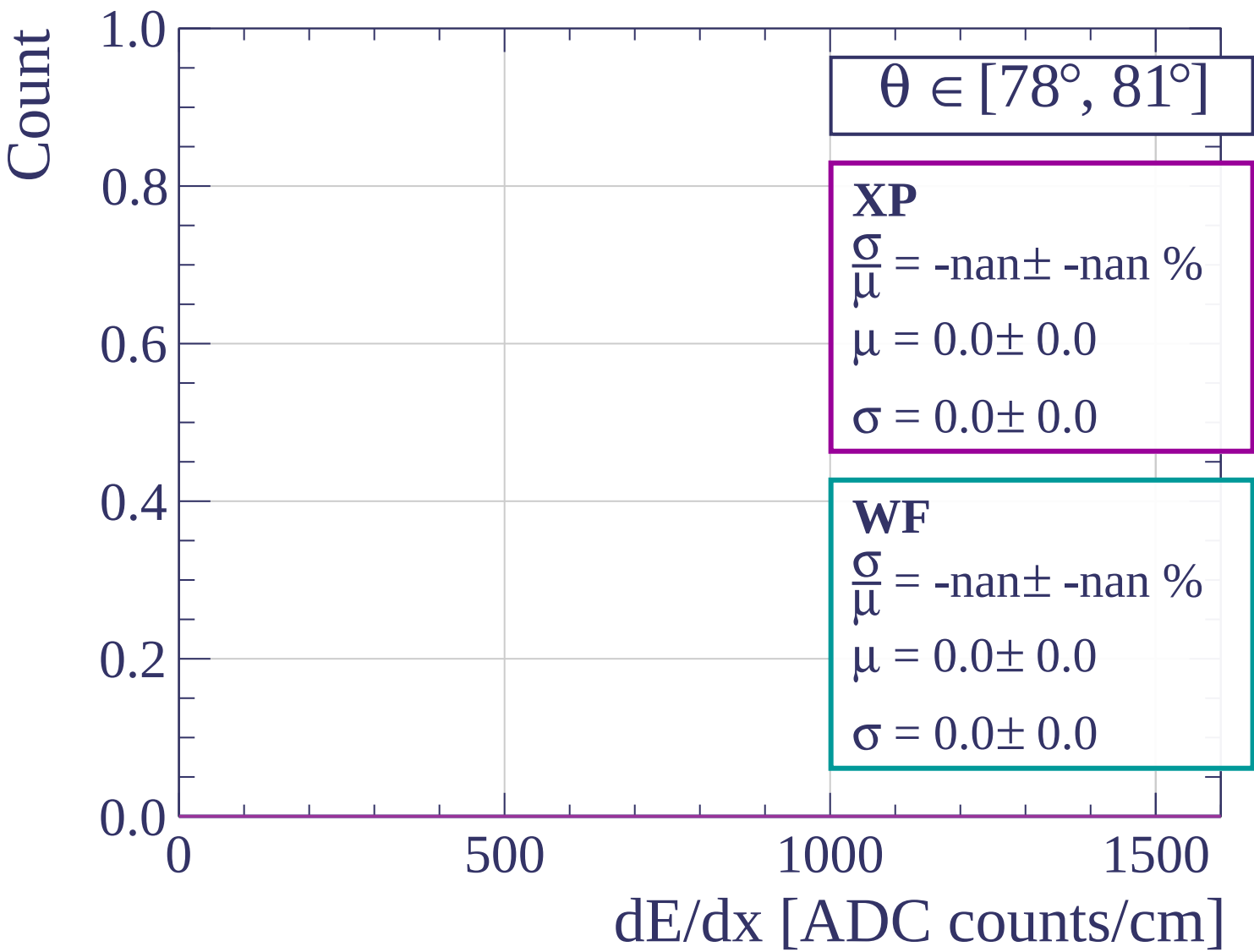


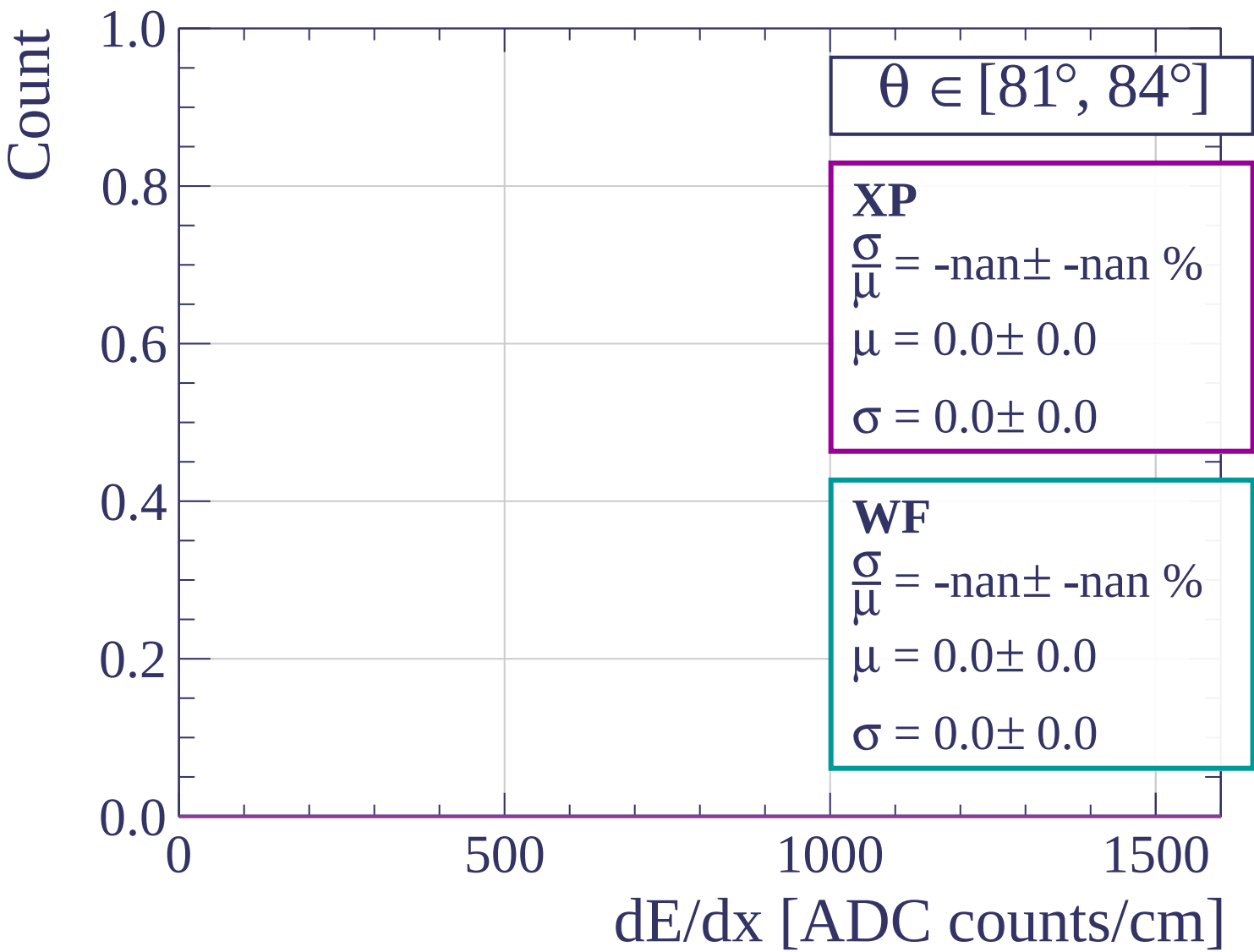


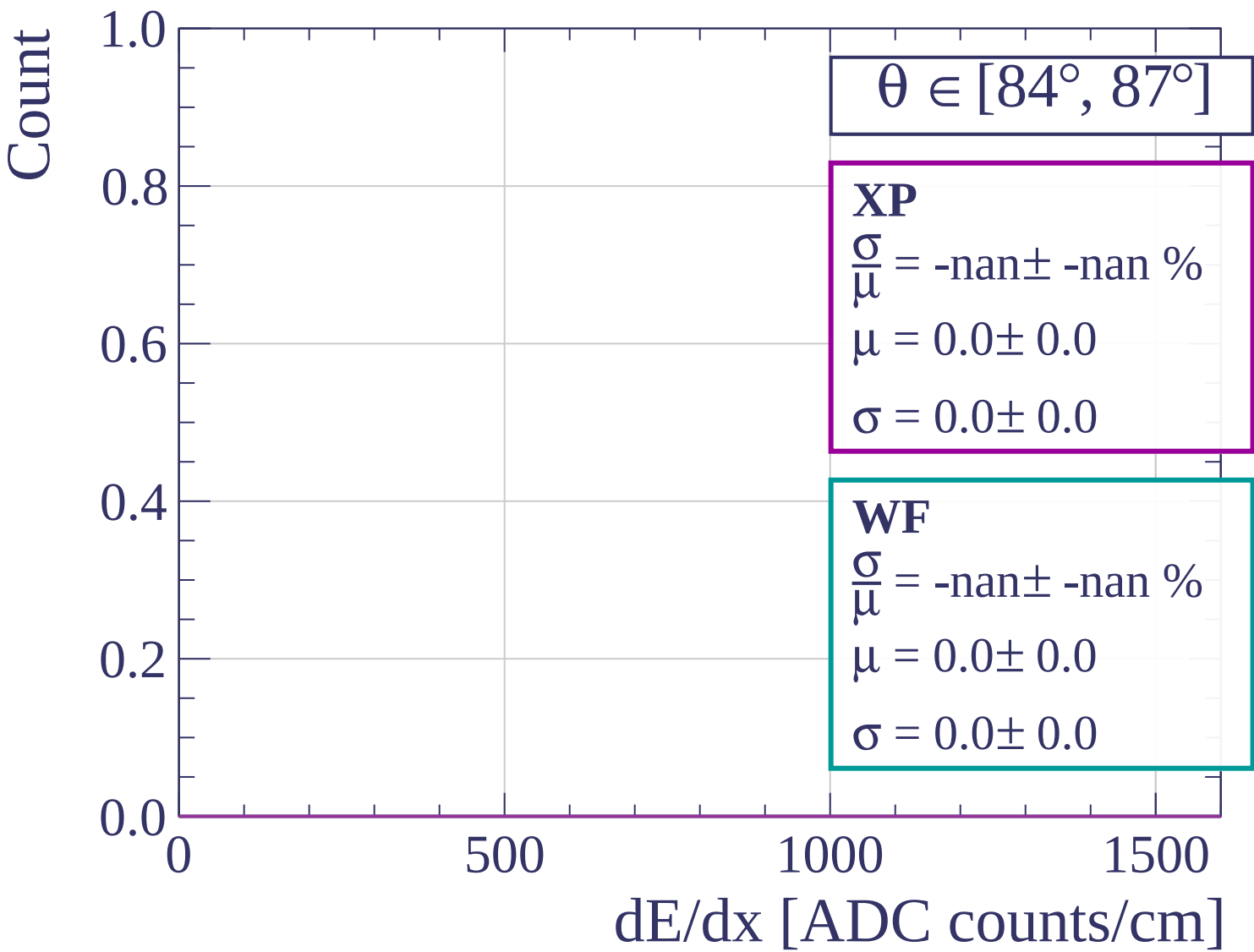


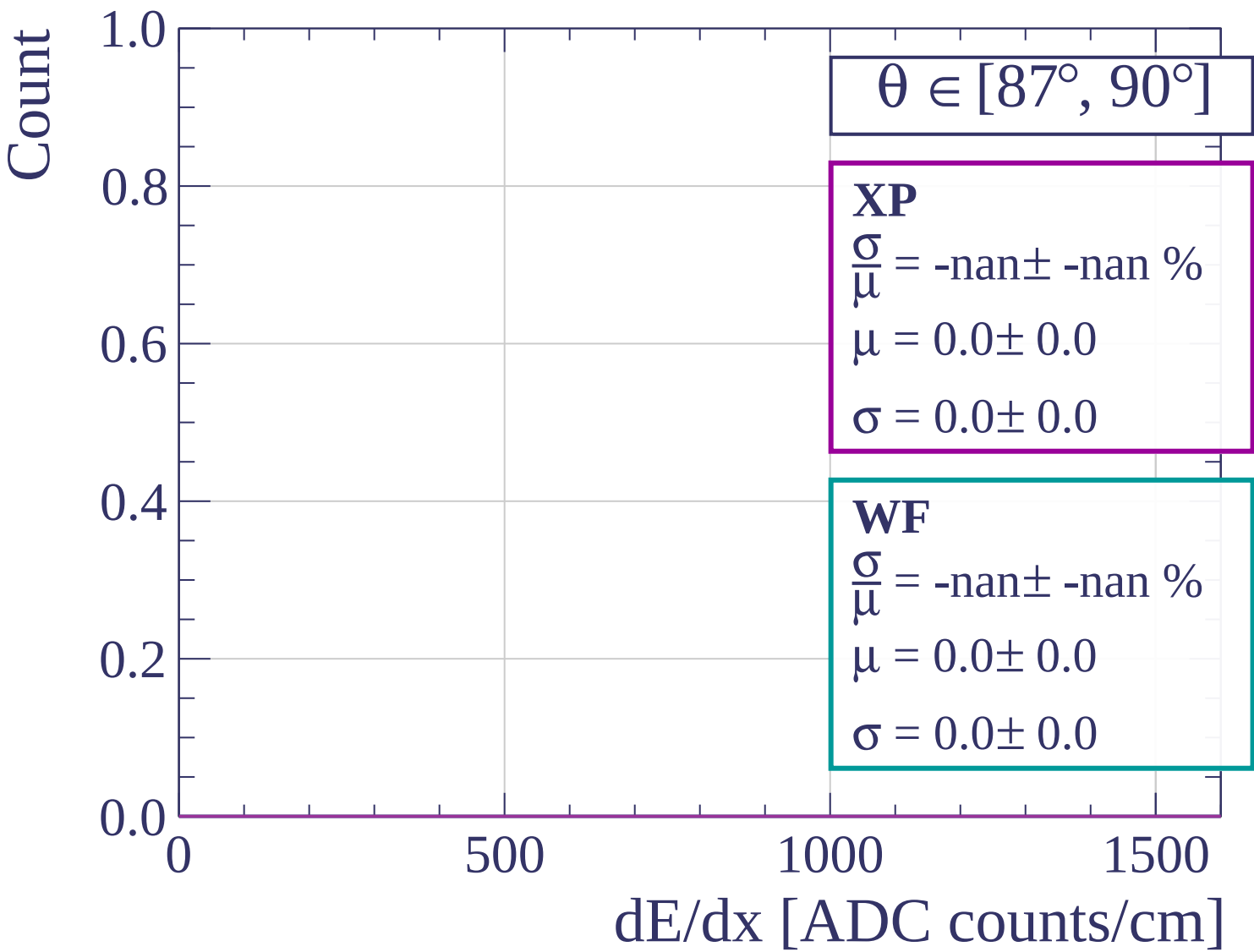




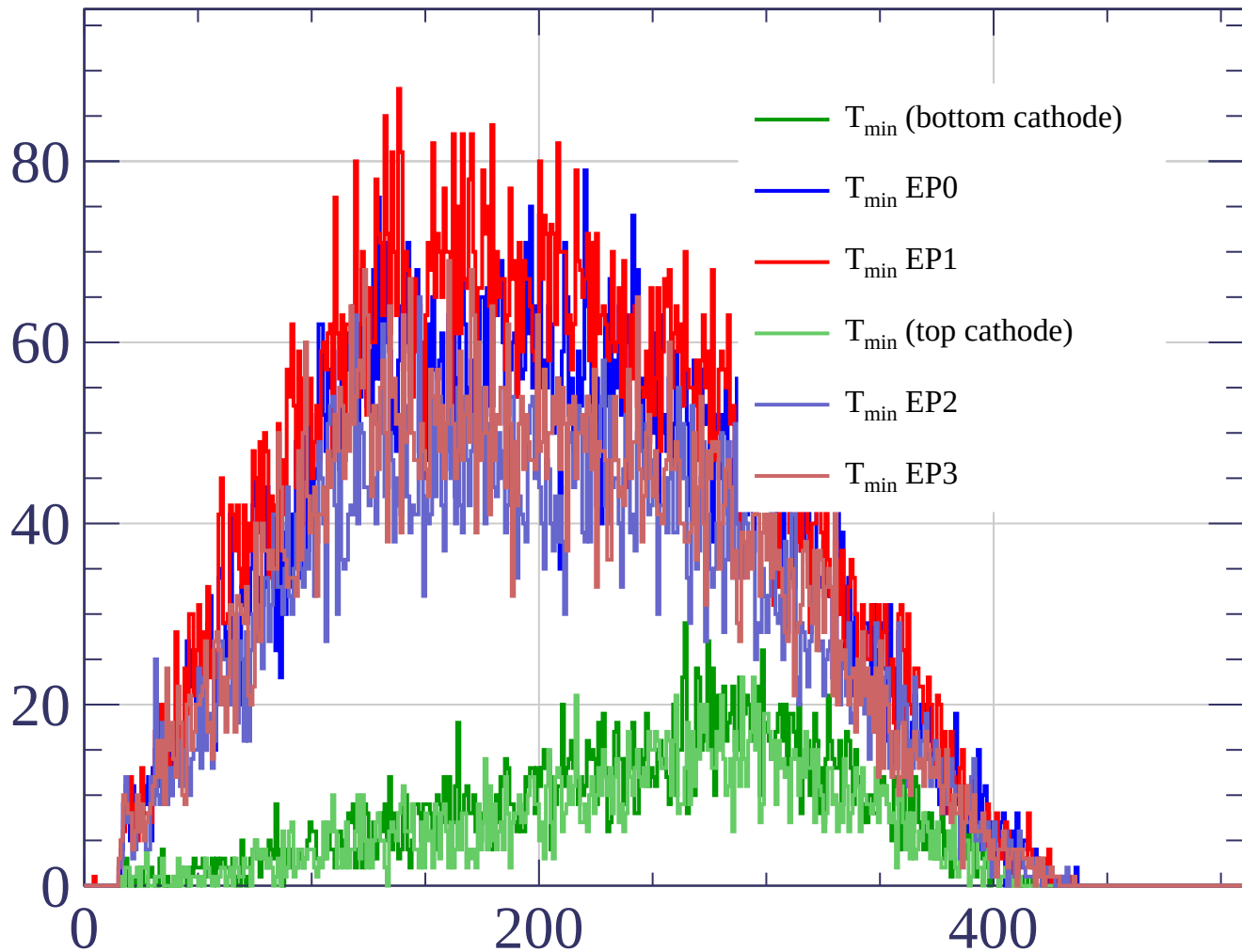








Count



Count

